GROWTH CURVE MODELING

GROWTH CURVE MODELING

Theory and Applications

MICHAEL J. PANIK
Department of Economics
University of Hartford
West Hartford, Connecticut

WILEY

Published by John Wiley & Sons, Inc., Hoboken, New Jersey.
Published simultaneously in Canada.

Library of Congress Cataloging-in-Publication Data:

Panik, Michael J.
Growth curve modeling : theory and applications / Michael J. Panik.
pages cm
Includes bibliographical references and index.
ISBN 978-1-118-76404-6 (cloth)
1. Mathematical statistics. 2. Time series analysis. 3. Regression analysis.
4. Multivariate analysis. I. Title.
QA276.P2243 2013
519.5–dc23

2013026535

Printed in the United States of America

10 9 8 7 6 5 4 3 2 1

In memory of Michael Christopher Duffy

CONTENTS

Preface **xiii**

1 Mathematical Preliminaries **1**

1.1 Arithmetic Progression, 1
1.2 Geometric Progression, 2
1.3 The Binomial Formula, 4
1.4 The Calculus of Finite Differences, 5
1.5 The Number *e*, 9
1.6 The Natural Logarithm, 10
1.7 The Exponential Function, 11
1.8 Exponential and Logarithmic Functions: Another Look, 13
1.9 Change of Base of a Logarithm, 14
1.10 The Arithmetic (Natural) Scale versus the Logarithmic Scale, 15
1.11 Compound Interest Arithmetic, 17

2 Fundamentals of Growth **21**

2.1 Time Series Data, 21
2.2 Relative and Average Rates of Change, 21
2.3 Annual Rates of Change, 25
2.3.1 Simple Rates of Change, 25
2.3.2 Compounded Rates of Change, 26
2.3.3 Comparing Two Time Series: Indexing Data to a Common Starting Point, 30
2.4 Discrete versus Continuous Growth, 32

2.5 The Growth of a Variable Expressed in Terms of the Growth of its Individual Arguments, 36
2.6 Growth Rate Variability, 46
2.7 Growth in a Mixture of Variables, 47

3 Parametric Growth Curve Modeling 49

3.1 Introduction, 49
3.2 The Linear Growth Model, 50
3.3 The Logarithmic Reciprocal Model, 51
3.4 The Logistic Model, 52
3.5 The Gompertz Model, 54
3.6 The Weibull Model, 55
3.7 The Negative Exponential Model, 56
3.8 The von Bertalanffy Model, 57
3.9 The Log-Logistic Model, 59
3.10 The Brody Growth Model, 61
3.11 The Janoschek Growth Model, 62
3.12 The Lundqvist–Korf Growth Model, 63
3.13 The Hossfeld Growth Model, 63
3.14 The Stannard Growth Model, 64
3.15 The Schnute Growth Model, 64
3.16 The Morgan–Mercer–Flodin (M–M–F) Growth Model, 66
3.17 The McDill–Amateis Growth Model, 68
3.18 An Assortment of Additional Growth Models, 69
3.18.1 The Sloboda Growth Model, 71
Appendix 3.A The Logistic Model Derived, 71
Appendix 3.B The Gompertz Model Derived, 74
Appendix 3.C The Negative Exponential Model Derived, 75
Appendix 3.D The von Bertalanffy and Richards Models Derived, 77
Appendix 3.E The Schnute Model Derived, 81
Appendix 3.F The McDill–Amateis Model Derived, 83
Appendix 3.G The Sloboda Model Derived, 85
Appendix 3.H A Generalized Michaelis–Menten Growth Equation, 86

4 Estimation of Trend 88

4.1 Linear Trend Equation, 88
4.2 Ordinary Least Squares (OLS) Estimation, 91
4.3 Maximum Likelihood (ML) Estimation, 92
4.4 The SAS System, 94
4.5 Changing the Unit of Time, 109
4.5.1 Annual Totals versus Monthly Averages versus Monthly Totals, 109
4.5.2 Annual Totals versus Quarterly Averages versus Quarterly Totals, 110

4.6 Autocorrelated Errors, 110
4.6.1 Properties of the OLS Estimators when ε Is AR(1), 111
4.6.2 Testing for the Absence of Autocorrelation: The Durbin–Watson Test, 113
4.6.3 Detection of and Estimation with Autocorrelated Errors, 115
4.7 Polynomial Models in t, 126
4.8 Issues Involving Trended Data, 136
4.8.1 Stochastic Processes and Time Series, 137
4.8.2 Autoregressive Process of Order p, 138
4.8.3 Random Walk Processes, 141
4.8.4 Integrated Processes, 145
4.8.5 Testing for Unit Roots, 146
Appendix 4.A OLS Estimated and Related Growth Rates, 158
4.A.1 The OLS Growth Rate, 158
4.A.2 The Log-Difference (LD) Growth Rate, 161
4.A.3 The Average Annual Growth Rate, 161
4.A.4 The Geometric Average Growth Rate, 162

5 Dynamic Site Equations Obtained from Growth Models 164

5.1 Introduction, 164
5.2 Base-Age-Specific (BAS) Models, 164
5.3 Algebraic Difference Approach (ADA) Models, 166
5.4 Generalized Algebraic Difference Approach (GADA) Models, 169
5.5 A Site Equation Generating Function, 179
5.5.1 ADA Derivations, 180
5.5.2 GADA Derivations, 180
5.6 The Grounded GADA (g-GADA) Model, 184
Appendix 5.A Glossary of Selected Forestry Terms, 186

6 Nonlinear Regression 188

6.1 Intrinsic Linearity/Nonlinearity, 188
6.2 Estimation of Intrinsically Nonlinear Regression Models, 190
6.2.1 Nonlinear Least Squares (NLS), 191
6.2.2 Maximum Likelihood (ML), 195
Appendix 6.A Gauss–Newton Iteration Scheme: The Single Parameter Case, 214
Appendix 6.B Gauss–Newton Iteration Scheme: The r Parameter Case, 217
Appendix 6.C The Newton–Raphson and Scoring Methods, 220
Appendix 6.D The Levenberg–Marquardt Modification/Compromise, 222
Appendix 6.E Selection of Initial Values, 223
6.E.1 Initial Values for the Logistic Curve, 224
6.E.2 Initial Values for the Gompertz Curve, 224
6.E.3 Initial Values for the Weibull Curve, 224
6.E.4 Initial Values for the Chapman–Richards Curve, 225

7 Yield–Density Curves 226

7.1 Introduction, 226
7.2 Structuring Yield–Density Equations, 227
7.3 Reciprocal Yield–Density Equations, 228
7.3.1 The Shinozaki and Kira Yield–Density Curve, 228
7.3.2 The Holliday Yield–Density Curves, 229
7.3.3 The Farazdaghi and Harris Yield–Density Curve, 230
7.3.4 The Bleasdale and Nelder Yield–Density Curve, 231
7.4 Weight of a Plant Part and Plant Density, 239
7.5 The Expolinear Growth Equation, 242
7.6 The Beta Growth Function, 249
7.7 Asymmetric Growth Equations (for Plant Parts), 253
7.7.1 Model I, 254
7.7.2 Model II, 255
7.7.3 Model III, 256
Appendix 7.A Derivation of the Shinozaki and Kira Yield–Density Curve, 257
Appendix 7.B Derivation of the Farazdaghi and Harris Yield–Density Curve, 258
Appendix 7.C Derivation of the Bleasdale and Nelder Yield–Density Curve, 259
Appendix 7.D Derivation of the Expolinear Growth Curve, 261
Appendix 7.E Derivation of the Beta Growth Function, 263
Appendix 7.F Derivation of Asymmetric Growth Equations, 266
Appendix 7.G Chanter Growth Function, 269

8 Nonlinear Mixed-Effects Models for Repeated Measurements Data 270

8.1 Some Basic Terminology Concerning Experimental Design, 270
8.2 Model Specification, 271
8.2.1 Model and Data Elements, 271
8.2.2 A Hierarchical (Staged) Model, 272
8.3 Some Special Cases of the Hierarchical Global Model, 274
8.4 The SAS/STAT NLMIXED Procedure for Fitting Nonlinear Mixed-Effects Model, 276

9 Modeling the Size and Growth Rate Distributions of Firms 293

9.1 Introduction, 293
9.2 Measuring Firm Size and Growth, 294
9.3 Modeling the Size Distribution of Firms, 294
9.4 Gibrat's Law (GL), 297
9.5 Rationalizing the Pareto Firm Size Distribution, 299
9.6 Modeling the Growth Rate Distribution of Firms, 300
9.7 Basic Empirics of Gibrat's Law (GL), 305
9.7.1 Firm Size and Expected Growth Rates, 305
9.7.2 Firm Size and Growth Rate Variability, 308

9.7.3 Econometric Issues, 310
9.7.4 Persistence of Growth Rates, 312
9.8 Conclusion, 313
Appendix 9.A Kernel Density Estimation, 314
9.A.1 Motivation, 314
9.A.2 Weighting Functions, 315
9.A.3 Smooth Weighting Functions: Kernel Estimators, 316
Appendix 9.B The Log-Normal and Gibrat Distributions, 322
9.B.1 Derivation of Log-Normal Forms, 322
9.B.2 Generalized Log-Normal Distribution, 325
Appendix 9.C The Theory of Proportionate Effect, 326
Appendix 9.D Classical Laplace Distribution, 328
9.D.1 The Symmetric Case, 328
9.D.2 The Asymmetric Case, 330
9.D.3 The Generalized Laplace Distribution, 331
9.D.4 The Log-Laplace Distribution, 332
Appendix 9.E Power-Law Behavior, 332
9.E.1 Pareto's Power Law, 333
9.E.2 Generalized Pareto Distributions, 335
9.E.3 Zipf's Power Law, 337
Appendix 9.F The Yule Distribution, 338
Appendix 9.G Overcoming Sample Selection Bias, 339
9.G.1 Selection and Gibrat's Law (GL), 339
9.G.2 Characterizing Selection Bias, 339
9.G.3 Correcting for Selection Bias: The Heckman (1976, 1979) Two-Step Procedure, 342
9.G.4 The Heckman Two-Step Procedure Under Modified Selection, 345

10 Fundamentals of Population Dynamics 352

10.1 The Concept of a Population, 352
10.2 The Concept of Population Growth, 353
10.3 Modeling Population Growth, 354
10.4 Exponential (Density-Independent) Population Growth, 357
10.4.1 The Continuous Case, 357
10.4.2 The Discrete Case, 359
10.4.3 Malthusian Population Growth Dynamics, 361
10.5 Density-Dependent Population Growth, 363
10.5.1 Logistic Growth Model, 364
10.6 Beverton–Holt Model, 371
10.7 Ricker Model, 374
10.8 Hassell Model, 377
10.9 Generalized Beverton–Holt (B–H) Model, 380
10.10 Generalized Ricker Model, 382

Appendix 10.A A Glossary of Selected Population Demography/Ecology Terms, 389
Appendix 10.B Equilibrium and Stability Analysis, 391
10.B.1 Stable and Unstable Equilibria, 391
10.B.2 The Need for a Qualitative Analysis of Equilibria, 392
10.B.3 Equilibria and Stability for Continuous-Time Models, 392
10.B.4 Equilibria and Stability for Discrete-Time Models, 394
Appendix 10.C Discretization of the Continuous-Time Logistic Growth Equation, 400
Appendix 10.D Derivation of the B–H S–R Relationship, 401
Appendix 10.E Derivation of the Ricker S–R Relationship, 403

Appendix A **405**

Table A.1 Standard Normal Areas (Z Is $N(0, 1)$), 405
Table A.2 Quantiles of Student's t Distribution (T Is t_v), 407
Table A.3 Quantiles of the Chi-Square Distribution (X Is χ^2_v), 408
Table A.4 Quantiles of Snedecor's F Distribution (F Is F_{v_1, v_2}), 410
Table A.5 Durbin–Watson DW Statistic—5% Significance Points d_L and d_U (n is the sample size and k' is the number of regressors excluding the intercept), 415
Table A.6 Empirical Cumulative Distribution of τ for $\rho = 1$, 419

References **420**

Index **431**

PREFACE

The concept of *growth* is all-pervasive. Indeed, issues concerning national economic growth, human population growth, agricultural/forest growth, the growth of firms as well as of various insect, bird, and fish species, and so on, routinely capture our attention. But how is such growth modeled and measured?

The objective of this book is to convey to those who attempt to monitor the change in some variable over time that there is no "one–size-fits-all" approach to growth measurement; a growth model useful for studying an agricultural crop will most assuredly not be appropriate for fishery management. And if, for instance, one is interested in calculating a growth rate for some time series data set, a decision has to be made as to whether or not one needs to determine a relative rate of growth, an average annual growth rate, an ordinary least squares growth rate, a geometric mean growth rate, among others. Moreover, the choice of a growth rate is subject to the idiosyncrasies of the data set itself, for example, we need to ask if the data series is trended or if it is stationary and if it is presented on an annual, a quarterly, or monthly basis. But this is not the whole story—we also need to ask if the appropriate growth curve should be linear, sigmoidal (S-shaped), with an upper asymptote, or, say, increases to a maximum and then decreases thereafter.

The aforementioned issues concerning the selection of a growth modeling methodology are of profound importance to those looking to develop sound growth measurement techniques. This book is an attempt to point them in the appropriate direction. It will appeal to students and researchers in a broad spectrum of activities (including business, government, economics, planning, medical research, resource management, among others) and presumes that the reader has had an elementary calculus course along with some exposure to basic statistical analysis. While derivations of virtually all of the major growth curves/models have been provided, they have been placed into end-of-chapter appendices so as not to interrupt the general flow of the material. Some important features of this book are: (i) in addition to detailed discussions of growth modeling/theory, the requisite mathematical and statistical apparatus needed to study the same is provided; (ii) SAS code (SAE/ETS 9.1, 2004) is given so that the reader can estimate their own specialized growth rates and curves; and (iii) an assortment of important applications are supplied.

Looking to specifics:

Chapter 1: This chapter reviews some mathematical preliminaries such as arithmetic and geometric progressions, finite differences, the logarithmic and exponential functions, and compound interest.

Chapter 2: This chapter introduces the fundamentals of growth: relative and average rates of change; discrete versus continuous growth; compounded rates of change; growth rate variability; growth in a mixture of variables; comparing time series; and the growth of a variable in terms of its components.

Chapter 3: This chapter presents a detailed look at some of the most popular growth curves: linear, logarithmic, reciprocal, logistic, Gompertz, Weibull, negative exponential, von Bertalanffy, Richards, log-logistic, Brody, along with many other forms. Derivations are in appendices.

Chapter 4: Trend estimation is the focus of this chapter. This chapter involves fitting linear as well as nonlinear trend models and dealing with autocorrelated errors, trended data, integrated processes, and testing for unit roots.

Chapter 5: This chapter presents dynamic site equations for forest growth modeling. Approaches included are base-age invariant, algebraic difference, generalized algebraic difference, and grounded generalized algebraic differences.

Chapter 6: This chapter deals with the estimation of intrinsically nonlinear regression equations via nonlinear least squares and maximum likelihood. Various iteration schemes are explored and SAS is utilized to generate nonlinear parameter estimates.

Chapter 7: The subject matter herein is the study of yield–density relationships for plants and plant parts. In particular, the reciprocal yield-density equations of Shinozaki and Kira, Holliday, Farazdaghi and Harris, and Bleasdale and Nelder are explored while some of the more modern specifications such as the expolinear, beta, and asymmetric growth functions are treated in detail.

Chapter 8: This chapter deals with nonlinear mixed effects models with repeated measurements data. Covers the rudiments of experimental design and introduces a hierarchical (staged) model and its applications.

Chapter 9: This chapter addresses issues concerning the size and growth distributions of firms. Gibrat's law is thoroughly developed, and its empirical underpinnings and tests thereof are treated in great detail. In particular, a whole assortment of specialized appendices covers the mathematical and statistical foundations for this area of analysis.

Chapter 10: The focus here is on population dynamics. Both discrete and continuous density-independent as well as density-dependent models are addressed. Malthusian and logistic population dynamics are covered along with the models of Beverton and Holt, Ricker and Hassell, and generalized Beverton and Holt and Ricker growth equations are also considered. In addition, Allee effects, the determination of equilibrium or fixed points, and tests for the stability of the same are treated throughout.

Although this project was initiated while the author was teaching at the University of Hartford, West Hartford, CT, the manuscript was completed over a number of years during which the author was Visiting Professor of Mathematics at Trinity College, Hartford, CT. A sincere thank you goes to my colleague Farhad Rassekh at the University of Hartford for all of our illuminating discussions concerning growth issues and methodology. His support and encouragement is greatly appreciated. I also wish to thank Paula Russo of Trinity College for allowing me to avail myself of the resources of the Mathematics Department.

A special thank you goes to Alice Schoenrock for all of her excellent work during the various phases of the preparation of the manuscript. Her timely response to a whole list of challenges is most admirable.

An additional note of appreciation goes to Susanne Steitz-Filler, Editor, Mathematics and Statistics, at John Wiley & Sons, for her professionalism, vision, and effort expended in the review and approval processes.

1

MATHEMATICAL PRELIMINARIES

1.1 ARITHMETIC PROGRESSION

We may define an *arithmetic progression* as a set of numbers in which each one after the first is obtained from the preceding one by adding a fixed number called the *common difference*. Suppose we denote the common difference of an arithmetic progression by d, the first term by a_1, …, and the nth term by a_n. Then the terms up to and including the nth term can be written as

$$a_1, a_1 + d, a_1 + 2d, \dots, a_1 + (n-1)d(= a_n). \tag{1.1}$$

If S_n denotes the sum of the first n terms of an arithmetic progression, then

$$S_n = a_1 + (a_1 + d) + (a_1 + 2d) + \cdots + (a_1 + (n-2)d) + (a_1 + (n-1)d). \tag{1.2}$$

If the n terms on the right-hand side of Equation 1.2 are written in reverse order, then S_n can also be expressed as

$$S_n = (a_1 + (n-1)d) + (a_1 + (n-2)d) + \cdots + (a_1 + 2d) + (a_1 + d) + a_1. \tag{1.3}$$

Upon adding Equations 1.2 and 1.3, we obtain

$$2S_n = n(2a_1 + (n-1)d)$$

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

or

$$S_n = \frac{n}{2}(2a_1 + (n-1)d) = \frac{n}{2}(a_1 + a_n). \tag{1.4}$$

EXAMPLE 1.1 Given the arithmetic progression −3, 0, 3, …, determine the 50th term and the sum of the first 100 terms. For $a_1 = -3$, the second term (0) minus the first term is $0 - (-3) = 3 = d$, the common difference. Then, from Equation 1.1,

$$a_{50} = -3 + (50-1)3 = 144;$$

and, from Equation 1.4,

$$S_{100} = \frac{100}{2}(2 \cdot 3 + (100-1)3) = 15{,}150.$$

■

1.2 GEOMETRIC PROGRESSION

A *geometric progression* is any set of numbers having a *common ratio*; that is, the quotient of any term (except the first) and the immediately preceding term is the same. Suppose we represent the common ratio of a geometric progression by r, the first term by $a_1(\neq 0), \ldots$, and the nth term by a_n. Then the terms up to and including the nth term are

$$a_1, a_1r, a_1r^2, \ldots, a_1r^{n-1}(= \mathrm{a}_n). \tag{1.5}$$

(Note that, as required,

$$\frac{\text{3rd term}}{\text{2nd term}} = \frac{a_1r^2}{a_1r} = r, \frac{\text{4th term}}{\text{3rd term}} = \frac{a_1r^3}{a_1r^2} = r, \text{etc.})$$

If the sum of the first n terms of a geometric progression is denoted as S_n, then

$$S_n = a_1 + a_1r + a_1r^2 + \cdots + a_1r^{n-1}. \tag{1.6}$$

Using Equation 1.6, let us form

$$rS_n = a_1r + a_1r^2 + a_1r^3 + \cdots + a_1r^n \tag{1.7}$$

so that, upon subtracting Equation 1.7 from Equation 1.6, we obtain

$$S_n - rS_n = a_1 - a_1r^n$$

or

$$S_n = \frac{a_1 - a_1r^n}{1-r} = \frac{a_1 - ra_n}{1-r}, \qquad r \neq 1. \tag{1.8}$$

EXAMPLE 1.2 Given the geometric progression 1/2, 3/4, 9/8, …, determine the sixth term and the sum of the first nine terms. For $a_1 = 1/2$, the second term (3/4) divided by the first term (1/2) is $(3/4)/(1/2) = 3/2 = r$, the common ratio. Then, from Equation 1.5,

$$a_6 = \frac{1}{2}\left(\frac{3}{2}\right)^5 = \frac{243}{64};$$

and, from Equation 1.8,

$$S_9 = \frac{(1/2)-(1/2)(3/2)^9}{1-(3/2)} = \frac{19{,}171}{512}.$$

■

Suppose we have a geometric progression with infinitely many terms. The sum of the terms of this type of geometric progression, in which the value of n can increase without bound, is called a *geometric series* and has the form

$$S = a_1 + a_1 r + a_1 r^2 + \cdots + a_1 r^{n-1} + \cdots. \tag{1.9}$$

If we again designate the sum of the first n terms in Equation 1.9 as S_n (here S_n is called a *finite partial sum* of the first n terms) or Equation 1.6, then, via Equation 1.8,

$$S_n = \frac{a_1}{1-r} - \frac{a_1 r^n}{1-r}. \tag{1.10}$$

If $|r| < 1$, then the second term in the difference on the right-hand side of Equation 1.10 decreases to zero as n increases indefinitely ($r^n \to 0$ as $n \to \infty$). Hence,

$$S = \lim_{n\to\infty} S_n = \frac{a_1}{1-r}, \quad |r| < 1. \tag{1.11}$$

Thus, the geometric series S is said to *converge* to the value $a_1/(1-r)$. If $|r| > 1$, the finite partial sums S_n do not approach any limiting value—the geometric series S does not converge; it is said to *diverge* since $|r^n| \to \infty$ as $n \to \infty$.

EXAMPLE 1.3 Given the geometric progression

$$1, \frac{1}{3}, \frac{1}{9}, \ldots, \frac{1}{3^{n-1}}, \ldots,$$

does the geometric series

$$S = 1 + \frac{1}{3} + \frac{1}{9} + \cdots + \frac{1}{3^{n-1}} + \cdots$$

converge? If so, to what value? Given $r = 1/3$, the nth finite partial sum is

$$S_n = 1 + \frac{1}{3} + \frac{1}{9} + \cdots + \frac{1}{3^{n-1}}$$

and, via Equation 1.10,

$$S_n = \frac{1}{1-(1/3)} - \frac{1 \cdot (1/3)^n}{1-(1/3)}.$$

Then

$$S = \lim_{n \to \infty} S_n = \frac{1}{1-(1/3)} = \frac{3}{2}.$$

■

1.3 THE BINOMIAL FORMULA

Suppose we are interested in finding $(a+b)^n$, where n is a positive integer. According to the *binomial formula*,

$$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{2 \cdot 3}a^{n-3}b^3 + \cdots + b^n, \tag{1.12}$$

with the coefficients of the terms on the right-hand side of Equation 1.12 termed *binomial coefficients* corresponding to the exponent n. For instance, from Equation 1.12,

$$(a+b)^5 = a^5 + 5a^4b + \frac{5 \cdot 4}{2}a^3b^2 + \frac{5 \cdot 4 \cdot 3}{2 \cdot 3}a^2b^3 + \frac{5 \cdot 4 \cdot 3 \cdot 2}{2 \cdot 3 \cdot 4}ab^4 + b^5.$$

Note that, in general:

1. There are $n + 1$ terms in the binomial expansion of $(a + b)^n$.
2. The exponent of a decreases by 1 from term to term, while the exponent of b increases by 1 from term to term, and the sum of the exponents of a and b is n.
3. The coefficients of the terms equidistant from the ends of the binomial expansion are equal.

A glance back at Equation 1.12 reveals that the $(r+1)$st term in the binomial expansion of $(a+b)^n$ is

$$\binom{n}{r} a^{n-r} b^r = \frac{n!}{r!(n-r)!} a^{n-r} b^r, \qquad r = 0, 1, \ldots, n.^1 \tag{1.13}$$

That is, for

$$r = 0, \quad \frac{n!}{0!n!} a^n b^0 = a^n;$$

$$r = 1, \quad \frac{n!}{1!(n-1)!} a^{n-1} b;$$

$$r = 2, \quad \frac{n!}{2!(n-2)!} a^{n-2} b^2; \text{ etc.}$$

If, as in the preceding text, $n=5$, then the preceding three binomial expansion terms are

$$(r = 0)\, a^5;$$

$$(r = 1) \quad \frac{5!}{1!4!} a^4 b = \frac{5 \cdot 4!}{4!} a^4 b = 5a^4 b;$$

$$(r = 2) \quad \frac{5!}{2!3!} a^3 b^2 = \frac{5 \cdot 4 \cdot 3!}{2!3!} = 10a^3 b^2; \text{ etc.}$$

Given Equation 1.13, we can now write the general binomial expansion formula as

$$\begin{aligned} (a+b)^n &= \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r \\ &= \sum_{r=0}^{n} \frac{n!}{r!(n-r)!} a^{n-r} b^r. \end{aligned} \tag{1.14}$$

1.4 THE CALCULUS OF FINITE DIFFERENCES

Suppose that the real-valued function $y=f(x)$ is defined on an interval containing x and Δx (i.e., x has been increased by an amount Δx). Since the *difference interval* Δx is generally a constant, we may simply denote this constant as h. Then the *difference operator* Δ applied to $f(x)$ is defined as

$$\Delta f(x) = f(x+h) - f(x). \tag{1.15}$$

Furthermore, while h may be any constant value, it is usually the case that $h=1$. Hence, in what follows, the interval of differencing in x is unity. Thus, Equation 1.15 becomes

[1] Here $r!$ is called *r factorial* and is calculated as $r! = r(r-1)(r-2)\cdots 3 \cdot 2 \cdot 1$ with $0! \equiv 1$.

$$\Delta f(x) = f(x+1) - f(x). \tag{1.15.1}$$

Given Equation 1.15.1, it is readily verified that:

1. $\Delta[cf(x)] = c\Delta f(x)$, c an arbitrary constant. (1.16)

Clearly

$$\begin{aligned}\Delta[cf(x)] &= cf(x+1) - cf(x)\\ &= c[f(x+1) - f(x)] = c\Delta f(x).\end{aligned}$$

For real-valued functions $f(x)$ and $g(x)$ both defined over an interval containing x and $x+1$,

2. $\Delta\left[f(x) \pm g(x)\right] = \Delta f(x) \pm \Delta g(x).$ (1.17)

Here

$$\begin{aligned}\Delta[f(x) \pm g(x)] &= [f(x+1) \pm g(x+1)] - [f(x) \pm g(x)]\\ &= [f(x+1) - f(x)] \pm [g(x+1) - g(x)]\\ &= \Delta f(x) \pm \Delta g(x).\end{aligned}$$

(Note that if c_1 and c_2 are arbitrary constants, then $\Delta[c_1 f(x) \pm c_2 g(x)] = c_1\Delta f(x) \pm c_2\Delta g(x)$.)

3. $\Delta\left[f(x)g(x)\right] = g(x)\Delta f(x) + f(x)\Delta g(x) + \Delta f(x)\Delta g(x).$ (1.18)

We first find

$$\Delta\left[f(x)g(x)\right] = f(x+1)g(x+1) - f(x)g(x).$$

If we now add and subtract $f(x)g(x+1)$ on the right-hand side of the previous expression, then we obtain

$$\begin{aligned}\Delta[f(x)g(x)] &= f(x+1)g(x+1) + f(x)g(x+1) - f(x)g(x+1) - f(x)g(x)\\ &= f(x)[g(x+1) - g(x)] + g(x+1)[f(x+1) - f(x)]\\ &= f(x)\Delta g(x) + g(x+1)\Delta f(x).\end{aligned}$$

Substituting $g(x+1) = g(x) + \Delta g(x)$ into the preceding expression yields Equation 1.18.

4. $\Delta\left(\dfrac{f(x)}{g(x)}\right) = \dfrac{g(x)\Delta f(x) - f(x)\Delta g(x)}{g(x)g(x+1)}, g(x)g(x+1) \neq 0.$ (1.19)

To see this, let

$$\Delta\left(\frac{f(x)}{g(x)}\right)=\frac{f(x+1)}{g(x+1)}-\frac{f(x)}{g(x)}=\frac{g(x)f(x+1)-f(x)g(x+1)}{g(x+1)g(x)}$$
$$=\frac{g(x)(\Delta f(x)+f(x))-f(x)(\Delta g(x)+g(x))}{g(x)g(x+1)}$$
$$=\frac{g(x)\Delta f(x)-f(x)\Delta g(x)}{g(x)g(x+1)}.$$

5. $\Delta a^x = a^x(a-1).$ (1.20)

Here

$$\Delta a^x = a^{x+1}-a^x = a^x(a-1).$$

6. $\Delta \log_a x = \log_a\left(1+\frac{1}{x}\right).$ (1.21)

We simply set

$$\Delta \log_a x = \log_a(x+1)-\log_a x = \log_a\left(\frac{x+1}{x}\right)$$
$$=\log_a\left(1+\frac{1}{x}\right).$$

7. $\Delta e^x = e^x(e-1).$ (1.22)

Set

$$\Delta e^x = e^{x+1}-e^x = e^x(e-1).$$

8. $\Delta x^n = nx^{n-1}+\binom{n}{2}x^{n-2}+\cdots+\binom{n}{n-1}x+1$, n a positive integer. (1.23)

We first find

$$\Delta x^n = (x+1)^n - x^n.$$

Then from the binomial expansion formula (Eq. 1.14) applied to $(x+1)^n$, we have

$$\Delta x^n = \sum_{r=0}^{n}\binom{n}{r}x^{n-r}-x^n=\sum_{r=1}^{n}\binom{n}{r}\ x^{n-r}$$

or Equation 1.23.

Given the real-valued function $f(x)$, we can, via the difference operator Δ, define a new function $\Delta f(x)$. If we apply the operator Δ to this new function $\Delta f(x)$, then we obtain the *second difference of f(x)* as the difference of the first difference or

$$\begin{aligned}\Delta^2 f(x) &= \Delta(\Delta f(x)) \\ &= \Delta[f(x+1)-f(x)] \\ &= \Delta f(x+1)-\Delta f(x).\end{aligned} \tag{1.24}$$

Similarly, the *third difference of f(x)*, which is the difference of the second difference, is

$$\Delta^3 f(x) = \Delta(\Delta^2 f(x)). \tag{1.25}$$

In general, the *nth difference of* $f(x)$, which is the difference of the $(n-1)$st difference of $f(x)$, is

$$\Delta^n f(x) = \Delta(\Delta^{n-1} f(x)), \quad n = 2,3,4,\ldots. \tag{1.26}$$

EXAMPLE 1.4 Given the real-valued function $y=f(x)=x^3+2x^2$, find $\Delta^3 f(x)$. First,

$$\Delta f(x) = (x+1)^3 + 2(x+1)^2 - x^3 - 2x^2.$$

Then, via the binomial expansion formula,

$$\Delta f(x) = x^3 + 3x^2 + 3x + 1 + 2(x^2 + 2x + 1) - x^3 - 2x^2 = 3x^2 + 7x + 3.$$

Next,

$$\begin{aligned}\Delta^2 f(x) &= \Delta(\Delta f(x)) = \Delta(3x^2 + 7x + 3) \\ &= 3(x+1)^2 + 7(x+1) + 3 - 3x^2 - 7x - 3 \\ &= 6x + 10.\end{aligned}$$

Finally,

$$\begin{aligned}\Delta^3 f(x) &= \Delta(\Delta^2 f(x)) = \Delta(6x+10) \\ &= 6(x+1) + 10 - 6x - 10 = 6.\end{aligned}$$

■

The preceding example problem serves as a nice lead-in to the following result:

9. Let $f(x)$ be a polynomial of degree n in x or $f(x)=a_0+a_1x+a_2x^2+\cdots+a_nx^n$, where the a_j, $j=0, 1, \ldots, n$, are arbitrary constants and $a_n \neq 0$. Then the nth difference of $f(x)$ is the constant function $\Delta^n f(x)=n!\,a_n$, and all succeeding differences vanish or $\Delta^p f(x)=0$, $p>n$.

To see this we have, from property or result no. 2 earlier,

$$\Delta f(x) = a_0 \Delta 1 + a_1 \Delta x + a_2 \Delta x^2 + \cdots + a_n \Delta x^n. \tag{1.27}$$

By property no. 8, Δ operating on x^n renders a finite number of terms with $n-1$ as the highest power of x. Applying this observation to Equation 1.27 enables us to conclude that Δ operating on a polynomial of degree n results in a polynomial of degree $n-1$. In a similar vein, $\Delta^2 f(x)$ will be a polynomial of degree $n-2$, and $\Delta^n f(x)$ will thus be a polynomial of degree 0 (i.e., a constant). Moreover, for $p>n$, Δ^p applied to a constant must be zero.

1.5 THE NUMBER *e*

Let us consider the *sequence* (an ordered countable set of numbers not necessarily all different) $x_1, x_2, \ldots, x_n, \ldots$, where

$$x_n = \left(1 + \frac{1}{n}\right)^n. \tag{1.28}$$

If we expand the right-hand side of Equation 1.28 by the binomial formula (Eq. 1.14), then

$$\begin{aligned}\left(1+\frac{1}{n}\right)^n &= 1 + \frac{n}{1!}\frac{1}{n} + \frac{n(n-1)}{2!}\frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!}\frac{1}{n^3} \\ &\quad + \cdots + \frac{n(n-1)\cdots 2\cdot 1}{n!}\frac{1}{n^n} \\ &= 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n}\right) + \frac{1}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) \\ &\quad + \cdots + \frac{1}{n!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{n-1}{n}\right).\end{aligned} \tag{1.29}$$

Suppose we now replace n by $n+1$ in Equation 1.28 so as to obtain

$$x_{n+1} = \left(1 + \frac{1}{n+1}\right)^{n+1}. \tag{1.30}$$

Again using the binomial expansion formula,

$$\begin{aligned}\left(1+\frac{1}{n+1}\right)^{n+1} &= 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n+1}\right) + \frac{1}{3!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right) \\ &\quad + \cdots + \left[\frac{1}{(n+1)!}\left(1-\frac{1}{n+1}\right)\cdots\left(\frac{2}{n+1}\right)\left(\frac{1}{n+1}\right)\right].\end{aligned} \tag{1.31}$$

A term-by-term comparison of Equations 1.29 and 1.31 reveals that x_{n+1} is always larger than x_n. In fact, Equation 1.31 has one more term than Equation 1.29. Hence, $x_{n+1} > x_n$; that is, the sequence of values specified by Equation 1.28 is strictly monotonically increasing.

Next, looking to the expansion of x_n (Eq. 1.29), we see that

$$x_n < 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} < 1 + 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = y_n.$$

Since y_n is a geometric progression (the common ratio $r = 1/2$), we have

$$x_n < 1 + \frac{1-(1/2)^{n+1}}{1-(1/2)} < 1 + \frac{1}{1/2} = 3.$$

Hence, Equation 1.28 is bounded from above. And since any monotone bounded sequence has a limit, we can denote the limit of Equation 1.28 as

$$\begin{aligned} \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} = e. \end{aligned} \tag{1.28.1}$$

To five decimal places, $e = 2.71828$.

1.6 THE NATURAL LOGARITHM

We may define the *natural logarithm* of x, for positive x, as

$$F(x) = \ln x = \int_1^x \frac{1}{r}\,dr, \quad x > 0 \tag{1.32}$$

(see Fig. 1.1). For $x = 1$, obviously $\ln 1 = 0$; and for $x < 1$,

$$\ln x = -\int_x^1 \frac{1}{r}\,dr.$$

Given $F(x) = \ln x$, it follows that $F'(x) = d\ln x/dx = 1/x$.

Looking to the graph of the *logarithmic function* $y = \ln x$ (Fig. 1.2a), we see that $\ln x$ is continuous, single valued, and monotonically increasing with $dy/dx = 1/x > 0$, while $d^2y/dx^2 = -1/x^2 < 0$. Since $\ln 1 = 0$, the curve passes through the point (1,0). Moreover, $\ln x \to +\infty$ as $x \to +\infty$; $\ln x = -\ln 1/x \to -\infty$ as $x \to 0+$.

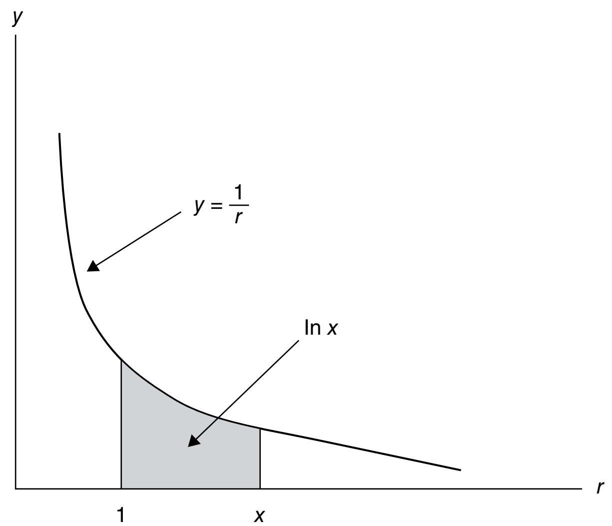

FIGURE 1.1 The natural logarithm of x.

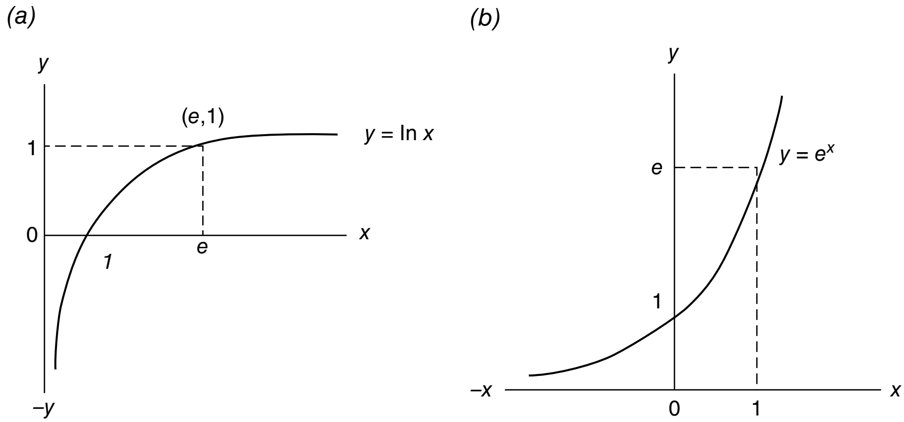

FIGURE 1.2 (**a**) Logarithmic function and (**b**) Exponential function.

1.7 THE EXPONENTIAL FUNCTION

Given the logarithmic function $y = \ln x$, if $x = e$, then $\ln e = 1$ (Fig. 1.2a). Hence, the value of x for which $\ln x = 1$ is e. Also, $\ln e^n = n \ln e = n$. Thus, the number whose natural logarithm is n is e^n so that the anti-natural logarithm of n is e^n or the *antilogarithm* is the inverse of the logarithm.

This said, given the logarithmic function $y = \ln x$, its inverse function is

$$y = \text{inverse natural logarithm of } x$$

or

$$x = \ln y, \tag{1.33}$$

where $x > 0$ for $y > 1$, $x = 0$ for $y = 1$, and $x < 0$ for $y < 1$.

Thus, there exists a one-to-one correspondence between the sets $Y = \{y | y > 0\}$ and $X = \{x | -\infty < x < +\infty\}$. In this regard, we can define y (>0) as a real-valued function of x for $-\infty < x < +\infty$. Hence,

$$y = e^x \tag{1.34}$$

is equivalent to Equation 1.33 and is called the *exponential function*. So with the logarithmic function continuous, single valued, and monotonically increasing, it follows that the exponential function, its inverse, exists and has the same exact properties. In sum,

$$y = e^x \text{ if and only if } x = \ln y.$$

Hence, e^x, defined for all real x, is that positive number y whose natural logarithm is x (Fig. 1.2b).

Some useful relationships between the exponential function and the (natural) logarithmic function are:

$$\ln e^x = x \ln e = x \qquad e^{(\ln x)^2} = x^{\ln x}$$

$$\ln e^{-x} = \ln(1/e^x) = -x \qquad e^{(a \ln x)^2} = x^{a^2 \ln x}$$

$$e^{\ln x} = x \qquad e^{(a \ln x)(b \ln y)} = y^{ab \ln x} = x^{ab \ln y}$$

$$e^{-\ln x} = \frac{1}{e^{\ln x}} = \frac{1}{x}$$

Moreover,

if $\ln y = \ln a + w(\ln r - \ln s)$, then

$$y = a\left(\frac{r}{s}\right)^w;$$

if $\ln y = a + w(r - s)$, then

$$y = e^a\left(\frac{e^r}{e^s}\right)^w;$$

and if $\ln y = a + w(\ln r - s)$, then

$$y = e^a\left(\frac{r}{e^s}\right)^w.$$

1.8 EXPONENTIAL AND LOGARITHMIC FUNCTIONS: ANOTHER LOOK

For the exponential function (Eq. 1.34), e served as the (fixed) *base* of this expression. Moreover, the unique inverse of Equation 1.34 is the logarithmic function $y = \ln x$, where it is to be implicitly understood that "y is the natural logarithm of x to the base e." However, other bases can be used.

Specially, let us alternatively specify an *exponential function* of x as

$$y = b^x, b > 1, \tag{1.35}$$

where b is the (fixed) *base* of the function. The base b will be taken to be a number greater than unity (since any positive number (y) can be expressed as a power (x) of a given number (b) greater than unity). Hence, Equation 1.35 is a continuous single-valued function, which is monotonically increasing for $-\infty < x < +\infty$ (Fig. 1.3a).

Since Equation 1.35 is continuous and single valued, it has a unique inverse called the *logarithmic function*

$$x = \log_b y, \quad b > 0 \text{ and } b \neq 1, \tag{1.36}$$

read "x is the logarithm of y to the base b" (Fig. 1.3b). In this regard, a number x is said to be the *logarithm* of a positive real number y to a given base b if x is the power to which b must be raised in order to obtain y. Hence,

$$y = b^x \text{ if and only if } x = \log_b y.$$

(Thus, $\log_5 25 = 2$ since $5^2 = 25$; and $\log_2 8 = 3$ since $2^3 = 8$.)

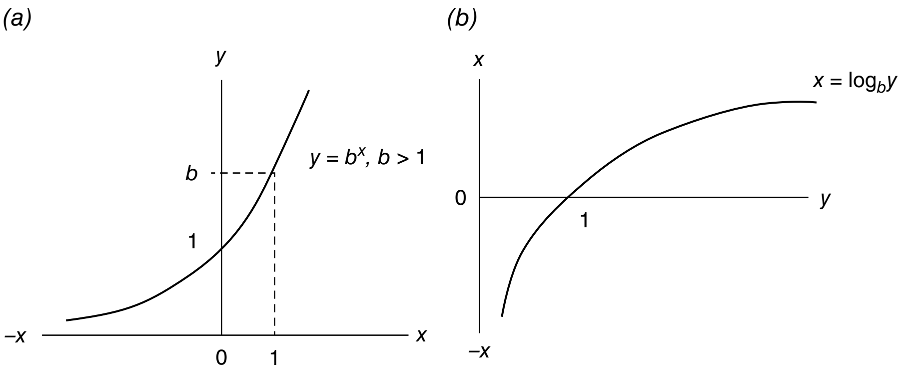

FIGURE 1.3 (**a**) Exponential function (fixed base b) and (**b**) Logarithmic function (fixed base b).

Clearly a logarithm is simply a fancy way to write an exponent. Note that $\log_b y$ is not defined for negative values of y or zero; that is, if $0<y<1$, then $\log_b y < 0$; if $y=1$, then $\log_b 1=0$; and if $y>1$, then $\log_b y>0$.

Some useful properties of logarithms are

base b	***base e***
1. $\log_b uv=\log_b u+\log_b v$	1. $\ln uv=\ln u+\ln v$
2. $\log_b \dfrac{u}{v} = \log_b u-\log_b v$	2. $\ln\dfrac{u}{v} = \ln u-\ln v$
3. $\log_b u^n=n\log_b u$	3. $\ln u^n=n\ln u$
4. $\log_b \dfrac{1}{p} = -\log_b p$	4. $\ln\dfrac{1}{p} = -\ln p$

Also, some useful differentiation formulas are

1. $\dfrac{da^x}{dx} = a^x \ln a$

2. $\dfrac{d\ln x}{dx} = \dfrac{1}{x}$

3. $\dfrac{d\log_b x}{dx} = \dfrac{1}{x}\log_b e$

4. $\dfrac{de^x}{dx} = e^x$

5. $\dfrac{de^{cx}}{dx} = ce^{cx}$, c a constant

1.9 CHANGE OF BASE OF A LOGARITHM

Our goal is to develop a method for transforming the logarithm of x to the base b to the logarithm of x to the base a. To this end, we know from the preceding discussion of logarithms that $y=\log_a x$ is equivalent to $x=a^y$. Let us now take the logarithm of this latter expression with respect to the base b; that is,

$$\log_b x = y\log_b a$$

or

$$y = \frac{\log_b x}{\log_b a} = \log_a x. \tag{1.37}$$

Here Equation 1.37 will be termed our *Change of Base Rule*: the logarithm of x to the base a is the logarithm of x to the base b divided by logarithm of a to the base b.

It is well known that a *common logarithm* is taken to the base 10 ($\log_{10}100=2$ since $10^2=100$), while a *natural logarithm*, as defined earlier, is taken to the base $e=2.71828$ ($\ln 20.08554=3$ since $e^3=20.08554$). We may employ Equation 1.37 to facilitate the conversion between them, that is,

1. To convert from common to natural logarithms,

$$\ln x = \frac{\log_{10} x}{\log_{10} e} = 2.303 \log_{10} x.$$

2. To convert from natural to common logarithms,

$$\log_{10} x = \frac{\ln x}{\ln 10} = 0.4343 \ln x.$$

1.10 THE ARITHMETIC (NATURAL) SCALE VERSUS THE LOGARITHMIC SCALE

Suppose the observations x_i, $i=1,2, \ldots$, on a variable X are measured relative to some specific scale and appear as points on the X-axis, with distances along this axis taken from some specific base or reference point. Now:

1. If distance along the X-axis is taken to be equal (or proportional) to the actual value of the X point plotted, then the observations on X are measured on an *arithmetic scale* (for $X=x_r$, the distance between the base of 0 and the plotted point is x_r units).
2. If an X value is plotted at a distance along the X-axis, which is equal (or proportional) to its logarithm (to, say, the base 10), then the observations on X are measured on a *logarithmic scale* (for $X=x_r$, the distance between the base value and the plotted point is $\log_{10} x_r$ units).

Looking to Figure 1.4, we see that on an arithmetic scale, either (i) the points appear at equal distances from each other (scale a) or (ii) the points appear at increasing distances from each other (scale b).

But as Figure 1.5 reveals, (i) taking the base 10 logarithms of the values on arithmetic scale a of Figure 1.4 produces a sequence of points exhibiting decreasing distances from each other (scale a′), or (ii) taking base 10 logarithms of the values on

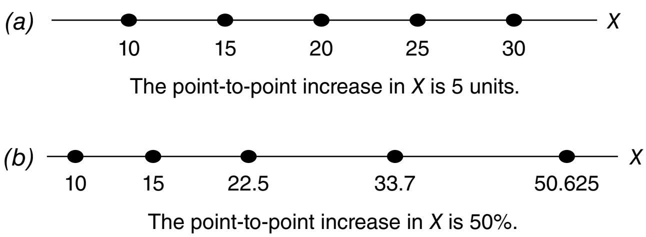

FIGURE 1.4 Arithmetic scale.

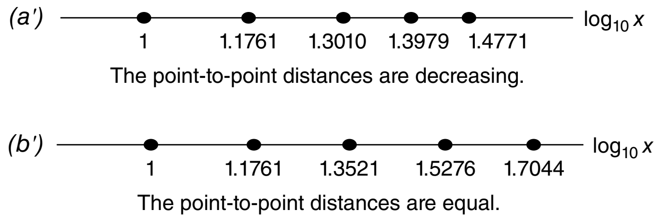

FIGURE 1.5 Logarithmic scale.

arithmetic scale b of Figure 1.4 renders a sequence of points that are located at equal distances from each other (scale b′).

If on an arithmetic scale a sequence of X values exhibits equal point-to-point *decreases* (e.g., consider 50, 40, 30, 20, 10), then the corresponding base 10 logarithmic scale displays values at increasing distances (to the left). And if the X values decrease by a fixed percentage on an arithmetic scale (e.g., for a 20% decrease we get 50, 40, 32, 25.6, 20.48), then the corresponding base 10 logarithmic values display equal point-to-point distances. (The reader is asked to verify these assertions for the given sets of arithmetic values.)

On the basis of the preceding discussion, it is evident that equal point-to-point distances on an arithmetic scale indicate *equal absolute changes* in a variable X; but equal point-to-point distances on a (base 10) logarithmic scale reflect equal proportional or percentage changes in X. For instance, if x_1,x_2, and x_3 are values of X plotted at equal distances on an arithmetic scale, then X increases by equal absolute amounts since $x_3-x_2=x_2-x_1$. However, if the (base 10) logarithms of these X values are plotted at equal distances on a logarithmic scale, then X increases by *equal proportional amounts* since $\log_{10}x_3-\log_{10}x_2=\log_{10}x_2-\log_{10}x_1$ or $\log_{10}(x_3/x_2)-\log_{10}(x_2/x_1)$ or $x_3/x_2=x_2/x_1$.

Suppose that instead of dealing with a single variable X, we introduce a second variable Y and posit a functional relationship between them of the form $y=f(x)$, where f is a rule or law of correspondence (i.e., a mapping), which associates with each admissible value x of X a unique admissible value y of Y. Then in terms of our measurement scales, we note that:

1. If absolute changes in the variables are of interest, then we can model y as a *linear* function of x or $y=a+bx$, where a is the vertical intercept and b is the slope ($b=\Delta y/\Delta x$). Hence, the absolute change in y is always the same constant proportion (b) of the absolute change in x.
2. If proportional or percentage increases in y (or the rate of growth in y) are of interest as x (measured on an arithmetic scale) increases in value, then we can model y as an *exponential function* of x or $y=ab^x$. Thus, $\log_{10}y=\log_{10}a+(\log_{10}b)x$, where $\log_{10}a$ is the vertical intercept and $\log_{10}b$ is the slope. Since only y is measured on a logarithmic scale, this relationship is termed a *semilogarithmic function* of x and, with $\log_{10}b$ constant, is linear in form or plots as a straight

line. In this circumstance, as x varies over a given interval, $\log_{10}y$ increases by equal increments; that is, y exhibits equal proportional or percentage increases in its value.

3. If proportional or percentage changes in both x and y are of interest, then we can model y as a *power function* of x or $y = ax^b$. Now $\log_{10}y = \log_{10}a + b\log_{10}x$, where $\log_{10}a$ is the vertical intercept, b is the slope, and both x and y are measured on a logarithmic scale. With both variables measured on a logarithmic scale, this relationship is referred to as a *double-logarithmic function*. Here proportional or percentage changes in y are explained by proportional or percentage changes in x, and if equal percentage changes in x precipitate equal percentage changes in y, then, with b constant, this function plots as a straight line. Here

$$b = \frac{\Delta \log_{10} y}{\Delta \log_{10} x}.$$

1.11 COMPOUND INTEREST ARITHMETIC

Suppose a principal amount of \$100.00 is invested and accumulates at a *compound interest rate* of 5% per year and interest is declared yearly. Then the following time profile of accumulation emerges:

Start year 1 $\quad$ \$100.00 invested

End year 1 $\quad$ $\$100.00 + \underbrace{\$100.00(0.05)}_{\text{interest}} = \$100.00(1.05) = \$105.00$

End year 2 $\quad$ $\$105.00 + \underbrace{\$105.00(0.05)}_{\text{interest}} = \$105.00(1.05)$

$= \$100.00(1.05)(1.05)$

$= \$100.00(1.05)^2 = \$110.25$

End year 3 $\quad$ $\$110.25 + \underbrace{\$110.25(0.05)}_{\text{interest}} = \$110.25(1.05)$

$= \$100.00(1.05)^2(1.05)$

$= \$100.00(1.05)^3 = \$115.7625$

⋮

etc.

In general, after t time periods or years, the accumulated amount at compound interest with annual compounding is

$$A_t = P(1 + r)^t, \tag{1.38}$$

where P is the principal invested, $100r\%$ is the yearly interest rate, and t indexes time in years. (In the preceding example, $P = \$100.00$ and $r = 0.05$.) So for, say, $t = 10$, $A_{10} = 100(1+0.05)^{10} = 100(1.62889) = \$162.889$. As this example problem reveals, the nature of compound interest is that, over the entire investment period, the interest itself earns interest.

What if interest is added twice a year rather than just once at the end of each year? Since the yearly interest rate is 5%, it follows that the half-yearly rate must be 2.5% so that 2.5% is added in each first half year and 2.5% is added in each second half year. So if a principal of \$100.00 is invested and accumulates at a compound interest rate of 2.5% per half year and interest is declared at the end of each half year, then the revised time profile is:

Start year 1 — \$100.00 invested

End year 1 — $\underbrace{\$100.00(1.025)}_{\text{first half year}} \times \underbrace{(1.025)}_{\text{second half year}} = \$100.00(1.025)^2 = \$105.0625$

End year 2 — $\underbrace{\$105.0625(1.025)}_{\text{first half year}} \times \underbrace{(1.025)}_{\text{second half year}} = \$105.0625(1.025)^2$

$= \$100.00(1.025)^4$

$= \$110.38129$

End year 3 — $\underbrace{\$110.38129(1.025)}_{\text{first half year}} \times \underbrace{(1.025)}_{\text{second half year}} = \$110.38129(1.025)^2$

$= \$100.00(1.025)^6$

$= \$115.96934.$

⋮

etc.

To summarize, if interest is declared half-yearly, P is the principal, and the yearly interest rate is $100r\%$, then after t years the accumulated amount is

$$A_{2,t} = P\left(1+\frac{r}{2}\right)^{2t}.$$

Again taking $t = 10$, $A_{2,10} = 100(1+0.025)^{20} = \$163.86144$. A comparison of $A_{10} = \$162.89$ with $A_{2,10} = \$163.86$ reveals that the more frequently interest is added, the larger is the accumulated amount at the end of a given period. In general, the *accumulated amount at compound interest with interest declared j times a year* is

$$A_{j,t} = P\left(1+\frac{r}{j}\right)^{jt}. \tag{1.39}$$

A moment's reflection concerning the structure of Equation 1.39 reveals that investment growth over time behaves as a geometric progression; that is, each amount is a fixed multiple $(1+(r/j))^j$ of the previous period's amount. That is, the sequence of terms of this geometric progression is:

$$P, P\left(1+\frac{r}{j}\right)^{j}, P\left(1+\frac{r}{j}\right)^{2j}, P\left(1+\frac{r}{j}\right)^{3j}, \ldots.$$

Hence, the growth process represented by Equation 1.39 can be expressed as the exponential function

$$A_{j,t} = Pb^{jt},\ b = 1+\frac{r}{j} \tag{1.40}$$

and is referred to as a *compound interest growth curve* (alternatively called a *geometric or exponential growth curve*). Transforming to logarithms gives

$$\log_{10} A_{j,t} = \log_{10} P + \log_{10}\left(1+\frac{r}{j}\right)(jt). \tag{1.41}$$

Clearly this semilogarithmic expression plots as a straight line with vertical intercept $\log_{10}P$ and (constant) slope $\log_{10}(1+(r/j))$. Obviously the magnitude of the slope depends upon r and j. In this regard, r/j is the proportionate rate of change in $A_{j,t}$ per unit period of time (i.e., per year if $j=1$, per half year if $j=2$, per quarter if $j=4$). Note that the independent variable on the right-hand side of Equation 1.41 is jt; it represents the total number of subperiods j within a year times the number of years t.

For instance, if $j=4$, then $jt=4t$ represents the total number of quarters spanned by the entire accumulation period; that is, if $t=1$, $4t=4$ spans one year; if $t=2$, $4t=8$ spans a two-year time interval.

A special case of Equation 1.41 is, from Equation 1.38,

$$\log_{10} A_t = \log_{10} P + \log_{10}(1+r)t, \tag{1.41.1}$$

where r is the proportionate rate of growth in A_t per unit of time.

Given Equation 1.39, let us assume that j increases without limit or, equivalently, that the compounding or conversion periods become shorter and shorter. In this instance the term $(1+(r/j))$ in Equation 1.39 is replaced by e, and we consequently have what is termed the case of *continuous compounding* or *continuous conversion*.

To see this, let us rewrite Equation 1.39 as

$$P\left[\left(1+\frac{r}{j}\right)^{j/r}\right]^{rt} = P\left[\left(1+\frac{1}{n}\right)^{n}\right]^{rt},$$

where $n=j/r$. Now, as the number of compounding or conversion periods $j \to +\infty$, it follows that $n \to +\infty$. Hence, via Equation 1.28.1,

$$\lim_{n\to\infty} P\left[\left(1+\frac{1}{n}\right)^n\right]^{rt} = Pe^{rt} = A.^2 \tag{1.42}$$

Here Equation 1.42 depicts a *natural exponential growth curve and represents the accumulated amount at the end of t years if the principal (P) grows at an exponential rate of 100r% per year or is compounded continuously at 100r% per year.* For instance, if we again take $P=\$100$, $r=0.05$, and $t=10$, then $A=100e^{0.05(10)}=\$164.872$.

An assortment of points concerning Equation 1.42 merits our attention. First, the variable t in Equation 1.39 is discontinuous since interest is declared at specific (discrete) intervals over the investment period. However, as $j\to+\infty$ and interest is declared with increasing frequency, t tends to become a continuous variable in Equation 1.42. Second, it is instructive to view the term $\left(1+\frac{1}{n}\right)^n$ in Equation 1.42 as the yearly accumulated amount of \$1.00 invested when interest is compounded at 100% per annum and declared n times over the year. Then as n increases without bound,

$$\lim_{n\to\infty} 1\cdot\left(1+\frac{1}{n}\right)^n = \$e.$$

So as compound interest is declared more and more frequently, the \$1.00 invested at 100% interest approaches \$$e$ at the end of a year. Third, transforming Equation 1.42 to logarithms yields

$$\ln A = \ln P + rt, \tag{1.43}$$

which plots as a straight line with vertical intercept ln P and (constant) slope r. Finally, if an amount is compounded yearly at $100r\%$ (e.g., see Eq. 1.38) and that same amount is compounded continuously at $100g\%$ and $100r\%$ and $100g\%$ are equivalent interest rates, then obviously $g=\ln(1+r)$.

[2] It is instructive to view the derivation of this expression in an alternative light. To this end, let us write Equation 1.39 as

$$P\left[\left(1+\frac{r}{j}\right)^{j/r}\right]^{rt} = P\left[(1+x)^{1/x}\right]^{rt},$$

where $x=r/j$. Then applying the binomial formula 1.12 to the term in square brackets yields

$$P\left[1^{1/x}+\frac{1}{x}\cdot 1^{\frac{1}{x}-1}x+\frac{\frac{1}{x}\left(\frac{1}{x}-1\right)}{2!}\cdot 1^{\frac{1}{x}-2}x^2+\cdots\right]^{rt} = P\left[1+1+\frac{1-x}{2!}+\frac{(1-x)(1-2x)}{3!}+\cdots\right]^{rt}.$$

As $j\to+\infty$, it follows that $x\to 0$ so that

$$\lim_{x\to\infty} P[.]^{rt} = P\left[1+1+\frac{1}{2!}+\frac{1}{3!}+\cdots\right]^{rt} = Pe^{rt}$$

via Equation 1.28.1.

2

FUNDAMENTALS OF GROWTH

2.1 TIME SERIES DATA

We may view a *time series* as a set of observations Y_t, $t = 0, 1, 2, \ldots, n$, on a variable Y that are indexed in order of time; that is, the Y_t's are measured at different time points or time intervals. Since t is the time index, if our data series starts in, for instance, 1980 and we have observations going to 2005 in one-year intervals, then we can assign a sequence of numbers to order the data points. In this regard, let us designate 1980 as the *origin* and assign it the value "0." Then 1981 is assigned the value 1, 1982 is given the value 2, and so on. Hence, we end up with the sequence of numbers 0, 1, 2, …, 25 as the set of observations on the time variable t. The convenience of this numerical assignment scheme for representing a sequence of years (or weeks, months, etc.) will become evident later on.

2.2 RELATIVE AND AVERAGE RATES OF CHANGE

Given the time series Y_t, $t = 0, 1, \ldots, n$, let us define the *relative rate of change in Y between periods t − 1 and t* as

$$\frac{\Delta Y_t}{Y_{t-1}} = \frac{Y_t - Y_{t-1}}{Y_{t-1}} = \frac{Y_t}{Y_{t-1}} - 1 = R_t, \quad t = 1, 2, \ldots, n. \tag{2.1}$$

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

TABLE 2.1 Relative Growth Rates in *Y* and Growth Relatives

Period t	R_t	Growth relatives
1	$R_1 = \frac{Y_1}{Y_0} - 1$	$\frac{Y_1}{Y_0}$
2	$R_2 = \frac{Y_2}{Y_1} - 1$	$\frac{Y_2}{Y_1}$
3	$R_3 = \frac{Y_3}{Y_2} - 1$	$\frac{Y_3}{Y_2}$
⋮	⋮	⋮
n	$R_n = \frac{Y_n}{Y_{n-1}} - 1$	$\frac{Y_n}{Y_{n-1}}$

If Y_0 denotes the value of Y at the beginning of period 1 and Y_t represents the value of Y at the end of period t, $t = 1, \ldots, n$, then the sequence of relative rates of growth over the entire time span appears in Table 2.1.

Moreover, the ratios Y_t/Y_{t-1} $(=R_t + 1)$, $t = 1, 2, \ldots, n$, are termed *growth relatives* and, from Table 2.1, appear as

$$\frac{Y_t}{Y_{t-1}}: \frac{Y_1}{Y_0}, \frac{Y_2}{Y_1}, \frac{Y_3}{Y_2}, \ldots, \frac{Y_n}{Y_{n-1}}. \tag{2.2}$$

EXAMPLE 2.1 Given the Y values appearing in Table 2.2, determine the sequence of growth relatives along with the period-to-period relative rates of growth in Y. Let $Y_0 = 200$. For instance, Y experiences almost a 17% rate of growth between periods 3 and 4 since $R_4 = 0.1666$ or $100(0.1666)\% = 16.66\%$. ■

What is average rate of growth in Y over the four periods given in the preceding example problem? At first blush one might be tempted to take the simple arithmetic average of the R_t's so as to obtain

$$\bar{R}_t = \frac{0.25 + (-0.16) + 0.4286 + 0.1666}{4} = \frac{0.6852}{4} = 0.1713$$

or 17.13%. Interestingly enough, this calculation would be misleading. The arithmetic average is the incorrect average to be used when it comes to finding the average rate of growth; the appropriate average rate of growth over the four periods is given by the *geometric mean*

TABLE 2.2 Growth Relatives and Relative Rates of Growth

Period	Y_t	Y_t/Y_{t-1}	$R_t = \frac{Y_t}{Y_{t-1}} - 1$
1	250	$\frac{Y_1}{Y_0} = \frac{250}{200} = 1.25$	0.25
2	210	$\frac{Y_2}{Y_1} = \frac{210}{250} = 0.84$	−0.16
3	300	$\frac{Y_3}{Y_2} = \frac{300}{210} = 1.4286$	0.4286
4	350	$\frac{Y_4}{Y_3} = \frac{350}{300} = 1.1666$	0.1666

$$\begin{aligned} \mathrm{GM} &= \left[(R_1+1)(R_2+1)\cdots(R_n+1)\right]^{1/n} - 1 \\ &= \left[\frac{Y_1}{Y_0}\cdot\frac{Y_2}{Y_1}\cdot\ldots\cdot\frac{Y_n}{Y_{n-1}}\right]^{1/n} - 1.^{1} \\ &= \left(\frac{Y_n}{Y_0}\right)^{1/n} - 1 \end{aligned} \tag{2.3}$$

EXAMPLE 2.2 Given the Y values presented in Table 2.2, find the appropriate average rate of growth or average percentage change in Y over the indicated four periods. We now determine

$$\begin{aligned} \mathrm{GM} &= \left[(R_1+1)(R_2+1)(R_3+1)(R_4+1)\right]^{1/4} - 1 \\ &= \left[(1.25)(0.84)(1.4286)(1.1666)\right]^{1/4} - 1 \\ &= (1.749935)^{0.25} - 1 = (1.75)^{0.25} - 1 \\ &= 1.150153 - 1 = 0.150153 \end{aligned}$$

[1] In general, the *geometric mean* of the n positive quantities $X_1, X_2, \ldots, X_n$ is the nth root of the product of the n items or

$$\mathrm{GM} = \sqrt[n]{X_1 \cdot X_2 \cdot \ldots \cdot X_n} = (X_1 \cdot X_2 \cdot \ldots \cdot X_n)^{1/n}. \tag{2.4}$$

(If at least one of the X_i's is 0 or negative, then the geometric mean is undefined.) Since

$$\log_{10} \mathrm{GM} = \frac{1}{n}\sum_{i=1}^{n} \log_{10} X_i, \tag{2.4.1}$$

the right-hand side of this expression is the arithmetic average of the logarithms of the X_i's.

or

$$\mathrm{GM} = \left(\frac{Y_n}{Y_0}\right)^{1/4} - 1 = \left(\frac{350}{200}\right)^{0.25} - 1$$
$$= (1.75)^{0.25} - 1 = 1.150153 - 1 = 0.150153.$$

Hence, the four-period average rate of growth in the Y series is about 15.02%; that is, between periods one and four, Y increased in value from $Y_0 = 200$ to $Y_4 = 350$, an average increase of 15.02%. ■

Let us rewrite Equation 1.38 or $A_t = P(1 + r)^t$ as

$$Y_t = Y_0 (1 + r)^t.$$

Then

$$\frac{Y_t}{Y_0} = (1 + r)^t \qquad \text{or} \qquad \left(\frac{Y_t}{Y_0}\right)^{1/t} = 1 + r$$

so that

$$r = \left(\frac{Y_t}{Y_0}\right)^{1/t} - 1. \tag{2.5}$$

Hence, the average rate of growth found by taking the geometric mean (Eq. 2.3) is simply the (annual) rate of interest in the compound interest formula (Eq. 1.38); that is, average growth over the four-period span is equivalent to compound interest growth. (So if the principal is 200 and $r = 0.150153$, then $Y_4 = 200\ (1.150153)^4 = 200\ (1.749998) = 350$ as expected.)

EXAMPLE 2.3 Suppose Y assumes the values 5, 7, 10, 11, and 20%. To average this set of percents, let us find

$$\mathrm{GM} = (5 \cdot 7 \cdot 10 \cdot 11 \cdot 20)^{1/5} = (77{,}000)^{0.20} = 9.49.$$

Note that $\mathrm{GM} < \bar{Y} = (5 + 7 + 10 + 11 + 20)/5 = 53/5 = 10.6$ since the geometric mean is not so severely affected by extremes as the arithmetic mean. ■

EXAMPLE 2.4 Suppose the value of Y in 1990 was $Y_{1990} = 1000$ and its value in 2000 was $Y_{2000} = 2500$. What is the average rate of growth or average percentage increase in Y between 1990 and 2000 (here $n = 11$)? We need to find

$$\mathrm{GM} = \left(\frac{Y_{2000}}{Y_{1990}}\right)^{1/(n-1)} - 1 = \left(\frac{2500}{1000}\right)^{0.10} - 1 = 0.095958.$$

So between 1990 and 2000, Y increased by approximately 9.6% on average. ■

In view of Example 2.4, let us modify Equation 2.3 slightly to consider the instance when we are faced with finding the average rate of growth in Y over the time span $t = 1, \ldots, n$. Then

$$\mathrm{GM} = \left(\frac{Y_n}{Y_1}\right)^{1/(n-1)} - 1. \tag{2.3.1}$$

In a similar vein, the compound interest rate of growth is determined by compounding over $t - 1$ periods; that is, from $Y_t = Y_1(1+r)^{t-1}$, Equation 2.5 becomes

$$r = \left(\frac{Y_t}{Y_1}\right)^{1/(t-1)} - 1. \tag{2.5.1}$$

2.3 ANNUAL RATES OF CHANGE

2.3.1 Simple Rates of Change

We noted in Equation 2.1 earlier that the relative rate of change in Y between periods $t - 1$ and t is

$$\frac{Y_t}{Y_{t-1}} - 1$$

or, in percent terms,

$$\left[\frac{Y_t}{Y_{t-1}} - 1\right] \times 100.$$

In this regard, the *percent change from a year ago in Y* is the percent change from the same period in the previous year. Hence, the *percent change from a year ago in Y between quarter t – 4 and the current quarter t* is

$$\left[\frac{Y_t}{Y_{t-4}} - 1\right] \times 100, \tag{2.6}$$

while the *percent change from a year ago in Y between month t – 12 and the current month t* is

$$\left[\frac{Y_t}{Y_{t-12}} - 1\right] \times 100. \tag{2.7}$$

Next, for consecutive quarters, the *percent change at an annual rate in Y between the previous quarter t – 1 and the current quarter t* is the quarterly percentage change multiplied by 4 or

$$\left[\frac{Y_t}{Y_{t-1}}-1\right]\times 100\times 4=\left[\frac{Y_t}{Y_{t-1}}-1\right]\times 400, \tag{2.8}$$

and, for consecutive months, the *percent change at an annual rate in Y between the previous month t – 1 and the current month t* is the monthly percentage change multiplied by 12 or

$$\left[\frac{Y_t}{Y_{t-1}}-1\right]\times 100\times 12=\left[\frac{Y_t}{Y_{t-1}}-1\right]\times 1200. \tag{2.9}$$

2.3.2 Compounded Rates of Change

Let us now compound the simple quarterly or monthly percent changes so as to express them as annual growth rates. In this regard, by the compounded annual rate of change in a series, we mean its growth rate over an entire year if the same simple percent change continued for four quarters or 12 months. More specifically, let us define the compound annual rate of change in Y between the previous quarter $t – 1$ and the current quarter t as

$$\left[\left(\frac{Y_t}{Y_{t-1}}\right)^4-1\right]\times 100, \quad \text{[formula for annualizing quarterly data]} \tag{2.10}$$

while the *compound annual rate of change in Y between the previous month t – 1 and the current month t* is

$$\left[\left(\frac{Y_t}{Y_{t-1}}\right)^{12}-1\right]\times 100, \quad \text{[formula for annualizing monthly data]} \tag{2.11}$$

For instance, we can easily rationalize Equation 2.10 as follows. Let us rewrite Equation 1.38 as

$$Y_t = Y_{t-1}(1+r)^t, \tag{2.12}$$

where Y_t is the value of Y in the current period (t), Y_{t-1} is the value of Y in the previous period ($t – 1$), and $100r$ % is the (compound) annual rate of change in Y. Given that r represents annual compounding, Equation 2.12 becomes, for quarterly data,

$$Y_t = Y_{t-1}(1+r)^{1/4}.$$

Then solving this expression for r enables us to write the compound annual rate of change in Y between the current quarter t and the previous quarter $t-1$ as

$$r=\left(\frac{Y_t}{Y_{t-1}}\right)^4-1.$$

In a similar vein, for monthly data, Equation 2.12 becomes

$$Y_t=Y_{t-1}(1+r)^{1/12},$$

where again $100r$ % is the (compound) annual rate of change. Then the compound annual rate of change in Y between the current month t and the previous month $t-1$ is

$$r=\left(\frac{Y_t}{Y_{t-1}}\right)^{12}-1.$$

EXAMPLE 2.5 For quarterly observations on Y, suppose $Y_{t-1}=100$ (the first quarter, say, begins with $Y=100$) and $Y_t=105$ (the second quarter begins with $Y=105$). Then, from Equation 2.10,

$$r=\left(\frac{105}{100}\right)^4-1=1.21551-1=0.21551.$$

Hence, the compound annual rate of change in Y between the current quarter and the previous quarter is 21.5515%; that is, 21.551% is the growth rate over the entire year if the same simple percent (21.551%) continued over all four quarters. Hence, annual growth (from the beginning of the first quarter through the end of the fourth quarter) at 21.551% yields

$$Y_t=Y_{t-1}(1+r)=100(1+0.21551)=121.551.$$

But this is the accumulated amount we would expect at the end of a year if the Y series started at $Y_0=100$ and grew at the compound annual rate of 21.551% from quarter to quarter:

Quarter 1: $Y_1=100(1+0.21551)^{0.25}=105.$
Quarter 2: $Y_2=105(1+0.21551)^{0.25}=110.25009.$
Quarter 3: $Y_3=110.25009(1+0.21551)^{0.25}=115.76259.$
Quarter 3: $Y_4=115.76259(1+0.21551)^{0.25}=121.55072.$ ■

For year-to-date calculations on quarterly data, the *annualized year-to-q percent change* is obtained from the formula

$$\left[\left(\frac{Y_{t,q}}{Y_{t-1,4}}\right)^{4/q} - 1\right] \times 100, \quad q = 1,2,3,4, \tag{2.10.1}$$

where $Y_{t-1,4}$ is the value of Y in the fourth quarter of year $t-1$, q is the number of quarters under consideration for year t, and $Y_{t,q}$ is the value of Y in the qth quarter of year t. And for year-to-date calculations for monthly data, the *annualized year-to-m percent change* is obtained via the formula

$$\left[\left(\frac{Y_{t,m}}{Y_{t-1,12}}\right)^{12/m} - 1\right] \times 100, \quad m = 1,2,\ldots,12, \tag{2.11.1}$$

where $Y_{t-1,12}$ is the value of Y in December of year $t-1$, m is the number of the month under consideration for year t, and $Y_{t,m}$ is the Y value in the mth month of year t.

EXAMPLE 2.6 Table 2.3 presents the monthly values of a variable Y from the last month (December) of 2008 up through the first eight months of 2009. As column three of this table reveals, Y grew by 14.93% during the first six months of 2009. Then in July and August of 2009, Y grew by 3.12% and 2.78% respectively. Is Y growth in July and August above or below the rate exhibited during the first six months of 2009?

To answer this question, we need to annualize these growth rates; that is, the growth rates need to be adjusted to reflect the amount Y would have changed over the course of a year if it had continued to grow at the given rate. We note first that the July growth rate is calculated, via Equation 2.1, as

TABLE 2.3 Monthly Percent Changes in Y_t

① Month	② Y_t	③ Monthly percent change (not annualized)	④ Monthly percent change (annualized)
December	653.7	n.a.	n.a.
January	663.2	1.45	18.90
February	670.1	1.04	13.23
March	693.4	3.48	50.70
April	714.6	3.06	43.53
May	740.1	3.56	52.31
June	751.3	1.51	19.75
July	774.7	3.12	44.49
August	796.2	2.78	38.88
December to June	n.a	14.93	32.09

$$\left[\left(\frac{774.7}{751.3}\right)-1\right]\times 100 = 3.12,$$

where 774.7 is the value of *Y* in July and 751.3 is the *Y* value for June. Similarly, the August growth rate is calculated as

$$\left[\left(\frac{796.2}{774.7}\right)-1\right]\times 100 = 2.78.$$

Also, the simple percent change from December 2008 to June 2009 is

$$\left[\left(\frac{751.3}{653.7}\right)-1\right]\times 100 = 14.93.$$

(Here we are applying a slight modification of Equation 2.1 or

$$\left[\left(\frac{Y_t}{Y_{t-m}}\right)-1\right]\times 100, \qquad (2.1.1)$$

where Y_{t-m} is the level of *Y* recorded *m* periods prior to period *t*. For this calculation, $m = 6$.)

In order to compare these three growth rates, we need to annualize them. An application of Equation 2.11 for July gives

$$\left[\left(\frac{774.7}{751.3}\right)^{12}-1\right]\times 100 = 44.49$$

(see column four of Table 2.3). Thus, 44.49% is the amount *Y* would have increased for the entire year if it had grown at the monthly rate of 3.12% for all 12 months. A second application of this formula for August yields

$$\left[\left(\frac{796.2}{774.7}\right)^{12}-1\right]\times 100 = 38.88.$$

Here 38.88% is the amount *Y* would have increased for the entire year if it had expanded at the monthly rate of 2.78% over the whole 12-month period. To obtain the annualized growth rate from December 2008 to June 2009, let us use Equation 2.11.1 to find

$$\left[\left(\frac{751.3}{653.7}\right)^{12/6}-1\right]\times 100 = 32.09.$$

This year-to-$m(=6)$ rate depicts the amount Y would have increased for the whole year if it had continued to grow at the pace experienced between January and June. On the basis of these annualized growth rates, we see that growth in Y for both the months of July and August is above the rate experienced in the first six months of 2009 (even though the column three entries in Table 2.3 could possibly lead one to conclude otherwise). ■

2.3.3 Comparing Two Time Series: Indexing Data to a Common Starting Point

In order to compare the performance of two time series data sets, say X_t and Y_t, we need to index the data to a common starting point. Hence, the initial values of X and Y must be set equal to each other so that differences over time between the X and Y variables can be highlighted. Once indexing to a common starting point is accomplished, we can readily determine their rates of growth, especially when the X_t and Y_t series are stated in different units. For instance, suppose Table 2.4 houses the values of the two time series data sets X_t and Y_t from 1998 to 2008. The time paths of these variables are depicted in Figure 2.1.

To index a set of time series data values, the observations must be made equal to each other at some given starting date. Typically, this value is 100. From this initial point onwards, every value is normalized to the starting value so as to preserve the same percentage value as in the original or indexed series. Subsequent values are then determined so that percentage changes in the indexed time series are coincident with those for the unindexed series. In sum, *indexing* involves modifying two (or more) time series data sets so that the resulting or indexed series start at the same value and change at the same rate as the unmodified or unindexed series.

So for data series X_t in Table 2.4, the indexed X_t's are calculated as

TABLE 2.4 Original and Indexed Time Series Data Sets X_t, Y_t

Year	X_t	Y_t	Indexed values of $X : X'_t$	Indexed values of $Y : Y'_t$
1998	57	420	100.00	100.00
1999	60	424	105.26	100.95
2000	65	430	114.04	102.38
2001	70	431	122.81	102.62
2002	73	432	128.07	102.86
2003	77	434	135.08	103.33
2004	78	435	136.84	103.57
2005	84	440	147.36	104.76
2006	86	446	150.88	106.19
2007	88	450	154.38	107.14
2008	94	452	164.91	107.62

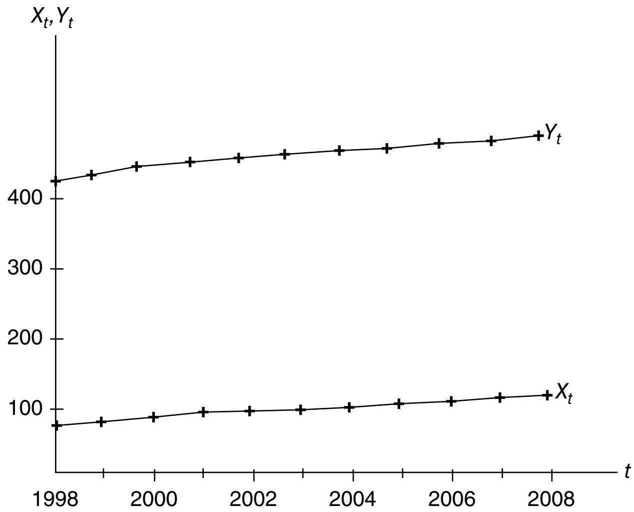

FIGURE 2.1 Time series observations on X,Y.

$$X'_t = \left(\frac{X_t}{X_0}\right) \times 100, \tag{2.2.1}$$

where X'_t denotes the indexed value of X_t and X_0 is the initial value of X. For the Y data set, the indexed Y_t's are determined as

$$Y'_t = \left(\frac{Y_t}{Y_0}\right) \times 100, \tag{2.2.2}$$

with Y'_t denoting the indexed value of Y and Y_0 representing the initial value of Y. The indexed values for both variables X and Y appear in Table 2.4 (here $X_0 = 57$ and $Y_0 = 420$).

What sort of information can be garnered from the indexed X and Y time series data? Between 1998 and 1999, the unindexed variable X_t increased from 57 to 60 or 5.26%, while the indexed variable X'_t also increased by 5.26% (i.e., by $105.26 - 100 = 5.26$). For this same time period, the unindexed variable Y_t increased from 420 to 424 or by 0.95%, while its indexed counterpart Y'_t also increased by 0.95% (i.e., by $100.95 - 100 = 0.95$).

What about the change in the X series from 1998 to, say, 2002? Here X_t increases by 28.07% ($128.07 - 100 = 28.07$); and between 1998 and 2002, the Y series increased by 2.86% ($102.86 - 100 = 2.86$). Additionally, for the entire 1998–2008 time span, the X data series grew by 64.91% ($164.91 - 100 = 64.91$), while the Y data series grew by only 7.62% (or $107.62 - 100 = 7.62$) (Fig. 2.2).

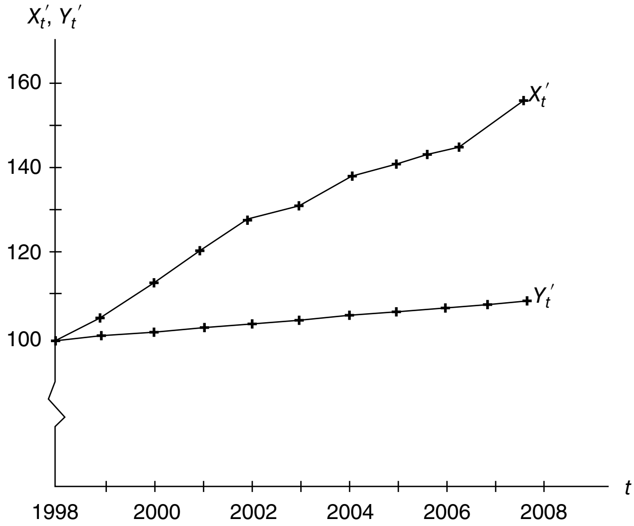

FIGURE 2.2 Time series observation on X_t', Y_t'.

2.4 DISCRETE VERSUS CONTINUOUS GROWTH

Suppose time t is measured in discrete units or intervals (e.g., the observations on some variable are made yearly, monthly, etc.). Then a constant growth series for a variable Y can be modeled as

$$Y_t = Y_0(1+r)^t, \tag{2.13}$$

where r is the constant proportionate rate of growth in Y per unit of time; that is, from Equation 2.13, the period-to-period or relative growth rate (RGR) is

$$R_t = \frac{Y_t}{Y_{t-1}} - 1 = \frac{Y_0(1+r)^t}{Y_0(1+r)^{t-1}} - 1 = (1+r) - 1 = r,$$

the constant proportionate (or compound interest) rate of growth.

Transforming Equation 2.13 to logarithms gives

$$\log_{10} Y_t = \log_{10} Y_0 + \log_{10}(1+r)t, \tag{2.13.1}$$

where the slope of this linear equation is $\log_{10}(1+r)$. Clearly this slope coefficient can be estimated (via ordinary least squares (OLS)) and denoted as

$$\widehat{\log_{10}(1+r)} = \log_{10}(1+\hat{r}) = b.$$

Then to find $\hat{r}$, set $1+\hat{r}=10^b$ or

$$\hat{r}=10^b-1. \tag{2.14}$$

If t is measured in one-year increments, then the estimated rate of growth in Y is $100\hat{r}\%$ per annum. Hence, the quarterly rate of growth must be

$$(1+\hat{q})^4=10^b=1+\hat{r}$$

or

$$\hat{q}=(1+\hat{r})^{1/4}-1=10^{b/4}-1 \tag{2.15}$$

(the estimated quarterly growth rate is $100\hat{q}\%$), and the monthly rate of growth is thus

$$(1+\hat{m})^{12}=10^b=1+\hat{r}$$

or

$$\hat{m}=(1+\hat{r})^{1/12}-1=10^{b/12}-1 \tag{2.16}$$

(the estimated monthly growth rate is $100\hat{m}\%$).

How many years will it take for Y_t to double in value? From Equation 2.13, set

$$\frac{Y_t}{Y_0}=(1+r)^t=2.$$

Then

$$t\log_{10}(1+r)=\log_{10}2$$

or the *doubling time* is

$$t=\frac{\log_{10}2}{\log_{10}(1+r)}=\frac{0.301}{\log_{10}(1+r)}. \tag{2.17}$$

So if, for instance, Y grows at 5% per annum, its doubling time is $t=0.301/\log_{10}(1.05)=0.301/0.02119=14.2$ years.

Next, suppose time t is a continuous variable. Then a constant growth series for Y can be represented (under continuous compounding) by the *exponential* or *semilogarithmic model*

$$Y_t=e^{\alpha+\beta t}=Y_0e^{\beta t}, \tag{2.18}$$

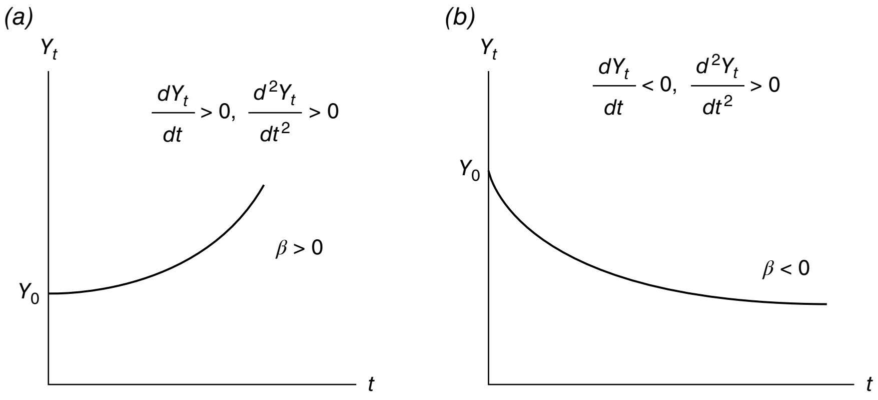

FIGURE 2.3 Semilogarithmic models: (**a**) Strictly increasing; (**b**) Strictly decreasing.

where, for $t = 0$, $Y_0 = e^{\alpha} =$ constant and

$$\frac{dY_t / dt}{Y_t} = \frac{\beta Y_0 e^{\beta t}}{Y_t} = \beta = \text{constant}$$

(Fig. 2.3) is the proportionate rate of change in Y_t per unit change in t (or the instantaneous rate of growth of Y at time t), while the relative rate of change in Y between periods $t - 1$ and t is

$$R_t = \frac{Y_t}{Y_{t-1}} - 1 = \frac{Y_0 e^{\beta t}}{Y_0 e^{\beta(t-1)}} - 1 = \frac{1}{e^{-\beta}} - 1 = e^{\beta} - 1 = \text{constant}. \tag{2.19}$$

(Compare this expression with Eq. 2.14.) A couple of additional points concerning the exponential growth rate β are in order. Using Equation 2.18 and the notion that the logarithm of Y_t grows at a constant rate, we have

$$\frac{Y_t}{Y_{t-1}} = \frac{Y_0 e^{\beta t}}{Y_0 e^{\beta(t-1)}} = e^{\beta}$$

or

$$\ln\left(\frac{Y_t}{Y_{t-1}}\right) = \ln Y_t - \ln Y_{t-1} = \beta.$$

Hence, the period-to-period difference in the logarithms of the Y series renders the exponential growth rate β. Also, using Equation 2.18, it is easily demonstrated that

$$\frac{d \ln Y_t}{dt} = \beta.$$

Transforming Equation 2.18 to (base 10) logarithms yields

$$\begin{aligned}\log_{10} Y_t &= \log_{10} Y_0 + (\beta \log_{10} e)t \\ &= \log_{10} Y_0 + (0.4343\beta)t,\end{aligned} \tag{2.20}$$

where the slope of this linear equation is equal to 0.4343β. If OLS is used to fit Equation 2.20 to a data set, then an estimate of the slope is

$$\hat{\beta} \log_{10} e = 0.4343\hat{\beta} = b$$

so that

$$\hat{\beta} = \frac{b}{0.4343}. \tag{2.21}$$

Suppose we set $\hat{\beta} \log_{10} e = \log_{10} (1+\hat{r})$ (the slopes of Eqs. 2.13.1 and 2.20 are equal). Then

$$\hat{\beta} = \frac{\log_{10}(1+\hat{r})}{\log_{10} e} = \ln(1+\hat{r})$$

so that $\hat{\beta}$ represents the rate at which continuous compounding yields the same growth in Y as a single discrete increment at rate $\hat{r}$. Note also that the continuous analogs of Equations 2.15 and 2.16 are $\ln \hat{q}$ and $\ln \hat{m}$, respectively.

Under a regime of continuous growth, we can also consider the issue of how long it will take for Y_t to double in value. From Equation 2.18, set

$$\frac{Y_t}{Y_0} = e^{\beta t} = 2.$$

Then

$$\beta t = \ln 2$$

or the *doubling time* is

$$t = \frac{\ln 2}{\beta} = \frac{0.693}{\beta}. \tag{2.22}$$

So if, for example, Y grows at 10% per year, its doubling time turns out to be 6.93 years.

In general, for t a continuous variable, the instantaneous *absolute growth rate* (AGR) in Y is

$$\text{ARG} = \frac{dY_t}{dt}, \tag{2.23}$$

while the *mean absolute growth rate over the time internal from t_1 to t_2* is

$$\begin{aligned}\overline{\text{ARG}} &= \frac{1}{t_2 - t_1}\int_{t_1}^{t_2}\frac{dY_t}{dt}\,dt = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} dY_t \\ &= \frac{1}{t_2 - t_1}\left(Y_t\Big]_{t_1}^{t_2}\right) = \frac{Y_2 - Y_1}{t_2 - t_1}.\end{aligned} \tag{2.24}$$

Next, for t again taken to be a continuous variable, the instantaneous RGR in Y is

$$\text{RGR} = \frac{1}{Y_t}\frac{dY_t}{dt}, \tag{2.25}$$

and the *mean relative growth rate over the time interval from t_1 to t_2* is

$$\begin{aligned}\overline{\text{RGR}} &= \frac{1}{t_2 - t_1}\int_{t_1}^{t_2}\frac{1}{Y_t}\frac{dY_t}{dt}\,dt = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2}\frac{dY_t}{Y_t} \\ &= \frac{1}{t_2 - t_1}\left(\ln Y_t\Big]_{t_1}^{t_2}\right) = \frac{\ln Y_2 - \ln Y_1}{t_2 - t_1}.\end{aligned} \tag{2.26}$$

2.5 THE GROWTH OF A VARIABLE EXPRESSED IN TERMS OF THE GROWTH OF ITS INDIVIDUAL ARGUMENTS

Suppose

$$Y = f(X_1, X_2, \ldots, X_n), \tag{2.27}$$

where Y and the X_i's, $i = 1, \ldots, n$, are time dependent and time varies continuously. Then at an instant t, the AGR *in Y* is

$$\frac{dY}{dt} = \frac{\partial f}{\partial X_1}\frac{dX_1}{dt} + \frac{\partial f}{\partial X_2}\frac{dX_2}{dt} + \cdots + \frac{\partial f}{\partial X_n}\frac{dX_n}{dt} = \sum_{i=1}^{n}\frac{\partial f}{\partial X_i}\frac{dX_i}{dt}. \tag{2.28}$$

At time t, let the X_i *relatives* be defined as

$$\theta_i = \frac{X_i}{Y}, \quad i = 1, \ldots, n. \tag{2.29}$$

Then from Equations 2.28 and 2.29, the RGR *in Y* is

$$\begin{aligned}R &= \frac{dY}{dt}\frac{1}{Y} = \sum_{i=1}^{n}\frac{\partial f}{\partial X_i}\frac{dX_i}{dt}\frac{1}{Y}\frac{X_i}{X_i} \\ &= \sum_{i=1}^{n}\frac{\partial f}{\partial X_i}\theta_i R_i,\end{aligned} \tag{2.30}$$

where

$$R_i = \frac{dX_i}{dt}\frac{1}{X_i}$$

is the RGR *in* X_i. As Equation 2.30 reveals, the RGR of $Y(R)$ can be written at any time t as the weighted sum of the RGRs of the X_i's (the R_i's), where the weights are determined as

$$w_i = \frac{\partial f}{\partial X_i}\theta_i, \quad i = 1,\ldots,n. \tag{2.31}$$

Next, the AGR *of the* ith X_i *relative* θ_i at a given time t can be written as the product of the ith X_i relative and the difference between the RGR of the X_i relative and the RGR of Y. To see this, let

$$\begin{aligned}\frac{d\theta_i}{dt} &= \frac{Y(dX_i/dt) - X_i(dY/dt)}{Y^2} = \frac{1}{Y}\frac{dX_i}{dt} - \theta_i R \\ &= \frac{1}{Y}\frac{dX_i}{dt}\frac{X_i}{X_i} - \theta_i R = \theta_i(R_i - R).\end{aligned} \tag{2.32}$$

Then the RGR *of the* X_i *relative* θ_i is

$$\frac{d\theta_i}{dt}\frac{1}{\theta_i} = R_i - R. \tag{2.33}$$

Note that Equation 2.33 can be expressed in terms of the RGRs R_i as follows. Substituting Equation 2.30 into Equation 2.33 renders

$$\begin{aligned}\frac{d\theta_i}{dt}\frac{1}{\theta_i} &= R_i - \sum_{i=1}^{n}\frac{\partial f}{\partial X_i}\theta_i R_i \\ &= R_i - \sum_{i=1}^{n} w_i R_i.\end{aligned} \tag{2.33.1}$$

And from Equations 2.31 and 2.32, the AGR *of* w_i at the time t is

$$\begin{aligned}\frac{dw_i}{dt} = \frac{d((\partial f/\partial X_i)\theta_i)}{dt} &= \frac{\partial^2 f}{\partial X_i^2}\frac{dX_i}{dt}\theta_i + \frac{\partial f}{\partial X_i}\frac{d\theta_i}{dt} \\ &= \frac{\partial^2 f}{\partial X_i^2}\frac{dX_i}{dt}\theta_i + \frac{\partial f}{\partial X_i}[\theta_i(R_i - R)],\end{aligned} \tag{2.34}$$

and the RGR *of* w_i is

$$\frac{dw_i}{dt}\frac{1}{w_i} = \frac{\partial^2 f}{\partial X_i^2}\frac{dX_i / dt}{\partial f / \partial X_i} + R_i - R. \tag{2.35}$$

Looking to specifics, if Equation 2.27 involves Y expressible as a linear combination of the X_i's, $i = 1, \ldots, n$, or

$$Y = \sum_{i=1}^{n} \alpha_i X_i, \quad \alpha_i \text{ constant for all } i, \tag{2.27.1}$$

or

$$1 = \sum_{i=1}^{n} \alpha_i \frac{X_i}{Y} = \sum_{i=1}^{n} \alpha_i \theta_i = \sum_{i=1}^{n} w_i, \tag{2.27.2}$$

where $w_i = \alpha_i \theta_i$, then Equation 2.28 becomes

$$\frac{dY}{dt} = \sum_{i=1}^{n} \alpha_i \frac{dX_i}{dt};$$

Equation 2.30 becomes

$$R = \sum_{i=1}^{n} \alpha_i \theta_i R_i; \tag{2.30.1}$$

Equation 2.32 simplifies to

$$\begin{aligned}
\frac{d\theta_i}{dt} &= \theta_i \left(R_i - \sum_{j=1}^{n} \alpha_j \theta_j R_j \right) \\
&= \theta_i \left(R_i - \sum_{j=1}^{n-1} \alpha_j \theta_j R_j - \alpha_n \theta_n R_n \right) \\
&= \theta_i \left(R_i - \sum_{j=1}^{n-1} \alpha_j \theta_j R_j - R_n \left(1 - \sum_{j=1}^{n-1} \alpha_j \theta_j \right) \right) \quad \text{(from Eq. 2.27.2)} \qquad (2.32.1) \\
&= \theta_i \left(R_i - R_n - \sum_{j=1}^{n-1} \alpha_j \theta_j R_j + R_n \sum_{j=1}^{n-1} \alpha_j \theta_j \right) \\
&= \theta_i \left(R_i - R_n + \sum_{j=1}^{n-1} (R_n - R_j) \alpha_j \theta_j \right), i \neq n;
\end{aligned}$$

and Equation 2.34 becomes

$$\begin{aligned}
\frac{dw_i}{dt} &= \alpha_i\theta_i(R_i - R) = \alpha_i\theta_i\left(R_i - \sum_{j=1}^{n}\alpha_j\theta_j R_j\right) \\
&= \alpha_i\theta_i\left(R_i - \sum_{j=1}^{n-1}\alpha_j\theta_j R_j - \alpha_n\theta_n R_n\right) \\
&= \alpha_i\theta_i\left(R_i - \sum_{j=1}^{n-1}\alpha_j\theta_j R_j - R_n\left(1 - \sum_{j=1}^{n-1}\alpha_j\theta_j\right)\right) \\
&= \alpha_i\theta_i\left(R_i - R_n - \sum_{j=1}^{n-1}\alpha_j\theta_j R_j + R_n\sum_{j=1}^{n}\alpha_j\theta_j\right) \\
&= \alpha_i\theta_i\left(R_i - R_n + \sum_{j=1}^{n-1}(R_j - R_n)\alpha_j\theta_j\right) \\
&= \alpha_i\frac{d\theta_i}{dt}, i \neq n.
\end{aligned} \tag{2.34.1}$$

Suppose that now each X_i exhibits exponential growth or

$$X_i = a_i e^{R_i t}, \quad i = 1, \ldots, n, \tag{2.36}$$

where

$$\frac{dX_i}{dt} = a_i R_i e^{R_i t}, \quad i = 1, \ldots, n,$$

and thus

$$\frac{dX_i}{dt}\frac{1}{X_i} = R_i, \quad i = 1, \ldots, n.$$

Now, if Y itself exhibits exponential growth, then, from Equation 2.27.1,

$$Y = \sum_{i=1}^{n}\alpha_i a_i e^{R_i t} = ae^{Rt}, \tag{2.37}$$

where

$$\frac{dY}{dt}\frac{1}{Y} = R.$$

Given Equation 2.36, in order for Equation 2.37 to hold, it must be the case that

$$R_1 = R_2 = \cdots = R_n = R. \tag{2.38}$$

To verify this, Equation 2.30.1 given Equation 2.38 simplifies to

$$R = \sum_{i=1}^{n} \alpha_i \theta_i R_i = R \sum_{i=1}^{n} \alpha_i \theta_i = R(1) = R$$

via Equation 2.27.2. In sum, Y cannot grow exponentially unless all the RGRs of the individual X_i's (the R_i's) are equal with the common value R.

However, if the RGRs of the X_i's are unequal, then the RGR of Y always follows a logistic function (see Eq. 3.6). To see this, let us, for convenience, examine the two-component version of Equation 2.34.1 or

$$\begin{aligned} \frac{dw_1}{dt} &= \alpha_1\theta_1(R_1 - R_2 + (R_2 - R_1)\alpha_1\theta_1) \\ &= \alpha_1\theta_1(R_1 - R_2 - (R_1 - R_2)\alpha_1\theta_1) \\ &= \alpha_1\theta_1[(1-\alpha_1\theta_1)(R_1 - R_2)] \end{aligned} \tag{2.39}$$

or, for $w_1 = \alpha_1\theta_1$,

$$\frac{dw_1}{w_1(1-w_1)} = (R_1 - R_2)\,dt,$$

where R_1 and R_2 are constants. Upon integrating this expression, we obtain the logistic function, that is,

$$-\ln\left(\frac{1-w_1}{w_1}\right) = (R_1 - R_2)t + \ln C \quad (\ln C \text{ constant})$$

and, for $c = C^{-1}$,

$$w_1 = \left(1 + ce^{(R_2 - R_1)t}\right)^{-1}. \tag{2.40}$$

From Equations 2.27.2 and 2.30.1,

$$\begin{aligned} R &= \sum_{i=1}^{2} \alpha_i\theta_i R_i = \sum_{i=1}^{2} w_i R_i \\ &= w_1 R_1 + w_2 R_2 = w_1 R_1 + (1-w_1)R_2 \\ &= R_2 - w_1(R_2 - R_1). \end{aligned} \tag{2.41}$$

Upon substituting Equation 2.40 into Equation 2.41, we get

$$R = R_2 - (R_2 - R_1)\left(1 + ce^{(R_2 - R_1)t}\right)^{-1}. \tag{2.42}$$

Since the second term on the right-hand side of Equation 2.42 is logistic in form, we see that the RGR in Y behaves according to the logistic function (Causton and Venus, 1981). So if $R_2 < R_1$, then the aforementioned second term is negative (R increases from a lower asymptote of R_2 to an upper asymptote of R_1), and if $R_2 > R_1$, the aforementioned term is positive (R increases from a lower asymptote of R_1 to an upper asymptote of R_2).

EXAMPLE 2.7 Suppose an economy has an *aggregate production function* (a functional relationship between a maximal flow of output (Y) resulting from a given flow of labor (L) and capital (K) services in production) of the form

$$Y(t) = A(t)F[L(t), K(t)], \tag{2.43}$$

where $A(t)$ depicts (neutral) technical progress and it is assumed that:

1. The inputs are essential or $F(0,0) = F(0,K) = F(L,0)$.
2. The input marginal products are positive or $F_L > 0$, $F_K > 0$.
3. Input marginal products are decreasing or $F_{LL} < 0$, $F_{KK} < 0$ (i.e., we have diminishing returns to the labor and capital inputs).
4. $dA/dt > 0$ or technical progress has a positive effect on output that is independent of the labor and capital inputs.

From Equation 2.43, we first determine that

$$\frac{dY}{dt} = \frac{dA}{dt}F + A(t)\left(F_L \frac{dL}{dt} + F_K \frac{dK}{dt}\right)$$

and thus

$$\begin{aligned}\frac{dY}{dt}\frac{1}{Y} &= \frac{dA}{dt}\frac{F}{Y} + \frac{A(t)}{Y}\left(F_L \frac{dL}{dt} + F_K \frac{dK}{dt}\right)\\ &= \frac{dA}{dt}\frac{1}{A} + \frac{1}{F}F_L\frac{dL}{dt} + \frac{1}{F}F_K\frac{dK}{dt}\\ &= \frac{dA}{dt}\frac{1}{A} + \frac{F_L}{F}\frac{dL}{dt}\frac{L}{L} + \frac{F_K}{F}\frac{dK}{dt}\frac{K}{K}\end{aligned}$$

or

$$R_Y = R_A + w_L R_L + w_K R_K, \tag{2.44}$$

where $w_L = F_L\theta_L, w_K = F_K\theta_K$ and $\theta_L = L/Y$ and $\theta_K = K/Y$ are the labor and capital relatives, respectively. Here R_Y, R_A, R_L, and R_K are the respective RGRs of output, technical progress, labor, and capital. Clearly Equation 2.44 indicates that the RGR of output is a weighted sum of the RGRs of technical progress ($w_A = 1$), labor, and capital. Moreover, the weights w_L and w_K are the output elasticities of the labor and capital inputs, respectively.

Next, we may express (via Eq. 2.33.1) the RGR of the L relative θ_L as

$$\begin{aligned}\frac{d\theta_L}{dt}\frac{1}{\theta_L} &= R_L - R_A - w_L R_L - w_K R_K \\ &= (1-w_L)R_L - R_A - w_K R_K,\end{aligned} \tag{2.45}$$

and the RGR of the K relative θ_K is

$$\begin{aligned}\frac{d\theta_K}{dt}\frac{1}{\theta_K} &= R_K - R_A - w_L R_L - w_K R_K \\ &= (1-w_K)R_K - R_A - w_L R_L.\end{aligned} \tag{2.46}$$

And from Equation 2.35, the RGRs of w_L and w_K are, respectively,

$$\frac{dw_L}{dt}\frac{1}{w_L} = F_{LL}\frac{dL/dt}{F_L} + R_L - R; \tag{2.47a}$$

$$\frac{dw_K}{dt}\frac{1}{w_K} = F_{KK}\frac{dK/dt}{F_K} + R_K - R. \tag{2.47b}$$

For the sake of specificity, let us assume that the aggregate production function has the *Cobb–Douglas* form

$$Y(t) = A(t)K(t)^{\alpha} L(t)^{1-\alpha}, \tag{2.48}$$

where $A(t)=A(0)e^{gt}$ is the level of (neutral) technical progress in period t (here $(1/A)\ dA/dt=R_A=g\,(>0)$ is constant or technology growth is exogenous) and $1-\alpha=w_L$ and $\alpha=w_K$ are the output elasticities with respect to L and K, respectively. Since $\alpha+(1-\alpha)=1$, we have constant returns to scale in production or $F(\lambda L,\lambda K)=\lambda F(L,K)$.

From Equations 2.44 and 2.48, we have

$$R_Y = g + (1-\alpha)R_L + \alpha R_K; \tag{2.49}$$

from Equations 2.44–2.46 and 2.48,

1. $\dfrac{d\theta_L}{dt}\dfrac{1}{\theta_L} = \alpha(R_L - R_K) - g,$
2. $\dfrac{d\theta_K}{dt}\dfrac{1}{\theta_K} = (1-\alpha)(R_K - R_L) - g;$

and from Equations 2.44, 2.47, and 2.48,

1. $\dfrac{dw_L}{dt}\dfrac{1}{w_L} = g - \alpha R_K,$
2. $\dfrac{dw_K}{dt}\dfrac{1}{w_K} = g - (1-\alpha)R_L.$

While the aforementioned R_Y is the relative rate of growth in output Y, we need to ultimately specify how the time path of Y is determined for a given (albeit stylized) economy. In particular, we shall examine the structure of the *Solow–Swan (neoclassical) growth model*. To this end, let us assume the following: there is no government or international trade (we posit a closed economy); technology is constant; there is full employment; the initial values of capital and labor, K_0 and L_0, respectively, are given; we have a constant savings rate $s = S(t)/Y(t)$ (S is aggregate savings) and a constant depreciation rate d; and the rate of growth of labor is $(1/L)dL/dt = R_L = n =$ constant.

Let us write Equation 2.48 as

$$y(t) = f(k(t)) = A(t)k(t)^{\alpha}, \quad \alpha < 1, \tag{2.48.1}$$

where $y(t) = Y(t)/L(t)$ is output per worker and $k(t) = K(t)/L(t)$ denotes capital per worker. Since the savings function $I(t) = sY(t) = S(t)$, the capital accumulation equation can be written as $dK/dt = sY(t) - dK(t)$ (the change in net capital stock equals gross investment (= savings) less depreciation). Then

$$\frac{dK/dt}{K(t)} = s\frac{Y(t)}{K(t)} - d = s\frac{y(t)}{k(t)} - d. \tag{2.50}$$

And since

$$\frac{dk}{dt}\frac{1}{k} = \frac{dK}{dt}\frac{1}{K} - \frac{dL}{dt}\frac{1}{L} = \frac{dK}{dt}\frac{1}{K} - n, \tag{2.51}$$

it follows from Equations 2.48.1, 2.50, and 2.51 that

$$\frac{dk}{dt}\frac{1}{k} + n = s\frac{y(t)}{k(t)} - d$$

or

$$\frac{dk}{dt} = sAk^{\alpha} - (d+n)k, \tag{2.52}$$

the *fundamental differential equation of the Solow–Swan growth model*.

Figure 2.4 provides a graphical description of the workings of the Solow–Swan model. At point A, $dk/dt = 0$ or $sAk^{\alpha} = (d+n)k$. Hence, a unique positive *steady-state* level of capital per worker k^* is determined. Moreover, this equilibrium solution is stable in that (i) if $sAk^{\alpha} > (d+n)k$ (the case at, say, k_1), then investment per worker exceeds the amount needed to account for depreciation and population growth so that $\Delta k > 0$ and the system moves to k^*; and (ii) if $sAk^{\alpha} < (d+n)k$ (the case at, say, k_2), then capital per worker falls ($\Delta k < 0$) since investment per worker is not high enough to offset population growth and depreciation and thus a movement back to k^* is warranted. Thus, the Solow–Swan growth model implies that there is no growth in the steady state; there is only positive or negative growth along the transition path (Eq. 2.52). (A more detailed account of equilibrium and stability analysis is provided in Appendix 10.B.)

Given that the equilibrium or steady-state capital to labor ratio k^* is determined where $dk/dt = 0$, what are the particular values of k^* and y^*? To find k^*, let us solve

$$0 = sA(k^*)^{\alpha} - (d+n)k^*$$

so as to obtain

$$k^* = \left(\frac{sA}{d+n}\right)^{1/(1-\alpha)}.$$

Then

$$y^* = A(k^*)^{\alpha} = \left(\frac{sA}{d+n}\right)^{\alpha/(1-\alpha)}.$$

And if we express total consumption as $C = Y - S$, then consumption per worker is $c = C/L = Y/L - S/L$ *or* $c = y - sy$. Hence, at k^*, steady-state consumption per worker is $c^* = (1-s)y^*$ (distance AB in Fig. 2.4). Looked at in another fashion, c^* is the output left over after providing for the investment that maintains the steady state. Hence, distance AB is also

$$c^* = y^* - (d+n)k^*.$$

The Solow–Swan model operates to put the economy on a *balanced growth path* (BGP)—a long-run trajectory on which output per worker (y), capita per worker (k),

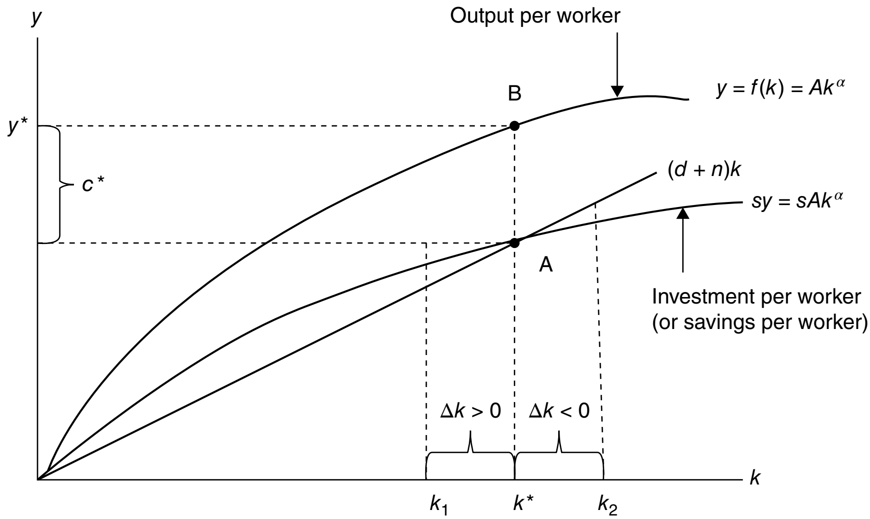

FIGURE 2.4 A locally stable steady-state solution.

and consumption per worker (c) all grow at constant (and possibly different) relative rates over time. (In fact, the "steady state" is a BGP with a zero growth rate $dk/dt = 0$.) To see this, let's start with Equation 2.52 or

$$R_k = \frac{dk/dt}{k} = s\frac{y}{k} - (d+n).$$

Along a BGP, R_k is constant. Hence, y/k must be constant or $R_y = R_k$. Moreover, from Equation 2.49 and the fact that $R_k = R_K - R_L$ and $R_y = R_Y - R_L$,

$$R_Y - R_L = g + \alpha(R_K - R_L)$$

or $R_y = g + \alpha R_k$. And since we have just demonstrated that $R_y = R_k$, it follows that the preceding expression can be written as $R_y = g/(1-\alpha)$; that is, only g and α can affect the relative rate of growth of output per worker along the BGP.

The preceding discussion has pointed to two sources of increases in output per worker y: technical progress, and increases in capital per worker k (called capital deepening). Moreover, in a Solow–Swan world, the economy tends to converge over time to a steady-state or constant-output growth path. And as indicated earlier, output growth is constant if capital growth is constant. But capital growth will be constant if Y/K is constant or if $R_k = R_y$ so that the steady-state growth rate is $R_Y = g/(1-\alpha) + n$ or $R_y = g/(1-\alpha)$. However, this model illustrates a very important point about the long-run growth of an economy: investment in capital cannot drive long-run growth in output per worker; the economy needs growth in technology (A) to offset diminishing returns to capital. That is, from $y = Ak^{\alpha}$, $\alpha < 1$, the marginal product of k is $MP_k = dy/dk = \alpha A k^{\alpha-1}$. So if A increases, then MP_k does likewise, and thus enhanced technology has a positive long-run impact on the RGR of y. This is realized by an upward shift in the y curve (and consequently in the sy curve) in Figure 2.4. This leads to a higher level of k^* and thus of y^* (equilibrium point A slides upwards along the $(d + n)k$ line to a new set of equilibrium values of k and y).

We noted earlier that steady-state consumption per worker is $c^* = y^* - sy^* = y^* - (d + n)k^*$. In addition, the optimal or *golden rule* (GR) *savings ratio* is the value of s, s_{GR}, which maximizes consumption per worker. (After all, an important issue for any society to consider is the determination of the amount it should save to achieve maximum welfare.) That is, to determine s_{GR}, let us solve the problem

$$\max_k c^* = \max_k \{y^* - (d+n)k^*\}.$$

To this end, we set

$$\frac{dy^*}{dk^*} - (d+n) = MP_k^* - (d+n) = 0$$

(assuming that the second-order conditions for a maximum are satisfied at k^*) or $MP_k^* = dy^*/dk^* = d + n$, the so-called GR *of capital accumulation*.

For the Cobb–Douglas case, $y = Ak^{\alpha}$, $\alpha < 1$, and thus $MP_k^* = \alpha A(k^*)^{\alpha-1}$. Since the steady state is where $sy^* = (d + n)k^*$, $d + n = s(y^*/k^*)$ and thus we get

$$\alpha A(k^*)^{\alpha-1} = s(y^*/k^*).$$

Then

$$s_{\mathrm{GR}} = \frac{\alpha A(k^*)^{\alpha-1} k^*}{y^*} = \frac{\alpha A(k^*)^{\alpha}}{A(k^*)^{\alpha}} = \alpha;$$

that is, the optimal savings ratio is the GR savings ratio; it equals the elasticity of y with respect to k or α when the economy exhibits steady-state growth and the production technology is Cobb–Douglas. ■

2.6 GROWTH RATE VARIABILITY

We previously defined the instantaneous growth rate in the variable Y at time t (in, say, years) as $(1/Y_t)\, dY_t/dt$ (Eq. 2.25). Let us set

$$\frac{1}{Y_t}\frac{dY_t}{dt} = r(t), \tag{2.53}$$

where $r(t)$ indicates that this growth rate varies with t. We also found that the change in Y over the time period from t_1 to t_2 could be calculated as

$$\begin{aligned} \int_{t_1}^{t_2} r(t)\,dt &= \int_{t_1}^{t_2} \frac{d\ln Y_t}{dt} dt = \ln Y_t \big]_{t_1}^{t_2} \\ &= \ln Y_2 - \ln Y_1 = \ln\left(\frac{Y_2}{Y_1}\right) \end{aligned} \tag{2.54}$$

or

$$e^{\int_{t_1}^{t_2} r(t)\,dt} = \frac{Y_2}{Y_1}. \tag{2.55}$$

Thus, the Y growth relative over the time interval $[t_1, t_2]$, Y_2/Y_1, is the exponential of the sum of the growth rates that obtained over that period.

If both sides of Equation 2.54 are divided by $t_2 - t_1$, then, as in Equation 2.26, the mean value of the instantaneous growth rate over $[t_1, t_2]$ is

$$\bar{r} = \frac{\int_{t_1}^{t_2} r(t)\,dt}{t_2 - t_1} = \frac{\ln(Y_2/Y_1)}{t_2 - t_1}. \tag{2.54.1}$$

As this expression reveals, the impact on Y of a varying rate of growth $r(t)$ is the same as if the mean rate $\bar{r}$ applied at each moment over the time interval $[t_1, t_2]$ since, from Equation 2.55, $Y_2 = Y_1 e^{\bar{r}(t_2-t_1)}$.

Since the rate of increase in Y, $r(t)$, can change over time, let us examine a few special cases for this expression on the interval $[t_1, t_2]$:

1. $r(t) = r^* =$ constant over $[t_1, t_2]$. Then Equation 2.55 simplifies to

$$Y_2 = Y_1 e^{\int_{t_1}^{t_2} r^* dt} = y_1 e^{r^*(t_2-t_1)} \tag{2.56}$$

(Y increases exponentially in $(t_2 - t_1)$). Let us rewrite this expression so as to obtain

$$r^* = \frac{\ln(Y_2/Y_1)}{t_2 - t_1}; \tag{2.57}$$

that is, if the instantaneous growth rate is constant over the time interval $[t_1, t_2]$, then its value r^* is obtained simply by knowing the value of Y at the beginning and at the end of the interval. Note that Equations 2.54.1 and 2.57 yield the same result. Why?

2. $r(t) = \alpha t$ over $[t_1, t_2]$, α a constant. Then

$$Y_2 = Y_1 e^{\int_{t_1}^{t_2} \alpha t dt} = Y_1 e^{\frac{\alpha}{2}\left(t_2^2 - t_1^2\right)}$$

(Y increases exponentially in $t_2^2 - t_1^2$ for $\alpha > 0$) and thus

$$\frac{\ln(Y_2/Y_1)}{t_2 - t_1} = \frac{\alpha}{2}\frac{t_2^2 - t_1^2}{t_2 - t_1}. \tag{2.58}$$

3. $r(t) = \alpha/t$ over $[t_1, t_2]$, α a constant. Now

$$Y_2 = Y_1 e^{\int_{t_1}^{t_2} (\alpha/t dt)} = Y_1 e^{\alpha \ln(t_2/t_1)} = Y_1 (t_2/t_1)^{\alpha}$$

(Y increases with t_2/t_1 for $\alpha > 0$; and for $\alpha = 1$, Y increases linearly with t_2/t_1). Then

$$\frac{\ln(Y_2/Y_1)}{t_2 - t_1} = \frac{\alpha \ln(t_2/t_1)}{t_2 - t_1}. \tag{2.59}$$

2.7 GROWTH IN A MIXTURE OF VARIABLES

Up to this point in our discussion of growth rates, we have focused exclusively on the instantaneous rate of growth of a single variable Y, which we denoted as $(1/Y_t)\, dY_t/dt$. However, suppose we have a mixture of variables Y_i, $i = 1, \ldots, k$, where the ith variable grows at a rate r_i from an initial value of, say, $Y_i(0)$ (this notation avoids double subscripts) to a total level at time t of $Y_i(t) = Y_i(0)e^{r_i t}$. Then the aggregate level reached by all variables in combination at time t is

$$Y(t) = \Sigma_i Y_i(0)e^{r_i t}. \tag{2.60}$$

Given $Y(t)$, let us denote its rate of increase as

$$\overline{r}_w = \frac{dY(t)}{dt}\frac{1}{Y(t)} = \frac{\Sigma_i Y_i(0) r_i e^{r_i t}}{\Sigma_i Y_i(0)e^{r_i t}} = \frac{\Sigma_i r_i Y_i(t)}{\Sigma_i Y_i(t)}. \tag{2.61}$$

Here $\overline{r}_w$ is the weighted mean of the r_i's, where the weights are the quantities $Y_i(t)/\Sigma_i Y_i(t)$ at time t. Clearly the weight for the ith variable Y_i is its proportion of the total level reached by all variables in combination or of $\Sigma_i Y_i(t)$.

Let us now consider the rate of change in the weighted mean $\overline{r}_w$. Under an application of the quotient rule of the calculus,[2] we have

$$\begin{aligned} \frac{d\overline{r}_w}{dt} &= \frac{\Sigma_i Y_i(0) r_i^2 e^{r_i t}}{\Sigma_i Y_i(0)e^{r_i t}} - \left(\frac{\Sigma_i Y_i(0) r_i e^{r_i t}}{\Sigma_i Y_i(0)e^{r_i t}}\right)^2 \\ &= \frac{\Sigma_i r_i^2 Y_i(t)}{\Sigma_i Y_i(t)} - \overline{r}_w^2. \end{aligned} \tag{2.62}$$

Thus, the rate of change in the weighted mean rate of change at time t equals the difference between the weighted mean square of the r_i's and the square of the weighted mean of the r_i's, where again the weights are $Y_i(t)/\Sigma_i Y_i(t)$. But this expression is the weighted mean square deviation from the weighted mean of the r_i's[3] or

$$\frac{d\overline{r}_w}{dt} = \sigma^2(t), \tag{2.63}$$

the variance of the weighted r_i's. Since $\sigma^2(t) \geq 0$, it follows that $\overline{r}_w$ is monotonically increasing over time. Note that $\sigma^2(t) = 0$ if the r_i's all have the same value. Then $d\overline{r}_w / dt = 0$ so that $\overline{r}_w$ must be a constant.

[2] For both $u(t)$ and $v(t)$ taken to be differentiable functions of t,

$$\frac{d(u/v)}{dt} = \frac{v(du/dt) - u(dv/dt)}{v^2} = \frac{1}{v}\frac{du}{dt} - \frac{u}{v^2}\frac{dv}{dt}.$$

[3] Let a variable X have values, $X_1, \ldots, X_n$. Then for weights w_i, $i = 1, \ldots, n$, the weighted mean square deviation from the weighted mean of the X_i's, $i = 1, \ldots, n$, is

$$\frac{\Sigma_i w_i X_i^2}{\Sigma w_i} - \left(\frac{\Sigma_i w_i X_i}{\Sigma w_i}\right) = \text{variance of the weighted } X_i\text{'s}.$$

If the X_i's are not weighted, then the (ordinary) variance of the X_i's is the familiar expression

$$\frac{\Sigma_i X_i^2}{n} - \overline{X}^2.$$

3

PARAMETRIC GROWTH CURVE MODELING

3.1 INTRODUCTION

The preceding chapter considered the computation of a variety of growth rates for a generic variable measured over time, with time either expressed in terms of discrete units or treated as a continuous variable. Also included was a very special type of parametric growth model that exhibited a constant relative rate of growth, namely, the exponential growth model. In this chapter we shall explore a whole host of alternative growth models that have been employed to study growth behavior in diverse fields such as forestry, agriculture, biology, engineering, and economics, to name but a few.

While linear or exponential growth may at times be appropriate, we shall, for the most part, concentrate on sigmoidal (S-shaped) growth curves. In this regard, some of the more common parametric growth models covered herein are:

Linear	Janoschek
Logarithmic reciprocal	Lundqvist–Korf
Logistic	Hossfeld
Gompertz	Stannard
Weibull	Schnute
Negative exponential	Morgan–Mercer–Flodin (M–M–F)
von Bertalanffy	McDill–Amateis

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

Chapman–Richards (C–R)	Levakovic (I, III)
Log logistic	Yoshida (I)
Brody	Sloboda

Although this list is by no means exhaustive, it gives a very good account of the mainstream types of growth models, which have become popular over the recent past.

We have referred to the aforementioned growth models as being "parametric" in nature. This is because these functions (and their properties) have been defined in terms of a set of parameters, which describe (either separately or in combination) their fundamental characteristics. Such parameters represent or relate to an *asymptote*, an *intercept on the Y-axis*, the *rate at which the response variable Y changes* from some starting or initial value to its terminal or final value, and additional parameters as required for suitable *flexibility in modeling*.

3.2 THE LINEAR GROWTH MODEL

Let us assume that the series of Y values is generated by the *linear model*

$$Y_t = Y_0 + \beta t,\ t = 1, 2, \ldots, n, \tag{3.1}$$

where Y_0 is the value of Y at the beginning of period 1 and β(=constant) is the slope (Fig. 3.1). Then the sequence of Y_t values and the relative rates of change in Y between periods $t-1$ and t (the R_t's) are provided in Table 3.1.

Note that for Y_0 positive and for increasing t, if $\beta > 0$, then the R_t's steadily decrease in value; and if $\beta < 0$, then the R_t's steadily increase. So even if the slope of Equation 3.1 is constant, the period-to-period relative growth rates determined from this expression are monotonic increasing or decreasing, depending on the sign of the slope.

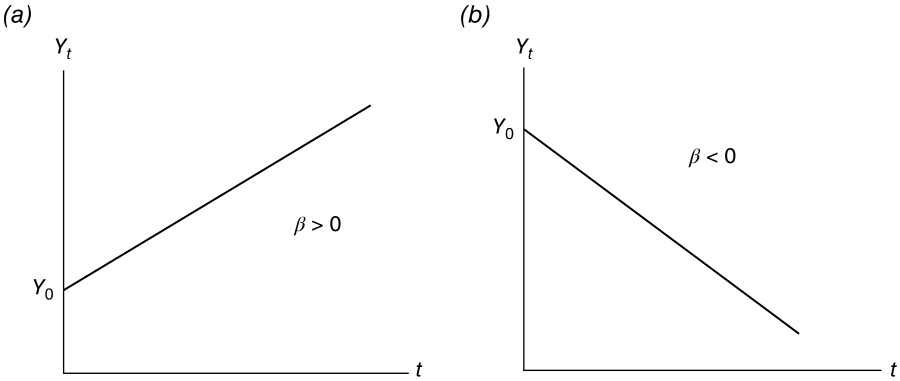

FIGURE 3.1 Linear models: (**a**) Positive slope; (**b**) Negative slope.

TABLE 3.1 Relative Growth Rates in *Y*

Period t	Y_t	R_t
1	$Y_1 = Y_0 + \beta$	$R_1 = \frac{Y_1 - Y_0}{Y_0} = \frac{\beta}{Y_0}$
2	$Y_2 = Y_0 + 2\beta$	$R_2 = \frac{Y_2 - Y_1}{Y_1} = \frac{\beta}{Y_0 + \beta}$
3	$Y_3 = Y_0 + 3\beta$	$R_3 = \frac{Y_3 - Y_2}{Y_2} = \frac{\beta}{Y_0 + 2\beta}$
.	.	.
.	.	.
.	.	.
n	$Y_n = Y_0 + n\beta$	$R_n = \frac{Y_n - Y_{n-1}}{Y_{n-1}} = \frac{\beta}{Y_0 + (n-1)\beta}$

3.3 THE LOGARITHMIC RECIPROCAL MODEL

Suppose that the series of values of a variable Y is generated by the *logarithmic reciprocal model*

$$Y_t = e^{\alpha - \beta/t}, \quad t \neq 0, \tag{3.2}$$

or, upon transforming to logarithms,

$$\ln Y_t = \alpha - \frac{\beta}{t}, \quad t \neq 0, \tag{3.2.1}$$

where $Y_t \to 0$ as $t \to +0$ (i.e., this expression is right-continuous at the origin) so that this function is defined as zero for $t=0$.

What about the general shape of the logarithmic reciprocal function? Given Equation 3.2, it is readily demonstrated that

$$\frac{dY_t}{dt} = \left(\frac{\beta}{t^2}\right) Y_t, \tag{3.3a}$$

$$\frac{d^2Y_t}{dt^2} = \frac{\beta}{t^3}\left(\frac{\beta}{t} - 2\right) Y_t. \tag{3.3b}$$

Since $d^2Y_t/dt^2 = 0$ for $t = \beta/2$, it follows that Equation 3.2 has a point of inflection[1] at $t_{\text{inf}} = \beta/2$. Substituting $t = \beta/2$ into Equation 3.2 yields $Y_{t_{\text{inf}}} = e^{\alpha-2} = e^{\alpha}e^{-2} = 0.1353e^{\alpha}$ (Fig. 3.2). Moreover, for $t < \beta/2$, $d^2Y_t/dt^2 < 0$ (the slope increases with t so that the

[1] A *point of inflection* is a point where a curve crosses over its tangent line and changes the direction of its concavity from upwards to downwards, or vice versa. In this regard, for $Y = f(t)$, if $f''(t_0) = 0$ and $f'''(t_0) \neq 0$, then f has a point of inflection at $t = t_0$ (a sufficient condition).

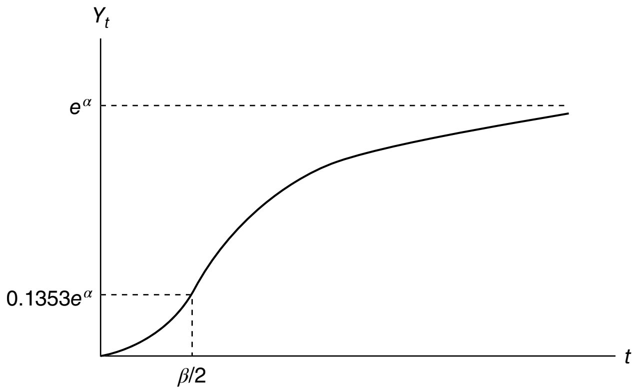

FIGURE 3.2 Logarithmic reciprocal model.

curve is concave upwards); and for $t>\beta/2$, $d^2Y_t/dt^2>0$ (the slope decreases as t increases so that the curve is concave downwards). Hence, Equation 3.2 provides an S-shaped pattern to growth. Additionally,

$$\lim_{t\to\infty} e^{\alpha-\beta/t} = e^\alpha = Y_\infty$$

so that e^α is a horizontal (upper) asymptote for Y_t and represents a *saturation point* for growth in Y.

Next, the instantaneous rate of growth of Y_t at time t is

$$\frac{dY_t/dt}{Y_t} = \frac{\beta}{t^2}, \tag{3.4}$$

while the relative rate of change in Y between periods $t-1$ and t is

$$R_t = \frac{Y_t}{Y_{t-1}} - 1 = e^{\beta/t(t-1)} - 1 \tag{3.5}$$

(β is thus a rate of growth parameter). Note that neither of these growth rates is constant and both decrease as t increases. In fact, each rate of growth drops abruptly beyond very small values of t.

3.4 THE LOGISTIC MODEL

Let the series of values of a variable Y over time be determined by the *logistic model* (Verhulst, 1838)

$$Y_t = \frac{Y_\infty}{1+\alpha e^{-\beta t}}, \quad t \geq 0, \tag{3.6}$$

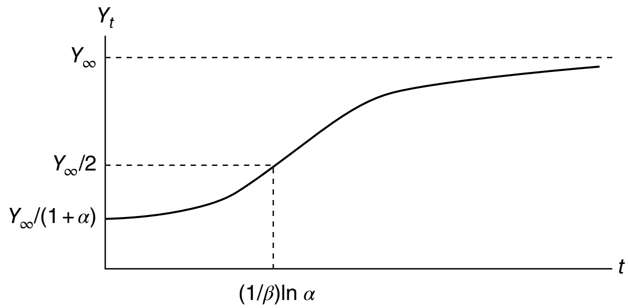

FIGURE 3.3 Logistic model.

where the parameters α, β (a growth rate parameter) and Y_∞ are all positive. At $t=0$, this curve starts out at $Y_0 = Y_\infty/(1+\alpha)$, and as $t \to \infty$, $Y_t \to Y_\infty$ (Y_∞ is the (upper) horizontal asymptote of the logistic function—it is termed the s*aturation* or *limit to growth parameter*).

To examine the shape of the logistic function, let us first find

$$\frac{dY_t}{dt} = \frac{\beta}{Y_\infty} Y_t \left(Y_\infty - Y_t\right), \tag{3.7a}$$

$$\frac{d^2Y_t}{dt^2} = \frac{\beta}{Y_\infty}\left(Y_\infty - 2Y_t\right)\frac{dY_t}{dt}. \tag{3.7b}$$

From Equation 3.7 it is easily seen that $d^2Y_t/dt^2 = 0$ for $Y_t = Y_\infty/2$ (half the saturation level). Then a substitution of this Y value into Equation 3.6 yields $t = \ln\alpha/\beta$, where it is assumed that $\alpha > 1$ so that $\ln\alpha > 0$. Hence, the logistic function has a point of inflection at $\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = (\ln\alpha/\beta, Y_\infty/2)$ and is symmetric about this point (Fig. 3.3). Also, for $t < \ln\alpha/\beta$, $d^2Y_t/dt^2 > 0$ (the curve is concave upwards); and for $t > \ln\alpha/\beta$, $d^2Y_t/dt^2 < 0$ (the curve is concave downwards). Here too Equation 3.6 exhibits an S-shaped pattern to growth.

Next, from Equations 3.7a and 3.7b,

$$\frac{dY_t/dt}{Y_t} = \frac{\beta}{Y_\infty}\left(Y_\infty - Y_t\right), \tag{3.8}$$

so that the instantaneous rate of growth of Y_t at time t is proportional to the amount by which Y_t falls short of the saturation parameter ceiling to Y growth $(Y_\infty - Y_t)$. Also, the relative rate of change in Y between periods $t-1$ and t is

$$R_t = \frac{Y_t}{Y_{t-1}} - 1 = \frac{e^{\beta-1}}{Y_\infty}\left(Y_\infty - Y_t\right). \tag{3.9}$$

Here also this growth rate is proportional to $Y_\infty - Y_t$.

A more detailed discussion of the logistic function (its derivation and comparison with the exponential or semilogarithmic function) is offered in Appendix 3.A.

3.5 THE GOMPERTZ MODEL

Suppose the time profile of a variable Y is specified by the *Gompertz* (1825) *model*

$$Y_t = Y_\infty e^{-\alpha e^{-\beta t}}, \quad t \geq 0, \tag{3.10}$$

where the parameters α, β (the rate of growth parameter) and Y_∞ are all positive. (A derivation of this expression is provided by Appendix 3.B.) For $t=0$, the initial value of Y is $Y_0 = Y_\infty e^{-\alpha}$, and as $t \to +\infty$, $Y_t \to Y_\infty$ (the upper limit (a horizontal asymptote) to growth). As was the case for the logistic function, Y_∞ is termed the *limit to growth parameter*.

To help ascertain the shape of the Gompertz function, we first look to

$$\frac{dY_t}{dt} = \alpha\beta e^{-\beta t} Y_t, \tag{3.11a}$$

$$\frac{d^2 Y_t}{dt^2} = \alpha\beta^2 e^{-\beta t}\left(\alpha e^{-\beta t} - 1\right) Y_t. \tag{3.11b}$$

From Equation 3.11b, it is readily seen that $d^2Y_t/dt^2 = 0$ at $t = \ln \alpha/\beta$. Then a substitution of this t value into Equation 3.10 gives $Y_t = 0.36788 Y_\infty$. Thus, the Gompertz function has a point of inflection at $\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = (\ln \alpha/\beta,\ 0.36788 Y_\infty)$ (see Fig. 3.4) and thus is S-shaped or sigmoidal.

Next, from Equation 3.11a, it follows that

$$\frac{dY_t / dt}{Y_t} = \alpha\beta e^{-\beta t}. \tag{3.12}$$

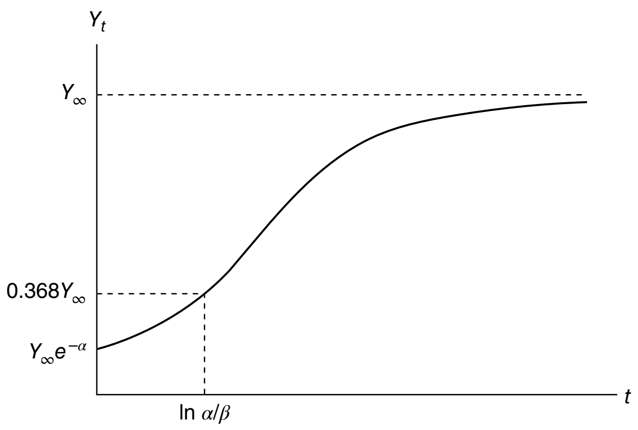

FIGURE 3.4 Gompertz model.

Thus, the instantaneous rate of growth of Y_t at time t is an exponentially decreasing function of time. Given Equation 3.12, it is easily demonstrated, via Equation 3.10, that

$$\frac{dY_t / dt}{Y_t} = \beta(\ln Y_\infty - \ln Y_t) \tag{3.13}$$

(we have a linear relationship between the instantaneous growth rate and $\ln Y_t$, with the said growth rate proportional to the amount by which $\ln Y_t$ falls short of $\ln Y_\infty$), while (again using Eq. 3.12)

$$\ln\left(\frac{dY_t / dt}{Y_t}\right) = \ln(\alpha\beta) - \beta t \tag{3.14}$$

(we have a linear relationship between the logarithm of the instantaneous growth rate and t). Also, the relative rate of change in Y between periods t and $t - 1$ is

$$R_t = \frac{Y_t}{Y_{t-1}} - 1 = \left(\frac{Y_t}{Y_\infty}\right)^{e^{\beta}-1} - 1. \tag{3.15}$$

3.6 THE WEIBULL MODEL

Let us assume that the series of values of a variable Y over time are generated by the *Weibull* (1951) *model*

$$Y_t = Y_\infty - \alpha e^{-\beta t^\gamma}, \quad t \geq 0, \tag{3.16}$$

where the parameters Y_∞, α, β (the growth rate parameter for a fixed γ) and γ (a shape parameter) are all positive. The source of this expression is a generalization (extension) of the Weibull cumulative distribution function

$$F(t;\alpha,\theta) = 1 - e^{-(t/\alpha)^\theta},$$

α and θ being parameters. Along with the introduction of additional parameters, "1" is replaced by Y_∞ as a less restrictive upper limit to growth; that is, $\lim_{t\to\infty} Y_t = Y_\infty$. Hence, Y_∞ is termed the *limit to growth parameter*. For $t=0$, the initial value of Y is $Y_0 = Y_\infty - \alpha$.

Looking to the first and second derivatives of Equation 3.16, we have, respectively,

$$\frac{dY_t}{dt} = \beta\gamma t^{\gamma-1}\left(Y_\infty - Y_t\right), \tag{3.17a}$$

$$\frac{d^2Y_t}{dt^2} = \beta\gamma t^{\gamma-1}\left[\left(\gamma-1\right)t^{-1}\left(Y_\infty - Y_t\right) - \frac{dY_t}{dt}\right]. \tag{3.17b}$$

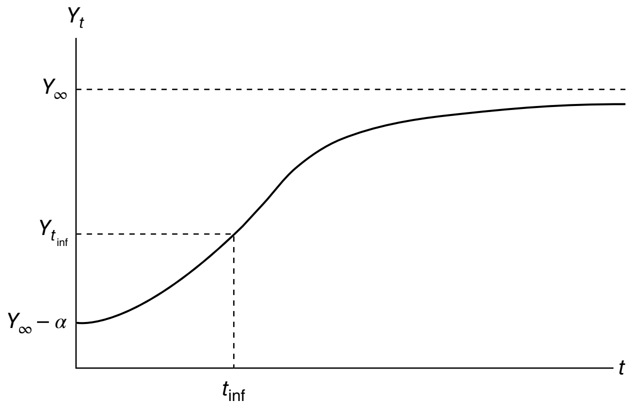

FIGURE 3.5 Weibull model.

Using Equation 3.17b, we can readily determine that $d^2Y_t/dt^2 = 0$ for $t = t_{\text{inf}} = [(\gamma - 1)/\beta\gamma]^{1/\gamma}$. Substituting this t value into Equation 3.16 gives $Y_t = Y_{t_{\text{inf}}} = Y_\infty - \alpha\exp[-(\gamma-1)/\gamma]$. Thus, the Weibull growth equation has a point of inflection at $\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right)$ and thus is S-shaped (Fig. 3.5).

From (3.17a) it follows that

$$\frac{dY_t/dt}{Y_t} = \beta\gamma t^{\gamma-1}\left(\frac{Y_\infty}{Y_t} - 1\right). \tag{3.18}$$

Hence, the instantaneous rate of growth in Y_t at time t approaches zero as $t \to +\infty$ (or as $Y_t \to Y_\infty$). And from Equation 3.18, we can write

$$\ln\left(\frac{dY_t/dt}{Y_t}\right) - \ln\left(\frac{Y_\infty}{Y_t} - 1\right) = \ln(\beta\gamma) + (\gamma - 1)t \tag{3.19}$$

(we have a linear relationship between the logarithm of the instantaneous growth rate less the logarithm of a type of "feedback term" $(\ln((Y_\infty/Y_t) - 1)$ and t). (To gain some insight into the role of a feedback term, see Equation 3.A.4).

3.7 THE NEGATIVE EXPONENTIAL MODEL

Suppose that the time profile of the values of a variable Y reflects the operation of the *negative exponential model*

$$Y_t = Y_\infty\left(1 - e^{-\beta t}\right), \quad t \geq 0, \tag{3.20}$$

where the parameters Y_∞ and β are both positive. (The derivation of Eq. 3.20 is provided in Appendix 3.C.) Here Y_∞ is the *limit to growth parameter* $\left(\lim_{t\to\infty} Y_t = Y_\infty\right)$, while β is the growth rate parameter. For $t = 0$, the initial value of Y is $Y_0 = 0$.

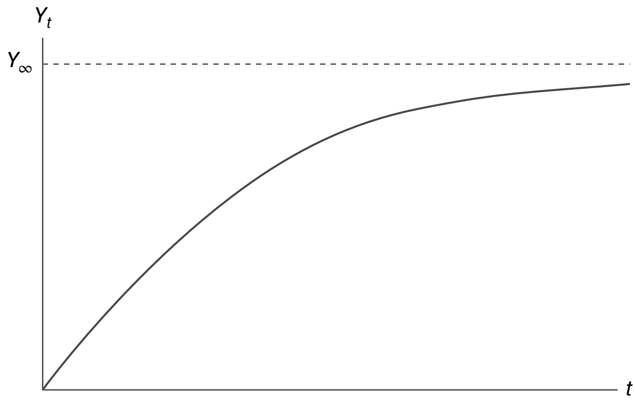

FIGURE 3.6 Negative exponential model.

Calculating the first and second derivatives of (3.20) yields

$$\frac{dY_t}{dt} = \beta(Y_\infty - Y_t); \tag{3.21a}$$

$$\frac{d^2Y_t}{dt^2} = -\beta\frac{dY_t}{dt} = -\beta^2(Y_\infty - Y_t). \tag{3.21b}$$

As Equation 3.21a and 3.21b reveals, as $t \to \infty$, the slope of the negative exponential function is positive and decreasing as $Y_t \to Y_\infty$ (Fig. 3.6).

Using Equation 3.21a, we can easily see that

$$\frac{dY_t / dt}{Y_t} = \beta\left(\frac{Y_\infty}{Y_t} - 1\right). \tag{3.22}$$

Thus, the instantaneous rate of growth in Y_t at time t approaches zero as $t \to \infty$ (or as $Y_t \to Y_\infty$). Note that this rate of growth is proportional to the "feedback term" $\frac{Y_\infty}{Y_t} - 1$.

3.8 THE VON BERTALANFFY MODEL

To describe a positive *net growth process* (pertaining to either length, weight, or size), von Bertalanffy (1957) asserted that the anabolic rate had to exceed the catabolic rate.[2] If the anabolic rate is a multiple of the kth power of Y_t and the catabolic

[2] We may view *anabolism* as constructive metabolism (requiring energy derived mainly from oxidation of organic compounds), while *catabolism* reflects destructive metabolism (living tissue is broken down into waste matter).

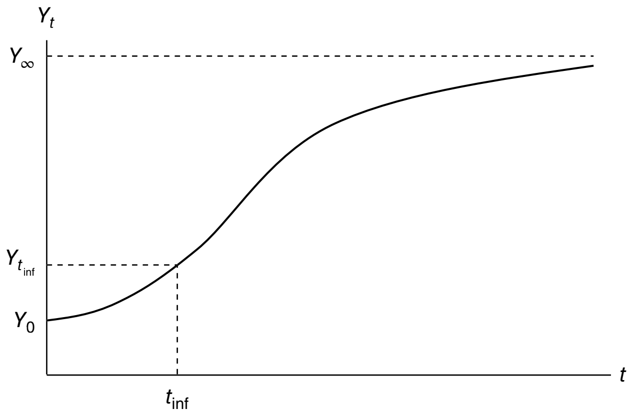

FIGURE 3.7 von Bertalanffy model.

rate is simply proportional to Y_t, then the *von Bertalanffy* (1957) *growth law* can be written as

$$\frac{dY_t}{dt} = \eta Y_t^k - \lambda Y_t, \tag{3.23}$$

where η and λ are constants. Once this expression is integrated (see Appendix 3.D), the *von Bertalanffy growth model* is of the form

$$Y_t = Y_\infty \left[1 - \beta e^{-\lambda(1-k)t}\right]^{1/(1-k)}, \tag{3.24}$$

where

$$Y_\infty = \lim_{t\to\infty} Y_t = (\eta / \lambda)^{1/(1-k)}$$

is the *limit to growth parameter*, $\beta = \alpha/\eta$ (α a constant of integration), λ is a growth rate parameter, $1 - k$ is a shape parameter, and the initial value of Y is

$$Y_0 = \left(\frac{\eta}{\lambda} - \frac{\alpha}{\lambda}\right)^{1/(1-k)}.$$

If $k > 1$ and both η and λ are negative, Equation 3.24 is sigmoidal with lower asymptote Y_0, upper asymptote Y_∞, and a point of inflection where $d^2Y_t/dt^2 = 0$ (Fig. 3.7). If $k < 1$ and both η and λ are positive, there is no lower asymptote. (An important variation of Eq. 3.24 is the *Chapman–Richards* (C–R) *growth model*, which is also considered in Appendix 3.D.)

von Bertalanffy determined empirically that $k = 2/3$ for a wide variety of animals (e.g., fisheries research tends to support this result). In general, to accommodate the possibility of exponential growth, von Bertalanffy allowed for $k \in [2/3, 1]$.

Given Equation 3.24, we can readily find

$$\frac{dY_t}{dt} = \beta\lambda Y_\infty^{1-k} e^{-\lambda(1-k)t} = \lambda Y_t\left[\left(\frac{Y_\infty}{Y_t}\right)^{1-k} - 1\right]; \tag{3.25a}$$

$$\begin{aligned}\frac{d^2Y_t}{dt^2} &= \beta\lambda Y_\infty^{1-k} e^{-\lambda(1-k)t} Y_t^k\left[\lambda(k-1) + kY_t^{-1}\frac{dY_t}{dt}\right]\\ &= \lambda\frac{dY_t}{dt}\left[k\left(\frac{Y_\infty}{Y_t}\right)^{1-k} - 1\right].\end{aligned} \tag{3.25b}$$

If we set $d^2Y_t/dt^2 = 0$, then it can be demonstrated that the point of inflection occurs at

$$\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = \left(\frac{1}{\lambda(k-1)}\ln\left(\frac{1-k}{\beta}\right), Y_\infty k^{1/(1-k)}\right).$$

Next, from Equation 3.25a, the instantaneous rate of growth in Y_t at time t is

$$\begin{aligned}\frac{dY_t/dt}{Y_t} &= \beta\lambda Y_\infty^{1-k} e^{-\lambda(1-k)t} Y_t^{k-1}\\ &= \lambda\left(Y_t^{k-1}Y_\infty^{1-k} - 1\right) = \lambda\left[\left(\frac{Y_\infty}{Y_t}\right)^{1-k} - 1\right];\end{aligned} \tag{3.26}$$

that is, this growth rate is proportional to the "feedback term" $(Y_\infty/Y_t)^{1-k} - 1$. It is interesting to note that the instantaneous rate of growth in Y_t is maximal at the point of inflection and, at this point, equals

$$\frac{dY_t/dt}{Y_t} = \frac{\lambda(1-k)}{k}.$$

3.9 THE LOG-LOGISTIC MODEL

If in the logistic function Equation 3.6 we replace t by $\ln t$, then we obtain the *log-logistic model*

$$Y_t = Y_\infty / \left(1 + \beta e^{-k\ln(t)}\right), \quad t > 0. \tag{3.27}$$

From Equation 3.27,

$$\frac{dY_t}{dt} = \frac{\beta k Y_t^2}{Y_\infty t e^{k\ln(t)}} \tag{3.28}$$

so that the instantaneous growth rate at time t can be written as

$$\frac{dY_t/dt}{Y_t} = \frac{\beta k}{t\left(\beta + e^{k\ln(t)}\right)} \tag{3.29}$$

(k is thus a rate of growth parameter for a fixed β) with

$$\lim_{t\to\infty} \frac{dY_t/dt}{Y_t} = 0.$$

Setting $d^2Y_t/dt^2 = 0$ enables us to obtain the coordinates of the point of inflection as

$$\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = \left(\left[\frac{1+k}{\beta(k-1)} \right]^{-1/k}, Y_\infty \left(\frac{k-1}{2k} \right) \right).$$

A popular application of the log-logistic function is in the fitting of dose–response curves. Specifically, one can determine the efficacy of a new "agonist" (a drug that causes a response) or the impact of some toxin (e.g., a weed-control agent). To this end, one should not simply apply small doses in small increments. It is typically the case that when researchers want to assess the impact of, say, some toxin, they increase doses logarithmically since, in this instance, the dose concentrations are equally spaced on the log scale.

For example, if dose concentration (in mg/kg) increments are displayed as

$$X: 1,\ 3.1623,\ 10,\ 31.623,\ 100,\ 316.228,\ 1000,\ 3162.28,\ 10{,}000,\ \ldots,$$

then the log (dose) values are

$$\log_{10}(X): 0,\ 0.5,\ 1,\ 1.5,\ 2,\ 2.5,\ 3,\ 3.5,\ 4,\ \ldots.$$

(Note that there is no dose of $X=0$; the actual dose is 1 mg/kg since $10^0 = 1$.) Hence, the log(dose)–response function serves to highlight the region of greatest interest wherein the percent of response increases at the highest rate. In fact, the dose response will reach the 50% level when log(dose) = log(ED_{50}), where ED_{50} is the "effective dose" at which 50% of the total response is achieved or $Y_\infty/2$.

If a lower asymptote or baseline response level (Y_L) is warranted, then Equation 3.27 can be reparameterized as

$$Y = Y_L + \frac{Y_\infty - Y_L}{1 + \beta e^{-k\ln(X)}}, \quad X > 0, \tag{3.30}$$

where now the effective dosage at which 50% of the total effect is attained is $(Y_\infty - Y_L)/2$.

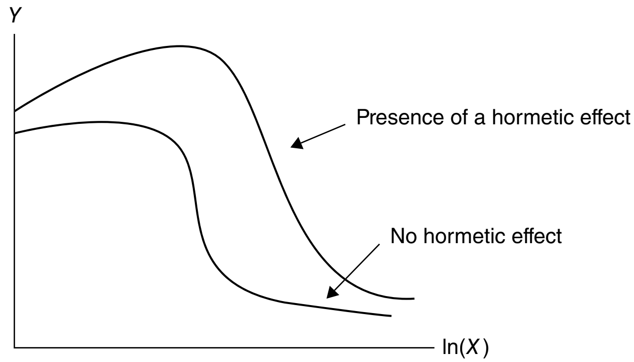

FIGURE 3.8 Presence of a hormetic effect.

Equation 3.30 may be modified further (Brain and Cousens, 1989) to capture "hormetic effects" (Fig. 3.8)[3] associated with low dosages as

$$Y = Y_L + \frac{Y_\infty - Y_L + \gamma X}{1 + \beta e^{-k \ln(X)}}, \quad X > 0, \tag{3.31}$$

where the parameter γ serves to measure the initial rate of increase at low dosages.

3.10 THE BRODY GROWTH MODEL

Brody (1945) partitions the sigmoidal process into two distinct phases:

Phase I. An expansionary phase that applies to a temporarily limited period of growth. For this phase Y_t is exponentially increasing or

$$Y_t = \alpha e^{kt}. \tag{3.32}$$

This phase I growth equation clearly has an infinite asymptote. Next follows:

Phase II. A declining phase in which Y_t is exponentially decreasing beyond the point where exponential growth ends. For this phase

$$Y_t = Y_\infty \left(1 - \beta e^{-kt}\right). \tag{3.33}$$

Since Y_t is either exponentially increasing via Equation 3.32 or exponentially decreasing (from Eq. 3.33), the Brody model lacks any point of inflection.

[3] Southman and Ehrlich (1943) (see also Schultz (1988) and Thimann (1956)) observed that subinhibitory levels of a toxin can produce stimulatory effects in organisms (e.g., increased growth in weeds can occur for subinhibitory doses of a herbicide). The presence of "hormesis" can raise the average response for low dosages above some control value.

3.11 THE JANOSCHEK GROWTH MODEL

A growth equation that is almost as flexible as the Richards growth function (Appendix 3.D) is the *Janoschek sigmoidal function* (Janoschek, 1957)

$$Y_t = Y_\infty \left(1 - \beta e^{-bt^c}\right), \quad c > 1, \tag{3.34}$$

where b is a growth parameter for a fixed c and c is a shape parameter. Given this expression,

$$\frac{dY_t}{dt} = -bct^{c-1} e^{-bt^c} \tag{3.35}$$

and thus, the instantaneous rate of growth at time t is

$$\frac{dY_t / dt}{Y_t} = bct^{c-1} \left(\frac{Y_\infty - 1}{Y_t} - 1 \right), \tag{3.36}$$

where $[(Y_\infty - 1)/Y_t - 1]$ serves as a type of "feedback term." Setting $d^2Y_t/dt^2 = 0$ enables us to find the coordinates of the point of inflection

$$\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = \left(\left(\frac{c-1}{cb} \right)^{1/c}, Y_\infty - \left(1 - e^{-(c-1)/c}\right) \right), \quad c > 1.$$

A modification of Equation 3.34 (Sager, 1984), for an initial value Y_0 greater than zero, can be written as

$$Y_t = Y_\infty - \left(Y_\infty - Y_0\right) e^{-kt^p}. \tag{3.37}$$

For $p < 1$, this function exhibits simple exponential growth; and for $p > 1$, sigmoidal growth occurs. Given Equation 3.37,

$$\frac{dY_t}{dt} = kp\left(Y_\infty - Y_0\right) t^{p-1} e^{-kt^p} \tag{3.38}$$

with the instantaneous growth rate at time t appearing as

$$\frac{dY_t / dt}{Y_t} = kpt^{p-1} \left(\frac{Y_\infty}{Y_t} - 1 \right). \tag{3.39}$$

Here too $(Y_\infty / Y_t - 1)$ represents a "feedback term." This modified Janoschek growth model has a point of inflection (which is optional) at

$$\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = \left(\left(\frac{p-1}{kp} \right)^{1/p}, Y_\infty - (Y_\infty - Y_0) e^{-\left(\frac{p-1}{p}\right)} \right), \quad p > 1.$$

3.12 THE LUNDQVIST–KORF GROWTH MODEL

Let us consider the *Lundqvist–Korf equation* (Korf, 1939; Lundqvist, 1957)

$$Y_t = Y_\infty e^{-kt^{-d}}, \tag{3.40}$$

where d denotes the shape parameter and k is a scale parameter. Then it is readily demonstrated that

$$\frac{dY_t}{dt} = kdt^{-d-1}Y_t \tag{3.41}$$

so that the instantaneous growth rate time t is

$$\frac{dY_t / dt}{Y_t} = kdt^{-d-1}. \tag{3.42}$$

And when $d^2Y_t/dt^2 = 0$, we can solve for the coordinates of the point of inflection as

$$\left(t_{\inf}, Y_{t_{\inf}}\right) = \left(\left(\frac{d+1}{kd}\right)^{-1/d}, Y_\infty e^{-(d+1)/d}\right).$$

Equation 3.40 is actually a generalization of the *Schumacher* (1939) *growth model*

$$Y_t = Y_\infty e^{bt^{-c}}. \tag{3.43}$$

3.13 THE HOSSFELD GROWTH MODEL

The *Hossfeld growth function* (Hossfeld, 1822) has the form

$$Y_t = Y_\infty \left(1 + b_1 t^{-b_2}\right)^{-1}, \quad b_1 > 1, \tag{3.44}$$

where

$$\frac{dY_t}{dt} = b_1 b_2 Y_t t^{-b_2 - 1}\left(1 + b_1 t^{-b_2}\right)^{-1}. \tag{3.45}$$

Given this latter expression, the instantaneous rate of growth at time t is

$$\frac{dY_t / dt}{Y_t} = b_1 b_2 \left(\frac{Y_t}{Y_\infty}\right) t^{-b_2 - 1}, \tag{3.46}$$

where b_2 serves as a growth rate parameter for a fixed b_1.

Additionally, setting $d^2Y_t/dt^2 = 0$ enables us to ultimately determine that Equation 3.44 has a point of inflection at

$$\left(t_{\inf}, Y_{t_{\inf}}\right) = \left(\left(\frac{1+b_2}{b_1\left(b_2-1\right)}\right)^{-1/b_2}, Y_\infty\left(\frac{b_2-1}{2b_2}\right)\right).$$

3.14 THE STANNARD GROWTH MODEL

The *Stannard growth function* (Stannard et al., 1985) has the form

$$Y_t = Y_\infty \left\{1 + e^{-[(\alpha+kt)/p]}\right\}^{-p}. \tag{3.47}$$

Given that

$$\frac{dY_t}{dt} = kY_t \left\{1 + e^{-[(\alpha+kt)/p]}\right\} e^{-[(\alpha+kt)/p]}, \tag{3.48}$$

we may easily express the instantaneous rate of growth at time t as

$$\frac{dY_t / dt}{Y_t} = k\left[1 - \left(\frac{Y_t}{Y_\infty}\right)^{\frac{1}{p}}\right], \tag{3.49}$$

where k serves as a rate of growth parameter given p and $[1-(Y_t/Y_\infty)^{1/p}]$ constitutes a "feedback term." And setting $d^2Y_t/dt^2 = 0$ enables us to determine the coordinates of the point of inflection as

$$\left(t_{\inf}, Y_{t_{\inf}}\right) = \left(\frac{p \ln p - \alpha}{k}, Y_\infty\left(\frac{p}{p+1}\right)^p\right).$$

3.15 THE SCHNUTE GROWTH MODEL

The Schnute (1981) study models the accelerated growth of a (fish) population by considering the relative growth rate of the relative growth rate. To this end, Schnute starts from the postulates:

1. The relative growth rate of a population Y_t is

$$k = \frac{d \ln Y_t}{dt} = \frac{dY_t / dt}{Y_t}. \tag{3.50}$$

2. The relative growth rate varies linearly with k and appears as

$$\frac{d\ln k}{dt} = \frac{dk/dt}{k} = -(a+bk). \tag{3.51}$$

Here a is a fixed growth rate and b is a shape parameter, which determines the point at which the initial acceleration of growth transitions to a slower growth pace.

Two forms of *Schnute's growth equation* now follow.
First,

$$Y_t = Y_\infty \left(1-\delta e^{-at}\right)^{1/b}, \tag{3.52}$$

with a representing a growth rate parameter and b serving as shape parameters. Then

$$\frac{dY_t}{dt} = \left(\frac{a\delta}{b}\right) Y_t e^{-at}\left(1-\delta e^{-at}\right)^{-1} \tag{3.53}$$

and thus, the instantaneous rate of growth at time t is

$$\frac{dY_t/dt}{Y_t} = \left(\frac{a\delta}{b}\right)\left(e^{at}-\delta\right)^{-1}. \tag{3.54}$$

Also, setting $d^2Y_t/dt^2=0$ enables us to solve for the coordinates of the point of inflection as

$$\left(t_{\inf}, Y_{t_{\inf}}\right) = \left(\frac{1}{a}\ln(\delta/b), Y_\infty(1-b)^{1/b}\right).$$

An alternative to Equation 3.52, which is useful for evolutionary computations, is obtained by coupling Equations 3.50 and 3.51 with the initial and terminal conditions

$$Y_1 = Y(t_1) \quad \text{and} \quad Y_2 = Y(t_2),$$

where t_1 and t_2 are the initial and final times and Y_1 and Y_2 are the initial and final population densities, respectively. Then it can be demonstrated that the system

$$\frac{dY_t}{dt} = kY_t, \quad \frac{dk}{dt} = -k(a+bk),$$

$$Y_1 = Y(t_1), \quad Y_2 = Y(t_2)$$

has solution

$$Y_t = \left[Y_1^b + \left(Y_2^b - Y_1^b\right)\frac{1-e^{-a(t-t_1)}}{1-e^{-a(t_2-t_1)}}\right]^{1/b}. \tag{3.55}$$

As $t \to +\infty$, Equation 3.55 approaches an upper asymptote when $a > 0$. (For details on the derivation of Equations 3.52 and 3.55, see Appendix 3.E.)

For Equation 3.55, the instantaneous rate of growth at time t is

$$\frac{dY_t / dt}{Y_t} = \frac{a}{b}\left[1 - \left(\frac{Y_1}{Y_2}\right)^b\right]\left(e^{a(t-t_1)} - 1\right)^{-1}, \tag{3.56}$$

and upon setting $d^2Y_t/dt^2 = 0$, we can determine that a point of inflection occurs at

$$\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = \left(t_1 - \frac{1}{a}\ln\left[b\left(\frac{Y_2^b - Y_1^b e^{-a(t_2-t_1)}}{Y_2^b - Y_1^b}\right)\right], (1-b)^{1/b}\left(\frac{Y_2^b - Y_1^b e^{-a(t_2-t_1)}}{1 - e^{-a(t_2-t_1)}}\right)^{1/b}\right),$$

save for the combination $a > 0$, $b > 1$.

Equations 3.52 and 3.55 are quite general and have some other well-known growth models subsumed within their structure. For instance, the following special cases emerge for selected values of the parameters a and b:

von Bertalanffy $(a > 0, b = 1)$
Richards $(a > 0, b < 0)$
Logistic $(a > 0, b = -1)$
Gompertz $(a > 0, b = 0)$
Exponential $(a < 0, b = 1)$

Note that the first four of these subcases are sigmoidal since $a > 0$.

3.16 THE MORGAN–MERCER–FLODIN (M–M–F) GROWTH MODEL

Two early models pertaining to biological kinetics were developed by Michaelis and Menten (1913) and Hill (1910, 1913). The *Michaelis–Menten equation*, which relates reaction velocity (Y) to substrate concentration (X) in an enzyme-catalyzed chemical reaction, is of the form

$$Y = -\frac{dX}{dt} = \frac{Y_\infty X}{k + X}, k > 0. \tag{3.57}$$

Here Equation 3.57 describes a law that governs the rate at which the body processes a drug and structurally represents a *hyperbolic saturation curve* since

$$\frac{dY}{dX} = \frac{kY_\infty}{(k+X)^2} > 0$$

and

$$\frac{d^2Y}{dX^2} = \frac{-2kY_\infty}{(k+X)^3} < 0.$$

(A generalization of the Michaelis–Menten equation is offered in Appendix 3.H.)

The *Hill equation*, which was used to describe the kinetics of the binding of oxygen to hemoglobin in respiratory physiology, appears as

$$Y = \frac{Y_\infty X^n}{k+X^n}, k > 0, \tag{3.58}$$

and depicts a *sigmoidal saturation function*, where k is a constant with the property that $Y = Y_\infty/2$ when $X = k^{1/n}$ and n (the *Hill coefficient*) is the kinetic order of the saturation function that describes its shape.

While Equations 3.57 and 3.58 were not originally applied in a growth context, several studies (Mercer et al., 1978; Morgan et al., 1975) used these equations as the basis for deriving a general saturation equation useful for modeling biological efficiency or for describing the ability of a nutrient to produce a response in humans or animals. An important feature of Equations 3.57 and 3.58 is that they pass through the origin. However, since most estimated nutrient–response curves have not been found to display this characteristic (such curves tend to an upper asymptote at high nutrient intake levels as well as exhibit a curvature as the asymptote is approached, which reflects either hyperbolic or sigmoidal behavior), M–M–F adopted two criteria that are to be satisfied by a general saturation function for modeling nutrient responses: (i) the equation must be able to display both hyperbolic and sigmoidal behavior, and (ii) the equation must be free to intersect the ordinate axis at any point dictated by the data. (Clearly the sigmoidal Hill equation (3.58) satisfies the first requirement since, for $n=1$, it reduces to the Michaelis–Menten hyperbolic equation (3.57). However, the second criterion is not met.)

Morgan–Mercer–Flodin thus modify Equation 3.58 by a simple translation of the ordinate axis, to wit, their general saturation function (which now satisfies both of the aforementioned criteria) takes the form

$$Y = \frac{ab + Y_\infty X^n}{b+X^n}, \tag{3.59}$$

where a is the ordinate intercept when $X=0$, n is a shape parameter (for $n=1$, Equation 3.59 is a rectangular hyperbola; when $a=0$, we get the Hill equation (3.58); and for $a=0$ and $n=1$, the Michaelis–Menten equation (3.57) obtains), and $b=(X_{0.50})^n$, with $X_{0.50}$ the value of X when Y is halfway to the maximum response, that is, a nutrient response level equal to $(Y_\infty + a)/2$.

For purposes of growth process modeling, a convenient alternative form of Equation 3.59 is

$$Y_t = Y_\infty - \frac{Y_\infty - \beta}{\left[1+(kt)^\delta\right]},^4 \tag{3.60}$$

where β depicts size or yield at $t=0$, k is the rate of growth, and δ is a parameter that controls the point of inflection. Given Equation 3.60,

$$\frac{dY_t}{dt} = (Y_\infty - Y_t)\frac{\delta k^\delta t^{\delta-1}}{\left[1+(kt)^\delta\right]} \tag{3.61}$$

with the instantaneous rate of growth at time t determined as

$$\frac{dY_t/dt}{Y_t} = \left(\frac{Y_\infty}{Y_t} - 1\right)\frac{\delta k^\delta t^{\delta-1}}{\left[1+(kt)^\delta\right]}, \tag{3.62}$$

where $(Y_\infty/Y_t - 1)$ constitutes a "feedback term." Setting $d^2Y_t/dt^2 = 0$ enables us to solve for the point of inflection

$$\left(t_{\inf}, Y_{t_{\inf}}\right) = \left(k^{-1}\left(\frac{\delta+1}{\delta-1}\right)^{1/\delta}, \frac{(\delta-1)Y_\infty + (\delta+1)\beta}{2\delta}\right).$$

3.17 THE MCDILL–AMATEIS GROWTH MODEL

McDill and Amateis (1992) develop a growth model of (pine) forest site quality that utilizes, as a measure of site quality, the values of site-specific parameters of a stand-height-growth equation, where observed "highest growth" refers to the height growth of a particular stand component chosen to assess "site quality." Ostensibly height growth is a good indicator of site quality, with site quality in turn reflecting the potential for a site to yield wood fiber/products.

The *McDill–Amateis height-growth function* contains two biologically meaningful parameters—a rate parameter and a maximum height parameter. Moreover, this expression also includes initial conditions (i.e., initial height at some predetermined initial age) that can either be treated as an additional (third) parameter to be estimated or specified from the data. The essential variables appearing in a height-growth equation are height (H), age (A), height growth (dH/dA), and asymptotic

[4] Let us rewrite Equation 3.60 as

$$Y_t = \frac{Y_\infty[1+(kt)^\delta] - (Y_\infty - \beta)}{[1+(kt)^\delta]} = \frac{\beta k^{-\delta} + Y_\infty t^\delta}{k^{-\delta} + t^\delta} = \frac{\alpha\beta + Y_\infty t^\delta}{\alpha + t^\delta},$$

where $\alpha = k^{-\delta}$. Clearly this expression coincides with Equation 3.59.

maximum height (H_∞) that trees on a given site can attain. In this regard, the McDill–Amateis stand-height-growth function has the form

$$H = \frac{H_\infty}{1-\left(1-\dfrac{H_\infty}{H_0}\right)\left(\dfrac{A_0}{A}\right)^a}, \tag{3.63}$$

where H_0 and A_0 serve as initial conditions. (Details on the derivation of this expression appear in Appendix 3.F.) For tree growth estimation, we require that $H_\infty > H_0 > 0$ and $a > 0$. Here height $H \to 0$ as age $A \to 0$, while $H \to H_\infty$ as $A \to +\infty$.

From Equation 3.63,

$$\frac{dH}{dA} = a\left(\frac{H}{A}\right)\left(1-\frac{H}{H_\infty}\right), \tag{3.64}$$

while the instantaneous rate of growth at age A is

$$\frac{dH/dA}{H} = \frac{a}{A}\left(1-\frac{H}{H_\infty}\right) \tag{3.65}$$

with $1 - H/H_\infty$ serving as a "feedback term." Next, setting $d^2H/dA^2 = 0$ enables us to obtain the coordinates of the point of inflection

$$\left(A_{\text{inf}}, H_{A_{\text{inf}}}\right) = \left(A_0\left(1-\frac{H_\infty}{H_0}\right)^{1/a}\left(\frac{1-a}{a+1}\right)^{1/a}, \frac{H_\infty}{2}\left(1-\frac{1}{a}\right)\right), \ a > 1.$$

For $a \le 1$, the growth equation is concave.

McDill and Amateis stipulate that their model works well for remeasurement data from permanent plots or for data that, in general, span a fairly long time period. As far as the initial conditions are concerned, the authors recommend using the data to set the initial conditions and treat measurement intervals as observations (i.e., each observation consists of an initial measurement (A_0, H_0) and a terminal measurement (A, H)). Hence, all measurements, save for the first and last, belong to two observations—as the initial measurement in one observation and as the ending measurement in another.

3.18 AN ASSORTMENT OF ADDITIONAL GROWTH MODELS

The following three growth functions (all modifications of the Hossfeld equation 3.44) have been found to be quite accurate in practice:

1. *The Levakovic I growth model: Levakovic's I growth equation* (1935) takes the form

$$Y_t = Y_\infty(1+bt^{-d})^{-c}. \tag{3.66}$$

Then

$$\frac{dY_t}{dt} = cbdY_\infty (1+bt^{-d})^{-c-1} t^{-d-1} \tag{3.67}$$

and thus, the instantaneous rate of change in Y_t at time t is

$$\frac{dY_t / dt}{Y_t} = cbdt^{-d-1} (Y_t / Y_\infty)^{1/c}. \tag{3.68}$$

Setting $d^2Y_t/dt^2 = 0$ enables as to solve for the point of inflection

$$\left(t_{\inf}, Y_{t_{\inf}}\right) = \left(\left(\frac{b(cd-1)}{d+1}\right)^{1/d}, Y_\infty \left(\frac{cd-1}{d(c+1)}\right)^{c} \right).$$

2. *The Levakovic III growth model*: The *Levakovic III growth equation* (1935) is a special case of Equation 3.66 and appears as

$$Y_t = Y_\infty (1+bt^{-2})^{-c}. \tag{3.69}$$

Then expressions for dY_t/dt, $(dY_t/dt)/Y_t$, and the point of inflection can be obtained by setting $d=2$ in Equation 3.66, in Equation 3.67, and in $\left(t_{\inf}, Y_{t_{\inf}}\right)$, respectively.

3. *The Yoshida I growth model: Yoshida's I growth equation* (1981) is structured as

$$Y_t = Y_\infty (1+bt^{-c})^{-1} + c, \tag{3.70}$$

where c is a parameter indicating initial size. Here

$$\begin{aligned} \frac{dY_t}{dt} &= bdY_\infty (1+bt^{-d})^{-2} t^{-d-1} \\ &= \frac{bd(Y_t - c)}{\left(1+bt^{-d}\right)} t^{-d-1}, \end{aligned} \tag{3.71}$$

with the instantaneous rate of growth at time t given by

$$\frac{dY_t / dt}{Y_t} = bd\left(1 - \frac{c}{Y_t}\right)\left(1+bt^{-d}\right) t^{-d-1}. \tag{3.72}$$

Equation 3.70 has a point of inflection at

$$\left(t_{\inf}, Y_{t_{\inf}}\right) = \left(\left(\frac{b(d-1)}{d+1}\right)^{\frac{1}{d}}, Y_\infty \left(\frac{2d}{d-1}\right) + c \right).$$

3.18.1 The Sloboda Growth Model

A growth function similar to the Gompertz equation (3.10) is the *Sloboda growth function* (1971a, b)

$$Y_t = Y_\infty e^{-\alpha e^{-\beta t^\gamma}}, \quad t \geq 0 \tag{3.73}$$

(note that this expression differs from the Gompertz equation by the presence of the additional parameter γ). Then

$$\frac{dY_t}{dt} = \alpha\beta\gamma Y_t t^{\gamma-1} e^{-\beta t^\gamma} \tag{3.74}$$

and thus, the instantaneous rate of change in Y_t at time t is

$$\frac{dY_t / dt}{Y_t} = \alpha\beta\gamma t^{\gamma-1} e^{-\beta t^\gamma}. \tag{3.75}$$

APPENDIX 3.A THE LOGISTIC MODEL DERIVED

For the simple exponential growth model, it is assumed that the instantaneous growth rate of a variable Y is a constant β,

$$\frac{dY_t / dt}{Y_t} = \beta, \quad \text{or} \quad \frac{dY_t}{dt} = \beta Y_t; \tag{3.A.1}$$

that is, the rate of change in Y is proportional to the current size of Y at time t, Y_t. Then

$$\int \frac{dY_t}{Y_t} = \int \beta dt$$

or

$$\ln Y_t = \beta t + \ln C \ (\ln C \text{ a constant})$$

and thus

$$Y_t = Ce^{\beta t}. \tag{3.A.2}$$

For $t = 0$, $Y_0 = C$ so that Equation 3.A.2 can be rewritten as

$$Y_t = Y_0 e^{\beta t}. \tag{3.A.3}$$

As Equation 3.A.3 reveals, there is no limit to growth; as $t \to +\infty$, it follows that $Y_t \to +\infty$ (Fig. 3.A.1a).

The so-called logistic law of growth states that, essentially, the rate of growth in a system or population Y may be limited. More specifically, Y grows exponentially at,

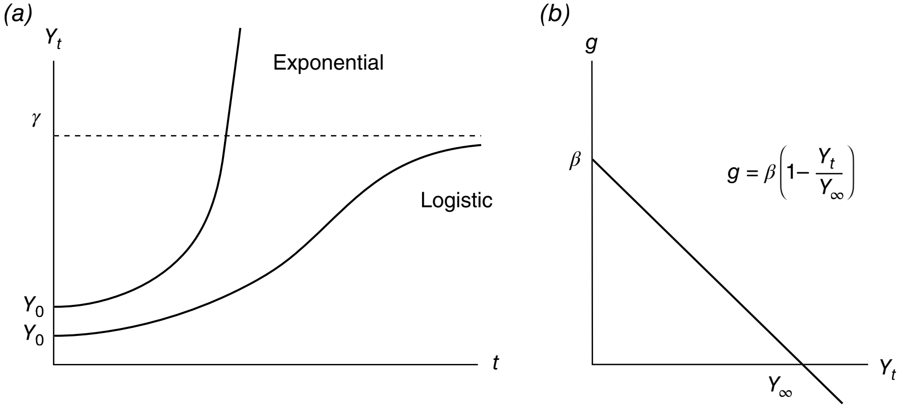

FIGURE 3.A.1 (**a**) Exponential versus logistic growth; (**b**) Feedback dependency.

say, rate g until an upper limit (the saturation parameter Y_∞) to the capacity of the system to grow is approached, at which point the growth rate slows, and the characteristic S-shaped or sigmoidal curve emerges. In this regard, let us assume that the growth rate g is dependent upon the *density of Y at time t* (Y_t/Y_∞) or

$$g = \beta\left(1 - \frac{Y_t}{Y_\infty}\right), \tag{3.A.4}$$

where $1-(Y_t/Y_\infty)$ is called a *feedback term*, the role of which is to account for the growth rate declining to zero as $Y_t \to Y_\infty$. That is, at low densities (Y_t is much smaller than Y_∞), g is maximal and close to β (Fig. 3.A.1b), and thus, the behavior of the logistic curve resembles that of an exponential growth curve; but at high densities (Y_t is near Y_∞), the feedback term slows growth until saturation is attained (when $Y_t = Y_\infty$), at which point the growth rate $g=0$. (If $Y_t > Y_\infty$, then the growth rate in Y turns negative.)

This said, we can now derive the logistic function as follows. Let us start with an exponential growth equation:

$$\frac{dY_t}{dt} = gY_t.$$

Then from Equation 3.A.4, the dynamics of Y growth is now characterized by the *logistic differential equation*

$$\frac{dY_t}{dt} = \beta Y_t\left(1 - \frac{Y_t}{Y_\infty}\right). \tag{3.A.5}$$

Let us set $x = Y_t/Y_\infty$. Then

$$\frac{dx}{dt} = \frac{1}{Y_\infty}\frac{dY_t}{dt}$$

and thus, Equation 3.A.5 can be rewritten as

$$\frac{dY_t / dt}{Y_\infty} = \beta \frac{Y_t}{Y_\infty}\left(1 - \frac{Y_t}{Y_\infty}\right)$$

or

$$\frac{dx}{dt} = \beta x(1-x). \tag{3.A.6}$$

Then the sequence of steps for solving Equation 3.A.6 is

$$\int \frac{dx}{x(1-x)} = \int \beta\, dt,$$

$$-\ln\frac{1-x}{x} = \beta t + \ln C,\ (\ln C \text{ a constant})$$

$$\ln\frac{1-x}{x} = -\beta t - \ln C,$$

$$\frac{1-x}{x} = \alpha e^{-\beta t},\ (\alpha = C^{-1}).$$

Then solving for x gives

$$x = \frac{1}{1+\alpha e^{-\beta t}}.$$

And since $x = Y_t/Y_\infty$, we ultimately obtain

$$Y_t = \frac{Y_\infty}{1+\alpha e^{-\beta t}},\quad t \geq 0 \tag{3.A.7}$$

(or Eq. 3.6). For $t=0$, $Y_0 = Y_\infty/(1+\alpha)$ so that $\alpha = (Y_\infty/Y_0) - 1$. (Note that a discrete version of Equation 3.A.7 is

$$Y_t = \frac{Y_\infty}{1+\alpha\beta^t}.) \tag{3.A.8}$$

Let us define the *growth time* (denoted Δt) as the length of the time interval during which Y progresses from 10% to 90% of the capacity limit Y_∞. To find Δt, let us first set

$$(Y_t =) 0.90 Y_\infty = \frac{Y_\infty}{1+\alpha e^{-\beta t}}.$$

Then solving for t yields

$$(t =) t_{0.90} = \frac{2.19723}{\beta} + \frac{\ln\alpha}{\beta}.$$

Next, from

$$0.10Y_\infty = \frac{Y_\infty}{1+\alpha e^{-\beta t}},$$

we find

$$t_{0.10} = -\frac{2.19722}{\beta} + \frac{\ln \alpha}{\beta}.$$

Hence,

$$\Delta t = t_{0.90} - t_{0.10} = \frac{4.39445}{\beta} + \frac{\ln(81)}{\beta}.$$

Additionally, the *midpoint* is the time t_m where $Y_{t_m} = Y_\infty / 2$—it is the time of most rapid increase in Y and occurs at the point of inflection so that $t_m = \ln \alpha / \beta$ (see Fig. 3.3).

The logistic model can be linearized by a change of variable. To this end, again set $x = Y_t / Y_\infty$. Then from Equation 3.A.7,

$$x = \frac{1}{1+\alpha e^{-\beta t}}, 1 - x = \frac{\alpha e^{-\beta t}}{1+\alpha e^{-\beta t}},$$

and thus

$$\frac{x}{1-x} = \alpha^{-1} e^{\beta t}.$$

Hence,

$$\begin{aligned} \ln \frac{x}{1-x} &= \ln(\alpha^{-1}) + \beta t \\ &= a + \beta t. \end{aligned} \tag{3.A.9}$$

APPENDIX 3.B THE GOMPERTZ MODEL DERIVED

Let us write what is called the *Gompertz differential equation* as

$$\frac{dY_t}{dt} = \beta Y_t \ln\left(\frac{Y_\infty}{Y_t}\right) \tag{3.B.1}$$

(see Equation 3.13 and its interpretation). Here the instantaneous rate of growth in Y is proportional to $\ln Y_\infty - \ln Y_t$. If we set $x = Y_t / Y_\infty$, then

$$\frac{dx}{dt} = \frac{1}{Y_\infty} \frac{dY_t}{dt}$$

and thus, Equation 3.B.1 can be rewritten as

$$\frac{dY_t / dt}{Y_\infty} = \beta \frac{Y_t}{Y_\infty} \ln\left(\frac{Y_\infty}{Y_t}\right)$$

or

$$\frac{dx}{dt} = \beta x \ln\left(\frac{1}{x}\right). \tag{3.B.2}$$

Then the solution of Equation (3.B.2) proceeds as follows:

$$\begin{aligned}
\int \frac{dx}{x\ln(1/x)} &= \int \beta\, dt, \\
-\ln\left(\ln\frac{1}{x}\right) &= \beta t + \ln C, \ (\ln C \text{ a constant}) \\
\ln\left(\ln\frac{1}{x}\right) &= -\beta t - \ln C, \\
\ln\left(\frac{1}{x}\right) &= \alpha e^{-\beta t}, \quad (\alpha = C^{-1}) \\
\ln x &= -\alpha e^{-\beta t}.
\end{aligned}$$

Solving for x gives

$$x = e^{-\alpha e^{-\beta t}}.$$

But since $x = Y_t/Y_\infty$, we ultimately obtain

$$Y_t = Y_\infty e^{-\alpha e^{-\beta t}}, \quad t \geq 0 \tag{3.B.3}$$

(or Eq. 3.10). For $t=0$, $Y_0 = Y_\infty e^{-\alpha}$ so that $\alpha = \ln(Y_\infty/Y_0)$. (The discrete version of Eq. 3.B.3 is

$$Y_t = Y_\infty \alpha^{\beta^t}.) \tag{3.B.4}$$

APPENDIX 3.C THE NEGATIVE EXPONENTIAL MODEL DERIVED

Suppose we posit that dY_t/dt is a linear function of Y_t or

$$\frac{dY_t}{dt} = \beta(Y_\infty - Y_t), \tag{3.C.1}$$

where β is constant and Y_∞ is the value of Y_t for which $dY_t/dt = 0$, that is, as $t \to \infty$, $Y_t \to Y_\infty$, and $dY_t/dt \to 0$. So if, for instance, we plot dY_t/dt against Y_t, then, as Equation 3.C.1

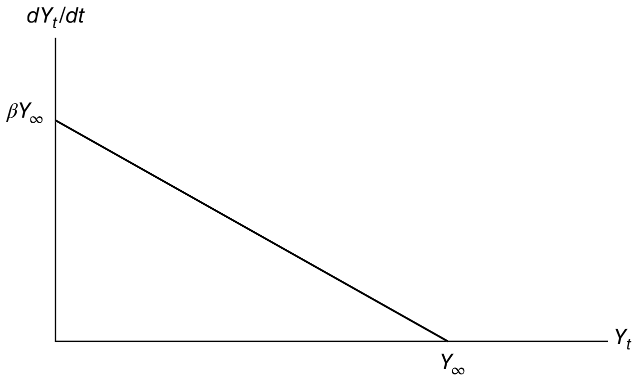

FIGURE 3.C.1 dY_t/d_t is a linear function of Yt.

reveals, we have a linear function with slope $-\beta$, which cuts the Y_t- axis at a point Y_∞ beyond which Y will not grow. Hence, Y_∞ is the asymptote of Y_t as $t \to \infty$ (Fig. 3.C.1).

Let us rewrite Equation 3.C.1 as

$$\frac{dY_t}{Y_\infty - Y_t} = \beta dt$$

so that

$$\int \frac{dY_t}{Y_\infty - Y_t} = \int \beta dt,$$

$$-\ln(Y_\infty - Y_t) = \beta t + \ln C, \text{ } (\ln C \text{ a constant})$$

$$\ln(Y_\infty - Y_t) = -\beta t - \ln C,$$

$$Y_\infty - Y_t = \alpha e^{-\beta t}, (\alpha = C^{-1})$$

and thus

$$Y_t = Y_\infty - \alpha e^{-\beta t}. \tag{3.C.2}$$

For $t=0$, $Y_0 = Y_\infty - \alpha = 0$ or $\alpha = Y_\infty$. Hence, Equation 3.C.2 becomes

$$Y_t = Y_\infty \left(1 - e^{-\beta t}\right). \tag{3.C.3}$$

Note that if we set $Y_0 = Y_\infty - \alpha$, then $\alpha = Y_\infty - Y_0$. Substituting this value for α into Equation 3.C.2 gives

$$Y_t = Y_\infty - (Y_t - Y_0)e^{-\beta t} = Y_\infty \left[1 - \left(1 - \frac{Y_0}{Y_\infty}\right)e^{-\beta t}\right] = Y_\infty \left(1 - \theta e^{-\beta t}\right), \quad t \geq 0, \tag{3.C.4}$$

the *monomolecular growth function*. For this growth equation,

$$\frac{dY_t}{dt} = Y_\infty \beta\theta e^{-\beta t} = \frac{Y_\infty \beta\theta}{e^{\beta t}}.$$

Since $\lim_{t\to\infty} dY_t / dt = 0$, the monomolecular function is continually decreasing; it has no point of inflection since $d^2Y_t/dt_2 = -Y_\infty\theta\beta^2 e^{-\beta t} < 0$ for all t. Additionally, the instantaneous growth rate is

$$\frac{dY_t / dt}{Y_t} = \frac{\beta\theta}{e^{\beta t} - \theta}$$

with

$$\lim_{t\to\infty} \frac{dY_t / dt}{Y_t} = 0.$$

APPENDIX 3.D THE VON BERTALANFFY AND RICHARDS MODELS DERIVED

Given the von Bertalanffy growth law

$$\frac{dY_t}{dt} + \lambda Y_t = \eta Y_t^k \tag{3.D.1}$$

(see Eq. 3.23),[5] set $x = Y_t^{1-k}$ so that

$$\frac{dx}{dt} = (1-k)Y_t^{-k} \frac{dY_t}{dt}$$

or

$$\frac{1}{1-k} Y_t^k \frac{dx}{dt} = \frac{dY_t}{dt}. \tag{3.D.2}$$

A substitution of Equation 3.D.1 into Equation 3.D.2 gives

$$\frac{1}{1-k}\frac{dx}{dt} = \eta - \lambda x,$$

$$\int \frac{dx}{\eta - \lambda x} = \int (1-k)dt,$$

[5] Equation 3.D.1 is a special case of the (Jakob) *Bernoulli differential equation* $y' + p(t)y = q(t)y^n$, n a constant (not necessarily an integer), with p and q given functions of t. If $n \neq 0$ or 1, the substitution $x = y^{1-n}$ (due to Leibnitz in 1696) linearizes the Bernoulli equation.

$$-\lambda^{-1}\ln(\eta-\lambda x)=(1-k)t+\ln C,\ (\ln C \text{ a constant})$$

$$\ln(\eta-\lambda x)=-\lambda(1-k)t-\lambda\ln C,$$

$$\eta-\lambda x=\alpha e^{-\lambda(1-k)t},\ (\alpha=C^{-\lambda})$$

$$x=\frac{\eta}{\lambda}-\frac{\alpha}{\lambda}e^{-\lambda(1-k)t}. \tag{3.D.3}$$

Then substituting Y_t^{1-k} for x in Equation 3.D.3 enables us to write

$$Y_t=\left[\frac{\eta}{\lambda}-\frac{\alpha}{\lambda}e^{-\lambda(1-k)t}\right]^{1/(1-k)}. \tag{3.D.4}$$

For $t=0$,

$$Y_0=\left[\frac{\eta}{\lambda}-\frac{\alpha}{\lambda}\right]^{1/(1-k)}$$

so that

$$\frac{\alpha}{\lambda}=\frac{\eta}{\lambda}-Y_0^{1-k}.$$

Hence, Equation 3.D.4 becomes

$$Y_t=\left[\frac{\eta}{\lambda}-\left(\frac{\eta}{\lambda}-Y_0^{1-k}\right)e^{-\lambda(1-k)t}\right]^{1/(1-k)}. \tag{3.D.5}$$

And since

$$\lim_{t\to\infty}Y_t=Y_\infty=\left(\frac{\eta}{\lambda}\right)^{1/(1-k)},$$

Equation 3.D.5 can be rewritten as

$$\begin{aligned}Y_t&=\left[Y_\infty^{1-k}-\left(Y_\infty^{1-k}-Y_0^{1-k}\right)e^{-\lambda(1-k)t}\right]^{1/(1-k)}\\&=Y_\infty\left[1-\beta e^{-\lambda(1-k)t}\right]^{\frac{1}{1-k}},\end{aligned} \tag{3.D.6}$$

the *von Bertalanffy growth equation*.

A modification of this function has been offered by Chapman (1961) and Richards (1959) (C–R). Specifically, C–R rewrite Equation 3.D.6 as

$$Y_t=Y_\infty\left[1-ae^{-\lambda t}\right]^m,\quad t\geq 0, \tag{3.D.7}$$

the so-called four-parameter C–R function. Here:

Y_∞ is an upper asymptote (the saturation level when $t \to \infty$).
a is essentially a position parameter (it locates the curve on the t-axis).
λ regulates the growth rate for a fixed value of m. It serves as a *maturity index*—a smaller (larger) λ indicates late (early) maturing.
m relates to curve shape—it allows a variable or optional point of inflection that is not a "fixed" proportion of Y_∞. For $m > 0$, the growth curve is S-shaped and the point of inflection is located at $\left(t_{\text{inf}}, Y_{t_{\text{inf}}}\right) = \left(\lambda^{-1}\ln(am), Y_\infty\left(\frac{m-1}{m}\right)^m\right)$.

If $m > 1$, the initial growth phase is exponential.
In addition, from (3.D.7), it can be shown that

$$\frac{dY_t}{dt} = m\lambda Y_t\left[\left(\frac{Y_\infty}{Y_t}\right)^{1/m} - 1\right]; \tag{3.D.8a}$$

$$\frac{dY_t/dt}{Y_t} = m\lambda\left[\left(\frac{Y_\infty}{Y_t}\right)^{1/m} - 1\right]. \tag{3.D.8b}$$

As Equation 3.D.8b reveals, the instantaneous rate of growth in Y_t at time t is proportional to the "feedback term" $(Y_\infty/Y_t)^{1/m} - 1$.

It is interesting to note that the C–R growth function is actually a generalization of some well-known growth curves, for example, for $m = -1$, we get the logistic function; for m near $\pm\infty$, the Gompertz function emerges; for $m = 3$, the von Bertalanffy equation results (see von Bertalanffy (1957)); and for $m = 1$, the (second) Brody function obtains.

It is important to mention that the original Richards (1959) growth equation represents a generalization of the logistic growth function (Eq. 3.A.7). That is, Richards generalized the logistic growth law (Eq. 3.A.5) by considering the modified growth equation (law)

$$\frac{dY_t}{dt} = \beta Y_t\left[1 - \left(\frac{Y_t}{Y_\infty}\right)^r\right] \tag{3.D.9}$$

(the growth rate is proportional to Y_t times a "feedback term") or

$$\frac{dY_t}{dt} = \beta Y_t + \eta Y_t^{r+1}. \tag{3.D.10}$$

Since Equation 3.D.10 is a Bernoulli differential equation (see footnote 5), it integrates to the *Richards growth function* (or *generalized logistic equation*)

$$Y_t = \frac{Y_\infty}{\left(1 + \alpha e^{-\beta r t}\right)^{1/r}}, \quad t \geq 0, \tag{3.D.11}$$

where $\alpha = (Y_\infty/Y_0)^r - 1$ and both β, $r > 0$.

Note that relative to the logistic equation 3.A.7, Equation 3.D.11 has an additional *shape parameter* r, which allows the shape of the sigmoid to be varied. In fact, varying r allows the point of inflection of the curve to be located at any value between the lower and upper asymptotes. Here too other growth curves are subsumed under Equation 3.D.11 for specific values of r, for example, when $r=1$, we get the standard logistic model; as $r \to 0$, the Richards curve tends towards the Gompertz curve; for $r=-1$, the monomolecular (or second Brody) case emerges; and for $r=-1/3$, the von Bertalanffy function obtains.

The preceding derivation of the von Bertalanffy growth equation was carried out in a purely mechanistic fashion. As an alternative to this approach, it is instructive to offer a derivation of this growth model based on physiological concepts (Beverton and Holt, 1957; von Bertalanffy, 1957). To this end, von Bertalanffy considers an individual organism as analogous to a reacting chemical system that obeys the *law of mass action*[6] and, consequently, classifies the physiological processes responsible for the mass of an organism at time t into those of anabolism (or synthesis) and catabolism (breakdown); that is, the rate of change in the body weight (w) of an organism can be written as

$$\frac{dw}{dt} = hw^n - kw^m, \tag{3.D.12}$$

where h and k are the coefficients of anabolism and catabolism, respectively, and the exponents n and m have a physiological basis. In this regard, how should these parameters be chosen? Suppose the rate of anabolism is taken as proportional to the resorption rate (the rate at which a "structure" is remodeled) of nutrients or to the magnitude of the resorbing surfaces, and suppose the rate of catabolism is specified as proportional to the total mass being broken down, with a constant percentage of body material being transformed to waste matter per unit time.

Given these considerations, Equation 3.D.12 can be rewritten as

$$\frac{dw}{dt} = hs - kw, \tag{3.D.13}$$

where:

h = rate of synthesis of mass per unit of resorbing surface
s = effective resorbing surface of the organism
k = rate of breakdown of mass per unit mass

If the organism is assumed to be growing isometrically and has a constant specific gravity, then we can express s and w in terms of the linear dimension l of the organism as $s = pl^2$ and $w = ql^3$ (p and q are constants). Hence, it follows that

$$\frac{dw}{dt} = \frac{d}{dt}\left(ql^3\right) = 3ql^2 \frac{dl}{dt}$$

[6] This law (an assertion about "kinetics") states that the rate of a given chemical reaction is proportional to the product of the activities (or molecular concentration) of the reactants (Guldberg and Waage, 1879).

and thus, from Equation 3.D.12,

$$\frac{dl}{dt} = \frac{hpl^2}{3ql^2} - \frac{kql^3}{3ql^2} = \frac{hp}{3q} - \frac{kl}{3}. \tag{3.D.14}$$

Setting $hp/3q = E$ and $k/3 = K$, Equation 3.D.14 becomes

$$\frac{dl}{dt} = E - Kl$$

or

$$\frac{dl}{E - Kl} = dt$$

with solution

$$l_t = \frac{E}{K} - \left(\frac{E}{K} l_0\right) e^{-Kt}, \tag{3.D.15}$$

where l_0 is the length of the organism at time $t = 0$. Note that as $t \to +\infty, l_t \to E/K = L_\infty$— the maximum possible length of the organism (under normal conditions). So as the organism ages, its length increases asymptotically to L_∞. Under this discussion, Equation 3.D.15 becomes

$$l_t = L_\infty - (L_\infty - L_0)e^{-Kt}. \tag{3.D.16}$$

Given Equation 3.D.16, the associated equation for weight (w) becomes, from $w = ql^3$,

$$w_t = \left\{W_\infty^{1/3} - \left(W_\infty^{1/3} - W_0^{1/3}\right)e^{-Kt}\right\}^3, \tag{3.D.17}$$

where W_∞ and W_0 are the weights corresponding to L_∞ and L_0, respectively. If we now set $W_t = 0$ and let $t = t_0$ in Equation (3.D.17), it follows that

$$W_0 = W_\infty \left(1 - e^{Kt_0}\right)^3.$$

Then a substitution of this expression into Equation 3.110 renders

$$w_t = W_\infty \left(1 - e^{-K(t - t_0)}\right)^3.$$

APPENDIX 3.E THE SCHNUTE MODEL DERIVED

Given Equations 3.50 and 3.51 or

$$k = \frac{1}{Y_t}\frac{dY_t}{dt} \quad \text{and} \quad \frac{1}{k}\frac{dk}{dt} = -(a + bk),$$

respectively, let us differentiate $dY_t/dt = kY_t$ with respect to t so as to obtain

$$\begin{aligned}
\frac{d^2Y_t}{dt^2} &= Y_t\frac{dk}{dt} + k\frac{dY_t}{dt} \\
&= Y_t\left[-k(a+bk)\right] + \frac{1}{Y_t}\left(\frac{dY_t}{dt}\right)^2 \\
&= Y_t\left[-\frac{1}{Y_t}\frac{dY_t}{dt}\left(a + b\frac{1}{Y_t}\frac{dY_t}{dt}\right)\right] + \frac{1}{Y_t}\left(\frac{dY_t}{dt}\right)^2 \\
&= -\frac{dY_t}{dt}\left(a + b\frac{1}{Y_t}\frac{dY_t}{dt}\right) + \frac{1}{Y_t}\left(\frac{dY_t}{dt}\right)^2 \\
&= \left[-a + (1-b)\frac{1}{Y_t}\frac{dY_t}{dt}\right]\frac{dY_t}{dt}.
\end{aligned} \tag{3.E.1}$$

Let us rewrite Equation 3.E.1 as

$$\frac{1}{dY_t/dt}\frac{d^2Y_t}{dt^2} = \left[-a + (1-b)\frac{1}{Y_t}\frac{dY_t}{dt}\right]. \tag{3.E.2}$$

Then integrating this expression yields

$$\begin{aligned}
\ln\left(\frac{dY_t}{dt}\right) &= -at + (1-b)\ln Y_t + \ln C_0, \ (\ln C_0 \text{ a constant}) \\
\frac{dY_t}{dt} &= C_0 e^{-at} Y_t^{1-b}, \\
Y_t^{b-1} dY_t &= C_0 e^{-at} dt, \\
\frac{1}{b}Y_t^b &= C_0\left(-\frac{1}{a}e^{-at}\right) + C_1 \ (C_1 \text{ a constant})
\end{aligned}$$

or

$$\begin{aligned}
Y_t^b &= -\frac{a}{b}C_0 e^{-at} + bC_1 \\
&= bC_1\left(1 - \frac{C_0}{aC_1}e^{-at}\right).
\end{aligned} \tag{3.E.3}$$

Then

$$Y_t = (bC_1)^{\frac{1}{b}}\left(1 - \frac{C_0}{aC_1}e^{-at}\right)^{1/b}. \tag{3.E.4}$$

Since

$$\lim_{t\to\infty} Y_t = (bC_1)^{1/b} = Y_\infty,$$

Equation 3.E.4 can be rewritten as

$$Y_t = Y_\infty (1-\delta e^{-at})^{1/b}, \tag{3.E.5}$$

where $\delta = C_0/aC_1$.

To derive Schnute (Eq. 3.55), let us employ Equation 3.E.3 to obtain

$$Y_1^b = bC_1 - \frac{b}{a}C_0 e^{-at_1},$$

$$Y_2^b = bC_1 - \frac{b}{a}C_0 e^{-at_2}$$

so that

$$Y_2^b - Y_1^b = -\frac{b}{a}C_0\left(e^{-at_2} - e^{-at_1}\right).$$

Then

$$\left(Y_2^b - Y_1^b\right)e^{at_1} = \frac{b}{a}C_0\left(1 - e^{-a(t_2-t_1)}\right)$$

and, via Equation 3.E.3,

$$\left(Y_t^b - Y_1^b\right)e^{at_1} = \frac{b}{a}C_0\left(1 - e^{-a(t-t_1)}\right).$$

Eliminating $\frac{b}{a}C_0$ from the latter two equations enables us to obtain

$$Y_t^b - Y_1^b + \left(Y_2^b - Y_1^b\right)\frac{1-e^{-a(t-t_1)}}{1-e^{-a(t_2-t_1)}}$$

or

$$Y_t = \left[Y_1^b + \left(Y_2^b - Y_1^b\right)\frac{1-e^{-a(t-t_1)}}{1-e^{-a(t_2-t_1)}}\right]^{1/b}. \tag{3.E.6}$$

APPENDIX 3.F THE MCDILL–AMATEIS MODEL DERIVED

McDill and Amateis derive their growth equation using *dimension analysis*, which ensures that the dimensions associated with the arguments on each side of a dimensionally compatible equation cancel. The process first identifies the relevant variables

TABLE 3.F.1 Variables and Dimensions

Variable	H	A	dH/dA	H_∞
Dimension	l	t	l/t	l

to be included in the model. Next, these variables are grouped into dimensionless products $\Pi_1, \Pi_2, \ldots, \Pi_k$, and the implicit function $f(\Pi_1, \Pi_2, \ldots, \Pi_k) = 0$ depicts all the dimensionally compatible equations involving the set of relevant variables.

The set of relevant variables and their associated dimensions appear in Table 3.F.1, where l = length and t = time. Then step two earlier has us form two dimensionless products:

$$\Pi_1 = \frac{dH}{dA}\frac{A}{H}, \Pi_2 = \frac{H}{H_\infty};$$

and the general function of these two dimensionless products is

$$f(\Pi_1, \Pi_2) = f\left(\frac{dH}{dA}\frac{A}{H}, \frac{H}{H_\infty}\right) = 0.$$

By virtue of the conditions underlying the implicit function theorem, there exists a function ϕ such that

$$\frac{dH}{dA}\frac{A}{H} = \phi\left(\frac{H}{H_\infty}\right). \tag{3.F.1}$$

How should ϕ be specified? The inclusion of the variable H_∞ in the growth equation mandates that the projected height growth of trees approaches zero as tree height H reaches H_∞. The simplest functional form of ϕ that exhibits this property is

$$\phi\left(\frac{H}{H_\infty}\right) = \left(1 - \frac{H}{H_\infty}\right); \tag{3.F.2}$$

that is, $\phi \to 0$ as $H/H_\infty \to 1$. Substituting Equation 3.F.2 into Equation 3.F.2 enables us to obtain

$$\frac{dH}{dA} = a\left(\frac{H}{A}\right)\left(1 - \frac{H}{H_\infty}\right). \tag{3.F.3}$$

Clearly height growth goes to zero as $H \to H_\infty$.

Let us rewrite Equation 3.F.3 as

$$\frac{dH}{H(H_\infty - H)} = \left(\frac{a}{H_\infty}\right)\frac{dA}{A}.$$

Then

$$-\frac{1}{H_\infty}\ln\left(\frac{H_\infty - H}{H}\right) = \left(\frac{a}{H_\infty}\right)\ln A + \ln C, \text{ (ln}C\text{ a constant)}$$

$$\left(\frac{H_\infty - H}{H}\right) = C^{-H_\infty} A^{-a},$$

$$H = \frac{H_\infty}{1 + C^{-H_\infty} A^{-a}}. \tag{3.F.4}$$

Evaluating Equation 3.F.4 at (A_0, H_0) yields

$$H_0 = \frac{H_\infty}{1 + C^{-H_\infty} A^{-a}},$$

from which we may obtain

$$C^{-H_\infty} = \left(\frac{H_\infty}{H} - 1\right) A_0^a.$$

Then a substitution of this expression into Equation 3.F.4 gives the final form of the McDill–Amateis height-growth equation or

$$H = \frac{H_\infty}{1 - \left(1 - \frac{H_\infty}{H_0}\right)\left(\frac{A_0}{A}\right)^a}.$$

APPENDIX 3.G THE SLOBODA MODEL DERIVED

Starting from the differential equation

$$\frac{dY_t}{dt} = \ln\left(\frac{Y_\infty}{Y_t}\right) b_1 Y_t t^{-b_2}, \tag{3.G.1}$$

let us rewrite Equation 3.G.1 as

$$\frac{dY_t}{\ln\left(Y_t / Y_\infty\right) Y_t} = -b_1 t^{-b_2}\, dt. \tag{3.G.2}$$

Then

$$\ln\left(\ln\left(\frac{Y_t}{Y_\infty}\right)\right) = \frac{-b_1}{-b_2 + 1} t^{-b_2 + 1} + \ln C, \text{ (ln } C \text{ a constant)}$$

and thus,

$$\ln\left(\frac{Y_t}{Y_\infty}\right) = Ce^{(b_1/(b_2-1))t^{-b_2+1}},$$

$$\frac{Y_t}{Y_\infty} = e^{Ce^{(b_1/(b_2-1))t^{-b_2+1}}},$$

or

$$Y_t = Y_\infty e^{Ce^{(b_1/(b_2-1))t^{-b_2+1}}}$$

$$= Y_\infty e^{-\alpha e^{-\beta t^\gamma}}, \tag{3.G.3}$$

where $-\alpha = C$, $-\beta = b_1/(b_2 - 1)$, and $\gamma = -b_2 + 1$.

APPENDIX 3.H A GENERALIZED MICHAELIS–MENTEN GROWTH EQUATION

In this section we consider a generalized Michaelis–Menten-type equation that exhibits a flexible functional form for describing (animal) growth and can produce sigmoidal and diminishing returns behavior in that it has a variable inflection point (Lopez et al., 2000). Assuming a closed system (no inputs or outputs), the increase in biomass (Y, kg) per unit of time (say, in weeks) is proportional to the substrate level S (kg) with proportionality factor μ (week $^{-1}$) or

$$\frac{dY}{dt} = \mu S. \tag{3.H.1}$$

Suppose μ changes with time according to

$$\mu = \frac{ct^{c-1}}{k^c + t^c}, \tag{3.H.2}$$

with c (dimensionless) and k (wk) being positive constants. (The conditions $c > 0$ and $k > 0$ ensure that $\mu > 0$, with k serving as the time when half-maximal growth is attained.) In Equation 3.H.2, μ can decrease continually for $c \leq 1$; it can increase to reach a maximum and then decrease again when $c > 1$.

A substitution of Equation 3.H.2 into Equation 3.H.1 gives

$$\frac{dY}{dt} = \mu S = \left(\frac{ct^{c-1}}{k^c + t^c}\right)(Y_\infty - Y),$$

where $S = Y_\infty - Y$. Then from this expression,

$$\int_{Y_o}^{Y} \frac{dY}{Y_\infty - Y} = \int_o^t \left(\frac{ct^{c-1}}{k^c + t^c}\right) dt$$

and, upon integrating,

$$-\ln(Y_\infty - Y)]_{Y_o}^{Y} = \ln(k^c + t^c)]_o^t,$$

$$-\ln(Y_\infty - Y) + \ln(Y_\infty - Y_o) = \ln(k^c + t^c) - \ln \mathrm{k}^c,$$

$$\frac{Y_\infty - Y_o}{Y_\infty - Y} = \frac{k^c + t^c}{k^c},$$

$$Y = \frac{Y_o k^c + Y_\infty t^c}{k^c + t^c}, \qquad (3.H.3)$$

the *generalized Michaelis–Menten growth function*.

This function has a point of inflection (t^*, Y^*), which occurs where $d^2Y/dt^2 = 0$ (provided $d^3Y/dt^3 \neq 0$), with

$$t^* = k\left(\frac{c-1}{c+1}\right)^{1/c}, \quad c > 1, \qquad (3.H.4)$$

and

$$Y^* = \left[\left(1 + \frac{1}{c}\right)\right]Y_o + \left(1 - \frac{1}{c}\right)Y_\infty / 2. \qquad (3.H.5)$$

Additionally:

1. When $c = 1$, we have $\mu = 1/(t + k)$, where k is the inverse of μ_{max}. Then

$$Y = \frac{Y_o k + Y_\infty t}{k + t}. \qquad (3.H.6)$$

Here Equation 3.H.6 is a rectangular hyperbola. And if $Y_o = 0$ in Equation 3.H.6, we obtain an expression of the form Equation 3.57, the Michaelis–Menten equation, with time replacing substrate concentration. The growth rate decreases continually and there is no point of inflection.

2. If μ = constant, then Equation (3.H.1) (with $S = Y_\infty - Y$) integrates to

$$Y = Y_\infty - \left(Y_\infty - Y\right)e^{-\mu t},$$

the *monomolecular growth function* (which decreases continually and has no point of inflection).

4

ESTIMATION OF TREND

4.1 LINEAR TREND EQUATION

Let us consider what we shall call a *purely mathematical (linear) model* involving Y as a function of time t or

$$Y = \beta_0 + \beta_1 t, \tag{4.1}$$

where β_0 is the (constant) *vertical intercept* and β_1 is the (constant) *slope*. Here β_0 is the value of Y when $t = 0$ while $\beta_1 = \Delta Y/\Delta t$ (=rise/run). Specifically, β_1 is the rate of change in Y per unit change in t. So if $\beta_1 = 5$, then when t increases by one unit, Y increases by five units; and if $\beta_1 = -3$, then as t increases by one unit, Y decreases by three units.

To obtain Equation 4.1, we need only two specific points in the (t, Y)-plane (Fig. 4.1). That is, $\beta_1 = \Delta Y/\Delta t = (Y_2 - Y_1)/(t_2 - t_1)$; and at, say, (t_2, Y_2), $\beta_0 = Y - \beta_1 t = Y_2 - \beta_1 t_2$. For instance, suppose $(t_1, Y_1) = (1, 2)$ and $(t_2, Y_2) = (4, 7)$. What is the equation of the line passing through these two points?

Clearly $\beta_1 = \Delta Y/\Delta t = (7-2)/(4-1) = 5/3$ while, at (t_1, Y_1), $\beta_0 = 2 - (5/3)(1) = 1/3$. Hence our particularization of Equation 4.1 is $Y = (1/3) + (5/3)t$. So when, $t = 0$, $Y = 1/3$; and, for $\beta_1 = 5/3$, when t increases by one unit, Y increases by 5/3 units.

What if we have more than two points in the (t, Y)-plane? Specifically, we may have a scatter of points such as the one depicted in Figure 4.2.

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

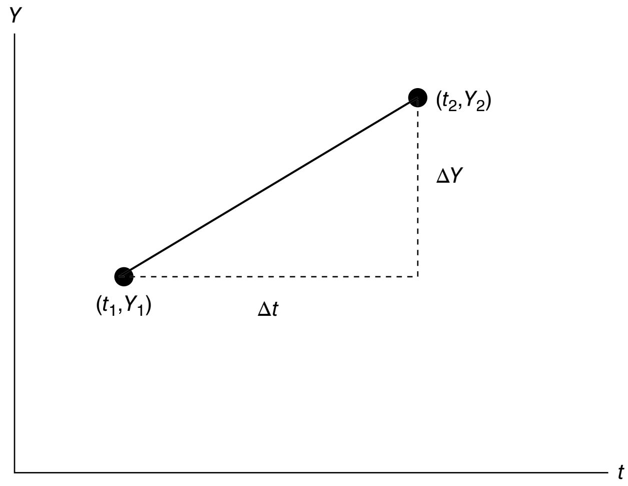

FIGURE 4.1 The line through the points (t_1,Y_1) and (t_2,Y_2).

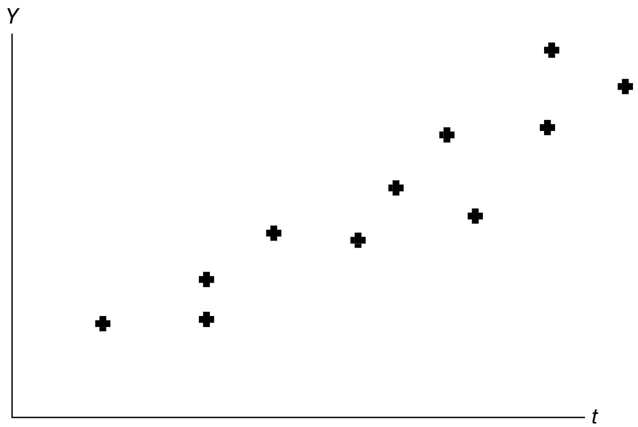

FIGURE 4.2 A scatter of points in the (t,Y)-plane.

In this circumstance, the above two-point procedure for determining Equation 4.1 is obviously no longer applicable. In this regard, we need to abandon the concept of a purely mathematical model and look to the development of a *statistical (linear) model*.

Why is it that the points illustrated in Figure 4.2 do not all lie on a straight line? The answer is that there are factors other than t that are operating to pull Y away from the line. Some of these factors cause Y to increase above the line while others cause Y to drop below that line. Can one readily predict how these extraneous factors will operate on Y? The answer is "no." So not only is Y determined by the *deterministic portion* $\beta_0+\beta_1 t$, it also has a *random portion* which we shall denote as ε. Hence our *statistical model* appears as

$$Y = \beta_0 + \beta_1 t + \varepsilon, \tag{4.2}$$

where the random error term ε is introduced to account for the net effect of all excluded variables or determinants of Y.

Since ε is a random variable, it has a probability distribution which is taken to satisfy the following set of assumptions (coupling Eq. 4.2 with these assumptions gives us what is termed the *strong classical linear regression model*):

1. ε_i is normally distributed for all i (since ε_i is determined by a whole host of factors, ostensibly working in opposite directions, we may expect small values of ε_i to occur more frequently than large ones.)
2. ε_i has a zero mean or $E(\varepsilon_i)=0$ for all i. (Positive deviations from $\beta_0+\beta_1 t$ are just as likely to occur as negative ones so that the random errors ε_i are distributed about a mean of zero.)
3. The errors ε_i are *homoscedastic* or $V(\varepsilon_i)=E(\varepsilon_i^2)-E(\varepsilon_i)^2=E(\varepsilon_i^2)=\sigma_\varepsilon^2=$ constant for all i. If $\sigma_\varepsilon^2 \neq$ constant, then errors are said to be *heteroscedastic*.
4. *Nonautocorrelation* of the ε_i's or $\mathrm{COV}(\varepsilon_i, \varepsilon_j)=E(\varepsilon_i, \varepsilon_j)=0$, $i\neq j$. (Successive ε_i's are uncorrelated.) In addition, t and ε_i are uncorrelated or $\mathrm{COV}(t_i, \varepsilon_j)=0$, $i\neq j$.
5. t is a nonrandom variable with finite variance.

Here assumptions 2 and 4 imply that the ε_i's are uncorrelated while assumptions 1, 2, and 4 imply that the ε_i's are independent. In addition, under assumption 1, we can conduct hypothesis tests and construct confidence intervals for β_0 and β_1.

Given assumptions 2 and 5, from Equation 4.2 we have

$$E(Y|t)=\beta_0+\beta_1 t, \tag{4.3}$$

which will serve as the *population regression line*. Hence Equation 4.2 can be rewritten as

$$Y=E(Y|t)+\varepsilon=\beta_0+\beta_1 t+\varepsilon. \tag{4.2.1}$$

How does Equation 4.3 compare with Equation 4.1? In Equation 4.1, Y equals $\beta_0+\beta_1 t$ exactly; in Equation 4.3, Y equals $\beta_0+\beta_1 t$ "on the average." Moreover, in Equation 4.3, β_0 is the average value of Y when $t=0$ or $\beta_0=E(Y|0)$; and β_1 is now the "average rate of change in Y per unit increase in t" or, stated alternatively, when t increases by one unit, Y changes by β_1 units "on the average."

When the values of β_0 and β_1 are estimated on the basis of observed data, then the *estimated (*or *sample) regression line*

$$\hat{Y}=\hat{\beta}_0+\hat{\beta}_1 t \tag{4.4}$$

serves as our proxy for the population regression line Equation 4.3, where $\hat{Y}$ is the fitted or estimated value of Y and $\hat{\beta}_0$ and $\hat{\beta}_1$ represent the estimated population parameters. (Think of $\hat{\beta}_0$ and $\hat{\beta}_1$ as arbitrary estimators of β_0 and β_1 respectively.) Since most (if not all) of the observed Y_i values will not lie exactly on the estimated regression line, the values of Y_i and $\hat{Y}_i$ differ. This difference will be denoted as $e_i = Y_i - \hat{Y}_i, i = 1,\ldots,n,$ and will be termed the *i*th *residual* or *deviation* from the estimated regression line. Clearly e_i serves as an estimates of the stochastic or unobserved disturbance ε_i.

4.2 ORDINARY LEAST SQUARES (OLS) ESTIMATION

The decision rule or criterion of goodness of fit to be employed in estimating β_0 and β_1 is depicted by the *Principle of Least Squares*: to obtain the line of best fit, choose $\hat{\beta}_0$ and $\hat{\beta}_1$ so as to minimize the sum of the squared deviations from the estimated regression line, i.e,

$$\min\left\{\sum_{i=1}^{n} e_i^2 = \sum\left(Y_i - \hat{Y}_i\right)^2 = \sum\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 t_i\right)^2 = F\left(\hat{\beta}_0, \hat{\beta}_1\right)\right\}.$$

(For convenience, the operator $\sum_{i=1}^{n}$ will at times be simplified to $\sum$, where it is to be understood that we always sum over all values of the i index, i.e., $i = 1,\ldots,n$.). Upon setting $\partial F / \partial\hat{\beta}_0 = \partial F / \partial\hat{\beta}_1 = 0$ (it is assumed that the second-order conditions for a minimum are satisfied), the resulting simultaneous linear equation system

$$\left.\begin{aligned}\sum e_i &= 0\\ \sum t_i e_i &= 0\end{aligned}\right\} \quad \text{or} \quad \begin{cases}(a) & n\hat{\beta}_0 + \hat{\beta}_1\sum t_i = \sum Y_i\\ (b) & \hat{\beta}_0\sum t_i + \hat{\beta}_1\sum t_i^2 = \sum t_i Y_i\end{cases} \tag{4.5}$$

yields the solution

$$\hat{\beta}_0 = \frac{\left(\sum Y_i\right)\left(\sum t_i^2\right) - \left(\sum t_i Y_i\right)\left(\sum t_i\right)}{n\sum t_i^2 - \left(\sum t_i\right)^2}, \quad \hat{\beta}_1 = \frac{n\sum t_i Y_i - \left(\sum t_i\right)\left(\sum Y_i\right)}{n\sum t_i^2 - \left(\sum t_i\right)^2}. \tag{4.6}$$

Here $\hat{\beta}_0$ and $\hat{\beta}_1$ are known as the *ordinary least squares (OLS) estimators* of β_0 and β_1, respectively. An important property of the least squares line of best fit $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 t$ is that it passes through the point of means $\left(\bar{t}, \bar{Y}\right)$ (Fig. 4.3).

What about the statistical properties of the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$? To answer this question we shall rely upon the *Gauss-Markov Theorem*: if assumptions 2–5 above hold, then, within the class of linear unbiased estimators of β_0 and β_1, the least-squares estimators have minimum variance, i.e., the least-squares estimators are *BLUE* (Best Linear Unbiased Estimators). Hence *BLUE* requires that: the estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ be expressible as *linear combinations* of the Y_i, $i = 1,\ldots,n$; that they be *unbiased*; and their *variances be smaller* than that of any alternative linear unbiased estimators of β_0 and β_1.

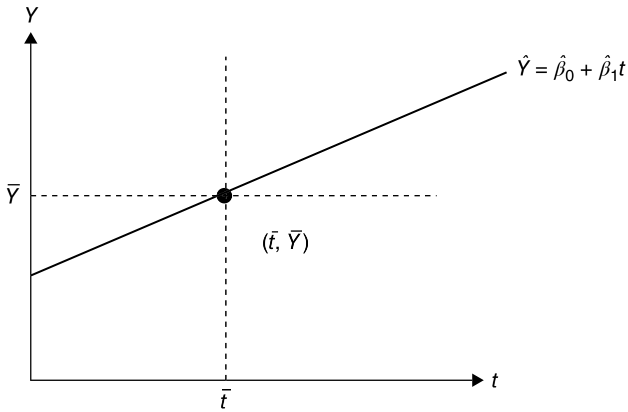

FIGURE 4.3 The OLS regression line passes through the point of means.

It is important to note that if we admit assumption 1 to our discussion, i.e., if we work with the strong classical linear regression model, then the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are maximum likelihood (ML) estimators of β_0 and β_1, respectively. It is the maximum likelihood estimation of β_0 and β_1 to which we now turn.

4.3 MAXIMUM LIKELIHOOD (ML) ESTIMATION

Given that the assumptions of the strong classical linear regression model hold so that the ε_i are independent and normally distributed, the joint probability density function of the ε_i's is provided by

$$\prod_{i=1}^{n} f\left(\varepsilon_i; 0, \sigma_\varepsilon^2\right) = \left(\sigma_\varepsilon^2 2\pi\right)^{-n/2} e^{-\frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^{n}\varepsilon_i^2}. \tag{4.7}$$

Since the Y_i's are normally distributed and independent (because the ε_i's are), and given that ε_i transforms linearly to Y_i, the likelihood function becomes, for $\varepsilon_i = Y_i - \beta_0 - \beta_1 t_i$, $i = 1, \ldots, n$,

$$\mathcal{L}\left(\beta_0, \beta_1, \sigma_\varepsilon^2; Y_1, \ldots, Y_n, n\right) = \left(\sigma_\varepsilon^2 2\pi\right)^{-n/2} e^{-\left(1/2\sigma_\varepsilon^2\right)\sum\left(Y_i - \beta_0 - \beta_1 t_i\right)^2}$$

or, in log-likelihood form,

$$\log \mathcal{L} = -\frac{n}{2}\log 2\pi - \frac{n}{2}\log \sigma_\varepsilon^2 - \frac{1}{2\sigma_\varepsilon^2}\sum\left(Y_i - \beta_0 - \beta_1 t_i\right)^2. \tag{4.8}$$

Differentiating Equation 4.8 partially with respect to the parameters β_0, β_1 and σ_ε^2 yields

$$\frac{\partial \log \mathcal{L}}{\partial \beta_0} = \frac{1}{\sigma_\varepsilon^2}\sum\left(Y_i - \beta_0 - \beta_1 t_i\right) \tag{4.9a}$$

$$\frac{\partial \log \mathcal{L}}{\partial \beta_1} = \frac{1}{\sigma_\varepsilon^2} \sum t_i \left(Y_i - \beta_0 - \beta_1 t_i\right) \tag{4.9b}$$

$$\frac{\partial \log \mathcal{L}}{\partial \sigma_\varepsilon^2} = -\frac{n}{2\sigma_\varepsilon^2} + \frac{1}{2\sigma_\varepsilon^4} \sum \left(Y_i - \beta_0 - \beta_1 t_i\right)^2 . \tag{4.9c}$$

Upon equating each of these expressions to zero (it is assumed that the second-order conditions for a maximum of $\log \mathcal{L}$ hold) and placing a tilde over each unknown parameters so as to indicate a maximum likelihood estimator of that parameter, we see that Equation 4.9a and b reduce to the system of least squares normal equations (Eq. 4.5) or

$$\begin{aligned} n\tilde{\beta}_0 + \tilde{\beta}_1 \sum t_i &= \sum Y_i \\ \tilde{\beta}_0 \sum t_i + \tilde{\beta}_1 \sum t_i^2 &= \sum t_i Y_i \end{aligned} \tag{4.10}$$

Hence the *maximum likelihood (ML) estimators* $\tilde{\beta}_0$ and $\tilde{\beta}_1$ of β_0 and β_1, respectively, are the same as the least squares estimators of these parameters. And from Equation 4.9c, we obtain the maximum likelihood estimator of σ_ε^2 or

$$\tilde{\sigma}_\varepsilon^2 = \frac{1}{n} \sum \left(Y_i - \tilde{\beta}_0 - \tilde{\beta}_1 t_i\right)^2 .$$

Since $\tilde{\beta}_0$ and $\tilde{\beta}_1$ coincide with the least squares estimators of the regression parameters, it follows that

$$\tilde{\sigma}_\varepsilon^2 = \frac{1}{n} \sum e_i^2 , \tag{4.11}$$

i.e., the maximum likelihood estimator of σ_ε^2 is the sample variance of the least squares residuals e_i.

In sum, for the strong classical linear regression model, the least squares estimators are equivalent to the maximum likelihood estimators of the regression intercept and slope. Moreover, while the least squares and maximum likelihood methods both provide us with formulas for the estimators for β_0 and β_1, the maximum likelihood routine goes a step further-it provides us with a formula for an estimator of σ_ε^2, namely Equation 4.11.

It was mentioned earlier that, by virtue of the Gauss-Markov theorem, the least squares estimators for β_0 and β_1 possess key finite-sample properties in that they are best linear unbiased estimators, i.e., out of the class of unbiased linear estimators for the regression parameters, the least squares estimators have minimum variance. Moreover, it can be shown that the least squares estimators are the most efficient estimators for β_0 and β_1 in that their variances satisfy the Cramer-Rao lower bounds for unbiased estimators. Furthermore, since the least squares estimators are minimum variance bound estimators, they are also sufficient estimators of the regression parameters.

Finally, the least squares estimators, being the same as the maximum likelihood estimators for β_0 and β_1, possess certain asymptotic or large-sample properties of a good estimator since, under certain (mild) regularity conditions imposed on the likelihood function, the maximum likelihood (and thus least squares) estimators are asymptotically unbiased, consistent, asymptotically efficient, and asymptotically normal.

4.4 THE SAS SYSTEM

In order to estimate the parameters of a linear trend equation such as Equation 4.1 (or Eq. 2.18), let us employ the SAS System. The following example problem involves the simple application of OLS.

EXAMPLE 4.1 Suppose we have $n = 18$ annual observation on the variable Y from 1990 to 2007, where Y depicts U.S. Gross Domestic Product (GDP) in billions of (chained) 2000 dollars (Table 4.1)[1]. To estimate both the linear and semilogarithmic trend equations, let us look to the following set of SAS code (Exhibit 4.1).

TABLE 4.1 Time Series Observations on *Y* (GDP)

Year	*Y*(GDP)	*t*
1990	7,112.5	1
1991	7,100.5	2
1992	7,336.6	3
1993	7,532.7	4
1994	7,835.5	5
1995	8,031.7	6
1996	8,328.9	7
1997	8,703.5	8
1998	9,066.9	9
1999	9,470.3	10
2000	9,817.0	11
2001	9,890.7	12
2002	10,048.8	13
2003	10,301.0	14
2004	10,675.8	15
2005	11,003.4	16
2006	11,319.4	17
2007	11,566.8	18

[1] For this data series as well as other macroeconomic variables see, for instance, the web site http://www.newyorkfed.org. This site also includes regional, governmental, and international data sets.

EXHIBIT 4.1 SAS Code for Estimating Linear and Semilogarithmic Trend Equations

```
data gdp1; ①
  input y @@; ②
lny=log(y); ③
t=_n_; ④
datalines;
7112.5 7100.5 7336.6 7532.7
7835.5 8031.7 8328.9 8703.5
9066.9 9470.3 9817.0 9890.7
10048.8 10301.0 10675.8 1103.4
11319.4 11566.8
proc print data=gdp1; ⑤
run;
axis2 logbase=e; ⑥
proc gplot data=gdp1; ⑦
plot y*t; ⑧
plot lny*t/haxis=axis1
                  vaxis=axis2; ⑨
run;
proc reg data =gdp1; ⑩
           ⑪    ⑫ ⑬    ⑭
model y=t/clb p clm cli;
plot y*t/conf; ⑮
plot y*t/pred; ⑯
model lny = t/clb p clm cli;  }
     plot lny*t/conf;         } ⑰
     plot lny*t/pred;         }
run;
```

① We name the data set *gdp*1.
② We name the input variable *y*. The double ampersand indicates use of the input statement without column locations.
③ We apply the natural logarithm transformation.
④ We apply the sequential period indicator to obtain values of time *t*.
⑤ We request a printout of the full data set by invoking the PRINT procedure.
⑥, ⑦, ⑧, ⑨ We invoke the GPLOT procedure so as to obtain graphs of *y* and ln*y* against *t*.

⑩ We invoke the REG procedure so as to obtain OLS estimates of the regression equations $Y=\beta_0+\beta_1 t$ and $\ln Y=\beta_0+\beta_1 t$. A slash separates each model statement from the request for regression ancillaries.
⑪ The *clb* option requests 95% confidence limits for each regression coefficient.
⑫ The *p* option requests, for each data point (t, Y), a print of the observed value Y_i of the dependent variable Y, the predicted value $\hat{Y}_i$, and the residual $e_i = Y_i - \hat{Y}_i$, $i=1,\ldots,n$.
⑬, ⑮ the *clm* option requests a 95% confidence interval for the mean of Y given t or $E(Y|t)=\beta_0+\beta_1 t$. The plot statement *plot y*t/conf* requests a plot of the resulting confidence band.
⑭, ⑯ the *cli* option requests a 95% prediction interval for Y_0 given t_0. The plot statement *plot y*t/pred* requests a plot of the resulting prediction band.
⑰ See items ⑪,…, ⑯.

The resulting output is presented in Table 4.2. Figure 4.4 and Figure 4.5 house the requested plots.

① We obtain a print of the Y, $\ln Y$ and t series.

② For Model 1 (Y is the dependent variable):

a. The regression mean square is MSR = SSR/1 = 37368853, the error mean square is MSE = SSE/$(n-2)$ = 17,146. Under $H_0: \beta_1 = 0$, the sample F value is MSR/MSE = 2179.40.

b. The p-value accompanying the calculated F value is very small, indicating a highly significant linear relationship between y and t (there is less than a 0.01% chance of finding a calculated F value at least as large as 2179.40 if $H_0: \beta_1 = 0$ is true).

c. R^2 = SSR/SST = 0.9927. Thus in excess of 99% of the variation in Y is explained by the linear influence of t.

③ For Model 1:

a. The estimated regression equation has the form $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 t = 6536.21 + 277.72t$. Here the average value of Y when $t = 0$ is 6536.21 while when t increases by one unit, Y increases by 277.72 units on the average.

b. The very small p-value accompanying the calculated t value for the estimated intercept indicates that the intercept is significantly different from zero (there is less than a 0.01% chance of finding a t value at least as large as 101.50 if $H_0: \beta_0 = 0$ is true). Similarly, the very small p-value supporting the calculated t value for the estimated slope reveals that the slope is significantly different from zero (there is less than a 0.01% chance of finding a t value at least as large as 46.68 if $H_0: \beta_1 = 0$ is true).

c. We may be 95% confident that $6399.70 \le \beta_0 \le 6672.71$; and we are likewise 95% confident that $265.11 \le \beta_1 \le 290.33$.

④ For Model 1:

a. For each data point (t, Y), the predicted value of Y is determined from the estimated regression equation, e.g., for observation number 2, the observed value of Y is $Y_2 = 7101$ while the predicted Y value is $\hat{Y}_2 = \hat{\beta}_0 + \hat{\beta}_1(2) = 6536.21046 + 277.72054(2) = 7092$.

b. These two columns provide lower and upper 95% confidence limits respectively for the mean of Y given t, e.g., for observation number 2, we may be 95% confident that the population average value of Y given $t = 2$ (or $E(Y_2 | 2) = \hat{\beta}_0 + \hat{\beta}_1(2)$) lies between 6977 and 7204 (Fig. 4.6), where $\hat{Y}_2 = 7092$ is an estimate of $E(Y_2|2)$.

c. These next two columns give lower and upper 95% prediction limits for Y_0 given t_0, e.g., for observation number 2, we may be 95% confident that $Y_0 = Y_2$ given $t_0 = 2$ (note that an estimate of Y_0 is $\hat{Y}_0 = \hat{Y}_2 = 7092$) lies between 6791 and 7392 (Fig. 4.7).

d. For any data point the residual is calculated as the difference between the observed Y value and the estimated or predicted value of Y or $e_i = Y_i - \hat{Y}_i$, e.g., for observation number 2, $e_2 = Y_2 - \hat{Y}_2 = 7101 - 7092 = 8.8485$.

TABLE 4.2 Output for Example 4.1

The SAS system
①

Obs	y	ln$y$	t
1	7,112.5	8.86961	1
2	7,100.5	8.86792	2
3	7,336.6	8.90063	3
4	7,532.7	8.92701	4
5	7,835.5	8.96642	5
6	8,031.7	8.99115	6
7	8,328.9	9.02749	7
8	8,703.5	9.07148	8
9	9,066.9	9.11239	9
10	9,470.3	9.15592	10
11	9,817.0	9.19187	11
12	9,890.7	9.19935	12
13	10,048.8	9.21521	13
14	10,301.0	9.24000	14
15	10,675.8	9.27573	15
16	11,003.4	9.30596	16
17	11,319.4	9.33427	17
18	11,566.8	9.35589	18

(*Continued*)

TABLE 4.2 Output for Example 4.1 (*Continued*)

The SAS System
The REG procedure
Model: MODEL1
Dependent variable: y
Number of observations read: 18
Number of observations used: 18

②
Analysis of variance

Source	DF	Sum of squares	Mean square	ⓐ F value	ⓑ Pr > F
Model	1	37,368,853	37,368,853	2179.40	<0.0001
Error	16	274,342	17,146		
Corrected total	17	37,643,196			

		ⓒ	
Root MSE	130.94426	R-square	0.9927
Dependent mean	9,174.55556	Adj R-square	0.9923
Coeff var	1.42725		

③
Parameter estimates

Variable	DF	ⓐ Parameter estimate	Standard error	t value	ⓑ Pr>|t|	ⓒ 95% confidence limits	
Intercept	1	6536.21046	64.39344	101.50	<0.0001	6399.70247	6672.71845
t	1	277.72054	5.94894	46.68	<0.0001	265.10935	290.33173

The REG procedure
Model: MODEL1
Dependent variable: y
④
Output statistics

Obs	Dependent variable	ⓐ Predicted value	Standard error mean predict	ⓑ 95% CL mean		ⓒ 95% CL predict		ⓓ Residual
1	7,113	6,814	59.2410	6,688	6,940	6,509	7,119	298.5690
2	7,101	7,092	54.2518	6,977	7,207	6,791	7,392	8.8485
3	7,337	7,369	49.4752	7,264	7,474	7,073	7,666	−32.7721
4	7,533	7,647	44.9791	7,552	7,742	7,354	7,941	−114.3926
5	7,836	7,925	40.8561	7,838	8,011	7,634	8,216	−89.3131
6	8,032	8,203	37.2304	8,124	8,281	7,914	8,491	−170.8337
7	8,329	8,480	34.2602	8,408	8,553	8,193	8,767	−151.3542
8	8,704	8,758	32.1279	8,690	8,826	8,472	9,044	−54.4748
9	9,067	9,036	31.0069	8,970	9,101	8,750	9,321	31.2047
10	9,470	9,313	31.0069	9,248	9,379	9,028	9,599	156.8842
11	9,817	9,591	32.1279	9,523	9,659	9,305	9,877	225.8636
12	9,891	9,869	34.2602	9,796	9,941	9,582	10,156	21.8431
13	10,049	10,147	37.2304	10,068	10,226	9,858	10,435	−97.7774
14	10,301	10,424	40.8561	10,338	10,511	10,134	10,715	−123.2980
15	10,676	10,702	44.9791	10,607	10,797	10,409	10,996	−26.2185
16	11,003	10,980	49.4752	10,875	11,085	10,683	11,276	23.6610
17	11,319	11,257	54.2518	11,142	11,372	10,957	11,558	61.9404
18	11,567	11,535	59.2410	11,410	11,661	11,231	11,840	31.6199
		Sum of residuals				0		
		Sum of squared residuals				27,4342		
		Predicted residual SS (PRESS)				35,9236		

(*Continued*)

TABLE 4.2 Output for Example 4.1 (*Continued*)

The REG procedure
Model: MODEL 2
Dependent variable: ln$y$
Number of observations read: 18
Number of observations used: 18

⑤ Analysis of variance

Source	DF	Sum of squares	Mean square	ⓐ F value	ⓑ Pr > F
Model	1	0.45399	0.45399	1768.55	<0.0001
Error	16	0.00411	0.00025670		
Corrected total	17	0.45810			
			ⓒ		
Root MSE		0.01602	R-square	0.9910	
Dependent mean		9.11157	Adj R-square	0.9905	
Coeff var		0.17584			

⑥ Parameter estimates

Variable	DF	ⓐ Parameter estimate	Standard error	t value	ⓑ Pr>|t|	ⓒ 95% confidence limits	
Intercept	1	8.82077	0.00788	1119.53	<0.0001	8.80407	8.83747
t	1	0.03061	0.00072789	42.05	<0.0001	0.02907	0.03215

The REG procedure
Model: MODEL2
Dependent variable: ln$y$
⑦
Output statistics

Obs	Dependent variable	ⓐ Predicted value	Standard error mean predict	ⓑ 95% CL mean		ⓒ 95% CL predict		ⓓ Residual
1	8.8696	8.8514	0.007249	8.8360	8.8667	8.8141	8.8887	0.0182
2	8.8679	8.8820	0.006638	8.8679	8.8961	8.8452	8.9188	–0.0141
3	8.9006	8.9126	0.006054	8.8998	8.9254	8.8763	8.9489	–0.0120
4	8.9270	8.9432	0.005504	8.9315	8.9549	8.9073	8.9791	–0.0162
5	8.9664	8.9738	0.004999	8.9632	8.9844	8.9382	9.0094	–0.007403
6	8.9912	9.0044	0.004555	8.9948	9.0141	8.9691	9.0397	–0.0133
7	9.0275	9.0350	0.004192	9.0262	9.0439	8.9999	9.0702	–0.007558
8	9.0715	9.0657	0.003931	9.0573	9.0740	9.0307	9.1006	0.005825
9	9.1124	9.0963	0.003794	9.0882	9.1043	9.0614	9.1312	0.0161
10	9.1559	9.1269	0.003794	9.1188	9.1349	9.0920	9.1618	0.0290
11	9.1919	9.1575	0.003931	9.1492	9.1658	9.1225	9.1925	0.0344
12	9.1994	9.1881	0.004192	9.1792	9.1970	9.1530	9.2232	0.0113
13	9.2152	9.2187	0.004555	9.2091	9.2284	9.1834	9.2540	–0.003502
14	9.2400	9.2493	0.004999	9.2387	9.2599	9.2137	9.2849	–0.009325
15	9.2757	9.2799	0.005504	9.2683	9.2916	9.2440	9.3158	–0.004197
16	9.3060	9.3105	0.006054	9.2977	9.3234	9.2742	9.3469	–0.004584
17	9.3343	9.3412	0.006638	9.3271	9.3552	9.3044	9.3779	–0.006881
18	9.3559	9.3718	0.007249	9.3564	9.3871	9.3345	9.4090	–0.0159
		Sum of residuals				0		
		Sum of squared residuals				0.00411		
		Predicted residual SS (PRESS)				0.00509		

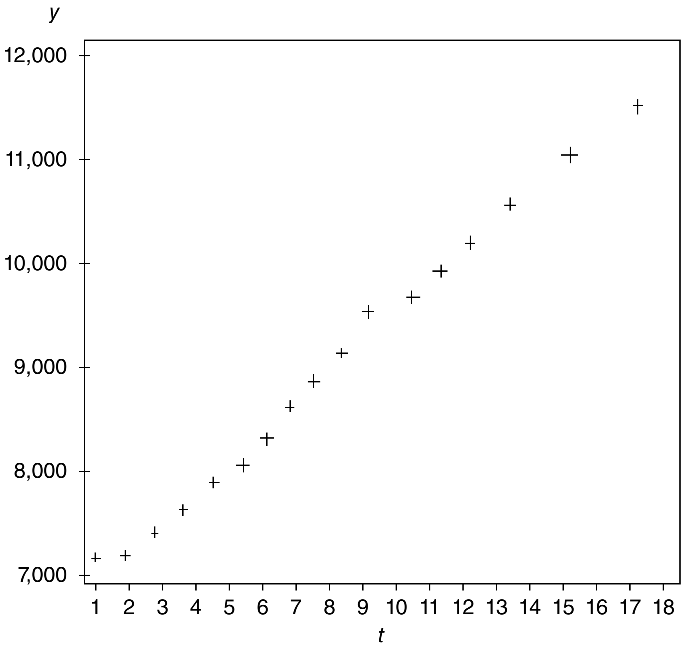

FIGURE 4.4 Plot of *Y* against time.

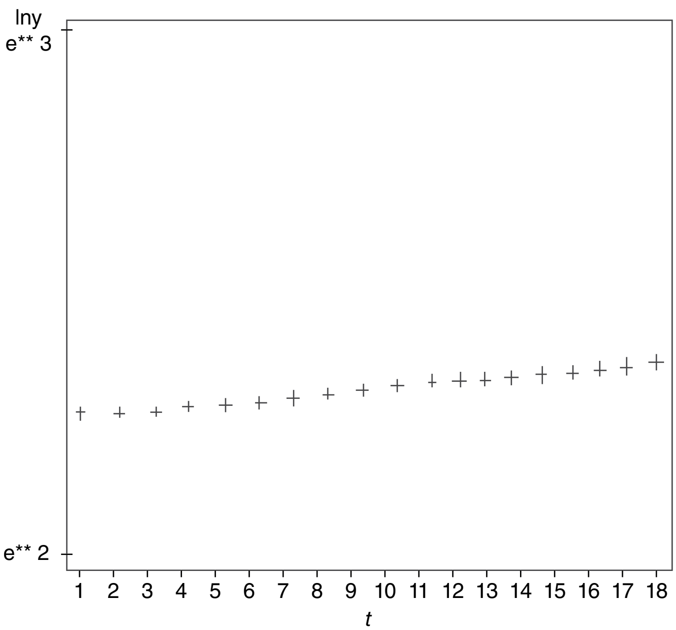

FIGURE 4.5 Plot of ln*Y* against time.

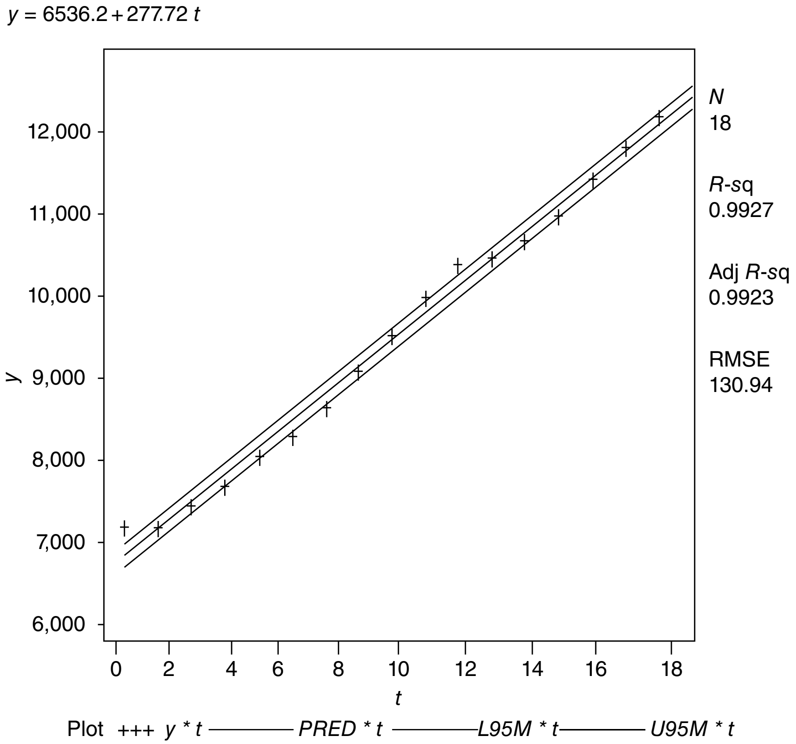

FIGURE 4.6 95% confidence limits for the mean Y given t.

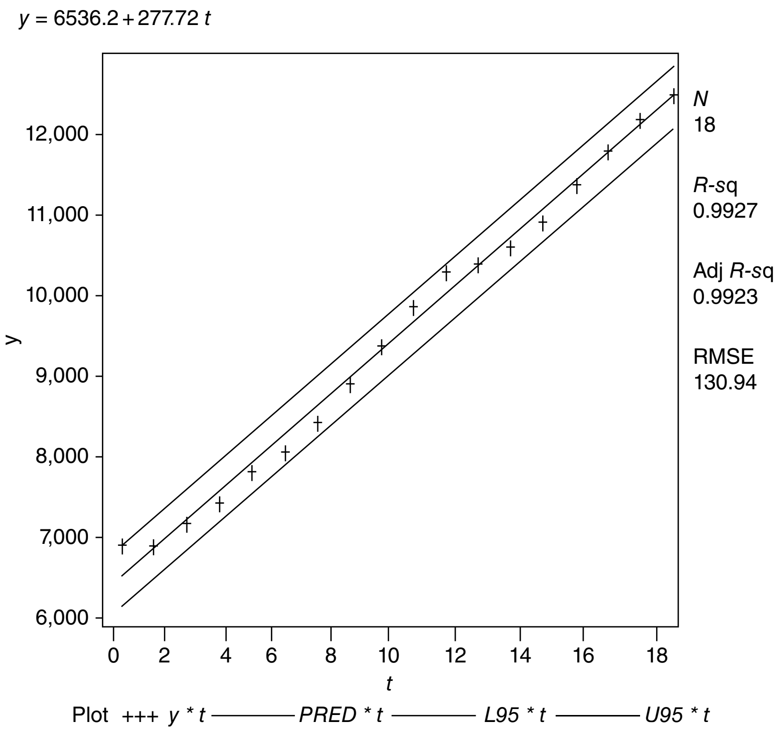

FIGURE 4.7 95% prediction limits for Y_o given t_0.

⑤ For Model 2 ($\ln Y$ is the dependent variable):

a. Here MSR = SSR/1 = 0.45399 and MSE = SSE/(n − 2) = 0.00025670. Under $H_0: \beta_1 = 0$, F = MSR/MSE = 1768.55.

b. Since the p-value associated with the preceding F value is very small, we may conclude that the linear relationship between $\ln Y$ and t is statistically significant (there is less than a 0.01% chance of finding a calculated F value at least as large as 1768.55 if $H_0: \beta_1 = 0$ is true).

c. R^2 = SSR/SST = 0.9910–in excess of 99% of the variation in $\ln Y$ is explained by the linear influence of t.

⑥ For Model 2:

a. The estimated regression equation is $\ln Y = \hat{\beta}_0 + \hat{\beta}_1 t = 8.82077 + 0.03061t$, where 8.82077 is the average value of $\ln Y$ when $t = 0$; and when t increases by one unit, $\ln Y$ increases by 0.03061 units on the average.

b. The small p-value associated with the calculated t value for the estimated intercept reveals that the intercept is significantly different from zero (there is less than a 0.01% chance of finding a t value at least as large as 1119.53 if $H_0: \beta_0 = 0$ is true). Also, the small p-value associated with the sample t value for the estimated slope reveals that the slope is significantly different from zero (there is less than a 0.01% chance of finding a t value at least as large as 42.05 if $H_0: \beta_1 = 0$ is true.)

c. We may be 95% confident that $8.80407 \le \beta_0 \le 8.83747$; and we are 95% confident that $0.02907 \le \beta_1 \le 0.03215$.

⑦ For Model 2:

a, d. As was the case for the linear trend model, the predicted values for the semilogarithmic case are determined from the estimated semilogarithmic trend equation, e.g., for, say, observation number 4, the observed $\ln Y$ is $\ln Y_4 = 8.9270$ and the predicted $\ln Y$ value is $\widehat{\ln Y_4} = \hat{\beta}_0 + \hat{\beta}_1(4) = 8.82077 + 0.03061(4) = 8.9432$. Then the residual at this data point is $e_4 = \ln Y_4 - \widehat{\ln Y} = 8.9270 - 8.9432 = -0.0162$.

b. For observation number 4, lower and upper 95% confidence limits for the mean value of $\ln Y$ given $t = 4$ are 8.9315 and 8.9549, respectively, i.e., we may be 95% confident that the population average value of $\ln Y$ given $t = 4$ (or $E(\ln Y\, 4) = \hat{\beta}_0 + \hat{\beta}_1(4)$) lies between these limits (Fig. 4.8), where $\widehat{\ln Y_4} = 8.9432$ is an estimate of $E(\ln Y|4)$.

c. Also, for observation number 4, lower and upper 95% prediction limits for $\ln Y_0 = \ln Y_4$ given $t_0 = 4$ are 8.9073 and 8.9791, respectively (Fig. 4.9), where an estimate of $\ln Y_0$ is $\widehat{\ln Y_0} = \widehat{\ln Y_4} = 8.9432$. Hence we may be 95% confident that $\ln Y_0 = \ln Y_4$ given $t_0 = 4$ lies between these limits.

It is evident that either the linear or semilogarithmic trend equation fits the data well. Which one of these equations should be used to estimate/predict GDP? The answer depends, of course, upon our objective. If we are interested in, say, predicting the *level* of Y in 2008, then either model can be used. In this instance $t = 19$ for the year 2008 so that, for Model 1,

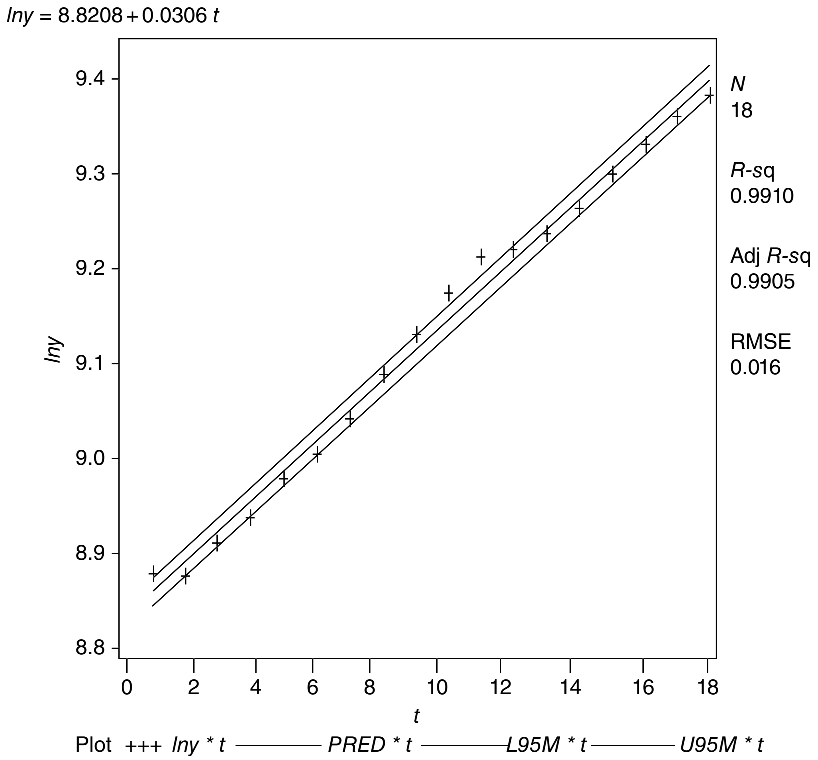

FIGURE 4.8 95% confidence limits for the mean of ln*Y* given *t*.

$$\hat{Y}_{2008} = 6536.21 + 277.72(19) = 11812.89;$$

and from Model 2,

$$\begin{aligned}\hat{Y}_{2008} &= e^{\hat{\beta}_0 + \hat{\beta}_1 t} = e^{8.82077+0.03061(19)} \\ &= 12116.943.\end{aligned}$$

(We shall try to resolve the obvious discrepancy between these two predicted values later on.) But if we are interested in estimating the average rate of growth or the compound interest rate of growth (under continuous compounding) over the sample period, then Model 2 should be used. For this model we found that $\hat{\beta}_1 = 0.03061$. Hence the average rate of growth in GDP over the period 1990 to 2007 is about 3% since $e^{\hat{\beta}_1} - 1 = e^{0.03061} - 1 = 0.031$. (This is easily verified since, via the geometric mean, the average rate of growth over the sample period is

$$GM = \left(\frac{Y_{2007}}{Y_{1990}}\right)^{1/18-1} - 1 = \left(\frac{11566.8}{7112.5}\right)^{1/17} - 1 = 0.029.)$$

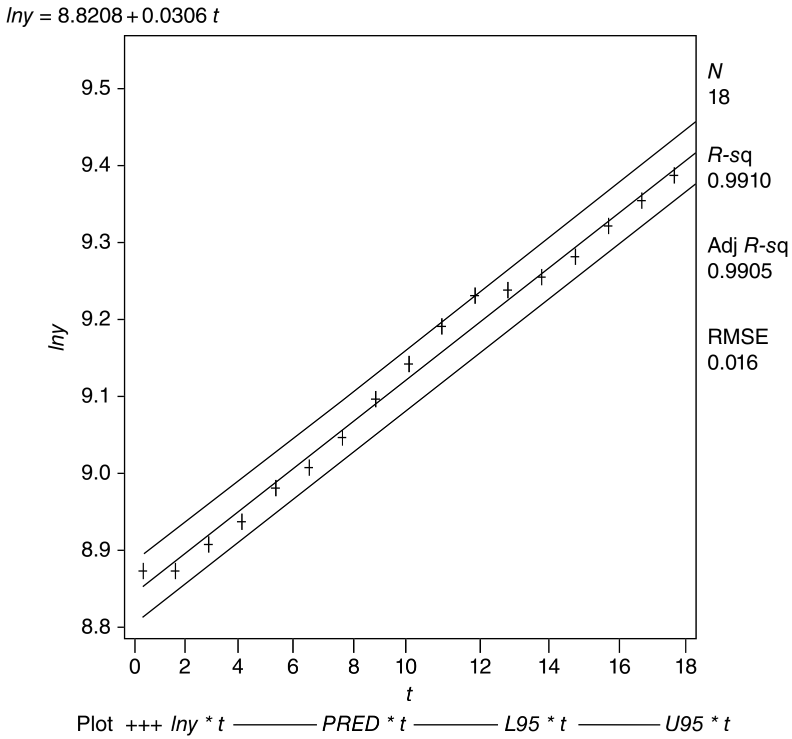

FIGURE 4.9 95% prediction limits for $\ln Y_0$ given t_0.

Given that $\hat{\beta}_1$ represents the OLS estimated slope for Model 2, we may term $e^{\hat{\beta}_1} - 1$ the *OLS estimated growth rate*. For additional details concerning this growth rate as well as the particulars of related growth rates, the reader can consult Appendix A to this chapter.

How should we interpret Figure 4.4 (with trend determined on an arithmetic scale) and Figure 4.5 (wherein trend is displayed using a semilogarithmic scale)? For period-to-period changes, a straight line on an arithmetic graph indicates a *constant amount of change*. Clearly the points in Figure 4.4 closely approximate a linear equation over the sample period. However, a straight line on a semilogarithmic graph indicates a *constant rate of change*. As Figure 4.5 reveals, a straight line provides an extremely good fit through the indicated set of points. So if the percentage change from period to period is constant, the straight line $\ln Y = \beta_0 + \beta_1 t$ with constant slope β_1 provides the best fit to the data. ■

When should a linear equation be used relative to a semilogarithmic equation when one wants to estimate the long-term trend in a variable? It is generally agreed that if the *first-differences between successive observations of a time series are approximately constant*, then a linear trend line is appropriate. (If the period-to-period changes are constant, in absolute value, then the Y series increases (or decreases) by approximately a constant amount each period.) To see this, let $Y_t = \beta_0 + \beta_1 t$.

Then

$$\begin{aligned}\Delta Y_t &= Y_t - Y_{t-1} = \beta_0 + \beta_1 t - \left[\beta_0 + \beta_1 (t-1)\right] \\ &= \beta_1 = \text{constant.}\end{aligned}$$

So for a linear trend equation, the first-differences are constant and equal to β_1.

However, a semilogarithmic trend equation with continuous compounding is useful for modeling time series data that are *changing (increasing or decreasing) at a constant rate*. In this case we concentrate on the first differences of the logarithms of the Y values between successive observations since the period-to-period differences in the logarithms of the Y series gives the exponential growth rate. If the said differences are approximately constant, then a semilogarithmic trend equation is called for. To verify this assertion, let $Y_t = e^{\beta_0 + \beta_1 t}$. Then

$$\begin{aligned}\Delta \ln Y_t &= \ln Y_t - \ln Y_{t-1} = \beta_0 + \beta_1 t - \left[\beta_0 + \beta_1 (t-1)\right] \\ &= \beta_1 = \text{constant.}\end{aligned}$$

EXAMPLE 4.2 Using the Example 4.1 data set, let us reestimate the semilogarithmic trend equation by using the specification $Y_t = \beta_0 \beta_1^t$. What, if anything, changes? We now want to fit the regression equation

$$\log_{10} Y_t = \log_{10} \beta_0 + \left(\log_{10} \beta_1\right) t.$$

In order to do so, a simple adjustment to Exhibit 4.1 can be made. Specifically, the natural logarithm ln$y$ = log(y) is replaced by log$y$ = log10(y) while in the model statement *lny = t/clb p clm cli* is replaced by *logy = t/clb p clm cli*. Once these adjustments are made and the program is run, the resulting (abbreviated) output is given in Table 4.3.

① For Model 3 (log$Y$ is the dependent variable):

a, b. The F and R^2 values are the same as those obtained when ln$Y$ was the dependent variable. Again the fit is highly significant.

② a. The estimated regression equation is

$$\begin{aligned}\widehat{\log_{10} Y} &= \widehat{\log_{10} \beta_0} + \left(\widehat{\log_{10} \beta_1}\right) t \\ &= 3.83081 + 0.01329t,\end{aligned}$$

where 3.83081 is the average value of $\log_{10} Y$ when t = 0; and when t increases by one unit, $\log_{10} Y$ increases by 0.01329 units on the average. Note that while the estimated slope of the regression line is $\widehat{\log_{10} \beta_1}$ = 0.01329, β_1 is the rate of increase in Y.

b. The t values obtained for this respecification of the semilogarithmic model are equal to those appearing above when ln$Y$ was the dependent variable (Model 2). Here too both the intercept and slope are significantly different from zero.

c. We may be 95% confident that $3.82356 \le \log_{10}\beta_0 \le 3.83806$; and we may be 95% confident that $0.01262 \le \log_{10}\beta_1 \le 0.01396$.

TABLE 4.3 Output for Example 4.2

The SAS system
The REG procedure
Model: MODEL 3
Dependent variable: log$y$
Number of observations read: 18
Number of observations used: 18

① Analysis of variance

Source	DF	Sum of squares	Mean square	ⓐ F value	Pr > F
Model	1	0.08563	0.08563	1768.55	<0.0001
Error	16	0.00077467	4.842E-05		
Corrected total	17	0.08640			

		ⓑ	
Root MSE	0.00969	R-square	0.9910
Dependent mean	3.95711	Adj R-square	0.9905
Coeff var	0.17584		

② Parameter estimates

Variable	DF	ⓐ Parameter estimate	Standard error	t value	ⓑ Pr>|t|	ⓒ 95% confidence limits	
Intercept	1	3.83081	0.00342	1119.53	<0.0001	3.82356	3.83806
t	1	0.01329	0.00031612	42.05	<0.0001	0.01262	0.01396

How are we to interpret these results? First, given $\widehat{\log_{10} Y} = 3.83081 + 0.01329t$, taking antilogs renders

$$\hat{Y}_t = \hat{\beta}_0 \hat{\beta}_1^{\,t} = 6673.451(1.0311)^t, \tag{4.12}$$

Where

$$\hat{\beta}_0 = 10^{3.83081} = 6673.451 \quad \text{and} \quad \hat{\beta}_1 = 10^{0.01329} = 1.0311.$$

(What are the 95% confidence limits for β_0 and β_1?) Clearly Equation 4.12 is similar to the compound interest formula $Y_t = \beta_0(1+r)^t$, where $\hat{r} = \hat{\beta}_1 - 1 = 0.0311$. Hence the constant average rate of growth in the Y series is $100\hat{r}\%$ or about 3% per annum. Note that this constant average rate of growth per annum is the same as that obtained above when continuous compounding was assumed. Next, the term $\log_{10}\beta_0$ is the logarithm of the trend at the origin. Thus its antilog is the trend value in natural numbers at the origin. ■

4.5 CHANGING THE UNIT OF TIME

In estimating the trend in the variable Y(GDP) we worked with the data in the form of annual totals. However, we may choose to express the data in the form of monthly averages or monthly totals (or in terms of quarterly averages or quarterly totals).

4.5.1 Annual Totals versus Monthly Averages versus Monthly Totals

Suppose we have a linear trend equation $Y_t = \beta_0 + \beta_1 t$ expressed in terms of *annual totals*. Then the *monthly average equation* is

$$Y_t = \frac{\beta_0}{12} + \frac{\beta_1}{12} t \tag{4.13}$$

(we divide β_0 and β_1 by 12 since the data are sums of 12 months), where $\beta_0/12$ is the *monthly level when* $t = 0$ and $\beta_1/12$ is the *annual increase of the monthly average value* (the annual change in monthly magnitudes).

Next, let us divide t by 12 in Equation 4.13 so that the time units will be in months as well (or simply divide β_1 by 12 again). Hence the *monthly total equation* is

$$Y_t = \frac{\beta_0}{12} + \frac{\beta_1}{12}\left(\frac{t}{12}\right) = \frac{\beta_0}{12} + \frac{\beta_1}{144} t, \tag{4.14}$$

where $\beta_1/144$ is the monthly increase of the monthly average value (the trend equation is now expressed in consecutive months).

For instance, from Example 4.1, the estimated annual total equation is $\hat{Y} = 6536.21 + 277.22t$. Then the monthly average equation is

$$\hat{Y} = \frac{6536.21}{12} + \frac{277.72}{12}t = 544.6842 + 23.1433t.$$

And the monthly equation is

$$\hat{Y} = \frac{6536.21}{12} + \frac{277.72}{144}t = 544.6842 + 1.9286t.$$

4.5.2 Annual Totals versus Quarterly Averages versus Quarterly Totals

Given the annual total linear trend equation $Y_t = \beta_0 + \beta_1 t$, *the quarterly average equation* is

$$Y_t = \frac{\beta_0}{4} + \frac{\beta_1}{4}t \tag{4.15}$$

(we divide β_0 and β_1 by 4 since the data are sums of four 3 month quarters), where $\beta_0/4$ is the *annual quarterly level when* $t = 0$ and $\beta_1/4$ is the *annual increase of the quarterly average value* (the annual change in quarterly magnitudes).

We next divide t by 4 in Equation 4.15 so that the time units are expressed in quarters as well (we simply divide β_1 by 4 again). Thus the *quarterly total equation* is

$$Y_t = \frac{\beta_0}{4} + \frac{\beta_1}{4}\left(\frac{t}{4}\right) = \frac{\beta_0}{4} + \frac{\beta_1}{16}t, \tag{4.16}$$

where $\beta_1/16$ is the quarterly increase of the quarterly average value (the trend equation is represented in consecutive quarters).

For example, given the annual total equation used in Section 4.5.1 or $Y_t = 6536.21 + 277.72t$, the quarterly average equation is

$$\hat{Y} = \frac{6536.21}{4} + \frac{277.72}{4}t = 1634.0525 + 69.43t;$$

and the quarterly equation is

$$\hat{Y} = \frac{6536.21}{4} + \frac{277.72}{16}t = 1634.0525 + 17.3575t.$$

4.6 AUTOCORRELATED ERRORS

In our estimation of the linear as well as the semilogarithmic trend equation we conveniently ignored a feature of time series data which can adversely impact the efficiency of our OLS estimates. Specifically, we are referring to the possibility that the error terms ε_i, $i = 1, \ldots, n$, might be autocorrelated. Let us now see exactly what the nature of this problem amounts to and, if autocorrelation is detected, how it is dealt with.

4.6.1 Properties of the OLS Estimators When ε Is AR(1)

Given the assumptions of the strong classical linear regression model (Section 4.1), $\mathrm{COV}(\varepsilon_i, \varepsilon_j) = E(\varepsilon_i \varepsilon_j) = 0$ for all $i \neq j$ (since $E(\varepsilon_i) = E(\varepsilon_j) = 0$). Moreover, since the ε_i's, $i = 1, \ldots, n$, are taken to be normally distributed, it follows that the zero covariance of ε_i and ε_j implies that these random error terms are also independent, i.e., the disturbance occurring at one data point is not associated with the disturbance occurring at any other such point. This independence property of the random error term is referred to as *nonautocorrelation*.

Under what circumstances can we expect the assumption of nonautocorrelation to be violated? It is typically the case that violations occur when regression estimates are made using time series data. In this regard, if the disturbance ε_i is associated with the disturbance ε_{i-s}, then the disturbances are said to be *autocorrelated* and

$$\mathrm{COV}\left(\varepsilon_i, \varepsilon_{i-s}\right) = E\left(\varepsilon_i \varepsilon_{i-s}\right) \neq 0, \quad i > s, \tag{4.17}$$

i.e., the effect of a myriad of random factors influencing the behavior of ε in period $i-s$ carry forward to affect the behavior of ε in period i.

To determine the impact of autocorrelation on our OLS regression results, let us be a bit more specific about the nature of Equation 4.17. In particular, what is our choice for s? It is usually the case that a *first-order (Markov) autoregressive scheme* (denoted as AR(1)) is posited or

$$AR(1): \varepsilon_i = \rho \varepsilon_{i-1} + u_i \tag{4.18}$$

for all i, where the parameter ρ is such that $|\rho| < 1$ and the new disturbance or random error term u_i (called an *innovation*) is taken to be *white noise* (it obeys the strong classical assumptions), i.e., u_i is normally and independently distributed with a zero mean, a constant variance σ_u^2, and is independent of ε_{i-1} or

1. u_i is $N(0, \sigma_u)$ for all i
2. $\mathrm{COV}\left(u_i, u_j\right) = E\left(u_i u_j\right) = \begin{cases} V\left(u_i\right) = \sigma_u^2, & i = s; \\ 0, & i \neq s \end{cases}$ (4.19)
3. $\mathrm{COV}(u_i, \varepsilon_{i-1}) = 0$ for all i.

(Note that if $\rho = 0$, $\varepsilon_i = u_i$ and $V\left(\varepsilon_i\right) = V\left(u_i\right) = \sigma_\varepsilon^2$ = constant for all i so that all of the basic assumptions of the strong classical linear regression model hold.) Here ρ is termed the *autoregression coefficient*—it represents the change in ε_i per unit change in ε_{i-1}. Additionally, Equation 4.18 implies that ε in any period i is equal to only a fraction ρ of the value of ε in the preceding period $i-1$ plus the contemporaneous effect of the random error term u_i.

Given that ε_i is AR(1) and Equation 4.19 holds, it can be demonstrated that

$$V(\varepsilon_i) = \sigma_\varepsilon^2 = \frac{\sigma_u^2}{1-\rho^2} \tag{4.20a}$$

$$\mathrm{COV}(\varepsilon_i, \varepsilon_{i-s}) = \rho^s \sigma_\varepsilon^2. \tag{4.20b}$$

As Equation 4.20b reveals,

$$\rho^s = \frac{\mathrm{COV}(\varepsilon_i, \varepsilon_{i-s})}{\sqrt{V(\varepsilon_i)}\sqrt{V(\varepsilon_{i-s})}}, \quad |\rho| < 1, \tag{4.21}$$

represents the coefficient of correlation between ε_i and ε_{i-s} since $\sigma_\varepsilon^2 = V(\varepsilon_i) = V(\varepsilon_{i-s})$. In particular, for $i=1$, $\mathrm{COV}(\varepsilon_i, \varepsilon_{i-1}) = \rho\sigma_\varepsilon^2$ and thus Equation 3.21 becomes

$$\rho = \frac{\mathrm{COV}(\varepsilon_i, \varepsilon_{i-1})}{\sigma_\varepsilon^2} = \frac{\mathrm{COV}(\varepsilon_i, \varepsilon_{i-1})}{\sqrt{V(\varepsilon_i)}\sqrt{V(\varepsilon_{i-1})}}, \quad |\rho| < 1, \tag{4.21.1}$$

the coefficient of correlation between successive values of the disturbance term or between ε_i and ε_{i-1}.

It is important to note that within the AR(1) process specified above there is no theoretical lower limit to the value of i, i.e., $i=-\infty$ is consistent with the beginning of this first-order autoregressive scheme. However, the first observation in our *finite* sample must be indexed at $i=1$. To take account of the effect of prior disturbances on ε_i, let us specify the *first* sample disturbance term as

$$\varepsilon_1 = \frac{u_1}{\sqrt{1-\rho^2}}, \tag{4.22}$$

with $\varepsilon_i = \rho\varepsilon_{i-1} + u_i$ for $i=2,3,\ldots$. By virtue of this specification of ε_1, all of the requisite features of ε_i, as specified by Equations 4.18 and 4.20 above, still hold, i.e., $E(\varepsilon_1)=0$, $V(\varepsilon_1) = \sigma_\varepsilon^2 = \sigma_u^2/(1-\rho^2)$, and $\mathrm{COV}(\varepsilon_1, \varepsilon_i) = \rho^{i-1}\sigma_\varepsilon^2$, $i=2,3,\ldots$. So given Equation 4.22, a sample of n data points is associated with exactly n random disturbance terms.

Let us now examine the properties of the OLS estimators $\hat{\beta}_1$ and $\hat{\beta}_0$ of β_1 and β_0 respectively when the random error term ε is autocorrelated. Suppose

$$Y_i = \beta_0 + \beta_1 t + \varepsilon_i, \quad i = 1,\ldots,n, \tag{4.23}$$

and that ε_i follows the first-order autoregressive scheme provided by Equation 4.18. Given this specification, we note first that the OLS estimators are still unbiased when the disturbances are AR(1). However, the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are *not BLUE* i.e., while unbiased, they are not best unbiased among all linear estimators of β_0 and β_1, and thus are inefficient. Moreover, it can be demonstrated that the OLS estimators in the presence of autocorrelated errors are, for large n, consistent although not asymptotically efficient.

This said, it should be evident that when the random disturbances are autocorrelated, the variances of the OLS estimators are biased and thus not appropriate for testing hypotheses about the regression parameters or computing confidence intervals for the same. Given $S_\varepsilon^2 = \sum e_i^2 / (n-2)$ (the usual OLS estimator of σ_ε^2), it can be verified that, for ε AR(1), $E\left(S_\varepsilon^2\right) < \sigma_\varepsilon^2$, i.e., the traditional OLS residual variance formula will understate the true error variance σ_ε^2. Clearly S_ε^2 is biased downward in the presence of autocorrelated disturbances. Moreover, this bias is transmitted to $V\left(\hat{\beta}_1\right)$. That is, it can also be verified that $E\left(S_{\hat{\beta}_1}^2\right) < V\left(\hat{\beta}_1\right)$ (the variance of $\hat{\beta}_1$ in the absence of autocorrelated error terms). We thus conclude that if the random error term ε is autoregressive, the usual OLS estimator of $V\left(\hat{\beta}_1\right)$ is negatively biased, i.e., the traditional OLS variance formula for $\hat{\beta}_1$ understates the true error variance. A similar conclusion holds for $V\left(\hat{\beta}_0\right)$.

Given the presence of this negative bias, any confidence interval estimates or hypothesis tests concerning β_0 and β_1 will be unreliable. In this regard, since the OLS standard errors of the estimated regression coefficients can be seriously understated for ε AR(1), any hypothesis tests concerning β_0 and β_1 will be biased (the calculated t statistics are inflated, thus overstating the statistical significance of the estimators of the regression parameters) and the confidence intervals for β_0 and β_1 will be narrower than they should be, thus inflating the precision of our estimates of these regression parameters.

4.6.2 Testing for the Absence of Autocorrelation: The Durbin–Watson Test

Having discussed the consequences of autocorrelated disturbances for OLS estimation of the regression parameters when the random error term ε is specified as AR(1), let us now look to our sample information in order to detect the presence of first-order autocorrelation. In particular, we shall work with the OLS residuals e_i so as to test the statistical significance of the AR(1) structure. To this end, we may test the null hypothesis that ε is *not* AR(1) or $H_0\!:\rho=0$, against any of the following alternative hypotheses:

CASE I	CASE II	CASE III
$H_0\!:\rho=0$ $H_1\!:\rho>0$ (positive autocorrelation)	$H_0\!:\rho=0$ $H_1\!:\rho<0$ (negative autocorrelation)	$H_0\!:\rho=0$ $H_1\!:\rho\neq 0$ (ε is autoregressive)

Our test statistic will be the *Durbin–Watson statistic* (DW)

$$DW = \frac{\sum_{i=2}^{n}\left(e_i - e_{i-1}\right)^2}{\sum_{i=1}^{n} e_i^2}, \quad 0 \le d \le 4, \tag{4.24}$$

where e_i is the usual OLS residual.

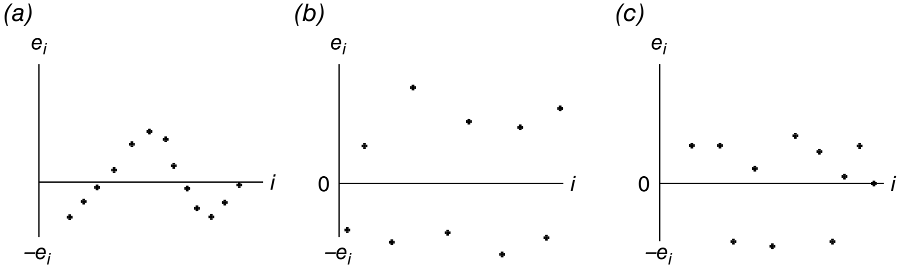

FIGURE 4.10 Types of first-order autocorrelation: (**a**) positive, (**b**) negative, and (**c**) random behavior.

It is instructive to consider the graphical behavior of the e_i's under positive or negative first-order autocorrelation. Given that the average of the e_i's is zero, these OLS residuals will be scattered about the horizontal (time)-axis. If there exists first-order positive autocorrelation, successive e_i's will tend to be close to each other (e_i tends to have the same sign as e_{i-1}), thus exhibiting runs above and below the horizontal-axis (Fig. 4.10a). In the presence of such runs $e_i - e_{i-1}$ will tend to be smaller in magnitude than e_i itself, thus leading (over all i) to a relatively small value of DW. If negative first-order autocorrelation occurs, successive e_i's tend to be on opposite sides of the horizontal-axis (Fig. 4.10b). Here $e_i - e_{i-1}$ will tend to be numerically larger than e_i, thus leading (over all i) to a relatively large DW value. If neither positive nor negative first-order autocorrelation occurs, then the e_i's are randomly distributed about the horizontal-axis and thus DW assumes an intermediate value (Fig. 4.10c).

Under $H_0: \rho = 0$, the exact sampling distribution of DW depends upon the values of the explanatory variable t. Since the critical values of DW are sensitive to the choice of the t's, obviously a single table covering all possible choices is not available. However, Durbin and Watson (1951) have shown that the actual sampling distribution of DW is bracketed by (it lies between) two limiting distributions, termed simply *lower* and *upper distributions*. For these lower and upper distributions Durbin and Watson establish lower (d_l) and upper (d_u) bounds for the critical values of DW which depend only upon the sample size (n) and the number of explanatory variables (k'). Then for a *Durbin–Watson (D–W) test* of $H_0: \rho = 0$ conducted at the $\alpha = P(\text{Type I Error}) = P(\text{TIE})$ level of significance, a set of decision rules for rejecting H_0 relative to H_1 consists of:

CASE I:
a. if $\text{DW} < d_l$, reject H_0 (ε is not AR(1)) in favor of H_1 (ε exhibits positive first-order autocorrelation);
b. if $\text{DW} > d_u$, do not reject H_0; and
c. if $d_l < \text{DW} < d_u$, the test is inconclusive.

CASE II:
a. if $\text{DW} > 4 - d_l$, reject H_0 (ε is not AR(1)) in favor of H_1 (ε exhibits negative first-order autocorrelation);
b. if $\text{DW} > 4 - d_u$, do not reject H_0; and
c. if $4 - d_u < \text{DW} < 4 - d_l$, the test is inconclusive.

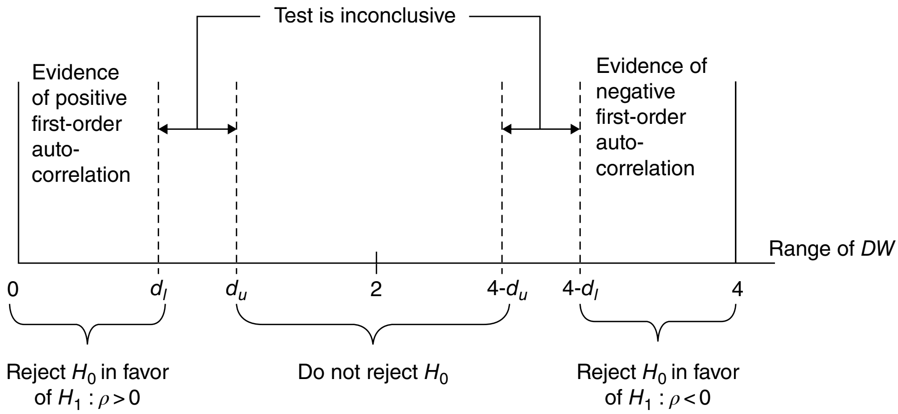

FIGURE 4.11 Range of the DW statistic.

CASE III. a. if $\mathrm{DW} < d_l$, or if $\mathrm{DW} > 4 - d_l$, reject H_0 (ε is not AR(1)) in favor of H_1 (ε is autoregressive);

b. if $d_u < \mathrm{DW} < 4 - d_u$, do not reject H_0; and

c. if $d_l \le \mathrm{DW} \le d_u$, or if $4 - d_u \le \mathrm{DW} \le 4 - d_l$, the test is inconclusive.

A description of the entire range of DW values (incorporating both regions of rejection and nonrejection of $H_0: \rho = 0$) is illustrated in Figure 4.11. The DW table exhibiting significance points of d_l and d_u for the 5% level of significance appears as Table A.5 of the Appendix.

We now turn to a few caveats pertaining to the conventional DW test:

1. a constant term must be included in the regression equation.
2. the explanatory variable t must be nonstochastic.
3. the size of the inconclusive region becomes fairly large for low degrees of freedom; and
4. the test can be performed only for a limited number of significance levels (typically 1% or 2.5% or 5%).

4.6.3 Detection of and Estimation with Autocorrelated Errors

In what follows we shall employ the SAS System to determine if the random error term ε is autocorrelated. (Remember that in the presence of autocorrelation the estimated standard errors of $\hat{\beta}_0$ and $\hat{\beta}_1$ are incorrect and thus the OLS estimates of β_0 and β_1 are inefficient.) If autocorrelation is detected, SAS will then be used to purge ε of this property so that efficient estimates of the regression parameters can be obtained.

To accomplish all this the *AUTOREG* procedure will be called upon. In general, this routine is used to estimate linear regression models for time-series data when the random errors are autocorrelated. It employs an autoregressive model to correct for the effects of autocorrelation. The procedure first tests for the presence of autocorrelation by calculating DW statistics and their associated p-values. By simultaneously estimating the regression parameters and the autoregressive model parameters, *AUOTREG* corrects the regression parameter estimates for the effects of autocorrelation.

The *AUTOREG* procedure produces the following sequence of general output features:

1. Statistics for the OLS residuals of the regression model ($Y=\beta_0+\beta_1 t+\varepsilon$).
2. OLS estimates of the regression parameters along with their standard errors and t values.
3. Estimates of autocorrelations determined from the OLS residuals.
4. Estimates of autoregression parameters $\rho_1, \rho_2, \ldots, \rho_s$ from the AR(s) specification $\varepsilon_i=\rho_1\varepsilon_{i-1}+\rho_2\varepsilon_{i-2}+\ldots+\rho_s\varepsilon_{i-s}+u_i$, where u_i is taken to be white noise. If in the MODEL statement the *backstep* option is used, only significant autoregressive parameters are shown. (This option requests that a stepwise autoregression be performed, i.e., in general, a high-order autoregression model having many lags is initially estimated and then, sequentially, autoregression parameters are removed until only those autoregression parameters having significant t values ($\alpha=0.05$) are retained.)
5. Maximum likelihood estimates of statistics for residuals (assuming that the maximum likelihood option is chosen).
6. Final estimates of the regression parameters and their standard errors recomputed under the assumption that the autoregression parameter estimates equal their population counterparts.

EXAMPLE 4.3 Using the Example 4.1 data set, let us: (i) determine if the OLS residuals generated from the linear and semilogarithmic trend equations exhibit a statistically significant autoregressive structure; and (ii) if a significant autoregressive structure is detected, reestimate the regression parameters and their standard errors.

Exhibit 4.2 specifies the SAS code supporting the implementation of *PROC AUTOREG*.

EXHIBIT 4.2 SAS Code for *PROC AUTOREG*

```
data gdp1;
  input y @@;
lny=log(y);
t=_n_;
datalines;
7112.5     7100.5 ...  11566.8
proc autoreg data=gdp1; ①
```

```
                     ②   ③       ④     ⑤
    model y=t/nlag=4 method=ml dw=4 dwprob;
    output out=new1 r=ry; ⑥
    data auto1; set new1; ⑦
run;
proc autoreg data=gdp1; ⑧
     model lny=t/nlag=4 method=ml dw=4 dwprob; ⑨
     output out=new2 r=rlny; ⑩
     data auto2; set new2; ⑪
run;
data auto3; merge auto1 auto2; ⑫
run;
proc plot data= auto3; ⑬
   plot ry*t= '*'; ⑭
    plot rlny*t= '*'; ⑮
run;
```

① We invoke the AUTOREG procedure.

② The MODEL statement option *nlag = 4* posits an AR(4) autoregressive model

$$\varepsilon_i = -\rho_1\varepsilon_{i-1} - \rho_2\varepsilon_{i-2} - \rho_3\varepsilon_{i-3} - \rho_4\varepsilon_{i-4} + u_i.$$

③ The maximum likelihood option *method = ml* is requested in the MODEL statement.

④, ⑤ The *dw = 4* and *dwprob* options in the MODEL statement request DW statistics for lags 1–4 and their *p*-values, respectively.

⑥ A data set *new1* is created that contains the residuals variable *ry* as well as all variables found in the data set *gdp1*.

⑦ For plotting purposes the data set *auto1* is formed which contains all variables found in data set *new1*.

⑧–⑪ See the discussion provided in ①–⑦ above.

⑫ We merge data sets auto1 and auto2 into data set auto3.

⑬–⑮ We invoke the plot procedure using data set auto3 and we plot the residuals *ry* and *r*ln*y* against *t*.

The output of this AUTOREG procedure is presented in Table 4.4, Figure 4.12, and Figure 4.13. Specifically:

Ⓐ ① Regress *R*-Square depicts the R^2 statistic for the transformed regression equation, i.e., it is a measure of goodness of fit of the systematic part of the regression model $(\beta_0+\beta_1 t)$ after the variables are adjusted for the effects of the estimated autocorrelation.

② Total *R*-Square is a measure of the contribution of both the systematic part of the regression model and past residual values in predicting the next

TABLE 4.4 Output for Example 4.3

The SAS system
The AUTOREG procedure
Ⓐ
Dependent variable: y

Ordinary least squares estimates

SSE	274,342.385	DFE	16
MSE	17,146	Root MSE	130.94426
SBC	230.234218	AIC	228.453475
① Regress R-square	0.9927	② Total R-square	0.9927

③ Durbin–Watson statistics

Order	DW	Pr < DW	Pr > DW
1	0.7587	0.0004	0.9996
2	1.6156	0.2055	0.7945
3	2.3977	0.8821	0.1179
4	2.4737	0.9529	0.0471

Note: Pr < DW is the p-value for testing positive autocorrelation, and Pr > DW is the p-value for testing negative autocorrelation.

④

Variable	DF	Estimate	Standard error	t value	Approx Pr > $\|t\|$
Intercept	1	6536	64.3934	101.50	<0.0001
t	1	277.7205	5.94891	46.681	<0.0001

⑤

Estimates of autocorrelations

Lag	Covariance	Correlation	−1 9 8 7 6 5 4 3 2 1 0 1 2 3 4 5 6 7 8 9 1
0	15,241.2	1.000000			********************	
1	6,955.4	0.456352			*********	
2	316.4	0.020758			
3	−5,689.2	−0.373276		*******		
4	−6,650.2	−0.436331		*********		

Preliminary MSE 9640.4

⑥

Estimates of autoregressive parameters

Lag	Coefficient	Standard error	t Value
1	−0.425680	0.285928	−1.49
2	0.031486	0.298507	0.11
3	0.304940	0.298507	1.02
4	0.137622	0.285928	0.48

Algorithm converged

⑦

Maximum likelihood estimates			
SSE	118,889.341	DFE	12
MSE	9,907	Root MSE	99.53615
SBC	228.448856	AIC	223.106625
Regress R-square	0.9911	Total R-square	0.9968

(*Continued*)

TABLE 4.4 Output for Example 4.3 (*Continued*)

The SAS system
The AUTOREG procedure
⑧
Durbin–Watson statistics

Order	DW	Pr < DW	Pr > DW
1	2.2028	0.5663	0.4337
2	1.5046	0.1420	0.8580
3	1.2905	0.1052	0.8948
4	1.0740	0.0727	0.9273

Note: Pr < DW is the p-value for testing positive autocorrelation, and Pr > DW is the p-value for testing negative autocorrelation.

⑨

Variable	DF	Estimate	Standard error	t value	Approx Pr>|t|
Intercept	1	6576	83.9831	78.30	<0.0001
t	1	274.164	7.8048	35.13	<0.0001
AR1	1	−0.8434	0.3016	−2.80	0.0161
AR2	1	0.1811	0.3423	0.53	0.6063
AR3	1	0.3925	0.3648	1.08	0.3031
AR4	1	−0.0468	0.2935	−0.16	0.8759

⑩
Autoregressive parameters assumed given

Variable	DF	Estimate	Standard error	t value	Approx Pr>|t|
Intercept	1	6576	79.9355	82.26	<0.0001
t	1	274.1638	7.5209	36.45	<0.0001

Ⓑ

Dependent variable: ln*y*

Ordinary least squares estimates			
SSE	0.00410724	DFE	16
MSE	0.0002567	Root MSE	0.01602
SBC	−94.074242	AIC	−95.854986
① Regress R-square	0.9910	② Total R-square	0.9910

③

Durbin–Watson statistics

Order	DW	Pr < DW	Pr > DW
1	0.6306	<0.0001	0.9999
2	1.2360	0.0450	0.9550
3	1.9175	0.5533	0.4467
4	2.1779	0.8423	0.1577

Note: Pr < DW is the p-value for testing positive autocorrelation, and Pr > DW is the p-value for testing negative autocorrelation.

④

Variable	DF	Estimate	Standard error	t value	Approx Pr > $\|t\|$
Intercept	1	8.8208	0.007879	1119.53	<0.0001
t	1	0.0306	0.000728	42.05	<0.0001

(*Continued*)

TABLE 4.4 Output for Example 4.3 (*Continued*)

The SAS system
The AUTOREG procedure
⑤
Estimates of autocorrelations

Lag	Covariance	Correlation	−1 9 8 7 6 5 4 3 2 1 0 1 2 3 4 5 6 7 8 9 1
0	0.000228	1.000000			********************	
1	0.000140	0.613575			************	
2	0.000064	0.281004			******	
3	−0.00002	−0.079747		**		
4	−0.00006	−0.244049		*****		

Preliminary MSE 0.000126
⑥
Estimates of autoregressive parameters

Lag	Coefficient	Standard error	t value
1	−0.648733	0.288435	−2.25
2	−0.063905	0.334526	−0.19
3	0.276228	0.334526	0.83
4	0.040786	0.288435	0.14

Algorithm converged
⑦
Maximum likelihood estimates

SSE	0.00161372	DFE	12
MSE	0.0001345	Root MSE	0.01160
SBC	−97.718939	AIC	−103.06117
Regress R-square	0.9814	Total R-square	0.9965

⑧

Durbin–Watson statistics

Order	DW	Pr < DW	Pr > DW
1	2.0489	0.4327	0.5673
2	1.5785	0.1826	0.8174
3	1.5073	0.2236	0.7764
4	1.2366	0.1429	0.8571

Note: Pr < DW is the *p*-value for testing positive autocorrelation, and Pr > DW is the *p*-value for testing negative autocorrelation.

⑨

Variable	DF	Estimate	Standard error	*t* Value	Approx Pr > |*t*|
Intercept	1	8.8309	0.0138	642.22	<0.0001
t	1	0.0297	0.001296	22.93	<0.0001
AR1	1	−0.9411	0.2846	−3.31	0.0063
AR2	1	0.0671	0.3447	0.19	0.8490
AR3	1	0.4452	0.3835	1.16	0.2683
AR4	1	−0.1155	0.3003	−0.38	0.7074

⑩

Autoregressive parameters assumed given

Variable	DF	Estimate	Standard error	*t* value	Approx Pr > |*t*|
Intercept	1	8.8309	0.0128	692.54	<0.0001
t	1	0.0297	0.001182	25.14	<0.0001

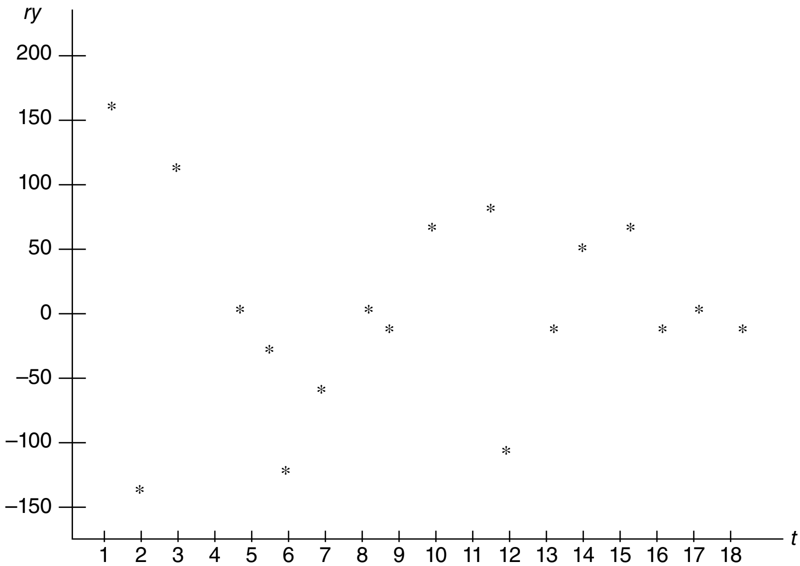

FIGURE 4.12 Residuals from the estimated linear trend equation versus time.

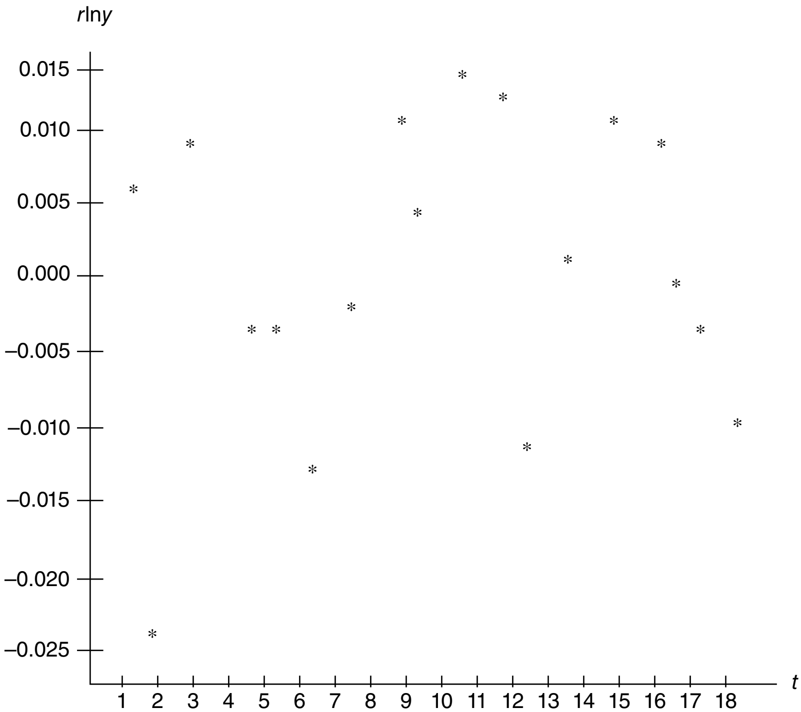

FIGURE 4.13 Residuals from the estimated semilogarithmic trend equation versus time.

successive value of the dependent variable Y. (Both Regress R-Square and Total R-Square have the same value when OLS estimation is used since no adjustment for autocorrelation has been undertaken.)

③ A glance at the set of DW test p-values indicates that the first-order autoregressive term indicates the presence of positive autocorrelation (p-value = 0.0004) while the fourth-order term provides evidence of negative autocorrelation (p-value = 0.0471). Strictly speaking, once autocorrelation of a specific order is detected, further testing at a higher order is not necessary. For instance, if the DW test indicates no first-order autocorrelation, then a second-order autocorrelation test should be conducted. And if no second-order autocorrelation is found, then perhaps a third-order test is called for, etc. But since the first-order test herein reveals the presence of positive autocorrelation, the second-, third- and higher-order tests are not warranted.

④ OLS parameter estimates are printed along with their t statistics and p-values.

⑤ Here the estimated autocorrelation coefficients, computed from the OLS residuals, are $\hat{\rho}_1 = 0.456$, $\hat{\rho}_2 = 0.021$, $\hat{\rho}_3 = -0.373$, and $\hat{\rho}_4 = -0.436$. As expected, $\hat{\rho}_1$ and $\hat{\rho}_4$ are fairly large in absolute value.

⑥ These are the preliminary estimates of the autoregressive parameters that are used as starting values for the iterative computation of maximum likelihood estimates.

⑦ The maximum likelihood estimate of Regress R-Square is lower than its OLS counterpart determined in ① above. This reflects the adjustment in the regression variables for the estimated autocorrelation. Also, the maximum likelihood estimate of the Total R-Square is higher than that obtained via OLS in ②. This reflects the improved fit obtained when past residuals are used to help predict the next Y value.

⑧ Under the adjustment for autocorrelation, none of the p-values for the new set of DW statistics used in testing for positive autocorrelation are below $\alpha = 0.05$. Hence autocorrelation is no longer a problem.

⑨ The values presented are maximum likelihood estimates for the autoregressive error model

$$
\begin{aligned}
&Y = \beta_0 + \beta_1 t + \varepsilon_i \\
&\varepsilon_i = -\rho_1\varepsilon_{i-1} - \rho_2\varepsilon_{i-2} - \rho_3\varepsilon_{i-3} - \rho_4\varepsilon_{i-4} + u_i \\
&u_i : N\left(0, \sigma_u^2\right), \operatorname{cov}\left(u_r, u_s\right) = 0, \quad r \neq s.
\end{aligned}
$$

Hence the estimated model has the form

$$
\begin{aligned}
&Y = 6576 + 274.1638t + \varepsilon_i \\
&\varepsilon_i = 0.8434\varepsilon_{i-1} - 0.1811\varepsilon_{i-2} - 0.3925\varepsilon_{i-3} + 0.0468\varepsilon_{i-4} + u_i,
\end{aligned}
$$

where the maximum likelihood estimate of σ_u^2 is, from ⑦, MSE = 9907. (Note that SAS writes the coefficient of ε_{i-r} in an autoregressive process as $-\rho_r$. This accounts for the minus sign on the estimated coefficient of ε_{i-1} so that $-\hat{\rho}_1 = -0.8434$.)

⑩ Efficient estimates for both β_0 and β_1 and their corresponding t statistics and p-values occupy the final portion of the AUTOREG output. Note that the OLS estimates of β_0 and β_1 are respectively $\hat{\beta}_0 = 6536$ and $\hat{\beta}_1 = 277.7205$ while the maximum likelihood estimates of these parameters adjusted for the effect of autocorrelation are $\tilde{\beta}_0 = 6576$ and $\tilde{\beta}_1 = 274.1638$ respectively. Also, as Figure 4.12 indicates, the residuals *ry* seem to be randomly distributed about zero.

Ⓑ ①, ② See Ⓐ above.

③ The p-values for the DW statistics associated with the first- and second-order autoregressive terms indicate the presence of positive autocorrelation. Hence only a first-order term is needed.

④–⑥ OLS parameter estimates, estimated autocorrelation coefficients, and preliminary estimates of these coefficients used for determining maximum likelihood results are respectively provided.

⑦ As expected, the maximum likelihood estimate of Regress R-Square decreases while estimated Total R-Square increases relative to their OLS counterparts.

⑧ None of the reported DW statistics are significant after the adjustment for autocorrelation.

⑨ The maximum likelihood estimates for the full autoregressive error model are provided. In addition, from ⑦, MSE = $\sigma_u^2 = 0.00013$.

⑩ Efficient estimates for the regression parameters along with their associated t and p-values constitute the final portion of the AUTOREG results. Given that the OLS estimates of β_0 and β_1 are respectively $\hat{\beta}_0 = 8.8208$ and $\hat{\beta}_1 = 0.0306$, we see that the maximum likelihood estimates for β_0 and β_1 obtained under an autocorrelation correction are $\tilde{\beta}_0 = 8.8309$ and $\tilde{\beta}_1 = 0.0297$ respectively. Here too the residuals *r*ln*y* display a random pattern about zero (Fig. 4.13). Note also that the maximum likelihood estimate $\tilde{\beta}_1$ of β_1 is also consistent with our previous geometric mean calculation of the average rate of growth of GDP over the sample period (GM = 0.029). ■

4.7 POLYNOMIAL MODELS IN *t*

We have been working with a linear (in the parameters) trend equation in which the regressor t was a *first-degree term*, i.e., for both the linear ($Y=\beta_0+\beta_1 t$) and semilogarithmic ($\ln Y=\beta_0+\beta_1 t$) cases t appears only to the first power. In general, the *degree*

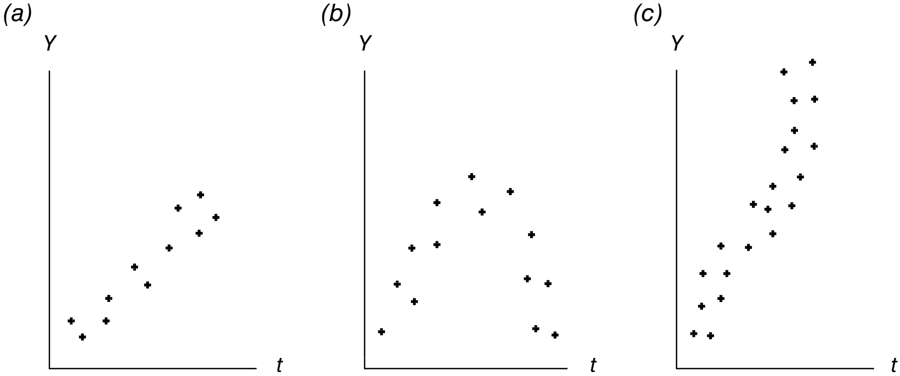

FIGURE 4.14 (**a**) Linear, (**b**) quadratic, and (**c**) cubic patterns of points.

of a polynomial model is the maximum power of any term in the model while the name given to a polynomial model reflects its highest degree.

While the fitting of first-degree trend models is quite common, it may be the case that nonlinearities in t need to be introduced in order to adequately model some specific phenomenon. For instance, suppose we have a set of n data points and the scatter of sample points is illustrated in Figure 4.14a. Clearly the visual pattern of these data points indicates that a *first-degree model* of the form

$$Y = \beta_0 + \beta_1 t + \varepsilon \tag{4.25}$$

is appropriate. However what if the scatter of data points displays a pattern such as in Figure 4.14b? In order to pick up the nonlinearity, we need to introduce into Equation 4.25 the second-degree term t^2. Hence we should fit a *second-degree polynomial* in t or

$$Y = \beta_0 + \beta_1 t + \beta_2 t^2 + \varepsilon. \tag{4.26}$$

And if the set of sample points displays a *point of inflection* (a point where a curve crosses over its tangent line and changes the direction of its concavity from upwards to downwards or vice versa), then, to model any such point (Fig. 4.14c), we need to introduce into Equation 4.26 the third-degree term t^3. We now need to fit a *third-degree polynomial* in t or

$$Y = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \varepsilon. \tag{4.27}$$

Polynomial expressions such as Equations 4.26 and 4.27 are termed *natural polynomials*. And as evidenced by the form of Equations 4.25–4.27, the degree of a polynomial in t is the degree of its term of highest degree.

For the latter two cases (Eqs. 4.26 and 4.27), in spite of the inclusion of second- and third-degree terms in t, each is characterized as a linear regression equation since each is still linear in the parameters. Moreover, if in the mth degree polynomial

TABLE 4.5 Time Series Observations on Y

Y	t
4	1
16	2
25	3
32	4
35	5
30	6
25	7
15	8
9	9
6	10
7	11
11	12
16	13
24	14
35	15

$$Y = \beta_0 + \sum_{j=1}^{m} \beta_j t^j + \varepsilon \tag{4.28}$$

we set $W_j = t^j$, $j = 1,\ldots,m$, then we obtain the more familiar first-degree structure $Y = \beta_0 + \sum_{j=1}^{m} \beta_j W_j + \varepsilon$ and thus OLS is applicable.

EXAMPLE 4.4 Table 4.5 presents $n = 15$ observations on the variables Y and time t. Does the cubic regression equation (4.27) serve as an adequate model for estimating the nonlinear trend in Y over time? The required SAS code is outlined in Exhibit 4.3.

EXHIBIT 4.3 SAS Code for Fitting a Cubic Polynomial

```
data cube;
  input y @@;
t=_n_; ①

tsq=t**2;
           ②
tcu=t**3;
datalines;
4 16 25 32 35 ... 35
proc reg data=cube;
```

① We apply the sequential period indicator to obtain values of time t.

② *tsq* and *tcu* represent the squared and cubed values of t respectively.

```
model y=t tsq tcu/ SS1 seqb
pcorr1; ③
```

③ We specify a cubic polynomial model in *t* and request sequential or added-in-order (Type I) sums of squares (*SS*1), sequential parameter estimates (*seqb*), and sequential or added-in-order partial correlation coefficients (*pcorr1*).

```
output out=q r=rq; ④
run;
data cube2; merge cube q; ⑤
run;
proc plot data=cube2; ⑥
  plot y*t
  plot rq*t='*'; ⑦
run;
```

④ The output data set *q* contains the collection of residual values *rq* obtained from the cubic regression fit.
⑤ We form data set cube2 by merging data sets cube and *q*.
⑥ We invoke the PLOT procedure using data values in cube2.
⑦ We request a plot of the *y* values and the residuals *rq* against time *t*.

The output generated from the preceding SAS program appears in Table 4.6 and in Figure 4.15 and Figure 4.16.

① To determine the significance of this third-degree polynomial in t (the *full model*), let us test $H_0: \beta_1 = \beta_2 = \beta_3 = 0$, against H_1: not all β_j's = 0, j = 1, 2, 3. Since the p-value for the sample F Statistic is less than 0.0001, we easily reject H_0 in favor of H_1 and conclude that a third-degree polynomial in t explains a statistically significant portion of the variation in Y. And with $R^2 = 0.9368$, about 93% of this variation is explained by the model. Also with $\sqrt{\text{MSE}} = 3.06$, we see that this third-degree polynomial fits the data reasonably well.

②, ③ The estimated third degree polynomial appears as

$$\begin{aligned}\hat{Y} &= \hat{\beta}_0 + \hat{\beta}_1 t + \hat{\beta}_2 t^2 + \hat{\beta}_3 t^3 \\ &= -17.9194 + 24.9842t - 3.8584t^2 + 0.1629t^3.\end{aligned}$$

The p-value associated with each estimated coefficient's t statistic is less than $\alpha = 0.05$, thus indicating that the intercept as well as each explanatory variable contributes significantly to explaining variation in Y, i.e., we reject $H_0: \beta_j = 0$, in favor of $H_1: \beta_j \neq 0$, j = 0, 1, 2, 3.

④–⑥ *TYPE* I or sequential sums of squares represent a partitioning of *SSR* into individual component sums of squares attributed to each regressor as it is introduced sequentially into the model (in the exact order specified in the

TABLE 4.6 Output for Example 4.4

The SAS system
The REG procedure
Model: MODEL 1
Dependent variable: y
Number of observations read: 15
Number of observations used: 15
①
Analysis of variance

Source	DF	Sum of squares	Mean square	F value	Pr > F
Model	3	1530.10034	510.03345	54.35	<0.0001
Error	11	103.23300	9.38482		
Corrected total	14	1633.33333			
Root MSE		3.06346	R-square	0.9368	
Dependent mean		19.33333	Adj R-square	0.9196	
Coeff var		15.84551			

Parameter estimates

Variable	DF	② Parameter estimate	Standard error	t value	③ Pr>|t|	④ Type I SS	⑤ Squared partial corr. type I
Intercept	1	−17.91941	4.15499	−4.31	0.0012	5606.66667	.
t	1	24.98424	2.17645	11.48	<0.0001	0.51429	0.00031487
tsq	1	−3.85840	0.31087	−12.41	<0.0001	10.35337	0.00634
tcu	1	0.16285	0.01280	12.72	<0.0001	1519.23268	0.93637

⑥

Sequential parameter estimates

Intercept	t	tsq	tcu
19.333333	0	0	0
19.676190	−0.042857	0	0
21.947253	−0.844409	0.050097	0
−17.919414	24.984241	−3.858400	0.162854

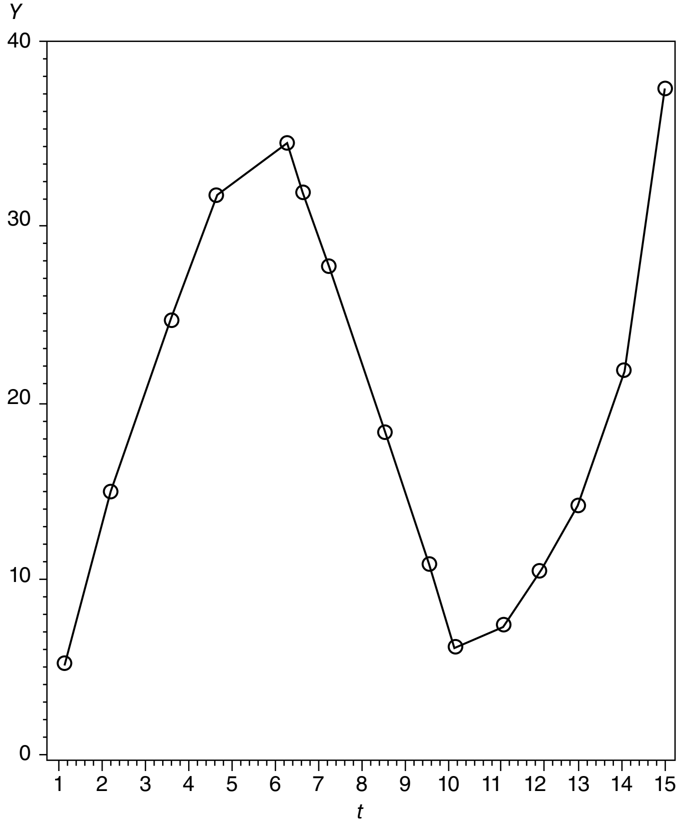

FIGURE 4.15 A plot of Y against time.

MODEL statement); it provides the contribution to *SSR* of each regressor as it is inserted, in turn, into the regression equation. Hence the *TYPE* I sum of squares is used to determine what order of polynomial is actually needed.

The various *TYPE* I sums of squares are:

$\text{SSR}(t|\text{intercept}) = 0.5143$—the regression sum of squares for a model containing only t and an intercept;

$\text{SSR}(tsq|\text{intercept}, t) = 10.3534$—the increment in the regression sum of squares due to the addition of tsq to a model already containing an intercept and t;

$\text{SSR}(tcu|\text{intercept}, t, tsq) = 1519.2326$ —the increase in the regression sum of squares due to the addition of tcu to a model already containing an intercept, t, and tsq.

Dividing each *TYPE* I sum of squares value by $\text{MSE}(full) = 9.3848$ (see ①) provides a set of F statistics that are used to determine if each additional

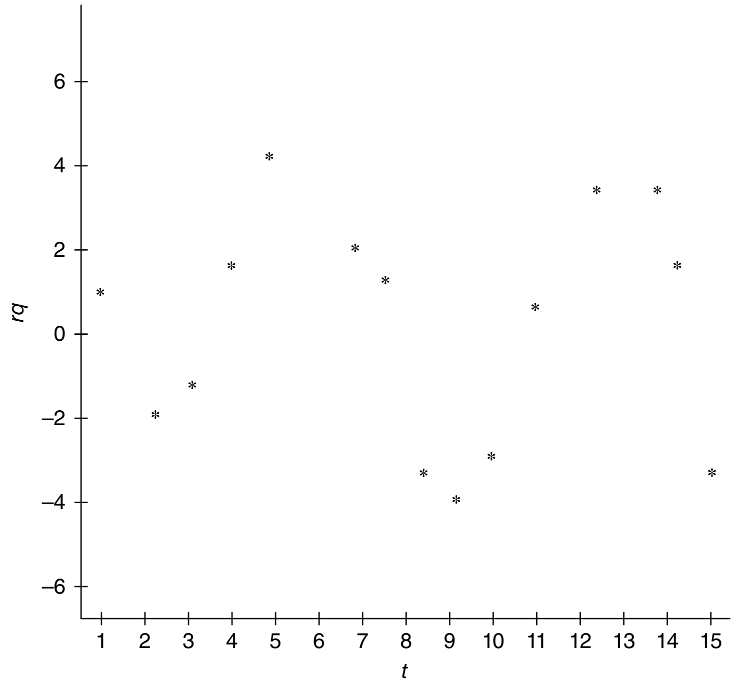

FIGURE 4.16 A plot of the residuals versus time.

regressor makes a statistically significant contribution to *SSR* and thus to explaining variation in *Y*. The sequential *F* values are:

$$F = \frac{\text{SSR}(t\,|\,\text{intercept})}{\text{MSE}(\mathit{full})} = \frac{0.5143}{9.3848} = 0.0548,$$

p-value > 0.10. A first-degree term is not warranted. Next,

$$F = \frac{\text{SSR}(tsq\,|\,\text{intercept}, t)}{\text{MSE}(\mathit{full})} = \frac{10.3534}{9.3848} = 1.1032,$$

p-value > 0.10. This result is not statistically significant—a second-degree term may not be appropriate. Finally,

$$F = \frac{\text{SSR}(tcu\,|\,\text{intercept}, t, tsq)}{\text{MSE}(\mathit{full})} = \frac{1519.2326}{9.3848} = 161.8822,$$

p-value < 0.0001. A third-degree term is justified. (Since the third-degree term is significant, *all* lower degree terms within the polynomial will be retained.)

Next, *pcorr1* determines the set of sequential or variables added-in-order squared partial correlation coefficients based on added-in-order *TYPE* I sums of squares:

$r^2_{yt} = 0.0003$—when t is added to a regression model which only displays an intercept, *SSE* decreases by 0.03%;
$r^2_{ytsq|t} = 0.063$—when *tsq* is added to a regression model already containing t, *SSE* decreases by 6.3%; and
$r^2_{ytcu|t,tsq} = 0.9364$—when *tcu* is introduced into a regression model which already contains t and *tsq*, *SSE* decreases by 93.64%.

The *SEQB* option prints the coefficients of the successive polynomials fitted by adding regressors in the order specified in the *MODEL* statement. If the preceding sequential F test revealed that a polynomial of degree lower than three was appropriate, then estimates of the coefficients of lower-degree polynomials are found in the various rows of ⑥:

Row 1—coefficients of the zero-order polynomial $\hat{Y} = \hat{\beta}_0 = 19.3333$
Row 2—coefficients of the first-order polynomial $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 t = 19.6762 - 0.0429t$
Row 3—coefficients of the second-order polynomial $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 t + \hat{\beta}_2 tsq = 21.9473 - 0.8444t + 0.0501tsq$ and
Row 4—coefficients of the full or third degree polynomial model (see ②).

⑦ Figure 4.15 depicts a plot of Y versus time while Figure 4.16 plots the residuals against time. Here the pattern appears to be fairly random and of constant variability, i.e., the errors are homoscedastic. ■

When should a linear, quadratic, or cubic trend equation be fit to a data set? We noted earlier that if the first-differences between successive observations of a time series are approximately constant ($\Delta Y_t = Y_t - Y_{t-1} = \beta_1 =$ constant), then a linear trend line ($Y_t = \beta_0 + \beta_1 t$) would provide a good fit to the data. But what if ΔY_t is not constant? How do we know if the quadratic or cubic case is appropriate? To answer this we need only look to the second and third differences between successive values of the time series. Specifically, let us specify the *second difference* as

$$\begin{aligned} \Delta^2 Y_t &= \Delta(\Delta Y_t) = \Delta(Y_t - Y_{t-1}) = \Delta Y_t - \Delta Y_{t-1} \\ &= Y_t - Y_{t-1} - (Y_{t-1} - Y_{t-2}) \\ &= Y_t - 2Y_{t-1} + Y_{-2}. \end{aligned} \tag{4.29}$$

(For the linear case, obviously $\Delta^2 Y_t = \beta_0 + \beta_1 t - 2[\beta_0 + \beta_1(t-1)] + \beta_0 + \beta_1(t-2) = 0$.)

Given the quadratic trend equation $Y_t = \beta_0 + \beta_1 t + \beta_2 t^2$, an application of Equation 4.29 gives us

$$\begin{aligned} \Delta^2 Y_t &= \beta_0 + \beta_1 t + \beta_2 t^2 - 2\left[\beta_0 + \beta_1(t-1) + \beta_2(t-1)^2\right] + \beta_0 + \beta_1(t-2) + \beta_2(t-2)^2 \\ &= 2\beta_2 = \text{constant}. \end{aligned}$$

So for the quadratic case, second differences are always constant.

Let us next define the *third difference* as

$$\begin{aligned}\Delta^3 Y_t &= \Delta\left(\Delta^2 Y_t\right) = \Delta\left(Y_t - 2Y_{t-1} + Y_{t-2}\right) \\ &= \Delta Y_t - 2\Delta Y_{t-1} + \Delta Y_{t-2} \\ &= Y_t - Y_{t-1} - 2\left(Y_{t-1} - Y_{t-2}\right) + Y_{t-2} - Y_{t-3} \\ &= Y_t - 3Y_{t-1} + 3Y_{t-2} - Y_{t-3}.\end{aligned} \tag{4.30}$$

Then applying Equation 3.30 to the cubic trend equation $Y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3$ renders

$$\begin{aligned}\Delta^3 Y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 - 3\left[\beta_0 + \beta_1 (t-1) + \beta_2 (t-1)^2 + \beta_3 (t-1)^3\right] \\ +3\left[\beta_0 + \beta_1 (t-2) + \beta_2 (t-2)^2 + \beta_3 (t-2)^3\right] \\ -\beta_0 - \beta_1 (t-3) - \beta_2 (t-3)^2 - \beta_3 (t-3)^3 = 6\beta_3 = \text{constant.}\end{aligned}$$

Hence, for the cubic case, third differences are always constant. In general, if the *n*th difference of a time series in approximately constant, then one should fit an *n*th order polynomial to the data set. Note that, in this circumstance, the (n + 1)st difference will always be approximately equal to zero.

EXAMPLE 4.5 Using the Example 4.4 data set, let us compare the time paths of the first- through the third-differences of this time series. The SAS code is provided in Exhibit 4.4.

EXHIBIT 4.4 SAS Code for Calculating Various Differences of the *Y* Series

```
data cube;
  input y @@;
dy = dif(y); ①
d2y = dif(dy); ②
d3y = dif(d2y); ③
t = _n_; ④
datalines;
4 16 25 32 35 30 25 15 9 6 7 11 16 24 35
proc print data = cube; ⑤
run;
symbol1 v=none   i=join l=1;
symbol2 v=none   i=join l=2; ⑥
symbol3 v=none   i=join l=3;
```

①–③ We request the first-through the third—differences of the *Y* variable. (Remember that $\Delta^n Y = \Delta(\Delta^{n-1} Y)$.)

④ We apply the sequential period indicator to obtain values of time *t*.

```
proc gplot data = cube; ⑦
plot dy*t d2y*t d3y*t/overlay; ⑧
run;
```

⑤ We request a print of the *Y* series along with all of the calculated differences.
⑥–⑧ Plots of the first-, second-, and third- differences are requested. The points will be joined by straight line segments and their individual graphs will appear in the same coordinate plane.

The output resulting from this SAS program appears as Table 4.7 and Figure 4.17. As this figure reveals, *dy* decreases and then increases (it obviously is not constant) while d^2y, although exhibiting values that are much less dispersed than those of *dy*, has essentially an upward trend. In turn, the d^3y values display a much flatter trajectory than those of d^2y, thus indicating that a cubic or third-degree polynomial is appropriate. ■

4.8 ISSUES INVOLVING TRENDED DATA

We noted earlier in this chapter that OLS could be used to estimate the parameters of a linear trend equation for time series data. However, as we shall now see, the OLS results are reliable if the data are trend free or stationary. In particular, specious or nonsense regression results can emerge when a variable Y_t is subject to stochastic trends. In what follows then, we shall focus on the detection and handling

TABLE 4.7 Output for Example 4.5

The SAS system					
Obs	*y*	*dy*	*d2y*	*d3y*	*t*
1	4	.	.	.	1
2	16	12	.	.	2
3	25	9	−3	.	3
4	32	7	−2	1	4
5	35	3	−4	−2	5
6	30	−5	−8	−4	6
7	25	−5	0	8	7
8	15	−10	−5	−5	8
9	9	−6	4	9	9
10	6	−3	3	−1	10
11	7	1	4	1	11
12	11	4	3	−1	12
13	16	5	1	−2	13
14	24	8	3	2	14
15	35	11	3	0	15

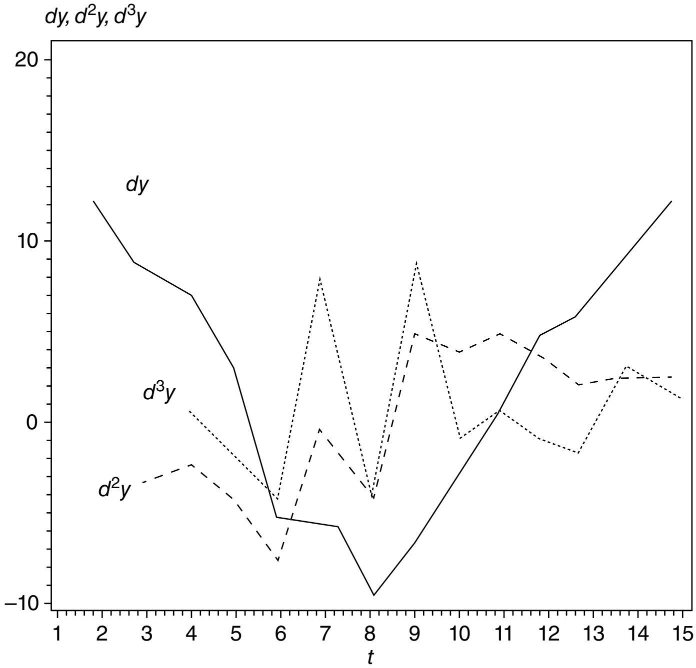

FIGURE 4.17 Plots of dy, d^2y, and d^3y against time.

of stochastic trends in a time series data set. But first some terminology and definitions.

4.8.1 Stochastic Processes and Time Series

A *stochastic process* $\{Y_t\}$ is a collection of real-valued random variables ordered or indexed by time. A (discrete) *time series* is a single realization of the stochastic process $\{Y_t\}$. The time series $\{Y_t\}$ is said to be *covariance (weakly) stationary* if

$$E(Y_t) = E(Y_{t+m}) = \mu = \text{constant}, \tag{4.31a}$$

$$V(Y_t) = V(Y_{t+m}) = \sigma_y^2 = \text{constant}, \tag{4.31b}$$

$$\mathrm{COV}(Y_t, Y_{t-\tau}) = \mathrm{COV}(Y_{t+m}, Y_{t+m-\tau}) = \gamma_\tau, \quad \tau = 1, 2, \ldots, \tag{4.31c}$$

where τ is the gap between the two time points $Y_t, Y_{t-\tau}$. For $\tau=0$, $\gamma_0 = V(Y_t)$. (Here $V(Y_t) = E\left[(Y_t - \mu)^2\right]$, with $\mathrm{COV}(Y_t,\ Y_{t-\tau}) = E\left[(Y_t - \mu)(Y_{t-\tau} - \mu)\right]$.) Thus a time series is covariance stationary if its mean, variance, and autocovariance are independent of time.

In what follows we shall use the terms stationary and covariance stationary interchangeably.

For a stationary series $\{Y_t\}$, the *autocorrelation function* between Y_t and $Y_{t-\tau}$ (or with lag τ) is defined as

$$\begin{aligned}\rho_\tau &= \mathrm{CORR}\left(Y_t, Y_{t-\tau}\right) = \frac{\mathrm{COV}\left(Y_t, Y_{t-\tau}\right)}{\sqrt{V\left(Y_t\right)}\sqrt{V\left(Y_{t-\tau}\right)}} \\ &= \frac{\mathrm{COV}\left(Y_t, Y_{t-\tau}\right)}{V\left(Y_t\right)} = \gamma_\tau/\gamma_0, \quad \tau = 1, 2, \ldots,\end{aligned} \tag{4.32}$$

since $V(Y_t) = V(Y_{t-\tau})$. The autocorrelation function indicates how much association or interdependence there is between neighboring data points in the $\{Y_t\}$ series. With γ_τ and γ_0 time independent, it follows that ρ_τ must be time independent (the autocorrelation between, say, Y_t and Y_{t-1} must be the same as that between Y_{t-m} and Y_{t-m+1}). Also, $\rho_0 = 1$.

The process $\{Y_t\}$ is said to be a *purely random* or *white noise process* if the Y_t's are independent and identically distributed with zero mean, constant variance, and zero autocovariance, i.e. for each value of t:

$$E\left(Y_t\right) = 0, \tag{4.33a}$$

$$V\left(Y_t\right) = E\left(Y_t^2\right) = \sigma_y^2, \tag{4.33b}$$

$$\mathrm{COV}\left(Y_t, Y_{t-\tau}\right) = E\left(Y_t Y_{t-\tau}\right) = 0 \text{ for any } \tau, \tag{4.33c}$$

and

$$\gamma_\tau = \begin{cases} \sigma_Y^2, & \tau = 0 \\ 0, & \tau \geq 1 \end{cases},$$

so that

$$\rho_\tau = \begin{cases} \sigma_Y^2, & \tau = 0; \\ 0, & \tau \geq 1. \end{cases}$$

Equation 4.33c implies that a white noise process has *no memory*.

4.8.2 Autoregressive Process of Order *p*

Let $\{\varepsilon_t\}$ constitute a white noise process. Then the time series $\{Y_t\}$ specified as

$$\begin{aligned}Y_t &= \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \ldots + \alpha_p Y_{t-p} + \varepsilon_t \\ &= \sum_{j=1}^{p} \alpha_j Y_{t-j} + \varepsilon_t, \quad t = 1, 2, \ldots,\end{aligned} \tag{4.34}$$

is termed an *autoregressive process of order p* and denoted AR(p) (it is autoregressive in the sense that Y_t is regressed on its own past values). For the AR(p) process to be stationary, it is necessary that $V(Y_t)$ is finite, where (it can be shown that)

$$V\left(Y_t\right) = \frac{\sigma_\varepsilon^2}{1 - \sum_{j=1}^{p} \alpha_j \rho_j}. \tag{4.35}$$

Two important special cases of Equation 3.34 are obtained for $p = 1, 2$. For $p = 1$, we have the AR(1) process

$$Y_t = \alpha Y_{t-1} + \varepsilon_t, \quad \alpha_1 \neq 0, \tag{4.36}$$

ε_t white noise. Then

$$E\left(Y_t\right) = \alpha_1 E\left(Y_{t-1}\right) + E\left(\varepsilon_t\right)$$

or $\mu = \alpha_1 \mu$. Because $\alpha_1 \neq 0$, it follows that $\mu = 0$. Also,

$$\begin{aligned} V\left(Y_t\right) = E\left(Y_t^2\right) &= E\left[\left(\alpha_1 Y_{t-1} + \varepsilon_t\right)^2\right] \\ &= \alpha_1^2 V\left(Y_t\right) + \sigma_\varepsilon^2 \end{aligned}$$

or

$$V\left(Y_t\right) = \frac{\sigma_\varepsilon^2}{1 - \alpha_1^2}.$$

For $V(Y_t)$ to be finite (or for AR(1) to be stationary) and nonnegative, $-1 < \alpha_1 < 1$. Looking to autocovariances, we first write

$$Y_{t-\tau} Y_t = \alpha_1 Y_{t-\tau} Y_{t-1} + Y_{t-\tau} \varepsilon_t$$

and thus

$$E\left(Y_{t-\tau} Y_t\right) = \alpha_1 E\left(Y_{t-\tau} Y_{t-1}\right) + E\left(Y_{t-\tau} \varepsilon_t\right).$$

For $\tau \geq 1$,

$$\gamma_\tau = \alpha_1 \gamma_{\tau-1}.$$

This recursive relation indicates that, for any $\tau \geq 1$, knowing $\gamma_{\tau-1}$ enables us to obtain γ_τ. For instance, given $\gamma_0 = V\left(Y_t\right) = \sigma_\varepsilon^2 / \left(1 - \alpha_1^2\right)$,

$$\gamma_1 = \alpha_1 \gamma_0 = \frac{\alpha_1 \sigma_\varepsilon^2}{1 - \alpha_1^2}, \quad \gamma_2 = \alpha_1 \gamma_1 = \frac{\alpha_1^2 \sigma_\varepsilon^2}{1 - \alpha_1^2}, \text{ etc.}$$

In general,

$$\gamma_\tau = \frac{\alpha_1^\tau \sigma_\varepsilon^2}{1-\alpha_1^2}, \quad \tau = 0, 1, 2, \ldots.$$

Also,

$$\rho_\tau = \frac{\gamma_\tau}{\gamma_0} = \alpha_1^\tau, \quad \tau = 0, 1, 2, \ldots.$$

Next, let $p = 2$. Then the AR(2) process is

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \varepsilon_t, \tag{4.37}$$

ε_t white noise. Then

$$E(Y_t) = \alpha_1 E(Y_{t-1}) + \alpha_2 E(Y_{t-2}) + E(\varepsilon_t)$$

or $\mu(1-\alpha_1-\alpha_2)=0$. For $\alpha_1-\alpha_2<1$, we have $\mu=0$ (for stationarity). To calculate autocovariances, we form

$$Y_{t-\tau}Y_t = \alpha_1 Y_{t-\tau}Y_{t-1} + \alpha_2 Y_{t-\tau}Y_{t-2} + Y_{t-\tau}\varepsilon_t \tag{4.38}$$

with

$$E(Y_{t-\tau}Y_t) = \alpha_1 E(Y_{t-\tau}Y_{t-1}) + \alpha_2 E(Y_{t-\tau}Y_{t-2}) + E(Y_{t-\tau}\varepsilon_t). \tag{4.38.1}$$

For $\tau \geq 1$,

$$\gamma_\tau = \alpha_1 \gamma_{\tau-1} + \alpha_2 \gamma_{\tau-2}$$

or

$$\rho_\tau = \frac{\gamma_\tau}{\gamma_0} = \alpha_1 \rho_{\tau-1} + \alpha_2 \rho_{\tau-2}, \quad \tau \geq 1$$

(since $\rho_{-\tau}=\rho_\tau$), a recursive relation called the *Yule-Walker equations* for obtaining ρ_τ, $\tau \geq 2$. That is, for $\rho_0=1$,

$$\rho_1 = \frac{\alpha_1}{1-\alpha_2}.$$

For $\tau=2$,

$$\rho_2 = \frac{\alpha_1^2}{1-\alpha_2} + \alpha_2.$$

For $\tau = 3$,

$$\rho_3 = \frac{\alpha_1^3 + 2\alpha_1\alpha_2 - \alpha_1\alpha_2^2}{1-\alpha_2}, \text{ etc.}$$

If $\tau = 0$ in Equation 4.38, Equation 4.39 becomes

$$\gamma_0 = \alpha_1\gamma_{-1} + \alpha_2\gamma_{-2} + \sigma_\varepsilon^2.$$

Dividing both sides of this expression by $\gamma_0(=V(Y_t))$ yields

$$1 = \alpha_1\rho_1 + \alpha_2\rho_2 + \frac{\sigma_\varepsilon^2}{\gamma_0}$$

or

$$V(Y_t) = \frac{\sigma_\varepsilon^2}{1-\alpha_1\rho_1 - \alpha_2\rho_2}.$$

Given the ρ_1 and ρ_2 values defined above, we alternatively have

$$V(Y_t) = \frac{(1-\alpha_2)\sigma_\varepsilon^2}{(1+\alpha_2)(1-\alpha_1-\alpha_2)(1+\alpha_1-\alpha_2)}.$$

For AR(2) to be stationary, $V(Y_t)$ must be finite and nonnegative. Hence each factor in the denominator of the preceding expression must be positive with $1-\alpha_2>0$ or

$$\begin{aligned} &\alpha_1 + \alpha_2 < 1 \\ &-\alpha_1 + \alpha_2 < 1 \\ &-1 < \alpha_2 < 1. \end{aligned}$$

4.8.3 Random Walk Processes

Let us consider the AR(1) process

$$Y_t = \alpha_1 Y_{t-1} + \varepsilon_t, \tag{4.39}$$

where $|\alpha_1| < 1$ ($\{Y_t\}$ is stationary) and $\{\varepsilon_t\}$ is white noise. A *random walk process* is a special case of Equation 4.39 in which $\alpha_1 = 1$. For convenience, let us modify our notation a bit and denote this particular process as

$$X_t = X_{t-1} + u_t, \tag{4.40}$$

where $\{u_t\}$ is white noise. Because the coefficient on X_{t-1} is unity, Equation 4.40 is also referred to as a *unit root process*. It is instructive to compare the properties of Equations 4.39 and 4.40. As we shall now see, Y_t in Equation 4.39 is stationary, while the unit root case depicted by X_t in Equation 4.40 is nonstationary.

Each of these autoregressive models can be transformed by successive back-substitutions to

$$\begin{aligned} &1.\ Y_t = \sum_{i=0}^{t-1} \alpha_1^i \varepsilon_{t-i}, \\ &2.\ X_t = \sum_{i=0}^{t-1} u_{t-i}, \end{aligned} \tag{4.41}$$

where the initial observation of each is assumed to be zero (e.g., $Y_0 = X_0 = 0$).

Note that each random shock u_{t-i} has a permanent or nondecaying effect on $\{X_t\}$ since its coefficient is unity. Looking to the statistical properties of these two time series specifications, we have:

$$\begin{aligned} &1.\ E(Y_t) = \sum_{i=0}^{t-1} \alpha_1^i E(\varepsilon_{t-i}) = 0, \\ &\quad E(X_t) = \sum_{i=0}^{t-1} E(u_{t-i}) = 0. \end{aligned}$$

$$\begin{aligned} &2.\ V(Y_t) = \sum_{i=0}^{t-1} \alpha_1^{2i} E(\varepsilon_{t-i}^2) = \frac{\sigma_\varepsilon^2}{1-\alpha_1^2}, \\ &\quad V(X_t) = \sum_{i=0}^{t-1} E(u_{t-i}^2) = t\sigma_u^2 \to \infty \quad \text{as} \quad t \to \infty. \end{aligned}$$

So while the means of Y_t, X_t are each zero, the variance of Y_t converges to a constant asymptotically while the variance of the random walk or unit root process increases as t increases. Moreover,

$$\begin{aligned} 3.\ \mathrm{COV}(Y_{t-\tau}, Y_t) &= E(Y_{t-\tau} Y_t) \\ &= E\left[\left(\sum\nolimits_{i=0}^{t-\tau-1} \alpha_1^i \varepsilon_{t-\tau-1}\right)\left(\sum\nolimits_{i=0}^{t-1} \alpha_1^i \varepsilon_{t-i}\right)\right] \\ &= \gamma_\tau^Y = \frac{\alpha_1^\tau \sigma_\varepsilon^2}{1-\alpha_1^2}, \\ \mathrm{COV}(X_{t-\tau}, X_t) &= E(X_{t-\tau} X_t) \\ &= E\left[\left(\sum\nolimits_{i=0}^{t-\tau-1} u_{t-\tau-1}\right)\left(\sum\nolimits_{i=0}^{t-1} u_{t-i}\right)\right] \\ &= \gamma_\tau^X = (t-\tau)\sigma_u^2. \end{aligned}$$

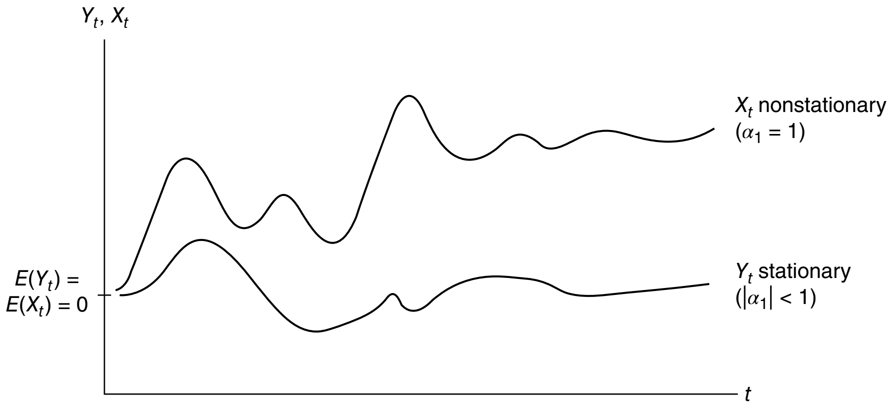

FIGURE 4.18 Stationary versus nonstationary processes.

Here too the autocovariance of Y_t converges asymptotically to a constant while the autocovariance of X_t is time dependent. Finally, looking to autocorrelations:

4. $\rho_\tau^Y = \gamma_\tau^Y / \gamma_0^Y = \alpha_1^\tau \to 0 \quad \text{as} \quad \tau \to \infty$
$\left(\text{with } |\alpha_1| < 1\right)$,
$$\rho_\tau^X = \gamma_\tau^X / \gamma_\tau^0 = \frac{t-\tau}{\tau}.$$

Hence, the autocorrelation function of X_t approaches zero only for very large values of τ.

The upshot of this discussion is that an AR(1) process with a unit root (a random walk process) is nonstationary. As Figure 4.18 reveals, the time path of stationary Y_t intersects its mean level at least once; Y_t converges towards its mean (it is said to be *mean-reverting*) while fluctuating randomly about the same. Contrary to the behavior of Y_t, X_t increases (decreases) systematically and is not mean-reverting as t increases. Interestingly enough, the random walk or unit root process (Eq. 4.40) can be made stationary by differencing, i.e., the variable $\Delta X_t = X_t - X_{t-1} = u_t$ is stationary about zero since $\{u_t\}$ is white noise. We shall elaborate on this point later on.

Before closing this section, let us consider two important modifications of Equations 4.39) (a stationary AR(1) process) and 4.40 (a nonstationary AR(1) process with a unit root). First, suppose we have the specification

$$Y_t = \mu + \alpha_1 Y_{t-1} + \varepsilon_t, \tag{4.42}$$

where $|\alpha_1| < 1$ and $\{\varepsilon_t\}$ is white noise. Here Equation 4.42 is a stationary AR(1) process with mean $E(Y_t) = \mu/(1-\alpha_1)$. But when $\alpha_1 = 1$ in Equation 4.42, we have (again modifying our notation a bit)

$$X_t = \mu + X_{t-1} + u_t, \tag{4.43}$$

where $\{u_t\}$ is white noise. This type of expression is termed a *random walk with nonzero* ($\mu \neq 0$) *drift*; it also constitutes a nonstationary AR(1) process with a unit root. Next, let

$$Y_t = \mu + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t, \tag{4.44}$$

where again $|\alpha_1| < 1$ with $\{\varepsilon_t\}$ white noise. Now $\{Y_t\}$ is a stationary AR(1) process about a linear trend (assuming $\lambda \neq 0$). But if $\alpha_1 = 1$, the appropriate modification of Equation 3.44 is

$$X_t = \mu + \lambda t + X_{t-1} + u_t, \tag{4.45}$$

with $\{u_t\}$ taken to be white noise. Here $\{X_t\}$ is a *random walk process about a time trend* ($\lambda \neq 0$); it too is nonstationary AR(1) with a unit root.

If from Equation 4.43 we form the difference $\Delta X_t = X_t - X_{t-1} = \mu + u_t$, then ΔX_t is stationary about μ since $\{u_t\}$ is white noise. Similarly, from Equation 4.45, the difference $\Delta X_t = X_t - X_{t-1} = \mu + \lambda t + u_t$ is stationary about a linear trend.

Let us elaborate briefly on the notion of *trend*. Specifically, a trend component of a time series $\{Y_t\}$ can be characterized as *deterministic* or *stochastic*. To see this, suppose a time series $\{Y_t\}$ always changes by a constant amount λ from period to period or $\Delta Y_t = \lambda$. Then the time path of Y_t can be specified as

$$Y_t = Y_0 + \lambda t, \tag{4.46}$$

where Y_0 is the initial value of Y_t (for $t = 0$). In this instance Y_t is said to have a nonstationary *deterministic linear time trend* component λt. If we now insert into Equation 4.46 the term $\varepsilon_t + \beta_1 \varepsilon_{t-1}$, $|\beta_1| < 1$, so that

$$Y_t = Y_0 + \lambda t + \varepsilon_t + \beta_1 \varepsilon_{t-1}, \tag{4.47}$$

then $\varepsilon_t + \beta_1 \varepsilon_{t-1}$ is stationary and Y_t only temporarily deviates from its trend by the amount $\varepsilon_t + \beta_1 \varepsilon_{t-1}$. Hence Equation 4.47 is termed a *trend-stationary model*; no permanent departures from trend are exhibited.

What if the inter-period change in Y_t is respecified as $\Delta Y_t = \lambda + \varepsilon_t$, where ε_t is white noise. Then the period-to-period change in Y_t is, on average, λ ($E(Y_t) = \lambda$ since $E(\varepsilon_t) = 0$). Now Y_t follows the time path

$$Y_t = Y_0 + \lambda t + \sum_{i=1}^{t} \varepsilon_i, \tag{4.48}$$

where again Y_0 is the starting value of Y_t. In addition to the deterministic trend term λt, model 4.48 entails a *stochastic trend* component $\sum_{i=1}^{t} \varepsilon_i$ consisting of cumulated shocks. And since each coefficient on ε_i is unity, each of these random shocks or disturbances has a permanent effect on Y_t. While Equation 4.48 is nonstationary (e.g., its mean is $E(Y_t) = Y_0 + \lambda t$), its first difference $\Delta Y_t = Y_t - Y_{t-1} = \lambda + \varepsilon_t$ is stationary.

As we shall see below, a nonstationary random walk or unit root process can be made stationary by differencing and will thus be termed a *difference stationary process*. We note briefly that if a nonstationary process exhibiting a deterministic trend can be made stationary by subtracting λt from Y_t, then that process is said to be *trend stationary*.

4.8.4 Integrated Processes

We indicated above that a time series $\{Y_t\}$ is (weakly) stationary if Equation 4.31 holds. Implicit in this definition is the notion that a stationary series cannot grow over time. However, the changes $\Delta Y_t = Y_t - Y_{t-1}$ in the series might be stationary. In this regard, the nonstationary variable Y_t is said to be *integrated of order one*, denoted $Y_t \sim I(1)$, if the changes in Y_t, ΔY_t, form a stationary series—in which case ΔY_t is said to be *integrated of order zero* and written $\Delta Y_t \sim I(0)$. Thus an *integrated series* with order of integration greater than zero is nonstationary. So if $Y_t \sim I(0)$, then $\{Y_t\}$ is a stationary series.

Let us briefly consider the principal differences between time series that are integrated of order zero versus integrated of order one. If $Y_t \sim I(0)$, then Y_t has a fixed mean and a constant (or bounded) variance. This type of series has a tendency to frequently return to, and intersect, its mean value. Moreover, a value of Y_t far from its mean has a tendency to be followed by values nearer to the mean. In this regard, the mean of the series is termed its *attractor*.

As far as $Y_t \sim I(1)$ is concerned, in the absence of drift, any such series has no attractor but has a variance that increases with time. Hence, in this instance, Y_t rarely returns to any specific level and tends to wander erratically. An $I(1)$ series will be relatively smooth when its time profile is compared to that of any $I(0)$ series and displays fluctuations that are less pronounced than those of an $I(0)$ series.

In general, a nonstationary series $\{Y_t\}$ which can be transformed to a stationary series by differencing d times in succession is said to be *integrated of order d* and denoted $Y_t \sim I(d)$. Hence $Y_t \sim I(d)$ is difference stationary if $\Delta^d Y_t \sim I(0)$. For example, if difference stationary $Y_t \sim I(2)$, then $\Delta^2 Y_t = \Delta(\Delta Y_t) \sim I(0)$ or $\Delta^2 Y_t$ is stationary. And if $Y_t \sim I(0)$, then Y_t is stationary and no differencing is needed. Moreover, if $Y_t \sim I(d)$, then $\Delta^b Y_t \sim I(d-b)$, $d > b > 0$. It is also true that if $Y_t \sim I(d)$, then $\alpha + \beta Y_t \sim I(d)$ where α and β are constants.

Looked at in another light, we can state that an *integrated series* accumulates past shocks or innovations, i.e., an integrated series is expressible as a sum. For example, we noted earlier that the random walk $Y_t = Y_{t-1} + \varepsilon_t$, $\{\varepsilon_t\}$ white noise, can be "integrated" or written as the sum of all previous errors or $Y_t = \sum_{j=0}^{\infty} \varepsilon_{t-j}$, with $Y_0 = 0$. Hence $Y_t \sim I(1)$ since old and new shocks have equal weight in determining Y_t so that Y_t obviously displays *permanent or long memory*. Hence this process was deemed nonstationary since its future time path depends on all past innovations; the said path is not wedded to some mean level to which it eventually reverts.

However, $\Delta Y_t = Y_t - Y_{t-1} = \sum_{j=0}^{\infty} \varepsilon_{t-j-1} = \varepsilon_t \sim I(0)$; that is, ΔY_t is stationary since it is white noise and thus exhibits *temporary or short memory*. So with a long-memory series, an "old" shock to the series has a noticeable impact on the current value of the series; for a short-memory series, an "old" shock to the series has little or no effect on the current level of the series.

In contrast to the random walk case, suppose $Y_t = \alpha_1 Y_{t-1} + \varepsilon_t$, $\{\varepsilon_t\}$ white noise. After integrating or summing, $Y_t = \sum_{j=0}^{\infty} \alpha_1^j \varepsilon_{t-j}, Y_0 = 0$. If $|\alpha_1| < 1$ (the stationarity condition), $\alpha_1^j \to 0$ as $j \to \infty$ so that $Y_t \sim I(0)$. Hence the effect of or level of memory associated with past shocks declines as time passes. But if $\alpha_1 = 1$, then $Y_t = \sum_{j=0}^{\infty} \varepsilon_{t-j} \sim I(1)$ since Y_t is now the sum of all past innovations.

4.8.5 Testing for Unit Roots

The Structure of the Dickey–Fuller (1979, 1981) Test Equation Based on our discussions in the preceding section, it should be evident that the nonstationarity of time series data precludes us from obtaining reliable regression parameter estimates. More specifically, nonstationary or unit root variables have means and variances that are time dependent so that any inferences based on OLS regression results are misleading.

To overcome this difficulty, we shall apply unit root tests to determine if our regression variables are stationary or nonstationary. Remember that the series $\{Y_t\}$ may be nonstationary because it has a deterministic trend or a stochastic trend. In this regard, to examine the issue of a variable's nonstationarity, we must address the question: should one detrend or difference a time series? The answer depends upon whether the time series is a trend-stationary or difference-stationary process. As explained earlier, a (deterministic) trend-stationary process has the form

$$Y_t = \gamma_0 + \gamma_1 t + \varepsilon_t, \tag{4.49}$$

whereas a difference stationary process is structured as

$$Y_t = \alpha_0 + Y_{t-1} + \varepsilon_t. \tag{4.50}$$

If $\{\varepsilon_t\}$ is an independent and identically distributed sequence of random variables, then Equation 4.50 constitutes a random walk with drift.

With both cases (Eqs. 4.49 and 4.50) subsumed under

$$\begin{aligned} Y_t &= \gamma_0 + \gamma_1 t + u_t \\ u_t &= \rho u_{t-1} + \varepsilon_t, \end{aligned} \tag{4.51}$$

where ε_t is taken to be a stationary random variable (it has a zero mean and a constant variance σ_ε^2), we can rewrite Equation 4.51 as

$$Y_t = \gamma_0 + \gamma_1 t + \rho u_{t-1} + \varepsilon_t. \tag{4.51.1}$$

Then substituting $Y_{t-1} = \gamma_0 + \gamma_1 (t-1) + u_{t-1}$ into Equation 4.51.1 yields

$$Y_t = \gamma_0 + \gamma_1 t + \rho\left[Y_{t-1} - \gamma_0 - \gamma_1 (t-1)\right] + \varepsilon_t. \tag{4.51.2}$$

If we now subtract Y_{t-1} from both sides of Equation 4.51.2, our final result is

$$\begin{aligned}\Delta Y_t &= \gamma_0 - \rho\gamma_0 + \rho\gamma_1 + \gamma_1 t - \rho\gamma_1 t + (\rho - 1)Y_{t-1} + \varepsilon_t \\ &= \beta_0 + \beta_1 t + (\rho - 1)Y_{t-1} + \varepsilon_t,\end{aligned} \tag{4.51.3}$$

where $\beta_0 = \gamma_0(1-\rho) + \rho\gamma_1$ and $\beta_1 = \gamma_1(1-\rho)$. In this regard, if $|\rho| < 1$, then Y_t is trend stationary; and if $\beta_1 = 0$ and $|\rho| = 1$ (the unit root case), then Y_t is difference stationary.

Let us rewrite Equation 4.51.3 as

$$\Delta Y_t = \beta_0 + \beta_1 t + \phi Y_{t-1} + \varepsilon_t, \tag{4.52}$$

where $\phi = \rho - 1$. Then to conduct a unit root test, set $H_0 : \phi = \rho - 1 = 0$ and $H_1 : \phi = \rho - 1 < 0$. However, since under the null hypothesis $Y_t \sim I(1)$, Equation 4.52 represents the regression of an $I(0)$ variable on an $I(1)$ variable so that the conventional t statistic does not have an asymptotic normal distribution and thus we cannot use the usual t distribution to test the null.

Nonstandard Critical Values To see exactly what sort of modified distribution should be used, we know that for observed

$$Y_t = \rho Y_{t-1} + \varepsilon_t, \qquad |\rho| < 1, \quad t = 1, 2, \ldots,$$

the OLS estimator for ρ is

$$\hat{\rho} = \frac{\sum_{t=2}^{n} Y_t Y_{t-1}}{\sum_{t=2}^{n} Y_{t-1}^2}$$

with

$$\left[\frac{n}{(1-\rho)^2}\right]^{1/2} (\hat{\rho} - \rho) \xrightarrow{\text{d}} N(0,1); \tag{4.53}$$

i.e., the quantity $\left[n/(1-\rho)^2\right]^{1/2} (\hat{\rho} - \rho)$ converges in distribution to a standard normal distribution function. However, for $H_0 : \rho = 1$ (versus $H_0 : \rho < 1$), Dickey and Fuller (1979) examine the distribution of $\hat{\rho}$ above under this null hypothesis and demonstrate that the statistic

$$n(\hat{\rho} - \rho) = n(\hat{\rho} - 1) \tag{4.54}$$

has a nonstandard limiting distribution (it is not normal or even symmetrical) and thus the usual t ratio is not t distributed so that critical values from conventional t tables are not applicable for hypothesis testing. In fact, tables of critical values for the limiting distribution of Equation 4.54 were computed by Dickey and Fuller under various assumptions (concerning the presence of drift, a deterministic trend and the behavior of ε_t). These new critical values will be termed *t-like critical values*.

To ascertain how Dickey–Fuller (unit root) tests are executed, it is readily seen that three separate regression models are nested in Equation 4.52:

$$\begin{aligned}
&1.\ \Delta Y_t = \phi Y_{t-1} + \varepsilon_t \\
&2.\ \Delta Y_t = \beta_0 + \phi Y_{t-1} + \varepsilon_t \\
&3.\ \Delta Y_t = \beta_0 + \beta_1 t + \phi Y_{t-1} + \varepsilon_t,
\end{aligned} \tag{4.55}$$

where $\phi = \rho - 1$. (Remember that if $\rho = 1$, then $\phi = 0$; and if $\rho < 1$, then $\phi < 0$.) In this regard, looking to Equation 4.55.1, to conduct a unit root test, set $H_0: \phi = 0$ (Y_t is nonstationary) versus $H_0: \phi < 0$ (Y_t is stationary) and let the conventional t statistic obtained from an OLS fit of Equation 4.55.1 be denoted as τ. However, τ does not follow the usual t distribution—the simulated asymptotic critical values of τ obtained under the null hypothesis that $\phi = 0$ are provided in Table A.6 (Fuller, 1976). It is important to note that these critical values are derived under the assumption that there is no drift ($\beta_0 = 0$) and no time trend ($\beta_1 = 0$). Hence the limiting distribution of the test statistic τ and the associated critical values are incorrect if these assumptions do not hold.

Dickey and Fuller actually derive a separate limiting distribution, and thus separate critical values, for the OLS t statistic under the null hypothesis that $\phi = 0$ for each of the two remaining regressions in Equation 4.55 (but under the assumption that Equation 4.55.1 generates the data). When Equation 4.55.2 is fit, the OLS t statistic for $\phi = 0$ is denoted τ_μ, and when Equation 4.55.3 is run, the OLS statistic for $\phi = 0$ is denoted as τ_τ. Selected critical values for these statistics are also presented in Table A.6. Since in each case $H_1: \phi < 0$, a calculated value smaller than the (negative) critical or tabular value would lead us to reject H_0 (a unit root) in favor of H_1 (stationarity).

The Augmented Dickey–Fuller Test Equation Thus far we have taken the ε_t's to be independent and identically distributed random variables. If this assumption is not warranted, then the limiting distributions and critical values obtained by Dickey and Fuller cannot be assumed to hold. However, these limiting distributions and critical values are tenable even when ε_t is autoregressive if what is called the *augmented Dickey–Fuller (ADF) regression* is run. That is, suppose the data-generating process is Equation 4.55.1 with $\phi = 0$ or $\rho = 1$ and that ε_t is a stationary autoregressive process of order p or

$$\begin{aligned}
\varepsilon_t &= \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \ldots + \theta_p \varepsilon_{t-p} + \eta_t \\
&= \sum_{i=1}^{p} \theta_i \varepsilon_{t-i} + \eta_t,
\end{aligned} \tag{4.56}$$

where $\{\eta_t\}$ constitutes a set of random variables which forms an independent and identically distributed process.

Consider the expression (Eq. 4.55.3), where $H_0: \phi = 0$ is to be tested against $H_1: \phi < 0$. Then via Equation 4.56,

$$\Delta Y_t = \beta_0 + \beta_1 t + \phi Y_{t-1} + \sum_{i=1}^{p} \theta_i \varepsilon_{t-i} + \eta_t, \tag{4.57}$$

which can be rewritten (since Equation 4.55.1 with $\rho = 1$ gives $\varepsilon_t = Y_t - Y_{t-1}$) as

$$\begin{aligned}\Delta Y_t &= \beta_0 + \beta_1 t + \phi Y_{t-1} + \theta_1 (Y_{t-1} - Y_{t-2}) \\ &\quad + \theta_2 (Y_{t-2} - Y_{t-3}) + \ldots + \theta_p (Y_{t-p} - Y_{t-p-1}) + \eta_t \\ &= \beta_0 + \beta_1 t + \phi Y_{t-1} + \sum_{i=1}^{p} \theta_i \Delta Y_{t-i} + \eta_t.\end{aligned} \tag{4.58}$$

Dickey and Fuller demonstrate that for this ADF regression the t statistic under $H_0: \phi = 0$ has the same nonstandard limiting distribution as does the τ_τ statistic. Thus the critical values for a significance test based on the ADF regression are identical to those associated with the statistic τ_τ.

What value should p assume in Equation 4.58? One can generally start with an *a priori* chosen upper level for p and then drop the last lagged difference term if it is not statistically significant (based on the ordinary t test). This process is repeated until the last lagged difference term is significant or until the residuals become white noise.

An important area of concern which emerges when testing for a unit root in a series is the issue of the *power of the Dickey–Fuller test* (i.e., the test's ability to detect a *false* null hypothesis). Specifically, the power of the Dickey–Fuller test is considered to be low. That is, since we are trying to ascertain whether a series is $I(1)$ (the null hypothesis) versus $I(0)$ (the alternative hypothesis), $I(0)$ alternatives in which the series may be *close to being* $I(1)$ but not exactly $I(1)$ are highly plausible since the true form of the data-generating process is unknown. Moreover, as evidenced by the Dickey–Fuller tables, the (simulated) critical values are sensitive to the specification of the test equation.

Suppose that in the execution of the Dickey–Fuller test we find that the null hypothesis of a unit root $H_0: \phi = \rho - 1 = 0$ cannot be rejected. In this circumstance it is possible that the order of integration of Y_t exceeds zero (or perhaps Y_t is not integrated at all). Hence our next step is to see if possibly $Y_t \sim I(1)$. Now, we know that if $Y_t \sim I(1)$, then $\Delta Y_t \sim I(0)$. Hence we must apply the Dickey–Fuller test to ΔY_t instead of to Y_t itself. To this end, our second round test equation is

$$\Delta(\Delta Y_t) = \Delta^2 Y_t = \phi \Delta Y_{t-1} + \varepsilon_t. \tag{4.59}$$

Again we are interested in the negativity of ϕ so that, under $H_0: \phi = 0$ (ΔY_t has a unit root) vs. $H_1: \phi < 0$ (ΔY_t is stationary), if we reject H_0, then ΔY_t is stationary or $\Delta Y_t \sim I(0)$ and thus $Y_t \sim I(1)$.

If the null hypothesis that ΔY_t has a unit root cannot be rejected, then we must conduct a test to determine if possibly $Y_t \sim I(2)$. For $Y_t \sim I(2)$, $\Delta Y_t \sim I(1)$ so that we must apply the Dickey–Fuller test to $\Delta^2 Y_t$. Thus the third round test equation appears as

$$\Delta(\Delta^2 Y_t) = \Delta^3 Y_t = \phi \Delta^2 Y_{t-1} + \varepsilon_t$$

We repeat this process until we determine the order of integration for Y_t or we conclude that $\{Y_t\}$ is not a difference stationary process.

The approach to unit root testing just described is termed a *bottom-up strategy*. Two obvious drawbacks associated with this testing methodology are: (i) it is possible that the Y_t series cannot eventually be transformed (by differencing) to stationarity; and (ii) *overdifferencing* may occur, in which case one reaches the erroneous conclusion that an order of integration higher that the true one exists. This latter outcome is detected by the emergence of high *positive* values of the Dickey–Fuller test statistic.

To overcome these difficulties, it has been suggested (Dickey and Pantula, 1987) that one might pursue a *general-to-specific testing strategy*—one starts from the highest reasonable or informed (suspected) order of integration (possibly $I(2)$) and then proceeds by *testing downwards* until stationarity obtains. A detailed account of this approach appears in the next section.

Testing Downwards for Unit Roots Consider the AR(2) model

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \varepsilon_t,$$

where ε_t is white noise. It can be demonstrated that this expression can be transformed to

$$\Delta^2 Y_t = \delta_1 \Delta Y_{t-1} + \delta_2 Y_{t-1} + \varepsilon_t. \tag{4.60}$$

Then the process of testing downwards for a unit root consists of the following steps:

1. Step 1: Test $H_0: Y_t \sim I(2)$ versus $H_1: Y_t \sim I(1)$. Under the null hypothesis of two unit roots, $\delta_1 = \delta_2 = 0$; under the alternative hypothesis, we may take $\delta_1 < 0$ and $\delta_2 = 0$. Clearly $\delta_2 = 0$ satisfies both the null and alternative hypotheses. Hence a Dickey–Fuller test of the null hypothesis is conducted by the OLS estimation of Equation 4.60 subject to $\delta_2 = 0$ or by estimating

$$\Delta^2 Y_t = \delta_1 \Delta Y_{t-1} + \varepsilon_t \tag{4.61}$$

   and determining if $\hat{\delta}_1$ is significantly negative by comparing the t statistic for $\hat{\delta}_1$ with the τ critical value taken from the Dickey–Fuller tables. If H_0 is rejected, proceed to step 2.
2. Step 2: Test $H_0: Y_t \sim I(1)$ versus $H_1: Y_t \sim I(0)$. Under this null hypothesis of a single unit root, we take $\delta_1 < 0$ and $\delta_2 = 0$, and under the alternative hypothesis, we must have $\delta_1 < 0$ and $\delta_2 < 0$ for the stationarity of Y_t. Equation 4.60 is then estimated via OLS, and the negativity of $\hat{\delta}_2$ is tested by comparing its t value to the Dickey–Fuller critical value τ. If $\hat{\delta}_2$ is not significantly negative, then the null hypothesis cannot be rejected. But if the null hypothesis is rejected, then we can conclude that $Y_t \sim I(0)$; that is, Y_t has no unit root and thus is stationary.
3. End.

TABLE 4.8 Time Series Observations on Y; Y_t Values, 1915–2001 (n = 87)

Year	Y_t	Year	Y_t	Year	Y_t
1915	69.0	1945	90.0	1975	97.3
1916	72.0	1946	91.1	1976	99.0
1917	75.0	1947	87.8	1977	101.0
1918	78.0	1948	89.5	1978	102.0
1919	84.0	1949	92.0	1979	98.7
1920	87.0	1950	92.2	1980	96.1
1921	91.0	1951	92.0	1981	95.3
1922	90.1	1952	91.0	1982	96.3
1923	89.0	1953	90.0	1983	98.0
1924	88.5	1954	88.0	1984	101.6
1925	88.0	1955	86.0	1985	102.0
1926	87.0	1956	84.5	1986	102.5
1927	85.3	1957	82.7	1987	100.0
1928	86.1	1958	80.0	1988	98.0
1929	88.3	1959	82.0	1989	93.0
1930	87.2	1960	83.1	1990	94.0
1931	86.6	1961	84.2	1991	94.2
1932	85.4	1962	84.1	1992	96.0
1933	85.1	1963	82.3	1993	98.0
1934	83.7	1964	80.8	1994	99.0
1935	85.3	1965	80.0	1995	100.0
1936	86.8	1966	82.0	1996	101.0
1937	89.2	1967	84.0	1997	102.0
1938	90.5	1968	86.0	1998	102.5
1939	92.1	1969	88.0	1999	103.0
1940	91.2	1970	90.0	2000	105.0
1941	90.0	1971	92.0	2001	108.0
1942	89.0	1972	94.0		
1943	88.2	1973	95.0		
1944	88.0	1974	96.0		

As a final point, we mention briefly that augmentation terms can be included in Equation 4.60 to address the issue of potential autocorrelation while a constant as well as a deterministic trend term can also be inserted into Equation 4.60 if warranted.

In the following two example problems we shall examine "automatic" methods for determining whether a time series $\{Y_t\}$ is difference stationary.

EXAMPLE 4.6 Given the time series dataset presented in Table 4.8 and illustrated in Figure 4.19, let us test the variable Y_t for the presence of a unit root. The SAS code appears as Exhibit 4.5.

EXHIBIT 4.5 SAS Code for %DFTEST

```
data dft;
  input year y;
datalines;
1915 69.0
1916 72.0
1917 75.0
  .    .
  .    .
  .    .
2000 105.0
2001 108.0
;
proc plot data=dft;
  plot y * year= '*';
run;
data dft; set dft;
%dftest (dft, y, trend=2, outstat=dfstat1); ①
%put p=&dftest; ②
proc print data=dfstat1;
run;
%dftest (dft, y, dif=(1), ar=1, outstat=dfstat2); ③
%put p=&dftest;
proc print data=dfstat2;
run;
```

① We invoke the *%DFTEST* macro. The two required arguments are: (i) the name of the SAS data set containing the time series variable to be tested for a unit root; and (ii) the name of the time series variable itself. The basic form of this macro is thus

`%dftest (dft,y` [, *options*]`);`

where the various options, which must be inserted after the two required arguments and separated by commas, are as follows:

DIF = (differencing-list)—specifies the degree of differencing to be applied to Y_t, e.g.,

DIF = (1)—requests ΔY_t; that is, the Y_t series is to be differenced once at lag 1.

DIF = (1, 1)—requests $\Delta^2 Y_t$; that is, the Y_t series is differenced once at lag 1 and then differenced again at lag 1.

DLAG = |1|2|4|12 —specifies the lag to be tested for a unit root (default is DLAG = 1).

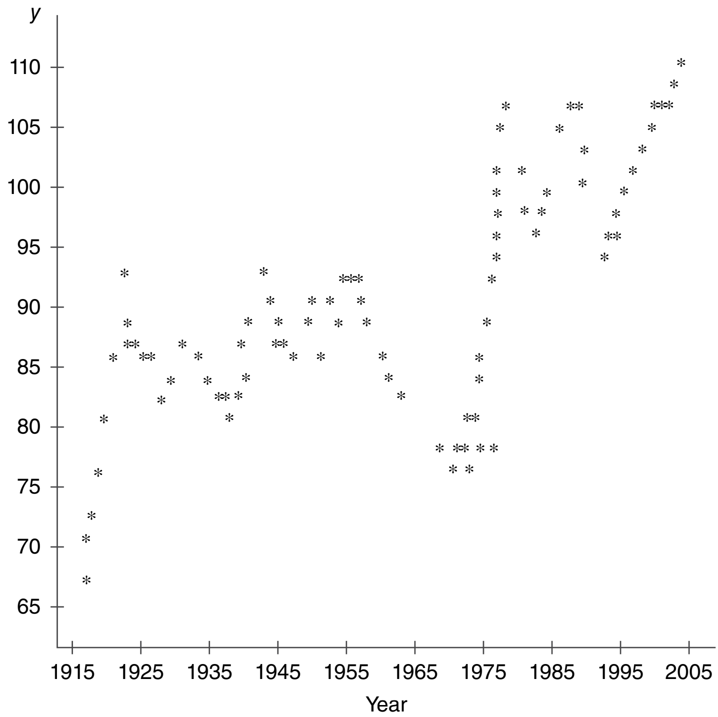

FIGURE 4.19 Time series observations on Y_t (1915–2001).

AR = *n*—specifies the order of the autoregressive model to be fit after any differencing is requested by the DIF = and DLAG = options (default is AR = 3).

OUT = SAS – data – set—writes residuals to an output data set.

OUTSTAT = SAS – data – set—writes the test statistic, parameter estimates, and other statistics to an output data set.

TREND = 0|1|2 —specifies the status of a deterministic linear time trend included in the model:

TREND = 0—assumes a zero mean and no linear time trend.

TREND = 1—includes an intercept term.

TREND = 2—includes an intercept and a linear time trend.

The default is TREND = 1.

This first test is applied to Y_t. The test equation includes an intercept, a linear trend term, and three autoregressive terms.

OUTSTAT = dfstat1 names the output data set.

② %PUT *p*= *&DFTEST* is the macro variable which defines the computed *p*-value for the Dickey–Fuller-type critical value.

③ The second test is applied to ΔY_t. The test equation includes an intercept term along with one autoregressive term.

Only an abridged version of the SAS output is presented below. Its salient features are:

TEST 1

$H_0: Y_t$ has a unit root; $H_1:$ not H_0.

TAU	_PVALUE_
−2.4412	0.35585

We cannot reject $H_0 - Y_t$ is nonstationary.

TEST 2

$H_0: \Delta Y_t$ has a unit root; $H_1:$ not H_0.

TAU	_PVALUE_
−4.4051	0.00062

We reject $H_0 - \Delta Y_t$ is stationary or $\Delta Y_t \sim I(0)$ and thus $Y_t \sim I(1)$; Y_t has a unit root and is difference stationary. ■

EXAMPLE 4.7 Our next approach for unit root detection is to use the STATIONARITY option in the ARIMA procedure. The Example 4.6 data series will be employed. The requisite SAS code is provided in Exhibit 4.6.

EXHIBIT 4.6 SAS Code for STATIONARITY Test

```
data dft;
  input year y;
datalines;
1915 69.0
1916 72.0
1917 75.0
.    .
.    .
.    .
2000 105.0
2001 108.0
;
proc arima data = dft; ①
       ②          ③                   ④
   identify var = y stationarity = (adf = (1));
run;
```

① We invoke the *ARIMA* procedure.

② The *IDENTIFY* statement reads the time series to be modeled, differences the series if necessary, and computes a variety of test statistics needed to help identify models to be estimated.

③ The VAR = option specifies the variable to be modeled. For the first test, VAR = *Y*.

```
proc arima data = dft;
              ⑤
identify var = y(1) stationarity = (adf = (1));
run;
```

④ The *STATIONARITY* = option performs unit root tests. We choose the augmented Dickey–Fuller test with one autoregressive term (ADF = (1)). (If, say, four autoregressive terms were warranted, then we would write ADF = (1, 2, 3, 4).)
⑤ We next test ΔY_t for a unit root by specifying *VAR* = Y(1), which renders the first difference of the Y_t series. (If $\Delta^2 Y_t$ were to be tested for a unit root, then we would write *VAR* = Y(1,1), which responds with the second difference of Y_t.)

The (condensed) output generated by these two stationarity tests appears in Table 4.9.

Ⓐ ① The residuals determined up to a specific lag are checked to see if they constitute a white noise process. Here we test H_0: all of the autocorrelations of the residual series up to a given lag are zero, against H_1 : not H_0. If we do not reject H_0 for all lags, then an ARIMA model is not appropriate for the Y_t series. (The *NLAG* = option controls the number of lags for which autocorrelations are calculated. The default is *NLAG* = 24.)

Since each of the *p*-values for the sample chi-square statistics is quite small, we reject H_0 in favor of H_1 and conclude that the Y_t series is stationary. Hence the current models do not adequately explain the Y_t series.

② Three separate augmented Dickey–Fuller test are performed using the following specifications: a zero mean; a single mean (or the presence of drift); and the presence of drift along with a deterministic linear time trend. In each case the *p*-value associated with the appropriate tau statistic is too large to reject the null hypothesis of a unit root. Thus the Y_t series is non-stationary.

Ⓑ ① Given that each of the computed chi-square values has a very small *p*-value, we reject the null hypothesis that the residuals of the ΔY_t process are white noise.

② The *p*-values associated with the tau statistics for the zero mean, single mean, and trend cases are each below 0.05, indicating that the rejection of the null hypothesis of a unit root for ΔY_t is warranted. Hence, we conclude that $\Delta Y_t \sim I(0)$ (ΔY is stationary) and thus $Y_t \sim I(1) - Y_t$ has a unit root and is difference stationary. ■

TABLE 4.9 Output for Example 4.7

The SAS system

Ⓐ

The ARIMA procedure

Name of variable: y

Mean of working series	90.57931
Standard deviation	7.514861
Number of observations	87

①

Autocorrelation check for white noise

To lag	Chi-square	DF	Pr > chi-square	Autocorrelations					
6	227.67	6	<0.0001	0.889	0.767	0.647	0.539	0.465	0.418
12	291.97	12	<0.0001	0.398	0.374	0.343	0.310	0.277	0.243
18	309.11	18	<0.0001	0.219	0.192	0.177	0.152	0.121	0.080

②

Augmented Dickey–Fuller unit root tests

Type	Lags	Rho	Pr < rho	Tau	Pr < tau	F	Pr > F
Zero mean	1	0.3459	0.7641	0.98	0.9119		
Single mean	1	−8.3147	0.1899	−1.88	0.3418	2.42	0.4623
Trend	1	−19.9672	0.0536	−3.02	0.1328	4.59	0.2726

Ⓑ

The ARIMA procedure

Name of variable = $y(1)$	
Period(s) of differencing	1
Mean of working series	0.453488
Standard deviation	1.876519
Number of observations	86
Observation(s) eliminated by differencing	1

①

Autocorrelation check for white noise

To lag	Chi-square	DF	Pr > Chi-square	Autocorrelations					
6	35.58	6	<0.0001	0.512	0.266	–0.007	–0.115	–0.202	–0.092
12	43.47	12	<0.0001	0.018	0.088	0.077	0.075	–0.084	–0.226
18	53.54	18	<0.0001	–0.224	–0.176	–0.086	0.051	0.054	–0.045

②

Augmented Dickey–Fuller unit root tests

Type	Lags	Rho	Pr < rho	Tau	Pr < tau	F	Pr > F
Zero mean	1	–37.0727	<0.0001	–4.28	<0.0001		
Single mean	1	–39.8110	0.0008	–4.41	0.0006	9.70	0.0010
Trend	1	–39.4048	0.0003	–4.34	0.0045	9.67	0.0010

APPENDIX 4.A OLS ESTIMATED AND RELATED GROWTH RATES

4.A.1 The OLS Growth Rate

An important consideration in the determination of the average rate of growth in a Y series over a number of years (indexed as t = 1, 2, …, n) is how the yearly weights are specified. In what follows we shall examine a variety of popular weighting schemes in order to ascertain their impact on the calculation of average growth rates (Mawson, 2002).

Given that compounding takes place over $t - 1$ years, the logarithmic transformation of the compound growth equation $Y_t = Y_1(1+r)^{t-1}$ enables us to obtain the regression equation

$$\begin{aligned}\ln Y_t &= \ln Y_1 + \ln(1+r)(t-1) + \varepsilon_t \\ &= \beta_0 + \beta_1 t + \varepsilon_t,\end{aligned} \tag{4.A.1}$$

where $\beta_0 = \ln Y_1 - \ln(1+r)$ and $\beta_1 = \ln(1+r)$. Then under the OLS estimation of this equation, $\hat{\beta}_1 = \ln(1+\hat{r})$ and thus the estimated compound rate of growth is $\hat{r} = e^{\hat{\beta}_1} - 1$. Let us examine $\hat{\beta}_1$ in greater detail.

From Equation 4.5 it is readily demonstrated that the OLS estimator $\hat{\beta}_1$ can be expressed in terms of deviations from the means of the variables t and $\ln Y_t$ as

$$\hat{\beta}_1 = \frac{\sum(t-\bar{t})\left(\ln Y_t - \overline{\ln Y_t}\right)}{\sum(t-\bar{t})^2}, \tag{4.A.2}$$

where $\bar{t} = (n+1)/2$.[2] Then Equation 4.A.2 simplifies to

$$\hat{\beta}_1 = \frac{\sum(t-\bar{t})\ln Y_t}{\sum(t-\bar{t})^2} = \sum_{t=1}^{n} W_t \ln Y_t, \tag{4.A.3}$$

Where

$$W_t = \frac{t-\bar{t}}{\sum_{t=1}^{n}(t-\bar{t})^2}. \tag{4.A.4}$$

[2] Note that, for $t = 1, 2, \ldots, n$,

$$\sum t = 1 + 2 + \ldots + n = \frac{n(n+1)}{2}$$

so that $\bar{t} = (n+1)/2$. Also,

$$\sum t^2 = 1 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

a result which will be utilized in what follows.

Since $\sum_{t=1}^{n}(t-\bar{t})=0$, if follows that $\sum_{t=1}^{n}W_t=0$.

Let us rewrite Equation 4.A.3 as

$$\begin{aligned}\hat{\beta}_1 &= \sum_{t=1}^{n} W_t \ln Y_t - \sum_{t=1}^{n} W_t \ln Y_1 \\ &= \sum_{t=1}^{n} W_t \left(\ln Y_t - \ln Y_1\right) = \sum_{t=2}^{n} W_t \left(\ln Y_t - \ln Y_1\right),\end{aligned} \tag{4.A.5}$$

since $W_1(\ln Y_1 - \ln Y_1) = 0$. Given that

$$\begin{aligned}\ln Y_t - \ln Y_1 &= \left(\ln Y_2 - \ln Y_1\right) + \left(\ln Y_3 - \ln Y_2\right) + \ldots + \left(\ln Y_t - \ln Y_{t-1}\right) \\ &= \sum_{s=2}^{t} \left(\ln Y_s - \ln Y_{s-1}\right),\end{aligned}$$

Equation 4.A.5 becomes

$$\begin{aligned}\hat{\beta}_1 &= \sum_{t=2}^{n} W_t \sum_{s=2}^{t} \left(\ln Y_s - \ln Y_{s-1}\right) \\ &= \sum_{s=2}^{n} \left(\sum_{t=s}^{n} W_t\right) \Delta \ln Y_s = \sum_{s=2}^{n} k_s \Delta \ln Y_s,\end{aligned} \tag{4.A.6}$$

where, from Equation 4.A.4, the weights are

$$k_s = \sum_{t=s}^{n} W_t = \frac{\sum_{t=s}^{n}(t-\bar{t})}{\sum_{t=1}^{n}(t-\bar{t})^2}. \tag{4.A.7}$$

So by virtue of Equation 4.A.6, we see that $\hat{\beta}_1$ is a weighted sum of the terms $\Delta \ln Y_s$, $s = 2, 3, \ldots, n$, where the weights are provided by Equation 4.A.7.

Let us write the numerator of Equation 4.A.7 as

$$\begin{aligned}\sum_{t=s}^{n}(t-\bar{t}) &= \sum_{t=1}^{n-(s-1)} \left[t + (s-1) - \bar{t}\right] \\ &= \sum_{t=1}^{n-(s-1)} t + \sum_{t=1}^{n-(s-1)} (s-1) - \sum_{t=1}^{n-(s-1)} \bar{t} \\ &= \frac{(n-s+1)(n-s+2)}{2} + (s-1)(n-s+1) - \frac{n+1}{2}(n-s+1) \\ &= \frac{1}{2}(n-s+1)(s-1).\end{aligned} \tag{4.A.8}$$

Also, the dominator of Equation 4.A.7 can be written as

$$\begin{aligned}\sum_{t=1}^{n}(t-\bar{t})^2 &= \sum_{t=1}^{n} t^2 - n\bar{t}^2 = \frac{n(n+1)(2n+1)}{6} - n\left(\frac{n+1}{2}\right) \\ &= \frac{1}{2}\left[\frac{n(n+1)(n-1)}{6}\right].\end{aligned} \tag{4.A.9}$$

A substitution of Equations 4.A.8 and 4.A.9 into Equation 4.A.7 yields

$$k_s = \frac{6(n-s+1)(s-1)}{n(n+1)(n-1)}, \quad s = 2,3,\ldots,n. \tag{4.A.10}$$

In view of Equation 3.A.10, the OLS estimator $\hat{\beta}_1$ becomes, from Equation 4.A.6,

$$\hat{\beta}_1 = \sum_{s=2}^{n} \left[\frac{6(n-s+1)(s-1)}{n(n+1)(n-1)} \right] \Delta \ln Y_s \tag{4.A.6.1}$$

and thus the *OLS estimated compound rate of growth* is $r_{\text{OLS}} = e^{\hat{\beta}_1} - 1$. Clearly, $\hat{\beta}_1$ is a weighted average of the $\Delta \ln Y_s$ terms since $\sum_{s=2}^{n} k_s = 1$.

Let us now consider a slight modification of Equation 4.A.6.1. Since $\Delta \ln Y_s = \ln Y_s - \ln Y_{s-1} \approx R_s = (Y_s - Y_{s-1})/Y_{s-1}$ (R_s is the relative growth rate between periods s and $s-1$), it follows from Equation 4.A.6.1 that

$$\hat{\beta}_1 = \sum_{s=2}^{n} \left[\frac{6(n-s+1)(s-1)}{n(n+1)(n-1)} \right] R_s. \tag{4.A.6.2}$$

Thus, the OLS estimate $\hat{\beta}_1$ (the growth rate for the entire sample period) is approximately equal to a weighted average of the annual relative growth rates R_s, where the weights attached to the annual growth rates form a quadratic expression in s, i.e., k_s in Equation 4.A.10 first increases with s until it reaches a maximum when $s = (n/2) + 1$ and then decreases symmetrically until $s = n$. Hence the overall OLS growth rate gives maximum weight to the relative growth rates R_s in the middle of the sample period, with lower weights attached to the R_s's appearing in the beginning and at the end of the entire growth period.[3]

[3] From the numerator of k_s in (4.A.10) let us form $f(s) = (n-s+1)(s-1) = -s^2 + (n+2)s - (n+1)$ (a quadratic in s). Then setting $f' = 0$ and solving for s yields $s^* = s = \frac{n}{2} + 1$ with $f''(s^*) < 0$ as required for a maximum of f. Then from (4.A.10), the weights rise in value, reach a maximum at s^* and then fall as $s = 2, 3, \ldots, n$:

$$\begin{aligned}
&s = 2, \; k_2 = \frac{6}{n(n+1)}, \\
&s = 3, \; k_3 = \frac{12(n-2)}{n(n+1)(n-1)}, \\
&\quad \vdots \\
&s = s^*, \; k_{s^*} = \frac{3n}{2(n+1)(n-1)}, \\
&\quad \vdots \\
&s = n-1, \; k_{n-1} = \frac{12(n-2)}{n(n+1)(n-1)}, \\
&s = n, \; k_n = \frac{6}{n(n+1)}.
\end{aligned}$$

4.A.2 The Log-Difference (LD) Growth Rate

Consider the LD regression model

$$\Delta \ln Y_t = \beta_1 + \varepsilon_t, \quad t = 1,\dots,n. \tag{4.A.11}$$

To estimate β_1 via OLS, let us minimize

$$g\left(\hat{\beta}_1\right) = \sum_{t=2}^{n} \varepsilon_t^2 = \sum_{t=2}^{n} \left(\Delta \ln Y_t - \hat{\beta}_1\right)^2,$$

where $\hat{\beta}_1$ is an arbitrary estimator for β_1. Setting $dg/d\hat{\beta}_1 = 0$ yields $\sum_{t=2}^{n}\left(\Delta \ln Y_t - \hat{\beta}_1\right) = 0$ or

$$\hat{\beta}_1 = \frac{1}{n-1}\sum_{t=2}^{n} \Delta \ln Y_t, \tag{4.A.12}$$

the log-difference regression slope.

How can we interpret this expression? From Equation 4.A.11, $\ln Y_t = \ln Y_{t-1} + \beta_1 + \varepsilon_t$ so that, for $t = 2,\dots,n$, we obtain the recursive system:

$$\begin{aligned}
\ln Y_2 &= \ln Y_1 + \beta_1 + \varepsilon_2, \\
\ln Y_3 &= \ln Y_2 + \beta_1 + \varepsilon_3 \\
&= \ln Y_1 + 2\beta_1 + \varepsilon_2 + \varepsilon_3, \\
\ln Y_4 &= \ln Y_3 + \beta_1 + \varepsilon_4 \\
&= \ln Y_1 + 3\beta_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4, \\
\ln Y_t &= \ln Y_1 + \beta_1 (t-1) + \sum\nolimits_{s=2}^{t} \varepsilon_s \\
&= \beta_0 + \beta_1 t + \sum\nolimits_{s=2}^{t} \varepsilon_s,
\end{aligned} \tag{4.A.13}$$

where $\beta_0 = \ln Y_1 - \beta_1$. Here Equation 4.A.13 mirrors Equation 4.A.1 except for the moving-average random error term in the former equation. Hence $\hat{\beta}_1$ in Equation 4.A.12 can also be interpreted as the total growth rate for the entire sample period. Clearly this growth rate is the simple average of the $n-1$ log-difference terms $\Delta \ln Y_t$. (We may view Equation 4.A.12 as the weighted average of the $\Delta \ln Y_t$ terms, where the weights $1/(n-1)$ are all the same.) Given this $\hat{\beta}_1$ value, the (estimated) compound rate of growth for the entire period is thus $r_{LD} = e^{\hat{\beta}_1} - 1$.

4.A.3 The Average Annual Growth Rate

It is readily seen that, by virtue of the argument underlying Equation 4.A.6.2, we can express Equation 4.A.12 as

$$r_{AAG} = \frac{1}{n-1}\sum_{t=2}^{n} R_t, \tag{4.A.12.1}$$

where $R_t (\approx \Delta \ln Y_t)$ is the relative growth rate between periods t and $t-1$. Hence Equation 4.A.12.1 represents the simple average of the annual growth rates and thus corresponds to the *annualized rate of growth* over the sample period. Since the same weight is given to the growth rate in each year, there occurs a perfect symmetry between positive and negative growth rates—if a negative growth rate occurs in any given year, it is completely offset by a positive growth rate of equal magnitude in another year.

4.A.4 The Geometric Average Growth Rate

Given the compound growth equation $Y_t = Y_1(1+r)^{t-1}$, let us rewrite this expression as

$$\begin{aligned}(1+r)^{t-1} &= \frac{Y_t}{Y_1} = \frac{Y_2}{Y_1}\frac{Y_3}{Y_2}\frac{Y_4}{Y_3}\cdots\frac{Y_{t-1}}{Y_{t-2}}\frac{Y_t}{Y_{t-1}} \\ &= \prod_{t=2}^{n}\left(\frac{Y_t}{Y_{t-1}}\right).\end{aligned} \tag{4.A.14}$$

Then

$$(1+r)^{t-1} = \prod_{t=2}^{n}\left(\frac{Y_t - Y_{t-1} + Y_{t-1}}{Y_{t-1}}\right) = \prod_{t=2}^{n}\left(R_t + 1\right)$$

or

$$1+r = \left[\prod_{t=2}^{n}\left(R_t+1\right)\right]^{1/(t-1)}, \tag{4.A.15}$$

where R_t is the relative growth rate between periods t and $t-1$. Hence $1+r$ equals the geometric average of the $R_t + 1$ terms. In this regard, let us represent

$$r_{\text{GM}} = \left[\prod_{t=2}^{n}\left(R_t+1\right)\right]^{\frac{1}{t-1}} - 1 \tag{4.A.16}$$

as the *geometric mean rate of growth* over the entire period. So with annual compounding, r_{GM} equals the rate at which Y_1 grows so that it equals Y_t in $t-1$ years. It should be obvious that under this growth scheme, r_{GM} ignores the movements in the values of Y between the beginning and end of the total growth period.

We noted above (Eq. 4.A.12) that

$$r_{\text{LD}} = e^{\hat{\beta}_1} - 1 = e^{\frac{1}{n-1}\sum_{t=2}^{n}\Delta\ln Y_t} - 1. \tag{4.A.17}$$

Since

$$\begin{aligned}\frac{1}{n-1}\sum_{t=2}^{n}\Delta\ln Y_t &= \frac{1}{n-1}\sum_{t=2}^{n}\left(\ln Y_t - \ln Y_{t-1}\right) \\ &= \frac{1}{n-1}\sum_{t=2}^{n}\ln\left(\frac{Y_t}{Y_{t-1}}\right) = \ln\left[\prod_{t=2}^{n}\left(\frac{Y_t}{Y_{t-1}}\right)\right]^{1/(n-1)},\end{aligned}$$

it follows from the discussion underlying the derivation of Equation 4.A.14 that

$$\begin{aligned} r_{\text{LD}} &= e^{\ln\left[\prod_{t=2}^{n}(Y_t/Y_{t-1})\right]^{1/(n-1)}} - 1 \\ &= \left[\prod_{t=2}^{n}\left(\frac{Y_t}{Y_{t-1}}\right)\right]^{1/(n-1)} - 1 = \left(\frac{Y_n}{Y_1}\right)^{1/(n-1)} - 1. \end{aligned}$$

Given that $t = 1, \ldots, n$, we are compounding over $n - 1$ periods, where Y_1 is the value of Y at the end of period 1.

To summarize, the four strategies for finding the average growth rate over some sample time period indexed by $t = 1, \ldots, n$ are:

1. OLS regression:

$$r_{\text{OLS}} = e^{\hat{\beta}_1} - 1, \quad \hat{\beta}_1 = \sum_{s=2}^{n} \frac{6(n-s+1)(s-1)}{n(n+1)(n-1)} \Delta \ln Y_t.$$

2. Log-difference regression:

$$r_{\text{LD}} = e^{\hat{\beta}_1} - 1, \quad \hat{\beta}_1 = \frac{1}{n-1}\sum_{s=2}^{n} \Delta \ln Y_t.$$

3. Simple average of the annual growth rates:

$$r_{\text{AAG}} = \frac{1}{n-1}\sum_{t=2}^{n} R_t,$$

where $R_t = (Y_t / Y_{t-1}) - 1$.

4. Geometric average growth:

$$r_{\text{GM}} = \left(\frac{Y_t}{Y_1}\right)^{1/(t-1)} - 1,$$

where, from our previous discussion, $r_{\text{GM}} = r_{\text{LD}} \approx r_{\text{AAG}}$.

Under what circumstances should these various growth rates be employed? If the $\ln Y_t$ series is $I(0)$ and thus does not contain a unit root (it is stationary about a deterministic trend), then the OLS approach to determining the average rate of growth of the Y_t series is appropriate. But if $\ln Y_t$ is $I(1)$, then the log-difference (LD) approach is preferred. (Note that since $r_{\text{LD}} = r_{\text{GM}} \approx r_{\text{AAG}}$, any of these latter three approaches can be utilized.)

5

DYNAMIC SITE EQUATIONS OBTAINED FROM GROWTH MODELS

5.1 INTRODUCTION

This chapter focuses primarily on the derivation of a variety of dynamic site equations, which constitute a critical component of forest growth and yield models, from the growth model-based equations introduced in Chapter 3.[1] In particular, we shall examine selected categories of site-quality models based on stand-height data for evaluating site productivity.

Three broad categories of site-quality models will be recognized:

1. Base-age-specific (BAS) models
2. Algebraic difference approach (ADA) models
3. Generalized algebraic difference approach (GADA) models

5.2 BASE-AGE-SPECIFIC (BAS) MODELS

Traditional height-based methods for assessing site quality employ *site-index curves* (i.e., site-index-based height vs. age curves), where each such curve defines an expected height–age relationship referenced by a specified base age. In fact, when

[1] Readers not familiar with some of the terminology used to discuss forest management issues are encouraged to review the key concepts offered in Appendix 5.A.

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

estimating the parameters of a size–age equation, it has typically been assumed that site index S equals the observed height value at the base age t_b or $S = H(t_b)$ and thus serves as a data-based measure of site quality. This assumption clearly results in so-called BAS parameter estimates.

Base-age-specific site models are essentially *static models*—they select a fixed base age and then identify the site curves by site-index class at this age. To see this, let us begin with a model of the form

$$H = f(t;\theta) + \varepsilon, \tag{5.1}$$

where H is average dominant height, t is stand age, θ is a vector of parameters, and ε is a random error term. Now, if some components of θ are expressed in terms of site index S or $\theta_k = \theta_k(S;\beta)$, β a vector of parameters, then Equation 5.1 can be rewritten as

$$H = f(t,S;\beta) + \varepsilon, \tag{5.1.1}$$

where S is included in the model as a *site-specific constant or regressor* and not a parameter. Given Equation 5.1.1, the *site-specific parameter* θ_k, defined in terms of site index S, is estimated on the basis of a single height–age point on a given plot.

For instance, suppose we seek to estimate the parameters of the equation

$$\ln H = \theta_{i1} + \theta_2 \left(\frac{1}{t}\right)^{\theta_3}, \qquad i = 1,\ldots,m, \tag{5.2}$$

where m is the number of sites. Here θ_{i1} is treated as a site-specific or local parameter, while θ_2 and θ_3 are global parameters common to each site. Let $\theta_{i1} = \theta_{i1}(S_i) = S_i$ for all i. Then the role of the S values in Equation 5.1.1 parallels that of the S_i's in Equation 5.2—they serve as site-specific terms in the model.

Under the Equation 5.1.1 specification, the parameter estimates are unique to the prespecified base age since S enters the model prior to estimation. In fact, the resulting site curves are relevant only for the fixed base age selected; the shape of the estimated site equations is not invariant under this (arbitrary) choice since the equation is forced through the preselected height–age point. Since the estimation of a site-specific parameter is based on a single stand-height value for S on a given plot, this observation is subject to both measurement and sampling error. Moreover, the original site-index specification will generally not equal $\hat{S} = \hat{H}(t_b)$ determined from Equation 5.1.1. This consequently affects the shape of the estimated site curve since the resulting parameter estimates are neither unbiased nor independent of the prespecified base age. In this regard, such estimates serve to mask distinct trends in data. For further details on BAS modeling (including the *parameter prediction method* (PPM)), see Clutter et al. (1983), Goelz and Burke (1992), and Huang (1997).

To avoid the problems inherent in the BAS methodology, Bailey and Clutter (1974) propose a modeling technique that admits the simultaneous estimation of site-specific or site-index parameters (the θ_k's) and the common or global model parameters. Their method, called the *algebraic difference approach* (ADA), utilizes *all* height–age observations for a plot when estimating site-specific parameters and is devoid of any preliminary selection of an arbitrary base age. With ADA, the parameter estimates are unbiased and stable regardless of base age. In fact, Bailey and Clutter (1974) introduce the concept of *base-age-invariant* (BAI) site curves, which means essentially that height predictions are unaffected by any arbitrary choices of base ages. The significance of the BAI concept is explored in the next section.

5.3 ALGEBRAIC DIFFERENCE APPROACH (ADA) MODELS

We closed the preceding section with the concept of BAI *site equations*: site curves are BAI if they are completely unaffected by all choices of base ages; that is, for a BAI site curve, any arbitrary height–age pair on the curve must define the exact same curve. Base-age-invariant equations also display the *path invariance property* (PIP): single-step or multistep height predictions must always result in the same values at a given final age.

Static equations such as Equation 5.1, which employ a single base age to define site variables, neither are BAI nor have the PIP. They thus lead to inconsistent site-index or height predictions. To avoid these problems, site models based upon dynamic site equations have been developed. Specifically, a *dynamic site equation* has the form

$$Y_t = f(t, t_0, Y_0), \tag{5.3}$$

where Y depicts height, t is time, and the combination (t_0, Y_0) represents a set of initial conditions. Since t_0 and Y_0 can assume any values, Equation 5.3 is truly a dynamic equation. And since Equation 5.3 contains a known value of the function at a known reference point (i.e., (t_0, Y_0)), it is termed *self-referencing*.

In terms of Equation 5.3, the BAI property refers to the notion that any number of points (t_0, Y_0) on a given site curve can be used to make predictions for a given age t and the predicted height Y will always be the same. Additionally, curves with the PIP inform us that predicting first from t_0 to t_1 and then from t_1 to t_2 renders the same height as that of the single-step prediction from t_0 to t_2.

In addition to formalizing the BAI concept, Bailey and Clutter (1974) present a technique for dynamic site curve derivation called the ADA for generating BAI site equations. The algebraic difference approach consists of two steps:

1. Identify a site-specific parameter in the base equation responsible for curve shape/behavior.
2. Replace this site-specific parameter in the base equation with its initial condition solution (t_0, Y_0). This enables us to solve for Y in terms of t, t_0, and Y_0.

Clearly this process eliminates the need to quantify site quality with site data prior to parameter estimation. More formally, let the base equation be written as

$$Y_t = f(t;\theta), \tag{5.4}$$

where θ is an n-vector of parameters $\theta_1, \ldots, \theta_n$. If, say, θ_n is deemed site specific and expressible as

$$\theta_n = g(t, Y, \theta_1, \ldots, \theta_{n-1}), \tag{5.5}$$

then, for (t_0, Y_0) a set of initial conditions, let

$$\theta_n^0 = g(t_0, Y_0, \theta_1, \ldots, \theta_{n-1}). \tag{5.5.1}$$

A substitution of θ_n^0 into the base equation (Eq. 5.4) yields

$$Y_t = h(t, t_0, Y_0) = f(t; \theta_1, \ldots, \theta_{n-1}, \theta_n^0). \tag{5.6}$$

Here Equation 5.6 represents a site-indexing equation that produces a site-index curve that is invariant under all choices of base age and thus eliminates the need for any preliminary specification of an arbitrary base age.

Previously, under BAS estimation, data was combined by plots; the basic observation was a plot's average height–age value, which was assumed to represent height growth on a single homogeneous site. Now, the BAI/ADA approach requires only that height–age data be grouped into unique site classes, with all of the observations on a plot used to determine site-specific parameters.

EXAMPLE 5.1 Consider the base equation

$$Y_t = a_i e^{bt^{-c}}, \quad i = 1, \ldots, m, \tag{5.7}$$

where the a_i's are site-specific parameters, b and c are global parameters, and there are m site plots. Let us use ADA to generate a dynamic site equation from Equation 5.7. To this end,

$$\begin{aligned} \ln Y_t &= \ln a_i + bt^{-c} \\ &= \alpha_i + bt^{-c} \end{aligned} \tag{5.7.1}$$

or

$$\alpha_i = \ln Y_t - bt^{-c}. \quad \text{[step 1 of ADA]} \tag{5.7.2}$$

At (t_0, Y_0), Equation 5.7.2 becomes

$$\alpha_i^0 = \ln Y_0 - bt_0^{-c}. \tag{5.8}$$

Then substituting Equation 5.8 into the base equation (Eq. 5.7.1) yields

$$\ln Y_t = \ln Y_0 + b(t^{-c} - t_0^{-c})$$

or

$$\begin{aligned} Y_t &= Y_0 e^{b(t^{-c} - t_0^{-c})} \\ &= h(t, t_0, Y_0). \quad \text{[step 2 of ADA]} \end{aligned} \tag{5.9}$$

(Note that when $t = t_0$, $Y_t = Y_0$.)

From Equation 5.7.1, the instantaneous rate of growth in Y at time t is

$$\frac{dY_t}{dt}\frac{1}{Y_t} = -cbt^{-c-1}. \tag{5.10}$$

It is apparent that this expression is independent of α_i for all i and that this relative rate of growth is constant across all sites at a given time. Hence, the curves obtained via system (Eq. 5.7.1) are vertically proportional and have the same shape; that is, site models based on Equation 5.7.1 would comprise a family of *anamorphic site curves* with variable asymptotes. ■

EXAMPLE 5.2 Let us rewrite Equation 5.7 as

$$Y_t = ae^{b_i t^{-c}}, \quad i = 1, \ldots, m, \tag{5.11}$$

where now the b_i's are the site-specific parameters. Again ADA will be utilized to generate a dynamic site equation. Given

$$\ln Y_t = \ln a + b_i t^{-c}, \tag{5.11.1}$$

we can readily obtain

$$b_i = (\ln Y_t - \ln a)t^c. \quad \text{[step 1 of ADA]} \tag{5.12}$$

At (t_0, Y_0), Equation 5.12 becomes

$$b_i^0 = (\ln Y_0 - \ln a)t_0^c. \tag{5.13}$$

A substitution of Equation 5.13 into the base equation (Eq. 5.11.1) renders

$$\ln Y_t = \ln a + (\ln Y_0 - \ln a)\left(\frac{t_0}{t}\right)^c$$

or

$$\begin{aligned} Y_t &= a\left(\frac{Y_0}{a}\right)^{(t_0/t)^c} \\ &= h(t,t_0,Y_0). \quad \text{[step 2 of ADA]} \end{aligned} \tag{5.14}$$

Given Equation 5.11.1, the instantaneous rate of growth in Y at time t is

$$\frac{dY_t}{dt}\frac{1}{Y_t} = -cb_i t^{-c-1}. \tag{5.15}$$

As this expression reveals, the relative rate of growth in Y is a function of both time and site since b_i is an argument on the right-hand side of Equation 5.15. Hence, the curves obtained from system (Eq. 5.11.1) are vertically nonproportional and thus exhibit different shapes; that is, site models established using Equation 5.11.1 constitute a family of *polymorphic site curves* with a single asymptote. ■

As these two examples have illustrated, BAI dynamic site equations derived via the ADA technique are either anamorphic with multiple asymptotes or simple polymorphic with a single asymptote, depending on whether the parameter selected to be site specific is an asymptote parameter or not. This is because the resulting dynamic equations were based on the response of a single parameter to site variation. In the next section we shall examine a new approach to dynamic site equation derivation, which is based on a multiparameter response to site variation. This is the GADA offered by Cieszewski and Bailey (2000).

5.4 GENERALIZED ALGEBRAIC DIFFERENCE APPROACH (GADA) MODELS

We noted earlier that site curves developed using ADA were BAI. Moreover, the application of ADA enabled us to derive site equations that were either anamorphic with variable asymptotes (such curves display the same shape for all sites, which, when scaled up or down, results in different asymptotes for different sites) or simple polymorphic with a single asymptote (these curves exhibit different shapes to the growth patterns for different sites), given that only a single site-specific parameter was expressible in terms of the initial conditions (t_0,Y_0).

If more than one parameter of a base equation is deemed site specific, can we develop site curves that exhibit polymorphism and variable asymptotes that change with site productivity? As we shall now see, the answer is in the affirmative. In fact, such curves will be termed *advanced polymorphic site curves*. The procedure that achieves this is the GADA of Cieszewski and Bailey (2000).

The advantage of GADA over ADA is that it generates dynamic site equations that are BAI and have the PIP, allows more than one parameter to be site specific, has the ability to produce dynamic site curves that are capable of exhibiting "concurrent" polymorphism and variable asymptotes, and exploits a variety of growth characteristics that operate across different sites. And like the dynamic site curves derived via ADA, GADA models predict height equal to site index at the reference age and have appropriate biological meaning.

In order to implement the GADA technique, Cieszewski and Bailey (2000) define a theoretical variable—called a *growth intensity factor* and denoted χ (chi)—that quantifies the dynamics uniquely associated with the growth characteristics (and thus site curve shape) of individual sites; it encompasses features such as climate, soil properties, water availability, different management regimes, and general ecological factors. It can be a variable (or a function of variables with new parameters), it is continuous and monotonic, and since it is an "unobservable" and "unmeasurable" independent variable, it is eventually replaced by the model's initial conditions. The role of χ is to specify how site-specific parameters change across different sites—it is used to eventually take account of site productivity variation.

This said, the steps needed to generate GADA site curves are:

1. Select a base equation.
2. Identify within the base equation any desired number of site-specific parameters. (The base equation is the same as in ADA: $Y_t=f(t;\theta_1,\ldots,\theta_n)$.)
3. Define unambiguously how the site-specific parameters vary across different sites by replacing them in the base equation with explicit functions of χ and new parameters. (Thus, the two-dimensional base equation expands into an explicit three-dimensional site equation $Y=f(t,\chi)$, which now describes cross-sectional and longitudinal changes.) Since χ is unobservable, we proceed to the last step.
4. Solve for the site-specific parameter χ in terms of the initial conditions (t_0,Y_0) so as to obtain the implicit function χ_0 and replace all χ's in the base equation with χ_0. We thus obtain a self-referencing dynamic site equation with implicitly defined initial conditions $Y_t=f(t,t_0,Y_0)$.

To amplify this discussion a bit and to highlight some of the salient features of GADA, let us consider the base equation

$$Y_t = f(t;\theta_1,\ldots,\theta_n). \tag{5.16}$$

Suppose θ_m through θ_n are taken to be site-specific parameters in Equation 5.16. Then, via step 3 earlier, let

$$\theta_i = g^i(\chi;\theta_{i1},\ldots,\theta_{ij_i}),\ i=m,\ldots,n, \tag{5.17}$$

and j_i is the number of parameters needed to fully specify θ_i. Then Equation 5.16 becomes

$$\begin{aligned} Y_t &= f(t,\chi) \\ &= f\left(t;\theta_1,\ldots,\theta_{m-1}, g^m\left(\chi;\theta_{m1},\ldots,\theta_{mj_m}\right),\ldots,g^n\left(\chi;\theta_{n1},\ldots,\theta_{nj_n}\right)\right). \end{aligned} \tag{5.16.1}$$

If Equation 5.16.1 can be solved for χ given the initial conditions (t_0,Y_0), we obtain

$$\chi_0 = h\left(t_0,Y_0,\theta_1,\ldots,\theta_{nj_n}\right) \tag{5.18}$$

and, when χ_0 is substituted into Equation 5.16.1 for χ, we have

$$\begin{aligned} Y_t &= f(t,t_0,Y_0) \\ &= f(t,\theta_1,\ldots,\theta_{m-1},\chi_0) \\ &= f\left(t,\theta_1,\ldots,\theta_{m-1},h\left(t_0,Y_0,\theta_1,\ldots,\theta_{nj_n}\right)\right). \end{aligned} \tag{5.19}$$

After eliminating any redundant parameters, Equation 5.19 becomes the GADA-formulated dynamic site equation with an implicitly defined initial condition

$$Y_t = f(t,t_0,Y_0) = f(t,t_0,Y_0,\theta_1,\ldots,\theta_k), \tag{5.19.1}$$

which typically has fewer or an equal number of parameters than Equation 5.16.

EXAMPLE 5.3 Given the base equation

$$Y_t = \theta_1 e^{\theta_2 t^{-\theta_3}}$$

or

$$\begin{aligned} \ln Y_t &= \ln\theta_1 + \theta_2 t^{-\theta_3} \\ &= \theta_1' + \theta_2 t^{-\theta_3}, \end{aligned} \tag{5.20}$$

let us derive a variety of dynamic site equations using GADA.

1. Suppose we rewrite Equation 5.20 as

$$\ln Y_t = \chi + \theta_2 t^{-\theta_3}. \tag{5.20.1}$$

Then an application of ADA with respect to χ in Equation 5.20.1 yields the anamorphic BAI site equation (Eq. 5.9). This completes the GADA procedure.

2. Let us rewrite Equation 5.20 as

$$\ln Y_t = \theta_1' + \chi t^{-\theta_3}. \tag{5.20.2}$$

An application of ADA to Equation 5.20.2 with respect to χ renders the BAI polymorphic site equation (Eq. 5.14). Again, this completes the GADA derivation.

3. In Equation 5.20, let $\theta_1' = \alpha + \beta\chi$. Then

$$\ln Y_t = \alpha + \beta\chi + \theta_2 t^{-\theta_3}. \tag{5.21}$$

Applying ADA to Equation 5.21 with respect to χ has us obtain, at (t_0, Y_0),

$$\chi_0 = \beta^{-1}(\ln Y_0 - \alpha - \theta_2 t_0^{-\theta_3}). \tag{5.22}$$

Upon substituting χ_0 into Equation 5.21, we have

$$\begin{aligned}\ln Y_t &= \alpha + \beta\chi_0 + \theta_2 t^{-\theta_3}\\ &= \ln Y_0 + \theta_2 (t^{-\theta_3} - t_0^{-\theta_3})\end{aligned}$$

or

$$Y_t = f(t, t_0, Y_0) = Y_0 e^{\theta_2 (t^{-\theta_3} - t_0^{-\theta_3})}$$

(the BAI anamorphic equation (Eq. 5.9)). Here too the application of GADA is complete.

4. Let us specify θ_2 in Equation 5.20 as $\theta_2 = \beta/\chi$ or

$$\ln Y_t = \theta_1' + \frac{\beta}{\chi} t^{-\theta_3}. \tag{5.23}$$

Applying ADA with respect to χ in Equation 5.32 has us first find, at (t_0, Y_0),

$$\chi_0 = \frac{\beta t_0^{-\theta_3}}{\ln Y_0 - \theta_1^{'}}. \tag{5.24}$$

Substituting χ_0 into Equation 5.23 gives

$$\ln Y_t = \theta_1' + (\ln Y_0 - \theta_1')\left(\frac{t_0}{t}\right)^{\theta_3}$$

or

$$Y_t = f(t, t_0, Y_0) = e^{\theta_1'}\left(\frac{Y_0}{e^{\theta_1'}}\right)^{(t_0/t)^{\theta_3}},$$

the BAI polymorphic equation (Eq. 5.14). The GADA derivation is complete.

5. In Equation 5.20, let us now deem θ_3 site specific so that

$$\ln Y_t = \theta_1' + \theta_2 t^{-\chi}. \tag{5.25}$$

If we apply ADA to Equation 5.25 with respect to χ, then, at (t_0, Y_0), we obtain

$$\chi_0 = \frac{-\ln((Y_0 - \theta_1')/\theta_2)}{\ln t_0}. \tag{5.26}$$

Substituting χ_0 into Equation 5.25 gives us a BAI polymorphic site equation of the form

$$\ln Y_1 = \theta_1' + \theta_2^{\,t\ln\left(\frac{Y_0-\theta_1'}{\theta_2}\right)\Big/\ln t_0}$$

or

$$Y_t = e^{\theta_1'+\theta_2 t^{\ln\left(\frac{Y_0-\theta_1'}{\theta_2}\right)\Big/\ln t_0}}.$$

This completes the GADA derivation. ■

At this point the reader may be asking: "What's the big deal about GADA?" Applying GADA gives us the same dynamic site equations previously obtained by ADA! The true advantage of GADA emerges when more than one parameter in a base equation is site specific. The next example addresses this issue.

EXAMPLE 5.4 Assuming that more than one parameter is needed to portray changes in curve shape across different sites, let us start with the base equation (Eq. 5.20) and posit that θ_1' and θ_2 are simultaneously site-specific parameters. Let us set $\theta_1' = \chi$ and $\theta_2 = \beta\chi$. (Here we are assuming concurrent polymorphism with varying asymptotes.) Hence,

$$\begin{aligned}\ln Y_t &= \chi + \beta\chi t^{-\theta_3} \\ &= \chi(1 + \beta t^{-\theta_3}).\end{aligned} \tag{5.27}$$

Applying GADA to Equation 5.27 with respect to χ has us form, at (t_0, Y_0),

$$\chi_0 = \frac{\ln Y_0}{1+\beta t_0^{-\theta_3}}. \tag{5.28}$$

Substituting Equation 5.28 into Equation 5.27 results in

$$\ln Y_t = \ln Y_0 \left(\frac{1+\beta t^{-\theta_3}}{1+\beta t_0^{-\theta_3}} \right)$$

or

$$Y_t = f(t, t_0, Y_0) = Y_0 e^{[(1+\beta t^{-\theta_3})/(1+\beta t_0^{-\theta_3})]},$$

a polymorphic BAI equation with variable asymptotes. ■

EXAMPLE 5.5 Let us determine a single BAI dynamic site equation that concurrently displays a polymorphism similar to that of Equation 5.14 and an anamorphism similar to Equation 5.9. To this end, we start with the two base equations (Eqs. 5.20.1 and 5.20.2, respectively), or

$$\ln Y_t = \chi + \theta_2 t^{-\theta_3},$$
$$\ln Y_t = \theta_1' + \chi t^{-\theta_3}.$$

If we simply add these base equations by sides, we obtain

$$2 \ln Y_t = \chi + \theta_2 t^{-\theta_3} + \theta_1' + \chi t^{-\theta_3}. \tag{5.29}$$

Solving for χ at (t_0, Y_0) yields

$$\chi_0 = \frac{2 \ln Y_0 - (\theta_1' + \theta_2 t_0^{-\theta_3})}{1+t_0^{-\theta_3}}. \tag{5.30}$$

Substituting χ_0 into Equation 5.29 gives us

$$\ln Y_t = \left(\frac{\theta_1'}{2} + \frac{\theta_2}{2} t^{-\theta_3} \right) + \left(\ln Y_0 - \frac{\theta_1'}{2} + \frac{\theta_2}{2} t_0^{-\theta_3} \right) \left(\frac{1+t^{\theta_3}}{1+t_0^{\theta_3}} \right) \left(\frac{t_0}{t} \right)^{\theta_3}.$$

■

EXAMPLE 5.6 Consider the *three-parameter Chapman–Richards base equation*

$$Y_t = a_1(1-e^{-a_2 t})^{a_3}. \tag{5.31}$$

Since this expression has been found to work extremely well in practice, we will use it to derive an assortment of dynamic BAI site equations by employing GADA. Keep in mind, however, that when only one parameter is taken to be site specific, ADA and GADA are equivalent techniques.

1. a_1 is site specific. Then

$$Y_t = \chi(1-e^{-a_2 t})^{a_3} \tag{5.32}$$

and, at (t_0, Y_0),

$$\chi_0 = Y_0(1-e^{-a_2 t_0})^{-a_3}.$$

A substitution of this expression into Equation 5.32 yields

$$Y_t = f(t, t_0, Y_0) = Y_0\left(\frac{1-e^{-a_2 t}}{1-e^{-a_2 t_0}}\right)^{a_3}. \tag{5.33}$$

2. a_2 is site specific. Hence,

$$Y_t = a_1(1-e^{-\chi t})^{a_3}. \tag{5.34}$$

At (t_0, Y_0),

$$\chi_0 = -\ln\left(1-(Y_0/a_1)^{1/a_3}\right)/t_0.$$

Inserting χ_0 into Equation 5.34 gives

$$Y_t = f(t, t_0, Y_0) = a_1\left(1-\left(1-\left(\frac{Y_0}{a_1}\right)^{1/a_3}\right)^{t/t_0}\right)^{a_3}. \tag{5.35}$$

3. a_3 is site specific. Now

$$Y_t = a_1(1-e^{-a_2 t})^{\chi} \tag{5.36}$$

and, at (t_0, Y_0),

$$\chi_0 = \frac{\ln Y_0 - \ln a_1}{\ln(1 - e^{-a_2 t_0})}.$$

Substituting this expression into Equation 5.36 renders

$$\begin{aligned} Y_t &= f(t, t_0, Y_0) \\ &= Y_0^{[\ln(1-e^{-a_2 t})/\ln(1-e^{-a_2 t_0})]} a_1^{1-[\ln(1-e^{-a_2 t})/\ln(1-e^{-a_2 t_0})]}. \end{aligned} \tag{5.37}$$

4. At times a fourth parameter a_4 is introduced into Equation 5.40; that is, the *four-parameter Chapman–Richards base equation* is

$$Y_t = a_1(1 - a_4 e^{-a_2 t})^{a_3}. \tag{5.31.1}$$

With a_4 site specific, Equation 5.29.1 appears as

$$Y_t = a_1(1 - \chi e^{-a_2 t})^{a_3} \tag{5.38}$$

and, at (t_0, Y_0),

$$\chi_0 = (1 - (Y_0 / a_1)^{1/a_3}) e^{a_2 t_0}.$$

Upon substituting χ_0 into Equation 5.38, we obtain

$$Y_t = f(t, t_0, Y_0) = a_1(1 - (1 - (Y_0 / a_1)^{1/a_3}) e^{-a_2(t - t_0)})^{a_3}. \tag{5.39}$$

In the four preceding derivations involving a single site-specific parameter, GADA was accomplished by simply applying ADA. Let us now consider the cases where two parameters in Equation 5.31 are taken to be site specific. In this instance, we are able to drive a model that exhibits polymorphism and variable asymptotes.

5. If a_2 is taken to be site specific in Equation 5.31, then neither a_1 nor a_3 can also be site specific along with a_2 (since, in this instance, we cannot get a closed-form solution for χ (Cieszewski, 2002)). Hence, we shall take a_1 and a_3 as site specific and thus dependent on χ. In what follows we shall reparameterize the base equation (Eq. 5.31) to a form that is convenient for algebraic manipulation. To this end, let $e^{a_1'}$ replace a_1 so that

$$Y_t = e^{a_1'}(1 - e^{-a_2 t})^{a_3}$$

or

$$\ln Y_t = a_1' + a_3 \ln(1 - e^{-a_2 t}). \tag{5.40}$$

Next, let us set

$$a_1' = \chi, a_2 = b_1, \text{ and } a_3 = b_2 + (b_3 / \chi)$$

which defines the GADA formulation

$$\ln Y_t = \chi + \big(b_2 + (b_3 / \chi)\big) \ln(1 - e^{-b_1 t}). \tag{5.41}$$

Then, from Equation 4.39,

$$\chi^2 + b_2(\ln(1 - e^{-b_1 t}) - \ln Y)\chi + b_3 \ln(1 - e^{-b_1 t}) = 0$$

is quadratic in χ. An application of the quadratic formula at (t_0, Y_0) gives us

$$\chi_0 = \frac{1}{2}\Big[\ln Y_0 - b_2 l_0 \pm ((\ln Y_0 - b_2 l_0)^2 - 4 b_3 l_0)^{1/2}\Big],$$

where $l_0 = \ln(1 - e^{-b_1 t_0})$. Selecting the root most likely to be real and positive (i.e., the one involving addition of the square root term) and substituting the associated χ_0 value into Equation 5.41 results in the BAI dynamic site equation

$$Y_t = f(t, t_0, Y_0) = Y_0 \left(\frac{1 - e^{-b_1 t}}{1 - e^{-b_1 t_0}} \right)^{(b_2 + b_3/\chi_0)}. \tag{5.42}$$

6. In Equation 4.29.1, let both a_2 and a_4 be site specific and set $a_1 = b_1$, $a_2 = \ln\chi$, $a_3 = b_3$, and $a_4 = b_2\chi$. Under this reparameterization, Equation 5.31.1 becomes

$$Y_t = b_1(1 - b_2 e^{-\ln \chi t})^{b_3}. \tag{5.43}$$

Solving Equation 4.41 for χ and evaluating this growth intensity factor at (t_0, Y_0) gives

$$\chi_0 = \left[\frac{1}{b_2} \left(1 - \left(\frac{Y_0}{b_1} \right)^{1/b_3} \right) \right]^{1/(1-t_0)}.$$

Substituting χ_0 into Equation 5.43 gives

$$Y_t = b_1\left(1 - b_2^{t/(1-t_0)}\left[1 - \left(\frac{Y_0}{b_1}\right)^{1/b_3}\right]^{(1-t)/(1-t_0)}\right)^{b_3}. \tag{5.44}$$

■

EXAMPLE 5.7 Consider the log-logistic base equation

$$Y_t = \frac{a_1}{1 + a_2 e^{-u}}, \tag{5.45}$$

where u is some linear combination of the logarithmic transformations of a set of independent variables. For simplicity, let $u = a_3 \ln t$. Then Equation 5.45 becomes the Hossfeld model (Eq. 3.66) or

$$Y_t = \frac{a_1}{1 + a_2 t^{-a_3}},\ a_3 > 1. \tag{5.45.1}$$

Following Cieszewski (2002), let $a_1 = b_1 + \chi$ and $a_2 = b_2/\chi$ with $a_3 = b_3$ or

$$Y_t = \frac{b_1 + \chi}{1 + (b_2/\chi)t^{-b_3}}. \tag{5.46}$$

Solving for χ at (t_0, Y_0) gives

$$\chi_0 = \frac{1}{2}\left[Y_0 - b_1 \pm ((Y_0 - b_1)^2 + 4b_2 Y_0 t_0^{-b_3})^{1/2}\right].$$

Selecting the root that has the greatest likelihood of being real and positive (the one involving $+\sqrt{.}$) and substituting the associated χ_0 term into Equation 5.46 yield

$$Y_t = \frac{b_1 + \chi_0}{1 + (b_2/\chi_0)t^{-b_3}}. \tag{5.47}$$

■

EXAMPLE 5.8 Suppose the parameter a_2 in Equation 5.31 is site specific. Then according to Equation 5.35, the GADA/ADA result is, hypothetically,

$$Y_2 = 53.49\left(1 - \left(1 - \left(\frac{Y_1}{53.49}\right)^{1/1.33}\right)^{t_2/t_1}\right)^{1.33}, \tag{5.48}$$

where Y_2 denotes predicted height at time/age t_2 and Y_1 is predictor height at time/age t_1.

To use Equation 5.48 to estimate average stand height (Y) for some derived age (t), given site index (S) and its associated base time (t_b), we simply substitute S for Y_1 and t_b for t_1 in Equation 5.48 so as to obtain

$$Y = 53.49\left(1-\left(1-\left(\frac{S}{53.49}\right)^{1/1.33}\right)^{t/t_b}\right)^{1.33}. \tag{5.48.1}$$

To estimate site index S at t_b, given stand height Y and age t, substitute S for Y_2 and t_b for t_2 in Equation 5.48. This results in the equation

$$S = 53.49\left(1-\left(1-\left(\frac{Y}{53.49}\right)^{1/1.33}\right)^{t_b/t}\right)^{1.33}. \tag{5.48.2}$$

For instance, let $t_b = 40$ years. Then Equation 5.48.2 becomes

$$S = 53.49\left(1-\left(1-\left(\frac{Y}{53.49}\right)^{1/1.33}\right)^{40/t}\right)^{1.33}, \tag{5.49}$$

and thus, upon rearranging Equation 5.49, we can readily obtain the system of height-growth curves

$$Y = 53.49\left(1-\left(1-\left(\frac{S}{53.49}\right)^{1/1.33}\right)^{t/40}\right)^{1.33}, \quad S = 10, 20, 30, 40, \ldots. \tag{5.50}$$

■

5.5 A SITE EQUATION GENERATING FUNCTION

Let us specify the *site equation generating function* as

$$Y_t = e^m T^b, \tag{5.51}$$

where m and b are estimable parameters of this *generic base function*, while specific base functions emerge from it for different assumed definitions of T (Cieszewski and Strub, 2006) (see Table 5.1).

TABLE 5.1 Different Definitions of T in Equation 5.51 Resulting in Specific Site Equations

Site model	T
Chapman–Richards	$T = 1 - e^{-at}$
Weibull	$T = 1 - e^{-t^{a}}$; $m = 0, b = 1$
Logistic	$T = \dfrac{1}{1 + e^{-at}}$
Schumacher	$T = e^{-1/t}$; $b = 1$
Gompertz	$T = e^{-e^{-at}}$
Log logistic	$T = \dfrac{1}{1 + e^{-a \ln t}}$
Korf	$T = e^{-1/t}$

5.5.1 ADA Derivations

Given the flexibility of Equation 5.51, various dynamic site equations can be derived from it. For instance, if in Equation 5.51 m is taken as site specific, then the ADA-derived anamorphic dynamic site equation that emerges is

$$Y_t = Y_0 \left(\frac{t}{t_0} \right)^b . \tag{5.52}$$

And if b is deemed site specific, then the resulting ADA-derived polymorphic (with a single asymptote) dynamic site equation has the form

$$Y_t = e^m t^{\ln(Y_0/e^m)/\ln t_0} . \tag{5.53}$$

As mentioned earlier in Section 5.3, ADA derivations are based upon the notion that a single parameter is site specific or responsive to variations in site productivity. Hence, ADA derivations usually result in dynamic models that are either anamorphic or polymorphic with a single asymptote. To derive models exhibiting both polymorphism and variable asymptotes, we noted in Section 5.4 that more than one parameter has to be site specific. In terms of Equation 5.51, this means that the site productivity responses in m and b must be modeled simultaneously via GADA.

5.5.2 GADA Derivations

To model mixed site productivity responses in parameters m and b of Equation 5.51, let us assume that variations in site productivity are accounted for by an unobserved growth intensity factor χ and that m and b are expressible in terms of χ. A variety of functional specifications of m and b in terms of χ now follow.

Linearity Suppose both m and b are linear functions of χ:

$$m = m_1 + m_2\chi,$$
$$b = b_1 + b_2\chi. \tag{5.54}$$

Then Equation 5.51 becomes

$$Y_t = e^{m_1 + m_2\chi} T^{b_1 + b_2\chi}$$

or

$$\ln Y_t = m_1 + m_2\chi + (b_1 + b_2\chi)\ln T. \tag{5.55}$$

Then at the self-referencing point or initial condition (T_0,Y_0),

$$\chi_0 = \frac{\ln Y_0 - m_1 - b_1 \ln T_0}{m_2 + b_2 \ln T_0}. \tag{5.56}$$

Since χ is arbitrary, and with an eye towards parsimony (getting rid of redundant parameters), let us consider the following particularizations of Equations 5.55 and 5.56:

1. (b_1=0, b_2=1—polymorphism).

$$Y_t = e^{m_1 + m_2\chi_0} T^{\chi_0}, \tag{5.57}$$

where

$$\chi_0 = \frac{\ln Y_0 - m_1}{m_2 + \ln T_0}. \tag{5.58}$$

2. (m_1=0, m_2=1—the asymptote varies with site productivity).

$$Y_t = e^{\chi_0} T^{b_1 + b_2\chi_0}, \tag{5.59}$$

$$\chi_0 = \frac{\ln Y_0 - b_1 \ln T_0}{1 + b_2 \ln T_0}. \tag{5.60}$$

Inverse Saturation Let us specify m and b as

$$m = m_1 + m_2\chi, \tag{5.61}$$

$$b = b_1 + \frac{b_2}{b_3 + \chi},$$

respectively. Then from Equation 5.51,

$$\ln Y_t = m_1 + m_2\chi + \left(b_1 + \frac{b_2}{b_3 + \chi}\right)\ln T$$

(a polymorphic equation with variable asymptotes) or

$$m_2\chi^2 + \left(-\ln Y_t + m_1 + m_2 b_3 + b_1 \ln T\right)\chi \\ + \left(-b_3 \ln Y_t + m_1 b_3 + b_1 b_3 \ln T + b_2 \ln T\right) = 0 \tag{5.62}$$

(a quadratic expression in χ). Given that there are two solutions to Equation 5.62, the root most likely to be positive at (T_0, Y_0) is

$$\chi_0 = \frac{-v_0 + [v_0^2 - 4m_2(-b_3 \ln Y_t + m_1 b_3 + b_1 b_3 \ln T + b_2 \ln T)]^{1/2}}{2m_2}, \tag{5.63}$$

where $\nu_0 = -\ln Y_0 + m_1 + m_2 b_3 + b_1 \ln T_0$. Here too parsimonious relationships will be defined by the following two cases:

1. ($b_2 = 1, b_3 = 0$—polymorphism, but the model may have a single asymptote if $m_2 = 0$).

$$Y_t = e^{m_1 + m_2\chi_0} T^{b_1 + \frac{1}{\chi_0}}, \tag{5.64}$$

where

$$\chi_0 = \frac{-v_0 + [v_0^2 - 4m_2 \ln T_0]^{1/2}}{2m_2} \tag{5.65}$$

and

$$v_0 = -\ln Y_0 + m_1 + b_1 \ln T_0.$$

2. ($m_1 = 0, m_2 = 1$—model may be anamorphic if $b_2 = 0$).

$$Y_t = e^{\chi_0} T^{b_1 + \frac{b_2}{\chi_0}}, \tag{5.66}$$

where

$$\chi_0 = \frac{-v_0 + [v_0^2 - 4(-b_3 \ln Y_0 + b_1 b_3 \ln T_0 + b_2 \ln T_0)]^{1/2}}{2} \tag{5.67}$$

and

$$v_0 = -\ln Y_0 + b_3 + b_1 \ln T_0.$$

Quadratic Suppose both m and b are quadratic functions of χ:

$$\begin{aligned} m &= m_1 + m_2\chi + m_3\chi^2, \\ b &= b_1 + b_2\chi + b_3\chi^2. \end{aligned} \tag{5.68}$$

Inserting these two expressions into Equation 5.51 gives

$$\ln Y_t = m_1 + m_2\chi + m_3\chi^2 + (b_1 + b_2\chi + b_3\chi^2)\ln T$$

or

$$(m_3 + b_3 \ln T)\chi^2 + (m_2 + b_2 \ln T)\chi + (m_1 + b_1 \ln T - \ln Y_t) = 0. \tag{5.69}$$

At (T_0, Y_0), the root of Equation 5.69 most likely to be positive is

$$\chi_0 = \frac{-v_0 + [v_0^2 - 4w_0(m_1 + b_1 \ln T_0 - \ln Y_0)]^{1/2}}{2w_0}, \tag{5.70}$$

where $\nu_0 = m_2 + b_2 \ln T_0$, $w_0 = m_3 + b_3 \ln T_0$. While a variety of specific cases involving parsimonious dynamic equations are subsumed in Equations 5.68–5.70, two such cases are the following (see Cieszewski and Strub (2006) for others):

1. ($b_1 = b_3 = 0, b_2 = 1$—the model is polymorphic without the quadratic or intercept term in b).

$$Y_t = e^{m_1 + m_2\chi_0 + m_3\chi_0^2} T^{\chi_0}, \tag{5.71}$$

with

$$\chi_0 = \frac{-v_0 + [v_0^2 - 4m_3(m_1 - \ln Y_0)]^{1/2}}{2m_3} \tag{5.72}$$

and

$$v_0 = m_2 + \ln T_0.$$

2. ($m_1 = m_3 = 0$, $m_2 = 1$—the model has variable asymptotes with quadratic site response in b).

$$Y_t = e^{\chi_0} T^{b_1 + b_2\chi_0 + b_3\chi_0^2}, \tag{5.73}$$

with

$$\chi_0 = \frac{-v_0 + [v_0^2 - 4w_0(b_1 \ln T_0 - \ln Y_0)]^{1/2}}{2w_0} \tag{5.74}$$

and

$$v_0 = 1 + b_2 \ln T_0,\; w_0 = b_3 \ln T_0.$$

EXAMPLE 5.9 Suppose from Table 5.1 we select $T = 1 - e^{-at}$ (the Chapman–Richards site model) in the Equation 5.57 and 5.58 specification. Then

$$T_0 = 1 - e^{-at_0},$$

$$\chi_0 = \frac{\ln Y_0 - m_1}{m_2 + \ln(1 - e^{-at_0})},$$

and

$$\begin{aligned} Y_t &= e^{m_1 + \frac{m_2(\ln Y_0 - m_1)}{m_2 + \ln(1 - e^{-at_0})}} (1 - e^{-at})^{\frac{\ln Y_0 - m_1}{m_2 + \ln(1 - e^{-at_0})}} \\ &= e^{m_1}[e^{m_2}(1 - e^{-at})]^{\frac{\ln Y_0 - m_1}{m_2 + \ln(1 - e^{-at_0})}}. \end{aligned}$$

■

5.6 THE GROUNDED GADA (g-GADA) MODEL

The grounded GADA (g-GADA) method can be thought of as a constrained form of GADA; it enables us to develop polymorphic BAI models with multiple asymptotes by designating the asymptote parameter (ultimately taken identically equal to the growth intensity factor χ) as site specific and letting one of the other parameters be expressed as a power function of χ. While the basic GADA approach requires an explicit solution for χ at an arbitrary reference point or initial condition (t_0, Y_0), g-GADA does not impose this requirement (Strand, 1964; Trorey, 1932; Wang et al., 2008). In fact, g-GADA allows χ to be estimated along with the other parameters of the model.

To see how the g-GADA technique works, let us start with ADA modeling. Given the base equation

$$Y_t = f(t; \boldsymbol{\theta}), \tag{5.75}$$

where Y_t is dominant height and θ is a vector of the n parameters $\theta_1, \ldots, \theta_n$, let χ (a component of θ) be local or site specific. Then the family of height curves can be written as

$$Y_t = f(t, \chi, \zeta) = f(t; \boldsymbol{\beta}), \tag{5.76}$$

where $\beta = (\chi, \zeta)$ and ζ is a vector of global or common parameters.

In GADA, more than one parameter is taken to be site specific. A key step in GADA is the expression of the site-specific parameters as functions of χ (χ is an unobserved measure of site quality). Once χ is explicitly solved for, it is expressed in terms of the initial condition (t_0,Y_0) (we thus get χ_0) and then χ_0 replaces χ in the base equation. So for a GADA model, let $\theta_k = \theta_k(\chi) = \theta_k(\chi;\beta_k)$, where k indexes the site-specific parameters and β_k is a parameter vector that specifies the dependency of θ_k on χ. Hence, the general GADA model has the form

$$Y_t = f(t;\theta) = f(t;(\psi(\chi,\beta),\zeta), \tag{5.77}$$

where $\theta = (\psi(\chi,\beta), \zeta)$, ψ is the site-specific subvector of θ depending on χ and the derived common parameter vector β (β, together with χ, replaces the site-specific θ_k parameters), and ζ is a vector of base equation global parameters.

What is the difference between specifications (Eq. 5.76) and (Eq. 5.77)? In Equation 5.76, χ is one of the parameters of the base model and is site specific; in Equation 5.77, χ is an unknown "nuisance" parameter, which might be inestimable.

To obtain the g-GADA form, let us modify Equation 5.77 so that χ is taken to be a particular site-specific parameter (as in ADA model (Eq. 5.76)). Then Equation 5.77 becomes

$$Y_t = f(t;\theta) = f(t;(\chi,\psi(\chi,\beta),\zeta)), \tag{5.78}$$

or the g-GADA model. Hence, χ is no longer a nuisance parameter—it can be estimated along with all other parameters.

In what follows we shall use as the base model the Chapman–Richards function

$$Y_t = a(1-e^{-bt})^c, \tag{5.79}$$

where the asymptote parameter a is taken to be site specific and thus a function of χ. In the g-GADA formulation of Equation 4.77, χ will be taken identical to the parameter a. (Remember that in GADA modeling, we typically set $a = e^{\chi}$. Also, for the GADA specification, one can use $c = c_1 + c_2/\chi$ (see Cieszewski, 2002).)

Since an explicit solution for χ is not required for constructing g-GADA models, one need only specify a flexible power function to characterize the dependency of parameters b and c in Equation 5.79 on the site-specific parameter χ; that is, we set

$$b = b_1\chi^{b_2} \quad \text{and} \quad c = c_1\chi^{c_2}. \tag{5.80}$$

This then renders the following two BAI g-GADA models:

$$Y_t = \chi(1-e^{-bt})^{c_1\chi^{c_2}}; \tag{5.81}$$

$$Y_t = \chi(1-e^{-b_1\chi^{b_2}t})^{c}. \tag{5.82}$$

APPENDIX 5.A GLOSSARY OF SELECTED FORESTRY TERMS

Growth: refers to the increase or increment in tree size over time.

Yield: refers to the amount of material (e.g., timber) available for harvest at a given point in time.

Site quality: refers to the capacity of a site to grow trees (or other vegetation). Site quality is also referred to as *site productivity*. Site quality is typically measured indirectly by selected site characteristics (e.g., soil, slope, precipitation, and elevation).

Site index (a proxy for site quality or site productivity): specific trees called *site trees* are measured to estimate site quality. In this regard, site index is taken to be the average height of dominant and codominant trees (they are in the upper canopy) of a given species in a stand on some site at some index age. In short, site index is equal to height at the index age (usually 25 or 50 years). So if a site has a site index of, say, 70 feet at 50 years, then it is expected that seedlings planted on that site today will be 70 feet high in 50 years. There is a strong correlation between site index and (timber) yield.

Index age (or simply *age*): may be age from seed, age since planting, or age since reaching breast height.

Site-index curves: serve to estimate site index from a sample of site trees. These curves describe a mathematical relationship between height, age, and site quality:

1. *Anamorphic site-index curves*: curves that are identically shaped since they are proportional to each other. Such curves exhibit a constant height–age relationship across the site spectrum.
2. *Polymorphic site-index curves*: these curves usually differ in shape and thus can depict different height–age relationships across different sites.

Site-index models: relate height, age, and site quality in (even-aged) single-species stands. The basic contention in such models is that better sites are assumed to follow height–age trajectories that are higher than those for poorer sites. An infinite number of such trajectories is determined by a one-parameter (θ) family of site curves or

$$H = f(t;\theta), \tag{5.A.1}$$

where H is height and time (t) represents age (Fig. 5.A.1a) (Garcia, 2004). Given a stand age and height, the site quality is assessed via the relative position of the curve that passes through that point. Conversely, for a fixed site-quality level $\overline{\theta}$, the curve $H = f(t;\overline{\theta})$ predicts future stand-height development.

Given Equation 5.A.1, we may view site index S as the curve height at a specific (arbitrary) base age t_b; that is, S is related to θ via substituting S for H and t_b for t in Equation 5.A.1 or

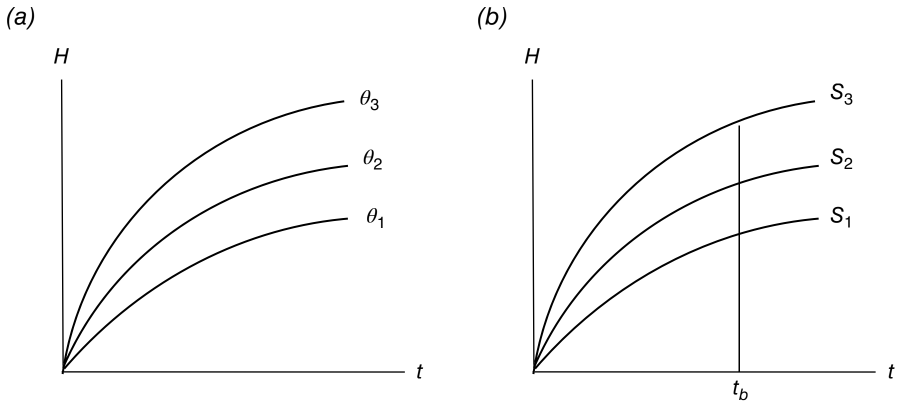

FIGURE 5.A.1 Parametric site curves. (**a**) A family of site curves labeled by a single parameter θ. (**b**) A family of site curves labeled by site index S for fixed t_b.

$$S = f(t_b;\theta) \tag{5.A.2}$$

(Fig. 5.A.1b).

In any particularization of Equation 5.A.1, there may be an additional parameter, say, θ', which is common to all stands. So while the site parameter θ varies from stand to stand or is site specific or *local*, the parameter θ' remains the same or is invariant across stands and is thus termed *global*. In this regard, let a specific form of Equation 5.A.1 appear as

$$H = \theta e^{-\theta'/t}, \tag{5.A.3}$$

where θ is taken to be local and θ' is specified as global. Alternatively, a site index S may be used as a local parameter instead of θ; that is, from Equation 5.A.2, let

$$S = \theta e^{-\theta'/t_b}. \tag{5.A.4}$$

Then solving Equation 5.A.4 for θ renders

$$\theta = Se^{\theta'/t_b},$$

and a substitution of this expression into Equation 5.A.3 yields

$$H = Se^{-\theta'(t^{-1}-t_b^{-1})}$$

(Garcia, 2006).

6

NONLINEAR REGRESSION

6.1 INTRINSIC LINEARITY/NONLINEARITY

In the classical linear regression model, the *multivariate regression hyperplane* has the form

$$Y_i = \beta_0 + \sum_{j=1}^{m} \beta_j X_{ij} + \varepsilon_i, \quad i = 1,\ldots,n, \tag{6.1}$$

where we have m predetermined explanatory variables or *regressors* $X_1, \ldots, X_m$ and ε is a random error term. Here Equation 6.1 depicts a *linear regression model* since it is linear in the parameters $\beta_0, \beta_1, \ldots, \beta_m$. It is assumed that $n > p = m + 1$ and no exact linear relationship exists between the X_j's, $j = 1, \ldots, m$.

For fixed X_j's, $j = 1, \ldots, m$, the *population regression hyperplane* is specified as the conditional mean of Y given the X_j's or

$$E\left(Y \middle| X_1, \ldots, X_m\right) = \beta_0 + \sum_{j=1}^{m} \beta_j X_j \tag{6.2}$$

given that $E(\varepsilon) = 0$. Given Equation 6.2,

$$\beta_0 = E\left(Y \middle| X_1 = 0, \ldots, X_m = 0\right); \tag{6.3}$$

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

that is, the population intercept is the mean of Y given that all of the X_j's are set equal to zero. Also,

$$\beta_j = \frac{\partial E\left(Y \middle| X_1,\ldots,X_m\right)}{\partial X_j}, \quad j = 1,\ldots,m,$$

is termed the jth *partial regression coefficient*; that is, as X_j increases by one unit, the average value of Y changes by β_j units, given that all remaining explanatory variables are held constant. Once the β_k's, $k=0, 1,\ldots,m$, are estimated from the sample information, we obtain the *sample regression hyperplane*

$$\hat{Y} = \hat{\beta}_0 + \sum_{j=1}^{m} \hat{\beta}_j X_j, \tag{6.4}$$

where $\hat{Y}$ is the estimated value of Y and the $\hat{\beta}_k$'s, $k=0, 1,\ldots,m$, represent the estimates of the population parameters. Then $e_i = Y_i - \hat{Y}_i$, $i = 1,\ldots,n$, denotes the ith *residual* or deviation from the sample regression hyperplane.

As mentioned earlier, the functional form of Equation 6.1 is *linear* since it is *linear in the parameters*. Hence, linearity does not necessarily refer to the relationship between Y and the X_j's or between the X_j's themselves, $j=1,\ldots,m$; but it does refer as to how the parameters appear in the regression equation. Since Equation 6.1 is linear in both the parameters and variables, it will be termed *fully* or *completely linear*.

However, to address the issue of any possible nonlinearity in the variables, suppose $W_1,\ldots,W_p$ is a set of p independent regressors; $\phi^1(W_1,\ldots,W_p)=\phi^1(.),\ldots,\phi^m(W_1,\ldots,W_p)=\phi^m(.)$ constitutes a set of m independent functions of the W_k's, $k=1,\ldots,p$; and $g(y)$ is a function of the explained variable Y. The regression equation

$$\begin{aligned} g(y) &= \beta_1\phi^1(.) + \beta_2\phi^2(.) + \cdots + \beta_m\phi^m(.) + \varepsilon \\ &= \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_m X_m + \varepsilon \end{aligned} \tag{6.5}$$

is *linear in the variables* $X_1=\phi^1(.), X_2=\phi^2(.),\ldots,X_m=\phi^m(.)$. In this regard, if the regression equation is linear in the parameters and nonlinear in the regressors but can be converted to a linear form by a suitable transformation or respecification of the regressors (such as in Eq. 6.5), then the regression equation is said to be *intrinsically linear in the variables*. For instance, $Y=\theta_1+\theta_2W^{-1}+\varepsilon$ is intrinsically linear in the variables since it is linear in the parameters, is nonlinear in W, and can be transformed to a linear form by setting $X=\phi(W)=W^{-1}$. Hence, $g(Y)=Y=\theta_1+\theta_2X+\varepsilon$ is now fully linear in the parameters and in the variables X and Y.

The preceding discussion of intrinsic linearity focused on nonlinearities in the variables and their respecification to achieve full or complete linearity. Let us now consider intrinsic linearity as it relates to the transformation or respecification of the parameters themselves.

In this regard, for a regression equation that is linear in the regressors, suppose $\theta_1,\ldots,\theta_r$ is a set of *elemental* or *restricted parameters* (i.e., parameters for which certain nonlinear relations among the regression coefficients must hold, e.g., one regression coefficient is the product or ratio of two other regression coefficients);

$\beta_1, \ldots, \beta_r$ is a set of *unrestricted parameters*; and each unrestricted parameter β_l is expressible as a one-to-one mapping or function of the restricted parameters $\theta_1, \ldots, \theta_r$ or $\beta_l = \gamma^l(\theta_1, \ldots, \theta_r) = \gamma^l(.), l = 1, \ldots, r$. Then the regression model

$$\begin{aligned} g(Y) &= \gamma^1(.)W_1 + \gamma^2(.)W_2 + \cdots + \gamma^r(.)W_r + \varepsilon \\ &= \beta_1 W_1 + \beta_2 W_2 + \cdots + \beta_r W_r + \varepsilon \end{aligned} \tag{6.6}$$

is *intrinsically linear in the restricted parameters* since it is fully linear in the unrestricted parameters.

If a regression equation is intrinsically linear in both the restricted parameters and regressors, then it is unambiguously said to be *intrinsically linear*. For example, $Y = \theta_1 W^{\theta_2} \eta$ is termed log linear since it can be written as linear in the logs of the variables or $\log Y = \log \theta_1 + \theta_2 \log W + \log \eta$. However, this logarithmic expression is not linear in the parameters since $\log \theta_1$ is not linear. If we set $g(y) = \log Y$, $X = \phi(W) = \log W, \beta_1 = \gamma^1(\theta_1) = \log \theta_1, \beta_2 = \gamma^2(\theta_2) = \theta_2$, and $\varepsilon = \log \eta$, then $g(Y) = \beta_1 + \beta_2 X + \varepsilon$ is fully linear in the parameters β_1 and β_2 and in the variables $g(Y)$ and X. Hence, the original expression involving Y and W is intrinsically linear.

Regression models that are nonlinear in the parameters as well as the variables are termed *intrinsically nonlinear*. For this class of functions, there is no transformation of variables that would induce linearity in the parameters. For example:

1. $Y = \theta_1 + (\theta_2 + W)^{-1} + \varepsilon$ is nonlinear in the parameters and with respect to W. No possible relabeling or respecification of the variables can lead to a fully linear regression model. Hence, this model is intrinsically nonlinear.
2. $Y = \theta_1 W_1^{\theta_2} W_2^{\theta_3} + \varepsilon$ is intrinsically nonlinear; no transformation of variables produces a regression equation that is fully linear in the parameters and variables.

6.2 ESTIMATION OF INTRINSICALLY NONLINEAR REGRESSION MODELS

When a regression equation is intrinsically nonlinear, estimates of the restricted parameters can be obtained by using either (i) the nonlinear least squares (NLS) (we minimize $\sum_{i=1}^n e_i^2$, which is a nonlinear function of the restricted parameters) or (ii) the method of maximum likelihood (ML) (we maximize the likelihood function of the sample with respect to the restricted parameters).

If NLS is used, then the estimates of the restricted parameters are the same as those obtained from OLS estimation applied to an unrestricted regression equation under exact identification (if the γ^l's, $l = 0, 1, \ldots, r$, are one-to-one, then the restricted parameters $\theta_1, \ldots, \theta_r$ are exactly identified in terms of the unrestricted parameters, $\beta_1, \ldots, \beta_r$, of the fully linear model (Eq. 6.6)). If the assumption that the ε_i's are independent and identically distributed $N\left(0, \sigma_\varepsilon^2\right)$ random variables is tenable, then ML estimation may be undertaken. In fact, if the maximization of the likelihood function is executed by minimizing $\sum_{i=1}^n e_i^2$, then the resulting estimates of the $\hat{\theta}$'s are the same as those obtained via NLS.

If the normality of the random error terms is not assumed, then the ML method is not applicable, and thus, NLS is the preferred method of estimating the θ's. Interestingly enough, even in the absence of the normality of ε, the asymptotic distribution of an NLS estimator is itself normal and has the same mean and variance as the ML estimator that would have obtained had normality of ε been presumed at the start.

In what follows we shall examine the salient features of both the NLS and the ML routine. And as we shall soon see, for each of these approaches to nonlinear estimation, the derived sets of normal equations tend to be highly nonlinear functions of the parameters and rather difficult to solve; indeed, the normal equations typically do not admit an explicit solution for the parameter estimates (the $\hat{\theta}$'s). Hence, the normal equations can only be solved for an approximation to the optimal set of parameter estimates using an iterative technique that starts from an initial best guess to the levels of those estimates.

6.2.1 Nonlinear Least Squares (NLS)

Let us start with a nonlinear regression model involving a single parameter θ. For regressors $X_1, \ldots, X_m$, let $\boldsymbol{X}_i = (X_{i1}, X_{i2}, \ldots, X_{im})$ and let $Y_i = f(\boldsymbol{X}_i, \theta) + \varepsilon_i$, $i = 1, \ldots, n$, where the ε_i's are independent and identically distributed random variables with a zero mean and constant variance σ_ε^2. The NLS estimator for θ is then the arbitrary estimator $\hat{\theta}$ that minimizes the residual sum of squares described by the *least squares function* $S(\hat{\theta}) = \sum_{i=1}^n e_i^2$; that is, we seek to

$$\min\left\{\sum_{i=1}^n e_i^2 = \sum_{i=1}^n \left(Y_i - f\left(\boldsymbol{X}_i, \hat{\theta}\right)\right)^2 = S(\hat{\theta})\right\}. \tag{6.7}$$

To this end, our first-order condition for a minimum is

$$\frac{dS(\hat{\theta})}{d\hat{\theta}} = -2\sum_{i=1}^n \left(Y_i - f\left(\boldsymbol{X}_i, \hat{\theta}\right)\right)\left(\frac{df(\boldsymbol{X}_i, \hat{\theta})}{d\hat{\theta}}\right) = 0$$

or

$$\sum_{i=1}^n \left(Y_i - f\left(\boldsymbol{X}_i, \hat{\theta}\right)\right)\left(\frac{df\left(\boldsymbol{X}_i, \hat{\theta}\right)}{d\hat{\theta}}\right) = 0. \tag{6.8}$$

So if we can find an estimator $\hat{\theta}$ that satisfies this equation, then $\hat{\theta}$ is termed an *NLS estimator of* θ; $\hat{\theta}$ thus yields the absolute smallest value (the global minimum) of $S(\theta)$. (Keep in mind, however, that multiple optimal solutions to Eq. 6.8—both local and global—can exist.) As mentioned earlier, this equation may be highly nonlinear, and thus, it may only be possible to obtain an approximation to $\hat{\theta}$ via a numerical technique. (See Appendix 6.A for details on the Gauss–Newton iterative method.)

EXAMPLE 6.1 Suppose $Y_i = f(X_i, \theta) + \varepsilon_i = \theta X_{i1} + \theta^{-1} X_{i2} + \varepsilon_i$, $i = 1, \ldots, n$. Then the NLS estimator for θ is the value $\hat{\theta}$, which, from Equation 6.7, minimizes

$$S(\hat{\theta}) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y_i - \hat{\theta} X_{i1} - \hat{\theta}^{-1} X_{i2}\right)^2.$$

Setting

$$\frac{dS(\hat{\theta})}{d\hat{\theta}} = 2\sum_{i=1}^{n} \left(Y_i - \hat{\theta} X_{i1} - \hat{\theta}^{-1} X_{i2}\right)\left(-X_{i1} + \hat{\theta}^{-2} X_{i2}\right) = 0$$

and simplifying yields

$$-\hat{\theta}^{-3} \sum X_{i2}^2 + \hat{\theta}^{-2} \sum X_{i2} Y_i + \hat{\theta} \sum X_{i2}^2 - \sum X_{i1} Y_i = 0.$$

Obviously a numerical routine needs to be employed in order to solve this expression for $\hat{\theta}$. ■

We next consider the case where the nonlinear regression model contains r parameters $\theta_1, \ldots, \theta_r$ that are to be estimated using NLS. Proceeding as in the earlier text, for regressors $X_1, \ldots, X_m$, again let $\boldsymbol{X}_i = (X_{i1}, X_{i2}, \ldots, X_{im})$, $i = 1, \ldots, n$, but now $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_r)$. Our regression equation is thus $Y_i = f(\boldsymbol{X}_i, \boldsymbol{\theta}) + \varepsilon_i$, $i = 1, \ldots, n$, where the ε_i constitute a set of independent and identically distributed random variables with a zero mean and constant variance σ_ε^2.

The least squares function is

$$S(\hat{\theta}) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y_i - f\left(\boldsymbol{X}_i, \hat{\theta}\right)\right)^2.$$

Again applying the principle of least squares, we choose the arbitrary estimators $\hat{\theta} = (\hat{\theta}_1, \ldots, \hat{\theta}_r)$ so as to minimize $\sum_{i=1}^{n} e_i^2$; that is, the NLS estimator for θ is the $\hat{\theta}$ for which

$$\min\left\{\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y_i - f\left(\boldsymbol{X}_i, \hat{\theta}\right)\right)^2 = S(\hat{\theta})\right\}. \tag{6.9}$$

Upon setting

$$\frac{\partial S(\hat{\theta})}{\partial \hat{\theta}_l} = 0, \quad l = 1, \ldots, r$$

(the first-order condition for a minimum), we get the system of *NLS normal equations*

$$\sum_{i=1}^{n} \left(Y_i - f\left(\boldsymbol{X}_i, \hat{\theta}\right)\right)\left(\frac{\partial f\left(\boldsymbol{X}_i, \hat{\theta}\right)}{\partial \hat{\theta}_l}\right) = 0, \quad l = 1, \ldots, r. \tag{6.10}$$

Then the estimator $\hat{\theta}$ that satisfies this equation system is the *NLS estimator for* $\boldsymbol{\theta}$—$S(\hat{\theta})$ is at its minimum (although multiple optimal solutions can occur).[1]

(Appendix 6.B considers the Gauss–Newton method for the *r* parameter case.)

EXAMPLE 6.2 Determine the set of NLS normal equations when

$$Y_i = f(\boldsymbol{X}_i, \theta) + \varepsilon_i = \theta_1 + \theta_2 X_i^{\theta_3} + \varepsilon_i, \quad i = 1,\ldots,n.$$

(Remember that $\frac{d}{dx}(a^x) = a^x \log_e a$.) Here

$$S(\hat{\theta}) = \sum_{i=1}^{n}\left(Y_i - \hat{\theta}_1 - \hat{\theta}_2 X_i^{\hat{\theta}_3}\right)^2$$

[1]In matrix terms, let

$$\underset{(n\times 1)}{\boldsymbol{Y}} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \underset{(n\times m)}{\boldsymbol{X}} = \begin{bmatrix} \boldsymbol{X}_1 \\ \vdots \\ \boldsymbol{X}_n \end{bmatrix}, \underset{(1\times m)}{\boldsymbol{X}_i} = \left(X_{i1}, X_{i2},\ldots, X_{im}\right),$$

$$\underset{(n\times 1)}{\boldsymbol{f}(\boldsymbol{X},\theta)} = \begin{bmatrix} f\left(\boldsymbol{X}_1,\theta\right) \\ \vdots \\ f\left(\boldsymbol{X}_n,\theta\right) \end{bmatrix}$$

with $\boldsymbol{Y} = \boldsymbol{f}(\boldsymbol{X}, \boldsymbol{\theta}) + \boldsymbol{\varepsilon}, \varepsilon' = (\varepsilon_1, \ldots, \varepsilon_n), E(\boldsymbol{\varepsilon}) = \boldsymbol{0}$ ($\boldsymbol{0}$ is the *null vector* containing all zeros), and $E(\varepsilon\varepsilon') = \sigma_\varepsilon^2 \boldsymbol{I}$. Set

$$\underset{(n\times 1)}{\boldsymbol{e}} = [\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \hat{\theta})] = \begin{bmatrix} \boldsymbol{Y}_1 - f\left(\boldsymbol{X}_1,\hat{\theta}\right) \\ \vdots \\ \boldsymbol{Y}_n - f\left(\boldsymbol{X}_n,\hat{\theta}\right) \end{bmatrix}$$

so that $S(\hat{\theta}) = \boldsymbol{e}'\boldsymbol{e} = [\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \hat{\theta})]'[\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \hat{\theta})] = \sum_{i=1}^{n}\left[\boldsymbol{Y} - f\left(\boldsymbol{X}_i,\hat{\theta}\right)\right]^2$. For

$$\underset{(n\times r)}{\frac{\partial \boldsymbol{f}(\boldsymbol{X},\hat{\theta})}{\partial \hat{\theta}}} = \frac{\partial\left(f\left(\boldsymbol{X}_1,\hat{\theta}\right),\ldots,f\left(\boldsymbol{X}_n,\hat{\theta}\right)\right)}{\partial\left(\hat{\theta}_1,\ldots,\hat{\theta}_r\right)}$$

$$= \begin{bmatrix} \partial f\left(\boldsymbol{X}_1,\hat{\theta}\right)/\partial\hat{\theta}_1 & \cdots & \partial f\left(\boldsymbol{X}_1,\hat{\theta}\right)/\partial\hat{\theta}_r \\ \vdots & & \vdots \\ \partial f\left(\boldsymbol{X}_n,\hat{\theta}\right)/\partial\hat{\theta}_1 & \cdots & \partial f\left(\boldsymbol{X}_n,\hat{\theta}\right)/\partial\hat{\theta}_r \end{bmatrix}$$

the $(n \times r)$ *Jacobian matrix* of $\boldsymbol{f}(\boldsymbol{X}, \hat{\theta})$ with respect to $\hat{\theta}$, the *NLS normal equations* are

$$\frac{\partial S(\hat{\theta})}{\partial \hat{\theta}} = \left(\frac{\partial \boldsymbol{f}(\boldsymbol{X}, \hat{\theta})}{\partial \hat{\theta}}\right)' \boldsymbol{e} = \boldsymbol{0}, \tag{6.10.1}$$

where a prime "′" denotes matrix transposition.

so that with

$$\frac{\partial S(\hat{\theta})}{\partial \hat{\theta}_1} = 2\sum_{i=1}^{n}\left(Y_i - \hat{\theta}_1 - \hat{\theta}_2 X_i^{\hat{\theta}_3}\right)(-1) = 0$$
$$\frac{\partial S(\hat{\theta})}{\partial \hat{\theta}_2} = 2\sum_{i=1}^{n}\left(Y_i - \hat{\theta}_1 - \hat{\theta}_2 X_i^{\hat{\theta}_3}\right)\left(-X_i^{\hat{\theta}_3}\right) = 0$$
$$\frac{\partial S(\hat{\theta})}{\partial \hat{\theta}_3} = 2\sum_{i=1}^{n}\left(Y_i - \hat{\theta}_1 - \hat{\theta}_2 X_i^{\hat{\theta}_3}\right)\left(-\hat{\theta}_2 X_i^{\hat{\theta}_3} \log X_i\right) = 0,$$

the system of NLS normal equations is

$$\begin{aligned}
&\sum_{i=1}^{n}\left(Y_i - \hat{\theta}_1 - \hat{\theta}_2 X_i^{\hat{\theta}_3}\right) = 0 \\
&\sum_{i=1}^{n}\left(Y_i X_i^{\hat{\theta}_3} - \hat{\theta}_1 X_i^{\hat{\theta}_3} - \hat{\theta}_2 X_i^{2\hat{\theta}_3}\right) = 0 \\
&\sum_{i=1}^{n}\left(\hat{\theta}_2 Y_i X_i^{\hat{\theta}_3} \log X_i - \hat{\theta}_1 \hat{\theta}_2 X_i^{\hat{\theta}_3} \log X_i - \hat{\theta}_2 X_i^{2\hat{\theta}_3} \log X_i\right) = 0.
\end{aligned} \tag{6.11}$$

In terms of Equation 6.10.1,

$$\frac{\partial \boldsymbol{f}(\boldsymbol{X},\hat{\theta})}{\partial \hat{\theta}} = \begin{bmatrix} \partial f\left(\boldsymbol{X}_1, \hat{\theta}\right)/\partial \hat{\theta}_1 & \partial f\left(\boldsymbol{X}_1, \hat{\theta}\right)/\partial \hat{\theta}_2 & \partial f\left(\boldsymbol{X}_1, \hat{\theta}\right)/\partial \hat{\theta}_3 \\ \vdots & \vdots & \vdots \\ \partial f\left(\boldsymbol{X}_n, \hat{\theta}\right)/\partial \hat{\theta}_1 & \partial f\left(\boldsymbol{X}_n, \hat{\theta}\right)/\partial \hat{\theta}_2 & \partial f\left(\boldsymbol{X}_n, \hat{\theta}\right)/\partial \hat{\theta}_3 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & X_1^{\hat{\theta}_3} & \hat{\theta}_2 X_1^{\hat{\theta}_3} \log X_1 \\ \vdots & \vdots & \vdots \\ 1 & X_n^{\hat{\theta}_3} & \hat{\theta}_2 X_n^{\hat{\theta}_3} \log X_n \end{bmatrix}.$$

Then the NLS normal equations in matrix form are

$$\begin{bmatrix} 1 & \cdots & 1 \\ X_1^{\hat{\theta}_3} & \cdots & X_n^{\hat{\theta}_3} \\ \hat{\theta}_2 X_1^{\hat{\theta}_3} \log X_1 & \cdots & \hat{\theta}_2 X_n^{\hat{\theta}_3} \log X_n \end{bmatrix} \begin{bmatrix} Y_1 - \hat{\theta}_1 - \hat{\theta}_2 X_1^{\hat{\theta}_3} \\ . \\ . \\ . \\ Y_n - \hat{\theta}_1 - \hat{\theta}_2 X_n^{\hat{\theta}_3} \end{bmatrix} = \mathbf{0}$$

or Equation 6.11. ■

6.2.2 Maximum Likelihood (ML)

For the nonlinear regression model $Y_i = f(\boldsymbol{X}_i, \theta) + \varepsilon_i$, $i = 1, \ldots, n$, suppose the ε_i are taken to be independent with $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$, $i = 1, \ldots, n$. Then the joint probability density function, when taken as a function of the arbitrary estimators $\tilde{\theta}$ and $\tilde{\sigma}_\varepsilon^2$ for θ and σ_ε^2, respectively, is termed the *likelihood function* for the sample and written as

$$\begin{aligned}\mathcal{L}(\tilde{\theta}, \tilde{\sigma}_\varepsilon^2, \boldsymbol{Y}; \boldsymbol{X}, n) &= (2\pi\tilde{\sigma}_\varepsilon^2)^{-n/2} e^{-(1/2\tilde{\sigma}_\varepsilon^2)\sum_{i=1}^n (Y_i - f(\boldsymbol{X}_i, \tilde{\theta}))^2} \\ &= (2\pi\tilde{\sigma}_\varepsilon^2)^{-n/2} e^{-(1/2\tilde{\sigma}_\varepsilon^2)S(\tilde{\theta})},\end{aligned} \tag{6.12}$$

where $\boldsymbol{Y}' = (Y_1, \ldots, Y_n)$, $\boldsymbol{X}' = (\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)$, and $S(\tilde{\theta})$ is the least squares function. For convenience, we shall work with the *log-likelihood function*

$$\log \mathcal{L} = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln \tilde{\sigma}_\varepsilon^2 - \frac{1}{2\tilde{\sigma}_\varepsilon^2} S(\tilde{\theta}). \tag{6.12.1}$$

Given Equation 6.12.1, the ML approach has us choose the estimator $\tilde{\theta}$ (the *ML estimator* of θ) for which the probability of obtaining the observed sample is greatest. In this regard, the first-order condition for $\log \mathcal{L}$ to attain a global maximum at $\theta = \tilde{\theta}$ is $\partial \log \mathcal{L}/\partial\theta\,|_{\theta=\tilde{\theta}} = 0$. Since the *likelihood equation* $\partial \log \mathcal{L}/\partial\theta$ will typically be highly nonlinear and not readily admit an explicit solution for $\tilde{\theta}$, an iterative method must be employed to approximate the value of $\tilde{\theta}$ (see Appendix 6.C).

It is important to note that (i) the ML method is a large-sample estimation method and (ii) if in the regression equation $Y_i = f(\boldsymbol{X}_i, \theta) + \varepsilon_i$ it is assumed that $\varepsilon_i \sim N\left(0, \sigma_\varepsilon^2\right), i = 1, \ldots, n$, then the ML estimator $\tilde{\theta}$ is identical to the NLS estimator $\hat{\theta}$ if the former is obtained by minimizing the least squares function $S(\hat{\theta})$.

To address this first point, let us determine, using Equation 6.12.1, the ML estimator of σ_ε^2, $\tilde{\sigma}_\varepsilon^2$, by solving $\partial \log \mathcal{L}/\partial\tilde{\sigma}_\varepsilon^2 = 0$ for $\tilde{\sigma}_\varepsilon^2 = S(\tilde{\theta})/n$. So for large n, $\tilde{\sigma}_\varepsilon^2 \approx SSE/(n-1)$.

Next, substituting $\tilde{\sigma}_\varepsilon^2 = S(\tilde{\theta})/n$ into Equation 6.12.1 results in the *condensed log-likelihood function*

$$\begin{aligned}\mathcal{L}^* &= \log \mathcal{L}(\tilde{\theta}, \tilde{\sigma}_\varepsilon^2; \boldsymbol{Y}, \boldsymbol{X}, n) \\ &= -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\left(\frac{S(\tilde{\theta})}{n}\right) - \frac{n}{2},\end{aligned}$$

which expresses $\log \mathcal{L}$ in terms of the argument $\tilde{\theta}$. Then setting $\partial\mathcal{L}^*/\partial\tilde{\theta} = 0$ implies $\partial S/\partial\tilde{\theta} = 0$, where this latter derivative condition is simply the first-order condition for minimizing $S(\theta)$ under NLS estimation. So in the normal error case, maximizing $\mathcal{L}^*$ is equivalent to minimizing the error sum of squares or least squares function $S(\theta)$.

For the r parameter case, let the regression equation appear as $Y_i = f(\boldsymbol{X}_i, \theta) + \varepsilon_i$, $i = 1, \ldots, n, \boldsymbol{\theta}' = (\theta_1, \ldots, \theta_r)$, or $\boldsymbol{Y} = \boldsymbol{f}(\boldsymbol{X}, \theta) + \varepsilon$, and let us write the least squares function as

$S(\tilde{\boldsymbol{\theta}}) = \boldsymbol{e}'\boldsymbol{e} = (\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \tilde{\boldsymbol{\theta}}))'(\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \tilde{\boldsymbol{\theta}}))$, where $\tilde{\boldsymbol{\theta}}$ is an arbitrary estimator for θ. For $\varepsilon \sim N(\mathbf{0}, \sigma_\varepsilon^2 \boldsymbol{I})$, the *likelihood function* is

$$\begin{aligned}\mathcal{L}(\tilde{\boldsymbol{\theta}}, \tilde{\sigma}_\varepsilon^2; \boldsymbol{Y}, \boldsymbol{X}, n) &= (2\pi\tilde{\sigma}_\varepsilon^2)^{-(n/2)} e^{-\left[(\boldsymbol{Y}-\boldsymbol{f}(\boldsymbol{X},\tilde{\boldsymbol{\theta}}))'(\boldsymbol{Y}-\boldsymbol{f}(\boldsymbol{X},\tilde{\boldsymbol{\theta}}))/2\tilde{\sigma}_\varepsilon^2\right]} \\ &= (2\pi\tilde{\sigma}_\varepsilon^2)^{-(n/2)} e^{-S(\tilde{\boldsymbol{\theta}})/2\tilde{\sigma}_\varepsilon^2}\end{aligned} \tag{6.13}$$

with $\tilde{\sigma}_\varepsilon^2$ serving as an arbitrary estimator of σ_ε^2. Then the log-likelihood function is written as

$$\log \mathcal{L} = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln \tilde{\sigma}_\varepsilon^2 - \frac{1}{2\tilde{\sigma}_\varepsilon^2} S(\tilde{\boldsymbol{\theta}}). \tag{6.13.1}$$

As indicated earlier, the ML method is concerned with finding an estimator of $\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}$, for which $\log \mathcal{L}$ attains a global maximum (we maximize the probability of obtaining the observed sample); and this maximum will occur where $\partial \log \mathcal{L}/\partial \boldsymbol{\theta}\big|_{\theta=\tilde{\theta}} = \mathbf{0}$. Here too the *likelihood equation* $\partial \log \mathcal{L}/\partial \tilde{\boldsymbol{\theta}}$ will be highly nonlinear and thus require an iterative method for approximating the components of $\tilde{\boldsymbol{\theta}}$ (see Appendix 6.C).

EXAMPLE 6.3 Using the data set appearing in Table 6.1, estimate the parameters of the following nonlinear regression model using the SAS procedure *NLIN*:

TABLE 6.1 Observations on the Variables Y_1, X_1, and X_2

Y	X_1	X_2
5	2	3
7	6	5
9	7	7
10	8	11
13	10	13
15	12	14
17	18	18
20	15	20
25	20	25
27	27	27
31	28	34
32	30	36
40	32	41
50	35	42
62	37	45
77	41	48
90	49	49
120	53	55
146	57	56
167	59	60

$$Y = \theta_1(e^{\theta_2 X_1} + e^{\theta_3 X_2}) + \varepsilon,$$

where $\varepsilon \sim N(\mathbf{0}, \sigma_\varepsilon^2 \boldsymbol{I})$. Exhibit 6.1 houses the SAS code.

EXHIBIT 6.1 SAS Code for Nonlinear Estimation

```
data nlinreg;
  input y x1 x2;
datalines;
  5    2     3
  7    6     5
  .    .     .
  .    .     .
  .    .     .
146   57    56
167   59    60
;      ①                ②
proc nlin data=nlinreg hougaard;
parms theta1=0.1 theta2=0.1 ③
  theta3=0.1;
model y = theta1*(exp(theta2*x1)
  + exp(theta3*x2)); ④
run;
```

① We invoke *PROC NLIN*—a procedure for estimating regression equations that are nonlinear in the parameters. The default solution is the Gauss–Newton iterative routine.

② We request the *Hougaard* option to obtain a skewness measure for the estimated parameters. Hougaard's measure of skewness, g_{1l}, $l = 1, \ldots, r$, is used to determine if an estimate is close to linear (i.e., has desirable statistical properties similar to the estimates obtained by fitting a linear regression model) or whether it exhibits marked nonlinearity. In this regard, (i) if $|g_{1l}| < 0.1$, the estimator $\hat{\theta}_l$ is very close to linear in its character; (ii) if $0.1 < |g_{1l}| < 0.25$, the estimator $\hat{\theta}_l$ is fairly close to linear; (iii) if $|g_{1l}| > 0.25$, evidence of moderate skewness to $\hat{\theta}_l$ emerges; and (iv) if $|g_{1l}| > 1$, there is a considerable departure in $\hat{\theta}_l$ from the linear behavior of unbiasedness, minimum variance, etc.

③ The *PARMS* statement is used to specify the names of the parameters and to set their starting values in order to initiate the iterative estimation process.

④ The *MODEL* statement explicitly specifies the full model in terms of the named parameters and variables.

The output generated by this SAS code appears in Table 6.2.

① Starting from the set of initial values specified in the PARMS statement, the process converges after 10 iterations to a stable set of parameter estimates $\hat{\theta}_1 = 2.9669$, $\hat{\theta}_2 = 0.0589$, and $\hat{\theta}_3 = 0.0531$.

TABLE 6.2 Output for Example 6.3

The SAS system
The NLIN procedure
Dependent variable: y
Method: Gauss–Newton

①
Iterative phase

Iter.	theta1	theta2	theta3	Sum of squares
0	0.1000	0.1000	0.1000	37,884.5
1	0.1303	0.0955	0.0955	37,064.6
2	0.1702	0.0911	0.0909	36,035.9
3	0.2733	0.0828	0.0819	35,723.8
4	0.3982	0.0772	0.0754	33,625.6
5	0.6803	0.0692	0.0657	30,666.4
6	1.6750	0.0562	0.0495	24,541.2
7	2.7796	0.0610	0.0552	416.4
8	2.9510	0.0590	0.0532	189.8
9	2.9665	0.0589	0.0531	189.7
10	2.9669	0.0589	0.0531	189.7
11	2.9669	0.0589	0.0531	189.7

Note: Convergence criterion met.

②
Estimation summary

Method	Gauss–Newton
Iterations	11
Subiterations	11
Average subiterations	1
R	1.413E-6
PPC (theta3)	4.914E-7
RPC (theta3)	7.177E-6
Object	4.53E-10
Objective	189.7281
Observations read	20
Observations used	20
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	③ F value	Approx Pr $>F$
Model	3	90,065.3	30,021.8	2690.01	<0.0001
Residual	17	189.7	11.1605		
Uncorrected total	20	90,255.0			

TABLE 6.2 Output for Example 6.3 (*Continued*)

The NLIN procedure

Parameter	④ Estimate	Approx std error	⑤ Approximate confidence	95% limits	⑥ Skewness
theta1	2.9669	0.2770	2.3825	3.5513	0.3535
theta2	0.0589	0.00178	0.0551	0.0627	–0.5518
theta3	0.0531	0.00499	0.0436	0.0625	–0.3624

Approximate correlation matrix

	theta1	theta2	theta3
theta1	1.0000000	–0.0005893	–0.8678772
theta2	–0.0005893	1.0000000	–0.4852048
theta3	–0.8678772	–0.4852048	1.0000000

② $R = 1.413E-6$ is a convergence measure that approaches zero as the value of the objective function approaches its optimum. PPC is a prospective change measure that indicates the maximum relative change in the parameter estimates when the parameter-change vector is computed for the next iteration. Hence, PPC($theta3$) $= 4.914E-7$ indicates that *theta*3 (it has the largest PPC value) would change by this relative amount if *PROC NLIN* were to perform an additional iteration. RPC is a retrospective change measure that indicates the maximum relative change in the parameters from the previous iteration step. Thus, RPC($theta3$) $= 7.177E-6$ reveals that *theta*3 (it displays the largest RPC value) changed by this amount relative to its value at the last iteration. *Object* measures the relative change in the objective function value between successive iterations. Hence, $Object = 4.53E-10$ indicates that the objective function value changed by this relative amount from the last iteration.

③ The large F value with p-value < 0.0001 indicates that the overall fit is quite good.

④, ⑤ The estimates of the parameters $\theta_1, \theta_2,$ and θ_3 are, respectively, $\hat{\theta}_1 = 2.9669$, $\hat{\theta}_2 = 0.0589$, and $\hat{\theta}_3 = 0.0531$. Since their associated 95% confidence limits do not contain zero, we can conclude that each of these parameters is significantly different from zero at the 5% level.

⑥ Looking to the Hougaard measure of skewness, we find that:

a. With $|g_{11}| = 0.35$, $\hat{\theta}_1$ is moderately skewed away from linear behavior.

b. Since $|g_{12}| = 0.55$, $\hat{\theta}_2$ is moderately skewed from linear behavior.

c. With $|g_{13}| = 0.36$, $\hat{\theta}_3$ is moderately skewed from linear behavior. ■

EXAMPLE 6.4 Let us reestimate the regression function appearing in Example 6.3 using the Table 6.1 data set. This time we shall employ a nonlinear regression routine from the SAS ETS program library. The requisite code is found in Exhibit 6.2.

EXHIBIT 6.2 SAS Code for Nonlinear Estimation (ETS)

```
data nlinreg;
  input y x1 x2;
datalines;
  5  2  3
  7  6  5
  9  7  7
 10  8 11
 13 10 13
 15 12 14
 17 18 18
 20 15 20
 25 20 25
 27 27 27
 31 28 34
 32 30 36
 40 32 41
 50 35 42
 62 37 45
 77 41 48
 90 49 49
120 53 55
146 57 56
167 59 60
; ①
proc model data = nlinreg;
  y = theta1*(exp(theta2*x1) + exp(theta3*x2)); ②
③ fit y start = (theta1  0.1  theta2  0.1  theta3  0.1)/fiml ④
      out = a outresid outpredict covb prl=both collin normal; ⑤
      test theta1  - (theta2/theta3) = 1;
      test theta2  -  theta3 = 0;  ⑥
proc print data = a;
run;
```

① The *MODEL* procedure is used to perform nonlinear regression analysis and is a much more general procedure than *NLIN*, which was used in the preceding example problem.

② We write out explicitly the complete regression equation (including all parameters and arithmetic operations) to be estimated save for the random error term.

③ We use a *FIT* statement to run our fully specified regression against the *nlinreg* data set. Starting values for the unknown parameters are also supplied.

④ We request that the full-information maximum likelihood (*FIML*) estimation technique be employed.

⑤ The *FIT* statement options used herein are:

- *out=a* (specifies the output data set *a* to house the residual and predicted values)
- *outresid* (writes the residual values to the output data set *a*)
- *outpredict* (writes the predicted values to the output data set *a*)
- *covb* (We request a print of the variance–covariance matrix of the estimated parameters.)
- *prl=both* (We request both Wald and likelihood ratio (LR) 95% profile likelihood confidence intervals on the estimated parameters. The default is 95% LR confidence intervals.)
- *collin* (We request a set of collinearity diagnostics.)
- *normal* (We request a test(s) of the regression residuals.)

⑥ The *TEST* option enables us to perform tests of nonlinear (as well as linear) hypotheses on the regression parameters estimated by the immediately preceding

FIT statement. Any number of individual *TEST* statements can be specified. The following options can also be used in any *TEST* statement: *WALD* (the Wald test (the default)), *LM* (the Lagrange multiplier test), *LR* (the *likelihood-ratio* test), or *ALL* (all three test are requested).

While not incorporated into the current SAS program, some additional options that can be employed within the *FIT* statement (after the slash) are:

outest = (writes the set of parameter estimates to a named data set)
method = marquardt (more on this later)
Breusch (performs the Breusch–Pagan test for heteroscedasticity)
corr (prints correlation matrices)
White (performs White's test for heteroscedasticity)
printall (provides all printing options)

Given the SAS code listed in Exhibit 6.2, the associated output generated appears in Table 6.3.

① The Gauss–Newton iteration routine is the default solution method.

② See the interpretation of these convergence criteria offered in Example 6.3.

③ A rule of thumb is that a condition index with a value of at least 30 indicates serious multicollinearity. The third condition index with a value of 21 reveals that a nontrivial amount of multicollinearity is present; there is one (modest) near-linear dependency in the data.

④,⑤ The R^2 value of 0.9957 indicates that the regression model has considerable explanatory power, while the t value for each estimated regression coefficient has a p-value < 0.0001. Hence, the parameter estimates $\hat{\theta}_1 = 2.9664$, $\hat{\theta}_2 = 0.0589$ and $\hat{\theta}_3 = 0.0530$ are highly significant.

⑥ We first test $H_0: \theta_1 - (\theta_2/\theta_3) = 1$. Given that the Wald test statistic $W = 12.20$ is highly significant (p-value < 0.0005), we reject H_0 in favor of H_1: not H_0. Next, we test $H_0: \theta_2 - \theta_3 = 0$. Now the Wald statistic $W = 1.48$ is not significant (p-value = 0.2241) so that we cannot reject H_0—θ_2 and θ_3 can be thought of as being (approximately) equal in value.

⑦ Both the Wald and LR 95% confidence intervals verify the results of the earlier t-tests for the estimated regression parameters—zero is not a member of any of these intervals. According to this set of results, each of our parameter estimates is significantly different from zero with 95% reliability. Note that a greater degree of precision is attained with the set of 95% LR confidence intervals relative to the 95% Wald confidence intervals (i.e., the LR intervals are narrower than the Wald intervals).

⑧ Maximum likelihood estimation requires that the sample size be large and the random errors $\varepsilon_i, i = 1, \ldots, n$, be normally distributed with a mean of zero and a

TABLE 6.3 Output for Example 6.4

The SAS system
The MODEL procedure

Model summary

Model variables	1
Parameters	3
Equations	1
Number of statements	5

Model variables	*y*		
Parameters (value)	theta1(0.1)	theta2(0.1)	theta3(0.1)
Equations	*y*		

The equation to estimate is
$y = F$(theta1, theta 2, theta3)

Note: At FIML Iteration 0 CONVERGE = 0.001 Criteria Met.

FIML estimation summary
Data set options
DATA = NLINREG
OUT = A

Minimization summary

Parameters estimated	3
Method	Gauss ①
Hessian	GLS
Covariance estimator	Cross
Iterations	0

②
Final convergence criteria

R	0.000266
PPC (theta3)	0.000132
RPC	.
Object	.
Trace (S)	9.486406
Gradient norm	0.010181
Log likelihood	−50.8774

Observations processed

Read	20
Solved	20

Collinearity diagnostics

			Proportion of variation		
Number	Eigenvalue	③ Condition number	theta1	theta2	theta3
1	2.960948	1.0000	0.0016	0.0049	0.0012
2	0.032560	9.5362	0.1146	0.8829	0.0283
3	0.006493	21.3550	0.8838	0.1121	0.9705

TABLE 6.3 Output for Example 6.4 (*Continued*)

Nonlinear FIML summary of residual errors

Equation	DF model	DF error	SSE	MSE	Root MSE	④ R-square	Adj R-square
y	3	17	189.7	11.1605	3.3407	0.9957	0.9952

⑤
Nonlinear FIML parameter estimates

Parameter	Estimate	Approx std error	t value	Approx Pr >|t|
theta1	2.966486	0.3197	9.28	<0.0001
theta2	0.058907	0.00145	40.59	<0.0001
theta3	0.053065	0.00448	11.85	<0.0001

⑥
Test results

Test	Type	Statistic	Pr > chi-square	Label
Test0	Wald	12.20	<0.0005	theta1 – (theta2/theta3) = 1
Test1	Wald	1.48	0.2241	theta2 – theta 3 = 0

⑦
Parameter Wald
95% confidence intervals

Parameter	Value	Lower	Upper
theta1	2.9665	2.3399	3.5931
theta2	0.0589	0.0561	0.0618
theta3	0.0531	0.0443	0.0618

Parameter LR
95% confidence intervals

Parameter	Value	Lower	Upper
theta1	2.9665	2.4826	3.5842
theta2	0.0589	0.0544	0.0620
theta3	0.0531	0.0430	0.0609

Number of observations		Statistics for system	
Used	20	Log likelihood	–50.8774
Missing	0		

⑧
Normality test

Equation	Test statistic	Value	Prob.
y	S–W W	0.98	0.8580
System	Mardia skewness	0.02	0.8877
	Mardia kurtosis	–0.60	0.5470

(*Continued*)

TABLE 6.3 Output for Example 6.4 (*Continued*)

⑧

Normality test

Equation	Test statistic	Value	Prob.
	Henze–Zirkler T	−1.84	0.0661

Covariance of parameter estimates

	theta1	theta2	theta3
theta1	0.1022124	−0.0001170	−0.0013479
theta2	−0.0001170	0.0000021	−0.0000005
theta3	−0.0013479	−0.0000005	0.0000201

The SAS system

Obs	_Estype_	_Type_	_Weight_	*y*	*x*1	*x*2
1	FIML	Predict	1	6.816	2	3
2	FIML	Residual	1	−1.816	2	3
3	FIML	Predict	1	8.092	6	5
4	FIML	Residual	1	−1.092	6	5
5	FIML	Predict	1	8.781	7	7
6	FIML	Residual	1	0.219	7	7
7	FIML	Predict	1	10.070	8	11
8	FIML	Residual	1	−0.070	8	11
9	FIML	Predict	1	11.260	10	13
10	FIML	Residual	1	1.740	10	13
11	FIML	Predict	1	12.251	12	14
12	FIML	Residual	1	2.749	12	14
13	FIML	Predict	1	16.275	18	18
14	FIML	Residual	1	0.725	18	18
15	FIML	Predict	1	15.751	15	20
16	FIML	Residual	1	4.249	15	20
17	FIML	Predict	1	20.815	20	25
18	FIML	Residual	1	4.185	20	25
19	FIML	Predict	1	26.984	27	27
20	FIML	Residual	1	0.016	27	27
21	FIML	Predict	1	33.459	28	34
22	FIML	Residual	1	−2.459	28	34
23	FIML	Predict	1	37.407	30	36
24	FIML	Residual	1	−5.407	30	36
25	FIML	Predict	1	45.668	32	41
26	FIML	Residual	1	−5.668	32	41
27	FIML	Predict	1	50.869	35	42
28	FIML	Residual	1	−0.869	35	42
29	FIML	Predict	1	58.539	37	45
30	FIML	Residual	1	3.461	37	45
31	FIML	Predict	1	71.084	41	48

TABLE 6.3 Output for Example 6.4 (*Continued*)

Obs	_Estype_	_Type_	_Weight_	y	$x1$	$x2$
32	FIML	Residual	1	5.916	41	48
33	FIML	Predict	1	93.135	49	49
34	FIML	Residual	1	−3.135	49	49
35	FIML	Predict	1	122.244	53	55
36	FIML	Residual	1	−2.244	53	55
37	FIML	Predict	1	143.124	57	56
38	FIML	Residual	1	2.876	57	56
39	FIML	Predict	1	167.474	59	60
40	FIML	Residual	1	−0.474	59	60

constant variance. Hence, it is important to test this normality assumption concerning the ε_i's. Let us focus on the Shapiro–Wilk (S–W) test. Here we test H_0: the random errors are normal versus H_1—not H_0. If the p-value associated with the S–W statistic is less than 0.05, then we reject H_0 in favor of H_1 and conclude that the observations have not been taken from a normal population; and if the p-value exceeds 0.05, we cannot reject H_0. Since for this test the S–W statistic equals 0.98 and the associated p-value = 0.8580, we cannot reject the null hypothesis of normality.

An important feature of PROC MODEL is the ability to incorporate nonlinear (as well as linear) restrictions on the model parameters into the estimation process associated with the current *FIT* statement. This is accomplished with the *RESTRICT* option, which can be written as:

```
restrict ______ = (or ≤, ≥, <, >)______;
```

Any number of individual *RESTRICT* statements can be specified. For instance, we can reestimate the current model by inserting the following *RESTRICT* option under the preceding *FIT* statement (after the slash):

```
restrict theta1* theta2+theta3=0.1;
```

The abbreviated SAS output results appear in Table 6.4.

① The Marquardt–Levenberg method is a modification of the basic linearization (Taylor's expansion) approach; it is a ridgelike procedure that is designed to reduce the residual sum of squares at each iteration step and is used in combination with the Gauss–Newton technique (see Bates and Watts (1988), Marquardt (1963), and Seber and Wild (1989)). For a discussion of the details of the Levenberg–Marquardt modification, see Appendix 6.D.

② The parameter estimate Restrict0 = 5.8028 pertains to the optimal value of the LM associated with the restriction or constraint imposed on the model parameters.

TABLE 6.4 Output for Example 6.4 (RESTRICT Option)

The SAS system
The MODEL procedure

Nonlinear FIML summary of residual errors

Equation	DF model	DF error	SSE	MSE	Root MSE	R-square	Adj R-square
y	2	18	478.8	26.6015	5.1577	0.9891	0.9885

Nonlinear FIML parameter estimates

Parameter	Estimate	Approx std err	t value	Approx Pr >|t|	Label
theta1	7.754618	0.6621	11.71	<0.0001	
theta2	0.051802	0.00162	31.98	<0.0001	
theta3	−0.30171	0.0221	−13.65	<0.0001	
②					
Restrict0	5.802838	2.8473	2.04	0.0374	theta1 × theta2 + theta3 = 0.1

Parameter Wald
95% confidence intervals

Parameter	Value	Lower	Upper
theta1	7.7546	6.4570	9.0523
theta2	0.0518	0.0518	0.0519
theta3	−0.3017	−0.3436	−0.2591

Note: Gauss method failed. Switching to Marquardt method. ①

Parameter LR
95% confidence intervals

Parameter	Value	Lower	Upper
theta1	7.7546	6.4728	9.1669
theta2	0.0518	0.0486	0.0551
theta3	−0.3017	−0.3465	−0.2569

Number of observations		Statistics for system	
Used	20	Log likelihood	−60.1348
Missing	0		

(If the constraint is binding or holds as an equality at the minimum of the error sum of squares, then Restrict0 > 0; and if the constraint is not binding at the said minimum, then Restrict0 = 0, and the constrained minimum of the likelihood function is the same as its unconstrained minimum.) Here the parameter restriction has a highly significant impact on the estimation results since the associated p-value = 0.0374. ■

TABLE 6.5 Data from Kreusler et al. (1879) on the Growth of "Badischer fruh" Maize in 1878

Date of harvesting	Day in the year (t)	Mean total dry weight per plant (g) (w)	Logarithm of mean total dry weight per plant (lw)
11 June	162	0.255	−1.366
18 June	169	0.308	−1.177
25 June	179	0.637	−0.451
2 July	183	2.319	0.841
9 July	190	4.654	1.538
16 July	197	9.019	2.199
23 July	204	20.001	2.996
30 July	211	34.557	3.542
6 August	218	57.587	4.053
13 August	225	70.095	4.249
20 August	232	85.165	4.445
27 August	239	111.649	4.715
3 September	246	124.760	4.826
10 September	253	121.990	4.804

EXAMPLE 6.5 A primary data set that is often used by researchers to describe and interpret the growth of the whole plant is that of Kreusler et al. (1879) and presented in Hunt (1982) (see Table 6.5). A plot of the logarithm of mean crop yield (total dry weight) per plant against day of the year appears in Figure 6.1. Given this data set, our objective is to estimate, via

PROC NLIN, the parameters of four sigmoidal growth functions, namely, the:

Logistic: $w = w_\infty/(1 + e^{\gamma-\beta t})$.
Gompertz: $w = w_\infty e^{-e^{\gamma-\beta t}}$.
Weibull: $w = w_\infty - \alpha e^{-\beta t^\delta}$.
Chapman–Richards: $w = w_\infty(1 - e^{\gamma-\beta t})^m$.

The requisite SAS code is provided in Exhibit 6.3.

EXHIBIT 6.3 SAS Code for Growth Curve Estimation

```
data kreu;
  input t w;
lw=log (w); ①
datalines;
162    0.255
169    0.305
.      .
.      .
.      .
253    121.990
;
```

① lw represents the natural logarithm of w.

② The plot procedure is used to generate Figure 6.1.

③ *Maxiter* = 1000 extends the number of iterations performed to 1000 (the default number of iterations is 100). *Noitprint* suppresses the iteration history.

```
proc plot data=kreu; ②
plot lw*t= '*';
proc nlin data=kreu hougaard maxiter=1000 noitprint; ③
      parms winf=4.5
            gamma =-1.2 ④
            beta=0.02;
      model lw=winf/(1+exp(gamma-beta*t)); ⑤
run;
proc nlin data=kreu hougaard maxiter=1000 noitprint;
      parms winf=4.5
            gamma=-1.2
            beta=0.02;
      model lw= winf*exp(-exp(gamma-beta*t)); ⑥
run;
proc nlin data=kreu hougaard maxiter=1000 noitprint;
      parms winf=4.5
            alpha=3.8
            beta=0.0015
            delta =1.5;
      model lw=winf-alpha*exp(-(beta*t)**delta); ⑦
run;
proc nlin data=kreu hougaard maxiter=1000 noitprint;
      parms winf=4.8
            gamma= 3.5
            beta= 0.02
            m = 1.7;
      model lw=winf *(1-exp(gamma-beta*t))**m; ⑧
run;
```

④ Initial values for the parameters are selected. (This is a critical step in the estimation process. For details on how initial values might be selected, see Appendix 6.E.)
⑤ Model statement for the logistic growth curve.
⑥ Model statement for the Gompertz growth curve.
⑦ Model statement for the Weibull growth curve.
⑧ Model statement for the Chapman–Richards growth curve.

The output generated by this SAS code is given in Table 6.6.

Ⓐ Output summary for logistic growth curve estimation.

① The overall fit is quite good, as evidenced by the significant F Statistic (the associated p-value is very low). (In fact, the F Statistics for the three remaining estimations are all significant as well.)

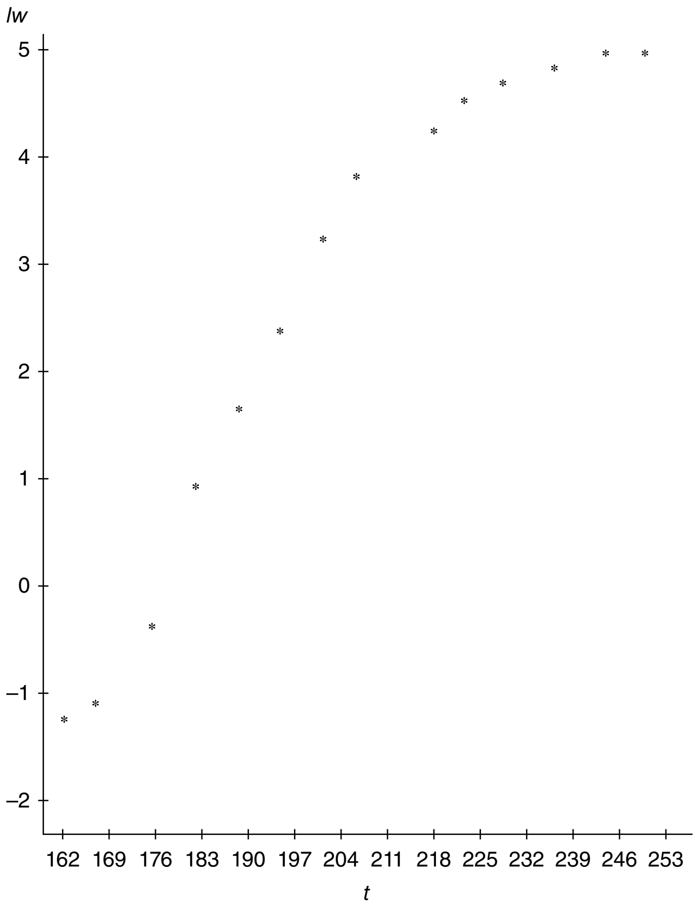

FIGURE 6.1 The SAS system. Plot of $lw \times t$. Data from Kreusler et al. (1879).

② The parameter estimates result in the logistic growth equation $w = \dfrac{4.6067}{1+e^{25.1873-0.1267t}}$.

③ None of the Wald approximate (asymptotic) 95% confidence intervals contain zero. Hence, each of the parameters w_∞, γ, and β is significantly different from zero at the 5% level.

④ (a) With $|g_{11}| = 0.3067, \hat{w}_\infty$ is moderately skewed away from linear behavior.
(b) With $|g_{12}| = 1.3617, \hat{\gamma}$ is highly skewed away from linear behavior.
(c) With $|g_{13}| = 1.3479, \hat{\beta}$ is also highly skewed away from linear behavior.

Ⓑ Output summary for Gompertz growth curve estimation.

① See comment A.1 in the preceding text.

② The parameter estimates render a Gompertz growth curve of the form $w = 4.8237e^{-e^{13.9012-0.0719t}}$.

TABLE 6.6 Output for Example 6.5

The SAS system
Ⓐ
The NLIN procedure
Dependent variable: *lw*
Method: Gauss–Newton
Iterative phase

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	26
Subiterations	9
Average subiterations	0.346154
R	7.185E-6
PPC (gamma)	7.177E-6
RPC (gamma)	0.00001
Object	3.2E-11
Objective	4.639434
Observations read	14
Observations used	14

The NLIN procedure
Estimation summary
Observations missing: 0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① *F* value	Approx Pr > *F*
Model	3	151.1	50.3678	119.42	<0.0001
Error	11	4.6394	0.4218		
Uncorrected total	14	155.7			

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
winf	4.6067	0.3189	3.9048	5.3085	0.3067
gamma	25.1873	8.1729	7.1988	43.1759	1.3617
beta	0.1267	0.0416	0.0352	0.2182	1.3479

Approximate correlation matrix

	winf	gamma	beta
winf	1.0000000	–0.4664748	–0.4863535
gamma	–0.4664748	1.0000000	0.9989618
beta	–0.4863535	0.9989618	1.0000000

TABLE 6.6 Output for Example 6.5 (*Continued*)

Ⓑ
The NLIN procedure
Dependent variable: *lw*
Method: Gauss–Newton
Iterative phase

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	33
Subiterations	10
Average subiterations	0.30303

The NLIN procedure
Estimation summary

R	9.054E-6
PPC (gamma)	8.702E-6
RPC (gamma)	0.000011
Object	3.16E-11
Objective	3.799256
Observations read	14
Observations used	14
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① F value	Approx Pr > F
Model	3	151.9	50.6479	146.64	<0.0001
Error	11	3.7993	0.3454		
Uncorrected total	14	155.7			

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
winf	4.8237	0.3994	3.9446	5.7028	0.7741
gamma	13.9012	4.3616	4.3014	23.5009	1.0610
beta	0.0719	0.0227	0.0219	0.1218	1.0278

Approximate correlation matrix

	winf	gamma	beta
winf	1.0000000	–0.7110618	–0.7292762
gamma	–0.7110618	1.0000000	0.9988905
beta	–0.7292762	0.9988905	1.0000000

(*Continued*)

TABLE 6.6 Output for Example 6.5 (*Continued*)

©
The NLIN procedure
Dependent variable: *lw*
Method: Gauss–Newton
Iterative phase

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	98
Subiterations	208
Average subiterations	2.122449
R	8.187E-7
PPC (delta)	3.642E-7
RPC (alpha)	9.742E-7
Object	1.02E-10
Objective	0.287621
Observations read	14
Observations used	14
Observations missing	0

Source	Sum of DF	Mean squares	① Squares	Approx *F* value	Pr > *F*
Model	3	66.8775	22.2925	775.07	<0.0001
Error	10	0.2876	0.0288		
Corrected total	13	67.1652			

The NLIN procedure

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
winf	4.7754	0.1142	4.5209	5.0300	0.3680
alpha	8.9370	1.0127	6.6806	11.1933	1.0253
beta	0.00525	0.000091	0.00505	0.00545	0.9816
delta	6.7182	0.9516	4.5978	8.8386	0.0745

Approximate correlation matrix

	winf	alpha	beta	delta
winf	1.0000000	0.5755010	0.2851921	−0.6152154
gamma	0.5755010	1.0000000	0.9350730	−0.9675236
beta	0.2851921	0.9350730	1.0000000	−0.8812987
delta	−0.6152154	−0.9675236	−0.8812987	1.0000000

TABLE 6.6 Output for Example 6.5 (*Continued*)

Ⓓ
The NLIN procedure
Dependent variable: *lw*
Method: Gauss–Newton
Iterative phase

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	823
Subiterations	1846
Average subiterations	2.243013
R	8.319E-6
PPC (m)	7.851E-6
RPC (m)	0.000042
Object	2.88E-12
Objective	0.280799
Observations read	14
Observations used	12
Observations missing	2

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	F value	Approx Pr>F
Model	4	152.2	38.0520	1084.10	<0.0001
Error	8	0.2808	0.0351		
Corrected total	12	152.5			

The NLIN procedure

② Parameter	Estimate	Approx std error	③ Approximate confidence	95% limits	Skewness
winf	5.2150	0.3653	4.3727	6.0574	.
gamma	7.2761	2.5333	1.4342	13.1179	.
beta	0.0413	0.0158	0.00489	0.0778	.
m	1.4630	1.1294	−1.1414	4.0674	.

Approximate correlation matrix

	winf	gamma	beta	*m*
winf	1.0000000	−0.9428442	−0.9316560	−0.7879877
gamma	−0.9428442	1.0000000	0.9978349	0.9132216
beta	−0.9316560	0.9978349	1.0000000	0.9374800
m	−0.7879877	0.9132216	0.9374800	1.0000000

③ None of the Wald approximate 95% confidence intervals contain zero, thus pointing to a statistically significant set of regression coefficients.

④ (a) With $|g_{11}| = 0.7741$, we see that $\hat{w}_\infty$ is moderately skewed away from linear behavior.

(b) Since $|g_{12}| = 1.0610$ and $|g_{13}| = 1.0278$, we see that $\hat{\gamma}$ and $\hat{\beta}$ are each highly skewed away from linear behavior.

Ⓒ Output summary for Weibull growth curve estimation.

① See comment A.1 in the preceding text.

② The resulting parametric form of the Weibull growth curve is

$$w = 4.7754 - 8.9370e^{-0.00525t^{6.7182}}.$$

③ Each of the Wald approximate 95% confidence intervals excludes zero. Hence, we have a statistically significant set of regression coefficients.

④ (a) Since $|g_{11}| = 0.3680$, we see that $\hat{w}_\infty$ is moderately skewed away from linear behavior.

(b) With $|g_{12}| = 1.0253$, we conclude that $\hat{\alpha}$ is highly skewed away from linear behavior.

(c) With $|g_{13}| = 0.9816$, we see that $\hat{\beta}$ is moderately skewed away from linear behavior.

(d) Since $|g_{14}| = 0.0745$, we conclude that $\hat{\delta}$ is very close to linear in character.

Ⓓ Output summary for Chapman–Richards growth curve estimation.

① See comment A.1 in the preceding text.

② The estimated form of the Chapman–Richards growth function is

$$w = 5.2150(1 - e^{7.2761 - 0.0413t})^{1.4530}.$$

③ Since the Wald approximate 95% confidence intervals for w_∞, γ, and β do not contain zero, we can conclude that these parameters are significantly different from zero (at the 5% level). However, the confidence interval for m (containing zero) reveals that this parameter is not statistically significant at the 5% level. ■

APPENDIX 6.A GAUSS–NEWTON ITERATION SCHEME: THE SINGLE PARAMETER CASE

Given the regression equation $Y_i = f(\mathbf{X}_i, \theta) + \varepsilon_i, i = 1, \ldots, n$, and *least squares function*, $S(\theta) = \sum_{i=1}^{n} (Y_i - f(\mathbf{X}_i, \theta))^2$, our objective is to find a $\theta = \hat{\theta}$ value that satisfies Equation 6.7. To this end, let us linearize or approximate $f(\mathbf{X}_i, \theta)$ at $\theta = \theta_0$ (our initial guess at $\hat{\theta}$) by the *first-order Taylor expansion*

$$f(\boldsymbol{X}_i,\theta) \approx f(\boldsymbol{X}_i,\theta_0) + \left.\frac{df(\boldsymbol{X}_i,\theta)}{d\theta}\right|_{\theta=\theta_0} (\theta-\theta_0) \tag{6.A.1}$$
$$= f(\boldsymbol{X}_i,\theta_0) + z_i(\theta_0)(\theta-\theta_0),$$

where $z_i(\theta_0) = (df(\boldsymbol{X}_i,\theta)/d\theta)_{\theta=\theta_0}$. Then the *linearized regression model* is

$$\begin{aligned} Y_i &= f(\boldsymbol{X}_i,\theta)+\varepsilon_i \\ &= f(\boldsymbol{X}_i,\theta_0) + z_i(\theta_0)(\theta-\theta_0)+\varepsilon_i, \quad i=1,\ldots,n, \end{aligned}$$

or

$$Y_i^0 = Y_i - f(\boldsymbol{X}_i,\theta_0) + z_i(\theta_0)\theta_0 = z_i(\theta_0)\theta + \varepsilon_i, \quad i=1,\ldots,n.$$

Hence, we need to find an OLS estimate for the slope θ of the linearized model

$$Y_i^0 = z_i(\theta_0)\theta + \varepsilon_i, \quad i=1,\ldots,n, \tag{6.A.2}$$

which satisfies

$$\min\left\{\sum_{i=1}^{n}(Y_i^0 - z_i(\theta_0)\theta)^2 = S(\theta)\right\}. \tag{6.A.3}$$

Let us denote the θ for which this minimum holds as

$$\theta_1 = (\boldsymbol{Z}(\theta_0)'\boldsymbol{Z}(\theta_0))^{-1}\boldsymbol{Z}(\theta_0)'\boldsymbol{Y}_0, \tag{6.A.4}$$

where $\boldsymbol{Z}(\theta_0)' = (z_1(\theta_0),\ldots,z_n(\theta_0))$ and $(\boldsymbol{Y}_0)' = (Y_1^0,\ldots,Y_n^0)$.

In terms of the components of Equation 6.A.4,

$$\theta_1 = \frac{\sum_{i=1}^{n} Y_i^0 z_i(\theta_0)}{\sum_{i=1}^{n} z_i(\theta_0)^2}. \tag{6.A.4.1}$$

Given this new (and ostensibly improved) $\theta=\theta_1$ value, we can now form a second linear approximation to $f(\boldsymbol{X}_i, \theta_0)$ (this time at θ_1), apply OLS so that $S(\theta)$ is minimized, and consequently obtain

$$\theta_2 = (\boldsymbol{Z}(\theta_1)'\boldsymbol{Z}(\theta_1))^{-1}\boldsymbol{Z}(\theta_1)'\boldsymbol{Y}_1, \tag{6.A.5}$$

where $\boldsymbol{Y}_1' = (Y_1', \ldots, Y_n')$ and $Y_i' = Y_i - f(\boldsymbol{X}_i, \theta_1) + z_i(\theta_1)\theta_1$, $i = 1, \ldots, n$. We continue this process and obtain, at the $(j+1)$st linear approximation,

$$\begin{aligned}\theta_{j+1} &= (\boldsymbol{Z}(\theta_j)'\boldsymbol{Z}(\theta_j))^{-1}\boldsymbol{Z}(\theta_j)'\boldsymbol{Y}_j \\ &= (\boldsymbol{Z}(\theta_j)'\boldsymbol{Z}(\theta_j))^{-1}\boldsymbol{Z}(\theta_j)'\left[\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \theta_j) + \boldsymbol{Z}(\theta_j)\theta_j\right] \\ &= \theta_j + (\boldsymbol{Z}(\theta_j)'\boldsymbol{Z}(\theta_j))^{-1}\boldsymbol{Z}(\theta_j)'(\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \theta_j)), \quad j = 0, 1, \ldots,\end{aligned} \tag{6.A.6}$$

where

$$\underset{(n\times 1)}{\boldsymbol{f}(\boldsymbol{X}, \theta_j)} = \begin{bmatrix} f(\boldsymbol{X}_1, \theta_j) \\ \cdot \\ \cdot \\ \cdot \\ f(\boldsymbol{X}_n, \theta_j) \end{bmatrix}.$$

For successive values of j, this procedure produces the sequence of OLS estimates $\theta_0, \theta_1, \theta_2, \ldots$.

Note that if process (Eq. 6.A.6) converges, then $\theta_{j+1} - \theta_j = 0$ and thus $\boldsymbol{Z}(\theta_j)'(\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \theta_j)) = 0$ or Equation 6.8 holds (our first-order condition for a minimum of $S(\theta)$). If $\theta_{j+1} - \theta_j \neq 0$, this iterative process is continued until $\left|(\theta_{j+1} - \theta_j)/\theta_j\right| < \delta$, $j = 0, 1, 2, \ldots$, where δ is a very small prespecified constant. It must be remembered that the process described by Equation 6.A.6 may not converge to $\hat{\theta}$ at all; and if convergence to $\hat{\theta}$ does occur, it may be quite slow, with successive solutions oscillating wildly. If convergence to a minimum obtains, this minimum may only be local in character and not correspond to the global minimum of $S(\theta)$. Hence, it is prudent to repeat the process using a few different initial values for θ. Finally, we note briefly that if the ε_i are independent and identically distributed with zero mean and constant variance σ_ε^2, then the *Gauss–Newton NLS estimator* θ_{j+1} is consistent and can be taken to have an approximate normal distribution with mean θ and variance $\hat{\sigma}_\varepsilon^2$ consistently estimated by

$$S_\varepsilon^2(\boldsymbol{Z}(\theta_{j+1})'\boldsymbol{Z}(\theta_{j+1}))^{-1}, \quad \text{where } S_\varepsilon^2 = \frac{S(\theta_{j+1})}{(n-1)}.$$

EXAMPLE 6.A.1 Given the regression equation $Y_i = \theta X_{i1} + \theta^{-1}X_{i2} + \varepsilon_i = f(\boldsymbol{X}_i, \theta) + \varepsilon_i$, $\boldsymbol{X}_i = (X_{i1}, X_{i2})$, $i = 1, \ldots, n$, let us employ the Gauss–Newton iteration routine specified by Equation 6.A.6. Let

$$z_i(\theta) = \frac{df(\boldsymbol{X}_i, \theta)}{d\theta} = X_{i1} - \theta^{-2}X_{i2}, \quad i = 1, \ldots, n.$$

Then

$$\boldsymbol{Z}(\theta)=\begin{bmatrix} z_1(\theta) \\ \vdots \\ z_n(\theta) \end{bmatrix}=\begin{bmatrix} X_{11}-\theta^{-2}X_{12} \\ \vdots \\ X_{n1}-\theta^{-2}X_{n2} \end{bmatrix},$$

$$\boldsymbol{Z}(\theta)'\boldsymbol{Z}(\theta)=\sum_{i=1}^{n}(X_{i1}-\theta^{-2}X_{i2})^2, \text{ and}$$

$$\boldsymbol{Z}(\theta)'(\boldsymbol{Y}-\boldsymbol{f}(\boldsymbol{X},\theta))=\sum_{i=1}^{n}(X_{i1}-\theta^{-2}X_{i2})(Y_i-\theta X_{i1}-\theta^{-1}X_{i2}),$$

where

$$\boldsymbol{Y}-\boldsymbol{f}(\boldsymbol{X},\theta)=\begin{bmatrix} Y_1-f(\boldsymbol{X}_1,\theta) \\ \vdots \\ Y_n-f(\boldsymbol{X}_n,\theta) \end{bmatrix}=\begin{bmatrix} Y_1-\theta X_{11}-\theta^{-1}X_{12} \\ \vdots \\ Y_n-\theta X_{n1}-\theta^{-1}X_{n2} \end{bmatrix}.$$

From Equation 6.A.6,

$$\theta_{j+1}=\theta_j+\frac{\sum_{i=1}^{n}(X_{i1}-\theta_j^{-2}X_{i2})(Y_i-\theta_j X_{i1}-\theta_j^{-1}X_{i2})}{\sum_{i=1}^{n}(X_{i1}-\theta_j^{-2}X_{i2})^2}, \quad j=0,1,2,\ldots,$$

with

$$\hat{\sigma}_\varepsilon=\frac{S(\theta_{j+1})}{(n-1)}=\frac{\sum_{i=1}^{n}(Y_i-\theta_{j+1}X_{i1}-\theta_{j+1}^{-1}X_{i2})^2}{n-1}. \qquad \blacksquare$$

APPENDIX 6.B GAUSS–NEWTON ITERATION SCHEME: THE *r* PARAMETER CASE

For the regression equation $Y_i=f(\boldsymbol{X}_i,\theta)+\varepsilon_i$, $i=1,\ldots n$, $\theta'=(\theta_1,\ldots,\theta_r)$, or $\boldsymbol{Y}=\boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta})+\boldsymbol{\varepsilon}$, let the *least squares function* appear as $S(\boldsymbol{\theta})=\boldsymbol{e}'\boldsymbol{e}=(\boldsymbol{Y}-\boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta}))'(\boldsymbol{Y}-\boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta}))$. Our goal is to determine a vector $\theta=\hat{\theta}$ for which Equation 6.9 is satisfied. We start by linearizing or approximating $\boldsymbol{f}(\boldsymbol{X},\theta)$ at $\theta=\theta_0$ (an initial guess at $\hat{\theta}$) via the *first-order Taylor expansion*

$$\boldsymbol{f}(\boldsymbol{X},\theta)\approx\boldsymbol{f}(\boldsymbol{X},\theta_0)+\boldsymbol{Z}(\theta_0)(\theta-\theta_0), \tag{6.B.1}$$

where the Jacobian matrix of $\boldsymbol{f}(\boldsymbol{X}, \theta)$ with respect to θ at θ_0 is

$$\underset{(n\times r)}{\boldsymbol{Z}(\theta_0)} = \frac{\partial \boldsymbol{f}(\boldsymbol{X},\theta)}{\partial \theta}\Big|_{\theta=\theta_0} = \begin{bmatrix} \frac{\partial f(\boldsymbol{X}_1,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_1}\Big|_{\theta_0} & \cdots & \frac{\partial f(\boldsymbol{X}_1,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_r}\Big|_{\theta_0} \\ \vdots & & \vdots \\ \frac{\partial f(\boldsymbol{X}_n,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_1}\Big|_{\theta_0} & \cdots & \frac{\partial f(\boldsymbol{X}_n,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_r}\Big|_{\theta_0} \end{bmatrix}.$$

Then the *linearized regression model* is

$$\begin{aligned} \boldsymbol{Y} &= \boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta}) + \varepsilon \\ &= \boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta}_0) + \boldsymbol{Z}(\boldsymbol{\theta}_0)(\boldsymbol{\theta}-\boldsymbol{\theta}_0) + \varepsilon \end{aligned}$$

or

$$\boldsymbol{Y}_0 = \boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta}_0) + \boldsymbol{Z}(\boldsymbol{\theta}_0)\boldsymbol{\theta}_0 = \boldsymbol{Z}(\boldsymbol{\theta}_0)\boldsymbol{\theta} + \varepsilon.$$

We thus need to determine the OLS estimate of $\boldsymbol{\theta}$ for the linearized regression model

$$\boldsymbol{Y}_0 = \boldsymbol{Z}(\boldsymbol{\theta}_0)\boldsymbol{\theta} + \varepsilon, \tag{6.B.2}$$

which satisfies

$$\min\left\{(\boldsymbol{Y}_0 - \boldsymbol{Z}(\boldsymbol{\theta}_0)\boldsymbol{\theta})'(\boldsymbol{Y}_0 - \boldsymbol{Z}(\boldsymbol{\theta}_0)\boldsymbol{\theta}) = S(\boldsymbol{\theta})\right\}. \tag{6.B.3}$$

Let us denote the θ for which this minimum is attained as

$$\boldsymbol{\theta}_1 = (\boldsymbol{Z}(\boldsymbol{\theta}_0)'\boldsymbol{Z}(\boldsymbol{\theta}_0))^{-1}\boldsymbol{Z}(\boldsymbol{\theta}_0)'\boldsymbol{Y}_0. \tag{6.B.4}$$

In general, we continue this process, and for the $(j+1)$st linear approximation of $\boldsymbol{f}(\boldsymbol{X}, \theta)$ at θ_j, the OLS estimate of θ_{j+1} is

$$\begin{aligned} \boldsymbol{\theta}_{j+1} &= (\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j))^{-1}\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Y}_j \\ &= \left(\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j)\right)^{-1}\boldsymbol{Z}(\boldsymbol{\theta}_j)'(\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta}_j) \\ &\quad + \boldsymbol{Z}(\boldsymbol{\theta}_j)\boldsymbol{\theta}_j) \\ &= \boldsymbol{\theta}_j + (\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j))^{-1}\boldsymbol{Z}(\boldsymbol{\theta}_j)'(\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta}_j)), \quad j = 0,1,\ldots. \end{aligned} \tag{6.B.5}$$

So for successive values of j, this procedure generates the sequence of OLS estimators $\boldsymbol{\theta}_0, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots$. When this process converges (or when $|(\theta_{l,j+1} - \theta_{l,j})/\theta_{l,j}| < \delta$, δ small, $l = 1, \ldots, r$), $\boldsymbol{\theta}_{j+1} - \boldsymbol{\theta}_j = 0$ and thus $\boldsymbol{Z}(\boldsymbol{\theta}_j)'(\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X}, \boldsymbol{\theta}_j)) = 0$ or Equation 6.10 holds (the first-order condition for a minimum of $S(\theta)$).

Since convergence to a global minimum may not occur, one should always try a few different starting points or initial values. In addition, if the ε_i are independent and identically distributed with a zero mean and constant variance σ_ε^2, then this *Gauss–Newton NLS estimator* $\boldsymbol{\theta}_{j+1}$ is consistent and can be taken to be approximately normally distributed with mean $\boldsymbol{\theta}$ and a variance–covariance matrix that is consistently estimated by $S_\varepsilon^2(\boldsymbol{Z}(\theta_{j+1})'\boldsymbol{Z}(\theta_{j+1}))^{-1}$, where $S_\varepsilon^2 = S(\theta_{j+1})/(n-r)$.

EXAMPLE 6.B.1 Given the regression equation $Y_i = \theta_1^2 X_{i1} + \theta_2^2 X_{i2} + \varepsilon_i = f(\boldsymbol{X}_i, \boldsymbol{\theta}) + \varepsilon_i, \boldsymbol{X}_i = (X_{i1}, X_{i2}), \theta' = (\theta_1, \theta_2), i = 1, \ldots, n,$ let us determine the Gauss–Newton iteration scheme represented by Equation 6.B.5. Given

$$\underset{(n\times 2)}{\boldsymbol{Z}(\theta)} = \begin{bmatrix} \dfrac{\partial f(\boldsymbol{X}_1,\theta)}{\partial\theta_1} & \dfrac{\partial f(\boldsymbol{X}_1,\theta)}{\partial\theta_2} \\ \dfrac{\partial f(\boldsymbol{X}_2,\theta)}{\partial\theta_1} & \dfrac{\partial f(\boldsymbol{X}_2,\theta)}{\partial\theta_2} \\ \vdots & \vdots \\ \dfrac{\partial f(\boldsymbol{X}_n,\theta)}{\partial\theta_1} & \dfrac{\partial f(\boldsymbol{X}_n,\theta)}{\partial\theta_2} \end{bmatrix} = \begin{bmatrix} 2\theta_1 X_{11} & 2\theta_2 X_{12} \\ 2\theta_1 X_{21} & 2\theta_2 X_{22} \\ \vdots & \vdots \\ 2\theta_1 X_{n1} & 2\theta_2 X_{n2} \end{bmatrix},$$

it follows that

$$\underset{(2\times 2)}{\boldsymbol{Z}(\theta)'\boldsymbol{Z}(\theta)} = \begin{bmatrix} 4\theta_1^2 \sum_{i=1}^{n} X_{i1}^2 & 2\theta_1\theta_2 \sum_{i=1}^{n} X_{i1}X_{i2} \\ 2\theta_2\theta_1 \sum_{i=1}^{n} X_{i2}X_{i1} & 4\theta_2^2 \sum_{i=1}^{n} X_{i2}^2 \end{bmatrix}$$

and

$$\underset{(2\times 1)}{\boldsymbol{Z}(\theta)'(\boldsymbol{Y} - \boldsymbol{f}(\boldsymbol{X},\theta))} = \begin{bmatrix} 2\theta_1 X_{11} & 2\theta_1 X_{21} \cdots 2\theta_1 X_{n1} \\ 2\theta_2 X_{12} & 2\theta_2 X_{22} \cdots 2\theta_2 X_{n2} \end{bmatrix} \begin{bmatrix} Y_1 - \theta_1^2 X_{11} - \theta_2^2 X_{12} \\ Y_2 - \theta_1^2 X_{21} - \theta_2^2 X_{22} \\ \vdots \\ Y_n - \theta_1^2 X_{n1} - \theta_2^2 X_{n2} \end{bmatrix}$$

$$= \begin{bmatrix} 2\theta_1 \sum_{i=1}^{n} X_{i1}(Y_i - \theta_1^2 X_{i1} - \theta_2^2 X_{i2}) \\ 2\theta_2 \sum_{i=1}^{n} X_{i2}(Y_i - \theta_1^2 X_{i1} - \theta_2^2 X_{i2}) \end{bmatrix},$$

where

$$\boldsymbol{Y}-\boldsymbol{f}(\boldsymbol{X},\boldsymbol{\theta})=\begin{bmatrix}Y_1-f(\boldsymbol{X}_1,\boldsymbol{\theta})\\ Y_2-f(\boldsymbol{X}_2,\boldsymbol{\theta})\\ \vdots\\ Y_n-f(\boldsymbol{X}_n,\boldsymbol{\theta})\end{bmatrix}=\begin{bmatrix}Y_1-\theta_1^2X_{11}-\theta_2^2X_{12}\\ Y_2-\theta_1^2X_{21}-\theta_2^2X_{22}\\ \vdots\\ Y_n-\theta_1^2X_{n1}-\theta_2^2X_{n2}\end{bmatrix}.$$

Then from Equation 6.B.5,

$$\boldsymbol{\theta}_{j+1}=\boldsymbol{\theta}_j+\begin{bmatrix}4\theta_1^2\sum_{i=1}^{n}X_{i1}^2 & 2\theta_1\theta_2\sum_{i=1}^{n}X_{i1}X_{i2}\\ 2\theta_2\theta_1\sum_{i=1}^{n}X_{i2}X_{i1} & 4\theta_2^2\sum_{i=1}^{n}X_{i2}^2\end{bmatrix}^{-1}$$

$$\times\begin{bmatrix}2\theta_1\sum_{i=1}^{n}X_{i1}(Y_i-\theta_1^2X_{i1}-\theta_2^2X_{i2})\\ 2\theta_2\sum_{i=1}^{n}X_{i2}(Y_i-\theta_1^2X_{i1}-\theta_2^2X_{i2})\end{bmatrix}.$$

■

APPENDIX 6.C THE NEWTON–RAPHSON AND SCORING METHODS

Given the regression equation $Y_i=f(\boldsymbol{X}_i, \boldsymbol{\theta})+\varepsilon_i, i=1,\ldots,n$, log-likelihood function $\log\mathcal{L}=-(n/2)\ln 2\pi-(n/2)\ln\sigma_\varepsilon^2-(1/2\sigma_\varepsilon^2)S(\theta)$, and *least squares function* $S(\theta)=\sum_{i-1}^{n}(Y_i=f(\boldsymbol{X}_i,\theta))^2$, our objective is to determine the value of $\boldsymbol{\theta},\tilde{\boldsymbol{\theta}}$, for which $(\partial\log\mathcal{L}/\partial\theta)_{\theta=\tilde{\theta}}=0$. For $\boldsymbol{\theta}=\boldsymbol{\theta}_0$, an initial (best guess) estimate of $\tilde{\boldsymbol{\theta}}$, let us construct a first-order Taylor expansion or linearization of $\partial\log\mathcal{L}/\partial\boldsymbol{\theta}$ at $\boldsymbol{\theta}_0$ as

$$\frac{\partial\log\mathcal{L}}{\partial\boldsymbol{\theta}}\approx\left.\frac{\partial\log\mathcal{L}}{\partial\boldsymbol{\theta}}\right|_{\theta=\theta_0}+\left.\frac{\partial^2\log\mathcal{L}}{\partial\boldsymbol{\theta}^2}\right|_{\theta=\theta_0}(\boldsymbol{\theta}-\boldsymbol{\theta}_0). \tag{6.C.1}$$

At the global maximum of $\log\mathcal{L}$, $\partial\log\mathcal{L}/\partial\theta=0$ and thus, from Equation 6.C.1,

$$(\boldsymbol{\theta}=)\boldsymbol{\theta}_1=\boldsymbol{\theta}_0-\left(\frac{\partial^2\log\mathcal{L}}{\partial\boldsymbol{\theta}^2}\right)_{\theta_0}^{-1}\left(\frac{\partial\log\mathcal{L}}{\partial\boldsymbol{\theta}}\right)_{\theta_0}. \tag{6.C.2}$$

In general, the $(j+1)$st iteration produces

$$\boldsymbol{\theta}_{j+1}=\boldsymbol{\theta}_j-\left(\frac{\partial^2\log\mathcal{L}}{\partial\boldsymbol{\theta}^2}\right)_{\theta_j}^{-1}\left(\frac{\partial\log\mathcal{L}}{\partial\boldsymbol{\theta}}\right)_{\theta_j},\quad j=0,1,2,\ldots. \tag{6.C.3}$$

This iteration technique is known as the *Newton–Raphson method*.

Let us term the quantity

$$G(\boldsymbol{\theta}_j) = \left(\frac{\partial \log \mathcal{L}}{\partial \boldsymbol{\theta}}\right)_{\theta_j} = \frac{-1}{2\sigma_\varepsilon^2}\left(\frac{\partial S}{\partial \boldsymbol{\theta}}\right)_{\theta_j} \tag{6.C.4}$$

the *score for* θ_j in the sample, while

$$I(\boldsymbol{\theta}_j) = -E\left[\left(\frac{\partial^2 \log \mathcal{L}}{\partial \boldsymbol{\theta}^2}\right)_{\theta_j}\right] = \frac{1}{\sigma_\varepsilon^2}\left(\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j)\right) \tag{6.C.5}$$

is called the *information on* $\boldsymbol{\theta}_j$ in the sample, where $\boldsymbol{Z}(\boldsymbol{\theta}_j)' = (z_1(\boldsymbol{\theta}_j), \ldots, z_n(\boldsymbol{\theta}_j))$ and $z_i(\boldsymbol{\theta}_j) = (df(\boldsymbol{X}_i, \boldsymbol{\theta})/d\theta)_{\theta=\theta_j}$. In this regard, for large n, the *method of scoring* has us replace $(\partial^2 \log \mathcal{L}/\partial \boldsymbol{\theta}^2)_{\theta_j}^{-1}$ by $-I(\boldsymbol{\theta}_j)^{-1}$ and $(\partial \log \mathcal{L}/\partial \boldsymbol{\theta})_{\theta_j}$ by $G(\boldsymbol{\theta}_j)$ to wit, Equation 6.C.3 becomes

$$\boldsymbol{\theta}_{j+1} = \boldsymbol{\theta}_j - \frac{1}{2}(\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j))^{-1}\left(\frac{\partial S}{\partial \theta}\right)_{\theta_j}, \quad j = 0, 1, 2, \ldots. \tag{6.C.6}$$

This process is continued until $\boldsymbol{\theta}_{j+1} - \boldsymbol{\theta}_j = 0$ or until $|(\boldsymbol{\theta}_{j+1} - \boldsymbol{\theta}_j)/\boldsymbol{\theta}_j| < \delta$, $j = 0, 1, 2, \ldots, \delta$ a small predetermined constant.

Given that the ε_i, $i = 1, \ldots, n$, are independent with $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$ for all i, the ML estimator $\tilde{\theta}$ is consistent and asymptotically normally distributed with the variance of $\tilde{\theta}$ consistently estimated by $I(\boldsymbol{\theta}_{j+1})^{-1} = \sigma_\varepsilon^2(\boldsymbol{Z}(\boldsymbol{\theta}_{j+1})'\boldsymbol{Z}(\boldsymbol{\theta}_{j+1}))^{-1}$, with $\tilde{\sigma}_\varepsilon^2 = S(\theta_{j+1})/n$.

We may extend Equation 6.C.3 to the r parameter case by writing $\boldsymbol{\theta}' = (\theta_1, \ldots, \theta_r)$,

$$\boldsymbol{\theta}_{j+1} = \boldsymbol{\theta}_j - \left(\frac{\partial^2 \log \mathcal{L}}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right)_{\theta_j}^{-1}\left(\frac{\partial \log \mathcal{L}}{\partial \boldsymbol{\theta}}\right)_{\theta_j}, \quad j = 0, 1, 2, \ldots. \tag{6.C.7}$$

This expression is known as the *generalized Newton–Raphson method*.

Let

$$G(\boldsymbol{\theta}_j) = \left(\frac{\partial \log \mathcal{L}}{\partial \boldsymbol{\theta}}\right)_{\theta_j} = \frac{-1}{2\sigma_\varepsilon^2}\left(\frac{\partial S}{\partial \boldsymbol{\theta}}\right)_{\theta_j} \tag{6.C.8}$$

represent the *score for* $\boldsymbol{\theta}_j$ in the sample, while

$$I(\boldsymbol{\theta}_j) = -E\left[\left(\frac{\partial^2 \log \mathcal{L}}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right)_{\theta_j}\right] = \frac{1}{\sigma_\varepsilon^2}\left(\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\theta_j)\right) \tag{6.C.9}$$

depicts the *information on* $\boldsymbol{\theta}_j$ in the sample. Then proceeding as in the earlier text, for large n, the *generalized method of scoring* has us substitute $-I(\boldsymbol{\theta}_{j+1})^{-1}$ and $G(\boldsymbol{\theta}_j)$ into Equation 6.C.7 so as to obtain

$$\boldsymbol{\theta}_{j+1} = \boldsymbol{\theta}_j - \frac{1}{2}\left[\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j)\right]^{-1}\left(\frac{\partial S}{\partial \boldsymbol{\theta}}\right)_{\boldsymbol{\theta}_j}, \quad j = 0, 1, 2, \ldots, \tag{6.C.10}$$

where

$$\underset{(n\times r)}{\boldsymbol{Z}(\boldsymbol{\theta}_j)} = \left.\frac{\partial f(\boldsymbol{X}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right|_{\theta=\theta_j} = \begin{bmatrix} \left.\dfrac{\partial f(\boldsymbol{X}_1, \boldsymbol{\theta})}{\partial \theta_1}\right|_{\theta_j} & \cdots & \left.\dfrac{\partial f(\boldsymbol{X}_1, \boldsymbol{\theta})}{\partial \theta_r}\right|_{\theta_j} \\ \vdots & & \vdots \\ \left.\dfrac{\partial f(\boldsymbol{X}_n, \boldsymbol{\theta})}{\partial \theta_1}\right|_{\theta_j} & \cdots & \left.\dfrac{\partial f(\boldsymbol{X}_n, \boldsymbol{\theta})}{\partial \theta_r}\right|_{\theta_j} \end{bmatrix}$$

and $(\partial S/\partial \boldsymbol{\theta})' = (\partial S/\partial \boldsymbol{\theta}_1, \ldots, \partial S/\partial \boldsymbol{\theta}_r)_{\theta_j}$. The iterations are continued until $\boldsymbol{\theta}_{j+1} - \boldsymbol{\theta}_j = 0$ or when $\left|(\theta_{l,j+1} - \theta_{lj})/\theta_{lj}\right| < \delta, \delta$ small, $l = 1, \ldots, r$.

For $\varepsilon \sim N(\mathbf{0}, \sigma_\varepsilon^2 \boldsymbol{I})$ and with the components of ε mutually independent, the ML estimator $\tilde{\theta}$ is consistent and asymptotically normal with the variance–covariance matrix of $\tilde{\theta}$ consistently estimated by $I(\boldsymbol{\theta}_{j+1})^{-1} = \tilde{\sigma}_\varepsilon^2\left[\boldsymbol{Z}(\boldsymbol{\theta}_{j+1})'\boldsymbol{Z}(\boldsymbol{\theta}_{j+1})\right]$, where $\tilde{\sigma}_\varepsilon^2 = S(\boldsymbol{\theta}_{j+1})/n$.

APPENDIX 6.D THE LEVENBERG–MARQUARDT MODIFICATION/COMPROMISE

If the columns of the Jacobian matrix of $f(\boldsymbol{X}, \boldsymbol{\theta})$ with respect to $\boldsymbol{X}$, $\boldsymbol{Z}(\boldsymbol{\theta})$, are highly collinear (there exist near-linear dependencies among the columns of $\boldsymbol{Z}(\boldsymbol{\theta})$), then the Gauss–Newton iterations can behave in a highly erratic fashion. Indeed, when $\boldsymbol{Z}(\boldsymbol{\theta})'\boldsymbol{Z}(\boldsymbol{\theta})$ exhibits near singularity ($\left|\boldsymbol{Z}(\boldsymbol{\theta})'\boldsymbol{Z}(\boldsymbol{\theta})\right| \approx 0$), the *parameter-change vector*

$$\begin{aligned} \Delta &= \boldsymbol{\theta}_{j+1} - \boldsymbol{\theta}_j \\ &= \left(\boldsymbol{Z}(\theta)'\boldsymbol{Z}(\theta)\right)^{-1}\boldsymbol{Z}(\theta)'\left(\boldsymbol{Y} - f(\boldsymbol{X}, \theta_j)\right) \end{aligned} \tag{6.D.1}$$

(see Eq. 6.B.5) can be quite large in magnitude, thus causing the parameter estimates to take on unrealistic values.

To mitigate against this possibility, Levenberg (1944) suggests that the *Gauss–Newton increment* (Eq. 6.D.1) be modified as

$$\Delta(k) = \left(\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j) + k\boldsymbol{I}_r\right)^{-1}\boldsymbol{Z}(\boldsymbol{\theta}_j)'\left(\boldsymbol{Y} - f(\boldsymbol{X}, \boldsymbol{\theta}_j)\right), \tag{6.D.2}$$

[Levenberg increment]

where k is a *conditioning factor* and $\boldsymbol{I}_r$ is an identity matrix of order r. As an alternative to Equation 6.D.2, Marquardt offers the modification

$$\Delta(k) = \left(\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j) + k\boldsymbol{D}\right)^{-1} \boldsymbol{Z}(\boldsymbol{\theta}_j)'\left(\boldsymbol{Y} - f(\boldsymbol{X},\boldsymbol{\theta}_j)\right), \tag{6.D.3}$$

[Marquardt increment]

where $\boldsymbol{D}$ is a diagonal matrix with entries consisting of the diagonal elements of $\boldsymbol{Z}(\boldsymbol{\theta}_j)'\boldsymbol{Z}(\boldsymbol{\theta}_j)$. (With $\boldsymbol{D}$ chosen in this fashion, the method is approximately invariant under the rescaling of $\boldsymbol{\theta}$.) As Equation 6.D.2 reveals, for the Levenberg increment, the Jacobian matrix $\boldsymbol{Z}(\boldsymbol{\theta})$ is changed to

$$\begin{bmatrix} \boldsymbol{Z}(\boldsymbol{\theta}) \\ \sqrt{k}\boldsymbol{I}_r \end{bmatrix};$$

and for the Marquardt increment, $\boldsymbol{Z}(\boldsymbol{\theta})$ is replaced by

$$\begin{bmatrix} \boldsymbol{Z}(\theta) \\ \sqrt{k}\boldsymbol{D}^{1/2} \end{bmatrix}.$$

When $\boldsymbol{D} = \boldsymbol{I}_r$, the *Levenberg–Marquardt direction* interpolates between the Gauss–Newton direction ($k \to 0$) and the direction of steepest descent ($k \to +\infty$)—the so-called *Levenberg–Marquardt compromise*.

APPENDIX 6.E SELECTION OF INITIAL VALUES

Example 6.5 addressed the task of estimating the parameters of the logistic, Gompertz, Weibull, and Chapman–Richards growth curves. Let us now consider how the initial set of parameter values might be determined. Before we consider each of these specific growth functions in turn, let us note first that:

1. The initial value of the asymptote parameter w_∞ can be readily approximated from a plot of the *lw* versus *t* series.
2. Good initial estimates for the other parameters can sometimes be obtained via a *grid search*—a process that offers PROC NLIN more than one starting value, for example, for the logistic equation, the parameter statement could read:

```
parms winf=4.5
gamma = −1.5 to 1 by 0.1
beta=0.01 to 0.03 by 0.01;
```

SAS calculates the initial residual sum of squares for all combinations of initial values and then begins the iterations with the best set.

3. At times initial estimates of some of the parameters may be obtained from the coordinates of the point of inflection of a growth curve, for example, for the Chapman–Richards growth function, the point of inflection appears as

$$\left(\beta^{-1}\ln(e^{\gamma}m),\, w_{\infty}\left(\frac{m-1}{m}\right)^{m}\right).$$

Once the asymptote w_{∞} is (graphically) approximated, a starting m value can be determined.

6.E.1 Initial Values for the Logistic Curve

From

$$w = \frac{w_{\infty}}{1+e^{\gamma-\beta t}}$$

we can readily obtain

$$\ln\left(\frac{w_{\infty}}{w}-1\right) = \gamma - \beta t.$$

Then regressing $\ln[(w_{\infty}/w)-1]$ on t can provide starting values for γ and β.

6.E.2 Initial Values for the Gompertz Curve

For

$$w = w_{\infty}e^{-e^{\gamma-\beta t}}$$

we can solve for

$$\ln\left[\ln\left(\frac{w_{\infty}}{w}\right)\right] = \gamma - \beta t.$$

Hence, regressing $\ln[\ln(w_{\infty}/w)]$ on t can produce reasonable starting values for γ and β.

6.E.3 Initial Values for the Weibull Curve

Given

$$w = w_{\infty} - \alpha e^{-\beta t^{\delta}},$$

set $t=0$. Then $w_0=w_\infty-\alpha$ so that $\alpha=w_\infty-w_0$. Next, solve for

$$\ln\left(\frac{w_\infty-w}{\alpha}\right)=-\beta t^{\delta}$$

and finally

$$\ln\left[-\ln\left(\frac{w_\infty-w}{\alpha}\right)\right]=\ln\beta+\delta\ln t.$$

Thus, regressing $\ln\{-\ln[(w_\infty-w)/\alpha]\}$ on $\ln t$ can render starting values for β and $\delta(\beta=e^{\widehat{\ln\beta}})$.

6.E.4 Initial Values for the Chapman–Richards Curve

From

$$w=w_\infty(1-e^{\gamma-\beta t})^m,$$

let us solve for

$$\ln\left[1-\left(\frac{w}{w_\infty}\right)^{1/m}\right]=\gamma-\beta t.$$

Hence, regressing $\ln[1-(w/w_\infty)^{1/m}]$ on t can yield starting values for γ and β. (Note that m can be obtained from the inflection point of the Chapman–Richards growth model.)

7

YIELD–DENSITY CURVES

7.1 INTRODUCTION

Yield–density models describe the functional relationship between the yield of a crop and density of planting. Such relationships are important for determining the optimum plant population level giving maximum yield per unit area.

Two broad forms of yield–density models have traditionally been studied, namely, those classified as either asymptotic or parabolic (Holliday, 1960). Let:

Y = crop yield (dry matter) per unit area

X = density (plant population in number of plants sown per unit area (this notion can also be stated in terms of spatial arrangement))

$w = Y/X$ = mean crop yield per plant[1]

If with increasing density Y rises to a maximum and then becomes relatively constant at high densities, then the yield–density relationship is said to be *asymptotic* (Fig. 7.1a). And if Y attains a maximum and then declines with increasing density, then the yield–density relationship is termed *parabolic* (Fig. 7.1b). While some yield–density equations involve a relationship between crop yield per unit area (Y) and plant density (X), the vast majority of yield–density models propose a functional relationship between mean yield per plant (w) and density (Fig. 7.1c).

[1] If all plants survive to harvest, then w is the mean yield (mean plant weight) at harvest. Otherwise, w represents total weight per unit area divided by density (X).

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

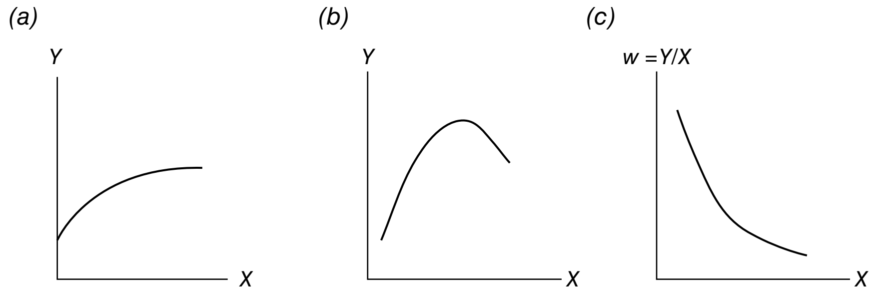

FIGURE 7.1 Yield–density functions: (**a**) asymptotic; (**b**) parabolic; and (**c**) decreasing mean yield.

7.2 STRUCTURING YIELD–DENSITY EQUATIONS

A variety of functional forms for yield–density equations have been offered as explanations for the connection between w and X. For instance, *exponential equations* such as

$$w = k10^{bX} \quad \text{or} \quad \log_{10} w = \log_{10} k + bX, b < 0 \tag{7.1}$$

(Duncan, 1958) or

$$w = ak^X \quad \text{or} \quad \ln w = \ln a + (\ln k)X \tag{7.2}$$

(Cramer and Jackobs, 1965) have been proposed. Also advanced are *geometric* or *power equations* that assume a linear relationship between the logarithm of w and the logarithm of X. For this specification,

$$wX^a = k \quad \text{or} \quad \ln w + a \ln X = \ln k \tag{7.3}$$

(Kira et al., 1953; Warne, 1951) (although Warne uses the logarithm of distance between plants in a row instead of ln X).

A drawback of Equation 7.3 is that this specification requires yield per plant to be increasing at high density levels, thus ruling out either asymptotic or parabolic behavior. The only circumstance in which an asymptotic shape emerges at high densities is when $a=1$—but this implies that yield $Y_\infty = w_\infty X = k =$ constant at all densities or that the *law of constant final yield* is operative.

The third broad class of yield–density models involves *reciprocal equations*—equations based on a relationship between the reciprocal of mean yield per plant ($1/w$) and density (X). This type of model has had the greatest success in describing yield–density relationships in terms of goodness of fit and biological significance. The following section discusses the development of reciprocal yield–density models in considerable detail. In particular, we shall explore the functional specifications offered by:

Shinozaki and Kira
Holliday

Farazdaghi and Harris
Bleasdale and Nelder

7.3 RECIPROCAL YIELD–DENSITY EQUATIONS

7.3.1 The Shinozaki and Kira Yield–Density Curve

Shinozaki and Kira (1956) were the first to propose a reciprocal yield–density relationship. Their model was derived from a logistic curve (of population growth) subject to the law of constant final yield. More specifically, Shinozaki and Kira invoke the following set of assumptions on the growth of a plant:

1. Plant growth can be described by the logistic differential equation

$$\frac{dY}{dt}\frac{1}{Y} = \lambda\left(1 - \frac{Y}{Y_\infty}\right), \tag{7.4}$$

where the growth (rate) coefficient λ is independent of X and both λ and Y_∞ are independent of t.

2. The law of constant final yield

$$Y_\infty = w_\infty X = k = \text{constant} \tag{7.5}$$

holds.

Given these assumptions, Appendix 7.A provides a derivation of the Shinozaki and Kira yield–density function, which appears as

$$\frac{1}{w} = a + bX, \quad a \text{ and } b \text{ parameters.} \tag{7.6}$$

From Equation 7.6, we can write

$$w = \frac{1}{a + bX} \tag{7.6.1}$$

so that

$$\frac{dw}{dX} = \frac{-b}{(a + bX)^2} < 0, b > 0$$

and $w \to a^{-1}$ as $X \to 0$. Also,

$$Y = \frac{X}{a + bX} = \frac{1}{(a / X) + b}$$

with $Y \to b^{-1}$ as $X \to \infty$. Hence, Equation 7.6 only describes an asymptotic yield–density relationship.

7.3.2 The Holliday Yield–Density Curves

When considering the fundamental relationship between crop yield and plant population, Holliday (1960) recognizes two distinct specifications—one in which yield depends upon a crop's growth in the reproductive phase (Engledow, 1926; Hudson, 1941) and another that results from the growth in the vegetative phase (Donald, 1951). For instance, in the former case and in the aggregate, grain yield for winter wheat is approximately quadratic and can be estimated by the equation

$$Y = \frac{Y_\infty X}{1 + Y_\infty bX + Y_\infty cX^2}, \tag{7.7}$$

where Y is yield per unit area and X depicts plant population, with Y_∞, a, b, and c parameters, or

$$\frac{1}{w} = a + bX + cX^2,\ a = Y_\infty^{-1} \tag{7.7.1}$$

(Holliday, 1960). From Equation 7.7.1,

$$Y = \frac{X}{a + bX + cX^2}. \tag{7.7.2}$$

Then setting $dY/dX = 0$ and solving for X renders the yield maximizing plant density level or $X_* = (a/c)^{1/2}$ (provided $d^2Y/dX^2 < 0$ at X_*). Substituting X_* into Equation 7.7.2 gives the maximum yield per unit area as

$$Y_* = \frac{(a/c)^{1/2}}{2a + b(a/c)^{1/2}}.$$

For the latter (vegetation phase) case, the relationship between, say, yield of potatoes (Davies, 1954) and plant population displays an asymptotic shape and can be defined by the expression

$$Y = \frac{Y_\infty X}{1 + Y_\infty bX}, \tag{7.8}$$

where Y is yield of dry matter per unit area and X is the number of plants per unit area, with Y_∞ and b parameters, or

$$\frac{1}{w} = a + bX,\ a = Y_\infty^{-1} \tag{7.8.1}$$

(Holliday, 1960). Note that this expression corresponds to the asymptotic yield–density equation of Shinozaki and Kira (Eq. 7.6).

7.3.3 The Farazdaghi and Harris Yield–Density Curve

Farazdaghi and Harris (1968) derived their yield–density equation from the same logistic curve as did Shinozaki and Kira. However, they modified the law of constant final yield to

$$Y_\infty = w_\infty X^\gamma = q = \text{constant}. \tag{7.9}$$

(Note that Eq. 7.9 is equivalent to Eq. 7.5 if $\gamma = 1$ at the end of the growing season, in which case $q = k$. In addition, Eq. 7.9 is valid for $X \geq X_s$, where X_s is the plant density at which competition for sunlight and growing space commences. For $X < X_s$, Y_∞ is independent of X.) Once Equation 7.9 is coupled with logistic growth behavior (see Appendix 7.B), the Farazdaghi and Harris yield–density curve appears as

$$\frac{1}{w} = a + bX^\gamma, \quad \text{with } a, b, \text{and } \gamma \text{ parameters.} \tag{7.10}$$

From Equation 7.10,

$$w = \frac{1}{a + bX^\gamma} \tag{7.10.1}$$

with

$$\frac{dw}{dX} = \frac{-b\gamma X^{\gamma-1}}{(a + bX^\gamma)^2} < 0,\ b > 0$$

and $w \to a^{-1}$ as $X \to 0$. In addition,

$$Y = \frac{X}{a + bX^\gamma} \tag{7.11}$$

attains a maximum at $X_* = [a/b(\gamma - 1)]^{1/\gamma}$ (provided $\gamma > 1$) with

$$Y_* = \gamma^{-1} b^{-(1/\gamma)} \left(\frac{\gamma - 1}{a} \right)^{1-(1/\gamma)}.$$

In sum, Equation 7.10 describes either asymptotic or parabolic yield–density behavior, depending on the value of γ. If $\gamma = 1$ (so that Eqs. 7.6, 7.8.1, and 7.10 are all equivalent), then asymptotic behavior holds, and if $\gamma > 1$, parabolic behavior emerges.

7.3.4 The Bleasdale and Nelder Yield–Density Curve

Bleasdale and Nelder (1960) offer a reciprocal equation which they derived from the generalized logistic growth curve of Richards (1959) under the assumption that the law of constant final yield holds. Specifically, their model has the form

$$\frac{1}{w^{\theta}} = a + bX^{\phi}, \quad \text{with } a, b, \theta, \text{and } \phi \text{ parameters} \tag{7.12}$$

(for a derivation of this expression, see Appendix 7.C). For

$$w = (a + bX^{\phi})^{-(1/\theta)}, \tag{7.13}$$

$$\frac{dw}{dX} = -b\left(\frac{\phi}{\theta}\right)\frac{X^{\phi-1}}{(a+bX^{\phi})^{(1/\theta)-1}} < 0, b > 0.$$

Now, depending on the relative magnitudes of θ and ϕ, Equation 7.12 represents a single generalized yield–density equation that can support either asymptotic or parabolic behavior. In this regard,

1. If $\theta = \phi$, then the yield–density relationship

$$\frac{1}{w^{\theta}} = a + bX^{\theta}, 0 < \theta < 1, \tag{7.12.1}$$

displays asymptotic behavior. Here

$$\frac{dw}{dX} = \frac{-bX^{\theta-1}}{(a+bX^{\theta})^{(1/\theta)+1}} < 0, b > 0,$$

and $w \to a^{-(1/\theta)}$ as $X \to 0$. Moreover,

$$Y = X(a + bX^{\theta})^{-(1/\theta)} \tag{7.14}$$

with $Y \to b^{-(1/\theta)}$ as $X \to \infty$.

2. If $\theta < \phi$, then we have parabolic yield–density behavior. That is, from Equation 7.13,

$$Y = X(a + bX^{\phi})^{-(1/\theta)} \tag{7.15}$$

attains a maximum at

$$X_* = \left[\frac{b}{a}\left(\frac{\phi}{\theta} - 1\right)\right]^{-(1/\phi)}$$

(provided $\theta < \phi$) with

$$Y_* = b^{-(1/\phi)} \left(\frac{\phi - \theta}{a\theta} \right)^{(1/\theta)-(1/\phi)}.$$

3. The instance where $\theta > \phi$ is of no particular consequence.

In practice, Bleasdale prefers to use a specialization of Equation 7.12 involving $\phi = 1$ or

$$\frac{1}{w^{\theta}} = a + bX, \tag{7.12.2}$$

the so-called *Bleasdale simplified equation*. When $\theta = 1$ in Equation 7.12.2 so that

$$\frac{1}{w} = a + bX, \tag{7.12.3}$$

we see that the law of constant final yield holds and thus yield per unit area (Y) displays asymptotic behavior with increasing plant density (X). (In this instance Eq. 7.12.3 is clearly equivalent to Eqs. 7.6, 7.8.1, and 7.10.) For $\theta < 1$, Equation 7.12.2 can model situations where the yield per unit area displays parabolic behavior with increases in plant density. To see this, let us rewrite Equation 7.12.2 as

$$Y = X(a + bX)^{-(1/\theta)}. \tag{7.16}$$

Then $dY/dX = 0$ and $d^2Y/dX^2 < 0$ at

$$X_* = \frac{a}{b} \left(\frac{\theta}{1 - \theta} \right), \ \theta < 1,$$

so that the maximum of Y is attained with

$$X_* = \frac{a}{b} \left(\frac{\theta}{1 - \theta} \right) \left(\frac{a}{1 - \theta} \right)^{-(1/\theta)}.$$

And if $\theta > 1$, yield per unit area behaves asymptotically as density increases since $Y \rightarrow b^{-(1/\theta)}$ as $X \rightarrow \infty$.

EXAMPLE 7.1 A data set that has proven useful for fitting reciprocal yield–density equations is that of I. S. Rogers and presented in Ratkowsky (Appendix 3.A, 1983) (see Table 7.1). A plot of w against X appears in Figure 7.2. Given this

TABLE 7.1 Onion Data (Brown Imperial Spanish) for Yield/Plant (w) versus Density (X) from Mount Gambier, South Australia[a]

X (plant/m^2)	w (g/plant)	X (plant/m^2)	w (g/plant)
20.64	176.58	55.66	95.52
26.91	159.07	59.35	94.94
26.91	122.41	59.72	119.28
28.02	128.32	63.04	93.64
32.44	125.77	67.09	85.73
34.28	126.81	68.93	89.26
35.76	147.77	69.30	88.55
36.49	117.29	73.36	76.81
38.71	133.49	80.73	76.63
39.44	128.87	89.58	90.53
39.81	110.04	95.47	71.28
40.92	111.15	98.05	56.61
42.76	134.12	98.42	75.09
43.50	99.94	102.48	65.26
45.34	128.70	105.80	64.48
45.71	152.17	106.53	61.84
46.82	100.36	108.75	65.19
47.18	123.32	115.38	57.10
47.92	114.44	150.77	52.68
48.66	131.27	152.24	47.01
53.45	115.12	155.19	44.28

[a]South Australian Department of Agriculture.

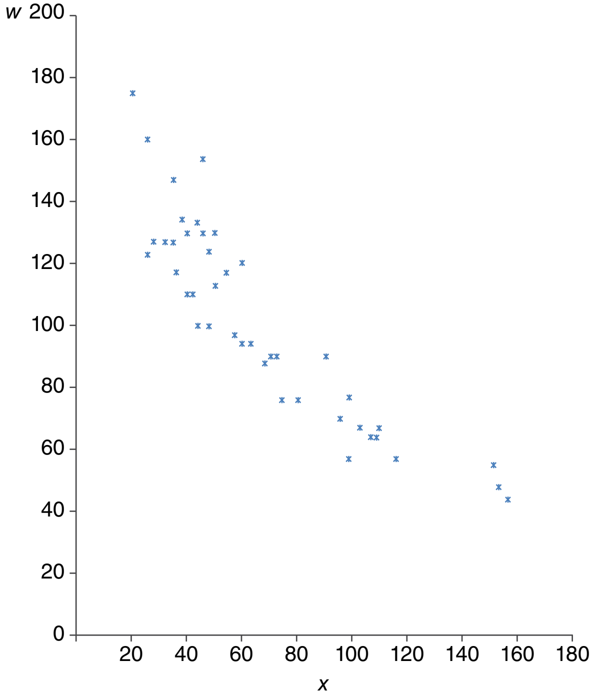

FIGURE 7.2 The SAS system. Plot of $w \times X$. Data from Ratkowsky (1983).

data set, our objective is to estimate, using PROC NLIN, the parameters of the four (reciprocal) yield–density functions:

Shinozaki–Kira (Eq. 7.6)—$w = (a + bX)^{-1}$.
Holliday (Eq. 7.7.1)—$w = (a + bX + cX^2)^{-1}$, $a = y_\infty^{-1}$.
Farazdaghi–Harris (Eq. 7.10)—$w = (a + bX^\gamma)^{-1}$.
Bleasdale–Nelder (Eq. 7.12.2)—$w = (a + bX)^\tau$, $\tau = -1/\theta$.

The requisite SAS code appears in Exhibit 7.1.

EXHIBIT 7.1 SAS Code for Yield–Density Curve Estimation

```
data yd;
   input x w;

datalines;
20.64 176.58
26.91 159.07
  .      .
  .      .
  .      .
15.19  44.28
;
proc plot data=yd; ①
     plot w*x='*';
proc nlin data=yd hougaard
  maxiter=1000 noitprint;
      parms a=0.005
           b=0.0002; ②
           z=a+b*x;
      model w=1/z; ③
run;
proc nlin data=yd hougaard
  maxiter=1000 noitprint;
      parms a=0.006
           b=0.00006
           c=0.0000003;
           z=a+b*x+c*(x**2);
      model w=1/z; ④
run;
proc nlin data=yd hougaard
  maxiter=1000 noitprint;
```

① The plot procedure is used to generate Figure 7.2.

② Initial values for the parameters are selected, and variable transformations for coding are made.

③ Model statement for the Shinozaki–Kira yield–density curve.

④ Model statement for the Holliday yield–density curve.

⑤ Model statement for the Farazdaghi–Harris yield–density curve.

⑥ Model statement for the Bleasdale–Nelder yield–density curve.

EXHIBIT 7.1 SAS Code for Yield–Density Curve Estimation (Cont'd)

```
        parms a=0.007
              b=0.00004
              gamma=1.4;
              z=a+b*(x**gamma);
        model w=1/z; ⑤
run;
proc nlin data=yd hougaard
  maxiter=1000 noitprint;
        parms a=0.05
              b=0.0006
              tau=-2;
              z=a+b*x;
        model w=z**tau; ⑥
run;
```

The output generated by this SAS code appears in Table 7.2.

Ⓐ Output summary for the Shinozaki–Kira yield–density function estimation.

① The overall fit is good, as revealed by the significant F Statistic (the associated p-value is very low). (In fact, the F Statistics for the three remaining estimations are highly significant as well.)

② The parameter estimates yield the Shinozaki–Kira yield–density equation

$$w = (0.00396 + 0.000103X)^{-1}.$$

③ Neither of the Wald approximate 95% confidence intervals contains zero. Hence, both the a and b parameters are significantly different from zero at the 5% level.

④ (i) With $|g_{11}| = 0.0197$, $\hat{a}$ is very close to linear in character.
(ii) With $|g_{12}| = 0.1107$, $\hat{b}$ is fairly close to linear in its behavior.

Ⓑ Output summary for the Holliday yield–density function estimation.

① See remark A.1 earlier.

② The parameter estimates render a Holliday yield–density curve of the form

$$w = (0.00471 + 0.000073X + 0.0000002587X^2)^{-1}.$$

③ Only the Wald 95% approximate confidence interval for c contains zero. Hence, the parameters a and b are significantly different from zero at the 5% level.

④ (i) With $|g_{11}| = 0.1697$ and $|g_{12}| = 0.1656$, we see that $\hat{a}$ and $\hat{b}$ are each fairly close to linear in character.
(ii) Since $|g_{13}| = 0.2706$, we see that $\hat{c}$ is moderately skewed away from linear behavior.

TABLE 7.2 Output for Example 7.1

The SAS system

Ⓐ

The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	5
Subiterations	1
Average subiterations	0.2
R	5.418E-7
PPC (*a*)	3.257E-7
RPC (*a*)	8.543E-6
Object	1.9E-10
Objective	6350.235
Observations read	42
Observations used	42
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① *F* value	Approx Pr > *F*
Model	2	469,402	234,701	1478.38	<0.0001
Error	40	6350.2	158.8		
Uncorrected total	42	475,753			

Parameter	② Estimate	Approx std error	③ Approximate 95% confidence limits		④ Skewness
a	0.00396	0.000384	0.00319	0.00474	0.0197
b	0.000103	8.719E-6	0.000086	0.000121	0.1107

Approximate correlation matrix

	a	*b*
a	1.0000000	−0.9168406
b	−0.9168406	1.0000000

Ⓑ

The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	5
R	8.696E-7
PPC (*b*)	2.247E-6
RPC (*b*)	0.000032

TABLE 7.2 Output for Example 7.1 (*Continued*)

Method	Gauss–Newton
Object	1.61E-10
Objective	6178.955
Observations read	42
Observations used	42
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① F value	Approx Pr > F
Model	3	469,574	156,525	987.94	<0.0001
Error	39	6179.0	158.4		
Uncorrected total	42	475,753			

Parameter	② Estimate	Approx std error	③ Approximate 95% confidence limits		④ Skewness
a	0.00470	0.000804	0.00308	0.00633	0.1697
b	0.000073	0.000031	0.000011	0.000134	−0.1656
c	2.587E-7	2.522E-7	−2.51E-7	7.689E-7	0.2706

Approximate correlation matrix

	a	b	c
a	1.0000000	−0.9646918	0.8668246
b	−0.9646918	1.0000000	−0.9546250
c	0.8668246	−0.9546250	1.0000000

©

The NLIN Procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	6
Subiterations	2
Average subiterations	0.333333
R	7.687E-6
PPC (b)	0.000063
RPC (b)	0.00051
Object	4.642E-9
Objective	6192.939
Observations read	42
Observations used	42
Observations missing	0

Note: An intercept was not specified for this model.

(*Continued*)

TABLE 7.2 Output for Example 7.1 (*Continued*)

Source	DF	Sum of squares	Mean square	① F value	Approx Pr > F
Model	3	469,560	156,520	985.68	<0.0001
Error	39	6192.9	158.8		
Uncorrected total	42	475,753			

Parameter	② Estimate	Approx std error	③ Approximate 95% confidence limits		④ Skewness
a	0.00512	0.000924	0.00325	0.00698	−0.9612
b	0.000024	0.000034	−0.00004	0.000093	4.0079
gamma	1.2947	0.2865	0.7151	1.8742	0.2071

Approximate correlation matrix

	a	b	gamma
a	1.0000000	−0.9585954	0.9418671
b	−0.9585954	1.0000000	−0.9979582
gamma	0.9418671	−0.9979582	1.0000000

Ⓓ

The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	6
R	1.364E-7
PPC (b)	2.051E-6
RPC (b)	0.000071
Object	1.61E-10
Objective	6178.907
Observations read	42
Observations used	42
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① F value	Approx Pr > F
Model	3	469,574	156,525	987.95	<0.0001
Error	39	6178.9	158.4		
Uncorrected total	42	475,753			

Parameter	② Estimate	Approx std error	③ Approximate 95% confidence limits		④ Skewness
a	0.0618	0.1489	−0.2395	0.3630	6.9383
b	0.000493	0.000575	−0.00067	0.00166	1.4845
Tau	−1.9249	1.6214	−5.2045	1.3547	−4.9251

TABLE 7.2 Output for Example 7.1 (*Continued*)

Approximate correlation matrix			
	a	*b*	tau
a	1.0000000	0.9966386	−0.9999145
b	0.9966386	1.0000000	−0.9975559
tau	−0.9999145	−0.9975559	1.0000000

Ⓒ Output summary for the Farazdaghi–Harris yield–density function estimation.

① See comment A.1 earlier.

② The resulting parametric form of the Farazdaghi–Harris yield–density curve is

$$w = \left(0.00512 + 0.000034X^{1.2947}\right)^{-1}.$$

③ Since the Wald 95% approximate confidence intervals for a and γ do not contain zero, we can conclude that these two parameters are significantly different from zero (at the 5% level). The confidence interval for b (containing zero) reveals that this parameter is not statistically significant at the 5% level.

④ (i) Since $|g_{11}| = 0.9612$, we see that $\hat{a}$ is moderately skewed away from linear behavior.

(ii) With $|g_{12}| = 4.0079$, it is evident that $\hat{b}$ is highly skewed away from linear behavior.

(iii) Since $|g_{13}| = 0.2017$, we see that $\hat{\gamma}$ is fairly close to linear in behavior.

Ⓓ Output summary for the Bleasdale–Nelder yield–density function estimation.

① See comment A.1 earlier.

② The parametric estimates imply a Bleasdale–Nelder yield–density expression of the form

$$w = (0.0618 + 0.000493X)^{-1.9249}.$$

(Note: $\tau = -1/\theta$ so that $\theta = 0.51948$.)

③ Each of the Wald 95% approximate confidence intervals contains zero, thus indicating that none of the regression parameters is significantly different from zero at the 5% level.

④ Since $|g_{1l}| > 1$ for $l = 1, 2, 3$, it follows that each of the parameter estimates $\hat{a}$, b, and $\hat{\tau}$ is highly skewed away from linear behavior. ■

7.4 WEIGHT OF A PLANT PART AND PLANT DENSITY

One of the main areas of application of reciprocal equations is in the development of a yield–density relationship for a plant part (e.g., seeds or leaves). Kira et al. (1956)

observed that the weight of a plant part could be related to the weight of the whole plant according to the *law of allometry*[2] (Huxley, 1924, 1972):

$$w_p = kw^{\alpha} \text{ or } \ln w_p = \ln k + \alpha \ln w, \ \alpha \text{ and } k \text{ parameters,} \tag{7.17}$$

where w_p is the weight of a plant part, w is plant weight, and α depicts the change in the proportion of the part to its total weight. Here Equation 7.17 will be used to describe the relative changes induced by density.

Following Bleasdale (1966a, 1967), let us rewrite Equation 7.17 as

$$w = kw_p^{\alpha} \tag{7.17.1}$$

and combine this expression with Equation 7.12.2 so as to obtain

$$\frac{1}{\left(kw_p^{\alpha}\right)^{\theta}} = a + bX$$

or

$$\frac{1}{w_p^{\theta_p}} = a_1 + b_1X, \tag{7.18}$$

where $\theta_p = \alpha\theta, a_1 = k^{\theta}\alpha$, and $b_1 = k^{\theta}b$. Equation 7.18 thus describes the way plant density affects the distribution of dry matter into plant parts.

Watkinson (1980) has reparameterized Equation 7.12.2 so as to make the parameters more biologically meaningful. That is, from Equation 7.12.2,

$$w^{-\theta} = a\left(1 + a^{-1}bX\right)$$

or

$$\begin{aligned} w &= a^{-(1/\theta)}\left(1 + a^{-1}bX\right)^{-(1/\theta)} \\ &= w_m\left(1 + a_1X\right)^{-b_1}, \end{aligned} \tag{7.19}$$

where $w_m = a^{-1/\theta}, a_1 = a^{-1}b$, and $b_1 = 1/\theta$. Here w is the mean yield per plant, X represents the density of survivors, w_m is the yield or dry matter production of an isolated plant, a_1 is the area required by a plant to achieve a yield of w_m, and b_1 describes the

[2] Certain biological attributes of organisms scale proportionately with body size; that is, the relationship between a physiological attribute and body size is the same regardless of the organism's size. However, it is often the case that this relationship changes relative to the size of the organism; for example, tree diameter increases with height, but at a greater-than-proportional rate. Both of these cases are examples of *allometric relationships*—the scaling relationship between some biological attribute and body size is dependent on the organism's size.

effectiveness with which resources are taken up from the area. Then from Equation 7.19, total plant yield in terms of surviving plant density is $Y = wX$ or

$$Y = \left[w_m \left(1 + a_1 X\right)^{-b_1} \right] X. \tag{7.20}$$

It is important to note that Equation 7.19 applies to "surviving densities" of plants. Hence, at high densities, considerations of plant mortality via self-thinning can be introduced by augmenting Equation 7.19 by the $-(3/2)$ *power law*:

$$w = cX^{-3/2}, \; c \text{ a constant,}^{3} \tag{7.21}$$

where w is the mean yield and X is the number of surviving plants, which truncates Equation 7.19 in a fashion such that densities higher than the point of intersection between Equations 7.19 and 7.21 cannot be attained because of the effects of self-thinning.

Given Equation 7.19, we can also employ Equation 7.17.1 to express the relationship between the weight of a plant part and the density of surviving plants as

$$w_p = \left(\frac{w_m}{k} \right)^{1/\alpha} \left(1 + a_1 X\right)^{-b_1/\alpha}$$

or, for $w_{mp} = (w_m/k)^{1/\alpha}$ and $b_1' = b_1/\alpha$,

$$w_p = w_{mp} \left(1 + a_1 X\right)^{-b_1'}. \tag{7.22}$$

Then the relationship between the total yield of a plant part Y_p and surviving population density X can be calculated from Equation 7.22 as $Y_p = w_p X$ or

$$Y_p = w_{mp} \left[\left(1 + a_1 X\right)^{-b_1'} \right] X. \tag{7.23}$$

Also from Equation 7.17.1, the self-thinning relationship for a plant part can be determined via Equation 7.22 as

$$kw_p^{\alpha} = cX^{-3/2}$$

[3] The phenomenon of *self-thinning* (density-dependent mortality) has been observed in high-density populations by Yoda et al. (1963). Specifically, for a wide range of plant species, it has been empirically determined that when a population of a particular density reaches the weight corresponding to that density on the line

$$\ln w = \ln c - \frac{3}{2} \ln X,$$

self-thinning will occur and subsequently stand development over time will be along this line. (Note that almost all data on self-thinning refers to shoot dry weight, but some evidence points to the $-(3/2)$ power law holding for total plant weight also.)

or

$$w_p = \left(\frac{c}{k}\right)^{1/\alpha} X^{-3/2\alpha}. \tag{7.24}$$

Hence, this equation predicts that self-thinning for a plant part follows a line of slope $-3/2\alpha$.

Given that X in Equation 7.19 represents the density of surviving plants, we can view this equation as providing a "static description" of the effects of plant competition. However, it is generally taken that both w_m and a are time dependent (actually, both of these terms tend to increase over time). This being the case, let us assume that the w_m time path can be described by the logistic equation

$$w_m = \frac{k}{1+ce^{-rt}} \tag{7.25}$$

and that a and w_m are related by the allometric equation

$$a = \alpha w_m^{\beta}, \tag{7.26}$$

with α and β predetermined (Li et al., 1996). Then a combination of Equations 7.19, 7.25, and 7.26 yields a "dynamic description" of the relationship between total yield per area at time t and density or

$$Y_t = \frac{kX}{\left(1+ce^{-rt}\right)\left\{1+\alpha\left[k/\left(1+ce^{-rt}\right)\right]^{\beta} X\right\}^{b}}. \tag{7.27}$$

7.5 THE EXPOLINEAR GROWTH EQUATION

The *expolinear growth equation* of Goudriaan and Monteith (1990) is designed to describe a continuous transition from exponential growth to full or linear growth[4] for crop stands in which dry matter typically increases at virtually a constant rate during the main growth stage. It was developed from evidence that uniform stands of vegetation accumulate dry matter at a rate closely related to the rate at which foliage intercepts or absorbs radiant energy and that, when light is limiting, the growth rate is proportional to intercepted radiation and consequently to an exponential function of leaf area.

As a process-based model for crop growth, the expolinear equation expresses the dry weight, the leaf area, the relative growth rate, and the *net assimilation rate*

[4] In the *exponential growth phase*, ground cover is still small—the space around plants is not fully occupied, and each new leaf contributes more to light being intercepted, thus reinforcing growth. There is no natural shading yet so that the contribution of a new leaf is identical to that of existing ones. In this phase the crop is still mostly vegetative and strongly dependent on radiation and temperature. In the *full growth phase*, the leaves start to encroach on or overshadow each other; the amount of light intercepted by new leaf area is diminished. The exponential phase has given way to a linear growth phase during which the bulk of dry matter formation occurs.

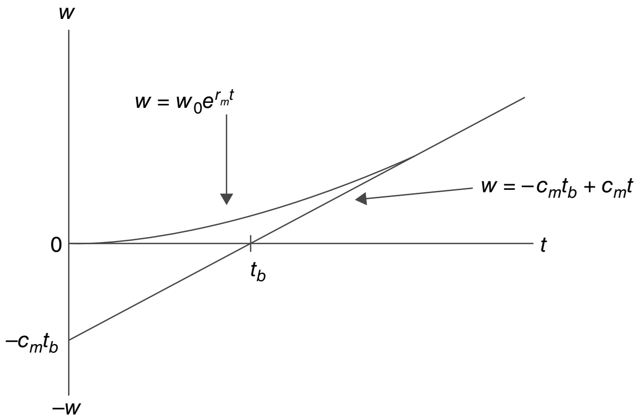

FIGURE 7.3 Expolinear growth.

(the rate of dry matter production per unit area of the leaves) as consistent functions of time by using the following *physiologically significant* parameters:

r_m—an initial maximum relative growth rate in the exponential phase
c_m—a maximum absolute growth rate in the linear phase
t_b—the time at which the crop stand effectively passes from exponential to linear growth

(t_b is termed "lost time" to signify the time lost by the crop when it is developing its canopy and not intercepting all the incident radiation.)

In this regard, the growth rate of a crop is initially exponential with a relative growth rate of r_m; subsequently, when the crop intercepts all the incident radiation, growth is linear at a maximum rate of c_m. Let w denote crop biomass (dry weight) at time t. Then during the exponential phase, $dw/dt = r_m w$, and during the linear phase, $dw/dt = c_m$. Hence, at the transition point between exponential and linear growth, we must have $r_m w = c_m$ so that the transition level of biomass is $w = c_m/r_m$. However, the transition is not abrupt. In fact, mutual shading can start early on in the growth process so that the transition from the exponential phase to the linear phase is gradual. And it is this gradual transition that is characterized by the expolinear equation. To see this, an integration of $dw/dt = r_m w$ yields $w = w_0 e^{r_m t}$ (w_0 is the initial biomass at time $t = 0$), and an integration of $dw/dt = c_m$ renders $w = c_m(t - t_b)$ (Fig. 7.3).

The set of working assumptions made by Goudriaan and Monteith (1990) consists of:

1. Exponential extinction of radiation (with a constant light extinction coefficient k). Hence, the fraction of incident light intercepted is

$$f = 1 - e^{-kL}, \tag{7.28}$$

where L is the *leaf area index* (leaf area per unit area of land—an index of mean crop leafiness) and kL represents an "effective leaf area" (an opaque horizontal area that intercepts the same amount of light as the actual leaf area).

2. A direct proportion between f and the actual growth rate of biomass or

$$\frac{dw}{dt} = c_m f. \tag{7.29}$$

3. A constant *leaf area ratio* (the ratio between leaf area and plant dry matter) that is further separated into the *leaf weight ratio* (the ratio of leaf weight to total plant weight in dry matter) p_1 and *specific leaf area* (the ratio between the leaf area and the leaf biomass) s. Then $p_1 s$, the ratio of the increments of leaf area and total biomass, remains constant during crop growth. Hence, the rate of formation of new leaf area (dL/dt) is connected to the growth rate of biomass via

$$\frac{dL}{dt} = p_1 s \frac{dw}{dt}, \tag{7.30}$$

where now s is the specific leaf area of "new leaves" and p_1 is the fraction of growth of total dry matter allocated to "new leaves." (Clearly s and p_1 represent incremental fractions of leaf area and leaf biomass, respectively.)

Substituting Equation 7.29 into Equation 7.30 yields

$$\frac{dL}{dt} = p_1 s c_m f. \tag{7.30.1}$$

For low ground cover (such as that found at the inception of growth), f can be approximated by kL so that Equation 7.30.1 becomes

$$\frac{dL}{dt} = p_1 s c_m kL \tag{7.31}$$

or

$$\frac{dL / dt}{L} = r_m = p_1 s c_m k = \text{constant}$$

and thus

$$p_1 s c_m = \frac{r_m}{k}. \tag{7.32}$$

Then the growth rate of leaf area L can be obtained by substituting Equation 7.32 into Equation 7.30.1 or

$$\frac{dL}{dt} = \left(\frac{r_m}{k}\right)\left(1 - e^{-kL}\right). \tag{7.30.2}$$

Once this expression is integrated and merged with dw/dt from Equation 7.29, we obtain, after a subsequent integration, the *expolinear growth equation*

$$w = \left(\frac{c_m}{r_m}\right)\ln\left[1 + e^{r_m(t-t_b)}\right]. \tag{7.33}$$

(For the derivation of Eq. 7.33, see Appendix 7.D.) Note that "lost time" t_b can be obtained from Equation 7.33 by setting $t=0$ or

$$w_0 = \left(\frac{c_m}{r_m}\right)\ln\left[1 + e^{-r_m t_b}\right]$$

or

$$t_b = -\frac{1}{r_m}\ln\left[e^{r_m w_0 / c_m} - 1\right]. \tag{7.34}$$

As Equation 7.33 reveals, the dry weight (w) of a crop stand can be specified as a function of its maximum growth rate (c_m) and its maximum relative growth rate (r_m), with its position on the time axis determined by the value of "lost time" (t_b). Then the *effective loss of biomass* ($c_m t_b$) can be written (via Eq. 7.34) as

$$c_m t_b = -\left(\frac{c_m}{r_m}\right)\ln\left[e^{r_m w_0 / c_m} - 1\right]. \tag{7.35}$$

A significant contribution of the expolinear function is that the growth of a crop stand can be divided into two phases: the first or exponential phase during which the growth rate increases from a very small value at emergence to a maximum rate (c_m) achieved soon after time t_b and a second (linear) phase during which the constant maximum growth rate c_m is maintained (assuming that environmental conditions remain constant). When *senescence* sets in and the linear growth phase ends, the growth rate decreases so that Equation 7.33 must be truncated when w reaches its maximum level w_m; that is, take

$$w = \min\left\{w_m, \left(\frac{c_m}{r_m}\right)\ln\left[1 + e^{r_m(t-t_b)}\right]\right\}. \tag{7.36}$$

And since in the linear phase we have $w = c_m(t - t_b)$, it follows that w_m is related to the time when growth stops by the expression $w_m = c_m(t - t_b)$; that is, growth terminates at the moment

$$t = t_g = t_b + w_m / c_m. \tag{7.37}$$

It was mentioned earlier that the expolinear growth function is designed to model a smooth transition from an early accelerated growth phase to a linear or constant growth phase. However, the life cycle of a plant or crop can be divided into three phases, where the third phase is a saturation phase for ripening (or senescence). To handle this third phase, Goudriaan and Monteith (1990) suggest truncating the curve at the maximum value of $w(w_m)$ (see Eq. 7.36). This truncation obviously reflects an abrupt transition from the linear to the saturation phase and results in no growth at all after the transition occurs.

To allow for a smooth transition to phase three, Goudriaan (1994) extends Equation 7.33 to an expolinear growth equation with two smooth transitions or to the form

$$w = \left(\frac{c_m}{r_m}\right)\ln\left[\frac{1+e^{r_m(t-t_b)}}{1+e^{r_m(t-t_g)}}\right],$$

where $t_g = t_b + w_m/c_m$ (see Eq. 7.37). This expression offers a symmetric sigmoidal pattern around time $t_b + w_m/2c_m$ (at this value of time, we have $d^2w/dt^2 = 0$ and dw/dt is at a maximum) and will be termed the *symmetric expolinear growth equation* (to distinguish it from its truncated counterpart).

EXAMPLE 7.2 We now turn to the estimation of the parameters of the expolinear growth equation (Eq. 7.33) or

$$w = \left(\frac{c_m}{c_r}\right)\ln\left[1+e^{r_m(t-t_b)}\right],$$

where w denotes crop yield (dry matter) at time t. The data series employed is presented in Table 7.3, with a plot of w against t provided in Figure 7.4.
The PROC NLIN code appears as Exhibit 7.2.

EXHIBIT 7.2 SAS Code for Expolinear Growth Curve Estimation

```
data dry;
     input t w;
datalines;
 1     1.2
 2     1.5
 .       .
 .       .
 .       .
12     16.4
;
proc plot data=dry;
     plot w*t="*";
run;
proc nlin data=dry hougaard maxiter=1000 noitprint;
        parms cm=0
              rm=1.5
              tb=2;
               z=1+exp(rm*(t-tb)); ①
              lz=log(z);
        model w=(cm/rm)*lz; ②
run;
```

① Initial values for the parameters are selected, and variable transformations for coding are made.

② Model statement for the expolinear growth equation.

TABLE 7.3 Dry Matter by Potato (*w*) on Sandy Loam Soil at the San Luis Valley of Colorado[a]

t (weeks)	*w* (mg/ha)
1	1.2
2	1.5
3	2.1
4	3.0
5	4.5
6	6.2
7	8.0
8	9.8
9	11.7
10	13.7
11	16.7
12	16.4

[a]Adapted from Soltanpour (1969).

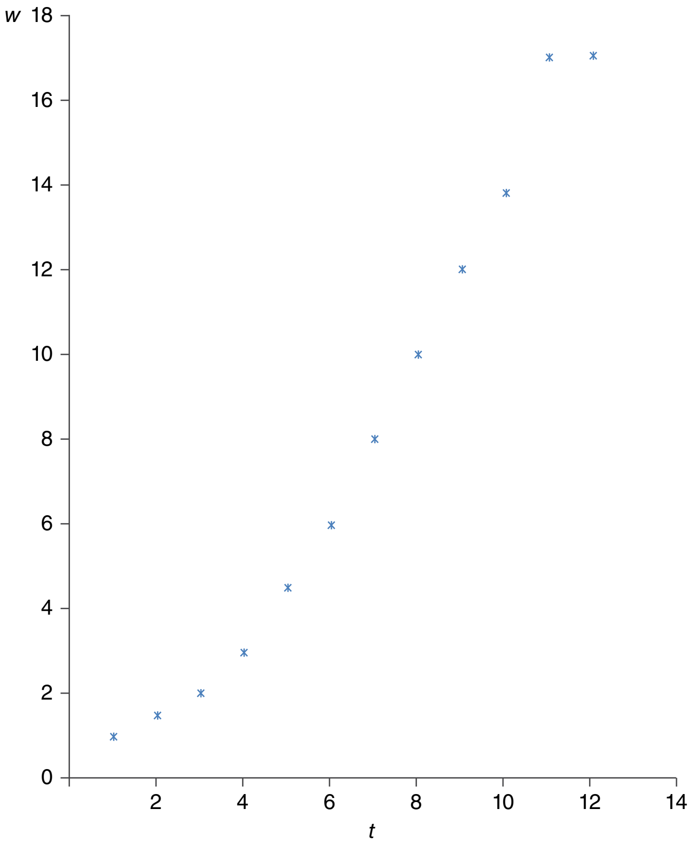

FIGURE 7.4 The SAS system. Plot of $w \times t$. Data from Soltanpour (1969).

TABLE 7.4 Output for Example 7.2

The SAS system
The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	10
Subiterations	2
Average subiterations	0.2
R	3.011E-6
PPC (*b*)	2.211E-6
RPC (*b*)	9.407E-6
Object	1.26E-10
Objective	1.26E-102.84225
Observations read	12
Observations used	12
Observations missing	0
Note: An intercept was not specified for this model.	0

Source	DF	Sum of squares	Mean square	① *F* value	Approx Pr > *F*
Model	3	1110.1	370.0	1171.75	<0.0001
Error	9	2.8422	0.3158		
Uncorrected total	12	1113.0			

Parameter	② Estimate	Approx std error	③ Approximate 95% confidence limits		④ Skewness
c_m	1.9379	0.1558	1.5884	2.2905	0.9256
r_m	0.6230	0.1630	0.2541	0.9918	1.3696
t_b	2.9902	0.5834	1.6704	4.3100	0.8103

Approximate correlation matrix

	c_m	r_m	t_b
c_m	1.0000000	−0.8045354	0.9746684
r_m	−0.8045354	1.0000000	−0.8653639
t_b	0.9746684	−0.8653639	1.0000000

The output generated by this SAS code is presented in Table 7.4.

① The small *p*-value associated with the sample *F* Statistic indicates that the model fits the data very well.

② The parameter estimates yield the expolinear growth function

$$w = 3.1106 \ln\left[1 + e^{0.6230(t-2.9902)}\right].$$

③ None of the Wald 95% approximate confidence intervals contains zero, thus indicating that all three parameters differ significantly from zero (at the 5% level).

④ (i) Since $|g_{11}|=0.9256$ and $|g_{13}|=0.8103$, we see that both $\hat{c}_m$ and $\hat{t}_b$ are moderately skewed away from linear behavior.

(ii) With $|g_{12}|=1.3696$, it is evident that $\hat{r}_m$ is highly skewed away from linear behavior. ■

7.6 THE BETA GROWTH FUNCTION

An equation developed by Yin et al. (2003) to describe a sigmoidal pattern of *determinate growth* (a zero growth rate at the beginning and at the end of a precisely specified growth period) is the *beta growth function*. Some salient features of this equation are:

1. It is sufficiently flexible in describing various asymmetric patterns of determinate sigmoidal growth, with the growth rate itself being a unimodal bell-shaped curve. (This result obtains because the beta probability density function represents a family of flexible asymmetric unimodal curves with two fixed end points.)
2. It predicts zero biomass at time zero.
3. It provides reasonable estimates of final crop weight and duration of the growth process.
4. It predicts a smooth transition to final or maximum crop weight (it has a nonlinear terminate segment, which displays a zero slope at the end of the growth period).

Modeling growth via the beta growth function is important since, with many annual agricultural crops determinate in nature, the traits of final weight and unambiguous length of the growing period can be quantified. In this regard, plant weight (w) as a function of time (t) is expressed in terms of the following parameters:

t_m—the time at which the maximum growth rate is achieved
t_e—the time at the end of growth
w_m—the maximum value of w (which is attained at t_e)

and appears as

$$w = w_m\left(1+\frac{t_e-t}{t_e-t_m}\right)\left(\frac{t}{t_e}\right)^{t_e/(t_e-t_m)}, \; 0 \le t_m < t_e, \tag{7.38}$$

the *beta growth function* (for a derivation of this expression, see the discussion underlying Eq. 7.E.10).

Equation 7.38 must satisfy the constraints (i) $w=0$ at the start of the growth period (at $t=0$) and (ii) $w=w_m$ when growth is terminated (at $t=t_e$). Clearly this equation holds for growth within the time period $0 \le t \le t_e$; otherwise set

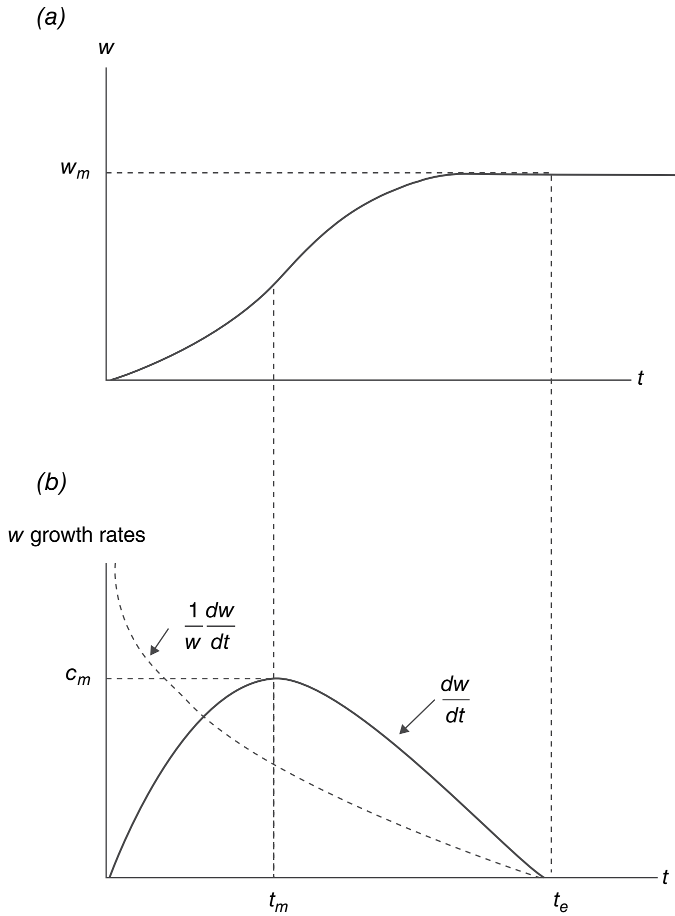

FIGURE 7.5 Beta growth function: (**a**) plant weight w with extension; (**b**) relative and absolute growth rates for w.

$$w = \begin{cases} 0, & t < 0; \\ w_m, & t > t_e. \end{cases}$$

If $t > t_e$, then the beta growth function is (Eq. 7.38) with the extension $w = w_m$ (Fig. 7.5a). Hence, the time path of w for the beta growth function starts at $w = 0$ for $t = 0$ and goes to w_m when $t = t_e$ with w remaining at w_m thereafter. Moreover, the transition in w to w_m is smooth since $dw/dt = 0$ at $t = t_e$.

Given that the beta growth rate function (or absolute growth rate dw/dt) is specified by Equation 7.E.4, it follows that the maximum growth rate c_m is determined from Equation 7.E.9 as

$$c_m = \left(\frac{2t_e - t_m}{t_e \left(t_e - t_m \right)} \right) \left(\frac{t_m}{t_e} \right)^{t_m/(t_e - t_m)} w_m, \tag{7.39}$$

while the relative growth rate

$$\frac{1}{w}\frac{dw}{dt}=\frac{(2t_e-t_m)(t_e-t)t_e}{(t_e-t_m)(2t_e-t_m-t)t} \tag{7.40}$$

is determined from Equations 7.38, 7.39, and 7.E.4 (see Fig. 7.5b). Thus, the relative growth rate for w is infinite at the start of the growth period and then declines monotonically as t increases and eventually reaches zero at time t_e. If $t > t_e$, the relative and absolute growth rates remain at zero under the restriction $w = w_m$.

The beta growth function (Eq. 7.38) always assumes an initial weight of zero. A more general beta function—one that has a small initial weight at emergence—appears as

$$w = w_b + (w_m - w_b)\left(1+\frac{t_e-t}{t_e-t_m}\right)\left(\frac{t-t_b}{t_e-t_b}\right)^{(t_m-t_b)/(t_e-t_m)}, t_b \le t_m \le t_e, \tag{7.41}$$

where w_b is the initial weight and t_b is the moment growth begins. (This equation can be obtained from the expression

$$dw = \int c_m\left(\frac{t_e-t}{t_e-t_m}\right)\left(\frac{t-t_b}{t_m-t_b}\right)^{(t_m-t_b)/(t_e-t_m)} dt$$

using the restriction that $w = w_b$ if $t = t_b$.) Equation 7.41 also predicts as sigmoidal growth pattern over the period $t_b \le t \le t_e$. Additionally, if $w_b \ne 0$, Equation 7.41 does not admit an infinite relative growth rate at time t_b.

To summarize:

1. The beta growth function (Eq. 7.38) generates a family of asymmetric growth curves over the time period $0 \le t \le t_e$ by varying t_m. It effectively represents a generalized polynomial equation.
2. Equation 7.38, unlike the truncated expolinear growth function, smoothly predicts w_m as the final weight at the termination of growth.
3. Equation 7.38 predicts a zero growth rate at both the start and end of growth.
4. The beta function can specify seed weight and its growth duration, thus allowing the full weight to be attained exactly within the growth period.

EXAMPLE 7.3 To estimate the parameters of the beta growth function (Eq. 7.38) or

$$w = w_m\left(1+\frac{t_e-t}{t_e-t_m}\right)\left(\frac{t}{t_e}\right)^{t_e/(t_e-t_m)}, 0 \le t_m < t_e,$$

let us employ the Example 7.2 data set (Table 7.3). The PROC NLIN code appears as Exhibit 7.3.

EXHIBIT 7.3 SAS Code for Beta Growth Curve Estimation

```
data drybeta;
       input t w;
datalines;
 1     1.2
 2     1.5
 .       .
 .       .
 .       .
12     16.4
;
proc plot data=drybeta;
     plot w*t='*';
run;
proc nlin data=drybeta hougaard maxiter=1000 noitprint;
        parms tm=0.5
              te=2
              wm=0.5;
              u=(te-t)/(te-tm); ①
              v=1+u;
              z=te/(te-tm);
              x=(t/te)**z;
        model w=wm*v*x; ②
run;
```

① Initial values for the parameters are selected, and variable transformations for coding are made.

② Model statement for the beta growth equation.

The output generated by this SAS code appears in Table 7.5.

① The small p-value for the computed F value indicates that the model fits the data very well.

② The estimated beta growth function appears as

$$w = 21.2075\left(1+\frac{16.7521-t}{8.5008}\right)\left(\frac{t}{16.7521}\right)^{1.97065}.$$

③ None of the Wald 95% approximate confidence intervals contains zero, thus indicating that all three parameters are significantly different from zero (at the 5% level).

④ (i) With $|g_{11}|=3.7289$, $|g_{12}|=3.4896$, and $|g_{13}|=3.9137$, we see that $\hat{t}_m$, $\hat{t}_e$, and $\hat{w}_m$ are highly skewed away from linear behavior. ■

TABLE 7.5 Output for Example 7.3

The SAS system
The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	10
Subiterations	2
Average subiterations	0.2
R	8.398E-6
PPC (*te*)	6.305E-6
RPC (*te*)	0.00002
Object	8.98E-10
Objective	3.44893
Observations read	12
Observations used	12
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① *F* value	Approx Pr > *F*
Model	3	1109.5	369.8	965.11	<0.0001
Error	9	3.4489	0.3832		
Uncorrected total	12	1113.0			

Parameter	② Estimate	Approx std error	③ Approximate 95% confidence limits		④ Skewness
tm	8.2513	1.2291	5.4708	11.0318	3.7289
te	16.7521	4.4527	6.6793	26.8249	3.4896
wm	21.2075	5.7141	8.2813	34.1337	3.9137

Approximate correlation matrix

	t_m	t_e	w_m
t_m	1.0000000	0.9484060	0.9751025
t_e	0.9484060	1.0000000	0.9916146
w_m	0.9751025	0.9916146	1.0000000

7.7 ASYMMETRIC GROWTH EQUATIONS (FOR PLANT PARTS)

The *asymmetric class of growth equations* is useful for modeling the time path of plant components (e.g., green leaf area, root length, and shoot weight) in that the said path normally exhibits an initial period of increase, attains a maximum, and then subsequently declines over the growing season (Werker and Jaggard, 1997). This behavior is due to the net effect of growth and senescence, with changes in the rates of growth and senescence over the growing season precipitated by changes in the environment or in the life cycle of the plant itself. The asymmetric models presented herein all have growth trajectories that rise and then fall during the growing season, and each trajectory has a turning point and a zero asymptote.

For a plant component Y, let us model the temporal dynamics of net growth as

$$\frac{dY}{dt} = \beta Y - \alpha Y = \mu Y, \tag{7.42}$$

where β and α are the relative rates of growth and senescence, respectively. Here $\mu = \beta - \alpha$ is the *net relative growth rate*.

7.7.1 Model I

We start with the linked differential equations

$$\frac{dY}{dt} = \mu Y; \tag{7.43a}$$

$$\frac{d\mu}{dt} = -k\left(\mu - \mu_{\min}\right). \tag{7.43b}$$

Here Equation 7.43b posits that μ decays exponentially at a constant rate k that is proportional to its distance from the final net relative growth rate $\mu_{\min}$ as t increases without bound. The solution to this system is (see Appendix 7.F)

$$\mu = \mu_{\min} + \left(\mu_0 - \mu_{\min}\right)e^{k(t-t_0)}; \tag{7.44a}$$

$$Y = Y_0 \exp\left\{\mu_{\min}\left(t - t_o\right) + \frac{\mu_0 - \mu_{\min}}{k}\left[1 - e^{-k(t-t_0)}\right]\right\}, \tag{7.44b}$$

where μ_0 and Y_0 are the initial relative growth rates in μ and Y at time t_0, respectively, and k is the speed at which μ_0 declines to $\mu_{\min}$ as $t \to \infty$. When $\mu_{\min} < 0$, Y has a *turning point* where $\mu = 0$, and this occurs at the point ($t_{\max}$, $Y_{\max}$) with coordinates

$$t_{\max} = t_0 - \frac{1}{k}\ln\left(\frac{-\mu_{\min}}{\mu_0 - \mu_{\min}}\right); \tag{7.45a}$$
$$\left(\text{set } \mu = 0 \text{ in Equation 7.44.a and solve for } t = t_{\max}\right)$$

$$Y_{\max} = Y_0 \exp\left\{\frac{\mu_0}{k} - \frac{\mu_{\min}}{k}\ln\left(\frac{-\mu_{\min}}{\mu_0 - \mu_{\min}}\right)\right\}. \tag{7.45b}$$
$$\left(\text{set } t = t_{\max} \text{ in Equation 7.44.b}\right)$$

Beyond the turning point, Y decays and tends to zero.

For purposes of parameter estimation, Werker and Jaggard (1997) eliminate μ_0 from Equations 7.44a and 7.44b and work with

$$\mu = \mu_{\min}\left(1 - e^{k(t - t_{\max})}\right); \quad (7.46a)$$

$$Y = Y_{\max}\exp\left\{\mu_{\min}\left(t - t_{\max}\right) - \frac{\mu_{\min}}{k}\left[1 - e^{-k(t - t_{\max})}\right]\right\}. \quad (7.46b)$$

In this instance $\mu_0 = \mu_{\min}\{1 - \exp[-k(t - t_{\max})]\}$.

7.7.2 Model II

In the second model, additional flexibility in the behavior of μ is introduced by rewriting Equations 7.43a and 7.43b as

$$\frac{dY}{dt} = \mu Y; \quad (7.47a)$$

$$\frac{d\mu}{dt} = -k\left(\mu - \mu_{\min}\right)\left(\frac{\mu_{\max} - \mu}{\mu_{\max} - \mu_{\min}}\right), \quad (7.47b)$$

where now $d\mu/dt$ is quadratic in μ. The solution to this linked system is (Appendix 7.F)

$$\mu = \mu_{\min} + \frac{\left(\mu_{\max} - \mu_{\min}\right)\left(\mu_0 - \mu_{\min}\right)e^{-k(t - t_0)}}{\left(\mu_{\max} - \mu_0\right) + \left(\mu_0 - \mu_{\min}\right)e^{-k(t - t_0)}}; \quad (7.48a)$$

$$Y = Y_0\left\{\left(\frac{\mu_{\max} - \mu_0}{\mu_{\max} - \mu_{\min}}\right)\exp\left[\frac{-\mu_{\min}k\left(t - t_0\right)}{\mu_{\max} - \mu_{\min}}\right] + \left(\frac{\mu_0 - \mu_{\min}}{\mu_{\max} - \mu_{\min}}\right)\exp\left[\frac{-\mu_{\max}k\left(t - t_0\right)}{\mu_{\max} - \mu_{\min}}\right]\right\}^{\frac{\mu_{\max} - \mu_{\min}}{-k}} \quad (7.48b)$$

with the additional parameter $\mu_{\max}$ allowing for a maximum relative growth rate, that is, $\mu \to \mu_{\max}$ as $t \to -\infty$ (in Model I, $\mu \to +\infty$ as $t \to -\infty$). Now the change in the relative growth rate is sigmoidal or S-shaped rather than exponential (as in Model I). In the presence of the additional parameter $\mu_{\max}$, $\mu_{\max}$ and $\mu_{\min}$ serve as theoretical limits on the relative growth rate of Y (μ), and the parameter k determines how fast the relative growth rate changes from $\mu_{\max}$ to $\mu_{\min}$. Here too, when $\mu_{\min} < 0$, Y has a turning point ($t_{\max}$, $Y_{\max}$) where $\mu = 0$, with

$$t_{\max} = t_0 - \frac{1}{k}\left[\ln\left(\frac{-\mu_{\min}}{\mu_0 - \mu_{\min}}\right) - \ln\left(\frac{\mu_{\max}}{\mu_{\max} - \mu_0}\right)\right]; \quad (7.49a)$$

$$Y_{\max} = Y_0 \exp\left[\frac{-\mu_{\min}}{k}\ln\left(\frac{-\mu_{\min}}{\mu_0-\mu_{\min}}\right)+\frac{\mu_{\max}}{k}\ln\left(\frac{\mu_{\max}}{\mu_{\max}-\mu_0}\right)\right]. \tag{7.49b}$$

Upon eliminating μ_0 from Equations 7.48a and 7.48b, we obtain

$$\mu = \mu_{\min} + \frac{\mu_{\min}}{(A-1)-A\,e^{-k(t-t_{\max})}}; \tag{7.50a}$$

$$Y = Y_{\max}\left[A\,e^{-(A-1)k(t-t_{\max})}-(A-1)e^{-Ak(t-t_{\max})}\right]^{-\mu_{\min}/k(A-1)}, \tag{7.50b}$$

where $A=\mu_{\max}/(\mu_{\max}-\mu_{\min})$.

7.7.3 Model III

Suppose we modify Model I by introducing a carrying capacity or maximum value term θ for Y in dY/dt such that $Y \to \theta$ for $\mu > 0$ (the carrying capacity θ has the effect of slowing down the growth rate in Y), that is,

$$\frac{dY}{dt} = \mu Y\left(\frac{\theta - Y}{\theta}\right); \tag{7.51a}$$

$$\frac{d\mu}{dt} = -k\left(\mu - \mu_{\min}\right). \tag{7.51b}$$

The solution to this system (Appendix 7.F) can be written as

$$Y = \frac{\theta}{1+\left(\left(\theta - Y_0\right)/Y_0\right)e^{-u}}, \tag{7.52}$$

where

$$u = \mu_{\min}\left(t-t_0\right)+\frac{\mu_0-\mu_{\min}}{k}\left(1-e^{-k(t-t_0)}\right). \tag{7.53}$$

When $\mu_{\min}<0$, Y has a turning point ($t_{\max}$, $Y_{\max}$) where $\mu=0$, with

$$t_{\max} = t_0 - \frac{1}{k}\ln\left(\frac{-\mu_{\min}}{\mu_0-\mu_{\min}}\right); \tag{7.54a}$$

$$Y_{\max} = \frac{\theta}{1+\left(\left(\theta - Y_0\right)/Y_0\right)e^{-u_{\max}}}, \tag{7.54b}$$

where

$$u_{\max} = -\frac{\mu_{\min}}{k}\left[\ln\left(\frac{-\mu_{\min}}{\mu_0-\mu_{\min}}\right)-\frac{\mu_0}{\mu_{\min}}\right]. \tag{7.55}$$

Eliminating μ_0 from Equations 7.52 and 7.53 renders

$$Y = \frac{\theta}{1 + \left(\left(\theta - Y_{\max} \right) / Y_{\max} \right) e^{-u'}}, \tag{7.56}$$

where

$$u' = \mu_{\min} \left(t - t_{\max} \right) - \frac{\mu_{\min}}{k} \left(1 - e^{-k\left(t - t_{\max}\right)} \right). \tag{7.57}$$

Some of the salient features of these models are:

1. When $\mu_{\min} = 0$ in Equations 7.44a and 7.44b (or in Model I), we obtain the Gompertz growth function (Eq. 3.10) and Y approaches the asymptote $Y_{\max}$.
2. When $\mu_{\min} = 0$ in Equations 7.48a and 7.48b (Model II), the Richards growth function (Eq. 3.D.7) emerges.
3. When $\mu_{\min} = 0$ in Equation 7.52 (Model III), the *Chanter growth function* (Appendix 7.G) obtains. (Models I, II, and III can thus be viewed as extensions of the Gompertz, Richards, and Chanter growth models—they differ by admitting an additional parameter ($\mu_{\min}$), which, following a sigmoidal rise in Y, determines the subsequent net rate of decline. This gives Models I, II, and III their unique turning points.)
4. An important distinction between Model I and Model II is found in their relative growth rates (μ): the time path for Model II is sigmoidal; for Model I it is exponential. Additionally, the parameter $\mu_{\max}$ exercises considerable control over both the speed and the timing in which μ changes from its initial value to its terminal value.
5. Models I and II display infinite relative growth rates for infinitesimally small values of Y. Model III differs from Models I and II as Y approaches θ, when the apparent growth and decay rates slow down, allowing for a more sustained period during which Y is near $Y_{\max}$.

APPENDIX 7.A DERIVATION OF THE SHINOZAKI AND KIRA YIELD–DENSITY CURVE

Shinozaki and Kira (1956) derived their yield–density model by combining a logistic differential equation of (population) growth

$$\frac{dY}{dt} \frac{1}{Y} = \lambda \left(1 - \frac{Y}{Y_\infty} \right), \tag{7.A.1}$$

where, at a given time t, Y is the plant dry weight, Y_∞ is the asymptote of Y, and λ is the growth coefficient (independent of plant density X), with the law of constant final yield

$$Y_\infty = w_\infty X = k = \text{constant}, \tag{7.A.2}$$

where k is the final yield per unit area (which is independent of X). From Equation 7.A.1, we can obtain

$$Y = \frac{Y_\infty}{1 + ce^{-\lambda t}} \tag{7.A.3}$$

(see the discussion underlying Equation 3.A.7), where c is a constant of integration. For $t = 0, Y_o = Y_\infty/(1 + c)$ or $c = (Y_\infty/Y_o) - 1$. Then from Equation 7.A.2, Equation 7.A.3 can be rewritten as

$$Y = \frac{w_\infty X}{1 + \left(\left(Y_\infty / Y_o\right) - 1\right)e^{-\lambda t}}$$

or

$$\begin{aligned}\frac{X}{Y} &= \frac{1 - e^{-\lambda t}}{w_\infty} + \frac{Y_\infty}{w_\infty Y_o} e^{-\lambda t} \\ &= \frac{1 - e^{-\lambda t}}{w_\infty} + \frac{w_\infty X}{w_\infty Y_o} e^{-\lambda t}\end{aligned}$$

or

$$\frac{1}{w} = a + bX, \tag{7.A.4}$$

where $w = Y/X$ and

$$a = \frac{1 - e^{-\lambda t}}{w_\infty}, \quad b = \frac{e^{-\lambda t}}{Y_o}.$$

APPENDIX 7.B DERIVATION OF THE FARAZDAGHI AND HARRIS YIELD–DENSITY CURVE

Farazdaghi and Harris (1968) combine the logistic growth law (Eq. 7.A.3) or

$$Y = \frac{Y_\infty}{1 + ce^{-\lambda t}},$$

where $c = (Y_\infty/Y_o) - 1$, with their modified law of constant final yield

$$Y_\infty = w_\infty X^\gamma = q = \text{constant} \tag{7.B.1}$$

to obtain

$$Y = \frac{w_\infty X^\gamma}{1 + \left(Y_\infty / Y_o - 1\right)e^{-\lambda t}}.$$

Then

$$\begin{aligned}\frac{X^{\gamma}}{Y} &= \frac{1-e^{-\lambda t}}{w_{\infty}} + \frac{Y_{\infty}}{w_{\infty} Y_{o}} e^{-\lambda t} \\ &= \frac{1-e^{-\lambda t}}{w_{\infty}} + \frac{w_{\infty} X^{\lambda}}{w_{\infty} Y_{o}} e^{-\lambda t}\end{aligned}$$

or

$$\frac{1}{w} = a + bX^{\gamma},$$

where $w = Y/X^{\gamma}$ and

$$a = \frac{1-e^{-\lambda t}}{w_{\infty}}, \qquad b = \frac{e^{-\lambda t}}{Y_{o}}.$$

APPENDIX 7.C DERIVATION OF THE BLEASDALE AND NELDER YIELD–DENSITY CURVE

Bleasdale and Nelder (1960) derive their yield–density model by coupling the Richards (1959) generalized logistic growth function (Eq. 3.D.11) or

$$Y = Y_{\infty}\left(1+\alpha e^{-\beta r t}\right)^{-1/r}, \alpha = \left(\frac{Y_{\infty}}{Y_{o}}\right)^{r} - 1, \text{ and } \beta, r > 0 \tag{7.C.1}$$

with an expression representing the law of constant final yield (Eq. 7.A.2) or

$$Y_{\infty} = w_{\infty} X = k = \text{constant}. \tag{7.C.2}$$

Then from Equations 7.C.1 and 7.C.2,

$$Y = \frac{w_{\infty} X}{\left[1+\left(\left(Y_{\infty}/Y_{o}\right)^{r}-1\right)e^{-\beta r t}\right]^{1/r}}$$

or

$$\begin{aligned}Y^{-r} &= w_{\infty}{}^{-r} X^{-r}\left[1+\left(\left(\frac{Y_{\infty}}{Y_{o}}\right)^{r}-1\right)e^{-\beta r t}\right] \\ &= w_{\infty}{}^{-r} X^{-r}\left[1+\left(\frac{w_{\infty} X}{Y_{o}}\right)^{r} e^{-\beta r t} - e^{-\beta r t}\right] \\ &= \left(\frac{1-e^{-\beta r t}}{w_{\infty}^{r}}\right) X^{-r} + \frac{e^{-\beta r t}}{Y_{o}^{\,r}}\end{aligned}$$

or

$$X^r Y^{-r} = \frac{1-e^{-\beta r t}}{w_\infty{}^r} + \left(\frac{e^{-\beta r t}}{Y_o{}^r}\right) X^r \tag{7.C.3}$$
$$= A + BX^r,$$

where

$$A = \frac{1-e^{-\beta r t}}{w_\infty{}^r}, \qquad B = \frac{e^{-\beta r t}}{Y_o^r}.$$

Upon setting $r = \theta$, we obtain the Bleasdale and Nelder family of yield–density curves

$$\frac{1}{w^\theta} = A + BX^\theta,\ A, B, \text{and } \theta \text{ parameters}, 0 < \theta < 1, \tag{7.C.4}$$

and $w = Y/X$. Under a reparameterization of Equation 7.C.4, we have

$$\frac{1}{w^\theta} = A + BX^\phi,\ \theta < \phi. \tag{7.C.4.1}$$

Under what circumstances is this reparameterization justified? If in Equation 7.C.1 we substitute a modified law of constant final yield

$$Y_\infty = w_\infty X^a = q = \text{constant}, \tag{7.C.2.1}$$

then

$$Y = \frac{w_\infty X^a}{\left[1 + \left(\left(Y_\infty / Y_o\right)^r - 1\right) e^{-\beta r t}\right]^{1/r}}.$$

From this expression we can readily obtain

$$Y^{-r} = w_\infty{}^{-r} X^{-ar}\left[1 + \left(\left(\frac{Y_\infty}{Y_o}\right)^r - 1\right) e^{-\beta r t}\right]$$
$$= w_\infty{}^{-r} X^{-ar}\left(1 - e^{-\beta r t}\right) + \frac{e^{-\beta r t}}{Y_0^r}$$

or

$$X^{ar} Y^{-r} = \frac{1-e^{-\beta r t}}{w_\infty{}^r} + \left(\frac{e^{-\beta r t}}{Y_o{}^r}\right) X^{ar}$$

or

$$\frac{1}{w^r} = A' + B' A^{ar}, \tag{7.C.5}$$

where $w = Y/X^a$ and

$$A' = \frac{1-e^{-\beta rt}}{w_\infty^{\ r}}, \quad B' = \frac{e^{-\beta rt}}{Y_o^{\ r}}.$$

If we set $r = \theta$ and $ar = \phi$, then Equation 7.C.5 becomes

$$\frac{1}{w^\theta} = A' + B'X^\phi, \theta < \phi. \tag{7.C.6}$$

Note that the restriction $\theta < \phi$ requires that $a > 1$.

APPENDIX 7.D DERIVATION OF THE EXPOLINEAR GROWTH CURVE

We determined in Section 7.5 that the growth rate of leaf area L could be expressed as

$$\frac{dL}{dt} = \left(\frac{r_m}{k}\right)\left(1-e^{-kL}\right), \tag{7.D.1}$$

where r_m and k are taken to be constant. Then Equation 7.D.1 can be rewritten as

$$\frac{kdL}{1-e^{-kL}} = r_m dt$$

or

$$\frac{ke^{kL}dL}{e^{kL}-1} = r_m dt$$

and thus

$$\ln\left(e^{kL}-1\right) = r_m t + \ln\left(e^{kL_o}-1\right),$$

$$e^{kL}-1 = \left(e^{kL_o}-1\right)e^{r_m t},$$

$$e^{kL} = 1 + \left(e^{kL_o}-1\right)e^{r_m t},$$

$$kL = \ln\left[1+\left(e^{kL_o}-1\right)e^{r_m t}\right],$$

$$L = \frac{1}{k}\ln\left[1+\left(e^{kL_o}-1\right)e^{r_m t}\right], \tag{7.D.2}$$

where L_o is the initial leaf area index value.

From Equations 7.29 and 7.D.2,

$$\begin{aligned}\frac{dw}{dt} &= c_m f = c_m\left(1-e^{-kL}\right)\\ &= c_m\left\{1-e^{-\ln\left[1+\left(e^{-kL_o}-1\right)e^{r_m t}\right]}\right\}\\ &= c_m\left[\frac{\left(e^{kL_o}-1\right)e^{r_m t}}{1+\left(e^{kL_o}-1\right)e^{r_m t}}\right].\end{aligned}$$

Then

$$dw = c_m\left[\frac{\left(e^{kL_o}-1\right)e^{r_m t}}{1+\left(e^{kL_o}-1\right)e^{r_m t}}\right]dt$$

and thus

$$w = c_m \ln\left[1+\left(e^{kL_o}-1\right)e^{r_m t}\right]\frac{1}{r_m}. \tag{7.D.3}$$

Note that since $f_o = 1-e^{-kL_o}$ or

$$e^{kL_o}-1 = \frac{f_o}{1-f_o},$$

it follows that Equation 7.D.3 can be rewritten as

$$w = \left(\frac{c_m}{r_m}\right)\ln\left[1+\left(\frac{f_0}{1-f_0}\right)e^{r_m t}\right]. \tag{7.D.4}$$

For t large, the increase in biomass occurs at the (approximate) rate c_m. If we omit the term “1” in Equation 7.D.4, w can be respecified as

$$w = \left(\frac{c_m}{r_m}\right)\ln\left(\frac{f_0}{1-f_0}\right)+c_m t$$

(see Fig. 7.3). When $w=0$, the intercept with the time-axis is

$$t = t_b = -\frac{1}{r_m}\ln\left(\frac{f_0}{1-f_0}\right) \tag{7.D.5}$$

or

$$\left(\frac{f_0}{1-f_0}\right) = e^{-r_m t_b}. \tag{7.D.6}$$

Then a substitution of Equation 7.D.6 into Equation 7.D.4 renders

$$w = \left(\frac{c_m}{r_m}\right) \ln\left[1 + e^{r_m(t-t_b)}\right], \tag{7.D.7}$$

the *expolinear growth equation*. Thus, the dry weight of a crop stand can be specified as a function of its maximum growth rate (c_m), its maximum relative growth rate (r_m), and a parameter indicating the position of the function on the time-axis (so-called "lost time" t_b). In this regard, the *effective loss of biomass* ($c_m\, t_b$) can be expressed (via Eqs. 7.32 and 7.D.5) as

$$c_m t_b = \frac{-\ln\left(f_0 / 1 - f_0\right)}{k p_1 s} \tag{7.D.8}$$

(see also Eq. 7.35).

APPENDIX 7.E DERIVATION OF THE BETA GROWTH FUNCTION

The *beta growth function* was first introduced by Yin et al. (1995) to describe the effects of the temperature (T) on the phasic rate of crop development. This function employs the cardinal temperatures: the base temperature (T_b), the optimum temperature (T_o), and the ceiling temperature (T_c). To model the relationship between the crop development rate (DR) and temperature, the beta function (generally used to represent a flexible family of nonsymmetric and unimodal probability density functions with two fixed end points and points of inflection on either side of the mode) was selected to smoothly describe the response of the DR to temperatures between T_b and T_c or

$$\mathrm{DR} = e^{\mu}\left(T - T_b\right)^{\alpha}\left(T_c - T\right)^{\beta}, \tag{7.E.1}$$

where μ, α, and β are parameters (values for T_b and T_c may or may not be given). Here α serves to determine the curvature of this relationship for $T<T_o$, while β determines the curvature of this expression when $T>T_o$. Clearly different combinations of α and β provide us with great flexibility in modeling DR as a function of temperature.

A glance back at Equation 7.E.1 reveals that the beta growth function becomes:

1. A linear model in T for $\alpha=1$ and $\beta=0$ or

$$\mathrm{DR} = e^{\mu}\left(T - T_b\right)$$

2. A power law when $\beta = 0$ or

$$\mathrm{DR} = e^{\mu}\left(T - T_b\right)^{\alpha}$$

3. A quadratic in T for $\alpha = \beta = 1$ or

$$\mathrm{DR} = e^{\mu}\left(T - T_b\right)\left(T_c - T\right)$$

4. A symmetric growth model when $\alpha = \beta$ or

$$\mathrm{DR} = e^{\mu}\left(T - T_b\right)^{\alpha}\left(T_c - T\right)^{\alpha}$$

Note that Equation 7.E.1 does not include T_o and the maximum DR as parameters, but estimates of T_o and maximum DR can be obtained as follows:

1. Setting

$$\frac{d\mathrm{DR}}{dT} = e^{\mu}\left(T - T_b\right)^{\alpha}\left(T_c - T\right)^{\beta}\left(\frac{\alpha}{T - T_b} - \frac{\beta}{T_c - T}\right) = 0$$

allows us to find the T value which makes $d\mathrm{DR}/dT$ vanish or

$$T_o = \frac{\alpha T_c + \beta T_b}{\alpha + \beta}. \tag{7.E.2}$$

2. Substituting T_o into Equation 7.E.1 yields the maximum value of DR (provided, of course, that $d^2\mathrm{DR}/dT^2 < 0$ at T_o) or

$$\mathrm{DR}_o = e^{\mu}\alpha^{\alpha}\beta^{\beta}\left(\frac{T_c - T_b}{\alpha + \beta}\right)^{\alpha+\beta}. \tag{7.E.3}$$

The importance of Equation 7.E.1 is that both low- and high-temperature effects are included in a single equation. Moreover, asymmetric responses in DR to temperature can be flexibly modeled along with any inflection in response for $T < T_o$ or $T > T_o$. (The temperature at which any point of inflection occurs can be found by setting $d^2\mathrm{DR}/dT^2 = 0$.) In this regard, it can be demonstrated that: (i) a point of inflection occurs in the range $T < T_o$ only if $\alpha > 1$ and $\beta < 1$; (ii) likewise a point of inflection occurs in the range $T > T_o$ only if $\beta > 1$ and $\alpha < 1$; and (iii) a point of inflection occurs in both ranges only if $\alpha > 1$ and $\beta > 1$.

To derive a new or alternative beta growth function, Yin et al. (2003) describe the time course of the crop's growth rate (assuming that the time at the beginning of the growth period (t_b) is zero) as

$$\frac{dw}{dt} = c_m\left(\frac{t_e - t}{t_e - t_m}\right)\left(\frac{t}{t_m}\right)^{t_m/(t_e - t_m)}, \tag{7.E.4}$$

where c_m is the maximum growth rate (achieved at time t_m). Then

$$\begin{aligned} w &= \int c_m \left(\frac{t_e - t}{t_e - t_m}\right)\left(\frac{t}{t_m}\right)^{t_m/(t_e-t_m)} dt^5 \\ &= \frac{c_m}{t_e - t_m}\left(\frac{1}{t_m}\right)^{t_m/(t_e-t_m)} \int (t_e - t)\, t^{t_m/(t_e-t_m)} dt. \end{aligned} \tag{7.E.5}$$

A substitution of Equation 7.E.7 into Equation 7.E.5 yields

$$\begin{aligned} w &= \frac{c_m}{t_e - t_m}\left(\frac{1}{t_m}\right)^{t_m/(t_e-t_m)} \left(\frac{t^{m+1}}{m+2}\right)(2t_e - t_m - t) \\ &= c_m t \left(\frac{2t_e - t_m - 1}{2t_e - t_m}\right)\left(\frac{t}{t_m}\right)^{t_m/(t_e-t_m)}. \end{aligned} \tag{7.E.6}$$

(The constant of integration is set to zero from the condition that $w=0$ at $t=0$.)

Let us set $t=t_e$ in Equation 7.E.8. Then from Equation 7.E.4,

$$\begin{aligned} w_m &= \int_0^{t_e} \frac{dw}{dt} dt \\ &= c_m t_e \left(\frac{t_e - t_m}{2t_e - t_m}\right)\left(\frac{t_e}{t_m}\right)^{t_m/(t_e-t_m)}. \end{aligned} \tag{7.E.7}$$

Then dividing Equation 7.E.8 by Equation 7.E.9 gives

$$\frac{w}{w_m} = \left(\frac{2t_e - t_m - t}{t_e - t_m}\right)\left(\frac{t}{t_e}\right)^{\frac{t_m}{t_e - t_m}+1},$$

[5] In general,

$$\int (a+bx)^n x^m dx = \frac{x^{m+1}(a+bx)^n}{m+n+1} + \frac{an}{m+n+1}\int x^m (a+bx)^{n-1} dx. \tag{7.E.8}$$

Set $a=t_e, b=-1, n=1, m=t_m/(t_e-t_m),\ m+1=t_e/(t_e-t_m)$.

Then Equation 7.E.6 becomes

$$\frac{t^{m+1}(t_e - t)}{m+2} + \frac{t_e}{m+2}\int t^m dt = \frac{t^{m+1}(t_e - t)}{m+2} + \frac{t_e}{m+2}\frac{1}{m+1}t^{m+1} = \frac{t^{m+1}}{m+2}(2t_e - t_m - t), \tag{7.E.9}$$

where $m+2=(2t_e-t_m)/(t_e-t_m)$.

and a reformulation of this expression renders

$$w = w_m\left(1+\frac{t_e-t}{t_e-t_m}\right)\left(\frac{t}{t_e}\right)^{t_e/(t_e-t_m)}, 0 \le t_m < t_e. \tag{7.E.10}$$

APPENDIX 7.F DERIVATION OF ASYMMETRIC GROWTH EQUATIONS

Model I

Consider the system

$$\frac{dY}{dt} = \mu Y; \tag{7.F.1a}$$

$$\frac{d\mu}{dt} = -k\left(\mu - \mu_{\min}\right). \tag{7.F.1b}$$

Let us rewrite Equation 7.F.1a as

$$\frac{dY}{Y} = \mu dt.$$

Then

$$\ln Y = \mu t + \ln C, \quad (\ln C \text{ constant})$$

$$Y = Ce^{\mu t}. \tag{7.F.2}$$

At the initial point (t_0, Y_0), $Y_0 = Ce^{\mu t_0}$ so that $C = Y_0 e^{-\mu t_0}$. Substituting this expression for C into Equation 7.F.2 yields

$$Y = Y_0 e^{\mu(t-t_0)}. \tag{7.F.3}$$

Rewriting Equation 7.F.1b as $d\mu/dt = a + b\mu$ (where $a = k\mu_{\min}$ and $b = -k$) leads to

$$\frac{d\mu}{a+b\mu} = dt,$$

$$\frac{1}{b}\ln(a+b\mu) = t + \ln C, \quad (\ln C \text{ constant})$$

$$\mu = \frac{1}{b}e^{bt}C^b - \frac{a}{b}. \tag{7.F.4}$$

For $t = t_0$,

$$\mu_0 = \frac{1}{b} e^{bt_0} C^b - \frac{a}{b}$$

and thus $C^b = e^{-bt_0}\left(b\mu_0 + a\right)$. A substitution of C^b into Equation 7.F.4 gives

$$\mu = e^{b(t-t_0)}\left(\mu_0 + \frac{a}{b}\right) - \frac{a}{b}.$$

From the definitions of a and b, we ultimately obtain

$$\mu = \mu_{\min} + \left(\mu_0 - \mu_{\min}\right) e^{-k(t-t_0)}. \tag{7.F.5}$$

Then a substitution of Equation 7.F.5 into Equation 7.F.3 yields Equation 7.44b.

Model II

Given the system

$$\frac{dY}{dt} = \mu Y; \tag{7.F.6a}$$

$$\frac{d\mu}{dt} = -k\left(\mu - \mu_{\min}\right)\left(\frac{\mu_{\max} - \mu}{\mu_{\max} - \mu_{\min}}\right), \tag{7.F.6b}$$

let us rewrite Equation 7.F.6b as

$$\frac{d\mu}{\left(\mu - \mu_{\min}\right)\left(\mu_{\max} - \mu\right)} = \left(\frac{-k}{\mu_{\max} - \mu_{\min}}\right) dt.$$

Then

$$\frac{1}{\mu_{\min} - \mu_{\max}} \ln\left(\frac{\mu_{\max} - \mu}{\mu - \mu_{\min}}\right) = \left(\frac{-k}{\mu_{\max} - \mu_{\min}}\right) t + \ln C, \quad (\ln C \text{ constant})$$

$$\left(\frac{\mu_{\max} - \mu}{\mu - \mu_{\min}}\right) = e^{kt} C^{\mu_{\min} - \mu_{\max}}. \tag{7.F.7}$$

Upon substituting (for $t = t_0$, $Y = Y_0$)

$$C^{\mu_{\min} - \mu_{\max}} = e^{-kt_0}\left(\frac{\mu_{\max} - \mu_0}{\mu_0 - \mu_{\min}}\right)$$

into Equation 7.F.7, we obtain

$$\left(\frac{\mu_{\max}-\mu}{\mu-\mu_{\min}}\right)=e^{k(t-t_0)}\left(\frac{\mu_{\max}-\mu_0}{\mu_0-\mu_{\min}}\right). \tag{7.F.8}$$

Then solving this expression for μ yields

$$\mu=\mu_{\min}+\frac{(\mu_{\max}-\mu_{\min})(\mu_0-\mu_{\min})e^{-k(t-t_0)}}{(\mu_{\max}-\mu_0)+(\mu_0-\mu_{\min})e^{-k(t-t_0)}}, \tag{7.F.9}$$

and substituting Equation 7.F.9 into Equation 7.F.3 gives us Equation 7.48b.

Model III

For the system

$$\frac{dY}{dt}=\mu Y\left(\frac{\theta-Y}{\theta}\right); \tag{7.F.10a}$$

$$\frac{d\mu}{dt}=-k(\mu-\mu_{\min}), \tag{7.F.10b}$$

let us rewrite Equation 7.F.10a as

$$\frac{dY}{Y(\theta-Y)}=\frac{\mu}{\theta}dt.$$

Then

$$-\frac{1}{\theta}\ln\left(\frac{\theta-Y}{Y}\right)=\frac{\mu}{\theta}t+\ln C,\quad (\ln C \text{ constant})$$

$$\frac{\theta-Y}{Y}=e^{\mu t}C^{-\theta},$$

$$Y=\frac{\theta}{1+e^{-\mu t}C^{-\theta}}. \tag{7.F.11}$$

At the point (t_0, Y_0),

$$C^{-\theta}=\left(\frac{\theta-Y_0}{Y_0}\right)e^{\mu t_0}.$$

Then substituting $C^{-\theta}$ into Equation 7.F.11 yields

$$Y=\frac{\theta}{1+\left((\theta-Y_0)/Y_0\right)e^{-\mu(t-t_0)}}. \tag{7.F.12}$$

A further substitution of Equation 7.F.5 into Equation 7.F.12 produces Equations 7.52 and 7.53.

APPENDIX 7.G CHANTER GROWTH FUNCTION

The Chanter growth function is essentially an amalgam of the logistic and Gompertz equations (Eqs. 3.6 and 3.10, respectively) (Chanter, 1976; France and Thornley, 1984). Let us modify the instantaneous rate of growth in dry matter Y_t, $(1/Y_t)\ (dY_t/dt)=\mu$, by two factors: (i) from the logistic, we use $1-(Y_t/Y_\infty)$, and (ii) from the Gompertz, we use $e^{-\beta t}$. Then

$$\frac{dY_t\,/\,dt}{Y_t}=\mu\left(1-\frac{Y_t}{Y_\infty}\right)e^{-\beta t}$$

or

$$\frac{dY_t}{Y_t\left(Y_\infty-Y_t\right)}=\frac{\mu}{Y_\infty}e^{-\beta t}. \qquad (7.G.1)$$

Upon integrating Equation 7.G.1, we obtain

$$-\frac{1}{Y_\infty}\ln\left(\frac{Y_\infty-Y_t}{Y_t}\right)=\frac{\mu}{Y_\infty}\left(-\frac{1}{\beta}e^{-\beta t}\right)+\ln C, \quad (\ln C\,\text{constant})$$

$$\left(\frac{Y_\infty-Y_t}{Y_t}\right)=\frac{e^{(\mu/\beta)e^{-\beta t}}}{C^{Y_\infty}},$$

$$Y_t=\frac{Y_\infty}{1+C^{-Y_\infty}e^{(\mu/\beta)e^{-\beta t}}}. \qquad (7.G.2)$$

For $t=0$,

$$Y_0=\frac{Y_\infty}{1+C^{-Y_\infty}e^{(\mu/\beta)}}$$

and thus

$$C^{-Y_\infty}=\left(\frac{Y_\infty}{Y_t}-1\right)e^{-(\mu/\beta)}. \qquad (7.G.3)$$

Then substituting Equation 7.G.3 into Equation 7.G.2 yields the *Chanter growth equation*

$$Y_t=\frac{Y_0Y_\infty}{Y_0+\left(Y_\infty-Y_0\right)e^{-(\mu/\beta)\left(1-e^{-\beta t}\right)}}. \qquad (7.G.4)$$

8

NONLINEAR MIXED-EFFECTS MODELS FOR REPEATED MEASUREMENTS DATA

8.1 SOME BASIC TERMINOLOGY CONCERNING EXPERIMENTAL DESIGN

We may view a *factor* as an independent variable that is related to or used to predict a response variable Y. The value or intensity setting assumed by a factor in an experiment is called its *level*. For instance, suppose we are interested in determining the *effect* of product price on sales. Here price is the factor. If the prices used are \$5.00, \$6.00, and \$7.00, then each of these prices is a level of the factor. In a single-factor study, a *treatment* corresponds to a factor level; that is, each price level is a separate treatment. Here treatment or level has an effect, and applying a different treatment typically has a different effect on the mean response of Y.

When we have multiple factors, the combinations of levels of the factors for which the response will be observed are "treatments." For example, suppose our objective is to investigate the effect of price level and type of packaging on the sales of some product. Now price (including the three levels: \$5.00, \$6.00, and \$7.00) and package type (involving the two levels: type A and type B) are factors. Then sales are recorded for each of the six price-packaging combinations or treatments.

A factor effect is termed *fixed* if the levels of the factor employed represent "all possible levels." That is, if type A or type B packaging are the only packaging options available and if conclusions about this experiment are restricted to these and only these two packaging types, then the packaging effects on sales are fixed. Factor effects are considered *random* if the levels utilized actually represent a random

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

sample taken from a larger set of possible levels. (The packaging effects would be considered random if there are potentially a large number of different packaging types and only two were sampled as representative of the entire population of packaging types.) Random effects arise from true random sampling and are assumed to depict random variables that have probability distributions with a zero mean and a finite variance. A design or model in which both fixed and random effects are present is termed *mixed*.

8.2 MODEL SPECIFICATION

This chapter considers the specification of a nonlinear mixed-effects model for repeated measurements data. Under *mixed-effects modeling*, individual responses follow a similar functional form but with parameters that vary across individuals. Hence, a mixed-effects model contains both fixed and random effects. With *repeated measurements data*, we observe a number of individuals repeatedly under differing experimental circumstances, where the individuals are assumed to be drawn randomly from a specific population.

Given a set of repeated measures on an individual or subject, we can identify two sources of variation in the data: random variation among observations associated with a given individual (*intraindividual variation*) and random variation occurring among individuals (*interindividual variation*).

8.2.1 Model and Data Elements

Given a set of repeated measures on response and predictor variables obtained from a sample of individuals, our objective is to fit a nonlinear model that simultaneously considers the overall nonlinear mean response as well as the variation within and between individuals.

The basic features of this model design are:

1. Repeat response measurements obtained from a random sample of different individuals (subjects, plots, etc.). Such responses are assumed to evolve continuously over time.
2. A nonlinear functional relationship between a response and a set of unknown parameters and at least one predictor variable, for each individual sampled.
3. Response configurations that are similar across individuals but may exhibit different parameter realizations for different individuals.
4. Within-individual variation that may display either a nonconstant variance or autocorrelation or both. Autocorrelation often emerges when observations within individuals are not randomly assigned to treatment levels (e.g., when using longitudinal data which, for instance, is ordered in time).
5. Interindividual variation between regression parameters. Such variation may be random, systematically tied to individual-specific traits, or a combination of both.

8.2.2 A Hierarchical (Staged) Model

Level 1 (Intraindividual Variation) It is in level 1 that we specify the mean and covariance structure for a given individual (Davidian and Giltinan, 1998; Lindstrom and Bates, 1990). Specifically, let Y_{ij} represent the jth response for the ith individual, $i=1,\ldots,m;\ j=1,\ldots,n_i$, with

$$Y_{ij} = f\left(\boldsymbol{\beta}_i, \boldsymbol{X}_{ij}\right) + e_{ij}, \tag{8.1}$$

where:

1. $\boldsymbol{X}_{ij}$ is a predictor vector (of covariates such as time, temperature, dose, plot size) for the jth response on the ith individual so that a total of $n=\sum_{i=1}^{m} n_i$ responses are observed.
2. f is a nonlinear regression function that characterizes systematic variation in the predictor variable Y; it models the relationship between Y and $\boldsymbol{X}$ and is assumed common to all individuals.
3. $\boldsymbol{\beta}_i$ is a $(p \times 1)$ vector of parameters for individual i, $i=1,\ldots,m$. While f is common to all individuals, $\boldsymbol{\beta}_i$ varies across individuals.
4. e_{ij} is a random error term with $E(e_{ij}|\boldsymbol{\beta}_i)=0$ for all i,j.
5. $E(Y_{ij}|\boldsymbol{\beta}_i)=f(\boldsymbol{\beta}_i, \boldsymbol{X}_{ij})$ for all i,j.

Let

$$\underset{(n_i\times 1)}{\boldsymbol{Y}_i} = \begin{bmatrix} Y_{i1} \\ Y_{i2} \\ \vdots \\ Y_{in_i} \end{bmatrix}, \ \underset{(n_i\times 1)}{\boldsymbol{e}_i} = \begin{bmatrix} e_{i1} \\ e_{i2} \\ \vdots \\ e_{in_i} \end{bmatrix}, \text{ and } \underset{(n_i\times 1)}{f_i\left(\boldsymbol{\beta}_i\right)} = \begin{bmatrix} f\left(\boldsymbol{\beta}_i, \boldsymbol{X}_{i1}\right) \\ f\left(\boldsymbol{\beta}_i, \boldsymbol{X}_{i2}\right) \\ \vdots \\ f\left(\boldsymbol{\beta}_i, \boldsymbol{X}_{in_i}\right) \end{bmatrix}.$$

Then the model for the ith individual appears as

$$\boldsymbol{Y}_i = f_i\left(\boldsymbol{\beta}_i\right) + \boldsymbol{e}_i, \ \boldsymbol{e}_i \sim N\left(\boldsymbol{0}, \sigma_e^2 \Gamma_i\right), \tag{8.2}$$

where $\mathrm{COV}\left(\boldsymbol{Y}_i \middle| \boldsymbol{\beta}_i\right) = \sigma_e^2 \Gamma_i$ is the $(n_i \times n_i)$ covariance matrix. Typically, $\boldsymbol{\Gamma}_i = \boldsymbol{I}_{ni}$ for all i.

Next:

6. The conditional distribution of $\boldsymbol{e}_i$ given $\boldsymbol{\beta}_i$ is

$$\boldsymbol{e}_i \middle| \boldsymbol{\beta}_i \sim N\left(\boldsymbol{0}, \boldsymbol{R}_i\left(\boldsymbol{\beta}_i, \boldsymbol{\eta}\right)\right), \tag{8.3}$$

where $\boldsymbol{R}_i(\boldsymbol{\beta}_i, \boldsymbol{\eta}) = \mathrm{COV}(\boldsymbol{e}_i|\boldsymbol{\beta}_i)$ denotes the common (across individuals) intraindividual variance structure that allows for variance heterogeneity and autocorrelation within individuals.

Level 2 (Interindividual Variation) It is in level 2 that we model variation in the regression parameters across individuals. Variation among individuals is modeled through the individual-specific regression parameters $\boldsymbol{\beta}_i$, where such variation depends on both systematic and random components. It is typically the case that differences in parameter values are due essentially to random variation rather than to differences in individual demographic or behavioral characteristics, especially if one's procedures are consistent across the various runs of the experiment. In this regard, let us assume that interindividual variation is attributable to unexplained factors or

$$\beta_i = \beta + \boldsymbol{u}_i, \quad i = 1, \ldots, m, \tag{8.4}$$

where $\boldsymbol{\beta}$ is a $(p \times 1)$ vector of fixed parameters, $\boldsymbol{u}_i$ is a $(p \times 1)$ vector of random effects, and

$$\boldsymbol{u}_i \sim N\left(\mathbf{0}, \sigma_u^2 \boldsymbol{D}\right), \tag{8.5}$$

with $\sigma_u^2 \boldsymbol{D}$ the $(p \times p)$ covariance matrix that quantifies the random interindividual variation. Given Equations 8.4 and 8.5, it follows that $E(\boldsymbol{\beta}_i) = \boldsymbol{\beta}$.

A generalization of $\boldsymbol{\beta}_i$ (Eq. 8.4) is provided by

$$\beta_i = \boldsymbol{A}_i \beta + \boldsymbol{B}_i \boldsymbol{u}_i, \quad i = 1, \ldots, m, \tag{8.6}$$

where $\boldsymbol{A}_i$ is an $(r \times p)$ *design matrix* for the fixed effects, $\boldsymbol{B}_i$ is an $(r \times p)$ *design matrix* for the random effects (if the $\boldsymbol{B}_i$ are equal for all i, then each individual has random effects drawn from the same distribution), and $\beta_i \sim N\left(\boldsymbol{A}_i \boldsymbol{\beta}, \sigma_u^2 \boldsymbol{D}\right)$. Given Equation 8.6, Equation 8.1 can be rewritten as

$$Y_{ij} = f\left(\boldsymbol{A}_i \boldsymbol{\beta} + \boldsymbol{B}_i \boldsymbol{u}_i, \boldsymbol{X}_{ij}\right) + e_{ij}, \quad i = 1, \ldots, m, \quad j = 1, \ldots, n_i. \tag{8.1.1}$$

Let us condense the m individual models by letting

$$\underset{(n\times 1)}{\boldsymbol{Y}} = \begin{bmatrix} \boldsymbol{Y}_1 \\ \boldsymbol{Y}_2 \\ \vdots \\ \boldsymbol{Y}_m \end{bmatrix}, \underset{(mr\times 1)}{\bar{\beta}} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}, \underset{(m\times 1)}{f\left(\bar{\beta}\right)} = \begin{bmatrix} f_1\left(\beta_1\right) \\ f_2\left(\beta_2\right) \\ \vdots \\ f_m\left(\beta_m\right) \end{bmatrix},$$

$$\underset{(mp\times mp)}{\bar{D}} = \operatorname{diag}(\boldsymbol{D}, \boldsymbol{D}, \ldots, \boldsymbol{D}), \underset{(mn_i \times mn_i)}{\bar{\Gamma}} = \operatorname{diag}(\Gamma_1, \Gamma_2, \ldots, \Gamma_m),$$

$$\underset{(mr\times mp)}{\boldsymbol{B}} = \operatorname{diag}(\boldsymbol{B}_1, \boldsymbol{B}_2, \ldots, \boldsymbol{B}_m),$$

$$\underset{(mrxp)}{A} = \begin{bmatrix} \boldsymbol{A}_1 \\ \boldsymbol{A}_2 \\ \vdots \\ \boldsymbol{A}_m \end{bmatrix}, \quad \text{and} \quad \underset{(mp\times 1)}{\boldsymbol{u}} = \begin{bmatrix} \boldsymbol{u}_1 \\ \boldsymbol{u}_2 \\ \vdots \\ \boldsymbol{u}_m \end{bmatrix}.$$

Then the *hierarchical global model* can be written as

$$\begin{aligned}\mathbf{Y}|\boldsymbol{u} &\sim N\left(f\left(\bar{\beta}\right),\sigma_e^2\Gamma\right),\\ \bar{\beta} &= \boldsymbol{A}\boldsymbol{\beta}+\boldsymbol{B}\boldsymbol{u},\\ \boldsymbol{u} &\sim N\left(\mathbf{0},\sigma_u^2\bar{\boldsymbol{D}}\right).\end{aligned} \tag{8.7}$$

8.3 SOME SPECIAL CASES OF THE HIERARCHICAL GLOBAL MODEL

1. Suppose $\boldsymbol{A}_i=\boldsymbol{I}_p$ and $\boldsymbol{B}_i=\boldsymbol{I}_p$ for all i so that, from Equation 8.6,

$$\beta_i = \boldsymbol{I}_p\beta+\boldsymbol{I}_p\boldsymbol{u}_i, \quad i=1,\ldots,m. \tag{8.8}$$

Hence, interindividual variation is due solely to unexplained phenomena, and all individuals experience random effects from the same distribution. Now

$$\underset{(mp\times p)}{\boldsymbol{A}} = \begin{bmatrix}\boldsymbol{I}_p\\ \boldsymbol{I}_p\\ \vdots\\ \boldsymbol{I}_p\end{bmatrix} \quad \text{and} \quad \underset{(mp\times mp)}{\boldsymbol{B}} = \begin{pmatrix} I_p & & \mathbf{0}\\ & \ddots & \\ \mathbf{0} & & I_p\end{pmatrix}.$$

In addition, let $\boldsymbol{\Gamma}_i=\boldsymbol{I}$ with $\boldsymbol{D}$ the same for each individual.

2. Up to this point we have assumed that all interindividual variation is attributable to only random effects. However, it may be the case that $\boldsymbol{\beta}_i$ is influenced by both systematic and random factors, where the former effect is due, conceivably, to weather conditions, natural variation among plots, and assorted group effects. For instance, suppose that the income growth of individual i varies according to sex (male or female), education level (high school diploma/certificate, 2-year college degree, four-year college degree, graduate degree), and numerous other factors that contribute to variation among individuals. In this regard, it is assumed that $\boldsymbol{\beta}_i$ is associated with a distribution having a mean dependent upon the sex–education level combination under which the individual was observed. Let the eight possible combination groups be indexed by k:

   $k=1$: male with high school diploma
   $k=2$: female with high school diploma
   $k=3$: male with two-year college degree
   $k=4$: female with two-year college degree, etc.

Then the regression parameter vector for individual i as a member of combination group k is

$$\beta_i = \beta^{(k)} + \boldsymbol{u}_i, \quad k = 1,\ldots,8, \tag{8.9}$$

where $\boldsymbol{\beta}^{(k)}$ is a vector of fixed parameters that serves as the mean of $\boldsymbol{\beta}_i$ for combination group k. Here too $\boldsymbol{D}$ is the same across individuals, thus indicating that the pattern of random variation in the $\boldsymbol{\beta}_i$ is the same across all combination groups. Now

$$\begin{aligned}\beta_i &= \boldsymbol{A}_i\beta + \boldsymbol{B}_i\boldsymbol{u}_i \\ &= \boldsymbol{A}_i\begin{bmatrix}\beta^{(1)} \\ \beta^{(2)} \\ \vdots \\ \beta^{(8)}\end{bmatrix} + \boldsymbol{I}_p\boldsymbol{u}_i, \quad i = 1,\ldots,m,\end{aligned} \tag{8.10}$$

where $\boldsymbol{\beta}$ is an $(8p \times 1)$ vector and

$$\begin{aligned} k = 1: &\quad \boldsymbol{A}_i = \left[\boldsymbol{I}_p \ \ \mathbf{0}\ldots\mathbf{0}\right], \\ k = 2: &\quad \boldsymbol{A}_i = \left[\mathbf{0} \ \ \boldsymbol{I}_p\ldots\mathbf{0}\right], \\ &\ \ \vdots \\ k = 8: &\quad \boldsymbol{A}_i = \left[\mathbf{0} \ \ \mathbf{0}\ldots\boldsymbol{I}_p\right]. \end{aligned}$$

Here $\boldsymbol{A}_i$ is a $(p \times 8p)$ systematic design matrix and $\mathbf{0}$ donates a $(p \times p)$ null matrix. In fact, if $\boldsymbol{\beta}_i$ varies systematically according to combination group, then $\boldsymbol{A}_i$ is always an *indicator matrix* containing 1 s and zeros. Then

$$\underset{(mp\times 8p)}{\boldsymbol{A}} = \begin{bmatrix}\boldsymbol{A}_1 \\ \boldsymbol{A}_2 \\ \vdots \\ \boldsymbol{A}_m\end{bmatrix} \quad \text{and} \quad \underset{(mp\times mp)}{\boldsymbol{B}} = \begin{pmatrix}\boldsymbol{I}_p & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \boldsymbol{I}_p\end{pmatrix}.$$

3. It is often the case that certain components of the $\boldsymbol{\beta}_i$ vector do not vary considerably across individuals; that is, estimates of one or more parameters may be quite consistent across individuals (plots, subjects, etc.). To model this assertion, suppose that in the preceding special case (#2), $p=4$. Then for an individual from combination group k, assume that elements of $\boldsymbol{\beta}_i=\boldsymbol{\beta}^{(k)}+\boldsymbol{u}_i$ are to appear as

$$\begin{aligned}\beta_{1i} &= \beta_1^{(k)} + u_{1i}, \\ \beta_{2i} &= \beta_2^{(k)}, \\ \beta_{3i} &= \beta_3^{(k)} + u_{3i}, \\ \beta_{4i} &= \beta_4^{(k)},\end{aligned} \tag{8.11}$$

where

$$\boldsymbol{\beta}^{(k)} = \begin{bmatrix} \beta_1^{(k)} \\ \beta_2^{(k)} \\ \beta_3^{(k)} \\ \beta_4^{(k)} \end{bmatrix} \text{ and } \boldsymbol{u}_i = \begin{bmatrix} u_{1i} \\ u_{2i} \\ u_{3i} \\ u_{4i} \end{bmatrix}, \quad \boldsymbol{u}_i \sim N\left(\mathbf{0}, \sigma_u^2 \boldsymbol{D}\right)$$

Here β_{2i} and β_{4i} are devoid of any random effects ($u_{2i} = u_{4i} = 0$) and consequently are the same across all individuals and equal to the fixed values $\beta_2^{(k)}$ and $\beta_4^{(k)}$, respectively. Then from (8.10),

$$\boldsymbol{\beta}_i = \boldsymbol{A}_i \boldsymbol{\beta} + \boldsymbol{B}_i \boldsymbol{u}_i,$$

where $\boldsymbol{A}_i$ is a (4×32) systematic design matrix and the random-effects design matrix

$$\boldsymbol{B}_i = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

so that

$$\boldsymbol{B}_i \boldsymbol{u}_i = \begin{bmatrix} u_{1i} \\ 0 \\ u_{3i} \\ 0 \end{bmatrix}$$

as required by (8.11).

8.4 THE SAS/STAT NLMIXED PROCEDURE FOR FITTING NONLINEAR MIXED-EFFECTS MODEL

PROC NLMIXED will be used to estimate the parameters of growth curves in which both fixed and random effects enter the response function in a nonlinear fashion. Such models are fit to repeated measurements data using maximum likelihood techniques. In particular, NLMIXED maximizes an approximation to the marginal likelihood function obtained by integrating the likelihood function over the random effects. Here approximations to the integral can be made by adaptive Gaussian quadrature (which employs empirical Bayes estimates of the random effects as the centering points for the quadrature and updates them at each iteration) or by a first-order Taylor expansion (about the origin), among others.

While a variety of optimization algorithms are available for maximizing the marginal likelihood function, the default is the (dual) quasi-Newton method. If the optimization process exhibits convergence to a stationary point, the resulting solution provides parameter estimates and their approximate standard errors.

EXAMPLE 8.1 The following data set (Table 8.1, the "orange tree data" from Draper and Smith (1981)) involves seven measurements of the tree trunk circumferences (in millimeters) on each of five orange trees. The observations are "balanced" in that measurements are taken on the same days for each tree, with the days marked consecutively, from January 1, 1969, to 1973. A graph of the data (Fig. 8.1) reveals that, for each of the five individual trees, the response Y follows a sigmoidal growth trend over time (t) and the only parameter that varies appreciably from tree to tree is $\boldsymbol{\beta}_1(=Y_{\text{inf}})$, the asymptote. Hence, only $\boldsymbol{\beta}_1$ will have a random component u_{1i}. Our objective is to fit a logistic, Gompertz, Weibull, and Chapman–Richards growth equation to this data set and to assess/compare the results. Remember that:

Logistic—$Y = \beta_1/\left(1+e^{\beta_2-\beta_3 t}\right)$.
Gompertz—$Y = \beta_1 e^{-e^{\beta_2-\beta_3 t}}$.
Weibull—$Y = \beta_1 - \beta_2 e^{-\beta_3 t^{\beta_4}}$.
Chapman–Richards—$Y = \beta_1\left(1-e^{\beta_2-\beta_3 t}\right)^{\beta_4}$.

Let us consider the logistic function in greater detail. In terms of our preceding notation for the hierarchical global model,

$$Y_{ij} = \frac{\beta_1 + u_{1i}}{1+e^{\beta_2-\beta_3 t_{ij}}} + e_{ij},$$

where:

Y_{ij} is the jth measurement on the ith tree.
t_{ij} is the jth day on which the ith tree is measured.
$\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ are fixed-effects parameters.

TABLE 8.1 Response *Y* versus Time *t*

	Response *Y* for tree number				
t	1	2	3	4	5
118	30	33	30	32	30
484	58	69	51	62	49
664	87	111	75	112	81
1004	115	156	108	167	125
1231	120	172	115	179	142
1372	142	203	139	209	174
1582	145	203	140	214	177

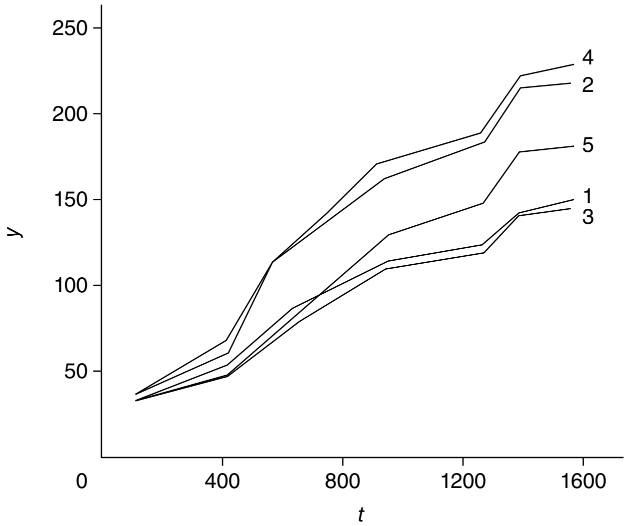

FIGURE 8.1 Orange tree data.

u_{1i} is a random-effects parameter with
$u_{1i} \sim N\left(0, \sigma_u^2 d\right)$, d a scalar.
$e_{ij} \sim N\left(0, \sigma_e^2\right)$ and independent of u_{1i}.

Here

$$\boldsymbol{\beta}_i = \boldsymbol{A}_i\boldsymbol{\beta} + \boldsymbol{B}_i\boldsymbol{u}_i = \boldsymbol{I}_3\begin{bmatrix}\beta_1\\ \beta_2\\ \beta_3\end{bmatrix} + \begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}u_{1i}\\ u_{2i}\\ u_{3i}\end{bmatrix}$$

$$= \begin{bmatrix}\beta_1 + u_{1i}\\ \beta_2\\ \beta_3\end{bmatrix},$$

$$\boldsymbol{u}_i = \begin{bmatrix}u_{1i}\\ u_{2i}\\ u_{3i}\end{bmatrix} \sim N\left(\boldsymbol{0}, \sigma_u^2\boldsymbol{D}\right), \boldsymbol{D} = \begin{bmatrix}d & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}, \text{ and}$$

$$\boldsymbol{e}_i \sim N\left(\boldsymbol{0}, \sigma_e^2\Gamma_i\right) = N(\boldsymbol{0}, \sigma_e^2\boldsymbol{I}).$$

A similar configuration holds for the three remaining growth functions.

The SAS code for estimating the parameters of these nonlinear mixed-effects models is provided by Exhibit 8.1.

EXHIBIT 8.1 SAS Code for Nonlinear Mixed Growth Model Estimation

```
data trees;
    ①
input tree t y;
datalines;
1     118     30
1     484     58
1     664     87
1    1004    115
1    1231    120
1    1372    142
1    1582     14
2     118     23
2     484     69
2     664    111
2    1004    156
2    1231    172
2    1372    203
2    1582    203
3     118     30
3     484     51
3     664     75
3    1004    108
3    1231    115
3    1372    139
3    1582    140
4     118     32
4     484     62
4     664    112
4    1004    167
4    1231    179
4    1372    209
4    1582    214
5     118     30
5     484     49
5     664     81
5    1004    125
5    1231    142
5    1372    174
5    1582    177
;
proc nlmixed data=trees;
parms beta1=150 beta2=10 beta3=-0.01
      s2u=120 s2e=20; ②
```

① Here *tree* is the subject variable that identifies observations belonging to the same tree. PROC NLMIXED assumes that the input data set is clustered according to the levels of the subject variable so that all observations from the same subject occur sequentially in the input date set.
② The PARMS statement identifies the unknown parameters and specifies their starting values. Also identified are the variance components s_u^2 and s_e^2 along with their starting values.
③ The BOUNDS statement specifies that s_u^2 cannot be negative and that s_e^2 must be positive.
④ The MODEL statement declares the probability model for the response variable *Y*, conditional on the random effects. The logistic mean model is specified along with normal errors having variance s_e^2. In general, one must specify a single dependent variable, a tilde (~), and then a distribution with its parameters. Valid distributions are:

Normal (*m, v*) specifies a normal distribution with mean *m* and variance *v*.
Binary (*p*) specifies a Bernoulli distribution with probability *p*.
Binomial (*m, p*) specifies a binomial distribution with *n* trials and probability *p*.

⑤ The RANDOM statement declares the normal distribution for the random coefficient u_1 with variance s_u^2. The SUBJECT = *tree* specification defines a variable that indicates when the random effect obtains new realizations. It thus changes according to the values of the *tree* variable.

```
  bounds s2u>=0, s2e>0; ③
  model y~normal((beta1+u1)/(1+exp(beta2-beta3*t)),
  s2e); ④
  random u1~normal(0,s2u)subject=tree; ⑤
run;
proc nlmixed data=trees;
  parms beta1=210 beta2=1.2 beta3=0.0011
     s2u=500 s2e=3000;
  bounds s2u>=0, s2e>0;
  model y~normal((beta1+u1)*exp(-exp(beta2-beta3*t)),
  s2e); ⑥
  random u1~normal(0,s2u))subject=tree;
run;
proc nlmixed data=trees;
  parms beta1=190 beta2=700 beta3=350 beta4=1
     s2u=1000 s2e=60;
  bounds s2u>=0, s2e>0;
  model y~normal((beta1+u1)-beta2*
  exp(-beta3*(t**beta4)), s2e); ⑦
  random u1~normal(0,s2u))subject=tree;
run;
proc nlmixed data=trees;
  parms beta1=190 beta2=700 beta3=350 beta4=1
     s2u=1000 s2e=60;
  bounds s2u>=0, s2e>0;
  model y~normal((beta1+u1)*(1-exp(beta2-beta3*t))**
  beta4)),s2e); ⑧
  random u1~normal(0,s2u))subject=tree;
run;
```

⑥ Model statement for the Gompertz growth curve.
⑦ Model statement for the Weibull growth curve.
⑧ Model statement for the Chapman–Richards growth curve.

The output generated by Exhibit 8.1 appears in Table 8.2. For each set of growth curve estimates, the *Iteration History* will be omitted.

Ⓐ Output summary for logistic growth curve estimation.

① The *Specification Table* lists some basic information about the particular mixed logistic model that has been specified.

② The *Dimensions Table* lists various “counts” related to the model. The items in this table serve as a check on the specification of the data set and model.

③ The *Parameters Table* houses the parameters to be estimated, their initial values, and the negative of the log-likelihood function at these initial values.

TABLE 8.2 Output for Example 8.1

The SAS system

Ⓐ

The NLMIXED procedure

①

Specifications

Data set	WORK.TREES
Dependent variable	*y*
Distribution for dependent variable	Normal
Random effects	u_1
Distribution for random effects	Normal
Subject variable	*tree*
Optimization technique	Dual quasi-Newton
Integration method	Adaptive Gaussian quadrature

②

Dimensions

Observations used	35
Observations not used	0
Total observations	35
Subjects	5
Max obs per subject	7
Parameters	5
Quadrature points	1

③

Parameters

beta1	beta2	beta3	*s2u*	*s2e*	NegLogLike
150	10	–0.01	120	20	14624.747

Note: GCONV convergence criterion satisfied.

④

Fit statistics

–2 log likelihood	264.5
AIC (smaller is better)	274.5
AICC (smaller is better)	276.6
BIC (smaller is better)	272.6

⑤

Parameter estimates

Parameter	Estimate	Standard Error	DF	*t* value	Pr > |*t*|	Alpha	Lower	Upper	Gradient
beta1	190.10	15.3928	4	12.35	0.0002	0.05	147.36	232.84	–7.56E-7
beta2	2.4122	0.1126	4	19.03	<0.0001	0.05	1.8296	2.4547	0.0003
beta3	0.002969	0.000233	4	12.72	0.0002	0.05	0.002321	0.003617	–0.05979

(*Continued*)

TABLE 8.2 Output for Example 8.1 (*Continued*)

Parameter	Estimate	Standard Error	DF	*t* value	Pr > |*t*|	Alpha	Lower	Upper	Gradient
s2u	976.44	0.01510	4	64,646.9	<0.0001	0.05	976.40	976.48	–1.26E-8
s2e	64.4467	16.6396	4	3.87	0.0179	0.05	18.2478	110.65	9.43E-7

Covariance matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	*s2u*	*s2e*
1	beta1	236.94	–0.3472	–0.00130	–0.2255	0.1168
2	beta2	–0.3472	0.01267	0.000021	0.000330	–0.00340
3	beta3	–0.00130	0.000021	5.447E-8	1.24E-6	–6.34E-6
4	s2u	–0.2255	0.000330	1.24E-6	0.000228	0.06090
5	s2e	0.1168	–0.00340	–6.34E-6	0.06090	276.88

Correlation matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	s2u	s2e
1	beta1	1.0000	–0.2004	–0.3630	–0.9701	0.000456
2	beta2	–0.2004	1.0000	0.8135	0.1940	–0.00182
3	beta3	–0.3630	0.8135	1.0000	0.3518	–0.00163
4	*s2u*	–0.00080	0.1940	0.3518	1.0000	0.2423
5	*s2e*	0.000456	–0.00182	–0.00163	0.2423	1.0000

Ⓑ

The NLMIXED procedure

Specifications

Data set	WORK.TREES
Dependent variable	*y*
Distribution for dependent variable	Normal
Random effects	*u*1
Distribution for random effects	Normal
Subject variable	*tree*
Optimization technique	Dual quasi-Newton
Integration method	Adaptive Gaussian quadrature

Dimensions

Observations used	35
Observations not used	0
Total observations	35
Subjects	5
Max obs per subject	7
Parameters	5
Quadrature points	1

TABLE 8.2 Output for Example 8.1 (*Continued*)

		Parameters			
beta1	beta2	beta3	*s2u*	*s2e*	NegLogLike
210	1.23	0.0019	500	3000	175.599515

Note: GCONV convergence criterion satisfied.

Fit statistics

–2 log likelihood	350.7
AIC (smaller is better)	360.7
AICC (smaller is better)	362.7
BIC (smaller is better)	358.7

①
Parameter estimates

Parameter	Estimate	Standard error	DF	*t* value	Pr > |*t*|	Alpha	Lower	Upper	Gradient
beta1	210.00	65.4686	4	3.21	0.0327	0.05	28.2294	391.77	–0.00328
beta2	0.9986	0.4092	4	2.44	0.0712	0.05	–0.1376	2.1348	–0.00089
beta3	0.001745	0.001052	4	1.66	0.1723	0.05	–0.00117	0.004665	–0.13039
s2u	500.00	1.4175	4	352.75	<0.0001	0.05	496.06	503.94	0.0004
s2e	3000.00	0.06405	4	46840.4	<0.0001	0.05	2999.82	3000.18	0.005042

Covariance matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	*s2u*	*s2e*
1	beta1	4286.14	–12.9114	–0.06250	–92.7985	–4.1931
2	beta2	–12.9114	0.1675	0.000325	0.2796	0.01263
3	beta3	–0.06250	0.000325	1.106E-6	0.001353	0.000061
4	s2u	–92.7985	0.2796	0.001353	2.0092	0.09078
5	s2e	–4.1931	0.01263	0.00061	0.09078	0.004102

Correlation matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	*s2u*	*s2e*
1	beta1	1.0000	–0.4819	–0.9078	–1.0000	–1.0000
2	beta2	–0.4819	1.0000	0.7558	0.4820	0.4820
3	beta3	–0.9078	0.7558	1.0000	0.9078	0.9078
4	*s2u*	–1.0000	0.4820	0.9078	1.0000	1.0000
5	*s2e*	–1.0000	0.4820	0.9078	1.0000	1.0000

©
The NLMIXED procedure
Specifications

Data set	WORK.TREES
Dependent variable	*y*
Distribution for dependent variable	Normal

(*Continued*)

TABLE 8.2 Output for Example 8.1 (*Continued*)

Data set	WORK.TREES
Random effects	*u*1
Distribution for random effects	Normal
Subject variable	*tree*
Optimization technique	Dual quasi-Newton
Integration method	Adaptive Gaussian quadrature

Dimensions

Observations used	35
Observations not used	0
Total observations	35
Subjects	5
Max obs per subject	7
Parameters	6
Quadrature points	1

Parameters

beta1	beta2	beta3	beta4	*s2u*	*s2e*	NegLogLike
150	10	0.01	10	120	20	2691.19756

Note: GCONV convergence criterion satisfied.

Fit statistics

–2 log likelihood	382.9
AIC (smaller is better)	394.9
AICC (smaller is better)	397.9
BIC (smaller is better)	392.6

①
Parameter estimates

Parameter	Estimate	Standard error	DF	*t* value	Pr > |*t*|	Alpha	Lower	Upper	Gradient
beta1	115.57	10.9041	4	10.60	0.0004	0.05	85.2925	145.84	–0.00004
beta2	10.0000	0.000136	4	73572.8	<0.0001	0.05	9.9996	10.0004	0
beta3	0.01000	0.001480	4	6.76	0.0025	0.05	0.005890	0.01411	0
beta4	10.0000	0.000136	4	73572.9	<0.0001	0.05	9.9996	10.0004	0
s2u	133.24	1183.27	4	0.11	0.9158	0.05	–3152.05	3418.53	0.001888
s2e	3228.79	787.88	4	4.10	0.0149	0.05	1041.28	5416.30	–8.08E-6

Covariance matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	beta4	*s2u*	*s2e*
1	beta1	118.90	–1.18E-6	–0.00001	–1.18E-6	–9.9928	1.485E-6

TABLE 8.2 Output for Example 8.1 (*Continued*)

Covariance matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	beta4	*s2u*	*s2e*
2	beta2	–1.18E-6	1.847E-8	2.012E-7	1.847E-8	0.1607	–0.00626
3	beta3	–0.00001	2.012E-7	2.191E-6	2.012E-7	1.7504	–0.06815
4	beta4	–1.18E-6	1.847E-8	2.012E-7	1.847E-8	0.1607	–0.00626
5	*s2u*	–9.9928	0.1607	1.7504	0.1607	1,400,129	–88,679
6	*s2e*	1.485E-6	–0.00626	–0.06815	–0.00626	–88,679	620,755

Correlation matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	beta4	*s2u*	*s2e*
1	beta1	1.0000	–0.00080	–0.00080	–0.00080	–0.00077	1.73E-10
2	beta2	–0.00080	1.0000	1.0000	1.0000	0.9993	–0.05843
3	beta3	–0.00080	1.0000	1.0000	1.0000	0.9993	–0.05843
4	beta4	–0.00080	1.0000	1.0000	1.0000	0.9993	0.05843
5	*s2u*	–0.00077	0.9993	0.9993	0.9993	1.0000	0.09512
6	*s2e*	1.73E-10	–0.05843	–0.05843	–0.05843	–0.09512	1.0000

Ⓓ

The NLMIXED procedure

Specifications

Data set	WORK.TREES
Dependent variable	*y*
Distribution for dependent variable	Normal
Random effects	*u*1
Distribution for random effects	Normal
Subject variable	*tree*
Optimization technique	Dual quasi-Newton
Integration method	Adaptive Gaussian quadrature

Dimensions

Observations used	35
Observations not used	0
Total observations	35
Subjects	5
Max obs per subject	7
Parameters	6
Quadrature points	1

Parameters

beta1	beta2	beta3	beta4	*s2u*	*s2e*	NegLogLike
190	110	0.97	2.61	45	2000	233.562581

Note: GCONV convergence criterion satisfied.

(*Continued*)

TABLE 8.2 Output for Example 8.1 (*Continued*)

Fit statistics	
–2 log likelihood	367.4
AIC (smaller is better)	379.4
AICC (smaller is better)	382.4
BIC (smaller is better)	377.1

①
Parameter estimates

Parameter	Estimate	Standard error	DF	t value	Pr > |t|	Alpha	Lower	Upper	Gradient
beta1	130.00	9.5237	4	13.65	0.0002	0.05	103.56	156.44	7.198E-6
beta2	110.00	7.5053	4	14.66	0.0001	0.05	89.1622	130.84	1.853E-7
beta3	0.9322	6.682E-6	4	139514	<0.0001	0.05	0.9322	0.9322	–0.00002
beta4	2.6581	0.002910	4	913.41	<0.0001	0.05	2.6500	2.6662	9.88E-12
s2u	116.97	300.22	4	0.39	0.7167	0.05	–716.56	950.50	0.000014
s2e	2019.23	517.45	4	3.90	0.0175	0.05	582.56	3455.91	–0.00005

Covariance matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	beta4	*s2u*	*s2e*
1	beta1	90.7014	0.000023	–1.15E-8	2.202E-9	0.1219	–6.89E-7
2	beta2	0.000023	56.3298	0.2387	0.02184	0.000926	–0.00575
3	beta3	–1.15E-8	0.2387	4.46E-11	0.000093	–4.7E-7	2.92E-6
4	beta4	2.202E-9	0.02184	0.000093	8.469E-6	0.001145	–0.00131
5	*s2u*	0.1219	0.000926	–4.7E-7	0.001145	90,129	–44,626
6	*s2e*	–6.89E-7	–0.00575	2.92E-6	–0.00131	–44,626	267,757

Correlation matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	beta4	*s2u*	*s2e*
1	beta1	1.0000	3.157E-7	–0.00018	7.944E-8	0.000043	–14E-11
2	beta2	3.157E-7	1.0000	1.0000	1.0000	4.108E-7	–1.48E-6
3	beta3	–0.00018	1.0000	1.0000	1.0000	–0.00023	0.000845
4	beta4	7.944E-8	1.0000	1.0000	1.0000	0.001310	–0.00087
5	*s2u*	0.000043	4.108E-7	–0.00023	0.001310	1.0000	–0.2873
6	*s2e*	–14E-11	–1.48E-6	0.000845	–0.00087	–0.2873	1.0000

④ The *Fit Statistics Table* displays the final maximum value of the log-likelihood function as well as the Akaike and Schwarz information criterion. These statistics can be used to compare different nonlinear mixed-effects model.

⑤ The *Parameter Estimates Table* lists the maximum likelihood estimates of the parameters, their approximate standard errors (determined from the final Hessian matrix), approximate t values, and approximate Wald 95% confidence intervals, with degrees of freedom equal to the number of subjects (trees) less the number of random effects. (Here d.f. = 5 − 1 = 4.) As this table reveals, $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ are highly significant as is the variance term s_u^2, thus indicating strong between-subject variability in the asymptote parameter $\boldsymbol{\beta}_1$. The final column of this table is the gradient vector (the vector of first partial derivatives of the log-likelihood function) evaluated at the optimal solution.

Each component of this vector (with the possible exception of the component associated with $\hat{\beta}_3$) seems to be quite small, thus indicating that the log-likelihood function has attained a stationary point.

Ⓑ Output summary for Gompertz growth curve estimation.

① The *Parameter Estimates Table* reveals that while the estimate for $\boldsymbol{\beta}_1$ is statistically significant, those for $\boldsymbol{\beta}_2$ and $\boldsymbol{\beta}_3$ are not. Moreover, since the estimate of the variance term s_u^2 is highly significant, we can conclude that there exists strong between-subject variability in the asymptote parameter $\boldsymbol{\beta}_1$.

A glance at the component of the gradient vector indicates that, save for the component associated with $\hat{\beta}_3$, stationarity has been realized.

Ⓒ Output summary for Weibull growth curve estimation.

① Estimates of the fixed parameters $\boldsymbol{\beta}_1,\ldots,\boldsymbol{\beta}_4$ are all statistically significant, while the estimate for s_u^2 is not. Hence, the fixed effects dominate and no between-subject variability in $\boldsymbol{\beta}_1$ is detected. In addition, all the components of the gradient vector of the log-likelihood function are small enough to indicate convergence to a stationary point.

Ⓓ Output summary for Chapman–Richards growth curve estimation.

① Here too we have statistically significant fixed effects (the p-values for the estimates of $\boldsymbol{\beta}_1,\ldots,\boldsymbol{\beta}_4$ are very low), while the estimate for s_u^2 is not significant—no between-subject variability in $\boldsymbol{\beta}_1$ is found.

On the basis of these results, it appears that the logistic growth curve provides the best fit to a repeated measures nonlinear mixed-effects growth model with fixed and random effects associated with the asymptote parameter $\boldsymbol{\beta}_1$. ■

The SAS code utilized in Example 8.1 does not constitute the whole story. Additional useful program elements are the following:

1. Options available in the PROC NLMIXED statement:

COV—requests the approximate covariance matrix for the parameter estimates.

CORR—requests the approximate correlation matrix for the parameter estimates.

HESS—requests the display of the final Hessian matrix.

MAXITER = i—specifies the maximum number of iterations in the optimization process.

METHOD = *value*—specifies the method for approximating the integral of the likelihood over the random effects. Valid values are:

FIRO—specifies the first-order method of Beal and Scheiner (1982).

GAUSS—specifies adaptive Gauss–Hermite quadrature (Pinheiro and Bates, 1995).

TECH = *value*—specifies the optimization technique.

CONGRA—performs a conjugate gradient optimization.

NMSIMP—performs a Nelder–Mead simplex optimization.

NEWRAP—performs a Newton–Raphson optimization combining line search with ridging.

NRRIDG—performs a Newton–Raphson optimization with ridging.

TRUREG—performs a trust region optimization.

QUANEW—performs a quasi-Newton optimization. This is the default optimization method.

The methods TRUREG, NEWRAP, and NRRIDG are suitable for small-scale problems. QUANEW is good for medium-sized problems. Large-scale problems are best handled by CONGRA.

UPD = *method*—specifies the update method for the quasi-Newton or conjugate gradient technique:

BFGS—performs the Broyden (1967) update of the Hessian.

DBFGS—performs the dual BFGS update of the Cholesky factor of the Hessian. This is the default update method.

DDFP—performs the dual Davidon (1959) update of the Cholesky factor of the Hessian.

DFP—performs the original DFD update of the inverse Hessian.

FR—performs the Fletcher–Reeves update (Fletcher, 1987).

CD—performs the conjugate-descent update (Fletcher, 1987).

PREDICT—enables one to construct predictions of an expression for each observation in the input data set. Predicted values are computed using the parameter estimates and the empirical Bayes estimates of the random effects. Results are placed in an output data set that is specified with the OUT = *option*.

For instance, following the RANDOM statement in Exhibit 8.1, we could put

```
PREDICT pred out=pred;
```

RANDOM—defines the random effects and their distribution. For multiple random effects, one should specify bracketed vectors for m and v, the latter consisting of the lower triangle of the random-effects variance–covariance matrix.

For instance, suppose in, say, the logistic function specified at the outset of Example 8.1 we posit that both $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ display mixed effects or

$$Y_{ij} = \frac{\beta_1 + u_{1i}}{1 + e^{(\beta_2 + u_{2i}) - \beta_3 t_{ij}}} + e_{ij}.$$

Then the RANDOM statement is written as

$$random\ u_1 u_2 \sim normal\left(\underset{(m)}{[0,0]}, \underset{(v)}{[s2u1, cu12, s2u2]}\right) \text{ subject=tree;}$$

Here v corresponds to

$$\begin{bmatrix} s_{u1}^2 & - \\ \text{cov}(u1,u2) & s_{u2}^2 \end{bmatrix}.$$

Alternatively, suppose that $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ all display mixed effects or

$$Y_{ij} = \frac{\beta_1 + u_{1i}}{1 + e^{(\beta_2 + u_{2i}) - (\beta_3 + u_{3i}) t_{ij}}} + e_{ij}.$$

Then the appropriate RANDOM statement appears as

$$random\ u_1 u_2 u_3 \sim normal\left(\underset{(m)}{[0,0,0]}, \underset{(v)}{\left[s2u1, cu12, s2u2, cu13, cu23, s2u3\right]}\right)$$
$$\text{subject=tree;}$$

Now v corresponds to

$$\begin{bmatrix} s_{u1}^2 & - & - \\ \text{cov}(u1,u2) & s_{u2}^2 & - \\ \text{cov}(u1,u3) & \text{cov}(u2,u3) & s_{u3}^2 \end{bmatrix}.$$

EXAMPLE 8.2 Using the "orange tree data" of Example 8.1, let us use PROC NLMIXED to estimate the parameters of the logistic growth function under the assumption that $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ have both fixed and random components. Here

$$Y_{ij} = \frac{\beta_1 + u_{1i}}{1 + e^{(\beta_2 + u_{2i}) - \beta_3 t_{ij}}} + e_{ij},$$

where:

$\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ are fixed-effects parameters.
u_{1i} and u_{2i} are random-effects parameters.
$e_{ij} \sim N\left(0, \sigma_e^2\right)$ and independent of u_{1i} and u_{2i}.

Now

$$\boldsymbol{\beta}_i = \boldsymbol{A}_i\boldsymbol{\beta} + \boldsymbol{B}_i\boldsymbol{u}_i = I_3\begin{bmatrix}\beta_1\\ \beta_2\\ \beta_3\end{bmatrix} + \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}u_{1i}\\ u_{2i}\\ u_{3i}\end{bmatrix}$$

$$= \begin{bmatrix}\beta_1 + u_{1i}\\ \beta_2 + u_{2i}\\ \beta_3\end{bmatrix},$$

$$\boldsymbol{u}_i = \begin{bmatrix}u_{1i}\\ u_{2i}\\ u_{3i}\end{bmatrix} \sim N\left(\boldsymbol{0}, \sigma_u^2\boldsymbol{D}\right), \boldsymbol{D} = \begin{bmatrix}d_{11} & d_{12} & 0\\ d_{21} & d_{22} & 0\\ 0 & 0 & 0\end{bmatrix}, \text{ and}$$

$$\boldsymbol{e}_i \sim N\left(\boldsymbol{0}, \sigma_e^2 I\right).$$

The SAS code for estimating the parameters of this model appears in Exhibit 8.2.

EXHIBIT 8.2 SAS Code for Logistic Mixed Parameter Estimation

```
data trees;
     input tree t y;
datalines;
1   118   30
1   484   58
⋮   ⋮     ⋮
5   1372  174
5   1582  177
;
proc nlmixed data=trees;
  parms beta1=190 beta2=2 beta3=0.003
      s2u1=100 cu12=0.1 s2u2=0.0005 s2e=100;
  bounds s2u1>=0, s2u2>=0, s2e>0;
  model y~normal((beta1+u1)/
  (1+exp((beta2+u2)-beta3*t)),s2e);
  random u1 u2~normal([0,0],
  [s2u1,cu12,s2u2])subject=tree;
run;
```

The output generated by this program appears in Table 8.3. The *Iteration History* is suppressed. Looking to the *Parameter Estimates Table*, we find that:

① 1. The p-values for the estimates of β_1, β_2, β_3, s_{u1}^2, and s_e^2 are all very small (each is less than 0.05), thus indicating that these estimates are all statistically significant.

2. Since the p-values for the estimates of $\boldsymbol{COV}(u1, u2)$ and s_{u2}^2 are quite large, it is evident that these estimates are not statistically significant. ■

TABLE 8.3 Output for Example 8.2

The SAS system
The NLMIXED procedure
Specifications

Data set	WORK.TREES
Dependent variable	*y*
Distribution for dependent variable	Normal
Random effects	*u*1 *u*2
Distribution for random effects	Normal
Subject variable	*tree*
Optimization technique	Dual quasi-Newton
Integration method	Adaptive Gaussian quadrature

Dimensions

Observations used	35
Observations not used	0
Total observations	35
Subjects	5
Max obs per subject	7
Parameters	7
Quadrature points	1

Parameters

beta1	beta2	beta3	s2u1	cu12	s2u2	s2e	NegLogLike
190	2	0.003	100	0.1	0.0005	100	146.969637

Note: GCONV convergence criterion satisfied.

Fit statistics

–2 log likelihood	262.9
AIC (smaller is better)	276.9
AICC (smaller is better)	281.1
BIC (smaller is better)	274.2

①
Parameter estimates

Parameter	Estimate	Standard error	DF	*t* value	Pr > |*t*|	Alpha	Lower	Upper	Gradient
beta1	189.27	16.4762	3	11.49	0.0014	0.05	136.84	241.71	–8.11E-7
beta2	2.1148	0.1207	3	17.53	0.0004	0.05	1.7308	2.4988	5.759E-6
beta3	0.002983	0.000226	3	13.22	0.0009	0.05	0.002265	0.003701	0.000147
*s*2u1	1171.12	0.02542	3	46064.1	< 0.0001	0.05	1171.04	1171.20	–0.00013
*cu*12	3.2986	2.6030	3	1.27	0.2945	0.05	–4.9853	11.5825	0.019647
*s*2*u*2	0.01221	0.02974	3	0.41	0.7091	0.05	–0.08245	0.1069	4.882563
*s*2*e*	59.4751	17.7657	3	3.35	0.0441	0.05	2.9368	116.01	–3.84E-7

(*Continued*)

TABLE 8.3 Output for Example 8.2 (*Continued*)

Covariance matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	s2u1	cu12	s2u2	s2e
1	beta1	271.46	0.3601	–0.00118	–0.1775	–2.4854	–0.01815	1.4509
2	beta2	0.3601	0.01456	0.000020	–0.00025	–0.01749	0.000041	–0.06136
3	beta3	–0.00118	0.000020	5.092E-8	2.928E-7	0.000091	9.591E-7	–0.00019
4	*s2u1*	–0.1775	–0.00025	2.928E-7	0.000646	–0.03528	–0.00046	0.3748
5	*cu12*	–2.4854	–0.01749	0.000091	–0.03528	6.7756	0.06533	–11.7765
6	*s2u2*	–0.01815	0.000041	9.591E-7	–0.00046	0.06533	0.000885	–0.2244
7	*s2e*	1.4509	–0.06136	–0.00019	0.3748	–11.7765	–0.2244	315.62

Correlation matrix of parameter estimates

Row	Parameter	beta1	beta2	beta3	s2u1	cu12	s2u2	s2e
1	beta1	1.0000	0.1812	–0.3187	–0.4238	–0.05795	–0.03704	0.004957
2	beta2	0.1812	1.0000	0.7219	–0.08168	–0.05569	0.01149	–0.02862
3	beta3	–0.3187	0.7219	1.0000	0.05104	0.1545	0.1429	–0.04707
4	*s*2u1	–0.4238	–0.08168	0.05104	1.0000	–0.5331	–0.6098	0.8297
5	*cu12*	–0.05795	–0.05569	0.1545	–0.5331	1.0000	0.8438	–0.2547
6	*s2u2*	–0.03704	0.01149	0.1429	–0.6098	0.8438	1.0000	–0.4246
7	*s2e*	0.004957	–0.02862	–0.04707	0.8297	–0.2547	–0.4246	1.0000

9

MODELING THE SIZE AND GROWTH RATE DISTRIBUTIONS OF FIRMS

9.1 INTRODUCTION

Firm growth/decline is an important consideration when discussing the dynamic behavior of an industry or even the aggregate economy. Whether one focuses on business capital formation or sales or net private-sector hirings, the determinants of firm growth are a key element in economic policy discussions, especially those pertaining to industrial concentration. Moreover, from a firm's viewpoint, the economic climate in which it operates will obviously impact its ability to prosper and grow; for example, issues pertaining to:

1. Uncertainty concerning how markets react to changes in technology
2. The intensity of competitive pressures in a dynamic market
3. Short-run adaptations to a changing and, at times, turbulent environment
4. Assessing and taking advantage of growth opportunities
5. Exercising its competitive advantages to the fullest extent possible

are a constant source of concern and managerial introspection.

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

9.2 MEASURING FIRM SIZE AND GROWTH

By far the most popular measure of firm size is "employment," although alternative measures of size such as "total sales" and "assets" have been used. Employment is attractive because it circumvents measurement problems connected with accounting for intangible assets, input price issues, and possible foreign currency adjustments.

As explained in Chapter 2, growth rates describe relative changes in the magnitude of a variable. So if the size (taken as employment level Y) of firm i at time t is denoted $Y_{i,t}$, then the period-to-period relative growth rate in $Y_{i,t}$ is

$$g_{i,t} = \frac{Y_{i,t} - Y_{i,t-1}}{Y_{i,t-1}} = \frac{Y_{i,t}}{Y_{i,t-1}} - 1. \tag{9.1}$$

Let's go a step further. For t a continuous variable, the instantaneous relative growth rate in Y is

$$\text{RGR} = \frac{1}{Y_t}\frac{dY_t}{dt};$$

and the mean relative growth rate over the time interval from t_1 to t_2 is

$$\overline{\text{RGR}} = \frac{\ln Y_2 - \ln Y_1}{t_2 - t_1}$$

(see Eq. 2.26). Now, if $t_2 - t_1 = 1$ (we have, say, yearly data), then we can write $\overline{\text{RGR}}$ as

$$g_{i,t} = \ln Y_t - \ln Y_{t-1}. \tag{9.2}$$

9.3 MODELING THE SIZE DISTRIBUTION OF FIRMS

A legitimate starting point for discussions on industrial structure is the aggregate *firm size distribution*. This type of distribution plots the values of a density estimator (such as a relative frequency) against a measure of firm size (e.g., number of employees or total sales) for a given industry at a specific point in time. When constructing a firm size distribution, it is important to "let the data speak for itself"; that is, no restrictions will be placed on a functional form of the size distribution. Indeed, one generally pursues a nonparametric approach to firm size density estimation. The tool employed for any such estimation is the kernel density estimator (KDE) (see Appendix 9.A for a discussion on kernel estimators). Once kernel estimates of the density of firm size have been obtained, one can choose an appropriate parametric functional form that best fits the empirical size distribution.

What sorts of parametric structures have been proposed to approximate the shape of the firm size distribution? (While the ensuing discussion is not meant to be exhaustive, it is sufficient to provide the reader with a feel for the types of theoretical distributions used.)

To answer this question, let us note first that a robust empirical regularity (some call it a "stylized fact"), which characterizes industrial structure and dynamics, is the observation that firm size distributions are positively or right skewed—their right-hand tails are heavy or elongated. Moreover, empirical evidence accumulated over the years indicates that the log-normal distribution or the Pareto distribution provides a good approximation to the firm size distribution. That is, for the log-normal case, the random variable Y has a *log-normal distribution* if $\ln Y$ has a normal distribution. The probability density function (pdf) of Y is then written as

$$g(y) = \frac{1}{\sigma y\sqrt{2\pi}} e^{-\frac{1}{2}((\ln y - \mu)/\sigma)^2}, \quad y > 0.$$

Here the mean and standard deviation of $\ln Y$ are μ and σ, respectively. In the Pareto (power-law) case, a random variable X has a *Pareto distribution* if its pdf is of the form

$$f(x) = \begin{cases} \dfrac{\alpha x_m^{\alpha}}{x^{\alpha+1}}, & x > x_m; \\ 0, & x \le x_m. \end{cases}$$

The mean and variance of the Pareto distribution are

$$\text{mean} = \frac{\alpha x_m}{\alpha - 1}, \quad \alpha > 1; \quad \text{variance} = \frac{x_m^2 \alpha}{(\alpha-1)^2(\alpha-2)}, \quad \alpha > 2.$$

(A much more detailed discussion of the log-normal distribution is provided by Appendix 9.B, Section 9.B.1. In particular, see the developments underlying Equations 9.B.1–9.B.5. Appendix 9.E, Section 9.E.1, houses an in-depth discussion of the Pareto distribution.) For instance, Figure 9.1 and Figure 9.2 provide examples of specific empirical firm size distributions.

In terms of an historical perspective, the first formal attempt to explain the size distribution of firms was undertaken by Gibrat (1931). He considered the size of French manufacturing firms in terms of employees and, upon observing a log-normal pattern to the firm size distribution (the logarithm of firm size is approximately normally distributed), proposed a model of firm growth capable of reproducing this pattern. That is, Gibrat (1931) proposed his *law of proportionate effect* (or simply *Gibrat's law* (GL)) that predicts that firm growth is a purely stochastic (or wholly random) phenomenon and that firm growth is consequently independent of firm size. (Since Gibrat's work is of paramount importance in that it provides a benchmark for empirical research into firm dynamics and growth, we shall develop his model in greater detail in the next section.)

Studies concerning the firm size distribution that emerged after Gibrat (1931) oftentimes presented varied results: Hart and Prais (1956) studied data on firms listed on the London Stock Exchange between 1885 and 1950 and concluded that the log-normal specification is correct; Simon and Bonini (1958), using data on large U.S.

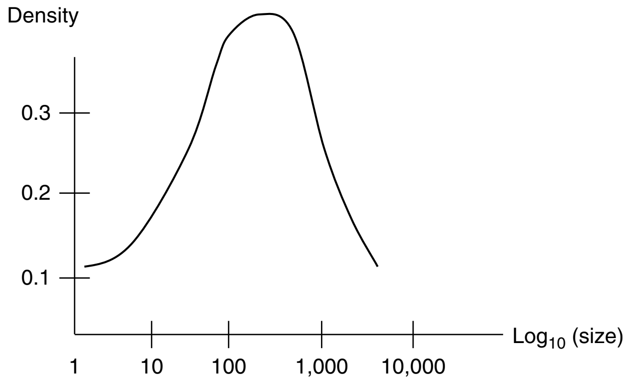

FIGURE 9.1 Firm size distribution (log-normal case). Kernel density estimate of firm size distribution based on employment data in Portuguese manufacturing, 1991, for publicly reporting firms. The approximation to a log-normal distribution is quite good since the density estimate is bell-shaped (Cabral and Mata, 2003).

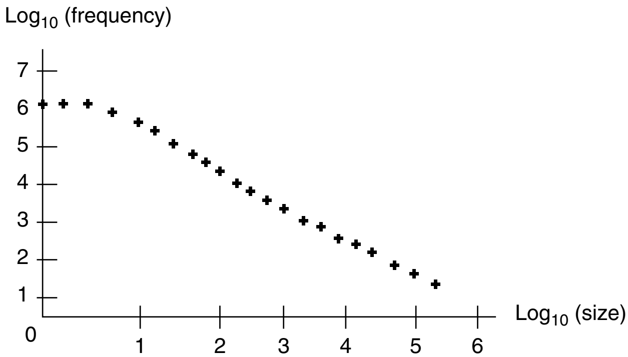

FIGURE 9.2 Firm size distribution (Pareto case). Size distribution of U.S. enterprises in 1998 (Census data) based on number of employees. The approximation to a Pareto (power-law) distribution is very good since the pattern of observations is markedly linear (Teitelbaum and Axtell, 2005).

firms, found that a log-normal curve generally fits well, although they view the log-normal distribution as a special case of the Yule distribution (for a discussion of the particulars of the Yule distribution, see Appendix 9.F); Steindl (1965) employs Austrian data and supports the Pareto distribution over the log-normal distribution since the former displayed superior performance in describing the upper tail of the size distribution; Ijiri and Simon (1977), using a sample of large (mostly publicly held) U.S. firms, similarly found that the Pareto distribution closely approximates the upper tail of the firm size distribution; Cabral and Mata (2003), using a comprehensive data set on Portuguese manufacturing firms operating in 1991, find that the firm size distribution is significantly right skewed, evolving over time to a more symmetric, stable, and approximately log-normal distribution; Santarelli and Lotti (2004) look at the evolution of the size distribution of new firms in Italian manufacturing and find that, over a period of five years, most of the distributions approach a log-normal distribution, although the more technically oriented industries approach the log-normal faster; Marsili (2005) found that, for the Dutch manufacturing sector, the Pareto curve provides a good fit for the upper tail of the firm size distribution, with the log-normal providing a better fit for the smaller firms in the distribution; Teitelbaum and Axtell (2005) studied data from the Census Bureau's "Statistics of U.S. Businesses" (1998–1999) and concluded that the firm size distribution was Pareto in form; Halkert (2006) concludes from data pertaining to the Chilean manufacturing census (1979–1996) that the firm size distribution follows a Pareto (power-law) distribution; and Growiec et al. (2008), using data from the worldwide pharmaceutical industry, conclude that the firm size distribution displays a log-normal shape with a Pareto (power-law) departure in the upper tail.

It should be evident from the preceding discussion that, while one can readily conclude that the firm size distribution is highly right skewed, there is no agreement on which skew distribution is most appropriate. As one might have also concluded, "the jury is still out." It may be the case that the functional form of the firm size distribution, which holds at the aggregate level, may not be appropriate when working with disaggregated industry data (see Quandt, 1966). Moreover, some features of the firm size distribution may turn out to be country specific (especially when one considers developed vs. developing economies). And it may even be the case that some sectors or industries do not display a good fit no matter what distribution is utilized.

This said, it may be the case that one needs to examine the firm size distribution in an evolutionary context; that is, while a wealth of empirical evidence indicates that the initial log-size distribution for start-up firms is markedly right skewed, it tends to become more symmetric with the passage of time, perhaps because new firms have been found to grow faster than more established and larger ones. In this light it has been argued that the log-normal distribution serves as the limiting size distribution to which a specific group of firms will converge (see Angelini and Generale, 2008; Cabral and Mata, 2003; Lotti and Santarelli, 2001).

9.4 GIBRAT'S LAW (GL)

The usual starting point for discussions on the growth of firms is Gibrat's "law of proportionate effect" (Gibrat, 1931). Gibrat formulated his "law"[1] of industrial dynamics in an effort to explain the observed skew shape of the size distribution of

[1] Incidentally, it may be informative to some to consider just what is meant by a "law" in this context and how it differs from a "theory." A theory attempts to explain "why" something happens. Hence, a theory is used to explain an observed phenomenon. Moreover, a theory consists of (1) definitions of the "variables" to be used, (2) "assumptions" that state the conditions under which the theory is relevant, (3) some "hypothesis" about how the variables are related, and (4) "predictions" (conditional statements of an "if" A, "then" B nature) that can be made from the assumptions underlying the theory and that are subject to tests against actual observed behavior. Now, if the theory holds up against the continued scrutiny of empirical evidence presented for many different venues, time periods, and data sets, then its status is elevated to that of a "law."

French manufacturing firms (Caves, 1998; Chesher, 1979; Coad, 2009; Geroski, 1995; Sutton, 1997).

Specifically, he assumed that the said distribution can be modeled by postulating that some unknown function of firm size X was normally distributed. Gibrat then conjectured that the simplest functional form that would fit the observed data points was $\ln X$.

Gibrat's law of proportionate effect implies that, in a process of growth, equal proportionate increments have the same chance of occurring in a specific time interval no matter what the current size of the firm is. Moreover, the law provides the requirement for log-normality; that is, firm size X at any step in the process is subject to the *law of proportionate effect* if the expected change in X is a random proportion of the previous value of X or

$$\Delta X_t = X_t - X_{t-1} = \varepsilon_t X_{t-1}, \tag{9.3}$$

where the ε_t's are "growth shocks" that amount to independent and identically distributed Gaussian random error terms. Thus, firm growth is posited as a stochastic process resulting from the interaction of many independent random factors. In this regard, firms within an industry draw growth rates from a distribution that is the same for all firms regardless of their current size or past history so that "growth rates are independent of firm size." As Mansfield (1962) notes, the probability of a given proportionate change in size over a specified period is the same for all firms in a particular industry, irrespective of their size at the beginning of the period. Hence, the law of proportionate effect operates via a simple stochastic process with Gaussian shocks that generates theoretical asymmetric log-normal firm size distributions.

To see exactly how Equation 9.3 leads to a log-normal firm size distribution, let us iterate this equation so as to obtain

$$X_t = (1+\varepsilon_t)X_{t-1} = X_0 \Pi_{s=1}^{t}(1+\varepsilon_s).$$

(Note that if we take $\ln X_t = \ln(1+\varepsilon_t) + \ln X_{t-1}$, we see that Equation 9.3 is a "random walk process" on a logarithmic scale.) Then

$$\ln X_t = \ln X_0 + \sum_{s=1}^{t} \ln(1+\varepsilon_s).$$

Let us approximate $\ln(1+\varepsilon_s)$ by ε_s (justified if we take short time periods so that ε_s is close to zero) to obtain

$$\ln X_t \approx \ln X_0 + \sum_{s=1}^{t} \varepsilon_s.$$

Moreover, $\ln X_0$ becomes insignificant as $t \to \infty$ so that we ultimately obtain

$$\ln X_t \approx \sum_{s=1}^{t} \varepsilon_s.$$

Now, since $\bar{\varepsilon}_t = \Sigma_{s=1}^{t}\varepsilon_s / t$, it follows that $\ln X_t = t\bar{\varepsilon}_t = t\bar{\varepsilon}$ since $\mathrm{E}(\varepsilon_t) = \bar{\varepsilon}$ for all t. Then, via the central limit theorem (CLT),

$$\sqrt{t}(\bar{\varepsilon} - \mu_\varepsilon)/\sigma_\varepsilon \to N(0,1) \quad \text{as } t \to \infty$$

(the normal distribution of $\ln X_t$). (A detailed discussion of this process appears in Appendices 9.B and 9.C.)

The importance of GL is that it represents the first formal dynamic model of firm growth and industrial structure; it provides a benchmark against which all alternative theories of firm growth and industrial development are compared. And for empirical purposes, it constitutes a useful null hypothesis against which firm growth can be compared. Sometimes researchers distinguish between two forms of GL: the weak case (only mean growth is independent of firm size) and the strong case (the growth rate of a firm is independent of its size, and the process yields a distribution of firm sizes that is log-normal).

9.5 RATIONALIZING THE PARETO FIRM SIZE DISTRIBUTION

The preceding section on Gibrat's law of proportionate effect provided the impetus for using the log-normal distribution to describe the size distribution of firms (Coad, 2008; Huberman and Adamic, 1999; Reed, 2001). While some researchers view GL as offering the best description of industrial development and firm dynamics, it is well known that the log-normal firm size distribution has its limitations: (i) the early studies that supported log-normality were typically restricted to large firms (see, for instance, Hart and Prais, 1956; Simon and Bonini, 1958); and (ii) under GL, all firms are drawn from the same pdf and, consequently, are, for a particular cohort, assumed to be of the same age. However, more recent studies involving data sets that contain smaller and younger firms along with larger and more established ones have found that the Pareto distribution provides a good approximation to the empirical firm size distribution (see, in particular, Axtell and Zipf, 2001; deWit, 2005). In this regard, although the log-normal and Pareto distributions closely explain most of the shape of the firm size distribution (deWit, 2005), the Pareto distribution is heavier in relation to the log-normal at the lower tail (where there appear a large number of very small firms) and decreases less rapidly than the log-normal over the upper tail (where a small number of large firms are found).

It is no surprise then that the Pareto and log-normal distributions exhibit a high degree of similarity when it comes to explaining the aggregate firm size distribution. In fact, as we shall now see, the Pareto firm size distribution can be obtained by:

1. Assuming a Gibrat growth process (Eq. 9.B.7) and its resulting log-normal firm size distribution (Eq. 9.B.5) (see also Equation 9.B.13)
2. Relaxing the assumption that all firms are of the same age

3. Specifying an exponential "firm age distribution" of the form

$$h(t) = \lambda e^{\lambda t}; \quad \text{and} \tag{9.4}$$

4. "Melding" the log-normal firm size distribution with the exponential firm age distribution[2]

Then, via Equation 9.6, let us integrate the log-normal distribution over the exponential distribution so as to obtain the Pareto distribution or

$$\begin{aligned} f(y) &= \int \lambda e^{\lambda t} \frac{1}{y\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{\ln y - \bar{\varepsilon} t}{\sigma\sqrt{t}}\right)^2} dt \\ &= cy^{-\beta}, \quad 1 \le \beta < +\infty, \end{aligned} \tag{9.5}$$

where

$$\beta = 1 - \frac{\bar{\varepsilon}}{\sigma^2} + \frac{(\bar{\varepsilon}^2 + 2\lambda\sigma^2)^{1/2}}{\sigma^2},$$

$$c = \lambda / \sigma((\bar{\varepsilon} / \sigma^2) + 2\lambda)^{1/2}.$$

In sum, if Gibrat's proportionate growth model with random shocks is "mixed" with an exponential firm age distribution, then a Pareto firm size distribution obtains.

9.6 MODELING THE GROWTH RATE DISTRIBUTION OF FIRMS

In Section 9.3 we indicated that the firm size distribution was consistently found to be right skewed. However, when it comes to growth rates, it is often the case that the *firm growth rate distribution* is "tent shaped." This distribution plots the values of a density estimator against a measure of firm growth (such as Equation 9.2) for a given industry at a particular point in time.

Early research on firm growth rate distributions reported that the empirical densities obtained exhibited heavy tails (see, for instance, Little, 1962). Stanley et al. (1996), using the Compustat database of U.S. manufacturing firms, discovered that

[2] To obtain a mixture of Equations 9.B.5 and 9.4 (Huberman and Adamic, 1999), let us apply the following rule: if the probability density function of a random variable Y, $p(y, t)$, depends on a parameter t, which itself is subject to the probability density function $h(t)$, then the probability density function of Y is provided by

$$f(y) = \int h(t)p(y,t)dt,; \tag{9.6}$$

that is, we integrate $p(y,t)$ over $h(t)$ to obtain $f(y)$.

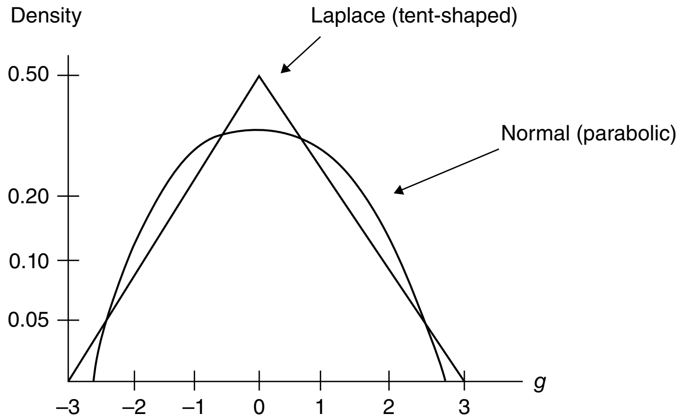

FIGURE 9.3 Comparison of the Laplace and normal distributions (semilog coordinates).

log firm growth rates are *(classical) Laplace distributed*. Specifically, let y_t denote the size of a firm at time t and let $g_t = \ln Y_t - \ln Y_{t-1}$ denote the log growth rate at time t (see Equation 9.2). Then across all firms and over time, the pdf of g is

$$f(g) = \frac{1}{\sqrt{2}\sigma} e^{-\left(\sqrt{2}|g-\bar{g}|\right)/\sigma}, \tag{9.7}$$

where $\bar{g}$ (a location parameter) is the average log growth rate and $\sigma(>0)$ is a scale parameter. (For additional details on the classical Laplace distribution, see Appendix 9.D, Section 9.D.1.)

It is well known that the Laplace distribution is heavy tailed when compared to a normal distribution with the same mean and standard deviation (Fig. 9.3; Teitelbaum and Axtell, 2005); and in the current context, this means that, in the Laplace case, there are many more firms that experience extreme (both positive and negative) growth rates relative to the normal or Gaussian case.

The findings of Stanley et al. (1996)—that the Laplace distribution of growth rates appears to be an extremely robust characterization of industrial dynamics that holds at various levels of aggregation—have been confirmed by a wide assortment of researchers (see, for instance, Fig. 9.4; Teitelbaum and Axtell, 2005). Indeed, Amaral et al. (1997) use data on U.S. manufacturing firms and obtain a tent-shaped distribution (using log-log plots) that closely resembles the Laplace distribution; Lee et al. (1998) calculated binned growth rates for firms (and countries' GDPs) and obtained similar regression lines in the log-log plane, thus pointing to the Laplace form; Bottazzi and Secchi (2001), employing data on the worldwide pharmaceutical industry, find that the growth rate distribution is close to the Laplace distribution; Bottazzi et al. (2002) conclude that the Laplace distribution is confirmed for Italian manufacturing; Bottazzi and Secchi (2003a), using the Compustat database for U.S. firms, find that the Laplace distribution is suitable for modeling growth rates at both the aggregate and disaggregate levels; Bottazzi and Secchi (2003b), studying Italian

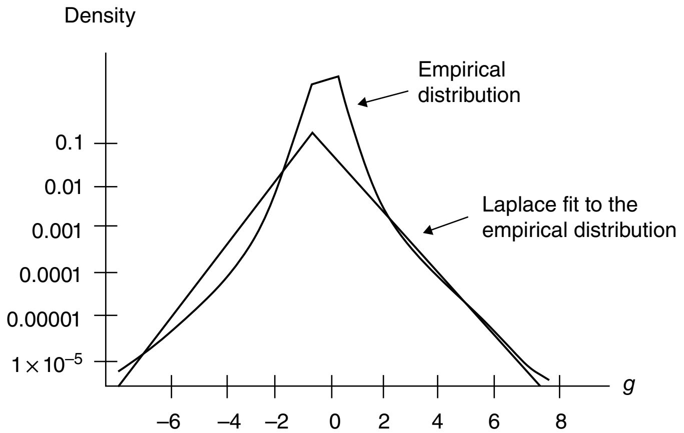

FIGURE 9.4 Comparison of the empirical distribution of log growth rates (Cefis data, 1998) with the Laplace distribution (semilog coordinates).

manufacturing data, find that the Laplace distribution fits well; Bottazzi and Secchi (2003b, 2005) confirm the use of the Laplace distribution in non-U.S. data sets as a useful representation of growth rate distributions; and Delli Gatti et al. (2005), by constructing a stochastic economy with simulated data, use the two so-called universal laws: (i) the distribution of firm's size is right skewed (and thus can be described by a power-law pdf); and (ii) the firm growth rate distribution follows a Laplace distribution, to show that, under very general conditions, (i) implies (ii)—the power-law distribution of firms' size is at the root of the Laplace distribution of growth rates. Similarly, Teitelbaum and Axtell (2005) utilized data from the Census Bureau's Statistics of U.S. Businesses (1998–1999) and concluded that growth rate dynamics overall and across industries are readily described by the Laplace distribution; Fu et al. (2005), using data from different levels of aggregation (including a pharmaceutical industry database (1994–2004), U.S. firms from 1973 to 2004 in the Compustat database, and growth rates of GDP for 195 countries (World Bank, 1960–2004)), find that the growth rate distribution is Laplacian in the body with power-law tails at all levels of aggregation; Halkert (2006) employs data from the Chilean manufacturing census from 1979 to 1996 and finds that tent-shaped (Laplacian) densities occur after conditioning on the age as well as the size and industry of the plant, with plant growth rates displaying fat tails; Bottazzi and Secchi (2006a) use the MICRO.1 databank (includes a large portion of Italian manufacturing) and find evidence to support the tent-shaped pattern for the firm growth rate distribution and also confirm the results of Stanley et al. (1996) and Bottazzi and Secchi (2005) that the shape is not the effect of aggregation; Bottazzi et al. (2008) studied growth rates for French manufacturing firms and found that the empirical distribution is tent shaped but had flatter tails than the Laplace distribution; and Pammolli et al. (2007), using the same data as Fu et al. (2005), develop a model to explain proportionate growth rates, with the model predicting, at any level of aggregation, that the pdf of growth rates is Laplace in the center, with power-law tails.

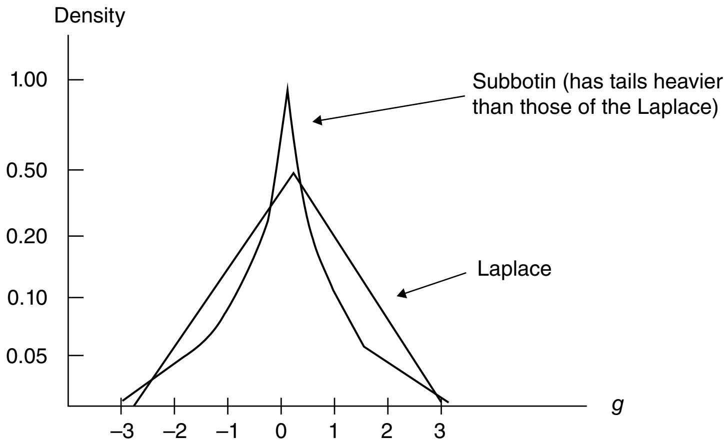

FIGURE 9.5 Comparison of the Laplace and Subbotin distributions.

The preceding discussion has uncovered some important considerations pertaining to the overall survival of firms. That is, since the Laplace distribution has much heavier tails relative to the normal distribution, we may conclude that some firms experience much stronger log growth rate fluctuations than would be the case if the economy were more Gaussian in nature. Moreover, if there is uncovered a significant departure from the Laplacian form, it is typically towards even heavier-tailed growth rate distributions and not towards normality. This said, some of the conclusions obtained from a few of the aforementioned studies have hinted at the possibility that a distribution with tails even heavier than those of the Laplace distribution should be used to model the log growth rate pdf (e.g., Bottazzi and Secchi, 2004; Bottazzi et al., 2008; Fu et al., 2005; and Pammolli et al., 2007).

If one is looking for a distribution that is tent shaped but with tails even heavier than those of the Laplace distribution, then the *Subbotin distribution* (Subbotin, 1923) meets these criteria. This distribution is a generalization of the Laplace distribution (as well as the log-normal distribution) and has a pdf of the form

$$f(g) = \frac{1}{2ab^{1/b}\Gamma\left(1+\frac{1}{b}\right)} e^{-\frac{1}{b}\left|\frac{g-\bar{g}}{a}\right|^{b}}, \quad -\infty < g < +\infty, \tag{9.8}$$

where b is a shape parameter (for $b=1$, the Subbotin distribution specializes to the Laplace distribution), a is a scale parameter, $\bar{g}$ is a location parameter, and $\Gamma(x)$ is the gamma function (of x) (see Appendix 9.D, Section 9.D.3, for additional details on the Subbotin distribution). (It is important to note that it is the value of the parameter b that dictates the heaviness of the tails—the larger the value of b, the lighter the tails.) In addition, a comparison between the Laplace and Subbotin pdf (for $a=1$) is presented in Figure 9.5 (Teitelbaum and Axtell, 2005).

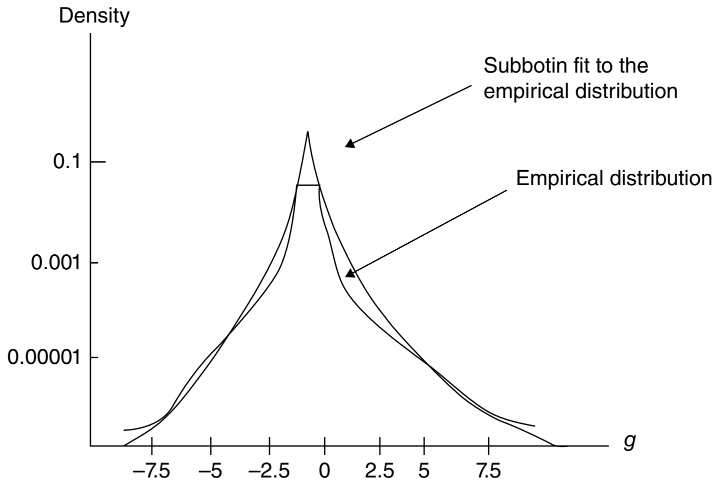

FIGURE 9.6 Comparison of the empirical distribution of log growth rates (U.S. Census data, 1998) with the Subbotin distribution ($b \approx 0.7$) having the same mean and standard deviation (semilog coordinates).

As indicated in Figure 9.6, Teitelbaum and Axtell (2005) fit a Subbotin distribution to (aggregate) Census data and get a fit better than the one obtained using the Laplace distribution; that is, the heavy tails reveal many large expansions and contractions. (They also showed that peaked and heavy-tailed growth rate distributions also emerged on an intra-industry basis.) A similar set of results was obtained by Bottazzi et al. (2011b). They explored the growth rate distribution of French manufacturing firms at the aggregate and disaggregate levels and found that the Subbotin distribution fits well; it produces a noticeable departure from the Laplace form in that the heavy tails obtained correspond to a high frequency of extreme events (firms undergo significant positive and negative changes in size).

There is some empirical evidence that the fit obtained by the Subbotin distribution can be improved upon by admitting an asymmetry to the firm growth rate distribution. In particular, Perline et al. (2006), using a set of comprehensive data (1998–2003) on U.S. businesses extracted from the Census database, found that, over longer time frames, business growth rate distributions become more asymmetric and trend slowly towards lighter tails and thus exhibit fewer extreme growth changes. That is, these authors find that a more flexible pdf called an *asymmetric Subbotin distribution* provides a very good approximation to U.S. business growth rate distributions (see Appendix 9.D, Section 9.D.3, for a discussion on this distribution) since it has a (skewness) parameter that allows it to transition from being very light tailed to very heavy tailed. In general, Perline et al. (2006) found that, due to growth rate asymmetries, negative growth rates seem to be more volatile than positive ones (see Fig. 9.7 for one of their asymmetric Subbotin fits (their Fig. 9.6.e) to the Census data).

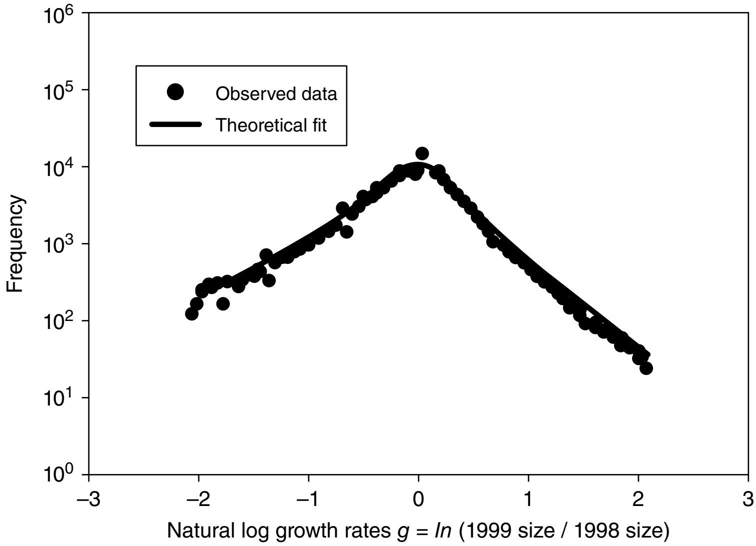

FIGURE 9.7 Subbotin fit to 1998–2003 log growth rates for 1998 establishment sizes of 32–63. *Source*: SBA Office of Advocacy Small Business Research Summary, No. 235, Dec. 2006.

9.7 BASIC EMPIRICS OF GIBRAT'S LAW (GL)

9.7.1 Firm Size and Expected Growth Rates

We noted earlier that, by way of the law of proportionate effect, the expected change in firm size X is a random proportion of the previous value of X or Equation 9.3 holds; that is, firm growth is a wholly stochastic process. We also noted that, by virtue of this process, firms within an industry draw growth rates from a distribution that is the same for all firms regardless of their current size or past history so that "firm growth rates are independent of firm size." To test this hypothesis of firm size evolution, let us start with the expression

$$X_{i,t} = X_{i,t-1}^{\beta} e^{a_i + \varepsilon_{i,t}}, \tag{9.9}$$

where the random error terms a_i and $\varepsilon_{i,t}$ are taken to be independent and identically distributed Gaussian. Then we may rewrite Equation 9.9 as

$$g_{i,t} = X_{i,t} / X_{i,t-1} = X_{i,t-1}^{\beta-1} e^{a_i + \varepsilon_{i,t}}$$

or

$$\begin{aligned} \ln g_{i,t} = \ln X_{i,t} - \ln X_{i,t-1} &= a_i + (\beta - 1) \ln X_{i,t-1} + \varepsilon_{i,t} \\ &= a_i + \gamma \ln X_{i,t-1} + \varepsilon_{i,t}. \end{aligned} \tag{9.10}$$

Clearly a test of GL hinges directly upon the behavior of the estimated slope coefficient $\hat{\gamma} = \hat{\beta} - 1$: (i) if $\hat{\gamma} = 0$ or $\hat{\beta} = 1$, the strong form of GL is confirmed (growth rates are independent of firm size); (ii) if $\hat{\gamma} < 0$, then $\hat{\beta} < 1$ and thus expected growth rates are higher for smaller firms (i.e., firm size experiences a "reversion to the mean"); and (c) if $\hat{\gamma} > 0$, then $\hat{\beta} > 1$ and thus expected growth rates are higher for larger firms.

It is important to note that the *weak version of GL* obtains if it is determined that there is no statistically significant relationship between growth and firm size ($\hat{\gamma}$ is "almost" zero or $\hat{\beta}$ is slightly greater than or slightly less than unity).

Equations 9.9 and 9.10 are the usual starting point for empirical investigations of GL. While many econometric issues are associated with the use of these equations (the topic of Section 9.7.3), the magnitude of the estimate of γ (or β) (and the statistical significance associated with the same) is key to determining the efficacy of GL.

Some early investigations of GL, which were based on samples of large and mature firms, tended to confirm the law (e.g., Hart and Prais (1956) studied large UK firms and found $\hat{\beta} > 1$, and Simon and Bonini (1958) considered large U.S. firms). This is in contrast to Mansfield (1962) who actually tested three different versions of GL for the U.S. steel, petroleum, and tire sectors:

M1. GL holds for all firms, including those that leave the industry during the period of study. (But since smaller firms are more likely to exit an industry relative to larger ones—exiting firms are imputed a proportionate growth rate of −1—this version of GL seems to be incapable of uncovering any relationship between firm growth and size since the probability of a firm exiting the industry is not independent of its size.)

M2. GL holds only for firms that survive over the entire period studied. (If survival is not independent of a firm's initial size—small firms are more likely to exit than large ones—then this version of the law can be affected by sample selection bias.)

M3. GL holds only for firms exceeding the minimum efficient size in the industry. (Such firms tend to be large surviving firms with "*J*-shaped" long-run average cost curves.)

He concludes that the law does not generally hold for any of these three cases since the variance of the observed growth rates decreases with size. Subsequent studies by and large support Mansfield's findings; that is, there exists a negative relationship between firm size and growth $(\hat{\beta} < 1)$ or that, on average, small firms tend to grow faster than large ones (e.g., Evans (1987a, b) and Hall (1987) worked with U.S. manufacturing data; Dunne et al. (1988, 1989) studied U.S. plant data; Dunne and Hughes (1994) and Hart and Oulton (1996) used a broad database of UK firms; Audretsch and Mahmood (1994) evaluated the post-entry performance of U.S. manufacturing firms started in 1976; Goddard et al. (2002) studied sizeable Japanese manufacturing firms; and Bottazzi and Secchi (2003a) dealt with relatively large U.S. manufacturing firms, among others).

Many of the more recent studies on GL include both small and large manufacturing firms in the sample. When this is done, the most common finding is that growth rates tend to be negatively related to firm size; that is, small firms tend to grow faster than large ones. Hence, the "broad-sample" empirical evidence does not support GL (e.g., Petrunia (2005) studied the manufacturing (and retail) sector of Canada; Esteves (2007) considered the manufacturing (and services) sector in Brazil; Bottazzi and Secchi (2005) dealt with the worldwide pharmaceutical sector; Audretsch and Elston (2002) presented results from German manufacturing; Laitinen (1999) analyzed Finnish firms; and Bottazzi et al. (2011b) studied French manufacturing firms). In addition, Teitelbaum and Axtell (2005), using a broad-based sample of firms taken from U.S. Census data from 1998 to 1999, concluded that for all industries analyzed, growth rates are unrelated to establishment size.

While GL is typically rejected for highly inclusive samples of firms, some studies have found that a weak version of this law does hold for subsamples of larger firms or for firms above a given size threshold (e.g., when Hart and Oulton (1996) tested firm size classes, they found that there was no relationship between size and growth for large firms). Similar conclusions were reached by Audretsch and Elston (2002) for German manufacturing, Geroski (1995) in a survey of firm growth dynamics, Esteves (2007) for the Brazilian manufacturing (and services) sector, and Bottazzi et al. (2008) for French manufacturing firms, among others.

An important question that naturally arises when testing GL is whether or not the growth experience of new small firms is different from that of older and more established larger firms. Lotti et al. (1999) studied new small Italian manufacturing firms and found that, in some selected industries, GL fails to hold in the years immediately after start-up. In subsequent years the patterns of growth of smaller firms do not differ significantly from those of larger ones, and thus GL is confirmed. So although an inverse relationship between size and growth emerged during early growth of start-up firms, most industries displayed convergence to a Gibrat-like pattern of growth over time. In another study (Lotti et al., 2003) of a cohort of new Italian firms, it was found that smaller firms initially grow faster than large ones, but it became more difficult to reject the hypothesis of the independence of growth and size as time passed. A similar set of results was obtained by Geroski and Gugler (2004) for large European firms, by Becchetti and Trovato (2002) for the Italian manufacturing sector, and by Geroski et al. (2003) and Cefis et al. (2007) for the worldwide pharmaceutical sector. In addition, Santarelli and Lotti (2004) focused on the size distribution of new firms in Italian industries and found that most of the distributions approach a log-normal distribution.

Other studies involving small business dynamics detected a negative relationship between firm size and expected growth rates (e.g., Johnson et al. (1999) and Phillips and Kirchhoff (1989) studied U.S. services data). However, Audretsch et al. (2004) studied small-scale Dutch services and determined that a weak version of GL could not be rejected—there was no statistically significant relationship detected between size and expected growth rates.

If the focus is entirely on small businesses (not necessarily start-ups), most studies find a negative relationship between expected growth and firm size. See, for instance, Segarra and Callejon (2002) for Spanish manufacturing and Almus and Nerlinger (2000) for German manufacturing.

By way of a brief summary of this section, a theme that consistently emerges is that: (i) GL has been rejected by recent studies using broad-sample data—small and younger firms display a higher growth rate than large and older firms so that firm growth does not appear to be independent of firm size—and (ii) GL is confirmed for subsamples of large well-established firms. And all this holds for manufacturing as well as for the services sector (with the exception of the Audretsch et al. (2004) finding for the Dutch hospitality sector).

9.7.2 Firm Size and Growth Rate Variability

It is a well-known empirical result that there exists a negative relationship between growth rate variance and firm size (for a couple of early observations of this phenomenon, see Meyer and Kuh (1957) and Hymer and Pashigan (1962)). (Evans (1987a) noted that this relationship holds over time for a high percentage of manufacturing industries and that there is some evidence that the sign of the size–variability relationship depends on the age and the number of plants in operation by the firm. See also Caves (1998) and Hart and Oulton (1996).)

What sort of model has been offered as an explanation for the relationship between the firm size and the variance of growth rates? Following Stanley et al. (1996) and Sutton (2001), for the set of firms with annual sales revenue within the interval $(X, X+\varepsilon)$, let us consider the variance σ^2 of the change in sales ΔX from one year to the next. We then ask: "What is the connection between $\sigma^2(\Delta X)$ and X?" The (usual) answer reported is a "power-law" relationship of the form

$$\sigma^2(\Delta X) = AX^{\beta}, \tag{9.11}$$

where β is the power-law coefficient. In terms of the proportionate growth rate $g=\Delta X/X$,

$$\begin{aligned} \sigma(g) &= \sigma\left(\frac{\Delta X}{X}\right) = \frac{\sigma(\Delta X)}{X} = A^{1/2}X^{\frac{\beta}{2}-1} \\ &= cX^{r}, \end{aligned} \tag{9.12}$$

where $c=A^{1/2}$ and $r=(\beta/2)-1$. Then a power-law relationship emerges if a plot of $\ln \sigma(g)=\ln c+r \ln X$ displays linearity. (Stanley et al. (1996) obtain $\hat{r} = -0.15$ using the Compustat, 1980–1998, database. Sutton (2001) reexamines this data set and finds that the power coefficient r fluctuates over time while falling in the range $-0.21 \le \hat{r} \le -0.15$.)

What is the link between the variance of growth rates and the firm size? Stanley et al. (1996) propose a model in which each firm consists of a collection of equal-sized business units and then entertains the possibility that the growth rates of these units are either statistically independent or correlated in some fashion. While independence is consistent with power-law behavior, in this circumstance r would have to equal −0.5 (Lee et al., 1998)—a value far below their reported level of −0.15. As an alternative justification of the link between firm size and growth rate variability, Stanley et al. (1996) offer a model of firm decision making, which might lead to the correlation of growth rates across a firm's business components. However, the question remains: "How does a power-law relationship emerge from a managerial model involving correlated changes in sales?" In fact, using the Stanley et al. (1996) data, Sutton (2001) finds that, while the growth rates of consistent businesses within a firm are positively correlated, the observed association is extremely weak and unstable.

As a follow-up to this finding, Sutton (2001) offers a so-called benchmark model of the firm based on a set of independent businesses that vary in size and provides a lower bound to the slope coefficient r in Equation 9.12. (Specifically, he hypothesizes that, for a randomly selected firm of size $X = \Sigma_i X_i$, where X (an integer) is annual sales value and X_i (an integer for all i) is annual sales revenue from business i, all partitions (X_i) of X are equiprobable. Then assuming that each business is affected by the same distribution of proportional shocks, it is demonstrated that the size–variance relationship displays power-law behavior with $-0.25 \leq \hat{r} \leq -0.21$, which corresponds to the lower bound of observed values.) Additionally, Sutton (2001) concludes that the presence of correlation in the growth rates of a firm's businesses will lead to a flattening of the expression provided in Equation 9.12.

The observation of a negative relationship between firm size and growth rate variability is essentially ubiquitous (e.g., see Amaral et al. (1997) and Bottazzi and Secchi (2003a) for results on the U.S. manufacturing sector; Lee et al. (1998) for a study of large U.S. manufacturing firms; the findings of Bottazzi and Secchi (2001), Matia et al. (2004), and Bottazzi and Secchi (2006b) for the worldwide pharmaceutical industry; the results of Petrunia (2005) for the retail and manufacturing sectors of Canada; the findings of Sutton (2002) for Japanese manufacturing data; and the results presented by Bottazzi et al. (2008) for French manufacturing firms). It is important to note, however, that Bottazzi et al. (2011a) fail to uncover any relationship between growth rate variability and firm size for Italian manufacturing. A similar conclusion was reached by Teitelbaum and Axtell (2005) for U.S. Census data—the variability in growth rates does not depend on establishment size.

While the notion that the relationship between the firm size (represented by sales revenue X) and the variance of firm growth rates follows an approximate power law $\sigma(X) = X^{-\beta(X)}$ (Stanley et al., 1996) has gained wide acceptance, Riccaboni et al. (2008) introduce a model of proportional growth, which leads these authors to question the appropriateness of the basic power-law specification. Treating firms as classes composed of various numbers of units having variable sizes and expressing firm sales as the sum of the sales of all their products, these authors derive a size–variance relationship from the model developed by Fu et al. (2005), which has a distribution that is Laplacian in the center and has power-law tails. This model was chosen by Riccaboni et al. (2008) because it accurately predicted the shape of the distribution of firm growth rates and the shape of the distribution of firm size. Hence, the Fu et al. (2005) model should have profound implications for the variance–size relationship of firms.

The main conclusion of Riccaboni et al. (2008) is that the size–variance relationship is not a true power law with a single well-defined exponent β, but, rather, undergoes a slow crossover from $\beta = 0$ for $X \to 0$ to $\beta = \frac{1}{2}$ for $X \to +\infty$. For a realistic set of model parameters, $\beta(X)$ is approximately constant and, depending on the average number of units in the firm, can vary from 0.14 to 0.20.

9.7.3 Econometric Issues

When it comes to addressing possible econometric problems associated with making regression parameter estimates from firm growth rate data, the "usual suspects" are heteroscedasticity and autocorrelation. With respect to heteroscedasticity, our findings in Section 9.7.2 pointed to a negative relationship between firm size and growth rate variance. In the light of this result, studies of GL from the 1980s on routinely corrected for heteroscedasticity (e.g., see White's model specification test (White, 1980)). If heteroscedasticity is detected, a variety of variance-stabilizing transformations are available (see Panik, 2009, pp. 156–161).

Dealing with autocorrelation is slightly more complicated. We noted in Chapter 4 that under the assumptions of the Classical Linear Regression Model, $COV(\varepsilon_{i,t}, \varepsilon_{i,t-s}) = 0$ for all $t > s$. If this condition does not hold, then the random disturbance terms ε_i, $i = 1, \ldots, n$, are autocorrelated; that is, the disturbance occurring at one data point is associated with the disturbance occurring at any other such point.

To test for the presence of autocorrelation and to examine how it might impact the efficacy of GL, let us posit a first-order autoregressive scheme

$$\varepsilon_{i,t} = \rho\varepsilon_{i,t-1} + \eta_{i,t}, \tag{9.13}$$

where $\eta_{i,t}$ is a nonautocorrelated random disturbance term (see Chapter 4, pp. 34–37). Additionally, from Equation 9.10,

$$\begin{aligned} \varepsilon_{i,t} &= \ln g_{i,t} - a_i - \gamma \ln X_{i,t-1}, \\ \varepsilon_{i,t-1} &= \ln g_{i,t-1} - a_i - \gamma \ln X_{i,t-2}. \end{aligned}$$

A substitution of these expressions into Equation 9.13 yields (after some rearrangement)

$$\begin{aligned} \ln g_{i,t} = a_i(1-\rho) + \rho \ln g_{i,t-1} + \gamma \ln X_{i,t-1} \\ - \gamma\rho \ln X_{i,t-2} + \eta_{i,t}. \end{aligned} \tag{9.14}$$

Once 9.14 is estimated, we may conclude that the law of proportionate effect holds only if $\hat{\gamma}\left(=\hat{\beta}-1\right)=0$ in Equation 9.10 and the disturbances $\varepsilon_{i,t}$ are nonautocorrelated over time.[3] Hence, GL is deemed valid if the joint null hypothesis

$$H_0(\text{law of proportionate effect in operation}): (\gamma, \rho) = (0, 0)$$

is not rejected in favor of the alternative

$$H_1(\beta \neq 1 \text{ and / or the } \varepsilon_{i,j}\text{'s are autocorrelated}): (\gamma, \rho) \neq (0, 0).$$

($\gamma=0$ is a necessary but not a sufficient condition for GL to hold so that GL can be rejected, due to the presence of autocorrelated disturbances, even when $\rho \approx 1$.)

The empirical results concerning growth rate autocorrelation are, at best, mixed. On the one hand, positive autocorrelation has been detected by Ijiri and Simon (1967) for large U.S. firms, Singh and Whittington (1975) for UK firms, Dunne and Hughes (1994) for large UK manufacturing firms, Chesher (1979) and Geroski et al. (1999) for UK quoted companies, Bottazzi and Secchi (2001) for worldwide pharmaceuticals, and Bottazzi and Secchi (2003a) for U.S. manufacturing, among others. On the other hand, negative autocorrelation has been uncovered by Goddard et al. (2002) for large Japanese firms, Bottazzi et al. (2011a) for Italian manufacturing, Bottazzi et al. (2011b) for French manufacturing, and Contini and Revelli (1989) for Italian manufacturing. Not surprisingly, a number of other studies have failed to find any significant growth rate autocorrelation (e.g., see Geroski and Mazzucato (2002) for the U.S. automobile industry and Lotti et al. (2003) for Italian manufacturing).

An interesting result that emerged from the study of French manufacturing firms for the period 1996–2002 by Coad (2007) was that the nature of annual growth rate

[3] From Equation 9.10,

$$g_{i,t} = e^{a_i+\varepsilon_{i,t}} X_{i,t-1}^{\beta-1},$$
$$\frac{X_{i,t}-X_{i,t-1}}{X_{i,t-1}} = e^{a_i+\varepsilon_{i,t}} X_{i,t-1}^{\beta-1},$$

and thus

$$X_{i,t}-X_{i,t-1} = \left(e^{ai+\varepsilon_{i,t}} X_{i,t-1}^{\beta-1}\right) X_{i,t-1} = h_{i,t} X_{i,t-1}.$$

If $\beta=1$ (the "strong" form of GL), then $h_{i,t} = e^{a_i+\varepsilon_{i,t}}$ can be viewed as the "proportion" in the Law of Proportionate Effect—the change in X at any step in the growth process is a random proportion of the previous period's value of X. Hence, $h_{i,t}$ and $X_{i,t-1}$ are independent and GL holds. But even if $\beta=1$, autocorrelated disturbances mean that $h_{i,t}$ and $X_{i,t-1}$ are dependent and GL will not hold. Remember that, according to GL, firm size at any point in time is the product of previous independent growth shocks. Clearly Gibrat's model is inappropriate if the assumption of nonautocorrelated growth rates is not supported by empirical evidence; that is, the presence of statistically significant autocorrelation prompts us to reject the proposition that growth is independent of size.

autocorrelation varies with firm size; on average, it is negative for small firms and positive for large ones. In the light of this result, Coad (2007) concludes that small and large firms probably operate on different time scales.

While heteroscedasticity and autocorrelation are important econometric issues to address when modeling firm growth behavior as it applies to the study of GL, an additional statistical consideration must be taken into account, namely, the concept of *sample selection bias* (Lotti et al., 2007; Santarelli et al., 2006). (See Appendix 9.G for details on the basics of a sample selection methodology.) Under this notion, one must consider both the market entry process and the role of survival/failure in determining the size composition of a population of firms over time. Once market dynamics and selection are taken into account (by considering entry and the presence of transient smaller firms), these authors find (consistent with previous studies) that, in general, GL must be rejected—small firms grow faster than their large counterparts. So although GL is rejected over the entire sample period, for surviving incumbent as well as newborn firms, a convergence towards Gibrat-like behavior over time is observed. Thus, market selection impacts the original population of firms, and the resulting industrial core (consisting of larger, well-established, and ostensibly efficient firms) tends towards a Gibrat-like pattern of growth.

Under a sample selection mechanism, a population of firms is chosen and GL is tested with sample attrition taken into account since a portion of firms in existence at the beginning of the period studied do not survive until the end of that period. In this regard, we must consider existing, start-up, and "fragile" firms (some of which will subsequently fail) that are subject to a selection process that occurs over the entire period as well as on a year-to-year basis, with the latter enabling us to consider what happens when the original population is gradually reshaped in favor of mature, efficient, and well-established firms.

As a result of market selection through failures, incumbent as well as newborn firms that survive tend to be larger and behave in accordance with GL; that is, relative size (and age) loses its relevance and growth patterns become consistent with the law of proportionate effect. So with the passage of time, the market "shakes out" a given subpopulation of firms, and the survivors (mature and larger firms) tend to display Gibrat-like growth. Hence, GL should be a way to describe the growth behavior of well-established business entities.

9.7.4 Persistence of Growth Rates

If firm growth can be modeled as a purely random process, then it follows that growth rates lack *persistence* over time; that is, fast growth in one period does not guarantee that a firm will be a top performer in the long run. In fact, significant growth events experienced by a firm in any one year are seldom repeated. It is the persistence of growth rates that violated GL, and attempts to investigate this law and to assess the influence of size on growth rates are frustrated by the fact that annual growth rates are autocorrelated. It was mentioned earlier that the growth of small firms is characterized by negative autocorrelation (and thus small-firm growth appears to be highly erratic) and the growth of large firms is subject to positive autocorrelation (so that large-firm growth is appreciably smoother).

If we consider the long-term growth experience of firms, it appears that the impact of autocorrelation on growth rates diminishes. Although controlling for growth rate autocorrelation may bring β closer to unity (it is still statistically different from 1), the significant negative relationship between firm size and growth rate behavior still persists.

9.8 CONCLUSION

Based on the wealth of empirical results presented earlier, it should be evident that if GL holds true:

1. Firms of different size classes display the same average proportionate growth rate.
2. The variance of growth rates is the same for all firm size classes.
3. Growth rates are not autocorrelated (i.e., no persistence of firm growth over time).

Antithetically, and consistent with the majority of the latest empirical findings, GL fails to hold if:

1. Smaller firms tend to have much higher expected growth rates relative to larger firms.
2. Variability of growth rates tends to be higher for small firms.
3. Growth rates are autocorrelated over time; that is, the assumption of no persistence of firm growth is violated. In fact, growth rates have exhibited positive autocorrelation for some industries and samples and negative autocorrelation for others.

What consensus concerning GL has emerged over the years? Many researchers feel that GL can be useful when applied to the evolution of market structure in that it explains, in stochastic terms, the observed skewed shape displayed by the size distribution of firms within most industries (Santarelli et al., 2006). In this regard, GL cannot be rejected if (i) firm growth is random and independent of initial size and (ii) the resulting firm size distribution is approximately log-normal. However, it can be rejected if a negative relationship exists between the initial size and the rate of firm growth.

Additionally, Audretsch and Elston (2002) offer some sage advice, specifically the caution "Don't ask whether Gibrat's Law holds or not." Ask, instead: "In what context is the empirical evidence compatible with Gibrat's Law, and in what context is it not?" Hence, the relationship between firm size and growth is dictated by context, which reflects the country (and its institutions), time period, and the particular industry sampled. Clearly market dynamics and selection are key players in any discussion of context.

At times the empirical results that emerge from the study of GL appear to be contradictory. However, the general view in this matter is that the contradictions are due to the systematic differences in the samples used; for example, early studies of GL typically used samples containing only large firms, and more recent studies employ samples containing a broad spectrum of firm sizes. In fact, for the most general sample of firms, GL does not hold. As stated earlier, industry context is critical—whether or not GL holds depends profoundly on the country-specific institutions, the sample period studied, and the industry chosen. All things considered, Gibrat's legacy may be described as:

1. GL is not valid in general or as a "universal law," as most empirical studies have revealed; it is rejected for the manufacturing sector in particular.
2. GL is more plausible for mature incumbent firms than for start-ups (for manufacturing, small firms tend to grow faster than large ones).
3. There is a convergence towards a Gibrat-like growth pattern over time.
4. Once market selection through failures is taken into account, the group of survivors tend to exhibit Gibrat-like growth behavior.
5. GL should not be taken as representative of general industrial dynamics, but it is a dynamic rule that can be used to model the growth behavior of large and mature firms.

(For an elaboration on these points, see Audretsch and Elston, 2002; Caves, 1998; Lotti et al., 2007; Santarelli et al., 2006; Sutton, 1997.)

APPENDIX 9.A KERNEL DENSITY ESTIMATION

9.A.1 Motivation

Suppose X is a continuous random variable with pdf $f(x)$. How can we estimate $f(x)$ from sample data? Let $X_1, \ldots, X_n$ constitute a collection of independent and identically distributed sample random variables with realizations $x_1, \ldots, x_n$. Suppose further that there are no restrictions on the form or shape of $f(x)$; that is, $f(x)$ does not belong to any specific parametric family of pdf(s). (*Parametric probability densities* have a fixed structure or functional form, and the parameters of this function constitute the only relevant information about the density; e.g., the normal density has a specific functional form and is completely specified by the parameters μ and σ.) In this regard, we shall pursue a nonparametric approach to the estimation of $f(x)$—$f(x)$ is estimated from the data set.

A well-known nonparametric estimator of a pdf is a *relative frequency histogram*, which provides us with a visualization of the pdf of X—it amounts to a set of vertical bars whose heights are proportional to the relative cell or class frequencies n_s/n and whose bases are the cell intervals I_s, where n_s is the number of x_i's falling into cell or class $I_s = (a_{s-1}, a_s]$, $s = 1, \ldots, d$, where $a_0 < min\,x_i \leq max\,x_i \leq a_d$. In order to construct a relative frequency histogram, no model assumptions need be made; that is, such items are nonparametric in nature. If the data points are assumed to be a random sample generated by X's pdf, then a relative frequency histogram can be viewed as a *density estimator*. That is,

$$f_n(x) = \frac{n_s}{n} \frac{1}{a_s - a_{s-1}}, \quad x \in I_s,\ s = 1, \ldots, d, \tag{9.A.1}$$

where $a_s - a_{s-1}$ is the length of I_s. It should be evident that the choice of the cells or classes (their number and width) has a marked effect on the shape and properties of $f_n(x)$.

9.A.2 Weighting Functions

A glance back at Equation 9.A.1 reveals that the cell relative frequencies n_s/n have been weighted by terms of the form $1/(a_s - a_{s-1})$, $s = 1, \ldots, d$. Let us now consider a more flexible weighting structure.

For $f(x)$, a pdf of the continuous random variable X,

$$P(x-h < X < x+h) = \int_{x-h}^{x+h} f(t)dt \approx 2hf(x), \tag{9.A.2}$$

for h small (Fig. 9.A.1). Hence,

$$f(x) \approx \frac{1}{2h} P(x-h < X < x+h). \tag{9.A.3}$$

If we now estimate $f(x) \approx \frac{1}{2h} P(x-h < X < x+h)$ by a sample relative frequency, then

$$f(x) \approx \frac{1}{2h} \frac{\text{Number of observations in } (x-h,\, x+h)}{n}. \tag{9.A.4}$$

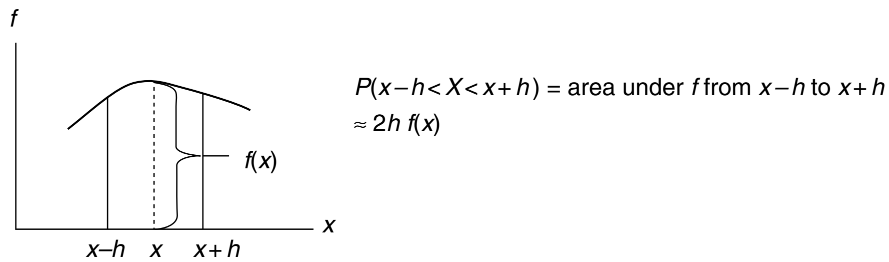

FIGURE 9.A.1 Probability that $X \varepsilon (x-h,\ x+h)$.

Clearly Equation 9.A.4 coincides with Equation 9.A.1 for $a_s - a_{s-1} = x + h - (x - h) = 2h$. Hence, the weighted histogram value $f_n(x)$ with small cell width serves as an estimator of the true or unknown pdf $f(x)$.

9.A.3 Smooth Weighting Functions: Kernel Estimators

Suppose that the underlying pdf $f(x)$ is a smooth function. In this instance the weighted histogram estimator $f_n(x)$ is inadequate for obtaining a "true" picture of $f(x)$. Hence, we need to employ a smooth density estimator to determine the density f. To impose smoothness on $f_n(x)$, let us consider a real-valued *kernel function* $k(x)$ satisfying $\int k(x)dx = 1$ (k is thus a pdf itself) that is symmetric, has a mean of zero, has unit variance, and descends smoothly to zero as $|x| \to +\infty$ (e.g., $k(x)$ might be $N(0,1)$ or standard normal). Then the univariate *kernel density estimator* (KDE) with kernel k and *bandwidth* (or *smoothing parameter*) $h > 0$ based on a random sample $X_1, \ldots X_n$ from $f(x)$ is

$$\hat{f}_n(x) = \frac{1}{nh}\sum_{i=1}^{n} k\left(\frac{x - X_i}{h}\right), \quad h > 0. \tag{9.A.5}$$

As this expression reveals, $\hat{f}_n(x)$ at x is obtained by summing (vertically) the contributions from individual standardized kernel densities $k[(x - X_i)/h]$ (with standard deviations h) centered at each X_i. Clearly those X_i's closest to x have the greatest impact on the magnitude of $\hat{f}_n(x)$.

Thus, KDEs smooth out the contribution of each observed data point over a suitably restricted or local neighborhood of that point. The contribution of data point X_i to the estimate at x depends on the distance $|x - X_i|$, and the extent of this contribution is dependent on the shape of the kernel function adopted and the bandwidth assigned to it. A KDE displays marked sensitivity to the choice of bandwidth h; that is, if h is too small, then the shape of $\hat{f}_n(x)$ is highly serrated with spiked peaks at the X_i's; and if h is too large, then $\hat{f}_n(x)$ tends to be oversmoothed. In this regard, KDEs are essentially generalized histograms—histograms that are smoothed out, do not have the features of the data set obscured by the choice of cells or classes, and might be more effective than a parametric function when the latter is fit to a density function displaying multiple modes.

Some popular/common kernel functions are as follows:
Gaussian:

$$k(x) = \frac{1}{\sqrt{2\pi}} e^{-\left(x^2/2\right)}, \quad x \in (-\infty, +\infty).$$

Quadratic:

$$k(x) = \begin{cases} \frac{3}{4}(1 - x^2), & |x| \le 1; \\ 0, & \text{otherwise.} \end{cases}$$

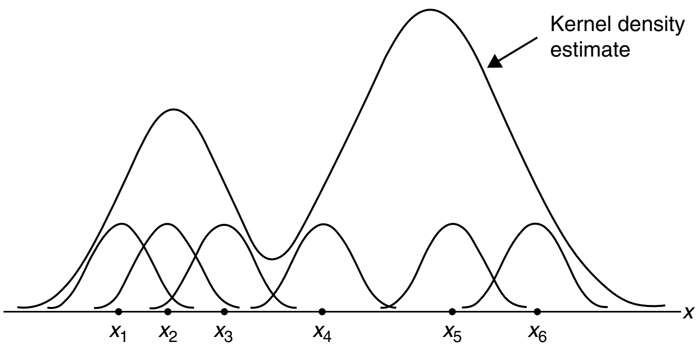

FIGURE 9.A.2 Gaussian kernel functions.

Triangular:

$$k(x) = \begin{cases} 1-|x|, & |x| \le 1; \\ 0, & \text{otherwise.} \end{cases}$$

Epanechnikov:

$$k(x) = \begin{cases} \dfrac{3}{4\sqrt{5}}\left(1-\dfrac{x^2}{5}\right), & |x| \le \sqrt{5}; \\ 0, & \text{otherwise.} \end{cases}$$

To see exactly how the KDE is developed, let us assume that we have the data points $X_1, \ldots, X_6$ (Fig. 9.A.2) and that, as a first step, we choose the Gaussian kernel function with bandwidth h (which determines the width of the kernel weighting function and thus the amount of smoothing) or

$$k(x,h) = \frac{1}{\sqrt{2\pi}h} e^{-(x^2/2h^2)}, \quad x \in (-\infty, +\infty).$$

Our next step has us center a Gaussian kernel with variance h^2 on each of the X_i's. The individual kernels are then vertically summed to make the KDE smooth.

EXAMPLE 9.A.1 Given the $n = 100$ observations appearing in Table 9.A.1, let us use univariate kernel density estimation (SAS PROC KDE) to approximate a hypothesized pdf.

PROC KDE uses a Gaussian density as the kernel, and its assumed variance determines the smoothness of the resulting estimate. Here the kernel is averaged across the observed data points to create a smooth approximation.

PROC KDE automatically selects the bandwidth and constructs a kernel density estimate once a bandwidth has been selected. The data are assigned to "bins" or

TABLE 9.A.1 Observations on X

35	60	20	25	35	25	60	80	30	65
85	75	45	45	55	45	70	65	55	85
55	95	60	35	85	80	30	50	90	40
65	20	75	55	65	65	50	25	40	60
110	45	30	65	25	35	65	35	60	60
40	60	50	85	45	55	85	55	75	70
55	75	65	40	80	70	35	85	40	75
70	95	80	90	60	90	55	65	60	75
90	100	105	70	100	40	70	95	70	90
40	105	110	60	100	90	95	90	40	60

discrete categories, and bandwidth selection is via a plug-in formula of Sheather and Jones (the default). The default number of grid points is used.

To provide the appropriate amount of smoothing, the *BWM*=option specifies a bandwidth multiplier used for each kernel density estimate. The default value is 1 (larger multipliers produce a smoother estimate, and smaller ones produce a rough or choppy estimate).

ODS Graphics must be enabled before requesting plots. By default, if ODS Graphics is enabled and the PLOTS=option is not specified, then the UNIVAR statement produces a histogram with an overlaid kernel density estimate.

The appropriate SAS code appears as Exhibit 9.A.1.

EXHIBIT 9.A.1 SAS Code for Univariate Kernel Density Estimation

```
data kde1; ①
  input x @@;
datalines;
35 60 20 25 35 … 90 95 90 40 60
;
ods graphics on; ②
proc kde data=kde1; ③
  univar x x (bwm=0.25) x (bwm=0.5)
      x (bwm=2);④
run;
ods graphics off;
```

① Data entered sequentially (starting with row 1 of Table 9.A.1).

② ODS Graphics enabled in order to obtain plots.

③ We invoke PROC KDE.

④ Four separate kernel density estimates are requested—the default (x) (with bandwidth 1) and those with bandwidths of 0.25, 0.5, and 2.

The output generated by this SAS program appears in Table 9.A.2 and Figure 9.A.3a to d.

TABLE 9.A.2 Output for Example 9.A.1

The SAS System
The KDE procedure

Inputs	
Data set	WORK.KDE1
Number of observations used	100
Variable	x
Bandwidth method	Sheather–Jones plug-in

Controls	x
Grid points	401
Lower grid *lim*it	20
Upper grid limit	110
Bandwidth multiplier	1

Inputs	
Data set	WORK.KDE1
Number of observations used	100
Variable	x
Bandwidth method	Sheather–Jones plug-in

Controls	x
Grid points	401
Lower grid limit	20
Upper grid limit	110
Bandwidth multiplier	0.25

Inputs	
Data set	WORK.KDE1
Number of observations used	100
Variable	x
Bandwidth method	Sheather–Jones plug-in

Controls	x
Grid points	401
Lower grid limit	20
Upper grid limit	110
Bandwidth multiplier	0.5

Inputs	
Data set	WORK.KDE1
Number of observations used	100
Variable	x
Bandwidth method	Sheather–Jones plug-in

(*Continued*)

TABLE 9.A.2 Output for Example 9.A.1 (*Continued*)

Controls	x
Grid points	401
Lower grid limit	20
Upper grid limit	110
Bandwidth multiplier	2

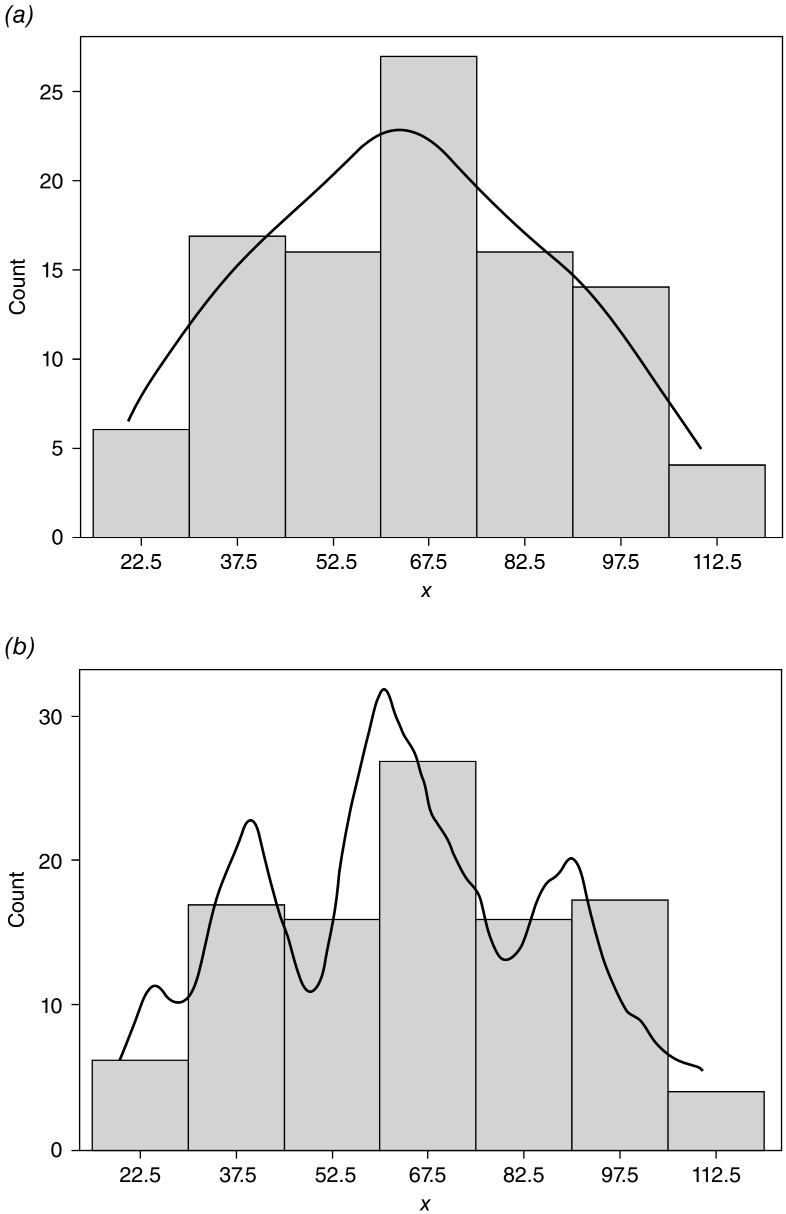

FIGURE 9.A.3 Kernel density estimate with bandwidth multipliers: (**a**) 1, (**b**) 0.25, (**c**) 0.5, and (**d**) 2.

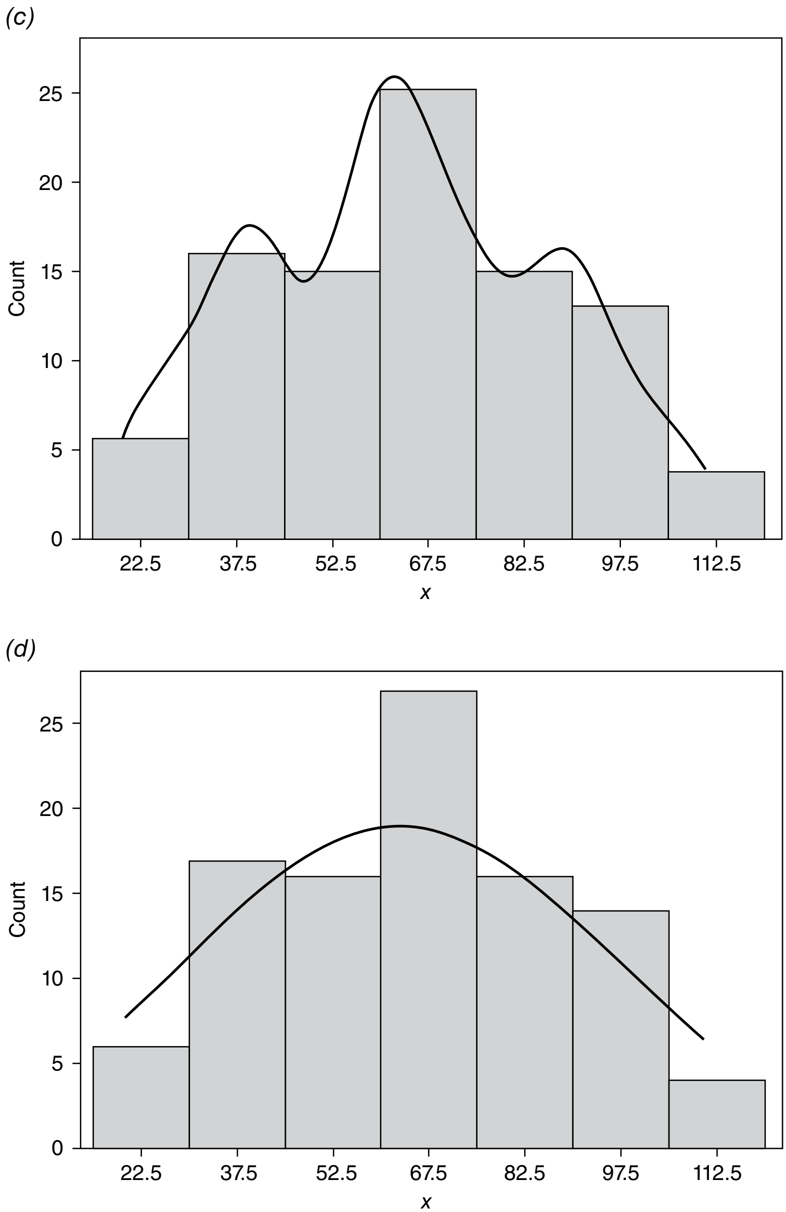

FIGURE 9.A.3 (*Continued*)

As these results indicate, probably the best kernel density estimate is provided by the one with a bandwidth multiplier of 0.5 (Fig. 9.A.3c). Bandwidth multipliers of 1 and 2 (Fig. 9.A.3a and d, respectively) result in oversmoothing, and specifying a bandwidth multiplier of 0.25 yields an undersmoothed (and sawtooth-like) kernel density estimate (Eq. A.3.b). ■

APPENDIX 9.B THE LOG-NORMAL AND GIBRAT DISTRIBUTIONS (AITCHISON AND BROWN, 1957; KALECKI, 1945)

9.B.1 Derivation of Log-Normal Forms

If a continuous random variable X is *normally distributed* with a mean of μ and a standard deviation of σ (written as $X \sim N(\mu,\sigma)$), then X's pdf is of the form

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < +\infty. \tag{9.B.1}$$

A special case of a $N(\mu,\sigma)$ distribution is the *standard normal distribution* (written as $Z=(X-\mu)/\sigma \sim N(0,1)$) with pdf

$$f(z;0,1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, \quad -\infty < z < +\infty, \tag{9.B.2}$$

where $z=(x-\mu)/\sigma$, $\sigma \neq 0$.

Let us now direct our attention to the notion of "functions of random variables." Specifically, given a continuous random variable X with pdf $f(x)$, suppose another random variable Y can be written in terms of X as $Y=u(X)$. How can we determine the pdf of Y? If we can find Y's cumulative distribution function

$$\begin{aligned} G(y) &= P(Y \le y) = P(u(x) \le y) \\ &= \int_{-\infty}^{y} f(x)dx = \int_{-\infty}^{u(x)} f(x)dx, \end{aligned} \tag{9.B.3}$$

then its pdf is conveniently given by

$$g(y) = G'(y) = f(u(x))u'(x) \tag{9.B.4}$$

(provided $u'(x)$ exists).

In this regard, let X be $N(\mu,\sigma)$ and let $y=e^X$. Then the cumulative distribution function of Y is

$$G(y) = P(Y \le y) = P\left(e^X \le y\right) = P(X \le \ln y), \quad y > 0.$$

So from Equation 9.B.3,

$$G(y) = \int_{-\infty}^{\ln y} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx, \quad y > 0.$$

Hence, the pdf of Y is

$$\begin{aligned} g(y) &= G'(y) = f(u(x))u'(x) = f(\ln\ y)\frac{1}{y} \\ &= \frac{1}{\sigma y\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\ln y-\mu}{\sigma}\right)^2}, \quad y > 0. \end{aligned} \tag{9.B.5}$$

Hence, Equation 9.B.5 represents the *log-normal distribution* in the random variable Y; that is, the random variable Y has a log-normal distribution if $X = \ln Y$ has a normal distribution. Stated alternatively, if X is a random variable with a normal distribution, then $Y = e^X$ has a log-normal distribution. The parameters of the log-normal distribution are μ and σ—the population mean of $\ln Y$ is μ, and the population standard deviation of $\ln Y$ is σ. In addition, let $X = \ln Y$ be $N(\mu,\sigma)$. Then the mean and variance of the log-normal variable $Y = e^X$ are, respectively,

$$E(Y) = E\left(e^X\right) = e^{\mu+\frac{1}{2}\sigma^2};$$
$$V(Y) = V(e^X) = e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right).$$

A specialization of Equation 9.B.5 is provided by the *Gibrat distribution* with pdf

$$g(y) = \frac{1}{y\sqrt{2\pi}} e^{-\frac{1}{2}(\ln\ y)^2}, \quad y > 0 \tag{9.B.6}$$

(see Fig. 9.B.1). Clearly Equation 9.B.6 (as well as Eq. 9.B.5) is skewed right.

Under what circumstances does a variable, say, X follow a log-normal distribution? X might be modeled as log-normal if it can be represented as the "product of a large number of positive independent random variables." Let us apply this observation to the derivation of the Gibrat distribution.

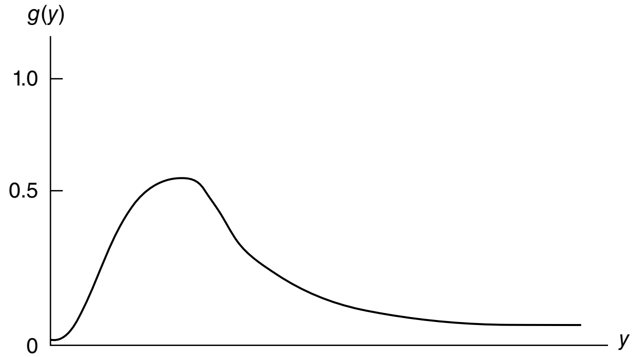

FIGURE 9.B.1 Log-normal pdf for $\mu = 0$, $\sigma = 1$.

Suppose x_t denotes, say, firm size at time t and ε_t is a random variable representing a multiplicative (stochastic) growth shock over the period from $t - 1$ to t; that is,

$$\Delta x_t = x_t - x_{t-1} = \varepsilon_t x_{t-1}. \tag{9.B.7}$$

Then

$$x_t = (1+\varepsilon_t)x_{t-1}. \tag{9.B.8}$$

But $x_{t-1} = \left(1+\varepsilon_{t-1}\right)x_{t-2}$, $x_{t-2} = \left(1+\varepsilon_{t-2}\right)x_{t-3}$, etc. So by successive substitution into Equation 9.B.8, we can rewrite this recursion formula as

$$x_t = \left(1+\varepsilon_t\right)\left(1+\varepsilon_{t-1}\right)\cdots\left(1+\varepsilon_2\right)\left(1+\varepsilon_1\right)x_0$$

or

$$\ln x_t = \ln x_0 + \sum_{s=1}^{t}\ln\left(1+\varepsilon_s\right). \tag{9.B.9}$$

Since $\ln(1+\varepsilon_s)$ can be approximated by ε_s, it follows that Equation 9.B.9 becomes

$$\ln x_t = \ln x_0 + \sum_{s=1}^{t}\varepsilon_s. \tag{9.B.10}$$

Now, as t increases without bound, $\ln x_0$ becomes more and more insignificant so that we ultimately obtain

$$\ln x_t \approx \sum_{s=1}^{t}\varepsilon_s, \tag{9.B.11}$$

with $\bar{\varepsilon}_t = \Sigma_{s=1}^{t}\varepsilon_s / t$.

Let us now invoke the (strong) *CLT*. Specifically, let $X_1, X_2, \ldots$ represent a sequence of independent and identically distributed random variables. For all i, $E(X_i) = \mu$ and $0 < \sigma_i^2 = \sigma^2 < +\infty$ (the finite variance condition). And for $\bar{X}_n = \Sigma_{i=1}^{n}X_i / n$, let $G_n(x)$ denote the cumulative distribution function of $\sqrt{n}\left(\bar{X}_n - \mu\right)/\sigma$. Then for any x, $-\infty < x < +\infty$,

$$\lim_{n\to\infty} G_n(x) = \int_{-\infty}^{x}\frac{1}{\sqrt{2\pi}}e^{\frac{1}{2}y^2}; \tag{9.B.12}$$

that is, $\sqrt{n}\left(\bar{X}_n - \mu\right)/\sigma$ has a limiting $N(0,1)$ distribution.

Now, from Equation 9.B.11, let $\ln\ x_t = t\bar{\varepsilon}_t = t\bar{\varepsilon}$ (since we require, via the CLT, that $E(\varepsilon_t) = \bar{\varepsilon}$ for all t). Let us define a new random variable

$$\frac{\ln\ x_t - t\mu_\varepsilon}{\sigma_\varepsilon\sqrt{t}} = \frac{t\bar{\varepsilon} - t\mu_\varepsilon}{\sigma_\varepsilon\sqrt{t}} = \sqrt{t}\left(\bar{\varepsilon} - \mu_\varepsilon\right)/\sigma_\varepsilon. \tag{9.B.13}$$

If $G_t(\ln\ x)$, $x>0$, denotes the cumulative distribution function of $\sqrt{t}\left(\bar{\varepsilon} - \mu_\varepsilon\right)/\sigma_\varepsilon$, then, via the CLT,

$$G(y) = \lim_{t\to\infty} G_t(\ln\ x) = \int_{-\infty}^{\ln x} \frac{1}{\sqrt{2\pi}} e^{\frac{1}{2}y^2}\, dy, \quad x > 0. \tag{9.B.14}$$

Then, as exhibited earlier, the random variable Y's pdf can be determined as

$$\begin{aligned} g(y) = G'(y) &= f(\ln\ x)\frac{1}{x} \\ &= \frac{1}{x\sqrt{2\pi}} e^{-\frac{1}{2}(\ln\ x)^2}, \quad x > 0, \end{aligned} \tag{9.B.15}$$

the *Gibrat distribution*—a normal distribution of the logarithm of X.

In sum, the log-normal distribution refers to the distribution of a random variable whose logarithm obeys the normal probability law; that is, the quantity

$$\frac{\Sigma_{s=1}^{t}\varepsilon_s - t\mu_\varepsilon}{\sqrt{t\sigma_\varepsilon^2}} \to N(0,1) \quad as\ \ t \to +\infty$$

so that the distribution of $\ln\ x_t$ approaches a normal limit for sufficiently large t or $\ln\ x_t \sim N\left(t\mu_\varepsilon, t\sigma_\varepsilon^2\right)$. Also, for large t,

$$\begin{aligned} &\text{a.} \quad \mu_x = E\left(x_t\right) = e^{t\left(\mu_\varepsilon + \sigma_\varepsilon^2/2\right)}; \\ &\text{b.} \quad \sigma_x^2 = V\left(x_t\right) = e^{2t\left(\mu_\varepsilon + \sigma_\varepsilon^2/2\right)}\left(e^{t\sigma_\varepsilon^2} - 1\right). \end{aligned} \tag{9.B.16}$$

9.B.2 Generalized Log-Normal Distribution

For continuous random variables X and Y, the *generalized log-normal distribution* or *Subbotin distribution* can be obtained as the distribution of $X=e^Y$, where Y follows the (symmetric) *generalized error distribution* (Subbotin, 1923; Vianelli, 1982a, b, 1983)

$$f(y) = \frac{1}{2r^{1/r}\sigma_r\Gamma\left(1+\frac{1}{r}\right)} e^{-|y-\mu|^r/r\sigma_r^r} \tag{9.B.17}$$

with $-\infty < y < +\infty$, $\Gamma(\alpha) = \int_0^{+\infty} y^{\alpha-1} e^{-y}\, dy$ the *gamma function* (of α), $\mu(-\infty < \mu < +\infty)$ the location parameter, $\sigma_r = [E(|y-\mu^r|)]^{1/r}$ the scale parameter, and $r(>0)$ the shape parameter. Then from Equation 9.B.17, the pdf of $X = e^Y$ is the *generalized log-normal pdf*:

$$f(x) = \frac{1}{2r^{1/r}\sigma_r \Gamma\left(1+\frac{1}{r}\right)} e^{-|\ln x - \mu|^r / r\sigma_r^r}, \tag{9.B.18}$$

where $-\infty < x < +\infty$, e^{μ} is a scale parameter and σ_r and r are shape parameters. For $r = 2$, $\ln x$ follows a normal distribution since $\Gamma(3/2) = \sqrt{\pi}/2$. In this instance σ_r is the standard deviation.

APPENDIX 9.C THE THEORY OF PROPORTIONATE EFFECT

Suppose the values of a variable X (>0) correspond to the outcomes of a random process at successive points of time and represent the joint effect of a large number of mutually independent causes (e.g., a growth process has this characteristic). Let us denote the initial value of X as X_o, its value at the jth step as X_j, and its final value after n steps as X_n. Suppose that at the jth step the change in X is a random proportion of some function ϕ of its previous value or (Aitchison and Brown, 1957; Kapteyn, 1903; Klein, 1962)

$$X_j - X_{j-1} = \varepsilon_j \phi\left(X_{j-1}\right), \tag{9.C.1}$$

where the random ε_j's are assumed to be mutually independent of each other and of the X_j's. Clearly the change in X at any step is not independent of the value already attained.

Let us concentrate on the special case where $\phi(X) = X$, that is, the change in X is a random proportion of its momentary value. Under this scenario, the *law of proportionate effect* can be framed as follows. Specifically, a variable X subject to a random process of change obeys the law of proportionate effect if the change in X at any step of the process is a random proportion of the preceding value of X. Under this restriction, Equation 9.C.1 reduces to

$$X_j - X_{j-1} = \varepsilon_j X_{j-1}. \tag{9.C.2}$$

To ascertain the implications of Equation 9.C.2, we need only look to the arguments underlying the development of Equations 9.B.7–9.B.11. Then, via the CLT (see Equations 9.B.12–9.B.15), we can conclude that a variable $\ln X_n$ subject to the law of proportionate effect has a limiting distribution that is normal as $n \to +\infty$ (or X_n is asymptotically log-normal as $n \to +\infty$).

We may view the outcomes X_n as constituting an ordered sequence of events over time. Under this (usual) interpretation, the greater the number of steps in the sequence (i.e., the longer the law operates), the greater the value of $\sigma^2_{X_n}$ (see Equation 9.B.16.b). But if the process generating the X_n's operates continually, then, under an extended application of the law, certain potential difficulties might obviously arise when $\sigma^2_{X_n} \to +\infty \ as \ n \to +\infty$. To see this, let us rewrite Equation 9.B.9 as

$$\begin{aligned} \ln\ x_t &= \ln x_o + \sum_{s=1}^{t} \ln\left(1+\varepsilon_s\right) \\ &= \ln x_o + \sum_{s=1}^{t} u_s, \end{aligned}$$

where $u_s = \ln(1+\varepsilon_s)$. Then, since the random u_s's are mutually independent,

$$V\left(\ln\ x_t\right) = \sum_{2=1}^{t} V\left(u_s\right).$$

So with the passage of time, the variance of $\ln x_t$ grows monotonically. The problem with this outcome is that it has not been observed in empirical applications (e.g., income inequality does not continually increase with the passage of time).

To address this shortcoming, we can replace Equation 9.C.2 by the alternative random process

$$X_j = X_{j-1}^{\sqrt{\rho}}\left(1+\varepsilon_j\right), \tag{9.C.3}$$

$$0 < \sqrt{\rho} < 1, \quad \text{or}$$

$$\ln\ X_j = \sqrt{\rho} \ln\ X_{j-1} + \ln\left(1+\varepsilon_j\right)$$

or, for convenience,

$$Y_j = \sqrt{\rho} Y_{j-1} + U_j. \tag{9.C.4}$$

Given this expression and an initial value Y_o, we can determine successive values of Y_j (see the process involving Equation 9.B.8–9.B.9) as

$$Y_j = \frac{1-\left(\sqrt{\rho}\right)^j}{1-\sqrt{\rho}} + \left(\sqrt{\rho}\right)^j Y_o + \sum_{s=1}^{t} \left(\sqrt{\rho}\right)^{t-s} u_s.$$

Then it can be shown that

$$V\left(Y_j\right) = \left(\frac{1-\rho^j}{1-\rho}\right) V(u)$$

so that, with $0 < \sqrt{\rho} < 1$,

$$\lim_{j \to +\infty} V\left(Y_j\right) = \frac{V(U)}{1-\rho}.$$

So under the reformulated process (Eq. 9.C.3), the variance of $\ln X_j$ does not become progressively larger as the number of periods increases.

APPENDIX 9.D CLASSICAL LAPLACE DISTRIBUTION

9.D.1 The Symmetric Case

A random variable X has a *classical Laplace* (μ,b) *distribution* if its (continuous and symmetric) pdf is

$$\begin{aligned} f\left(x;\mu,b\right) &= \frac{1}{2b} e^{|x-\mu|/b} \\ &= \frac{1}{2b}\begin{cases} e^{(x-\mu)/b}, & x \le \mu; \\ e^{-(x-\mu)/b}, & x > \mu, \end{cases} \end{aligned} \tag{9.D.1}$$

where $-\infty < x < +\infty$, μ is a location parameter ($-\infty < \mu < +\infty$), and b (>0) is a scale parameter. (Expression 9.D.1 also appears as $f = \left(1/\sqrt{2}b\right) exp\left\{-\sqrt{2}\,|x-\mu|/b\right\}, b > 0.$) In addition,

$$\begin{aligned} &E(X) = \text{median} = \text{mode} = \mu, \\ &V(X) = 2b^2,^4 \\ &\text{Coefficient of variation} = \sqrt{2}b/|\mu|, \\ &\text{Coefficient of skewness} = 0, \\ &\text{Excess kurtosis} = 3.^5 \end{aligned}$$

[4] In general, the nth central moment of Equation 9.D.1 is

$$E(X-\mu)^n = \begin{cases} 0, & \text{if } n \text{ is odd;} \\ b^n n!, & \text{if } n \text{ is even.} \end{cases}$$

[5] Remember that excess kurtosis is a measure of the peakedness and heaviness of the tails of a distribution. For a normal distribution, excess kurtosis is zero. Thus, the classical Laplace distribution has a greater degree of peakedness, and thus has heavier tails, than the normal distribution. (Having tails heavier than the normal distribution means that extreme values have a greater probability of occurrence in the Laplace case relative to the normal case.)

The cumulative distribution function associated with (Eq. 9.D.1) is

$$F(x) = \int_{-\infty}^{x} f(v)dv = \begin{cases} \frac{1}{2} e^{(x-\mu)/b}, & x \le \mu; \\ 1 - \frac{1}{2} e^{-(x-\mu)/b}, & x > \mu, \end{cases} \tag{9.D.2}$$

and thus the qth quantile of (Eq. 9.D.1) is given by

$$\xi_q = \begin{cases} \mu + b \ \ln(2q), & q \in \left(0, \frac{1}{2}\right]; \\ \mu - b \ \ln\left[2(1-q)\right], & q \in \left(\frac{1}{2}, 1\right). \end{cases} \tag{9.D.3}$$

For instance, the quartiles of the classical Laplace (μ,b) distribution are $Q_1 = \xi_{\frac{1}{4}} = \mu - b\ln(2)$, $Q_2 = \xi_{\frac{1}{2}} = \mu =$ median, and $Q_3 = \xi_{\frac{3}{4}} = \mu + b \ \ln(2)$.

As a special case of Equation 9.D.1, the *standard classical Laplace distribution* is

$$f(x;0,1) = \frac{1}{2} e^{-|x|} = \begin{cases} e^{x}, & x \le 0; \\ e^{-x}, & x > 0, \end{cases} \tag{9.D.4}$$

with

$$E(X) = \text{median} = \text{mode} = 0,$$
$$V(X) = 2$$

(see Fig. 9.D.1). The cumulative distribution function associated with Equation 9.D.4 is, from Equation 9.D.2,

$$F(x) = \begin{cases} \frac{1}{2} e^{x}, & x \le 0; \\ 1 - \frac{1}{2} e^{-x}, & x > 0. \end{cases} \tag{9.D.5}$$

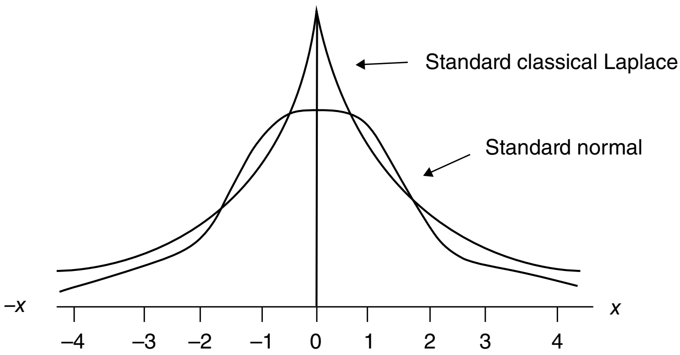

FIGURE 9.D.1 Comparison of the standard normal and standard classical Laplace distributions.

It is important to note that the classical Laplace (μ,b) distribution is also termed the *double exponential distribution* since it can be obtained by reflecting the *exponential distribution* $f(x;\mu,b)=(1/b)\, exp\{-(x-\mu)/b\}$, $0<x<+\infty$, $b>0$, about its mean.[6]

9.D.2 The Asymmetric Case

Our objective here is to introduce skewness into a symmetric distribution by converting its symmetric pdf into a skewed one via specifying inverse scale factors in the positive and negative orthants. To this end, a symmetric pdf f can be used to generate the following class of skewed probability densities indexed by a (scale invariant) parameter κ (>0) (Fernandez and Steel, 1998; Hinkley and Revankar, 1977; Kozubowski and Podgorski, 2003):

$$f(x;\kappa)=\frac{2\kappa}{1+\kappa^2}\begin{cases} f(\kappa x), & x\geq 0; \\ f\left(\dfrac{x}{\kappa}\right), & x<0. \end{cases} \tag{9.D.6}$$

In this regard, a *skewed classical Laplace* (μ,b,κ) *distribution* can be written as

$$f(x;\mu,b,\kappa)=\frac{\kappa}{b\left(1+\kappa^2\right)}\begin{cases} e^{(x-\mu)/b\kappa}, & x\leq\mu; \\ e^{-\kappa(x-\mu)/b}, & x>\mu. \end{cases} \tag{9.D.7}$$

[6] If (x, y) is a point in a plane, then its *reflection in the y-axis* is the point $(-x, y)$. Now, if each point on a curve is reflected in a line, the resulting curve is called the *reflection of the original curve in the line*. So if x is replaced by $-x$ in the equation $y=f(x)$, the resulting equation, $y=f(-x)$, is the *reflection of f in the y-axis*. For example, given the *standard exponential probability density function* $f(x)=e^{-x}$, $0<x<+\infty$, its reflection is $f(-x)=e^{x}$, $-\infty<x<0$, and thus, for $-\infty<x<+\infty$, we obtain Equation 9.D.4.

(Note that if $\kappa = 1$, Equation 9.D.1 obtains. And if $\kappa \neq 1$, $\mu = 0$, and $b = 1$, the resulting skewed Laplace pdf is standard.) Its associated cumulative distribution function appears as

$$F(x) = \frac{1}{1+\kappa^2} \begin{cases} \kappa^2 e^{(x-\mu)/b\kappa}, & x \le \mu; \\ 1+\kappa^2 - e^{-\kappa(x-\mu)/b}, & x > \mu. \end{cases} \quad (9.D.8)$$

An alternative specification of an *asymmetric* or *skewed Laplace distribution* appears as

$$f(x;\alpha,\beta,\mu) = \frac{\alpha\beta}{\alpha+\beta} \begin{cases} e^{-\alpha(x-\mu)}, & x \ge \mu; \\ e^{\beta(x-\mu)}, & x < \mu, \end{cases} \quad (9.D.9)$$

where the parameters $\alpha(>0)$ and $\beta(>0)$ are used to describe the left- and right-tail shapes, respectively. In this regard: (i) if $\alpha = \beta$, the distribution is symmetric (and the location parameter μ is the true mean); (ii) the distribution becomes more asymmetric as the values of α and β diverge from each other (thus, large α and β values result in a pointed distribution, while low α and β values result in a flat distribution); and (iii) for $\alpha > \beta$, the left tail is thinner than the right tail (and thus fewer population values appear to the left of μ relative to its right), with the opposite true when $\alpha < \beta$.

9.D.3 The Generalized Laplace Distribution

A *generalization of the Laplace distribution* (also called the *exponential power distribution*) or *Subbotin distribution* can be written as (Subbotin, 1923)

$$f_p\left(x;\mu,b_p,p\right) = \frac{1}{2p^{1/p} b_p \Gamma(1+(1/p))} e^{-|x-\mu|^p / p b_p^p}, \quad (9.D.10)$$

where $-\infty < x < +\infty$, $\Gamma(\alpha) = \int_0^{+\infty} y^{\alpha-1} e^{-y} dy$ is the gamma function of (α), μ is the location parameter, $b_p = [E(|X-\mu|^p)]^{1/p}$ is the scale parameter, and p is the shape parameter. For $p = 1$, the classical Laplace (μ,b) pdf obtains (since $\Gamma(2) = 1$). (A glance at Equation 9.B.17 reveals that the Subbotin distribution is a generalization of the log-normal, Laplace, and normal $(p = 2)$ distributions.)

An *asymmetric Subbotin distribution* (see Ayebo and Kozubouski, 2004) can be written as

$$f(x;p,\kappa,\mu,b) = \frac{p\kappa}{b\Gamma(1/p)(1+\kappa^2)} \begin{cases} e^{-(\kappa(x-\mu)/b)^p}, & x \ge \mu; \\ e^{-(|x-\mu|/b\kappa)^p}, & x < \mu, \end{cases} \quad (9.D.11)$$

where κ (>0) is a skewness parameter. For $\kappa = 1$, (Eq. 9.D.11) reduces to (Eq. 9.D.10), the (symmetric) Subbotin distribution; and for $\kappa = p = 1$, we obtain the (symmetric) Laplace distribution.

9.D.4 The Log-Laplace Distribution

Given $p = 1$ in (Eq. 9.D.10), the pdf of $U = e^X$ is the *log-Laplace pdf* (Kozubowski and Podgorski, 2000)

$$f(u;\mu,b) = \frac{1}{2ub} e^{-|\ln u - \mu|/b}, \tag{9.D.12}$$

where $-\infty < u < +\infty$, μ is the location parameter, and b is the distance to the median. So for $p = 1$ in Equation 9.D.10, $\ln u$ follows a Laplace distribution. In general, if X has a classical Laplace (μ,b) distribution, then $U = e^X$ has a log-Laplace distribution.

If X represents an asymmetric Laplace random variable with pdf (Eq. 9.D.9), then the variable $Y = e^X$ follows an asymmetric log-Laplace distribution with pdf

$$g(y) = \frac{1}{\delta}\frac{\alpha\beta}{\alpha+\beta}\begin{cases} \left(\dfrac{y}{\delta}\right)^{\beta-1}, & 0 < x < \delta; \\ \left(\dfrac{\delta}{y}\right)^{\alpha-1}, & x \geq \delta, \end{cases} \tag{9.D.13}$$

where $\delta = e^\mu$. Plotted on a log-log scale, (Eq. 9.D.13) has a distinct tent shape. The cumulative distribution function associated with Equation 9.D.13 is

$$G(x) = \begin{cases} 0, \quad x < 0; \\ \dfrac{\alpha}{\alpha+\beta}\left(\dfrac{x}{\delta}\right)^{\beta}, & 0 \leq x < \delta; \\ 1 - \dfrac{\beta}{\alpha+\beta}\left(\dfrac{\delta}{x}\right)^{\alpha}, & x \geq \delta. \end{cases}$$

APPENDIX 9.E POWER-LAW BEHAVIOR

A diverse assortment of man-made as well as natural phenomena exhibit a power-law distribution (e.g., incomes, city sizes, earthquake magnitudes, word frequencies, the number of visits to a website, firm sizes) (Adamic, 2012; Kleiber and Kotz, 2003; Newman, 2006; Reed, 2001, 2003a, b). In general, a variable X follows a *power law* when the probability of measuring a particular value of X varies inversely as a power of that value or $P(X = x) = \theta x^{-\alpha}$, θ and α parameters. Power-law behavior implies that items of small size are extremely common, whereas items having large size are extremely rare. An important feature of power-law regularity is that it can apply to event distributions that are unranked (Pareto's power law (Pareto, 1897)) or ranked (Zipf's power law (Zipf, 1949)). Let us consider each of these "laws" in turn. As we shall see later, these power laws constitute essentially two ways of looking at the same thing.

9.E.1 Pareto's Power Law

Suppose X is a random variable that follows a *Pareto distribution*, which is defined in terms of the probability that X exceeds x or

$$P(X \geq x) = \begin{cases} \left(\dfrac{x_m}{x}\right)^{\alpha}, & x \geq x_m; \\ 1, & x < x_m, \end{cases} \tag{9.E.1}$$

where the scale parameter $x_m (>0)$ is the minimum possible (effective) value for x and the shape parameter $\alpha > 0$. Here Equation 9.E.1 forms the basis of *Pareto's law* (PL) (which was originally stated in terms of incomes); that is, for large x, the percentage of incomes exceeding x is proportional to $x^{-\alpha}$ or

$$\ln P(X \geq x) = \ln C - \alpha \ln x, \tag{9.E.2}$$

where $C = x_m^{\alpha}$. Hence, the logarithm of the percentage of units with an income in excess of some value is a decreasing linear function of the logarithm of that value.

Given Equation 9.E.1, the Pareto cumulative distribution function is

$$P(X \leq x) = F(x) = \begin{cases} 1 - \left(\dfrac{x_m}{x}\right)^{\alpha}, & x \geq x_m; \\ 0, & x < x_m. \end{cases} \tag{9.E.3}$$

And upon differentiating $F(x)$ with respect to x, we obtain the *Pareto pdf*

$$f(x) = \begin{cases} \dfrac{\alpha x_m^{a}}{x^{\alpha+1}}, & x > x_m; \\ 0, & x \leq x_m, \end{cases} \tag{9.E.4}$$

which constitutes the *Pareto power-law function*—it represents the relative frequency of units (x) whose income is exactly x ($>x_m$). Note that as $x \to 0$, $f(x) \to +\infty$, and, as $x \to +\infty$, $f(x) \to 0$. In addition, for the Pareto power-law function,

$$\text{Mean} = \int_{x_m}^{+\infty} x f(x) dx = \frac{\alpha x_m}{\alpha - 1}, \quad \alpha > 1;$$

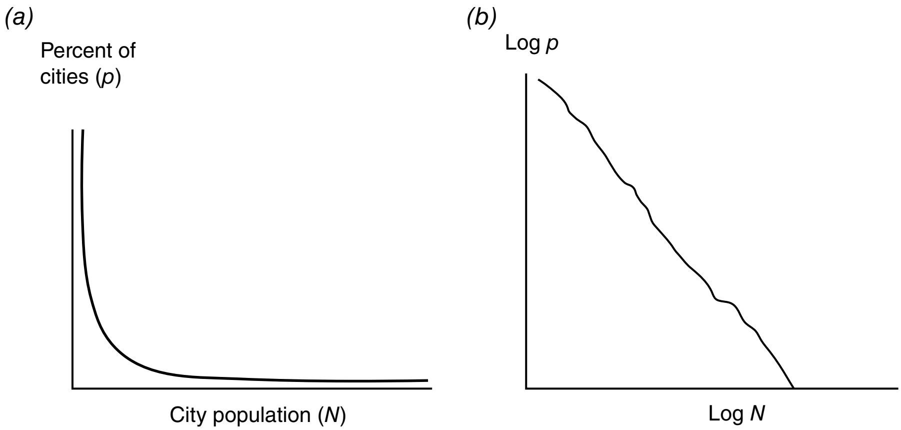

FIGURE 9.E.1 Percent distribution of U.S. cities (2000). (**a**) Percent of cities p vs. population N. (**b**) Log p vs. log N is linear.

$$\text{Median} = x_m 2^{1/\alpha} \quad \left(\text{obtained by solving} \int_{\text{median}}^{+\infty} f(x)dx = \frac{1}{2}\right); \quad \text{and} \tag{9.E.5}$$

$$\text{Variance} = \frac{x_m^2 \alpha}{(\alpha-1)^2(\alpha-2)}, \quad \alpha > 2.$$

Converting Equation 9.E.4 to logarithms yields

$$\ln f(x) = \ln \bar{C} - (\alpha+1)\ln x, \tag{9.E.4.1}$$

where $\bar{C} = \alpha x_m^{\alpha}$. Hence, on a log-log scale, we obtain a linear equation with slope $-(\alpha+1)$. That is, in terms of Equation 9.E.4.1, Pareto's power-law function stipulates that the logarithm of the percentage of units with an income equal to some value is a decreasing linear function of the logarithm of that value.

In fact, a log-log linear expression is an essential characteristic of any power-law function. For instance, if we construct a percent distribution of U.S. cities (2000 census) (Newman, 2006, Fig. 2), we find that this distribution is highly right skewed—while most of the distribution involves fairly small city sizes, there are a small number of cities with a size much larger than the typical value, which is reflected by the very long right tail of the distribution (Fig. 9.E.1a). Now, let us use logarithmic horizontal and vertical axes to replot the distribution. In this instance the percent distribution is closely plotted as a straight line (Fig. 9.E.1b).

EXAMPLE 9.E.1 When considering the distribution of wealth (W) of the citizenry of a country, an important question that arises is: "Where does the majority of the overall wealth lie?" (Here *wealth* is defined as aggregate net worth, which is the total value of an individual's asset holdings at current market prices less his/her debts or financial obligations.) Let us use Equation 9.E.5 to write the median level of wealth as

$x_m 2^{1/\alpha}$—this quantity divides the richer half of the population from the poorer half. How much of aggregate wealth itself lies in these two halves? To answer this, let us first determine (via Eq. 9.E.4) the proportion of wealth in the hands of the richer half as

$$\frac{\int_{\text{median}}^{+\infty} xf(x)dx}{\text{mean}} = 2^{-(\alpha-1)/\alpha}, \quad \alpha > 1. \tag{9.E.6}$$

This expression gives the fraction of wealth in the hands of the richest 50% of the population.

In general, the fraction of the population with a personal wealth equal to or in excess of x is given by P (Eq. 9.E.1), and the fraction of the total wealth enjoyed by that segment of the population is

$$W(x) = \frac{\int_x^{\infty} x' f(x)dx'}{\text{mean}} = \left(\frac{x_m}{x}\right)^{\alpha-1}, \quad \alpha > 1. \tag{9.E.7}$$

Upon eliminating x_m/x between Equations 9.E.1 and 9.E.7, we obtain an expression relating the fraction W of the aggregate wealth controlled by the fraction P of the richest people (if wealth is distributed according to the Pareto distribution with exponent α) or

$$W = P^{(\alpha-1)/\alpha}, \quad \alpha > 1. \tag{9.E.8}$$

An empirical regularity (curiosity) that oft emerges for many economies is the so-called 80/20 rule—about 80% (W=0.80) of a country's wealth is concentrated in the hand of the richest 20% (P=0.20) of the population. ■

A symmetric Pareto pdf is provided by

$$f(x) = \begin{cases} \left(\dfrac{\alpha x_m^{\alpha}}{2}\right)|x|^{-(\alpha+1)}, & |x| > x_m; \\ 0, & \text{otherwise.} \end{cases}$$

For $x > x_m$, this expression has a shape similar to that corresponding to Equation 9.E.4.

9.E.2 Generalized Pareto Distributions

A *generalized (three-parameter) Pareto cumulative distribution function* can be written as

$$F(x) = 1 - \left(1 + \frac{\xi(x-\mu)}{\sigma}\right)^{-1/\xi}, \quad \xi \neq 0, \tag{9.E.9}$$

where $\mu \in (-\infty, +\infty)$ is a location parameter, $\sigma(>0)$ is a scale parameter, and $\xi \in (-\infty, +\infty)$ is a shape parameter. This expression is defined for $x \geq \mu$ when $\xi \geq 0$; when $\xi < 0$, it holds for $x \leq \mu - \sigma/\xi$. Upon differentiating Equation 9.E.9 with respect to x, we obtain the *generalized Pareto pdf*

$$f(x) = \frac{\sigma^{1/\xi}}{(\sigma + \xi(x-\mu))^{(1/\xi)-1}}, \tag{9.E.10}$$

again for $x \geq \mu$ when $\xi \geq 0$ and $x \leq \mu - \sigma/\xi$ for $\xi < 0$.

From Equation 9.E.10, it can be shown that

$$\text{Mean} = \mu + \frac{\sigma}{1-\xi}, \quad \xi < 1;$$

$$\text{Median} = \mu + \frac{\sigma\left(2^{\xi} - 1\right)}{\xi}; \quad \text{and}$$

$$\text{Variance} = \frac{\sigma^2}{(1-\xi)^2(1-2\xi)}, \quad \xi < \frac{1}{2}.$$

An additional generalization of the Pareto distribution, which allows for a more flexible shape, admits the introduction of a logarithmic adjustment term (Ziebach, 2000). Specifically, the *log-adjusted Pareto cumulative distribution function* is

$$F(x) = 1 - \left(\frac{x_m}{x}\right)^{\alpha} \left(\frac{\ln\ x_m}{\ln\ x}\right)^{\beta}, \quad 1 < x_m \leq x, \tag{9.E.11}$$

where either $\alpha > 0$ and $\beta \geq -\alpha \ln x_m$ or $\alpha = 0$ and $\beta > 0$. The associated *log-adjusted Pareto pdf* appears as

$$f(x) = \frac{x_m^{\alpha}\left(\ln x_m\right)^{\beta}(\alpha \ln x + \beta)}{x^{\alpha+1}(\ln x)^{\beta+1}}, \quad 1 < x_m \leq x, \tag{9.E.12}$$

which is decreasing for $\beta \geq 0$. (Note that when $\beta = 0$, Equation 9.E.11 becomes Equation 9.E.3, while Equation 9.E.12 reduces to Equation 9.E.4.) This equation is unimodal when

$$-\alpha \ln x_m < \beta < -\alpha \ln x_m + \frac{1}{2}\left\{\left[(\ln x_m + 1)^2 + 4\alpha \ln x_m\right]^{1/2} - \left(\ln x_m + 1\right)\right\}.$$

Only if $\beta = -\alpha \ln x_m$ do we have $f(x_m) = 0$; otherwise, $f(x_m) > 0$. For $\alpha > 1$,

$$\text{mean} = \frac{\alpha x_m}{\alpha - 1} - \beta(\alpha-1)^{\beta-1} x_m{}^{\alpha}\left(\ln\ x_m\right)^{\beta} \Gamma\left(-\beta; (\alpha-1)\ln\ x_m\right).$$

Note that the first term on the right-hand side of this expression corresponds to the mean of Equation 9.E.4, while the second term describes the logarithmic adjustment due to the presence of β.

9.E.3 Zipf's Power Law

A rank-size distribution displays the magnitude (frequency) y of the value of a variable relative to its rank r and can be described by a rank-size power-law function of the form

$$r = Ay^{-b}, \tag{9.E.13}$$

with A and b serving as parameters. Converting this expression to logarithms gives

$$\ln r = \ln A - b \ln y. \tag{9.E.14}$$

Thus, on a log-log scale, Equation 9.E.14 amounts to a linear equation with slope $-b<0$. So for a rank-size distribution to exist, the logarithm of an item's rank ($\ln r$) must be a decreasing linear function of the logarithm of that item's size ($\ln y$) (Fig. 9.E.2a).

For instance, for many countries, the size distribution of cities (and of firms) obeys power laws or follows rank-size distributions; for example, suppose a variable depicts city size (y) as measured by city population. Suppose further that we order the cities in decreasing order of their size so as to obtain the rank of each city according to its size. Then a rank-size distribution for city size obtains if the relationship between the logarithms of these two variables is linear and displays a negative slope (see Brackman et al., 2001; Gabaix and Ioannides, 2004).

If in Equation 9.E.13 (or in Eq. 9.E.14) we have $b=-1$, then the resulting equation is known as *Zipf's power-law function*. Zipf (1949) suspected that, for many diverse populations, the size of the rth largest occurrence of an event is inversely proportional

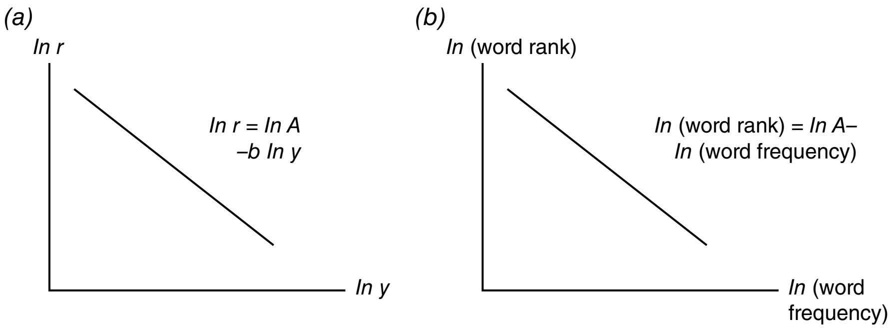

FIGURE 9.E.2 Rank-size distributions. (**a**) Rank-size power law function. (**b**) Zipf's law for language.

to its rank—the so-called *Zipf's law* (*ZL*). In fact, Zipf worked with language and observed that if he determined the absolute frequency of use of each word in a text and then rank ordered these words from most common to least common, the absolute frequency of any word is inversely proportional to its rank in the absolute frequency distribution. Thus, the most frequently used word will occur approximately twice as often as the second most frequently used word, which in turn occurs twice as often as the fourth most frequently used word, and so on. In terms of Equation 9.E.13, ZL holds if $b=1$ or

$$r = \frac{A}{y} \tag{9.E.13.1}$$

and thus, in terms of Equation 9.E.14,

$$\ln\ r = \ln\ A - \ln\ y. \tag{9.E.14.1}$$

Again, r is the rank of a word and y is its frequency (of use) value.

APPENDIX 9.F THE YULE DISTRIBUTION

Yule (1925) studied the behavior of a particular stochastic process (ultimately called a Yule process (YP)) that served as a mechanism for explaining the limiting power-law distribution of the number of species per genus of flowering plants. In particular, Yule determined that the process generated a distribution with a power-law tail.[7]

A *YP* relates to the growth in the number of species per genus in some higher-level taxon of plants or animals. New genera are added to (and never removed from) a taxon when a new species is sufficiently different from its predecessors so that it does not belong to any of the existing genera, and new species are added as old ones *speciate* or split in two. If we assume that a new species belongs to the same genus as their parent (save for those starting new genera), the probability that a new species is added to a new genus is taken to be proportional to the number of species already in the genus; that is, the stochastic rate at which new genera accumulate new species is linear in the number they already have.

In general, suppose species are added at a constant rate of m new species for each new genus, with each new genus starting out with k_o species, and, additionally, new species are added to genera at a rate proportional to the number k they already have (since each of the k species has the same chance per unit of time of speciating) plus a constant a ($>-k_o$). Thus, the number of genera increases steadily in a YP, as does

[7] *Taxonomy* is the branch of science dealing with the classification of plants or animals. A *taxon* is a branch of the evolutionary tree—a group of species all described by repeated *speciation* (the splitting of one species into another) from a common ancestor. Examples are genus, order, family, etc. Furthermore, a *genus* is a class, a category used to classify plants or animals that are similar in structure (the plural of genus is *genera*). A *species* is a distinct type of plant or animal. Hence, similar species form a genus or group of related plants or animals.

the number of species within each genus. Then it can be demonstrated that the fraction $p(k)$ of genera having k species has a long-run limiting value of

$$p(k) = \begin{cases} cB(k+a;\gamma), & k > k_o; \\ 0 & \text{otherwise,} \end{cases}$$

where $B(k+a;\gamma)$ is a beta function. Since the beta function behaves asymptotically as a power law $B(k;\gamma) \sim k^{-\gamma}$ for large values of k, it follows that the YP generates a long-tailed power-law distribution

$$p(k) \propto k^{-\gamma}, \gamma = 2 + \frac{k_o + a}{m}.$$

In sum, Yule (among others) verified that the distribution of the number of species in a genus (or other taxonomic group) appears to closely follow a power law.

APPENDIX 9.G OVERCOMING SAMPLE SELECTION BIAS

9.G.1 Selection and Gibrat's Law (GL)

One reason why GL might have been rejected by many previous empirical studies was the incomplete consideration given to the entry and selection process; that is, since a given population of firms has both start-ups and so-called "fragile" firms (those that are on the verge of failing), such firms tend to be small and thus are over-represented in the sample. (Remember that it was the smaller fast-growing firms that, in part, led to the rejection of GL.) After all, under selection by market forces, it is the surviving larger firms (incumbents and newborns that "made it") that tend to display a Gibrat-like growth pattern. How can we explicitly model the selection process?

9.G.2 Characterizing Selection Bias

Sample selection issues emerge when one considers models with *limited dependent variables* that involve data limited to a specific unobservable range. In this regard, the occurrence of such models may be due to (Davidson and MacKinnon, 1993; Greene, 1997; Heckman, 1979; Johnson and Kotz, 1970; Kmenta, 1986):

1. *Censoring*—occurs when no observations have been systematically excluded from the sample, but some information contained within them has been suppressed or is unobservable. Typically only the values of some dependent variable Y are unobservable in some range, while the regressors are all observed in that range. For instance, observations on dependent Y are not available beyond some threshold or cutoff value; for example, households with all income levels

are included in a random sample, but for those with incomes below the poverty level, the amount reported is always at the poverty level.

2. *Truncation*—occurs when not only are the values of the dependent variable in the unobservable range missing, but the corresponding values of the regressors are also missing in that range; all data points outside a given range are lost; for example, households with incomes at the $100,000 level and up are excluded from the sample. Hence, we are concerned with making inferences about a population from a sample drawn from a restricted portion of that population.
3. *Attrition*—occurs in, say, a longitudinal study when some of the items included in the sample at the beginning of the study become lost or opt out and do not remain in the sample until the end of the study; for example, we may be interested in the impact of a monthlong driver reeducation course on drunk-driving recidivism, but some of the participants drop out of the course before its completion.

To make this discussion a bit more concrete, let us consider the determinants of wage offers to married women in a somewhat idealized setting. In general, a married woman enters the labor force only if her unobserved "reservation wage" (the wage rate that will just coax an unemployed person to enter the labor force) is less than the market wage rate for her line of work. However, a sample of married women who are in the labor force will exclude those whose reservation wage exceeds the market wage rate. In fact, they will be excluded from the sample no matter how close their reservation wage happens to be to the market wage. Since obtaining data only for married women who work can be viewed as a nonrandom selection from the population, estimating the determinants of wages from this subpopulation can introduce a bias (one only has access to the wages observed for married women who work).

The preceding discussion depicts the standard case of *sample selection bias*—information on the dependent variable (wages) for part of the sample is missing. What we actually have here are two distinct processes that are in play simultaneously, and these processes can be depicted by two equations that are related to each other and that make up the *sample selection model*:

1. *Outcome equation*—used to examine the substantive question of interest (e.g., what are the determinants of wage offers for married women?)
2. *Selection equation*—used to detect selection bias and to correct the outcome equation for any such bias.

Let us now consider a sample selection model for explaining the wages of married women. The effect of some regressor X (household characteristics) on married women's wages W can be written as the (OLS) equation:

$$W_i = \beta X_i + \varepsilon_i, \quad \text{[outcome equation]} \tag{9.G.1}$$

where ε_i is a random error term. The basic selection problem arises, in part, because of the following *selection effect*: the sample consists only of married women who choose to work (these women may differ in important unmeasured ways from married women who do not work). The selection equation for entering the labor market might be

$$E_i^* = \gamma Z_i + u_i, \quad \text{[selection equation]} \tag{9.G.2}$$

where E_i^* (which is unobserved) is the difference between the market wage and the reservation wage, Z_i is a regressor or factor (such as education) that is known to influence a woman's decision to work, and the random error u_i is jointly normally distributed with ε_i and contains many unmeasured characteristics in the selection equation. We don't observe E_i^*; all we observe is a binary variable E_i (=1 if the married woman enters the labor force and = 0 otherwise). That is,

$$E_i = \begin{cases} 1, & E_i^* > 0; \\ 0, & E_i^* \le 0. \end{cases} \tag{9.G.3}$$

So for married women in the work force, W_i is the actual wage rate $\left(E_i^* > 0 \text{ and } E_i = 1\right)$; for those not working $E_i^* \le 0$ $\left(E_i = 0\right)$ and W_i is recorded as zero. In fact, when observations on a dependent variable are recorded only when another variable assumes values above or below some threshold level, the sample is said to be *sample selected.*

A second (and more subtle) *selection effect* emerges when some married women go to work in spite of the fact that their reservation wage exceeds the market wage rate. This is because, ostensibly, they feel the work is personally satisfying; they place a premium on some unmeasured variable that appears in u_i in Equation 9.G.2.

Suppose that the unmeasured factors that influence the selection equation are associated with the unmeasured factors that affect the outcome equation; that is, if the error term ε_i in the outcome equation (Eq. 9.G.1) is correlated with the error term u_i in the selection equation, then ε_i will not have zero mean and will be correlated with the regressors. In this circumstance the outcome equation cannot be consistently estimated without explicitly taking the selection equation into account. This is the essence of the *selection bias problem.*

Let's be a bit more specific about the assumptions on the distribution of, and connection between, the error terms in the selection and outcome equations. In particular, we posit a bivariate normal distribution with zero means and correlation ρ; that is,

$$(\varepsilon, u) \sim N(0, 0, \sigma_\varepsilon, \sigma_u, \rho_{\varepsilon u}) = N(0, 0, \sigma, 1, \rho)$$

and (ε, u) is independent of X and Z.

(Here the standard deviation of u is taken to be 1—a simplification that normalizes the variance of the error term in the probit regression model introduced later. Besides, only the sign of E_i^* is needed.)

9.G.3 Correcting for Selection Bias: The Heckman (1976, 1979) Two-Step Procedure

We now turn to the sample selection problem proper. We start with a set of conditional means. Specifically, let us take the expected value of (Eq. 9.G.1) conditional upon a married woman working and the values of X:

$$\begin{aligned} E\left(W_i \middle| W_i \text{ observed}\right) &= E\left(W_i \middle| E_i^* > 0\right) \\ &= E\left(\beta X_i + \varepsilon_i \middle| \gamma Z_i + u_i > 0\right) \\ &= \beta X_i + E\left(\varepsilon_i \middle| \gamma Z_i + u_i > 0\right) \\ &= \beta X_i + E\left(\varepsilon_i \middle| u_i > -\gamma Z_i\right). \end{aligned} \tag{9.G.4}$$

Now, if the random errors ε_i and u_i are independent, then the second term on the right-hand side of Equation 9.G.4 is zero, and thus the OLS regression of W_i on X_i will yield unbiased estimates of β. But if $\rho \neq 0$ (the error terms are correlated), then the preceding truncated mean does not equal βX_i, and thus we must explicitly take account of selection. (Regressing W on X for those married women employed means that we are not fitting the equation for the population as a whole. Married women who are employed will always tend to report higher wages relative to those that are unemployed since, for the latter group, $E_i^* \leq 0$.) Hence, our regression results will tend to be biased; that is, we encounter *sample selection bias*. Note also that from the expectation term on the right-hand side of Equation 9.G.4, u_i is truncated (bounded from below) as well. As Heckman (1979) noted, the difficulty with Equation 9.G.4 can be described as an "omitted variables problem"—the term $E(\varepsilon_i | u_i > -\gamma Z_i)$ is the "omitted variable" in Equation 9.G.4. Clearly an estimate of the omitted variable enables us to solve the problem of sample selection bias.

When ε_i and u_i are correlated,

$$E\left(\varepsilon_i \middle| u_i > -\gamma Z_i\right) = \sigma\rho\lambda_i\left(\alpha_i\right) = \beta_\lambda \lambda_i\left(\alpha_i\right),^{8} \tag{9.G.5}$$

where $\alpha_i = -\gamma Z_i$ is the standard normal (remember that $E(u_i) = 0$ and $\sigma_u = 1$), $\lambda_i(\alpha_i) = \phi(\gamma Z_i)/\Phi(\gamma Z_i)$ (the *inverse Mills ratio*), ϕ denotes the standard normal pdf, Φ represents the standard normal cumulative distribution function, and $\beta_\lambda = \sigma\rho$ is the regression coefficient on the selectivity regressor $\lambda_i(\alpha_i)$. In view of Equation 9.G.5, the wage or outcome equation regression model (Eq. 9.Ga) becomes, from Equation 9.G.4, the *augmented outcome equation*

$$\begin{aligned} W_i \middle| E_i^* > 0 &= E\left(W_i \middle| E_i^* > 0\right) + v_i \\ &= \beta X_i + \beta_\lambda \lambda_i\left(\alpha_i\right) + v_i, \end{aligned} \tag{9.G.6}$$

[8] See Greene (1997, pp. 974–978) and Davidson and MacKinnon (1993, pp. 542–545).

with v_i depicting a random error term. Under this adjustment, OLS regression using Equation 9.G.6 produces consistent estimates of β. If the term $\beta_\lambda \lambda_i(\alpha_i)$ is omitted from Equation 9.G.6 and the modified equation is estimated, then the resulting *omitted variable bias* ($\lambda_i(\alpha_i)$ is the omitted variable) translates to a *sample selection bias* (or, stated alternatively, the problem of the truncation of ε_i translates into a problem involving an omitted regressor).

The quantity λ_i is not observed. However, it can be consistently estimated by the following maximum likelihood function for the probit model. Specifically, the aforementioned selection model displays two types of observations: ones for which both W_i and E_i are observed to be zero and ones for which $W_i = \beta X_i + \varepsilon_i$ and $E_i = 1$. The likelihood function for E_i is the standard probit form

$$\mathfrak{L} = \prod_{E_i=1} \Phi\left(\gamma Z_i\right) \prod_{E_i=1} \left[1 - \Phi\left(\gamma Z_i\right)\right]$$

or, in log-likelihood terms,

$$\mathfrak{L}^* = \ln \mathfrak{L} = \underset{\substack{\text{[we sum over}\\ \text{those observations}\\ \text{selected into}\\ \text{the sample]}}}{\sum_{E_i=1} \ln \Phi\left(\gamma Z_i\right)} + \underset{\substack{\text{[we sum over}\\ \text{those observations}\\ \text{not selected]}}}{\sum_{E_i=0} \ln \Phi\left(-\gamma Z_i\right)}.^{9} \tag{9.G.7}$$

Then maximizing $\mathfrak{L}^*$ with respect to the unknown parameters renders consistent estimates of γZ_i and therefore of λ_i also. This now enables us to estimate the parameters of Equation 9.G.6 or to obtain $\hat{\gamma}$ and $\hat{\beta}_\lambda$ using those observations for which $E_i^* > 0$.

These comments may be consolidated in the *Heckman two-step procedure*, given that ε_i and u_i have mean zero and are independent of the regressors with $u_i \sim N(0,1)$:

Step I—use an ordinary probit model (Eqs. 9.G.7 or 9.G.7.1) to obtain a consistent estimate via maximum likelihood of the parameter γ of the selection equation. For each observation in the "selected sample," evaluate the selectivity regressor (the inverse Mills ratio)

$$\lambda_i = \frac{\phi\left(\gamma Z_i\right)}{\Phi\left(\gamma Z_i\right)}$$

at $\hat{\gamma}$.

[9] The log-likelihood function can alternatively be written as

$$\mathfrak{L}^* = \sum_i \left[E_i \ln \Phi\left(\gamma Z_i\right) + \left(1 - E_i\right) \ln \Phi\left(-\gamma Z_i\right)\right]. \tag{9.G.7.1}$$

Step II—estimate β and $\beta_\lambda = \sigma\rho$ in the augmented outcome equation

$$W \Big| E_i^* > 0 = \beta X_i + \beta_\lambda \frac{\phi(\hat{\gamma} Z_i)}{\Phi(\hat{\gamma} Z_i)} + v_i \tag{9.G.8}$$

by OLS using only the $W_i > 0$.

The parameter estimates from this two-step process are consistent and asymptotically normal.

In summary, to mitigate against any sample selectivity bias, the Heckman procedure estimates the outcome equation using only a subsample for which W is observed. It accomplishes this by using a two-step process. In Step I, we estimate a probit model for which the binary status of W (observed or not observed) depends on E. This enables us to estimate the inverse Mills ratio for each observation in the subsample for which W is observed (E_i and Z_i are observed for a random sample of married women, but W_i is observed only when $E_i = 1$). In Step II, we regress W on X and λ using the subsample for which W is observed.

Let us now consider a few important points pertaining to the Heckman procedure:

1. What are the *marginal effects* obtained from Equation 7.G.6? (Assume β and X_i are vectors of parameters and regressors, respectively.) The marginal effect of the regressor X_{ik} or W_i in the observed sample consists of two separate terms: (i) the "direct effect" of X_{ik} on the mean of W_i is simply β_k, and (ii) there is an "indirect effect" of X_{ik} on W_i if X_{ik} also appears in the selection equation; that is, X_{ik} changes not only the mean of W_i but also the probability that an observation is in the sample. Hence, it affects W_i through $\lambda_i(\alpha_i)$. Thus, the complete marginal effect of X_{ik} on W_i in the observed sample is

$$\frac{\partial E\left(W_i \Big| E_i^* > 0\right)}{\partial X_{ik}} = \beta_k - \gamma_k \sigma\rho\delta_i(\alpha_i), \tag{9.G.9}$$

   where $\delta_i(\alpha_i) = \lambda_i(\alpha_i)^2 - \alpha_i\lambda_i(\alpha_i)$. The upshot of all this is that if $\rho \neq 0$ and X_k appears in both the selection and outcome equations, then β_k by itself does not indicate the marginal impact of X_{ik} on W_i.
2. Since $\sigma \neq 0$, the ordinary t statistic for β_λ under $H_o: \beta_\lambda = 0$ can be used to test $H_o: \rho = 0$. So if β_λ is statistically significant, then we have selection bias. But if this estimated coefficient is not statistically significant, then there is no selection bias ("selectivity" is not an issue) as formulated by the selection equation, provided, of course, that this equation is correctly specified. Hence, testing the null hypothesis that the coefficient on λ_i is zero is equivalent to testing for sample selectivity.
3. If the selection equation does not perform very well (it is not very good at "selecting"), then the outcome equation will be estimated imprecisely.

4. For the Heckman equation to be correctly identified, there should be at least one regressor that appears in the selection equation but not in the outcome equation (the regressor affects selection but not the outcome).
5. The truncated nature of v_i means that heteroscedasticity will impact the Step II estimates—the standard errors will be biased and inconsistent. (To circumvent this problem, the outcome equation can be estimated via weighted least squares.)
6. Although taking account of sample selection bias using the Heckman two-step routine is known to be computationally efficient, the maximum likelihood method is more often than not used to estimate selection models. Although the two-step estimates are consistent, they are not asymptotically efficient even with the normality assumption. If maximum likelihood estimation is used in place of the Heckman technique, then the log-likelihood function for $W_i \middle| E_i^* > 0 = \beta X_i + \beta_\lambda \lambda_i(\alpha_i) + v_i$ is

$$\begin{aligned}\mathfrak{L}^* &= \sum_{E_i=0} \ln\left[1-\Phi(\gamma Z_i)\right] \\ &\quad + \sum_{E_i=1}\left\{\ln\,\phi\left(\frac{W_i-\beta X_i}{\sigma}\right) - \ln\sigma + \ln\Phi\left[\frac{\gamma Z_i + \rho\left((W_i-\beta X_i)/\sigma\right)}{\left(1-\rho^2\right)^{1/2}}\right]\right\}.\end{aligned} \tag{9.G.10}$$

9.G.4 The Heckman Two-Step Procedure Under Modified Selection

The preceding discussion of the Heckman two-step procedure explicitly used a selection equation (Eq. 9.G.2) to estimate the outcome equation (Eq. 9.G.1). If the factors affecting selection cannot be easily measured (or readily discerned), then an abbreviated selection device can be used. In this regard, suppose the outcome equation appears as

$$W_i^* = \beta X_i + \varepsilon_i^*, \tag{9.G.11}$$

where $W_i^* \sim N(\beta X_i, \sigma)$ and the random error term $\varepsilon_i^* \sim N(0,\sigma)$. For married women in the labor force, W_i^* is the actual wage rate; for those married women not working, W_i^* is obviously not observed and is recorded as zero. Instead of observing W_i^*, we actually observe

$$W_i = \begin{cases} W_i^*, & W_i^* > 0; \\ 0, & W_i^* \le 0. \end{cases} \tag{9.G.12}$$

Then from Equation 9.G.12, Equation 9.G.11 can be rewritten as

$$W_i = \beta X_i + \varepsilon_i. \tag{9.G.13}$$

Hence, W_i is truncated at zero and ε_i is truncated at $-\beta X_i$. Clearly the sample consists of two separate sets of observations: (i) the observations for which $W_i = 0$ (here the values of X_i are known and $W_i^* \le 0$) and (ii) the subset of observations for which the values of X_i and W_i are known (since $W_i^* > 0$).

Looking to the conditional expectation of W_i given $W_i^* > 0$, we have

$$\begin{aligned} E\left(W_i \middle| W_i^* > 0\right) &= \beta X_i + E\left(\varepsilon_i \middle| W_i^* > 0\right) \\ &= \beta X_i + E\left(\varepsilon_i \middle| \varepsilon_i^* > -\beta X_i\right). \end{aligned}$$

Then it can be shown that

$$E\left(\varepsilon_i \middle| \varepsilon_i^* > -\beta X_i\right) = \sigma\lambda_i,$$

where λ_i (the inverse Mills ratio) can be expressed as

$$\lambda_i\left(\alpha_i\right) = \frac{\phi\left(\alpha_i\right)}{\Phi\left(\alpha_i\right)},$$

where $\alpha_i = \beta X_i / \sigma$ is a standard normal variate. Since the mean of truncated ε_i is not zero, the *augmented outcome equation* for those observations for which $W_i^* > 0$ is

$$W_i \middle| W_i^* > 0 = \beta X_i + \sigma\lambda\left(\alpha_i\right) + v_i, \tag{9.G.14}$$

where v_i is a random error term and σ is the regression coefficient on the selectivity regressor $\lambda_i(\alpha_i)$. (If $\lambda_i(\alpha_i)$ is omitted from Equation 9.G.14, then we incur an *omitted variable bias*.) With OLS applied to Equation 9.G.14, β is estimated consistently.

Given that λ_i is unobserved, it too can be estimated consistently via the log-likelihood function $\mathfrak{L}^*$ for the binary variable

$$E_i = \begin{cases} 1, & W_i^* > 0; \\ 0, & W_i^* \le 0 \end{cases}$$

(here $E_i = 1$ if the married woman is in the work force and $E_i = 0$ otherwise), where

$$\mathfrak{L}^* = \Sigma_i \left[E_i \ln \Phi\left(\frac{\beta X_i}{\alpha}\right) + \left(1 - E_i\right) \ln \Phi\left(-\frac{\beta X_i}{\sigma}\right) \right]. \tag{9.G.15}$$

Maximizing $\mathfrak{L}^*$ enables us to estimate $\beta X_i/\sigma$, and thus λ_i, consistently. Then replacing λ_i *by* $\hat{\lambda}_i$ in Equation 9.G.14 and applying OLS to the resulting expression for those observations with $W_i^* > 0$ enables us to obtain an estimate of β, which is consistent and asymptotically normal.

The preceding estimation approach is summarized by the (modified) *Heckman two-step procedure*, given that $\varepsilon_i^* \sim N(0,\sigma)$ and ε_i^* is independent of X_i:

Step I—maximize the likelihood function (Eq. 9.G.15) to obtain a consistent estimate of $\beta X_i/\sigma$ and thus of the selectivity regressor $\lambda_i = \dfrac{\phi(\beta X_i/\sigma)}{\Phi(\beta X_i/\sigma)}$.

Step II—replace λ_i by $\hat{\lambda}_i$ and apply OLS to (Eq. 9.G.14) using only the $W_i>0$.

It was mentioned earlier that while the Heckman two-step estimates are consistent, they are not asymptotically efficient under the normality assumption. So in place of the Heckman routine, maximum likelihood could be used to estimate the selection model. In this regard, for the observations for which $W_i \le 0$, the likelihood function is $P(W_i^* \le 0) = P\left(\dfrac{\varepsilon_i}{\sigma} \le \dfrac{-\beta X_i}{\sigma}\right) = \Phi\left(\dfrac{-\beta X_i}{\sigma}\right)$; and for the observations for which $W_i>0$, the likelihood function has the customary form. Then for all n observations, the log-likelihood function appears as

$$\begin{aligned}\mathfrak{L}^* = \sum_{i=1}^{n} \Bigg\{ &(1-E_i)\ln \Phi\left(\frac{-\beta X_i}{\sigma}\right) \\ &+ E_i\left[-\frac{1}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(W_i - \beta X_i)^2\right]\Bigg\},\end{aligned} \tag{9.G.16}$$

where

$$E_i = \begin{cases} 1, & W_i > 0; \\ 0, & W_i = 0. \end{cases}$$

EXAMPLE 9.G.1 Let's use a sample selection routine to estimate a wage-offer function for married women. Our sample selection model appears as

$$\text{work} = \gamma_o + \gamma_1\,\text{ed} + \gamma_2\,\text{age} + \gamma_3\,\text{mrd} + \gamma_4\,\text{child} + u_i, \quad \text{(selection equation)}$$
$$\text{wage} = \beta_0 + \beta_1\,\text{ed} + \beta_2\,\text{age} + \varepsilon_i, \quad \text{(outcome equation)}$$

and the data set appears in Table 9.G.1. (Note that the selectivity variable is *work*, where *work* = 1 if the woman is employed and = 0 if not.) The requisite SAS code appears as Exhibit 9.G.1.

TABLE 9.G.1 Data for Wage-Offer Modeling

Age (years)	Education, *ed* (years)	Married, *mrd* (yes = 1, no = 0)	No. of children, *child*	Working, *work* (yes = 1, no = 0)	*Wage* ($/hour)
27	12	1	3	0	0
30	16	0	0	1	40
28	12	1	2	0	0
46	14	1	2	1	30
24	12	1	1	0	0
65	12	1	1	0	0
31	14	1	1	1	35
21	12	0	0	0	0
50	14	1	2	1	42
52	12	0	0	0	0
50	16	1	2	1	75
61	16	1	1	1	85
20	12	1	1	0	0
35	14	0	0	1	70
66	14	1	3	1	75
71	16	0	0	1	101
50	16	0	0	1	75
44	16	1	1	1	87
28	14	1	1	0	0
30	16	0	0	1	45
29	12	1	1	0	0
26	16	0	0	1	43
22	12	1	1	0	0
19	12	1	1	0	0
61	12	1	2	1	80
71	16	1	4	1	107
35	14	0	0	0	0
66	14	0	0	1	70
39	14	1	2	1	53
46	14	1	1	1	61

EXHIBIT 9.G.1 SAS Code for the Sample Selection Model

```
data womenwk1;
  input age ed mrd child work wage;
datalines;
27  12  1  3  0  0
30  16  0  0  1  40
.   .   .  .  .  .
.   .   .  .  .  .
.   .   .  .  .  .
46  14  1  1  1  61
;
proc qlim data = womenwk1; ①
  model work = ed age mrd child/discrete;②
  model wage = ed age/select (work = 1);③
run;
```

① We employ the data set womenwk1.

② We specify the selection equation with *work* designated as a discrete response variable.

③ We specify the outcome equation, with *work* chosen as the variable that selection is based upon.

The output generated by this selection routine appears in Table 9.G.2.

TABLE 9.G.2 Output for Example 9.G.2

The SAS system
The QLIM procedure
Summary statistics of continuous responses

							N Obs	
Variable	N	Mean	Standard error	Type	Lower bound	Upper bound	Lower bound	Upper bound
Wage	18	65.22222	22.804942	Regular				

①
Discrete response profile of work

Index	Value	Total frequency
1	0	12
2	1	18

Model fit summary

Number of endogenous variables	2
Endogenous variable	Work wage
Number of observations	30
Log likelihood	−71.35474
Maximum absolute gradient	42.04823
Number of iterations	118
Optimization method	Quasi-Newton
AIC	160.70947
Schwarz criterion	173.32025

Algorithm converged.

②

Parameter estimates

Parameter	DF	Estimate	Standard error	t value	Approx Pr > [t]	Parameter label
wage.Intercept	1	−81.327538	39.881218	−2.04	0.0414	
wage.ed	1	5.889096	2.514062	2.34	0.0192	
wage.age	1	1.213063	0.208566	5.82	<0.0001	
_Sigma.wage	1	12.696407	2.406821	5.28	<0.0001	
work.Intercept	1	−588.825928	.	.	.	
work.ed	1	37.087301	.	.	.	
work.age	1	1.929712	0	.	.	
work.mrd	1	−4.711566	483.239072	−0.01	0.9922	
work.child	1	14.895704	483.275574	0.03	0.9754	
_Rho	1	0.999990	0	.	.	
③						
Restrict1	−1	42.048232	.	.	[a]	0.999990 ≥ _Rho

[a] Probability computed using beta distribution.

① Selection is based on the discrete response variable, *work* (=1). Hence, the variable *wage* is observed for 18 married women who are working.

② The outcome equation appears to be highly significant (since all of the p-values for the estimated coefficients are below 0.05). In particular, the outcome equation regression coefficient _Sigma.wage on the selectivity regressor is highly significant, thus indicating that we have selectivity bias. Moreover, since $\rho \approx 1$, we see that the error random variables ε and u are essentially perfectly correlated.

③ Under the maximum likelihood estimation of this selection model, SAS imposes the restriction that $|\rho| \le 1$. ■

10

FUNDAMENTALS OF POPULATION DYNAMICS

10.1 THE CONCEPT OF A POPULATION

A *population* is essentially a group of individuals of the same species[1,2] living within a limited area. The term population can refer to a group of bacterial organisms, animals, plants, birds, fish, humans, etc. But whatever the group under discussion, it is, first and foremost, a collection of individuals; that is, each and every member of a population is considered to be an *individual*. In addition, for the sake of simplicity, we shall limit our discussion of populations to those composed of a *single species*—a separate and distinct biological group found within a given space. While structurally some populations can be quite complex, the simplest type of population is one with *nonoverlapping generations*—one generation dies out before the next generation is born (some fish species, seasonally breeding insects, and annual plants come to mind). In the light of this definition, it is the human population (among others) that exhibits *overlapping generations*.

Let the function $P(t)$ denote the size of a single-species population at time t. Here "size" may be the number of individuals or the collective biomass of the group. In fact, it can also be the average spatial density of a population or, for aquatic populations, the average concentration per unit of water.

It should be evident that the population of a biological species is affected by many factors such as birth and death rates, migration (immigration and emigration), the

[1] A *species* is a distinct group of related plants or animals that usually breed only among themselves.
[2] Readers not familiar with the terminology of population demographics can refer to Appendix 10.A.

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

environment, competition, and harvesting. In what follows we shall focus primarily on the average characteristics of populations, in particular on average birth and death rates, and we shall abstract from issues pertaining to migration.

Populations are not static. Even casual observation leads one to conclude that multiplicities of biological species are forever changing; some populations grow continually, while others flirt with extinction. Moreover, the sizes of some populations appear to stay remarkably constant for extended periods of time (even if they display sizable year-to-year oscillations about average size). And why do some populations cycle rapidly, while others cycle slowly? Clearly the growth pattern of a species will have important consequences for its survival and evolution.

It should be evident that there is a need to offer an explanation of the complexity and diversity found in the growth processes of biological populations. And this will be attempted through reliance upon the fundamental principles of dynamic systems that mirror ecological processes. Our modus operandi will be the use of population growth models, which will enable us to make precise theoretical arguments about the factors affecting the rate of change in population size. To accomplish this, it is first necessary to address the essentials of the general rules of change.

10.2 THE CONCEPT OF POPULATION GROWTH

Population growth generally refers to the change in a population over time; it represents the change in the number of individuals of any species using "per unit time" as a basis for measurement. In demography and ecology, the *population growth rate* (*PGR*) is the rate at which the number of individuals in a population changes per unit time, often expressed as a percentage of the number of individuals in the population at the beginning of that period. Thus,

$$PGR = \frac{\ln P(t_2) - \ln P(t_1)}{t_2 - t_1}, \quad t_2 > t_1, \tag{10.1}$$

(the mean relative growth rate over the time period from t_1 to t_2) so that

$$\% \text{ growth rate} = PGR \times 100.$$

If the time interval $t_2 - t_1$ is taken to be smaller and smaller, then, in the limit, PGR (Eq. 10.1) approaches the logarithmic derivative of the population function $P(t)$ or $PGR = (1/P(t))dP(t)/dt$.

Given that we are assuming a closed population, size can be changed only by birth and death processes. Hence, $dP(t)/dt = B(t) - D(t)$, where $B(t)$ is the number of births per unit time at time t and $D(t)$ is the number of deaths per unit time at time t. More often than not, births and deaths are expressed on a per capita basis. In this regard, we can rewrite the preceding equation as $(1/P(t))\ dP(t)/dt = b(t) - d(t)$, where $b(t) = B(t)/P(t)$ is the per capita birth rate (the average number of offspring born to one typical individual per unit time, at time t) and $d(t) = D(t)/P(t)$ is the per capita death (mortality) rate (the average number of deaths per unit time, at time t). Then $b(t) - d(t) = r(t)$ is the *net birth rate* or the *natural rate of growth*.

For a single biological species, we shall recognize two fundamental types of PGRs: (i) *unconstrained population growth*, where the per capita growth rates are affected only by environmental factors (here the PGR depends only upon the passage of time or is simply taken to be constant; an additional element of this case is that a species faces no resource limitations and is insulated from competition and predation), and (ii) *density-dependent population growth*, where the per capita birth and death rates are influenced by the population size itself.

Why is density-dependent population growth important? For many biological populations, available resources are limited (to varying degrees), and either directly or indirectly, the size of the population impacts the per capita birth and death rates. To address this feedback issues between the said PGRs and population size, all we need to do is make the birth and death rates at time t functions of the population size $P(t)$ at time t.

Populations change over time because they are generally subject to two types of dynamic processes: (i) *exogenous* changes, which are external to the system and thus are independent of the system or population level (exogenous changes to a system do not, in turn, impact the causal factor), and (ii) *endogenous* influences, which precipitate change in a dynamic variable and are, in turn, affected by those changes. Hence, endogenity implies that causal effects and system levels are linked together in a closed feedback loop, and the two reciprocative factors are deemed part of the same dynamic process; that is, endogenous effects occur whenever the level of a variable actually impacts itself, either directly or via another system element.

10.3 MODELING POPULATION GROWTH

Given the diversity of biological organisms (ranging in complexity from single-celled entities to various plants and animals and humans), a fundamental issue confronting population biologists and ecologists is the determination of the causes and consequences of the *regulation of population growth*. There is considerable agreement that populations: (i) are limited by factors such as the vicissitudes of natural phenomena; (ii) face restrictions in their food supply; (iii) are regulated by competition, predators, parasites, and diseases; and (iv) are self-regulated through cannibalism and territoriality. However, in order to make predictions about population size, we need to capture the essence of the observed patterns and processes that tend to impose regulation. And this can be done via the construction of models that seek to capture nature, albeit models that are oversimplifications and/or abstractions of reality.

We need some basic assumptions about the operating mechanisms and the environmental particulars at work that impact population growth. Specifically, we shall consider only *single-species models* (which involve one and only one biological organism) and ignore the effects of age structure. Moreover, such models will be constructed either in discrete time or in continuous time. For the *discrete-time* case, we consider a population with a fixed time interval between generations. (Discrete-time modes are alternatively called *metered models*—for each period of time, the population size P constitutes a single data point.) Here we measure time in units of generations, which may be one year, for example, salmon have an annual spawning season and thus one generation per year. In this regard, discrete-time models exhibit *nonoverlapping generations* (a generation dies out before the next succeeding generation is born). For the *continuous-time* case, generations overlap and birth can occur at any time, and the response of the population to external forces is instantaneous. For such models, we can record the population size at all times, rather than at specified intervals.

We now turn to the basic structure of continuous- and discrete-time models:

1. *Continuous-time model* has the general form

$$\frac{dP}{dt} = r(P)P = f(P), \quad 0 \le t < +\infty, \tag{10.2}$$

where $r(P)$ is termed the *growth function* (Fig. 10.1a). Here r is a continuous function of population size P and $dP/dt = 0$ when $P = 0$. Also, $r(P) = (1/P)\, dP/dt$ is taken to be the *instantaneous relative (per capita) growth rate* or *intrinsic growth rate*.

2. *Discrete-time model* has the general form

$$P_{t+1} = R(P_t)P_t = F(P_t), \tag{10.3}$$

where $R(P_t)$ is the *growth* or *fitness* or *reproduction function* (Fig. 10.1b) and P_{t+1}/P_t is termed the *per capita growth rate*. Short-term dynamics are incorporated in the term $R(P_t)$, while the model's long-term dynamics are described by the recurrence scheme (Eq. 10.3). To calculate the relative rate of change in P between periods t and $t+1$, we need only find

$$\frac{P_{t+1} - P_t}{P_t} = R(P_t) - 1. \tag{10.4}$$

Let's go a step further. Both continuous-time and discrete-time models may be either density independent or density dependent. A model is *density independent* if, in Equation 10.2, $r(P) = r =$ constant (the function $r(P)$ is independent of population size) or, in Equation 10.3, $R(P) = R =$ constant (the expression $R(P)$ is independent of population size). A model is density dependent if the per capita rate of growth determined from either Equations 10.2 or 10.3 depends on population size P.

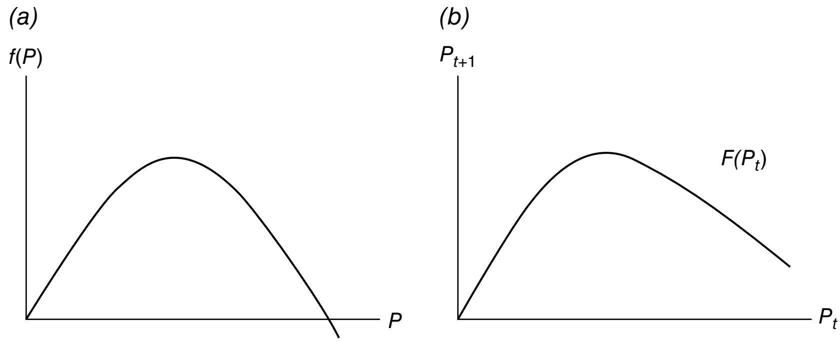

FIGURE 10.1 (**a**) Continuous-time growth function. (**b**) Discrete-time growth function.

We mentioned earlier that a major issue concerning population growth is its "regulation." In fact, such regulation occurs via density-dependent factors, which, because of resource limitations, set in after a critical population size P_c has been attained (since, say, $F(Pt)$ is taken to be a decreasing function of P at P_c) and which manifest themselves through two types of *intraspecific* (within the same species) *competition*: *scramble competition* and *contest competition*. To see how these two varieties of intraspecific competition differ in the way scarce resources are shared among individuals, let us assume that the discrete-time expression (Eq. 10.3) holds (we have a single-species, seasonal reproduction, and first-order feedback[3]). Then:

1. Scramble competition—under equal or random partitioning, resources are shared among the individuals of a population as long as they are available. Hence, there is a critical population size P_c above which the amount of resources is not enough to assure population survival.
2. Contest competition—here only the stronger individuals get the amount of resources they need to survive. If there are enough resources for all individuals, then population grows; otherwise, only the "winners" survive to reproduce.

The impact of density dependence on the regulation of population growth is a matter of degree; that is, the majority of researchers believe that most populations in nature are characterized by imperfect density dependence—density-dependent and density-independent factors work together to regulate biological populations. Hence, the rate of growth of a population relative to its size does not unequivocally imply any particular population regulation mechanism. In fact, the effect of density dependence can be viewed in essentially three separate ways: (i) the direct

[3] *Feedback* occurs whenever a change in the state of a dynamic variable is affected by the past values of that variable. So for population P, a density-induced feedback process appears as $\Delta P = F(P_{t-j})$, where P_{t-j} is population j generations in the past and F is a *feedback function*, which specifies how a population changes in response to its previous level.

impact of population size on the per capita growth rate, (ii) the impact of population size on the birth rate, and (iii) the impact of population size on survivorship (population size impacts mortality or the death rate) rather than on reproduction.

10.4 EXPONENTIAL (DENSITY-INDEPENDENT) POPULATION GROWTH

10.4.1 The Continuous Case

Let us return to the continuous-time density-independent per capita growth model presented in the preceding section. That is, Equation 10.2 becomes *continuous-time density-independent growth model*:

$$\frac{dP}{dt} = rP, \quad r \text{ constant.} \tag{10.5}$$

The instantaneous absolute growth rate is proportional to the existing population at time t, or the instantaneous relative (per capita) growth rate of the population is

$$\frac{dP/dt}{P} = r, \quad r \text{ constant.} \tag{10.6}$$

For a single-species biological population with no migration, we may interpret r in the following fashion. We first note that

$$\frac{dP}{dt} = \text{births} - \text{deaths.} \tag{10.7a}$$

If births and deaths are each taken to be proportional to P, then births $= \beta P$ and deaths $= \delta P$, where β is the constant per capita birth rate (the number of individuals or organisms produced per individual per unit time) and δ is the constant per capita death rate (the probability of dying or the mortality risk per individual per unit time).

Then

$$\frac{dP}{dt} = \beta P - \delta P = (\beta - \delta)P = rP, \tag{10.7b}$$

with $r = \beta - \delta$. Thus, r is the average per capita number of offspring minus the average per capita number of deaths per unit time—it will be termed the constant *net birth* (*reproduction*) *rate* or simply the *constant per capita growth rate* of the population.

The solution to the differential equation (Eq. 10.7b) is

$$P(t) = P_0 e^{rt}, \tag{10.8}$$

where, for $t=0$, $P(0)=P_0$ (see Equation 2.18 and Appendix 3.A).[4] What about the time path of a population that behaves according to Equation 10.8? It should be apparent that:

1. $r > 0$ (or $\beta>\delta$) implies that the population grows exponentially with $\lim_{t\to+\infty} P(t)=+\infty$ (the population size is unbounded).
2. $r < 0$ (or $\beta<\delta$) implies that the population declines exponentially with $\lim_{t\to+\infty} P(t)=0$ (the population faces extinction).

In what other context might an equation such as Equation 10.7b arise? Suppose $P(t)$ depicts the population (density) for, say, a microorganism (bacteria) at time t. Under bacterial cell division or splitting, let us assume that the birth rate of new organisms is proportional to the current level of organisms present. So if the population size at time t is P, then over a short time interval of duration Δt (>0) from t to $t + \Delta t$, the number of births is approximately $\beta P\Delta t$, where β is the (constant) per capita birth rate. In an analogous fashion, the number of deaths over the same time interval is approximately $\delta P\Delta t$, where δ is the (constant) per capita death rate. Then the net change in population size from time t to time $t + \Delta t$ is $P(t+\Delta t)-P(t)\approx(\beta-\delta)P(t)\Delta t$ or

$$\frac{P(t+\Delta t)-P(t)}{\Delta t} \approx (\beta-\delta)P(t), \tag{10.9}$$

where $\beta P(t)\Delta t-\delta P(t)\Delta t=(\beta-\delta)P(t)\Delta t$ is the net increment to the population (density) due to births and deaths during the time interval Δt. Then passing to the limit as $\Delta t\to 0$ in Equation 10.9 renders

$$\lim_{\Delta t\to 0}\frac{P(t+\Delta t)-P(t)}{\Delta t}=\frac{dp}{dt}=rP, \quad r=\beta-\delta,$$

where r is the microorganism net reproduction rate.

Given the preceding developments, it should be evident that the following two statements are equivalent:

1. The population P grows exponentially if its rate of growth is proportional to P itself.
2. The percentage or relative growth of P remains constant over time.

[4] An alternative representation of Equation 10.8 is

$$P(t)=P(t_0)e^{-r(t-t_0)},$$

where t_0 is any arbitrary (convenient) time point and $P(t_0)$ is the value of P at that time point.

In addition, a generalization of Equation 10.7.b is $dP/dt=rP+c$, where r and c are constant. The solution to this expression is a generalization of Equation 10.8 or

$$P(t)=\left(P_0+\frac{c}{r}\right)e^{rt}-\frac{c}{r}.$$

In fact, to determine if a given population exhibits exponential growth according to Equation 10.8, simply plot $\ln P$ versus time t; $\ln P = \ln P_0 + rt$ must be a linear semi-logarithmic function of t with slope r.

At this point in our discussion of the continuous-time density-independent growth model, it is instructive to review and scrutinize the set of key assumptions made. Specifically, it is assumed that:

1. Net reproduction occurs continuously (we have overlapping generations and no seasonality) and a constant growth rate $r > 0$.
2. All individuals or organisms are identical (no age structure) and develop independently of one another.
3. Environmental conditions are unrestricted and unchanging (no resource limitations).

It appears that these assumptions are highly restrictive in nature. However, the exponential model (Eq. 10.8) is very robust—it offers reasonable precision even if the aforementioned conditions are violated. In reality, individuals may differ in their age, ability to survive, and mortality risk. But if the population consists of a very large number of individuals, then their birth and death rates can be averaged to yield a fairly constant growth rate. Moreover, while Equation 10.8 might adequately describe short-term growth for small populations with a vast pool of resources, it is not useful for modeling the long-term growth of a population when resource limitations will most assuredly become a reality.

So when should an exponential model be applied? There is general agreement that Equation 10.8 is applicable in situations involving the growth of bacteria in an unlimited environment, the spreading of a computer virus, a small P_0 value and a short time horizon, small population sizes that are devoid of competitive pressures and density dependence (no interaction among individuals), the rate of decay of an antibiotic in the bloodstream, the growth of a pest population that has been introduced into an agricultural setting, and monitoring the number of new cases of an infectious disease at the beginning of an epidemic, among others.

10.4.2 The Discrete Case

We now look to the discrete-time density-independent per capita growth model offered in Section 10.3. Specifically Equation 10.3 becomes

discrete-time density-independent growth model:

$$P_{t+1} = RP_t, \quad R \text{ constant} \tag{10.10}$$

(the population in period $t+1$ is proportional to its level in the preceding period). In this discrete-time analogue to Equation 10.8, time t is measured in nonoverlapping generations.

How might an expression such as Equation 10.10 arise in practice? To answer this, let us assume that a population changes only by births and deaths and that births and deaths are each proportional to the previous year's population level P_t. Then number of births in generation $t+1 = BP_t$ (B is the constant per capita birth rate), and number of deaths in generation $t+1 = DP_t$ (D is the constant per capita death rate). Then the change in population (density) between generations t and $t+1$ (ostensibly successive generations are one year apart) is

$$\Delta P_t = P_{t+1} - P_t$$

or

$$\Delta P_t = \text{births} - \text{deaths} = BP_t - DP_t,$$

and thus

$$P_{t+1} - P_t = (B - D)P_t. \quad (10.11)$$

Then using this expression, the *relative rate of change in P between periods or generations t and* $t+1$ is

$$\frac{P_{t+1} - P_t}{P_t} = B - D, \quad (10.12)$$

where $B - D$ is the *net birth rate*. In addition, Equation 10.11 (or 10.12) implies the recursive scheme

$$P_{t+1} = (1 + B - D)P_t = RP_t, \quad R \text{ constant}, \quad (10.10)$$

where $R = 1 + B - D$ is the *net reproduction rate* or *intrinsic rate of change* in P.

As Equation 10.10 reveals, the size of the population for any generation is described by the sequence $\{P_t\}$, $t = 0, 1, 2, \ldots$, where P_0 serves as the initial population size and the *orbit of* P_0 is $\{P_t\}$, $t = 0, 1, 2, 3, \ldots$. Given P_0, the recursive equation (Eq. 10.10) can be solved algebraically for a unique solution as follows:

$$\begin{aligned}
&[\text{Generation } 1]\ P_1 = RP_0, \\
&[\text{Generation } 2]\ P_2 = RP_1 = R^2 P_0, \\
&\qquad \vdots \\
&[\text{Generation } n]\ P_n = R^n P_0.
\end{aligned}$$

In general,

$$P_t = R^t P_0, \qquad R \text{ constant}, \quad t = 1, 2, \ldots. \quad (10.13)$$

Clearly this expression represents geometric or exponential growth. What about the behavior of Equation 10.13 over successive generations? Two relevant cases present themselves:

1. If $|R| > 1$, then $P_t \to +\infty$ as $t \to +\infty$ so that population increases without bound.
2. If $|R| < 1$, then $P_t \to 0$ as $t \to +\infty$ so that population declines to zero.

A couple of additional points are in order. First, let us transform equation (Eq. 10.13) to logarithms so as to obtain

$$\ln P = \ln P_0 + (\ln R)t.$$

This expression represents a linear function of t with slope $\ln R$. Hence, a population having a linear semilogarithmic reproduction curve grows (or declines) exponentially or as a geometric series and is independent of population density. Next, we determined earlier that for the continuous-time version of exponential growth, the growth equation (Eq. 10.8) or $P(t) = P_0 e^{rt}$ was relevant. What is the connection between the per capita growth rate r and the discrete-time per capita growth rate R determined from Equation 10.13? One moment's reflection reveals that $R = e^r \approx 1 + r$.

10.4.3 Malthusian Population Growth Dynamics

Issues regarding exponential growth, while currently not a subject of vigorous debate, have a rich history connected with them. An early population growth model that received considerable attention was the one offered by Thomas Malthus (1798) to describe the time track of the population of Great Britain. Malthus assumed that the population at time $t+1$ is proportional to the population at time t or $P_{t+1} = RP_t$, $t = 0, 1, 2, \ldots$, where $R = 1 + B - D =$ constant is the natural growth rate. That is, any species can potentially increase in numbers by way of a geometric series; and if the species exhibits nonoverlapping generations and each individual or organism produces R offspring, then the population numbers in generations $t = 0, 1, 2, \ldots$ follow the pattern

$$\begin{aligned} P_1 &= RP_0, \\ P_2 &= RP_1 = R^2 P_0, \\ &\vdots \\ P_n &= R^n P_0, \text{ etc.} \end{aligned}$$

Hence, the values of the Malthusian linear reproduction function $P_{t+1} = RP_t$ grow as a geometric series with each successive generation (Fig. 10.2).

But this is not the whole story. Malthus contended that the growth of the human population is fundamentally different from the growth of the food supply needed to sustain it. So while the human population was growing in an exponential or geometric fashion, the food supply was only growing arithmetically or linearly. Malthus thus concluded that if left unchecked, it would simply be a matter of time before the world's

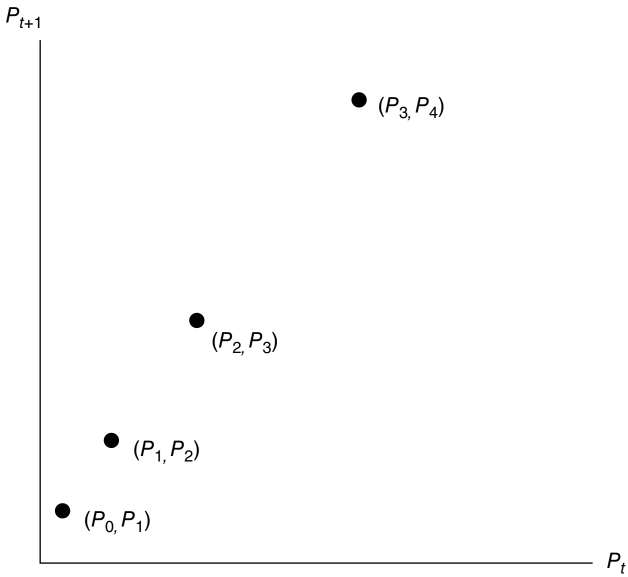

FIGURE 10.2 Linear reproduction function and exponential growth.

population would become too large to feed itself. This note of "doom and gloom" led Thomas Carlyle (1849) to refer to classical economics as the "dismal science."

We noted earlier that if $|R|>1$ in Equation 10.13, then $P_t \to +\infty$ as $t \to +\infty$; and if $|R|<1$, then $P_t \to 0$ as $t \to +\infty$. An alternative way of illustrating this conclusion is to examine the behavior of the linear reproduction curve $P_{t+1}=RP_t$ itself as $t \to +\infty$. Our approach will be to use *cobwebbing or geometric iteration*. In Figure 10.3a, let us graph the linear reproduction or *generating curve* $P_{t+1}=RP_t$, $R>1$, and a one-to-one *reference line* $P_{t+1}=P_t$ in the P_t, P_{t+1}-plane. We then start with P_0 on the horizontal axis, evaluate P_1 on the reproduction curve, and use the reference line to reflect this P_1 value back to the horizontal axis. The process is then repeated for increasing values of *t*. Clearly P_t will increase without bound. For Figure 10.3b, with $R<1$, we again start with P_0 on the horizontal axis, evaluate P_1 on the reproduction curve, and then use the reference line to reflect P_1 back to the horizontal axis. The process is repeated for increasing values of *t* and clearly P_t converges to zero.

This cobwebbing routine is part of a body of material called "equilibrium and stability analysis." That is, the *equilibrium* or *fixed point* $P=0$ in Figure 10.3 is termed *attracting* when $0<R<1$; it is said to be *repelling* when $R>1$. So for $0<R<1$, the equilibrium at $P=0$ is *stable*; when $R>1$, this equilibrium is *unstable*. (Readers interested in exploring issues pertaining to equilibrium and stability analyses in greater detail are directed to Appendix 10.B.)

There is a general consensus that the Malthusian model is a reasonable description of population growth only for the short term and for small populations with vast resources; it is not appropriate for describing long-term population growth. If a population faces resource limitations, the Malthusian model leaves much to be desired.

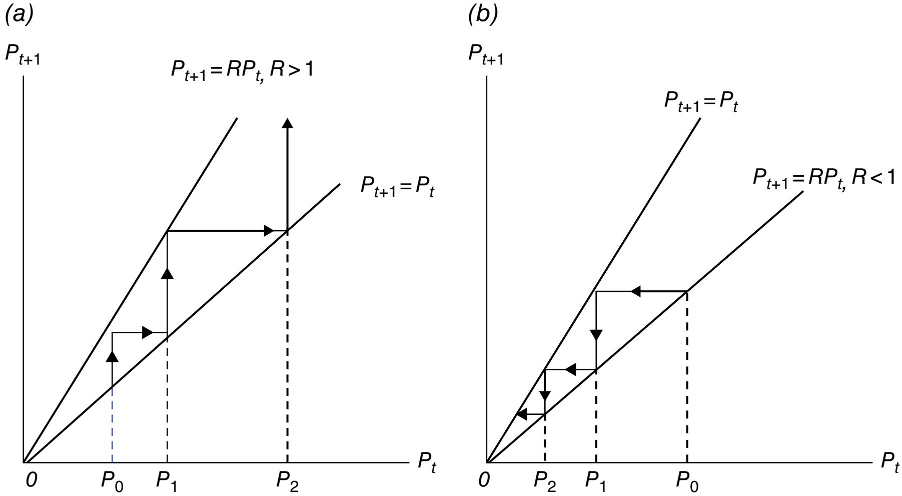

FIGURE 10.3 (**a**) If $R > 1$, P_t increases without bound. (**b**) if $R < 1$, P_t decreases to zero.

We noted earlier that as population increases, resources become more and more limited. Intraspecific competition becomes a reality, and density-dependent factors negatively impact the PGR. Hence, the assumption that the growth rate of a population is proportional to its size in perpetuity is unrealistic. Population growth rates tend to decrease over time even though exponential growth occurs at the outset. Growth rates tend to change with population size, approaching zero as the population gets larger and larger.

The preceding discussion has brought to the fore a very important conclusion—PGRs are not constant; they depend on population size or are density dependent. In this regard, the next and subsequent sections present a set of models that are much more realistic and sophisticated than the simple exponential growth models. Specifically, reproduction is taken to depend on the current population size. This then takes us back to the general continuous- and discrete-time models (Eqs. 10.2 and 10.3), respectively. In fact, since birth and death rates do not normally remain constant over time, we need to more adequately specify the principal feedback processes that affect these rates.

10.5 DENSITY-DEPENDENT POPULATION GROWTH

In this section we shall relax a key assumption of the Malthusian population growth model, namely, that reproduction per capita per unit time is constant. That is, the continuous-time Malthusian population growth model with $(1/P)dP/dt = r =$ constant is now replaced by the more general expression $(1/P)dP/dt = r(P)$ or $dP/dt = r(P)P$. Clearly reproduction per capita $r(P)$ depends on current population size and consequently is "density dependent."

For r positive and constant, the Malthusian model predicts exponential growth. However, exponential growth is not very realistic for most populations. Indeed, it is not an adequate description of the long-run dynamics of population growth. For one thing, the environment or *carrying capacity*[5] of the resource base is not unlimited so that, ostensibly, intraspecific competition (individuals compete for food, habitat, and other scarce resources) forces the population to approach some threshold or sustainable size. In this regard, what is needed are population growth models that cannot support exponential growth indefinitely. While some populations can grow exponentially at the outset, it is typically the case that their growth rates tend to decrease as population size increases. Hence, exponential behavior can only characterize population dynamics for short periods of time so that the assumption that the instantaneous absolute growth rate dP/dt is proportional to P (r is the constant of proportionality) is too heroic for longer time horizons.

Having made the case for the relevance of density-dependent population growth models, the remainder of this chapter focuses on continuous-time and discrete-time growth equations of the form $dP/dt = r(P)P = f(P)(dr/dP < 0)$ and $P_{t+1} = R(P_t)P_t = F(P_t)(dR/dP < 0)$, respectively. We start with the logistic case—the simplest form that satisfies, say, $dP/dt = r(P)P$. Other density-dependent growth models treated herein are those offered by Beverton and Holt, Ricker, Hassell, and the generalized Beverton and Holt and Ricker equations. As was the case earlier, we deal exclusively with an isolated population (no migration).

10.5.1 Logistic Growth Model

Continuous-Time Logistic

No population expands without limits—especially when its per capita rate of growth depends upon population density (Verhulst, 1838). When a population is far from its growth limit, it can possibly grow exponentially. But when nearing that limit, the PGR declines significantly and reaches zero when its limit has been attained. (When the limit is exceeded, the PGR is negative, and thus population numbers obviously decline.) This is essentially the line of reasoning used by Verhulst (1838). He proposed a long-run adjustment to exponential growth via a self-regulating mechanism (it compensates for overcrowding via intraspecific competition) that becomes operational when a population becomes “too large.” Here “too large” means that the population size approaches a “limiting population” that is determined by the carrying capacity of the environment. To model this phenomenon, let us modify r by multiplying it by an “overcrowding term” so that we obtain a density-dependent population growth equation. Specifically, let the intrinsic growth rate $(1/P)dP/dt$ be a decreasing function of P or $r(P) = r(1 - (P/K))$ (we have a constant linear decrease in r as population size increases), where r is the constant maximum intrinsic growth rate when $P = 0$ and K represents the limiting size or carrying capacity of the population (Fig. 10.4). (In fact, the logistic equation is the simplest population growth model that takes into account carrying capacity.)

[5] The carrying capacity of a biological species is the maximum size or “maximum load” of the species that its environment can sustain indefinitely without negatively impacting the species or its given resource base. For humans, more complex features such as adequate medical care, sanitation, and waste disposal are included as a part of the environmental landscape.

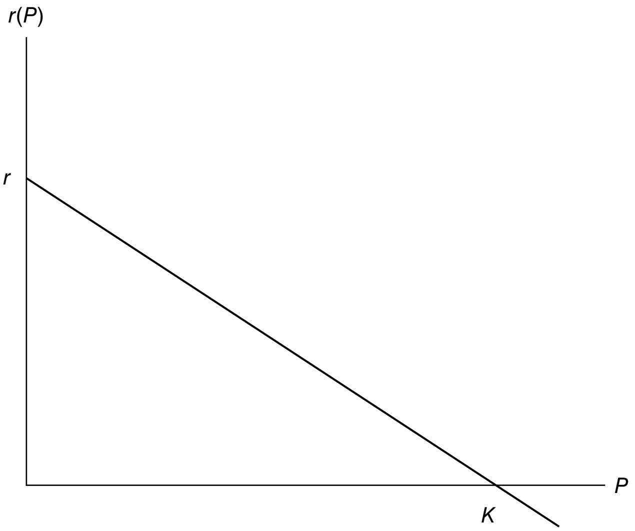

FIGURE 10.4 Reproduction per capita is density dependent.

So at low population densities ($P<K$), the PGR is maximal and is approximately r (the population grows exponentially and is not subject to intraspecific competition); the PGR declines as P increases (density-dependent factors negatively impact mortality, survival, and/or birth rates); and the growth rate equals zero when $P=K$. If $P>K$, population growth is negative.

Given the preceding choice for $r(P)$, the dynamics of resource-limited population growth can be described by the expression

$$\frac{dP}{dt} = r(P)P = rP\left(1-\frac{P}{K}\right). \tag{10.14}$$

(This equation has a Malthusian growth term and a term representing crowding effects.) The associated differential equation has the solution

$$P(t) = \frac{P_0 K}{P_0 + (K-P_0)e^{-rt}}, \quad t \geq 0 \tag{10.15}$$

with $P(0)=P_0$.[6] Let us term (Eq. 10.15) the *logistic growth function*. As Figure 10.5 reveals:

1. For $0<P_0<K$, population increases asymptotically to K (population follows a logistic curve).

[6] We previously expressed the logistic growth function as

$$Y_t = \frac{Y_\infty}{1+\alpha e^{-\beta t}}, \quad t \geq 0 \tag{3.6}$$

(see Equation 3.6 and Appendix 3.A). If we set $Y_t=P_t$, $\beta=r$, $Y_\infty=K$, and $\alpha=(Y_\infty/P_0)-1$, then Equation 3.6 readily transforms to Equation 10.15.

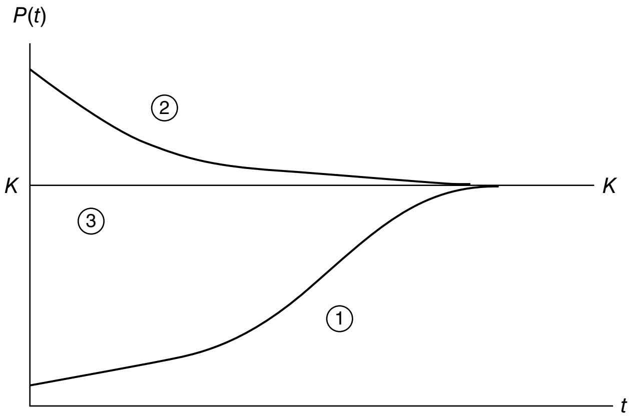

FIGURE 10.5 Population follows the logistic model.

2. For $P_0 > K$, population decreases asymptotically to K (for these first two cases, it is easily verified that $\lim_{t \to +\infty} P(t) = K$).
3. For $P_0 = K$, the population level $P(t) = K$ for all t.
4. For $P_0 = 0$, $P(t) = 0$ for all t.

A few comments pertaining to Equation 10.15 are in order:

1. The logistic model is useful for a population that has a tendency to grow from effectively zero up to some carrying capacity K.
2. The maximum possible rate of population growth is r, which is the net effect of reproduction and mortality. A high value of r is indicative of organisms that tend to reproduce rapidly; and those organisms with a low r value tend to reproduce slowly.
3. The carrying capacity term K has biological significance for populations exhibiting a well-defined social structure and a strong interaction among individuals (in terms of competition for territory, light, etc.) that influences their reproduction.
4. The logistic model combines two important density-dependent ecological processes: reproduction and competition.
5. The logistic model works well for populations of microorganisms (e.g., yeast grown in a restricted environment), collared doves in Great Britain (Hudson, 1972), and sheep ranching in Tasmania (Davidson, 1938).

The logistic model is not without its drawbacks:

1. Since r controls both population growth and decline, it is questionable that organisms with a low birth rate should also have the same low rate of death. So if, for instance, the birth rate is low and the death rate is high, then obviously the logistic model leaves much to be desired.

2. The logistic model does not indicate when a population might face extinction.
3. Large population levels can be associated with negative population growth (if $P > K$ in Fig. 10.4).
4. The logistic model ignores age structure; that is, the mortality rate may depend on age and the birth rate may depend on adult population size rather than on total population.
5. The logistic model ignores the issue of population integration, for example, mutualism and predation.

It should be apparent from the foregoing analysis of the logistic model that reproduction per capita $r(P) = r(1-(P/K))$ is an assumed form and not a consequence of any fundamental population growth law. No biological mechanism has been offered as an explanation for reproduction per capita being linear and density dependent. This said, we can still give legitimate demographic meaning to $r(P)$ by assuming that the vital rates (birth and mortality) are density dependent. To this end, let us write the per capita birth and death rates (b and d, respectively) as functions of population size P so that $(1/P)dP/dt = [b(P) - d(P)]$. Suppose both of these rates change linearly with population size so that $b(P) = b_0 + b_1 P$ *and* $d(P) = d_0 + d_1 P$. Then

$$\frac{dP}{dt} = (b_0 - d_0)P - (b_1 + d_1)P^2.$$

If we define the intrinsic growth rate as $r = b_0 - d_0$ and the carrying capacity as $K = r/(b_1 + d_1)$, then the preceding expression becomes Equation 10.14 or $dP/dt = rP(1-(P/K))$ as one might have anticipated.

We now turn to the task of finding any equilibrium points of the logistic density-dependent rate of population change $dP/dt = r(P)P = f(P) = rP(1-(P/K))$. If any such points are found, then we need to determine if these points are locally stable or unstable. We know from Section 10.B.3 that equilibria (P^*'s) are those P's that make $dP/dt = 0$ or

$$f(P^*) = rP^*\left(1 - \frac{P^*}{K}\right) = 0.$$

It is readily seen that this equality holds if either $P^* = 0$ or $P^*/K = 1$. Hence, we have two equilibrium values of $P: P_1^* = 0$ and $P_2^* = K$.

To check for stability (the reader is again referred to Section 10.B.3), we first find

$$f'(P) = r - 2r\frac{P}{K}.$$

Then:

1. $f'(P_1^*) = f'(0) = r > 0$ (the equilibrium at $P_1^* = 0$ is unstable).
2. $f'(P_2^*) = f'(K) = -r < 0$ (the equilibrium at $P_2^* = K$ is locally unstable).

Clearly these results are consistent with the behavior of Equation 10.15 as $t \to +\infty$ (see also Fig. 10.5).

One final point is in order. It has been observed (Allee, 1931; Courchamp et al., 2008; Odum and Allee, 1954; Lidicker, 2002, 2010) that in high-density populations, mutually positive cooperative effects were significant (they allowed a population to generate a high level of average well-being and to attain a higher stable equilibrium density level). By the same token, as a population declined in numbers, there could be insufficient membership to achieve the cooperative benefits. So while cooperation (or facilitation) could enhance population performance, it was subject to the risk that at low population densities, its loss could lead to an increased chance of extinction; that is, a declining population may even reach a density at which the per capita growth rate will be zero before population density itself reaches zero. In this instance, any further declines will cause the growth rate to become negative, and this will likely be followed by extinction.

In a societal setting, the intrinsic growth rate r might be negative at low population densities due to, say, the possibility of diminished reproductive opportunities (it may be difficult to find a mate) or to other factors; and it is again negative at higher densities because of intraspecific competition due to overcrowding. In this regard, an *Allee effect* or "underpopulation effect" implies a positive relationship between population density and the reproduction and survival of individuals—it arises when the intrinsic or per capita growth rate increases at low densities, reaches a maximum at an intermediate population density, and then declines in the direction of high population densities. Hence, there is a population density threshold (where the per capita growth rate equals zero) below which the population will decrease and possibly become extinct (see point A of Fig. 10.6).

The traditional approach to population dynamics stipulates that due to intraspecific competition for resources, a population will experience a reduced growth rate at high density levels and an increasing growth rate at lower ones. This is the message of the logistic model. Under the Alee effect, the reverse is true at low density levels (when $P<P^*$ in Fig. 10.6). So in a logistic world, $(1/P)dP/dt$ decreases with increasing P throughout the entire range of population sizes; but if the Allee effect is operative, $(1/P)dP/dt$ increases with increasing P over some limited range of population levels.

To model the Allee effect, let us perform a minor adjustment to the logistic per capita growth equation $r(P)=r(1-(P/K))$ so as to obtain the *sparsity-impacted logistic per capita growth equation*

$$g(P) = r(P)\left(\frac{P}{\theta}-1\right) = r\left(1-\frac{P}{K}\right)\left(\frac{P}{\theta}-1\right), \tag{10.16a}$$

where θ is termed *a sparsity parameter* $(0<\theta<K)$ and $((P/\theta)-1)$ is the *sparsity component* of $g(P)$. Setting $g(P)=0$ and solving (a quadratic equation) for the P's render $P=\theta$ and $P=K$ (Fig. 10.7). In addition, $g(P)$ attains a maximum at $P^*=(\theta+K)/2$ (where $g'(P^*)=0$ and $g''(P^*)<0$), $g(0)=-r$, and $g(P^*)=(r/4\theta K)(K-\theta)^2$.

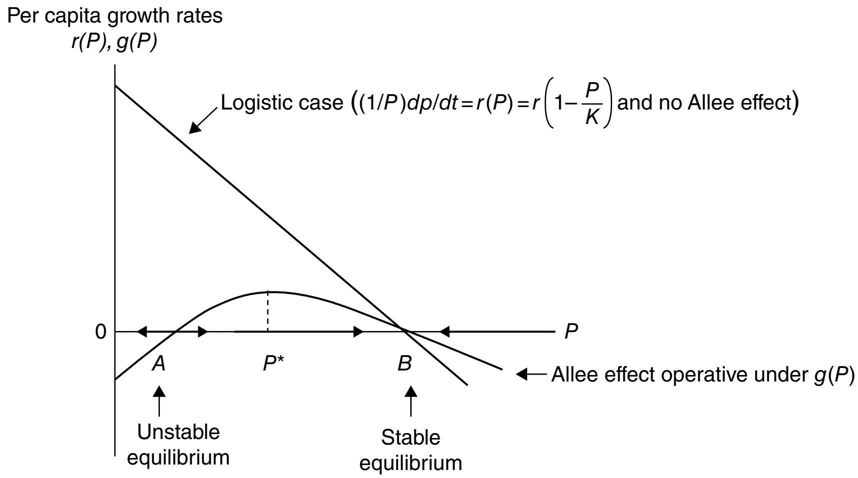

FIGURE 10.6 Allee effect: $g'(P)>0$ for $P<P^*$ and $g'(P)<0$ for $P>P^*$, where P^* is the population density for maximal reproduction and $(1/P)dP/dt = g(P)$.

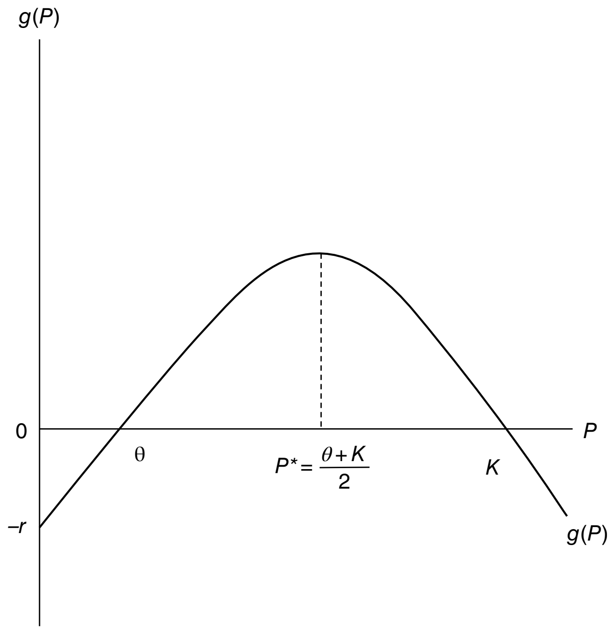

FIGURE 10.7 Sparsity-impacted logistic per capita growth curve.

To demonstrate that $P=\theta$ is an unstable equilibrium population level, we first determine, from $f(P)=Pg(P)=rP(1-(P/K))((P/\theta)-1)$, that $f(\theta)=0$ and $f'(\theta)=1-(\theta/K)$. For $\theta<K$, we find that $f'(\theta)>0$ so that the equilibrium population density level $P=\theta$ is unstable.

Discrete-Time Logistic Discrete-time difference equation models of population growth are of fairly recent vintage. Such models specify the per capita growth rate $P_{t+1}/P_t = R(P_t)$ in more realistic forms and tend to have a firmer biological grounding relative to their predecessor continuous-time models. In particular, discrete-time difference equations can be used to model species that exhibit little or no overlap of generations (e.g., it is not necessary to take account of age structure in modeling, say, annual plants or most insect populations). In fact, population change in nature typically occurs in a discontinuous fashion; that is, seasonality tends to dominate birth/growth processes.

An appropriate way to derive the discrete-time logistic equation is to replace the derivative dP/dt by a difference form with a time step Δt of unity. To this end, from Equation 10.14, set

$$\frac{P(t+\Delta t)-P(t)}{\Delta t} = rP(t)\left(1-\frac{P(t)}{K}\right)$$

or, for $\Delta t = 1$,

$$\begin{aligned} P(t+1) &= P(t) + rP(t)\left(1-\frac{P(t)}{K}\right) \qquad (10.16b) \\ &= (1+r)P(t)\left(1+\frac{r}{1+r}\frac{P(t)}{K}\right). \end{aligned}$$

Let us rescale Equation 10.16b by using $P(t) = ((1+r)/r)Kx(t)$. Hence, we obtain

$$P(t+1) = (1+r)\left(\frac{1+r}{r}\right)Kx(t)(1+x(t)). \qquad (10.16c)$$

Then $P(t+1) = ((1+r)/r)Kx(t+1)$ so that Equation 10.16c becomes

$$x(t+1) = r'x(t)(1-x(t))$$

or

$$x_{t+1} = r'x_t(1-x_t), \qquad (10.17)$$

where $r' = 1+r$. (Note that when P goes from 0 to K, x goes from 0 to $r/(1+r)$). While Equation 10.17 is one form of a discrete-time logistic growth model, other forms abound (see Appendix 10.C).

Let us rewrite Equation 10.17 as $F(x) = r'x(1-x)$. To determine any fixed or steady-state points of F, set $F(x) = x$ or $r'x(1-x) = x$. It is readily verified that the preceding equality holds for $x = 0$ and $x = (r'-1)/r' = r/(1+r)$. When $x = 0$, $P = ((1+r)/r)K(0) = 0$; and when $x = r/(1+r)$, $P = ((1+r)/r)K(r/(1+r)) = K$.

We next look to the stability of these x solutions. From $F = r'x(1-x)$, we obtain $F' = r' - 2r'x$. For $|F'(0)| = r' = 1+r > 1$, we see that $x = 0$ is an unstable fixed or equilibrium point. And for $|F'(r/(1+r))| = 1 - r < 1$, we see that $x = r/(1+r)$ is a locally stable fixed point.

The continuous-time logistic model, which is a simple extension of the Malthusian model of exponential population growth, improved on the latter by explicitly accounting for the effects of intraspecific competition and overcrowding that tend to limit population growth (the reproduction function turns negative for large populations). By the same token, in the discrete-time logistic case, large realized populations in one generation can return a negative population in the next generation. Since this type of outcome is unrealistic (it can also lead to severe fluctuations in population size), it is important to modify both versions of the aforementioned logistic model so that, for large population densities, even though there is a reduction in the growth rate, population should remain nonnegative. In addition, the logistic per capita growth rate declines in a linear fashion with increasing P. (Clearly this "linearity" specification should also be modified as well since, in nature, the relationship between the per capita growth rate and P is, more often than not, highly nonlinear.) So while most population growth models tend to be based on the logistic equation, some popular extensions of the logistic architecture are the subject matter of the ensuing sections.

10.6 BEVERTON–HOLT MODEL

The Beverton–Holt (B–H) (1957) model is a discrete-time equation used in fisheries management that describes a connection between "stocks" and "recruits"—the so-called *stock–recruitment* (*S–R*) *relationship*. We may view the *spawning stock* (*S*) or "stock" for short as the portion of the fish population wherein reproduction occurs. *Recruits* (*R*) are the entrants to the stock of fishes (the entry process itself is called *recruitment*). Recruitment can be defined in terms of (i) the age of the fish (e.g., fish surviving to age one year are said to have been recruited to age one year) and (ii) the size of the fish (fish that survive to a "catchable size" are said to have been recruited to the fishery).

Various types of *S–R* relationships may be considered. First, an *S–R* relation may be density *independent* ($R/S = a = constant$—the ratio of recruits to spawners is constant and independent of the number of spawners, where a is the density-independent parameter). Second, the connection between S and R may be described as *density dependent*. Such relationships may be specified as:

1. *Compensatory*—per capita recruitment declines with increasing size of the S.
2. *Overcompensatory*—per capita recruitment declines with "high" values of S (e.g., when disease outbreaks occur due to overcrowding).
3. *Depensatory*—per capita recruitment actually decreases when S falls below a certain level (the Allee effect).

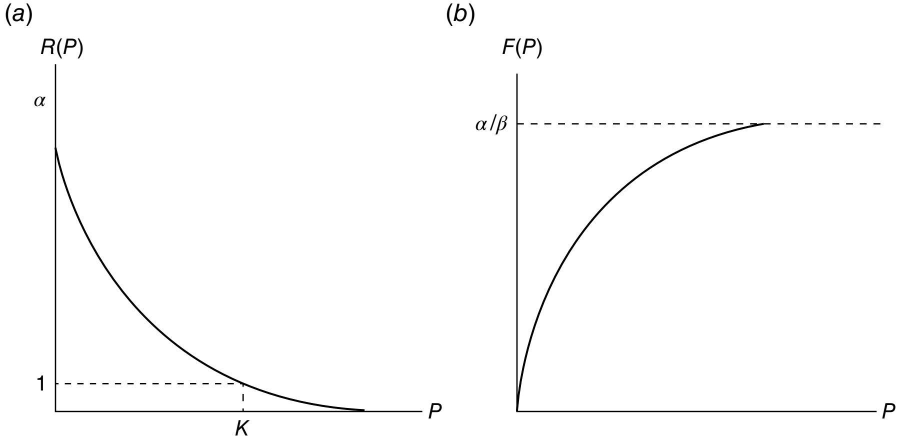

FIGURE 10.8 (**a**) B–H density-dependent fitness function. (**b**) B–H density-dependent generating function.

The B–H model is used to describe the population of fish species (e.g., Pacific salmon), which reproduce but once over their lifetime. In this regard, salmon populations display nonoverlapping generations, with the size of each generation a function of the size of the previous generation from which it was spawned. Hence, the model depicts the density-dependent recruitment of a population in which the resources are limited and unequally shared (contest competition). In considering density-dependent survival from one life stage to the next, Beverton and Holt assume that compensatory mortality depends on the density of a cohort at each point in time; that is, juvenile competition results in a mortality rate that is dependent on the number of fish alive in the cohort at any one time, and it is assumed that predators are always present. Hence, the B–H model is appropriate if there is an upper limit to the availability of food (or space) or if predators can readily adjust their actions to changes in prey availability.

What are the particulars of the B–H equation? Given the discrete-time generating function $P_{t+1} = R(P_t)P_t = F(P_t)$, suppose the fitness function $R(P_t)$ or per capita growth rate decreases as a rational function

$$R(P) = \frac{\alpha}{1+\beta P}, \quad \alpha, \beta > 0 \tag{10.18}$$

(Fig. 10.8a), where α is the growth rate and β serves as a measure of growth restriction. Stated alternatively, the number of R per spawner is a decreasing function of the number of spawners. (Note that the fitness function Equation 10.18 is an improvement over the logistic case since the former is always nonnegative.) Clearly $R'(P) = -\alpha\beta(1+\beta P)^{-2} < 0$. (This form chosen for the fitness function can be explained by the heavy cannibalism of juveniles by adults of the same species during a period of very short duration after birth.)

When the population size is very small ($P \approx 0$), the population grows exponentially since, in this instance, $R(0)=\alpha=$constant. In addition, suppose that as $R(P)$ decreases with P, the population reaches a threshold size K (its carrying capacity) and that $R(K)=1$. Then $R(K)=\alpha/(1+\beta K)=1$ or $\beta=(a-1)/K$. Hence, Equation 10.18 can be rewritten as

$$R(P_t)=\frac{P_{t+1}}{P_t}=\frac{\alpha}{1+((\alpha-1)/K)P_t}$$

or

$$P_{t+1}=\frac{\alpha P_t}{1+((\alpha-1)/K)P_t}, \quad t=0,1,2,\ldots, \tag{10.19}$$

the *B–H generating function* or the *B–H metered equation* (a formal derivation of the B–H S–R model is provided in Appendix 10.D).

To assess the properties of the B–H model, let us express its generating function as

$$F(P)=\frac{\alpha P}{1+\beta P}, \quad P>0$$

(Fig. 10.8b). Then:

1. Since $F'(P)=\alpha/(1+\beta P)^2>0$ for $P>0$, it is evident that F is a monotonic increasing function of P.
2. $\lim_{P\to+\infty}F(P)=\alpha/\beta$ (via l'Hôpital's rule). Hence, F has an upper asymptote or peak recruitment value of a/β.
3. Since $F''(P)=-2a\beta/(1+\beta P)^3$, it is readily seen that F is concave downwards for $\beta>0$.

To determine any equilibria for Equation 10.19, we can solve

$$P=\frac{\alpha P}{1+\beta P}, \quad \beta=\frac{\alpha-1}{K},$$

for the fixed or steady-state points $\bar{P}_1=0$ (the extinct equilibrium) and $\bar{P}_2=(\alpha-1)/\beta$ (the viable equilibrium). We next test these equilibrium points for stability:

1. $|F'(0)|=\alpha<1$ if $0<\alpha<1$. Thus, $\bar{P}_1=0$ is a locally stable equilibrium point if $0<\alpha<1$; it is unstable if $\alpha>1$.
2. $|F'((\alpha-1)/\beta)|=(1/\alpha)<1$ if $\alpha>1$. So if $\alpha>1$, $\bar{P}_2=(\alpha-1)/\beta>0$ is a locally stable equilibrium point.

(Given that $(\alpha-1)/\beta$ is a positive fixed point if $\alpha>1$, let us set $(\alpha-1)/\beta=K$. Then $\beta=(\alpha-1)/K$, and thus $F(P)=\alpha P/[1+(\alpha-1)P/K]$ so that Eq. 10.19 obtains.) In sum, if

$0<\alpha\leq 1$, then 0 is the only equilibrium point since $(\alpha-1)/\beta\leq 0$ (negative values of P are inadmissible). In addition, $\bar{P}_1=0$ is locally stable if $0<\alpha<1$. But if $\alpha>1$, then the B–H equation has two equilibrium points: $\bar{P}_1=0$ and $\bar{P}_2=(\alpha-1)/\beta>0$, with only the latter being locally stable.

We close this section with a couple of short commentaries. First, even though Equation 10.19 is nonlinear, it can be solved explicitly (via a change of variable $Y_t=P_t^{-1}$) to render the closed form specification for the time path of P or

$$P_t=\frac{KP_0}{P_0+(K-P_0)\alpha^{-t}}. \tag{10.20}$$

Second, we noted earlier that if depensatory effects (or Allee effects) exist, then per capita reproductive success declines at low population levels. In such circumstances, reduced mortality may be insufficient to allow for the recovery of the population. A modification of the B–HS–R function to include depensatory recruitment might appear as

$$R=\frac{\alpha S^{\delta}}{1+\beta S^{\delta}},\quad \alpha>0, \tag{10.21}$$

where the parameter δ controls the degree of depensation (Meyers et al., 1995). If $\delta=1$, then obviously there are no depensatory effects; and if $\delta>1$, Equation 10.21 is S-shaped.

10.7 RICKER MODEL

Like the B–H model of the preceding section, the Ricker (1954, 1958, 1975) model is an S–R model—it too represents the population of young fish (Pacific salmon), which have reached a catchable age (R) as a function of the spawning stock (S). The model provides a description of (i) an isolated single-species population, which only produces offspring at a specific time each year, and (ii) density-dependent survival from one life stage to the next. Here compensatory mortality is taken to depend on initial density and can be explained by (i) cannibalism of eggs and juveniles by adults and (ii) a limitation on food, with resources taken to be equally partitioned among the population (scramble competition). Clearly the number of starvation deaths is a function of the number of individuals sharing the food.

Looking to the specifics of the Ricker density-dependent S–R equation, suppose we posit that the number of R per spawner is a decreasing function of the number of spawners, or

$$\frac{R(S)}{S}=ae^{-bS}, \tag{10.22}$$

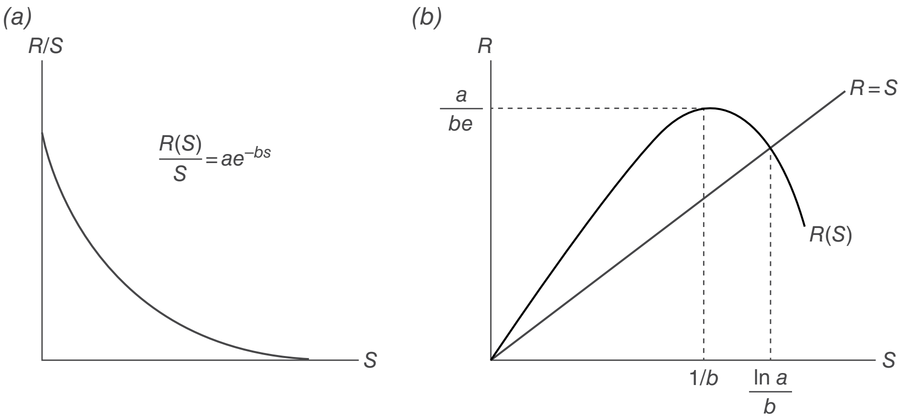

FIGURE 10.9 (**a**) Ricker density-dependent fitness function. (**b**) Ricker density-dependent generating function.

where the parameters a and b are density independent and density dependent, respectively (Fig. 10.9a). Then the *Ricker generating function* has the form

$$R(S) = aSe^{-bS} \tag{10.23}$$

(Fig. 10.9b). As this figure indicates, the peak recruitment level a/be occurs at an intermediate S value $1/b$. (Here $dR/dS=0$ at $S = 1/b$; and $d^2R/dS^2<0$ at $S=1/b$. Thus, R attains a strong global maximum with respect to S at $1/b$. In addition, $R(1/b)=a/be$.) A more formal derivation of the Ricker S–R relationship is offered in Appendix 10.E.

Note that if density dependence in the S–R equation (Eq. 10.23) does not exist, then $b=0$ and $R=aS$ or $R_{t+1}=S_{t+1}=aS_t$ (R_t recruits is equivalent to S_{t+1} parents or spawners). If $a>1$, then exponential growth occurs, especially when the spawning population is small. In the Ricker scheme of things, the exponential term aS_t is modified by the *mortality factor* e^{-bS_t}, which reflects predation and the negative effects of overcrowding, hence the rationale for Equation 10.22. In particular, the parameter b is thought to reflect the intensity of cannibalism and intraspecific competition. Note that the mortality factor causes reduction in the growth rate (it is the source of the curvature in Fig. 10.9a and b) but allows R_{t+1} to remain positive.

It should be apparent that R_t will be the next generation of stock that spawns at time $t+1$; that is,

$$\begin{aligned} R_1 &= S_1 = aS_0e^{-bS_0} \\ R_2 &= S_2 = aS_1e^{-bS_1} \\ &\ldots\ldots\ldots\ldots\ldots\ldots \\ R_{n+1} &= S_{n+1} = aS_ne^{-bS_n}. \end{aligned}$$

Hence, Equation 10.23 can be rewritten as the *Ricker generating function* or *metered equation*

$$R_{t+1} = S_{t+1} = aS_t e^{-bS_t}, \quad t = 0, 1, 2, \ldots. \tag{10.24}$$

Thus, the number of spawners determined from one generation to the next follows (Eq. 10.24).

While it may or may not be obvious, we can view the Ricker equation (Eq. 10.24) as an adaptation of the logistic equation $P_{t+1} = rP_t(1-(P_t/K))$; that is, the Ricker version of the latter is

$$P_{t+1} = P_t e^{r(1-(P_t/K))}, \tag{10.25a}$$

where r is the intrinsic growth rate. Let us rewrite this expression as

$$\begin{aligned} P_{t+1} &= P_t e^r e^{-(rP_t/K)} \\ &= aP_t e^{-bP_t}, \end{aligned} \tag{10.25b}$$

where $a = e^r$ and $b = r/K$. Clearly Equation 10.25b resembles Equation 10.24.

Let us now compare the Ricker model with the B–H model. Very generally, in the B–H case, above a certain level of S, there is no relationship between the stock and recruitment—as the parent stock increases without bound, the recruitment level approaches the asymptote α/β. In the Ricker formulation, the S–R equation exists for all sizes of the spawning stock, and there is an optional (maximal) stock size.

We may find any equilibrium points for Equation 10.23 by setting

$$R = F(S) = aSe^{-bs} = S.$$

(Note that along $R = S$, the dying S are exactly replaced by their young R.) The fixed or steady-state points, which obtain, are the extinct equilibrium $\bar{S}_1 = 0$ and the viable equilibrium $\bar{S}_2 = \ln a/b$ (Figure 10.9b). To test these equilibrium points for stability, we first find

$$F'(S) = ae^{-bS}(1 - bS).$$

Then:

1. $|F'(0)| = a < 1$ if $0 < a < 1$. Thus, $\bar{S}_1 = 0$ is a locally stable equilibrium point if $0 < a < 1$; it is unstable if $a > 1$.
2. $|F'((\ln a)/b)| = |1 - \ln a| < 1$ if $2 > \ln a > 0$ or $e^2 > a > 1$. So if $e^2 > a > 1$, $\bar{S}_2 = \ln a/b$ is a locally stable equilibrium point.

In sum, if $0 < a \leq 1$, then $\bar{S}_1 = 0$ is the only equilibrium point; and if $0 < a < 1$, then 0 is locally stable. But if $e^2 > a > 1$, the Ricker equation has two equilibrium points: $\bar{S}_1 = 0$ and $\bar{S}_2 = \ln a/b$, with only the latter equilibrium stock level being locally stable.

10.8 HASSELL MODEL

Hassell (1975) offers a discrete-time single-species growth relationship (which is particularly well suited to the study of insect populations) to explain the density-dependent feedback that often occurs to reduce actual growth rates. To explore Hassell's model, let us first consider some notation. Specifically, P_t denotes the (original) population density for generation t; P_S is the density of survivors in generation t (*mortality* is expressed as $\ln(P_t/P_S)$); and P_c is a threshold population density level below which the mortality becomes negative.

Given this terminology, the *Hassell generating function* may be expressed as

$$P_{t+1} = \frac{\lambda P_t}{(1+aP_t)^b}, \tag{10.26}$$

where $\lambda(>0)$ is the population (net) growth rate (the maximum reproductive potential of an individual in the absence of intraspecific competition or the intrinsic growth rate for "small" populations), $a(>0)$ scales the population size to the carrying capacity of the habitat or environment (a is used to define the threshold density $P_c = a^{-1}$), and $b(>0)$ describes the form and strength of intraspecific competition. Note that both a and b serve to define the inhibitive density-dependent feedback, which limits growth. It should be evident that the Hassell model is an extension of the B–H model (to get the latter, simply set $b = 1$) and a viable alternative to the logistic and Ricker models.

To exhibit the form of the density-dependent function from the model, let us express mortality ($\ln(P_t/P_S)$) in terms of the logarithm of population density ($\ln P_t$), that is, from Equation 10.26,

$$\frac{\lambda P_t}{P_{t+1}} = (1+aP_t)^b.$$

If we set $P_{t+1} = \lambda P_s$, then we can write the preceding expression as

$$\ln\frac{P_t}{P_s} = b\ln(1+aP_t), \tag{10.27}$$

with $\ln(P_t/P_s) \to 0$ when $P_t \to 0$ (see Fig. 10.10). Note that b is the constant slope of Equation 10.27 attained at high population densities (at lower population levels the slope is not constant).

An important property of Equation 10.26 is that varying the density-dependent parameter b describes the form of intraspecific competition for resources, which ranges

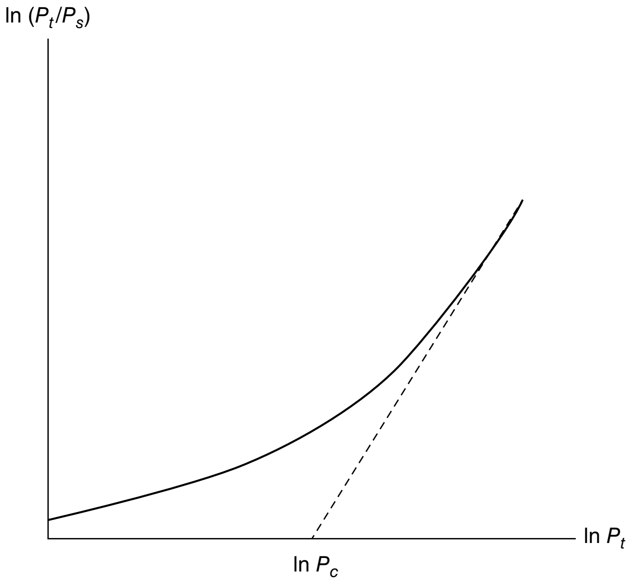

FIGURE 10.10 Mortality in terms of ln P_t.

from the extremes of "scramble competition" ($b \to +\infty$) to "contest competition" ($b \to 1$). In fact, the condition $+\infty > b > 1$ when $P_t > P_c$ depicts varying combinations of scramble and contest competition.

We next examine the equilibrium and stability properties of the Hassell model. Let us write the Hassell updating function as

$$P_{t+1} = F(P_t) = \frac{\lambda P_t}{(1 + aP_t)^b}$$

and set $P_{t+1} = P_t = \bar{P}$ so as to obtain

$$\bar{P} = \frac{\lambda \bar{P}}{\left(1 + a\bar{P}\right)^b}.$$

Clearly this equality holds if $\bar{P}_1 = 0$ (call it the trivial equilibrium) or if $\bar{P}_2 = a^{-1}\left(\lambda^{1/b} - 1\right)$. Here $\bar{P}_2$ is a nontrivial equilibrium point if $\lambda > 1$.

Looking to the stability of the Hassell model, let us first find

$$F' = \lambda \left[\frac{1 + aP(1-b)}{(1 + aP)^{b+1}} \right].$$

(Remember that for local stability at the fixed or equilibrium point $\bar{P}$, we require $|F'(\bar{P})| < 1$; $|F'(\bar{P})| > 1$ signals an unstable equilibrium at $\bar{P}$.) Then:

1. $|F'(0)| = \lambda < 1$ if $0 < \lambda < 1$. Thus, $\bar{P}_1 = 0$ is a locally stable equilibrium point if $0 < \lambda < 1$; it is unstable if $\lambda > 1$.
2. $\left|F'\left(\dfrac{\lambda^{1/b} - 1}{a}\right)\right| = \left|1 + b\left(\lambda^{-(1/b)} - 1\right)\right| < 1$ if $b\left(1 - \lambda^{-(1/b)}\right) < 2$. So if $b\left(1 - \lambda^{-(1/b)}\right) < 2$ and $\lambda > 1$, $\bar{P}_2 = a^{-1}\left(\lambda^{1/b} - 1\right)$ represents a (nontrivial) locally stable equilibrium point.

Let us modify the Hassell fitness function $P_{t+1}/P_t = R(P_t) = \lambda(1 + aP_t)^{-b}$ by including an Allee effect via a component of fitness that increases with population size. To this end, let us form a modified Hassell fitness function by multiplying $R(P_t)$ by the *Allee effect function*

$$G(P_t) = 1 - Ae^{-aP_t/\gamma}, \tag{10.28}$$

where A and γ are positive constants and $A \in [0,1]$ (Fowler and Ruxton, 2002). As required for an Allee effect, A and γ should be chosen so that $G(P_t)$ increases at a decreasing rate. Given Equation 10.28, the *adjusted Hassell fitness function*, which incorporates an Allee effect (along with the original intraspecific competitive effects), appears as

$$H(P_t) = G(P_t)R(P_t) = \frac{\lambda\left(1 - Ae^{-aP_t/\gamma}\right)}{\left(1 + aP_t\right)^b}. \tag{10.29}$$

What is the effect of the parameters A and γ on this adjusted Hassell equation? Specifically, Fowler and Ruxton (2002) observe:

1. In the limit as $A \to 0$, the Allee effect is negligible and thus Equation 10.29 reduces to $R(P_t)$.

   But as A increases in value, the magnitude of the Allee effect concomitantly increases. Additionally, for low levels of A, even though there is a nontrivial Allee effect that reduces per capita fitness as population size increases, this increasing population size will always be impacted more by competitive effects rather than by the Allee effect proper. Hence, per capita fitness always decreases with population size. In this instance maximum per capita fitness occurs when population size approaches zero. (This outcome was labeled a *component Allee effect* in Stephens et al. (1999).) However, if $A > \gamma b/(1 + \gamma b) = A^*$, then at low population levels, increasing population size increases per capita fitness since an increase in population size exerts a stronger influence in overcoming the Allee effect than on competition. As population size increases further, competitive effects dominate and per capita fitness decreases. In this circumstance, maximum per capita fitness occurs at a nonzero population level. (This outcome is termed a *demographic Allee effect* in Stephens et al. (1999).)

2. As γ increases, the range of population values over which the Allee effect has a significant influence also increases. By virtue of the condition $A > A^*$, a component Allee effect can always be translated into a demographic Allee effect by an appropriate increase in γ. But if $A < A^*$ (a low value of A leads to a component Allee effect), a sufficient increase in A will ultimately induce a demographic Allee effect; that is, there is always an assortment of A's for which $A > A^*$ no matter what values the other parameters assume.

A couple of final comments concerning Equation 10.29 are in order. First, the inclusion of an Allee effect in the basic Hassell fitness function never admits a stable equilibrium point when any such point would not exist in the absence of the Allee effect modification. When any nontrivial equilibrium population levels are obtained for Equation 10.29, they always occur at a lower population level than any equilibrium determined from the original Hassell fitness function. Second, the Allee effect has a stabilizing influence on the Hassell model; in fact, an increase in stability with increased competition occurs due to the interaction between competitive and Allee effects (Fowler and Ruxton, 2002).

10.9 GENERALIZED BEVERTON–HOLT (B–H) MODEL

The basic B–H equation (Eq. 10.19) is readily transformed to the *generalized B–H generating function* by modifying the former to read (Getz, 1996; Maynard-Smith and Slatkin, 1973; Schoombie and Getz, 1998)

$$P_{t+1} = F(P_t) = \frac{\lambda P_t}{1+(aP_t)^b}, \qquad b > 1, \quad t = 0, 1, 2, \ldots. \tag{10.30a}$$

To explore the properties of this expression, let us work with the *generalized B–H fitness function*

$$R(P) = \frac{\lambda}{1+(aP)^b}, \quad b > 1, \tag{10.30b}$$

where λ is the rate of increase in the absence of competition, a is a scaling parameter (note that (Eqs. 10.30a and 10.30b) are often written with K^{-1} substituted for a, where K is carrying capacity), and b describes the intensity of intraspecific competition. In this regard, b is termed an *abruptness parameter* in that it controls how rapidly density dependence sets in. So in the absence of competition, $b = 0$; under contest competition, $b \approx 1$, and under scramble competition, $b > 1$.

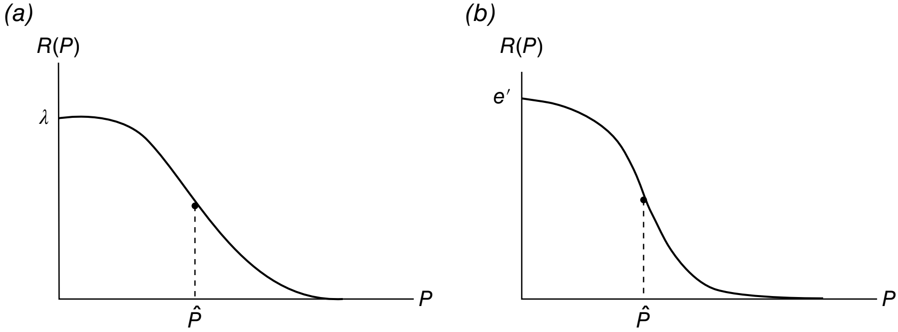

FIGURE 10.11 (**a**) Generalized B–H fitness function. (**b**) Generalized Ricker fitness function.

Looking to the properties of Equation 10.30b:

1. $R(0)=\lambda$.
2. $R'(P)=-\lambda b a^b P^{b-1}[1+(aP)^b]^{-2}<0$ and $R'(0)=0$ (the constraint $b>1$ ensures that $R'(0)=0$).
3. $R''(\hat{P})=0$ if $(b-1)-2ba^b\hat{P}^b\left[1+(a\hat{P})^b\right]^{-1}=0$ or $\hat{P}=\dfrac{1}{a}\left(\dfrac{b-1}{1+b}\right)^{1/b}$, $b>1$.

Provided $R'''\left(\hat{P}\right)\neq 0$, $\hat{P}$ corresponds to a point of inflection of R. Hence, the generalized B–H fitness function has a slope that is not a minimum at $P=0$ and has an inverted sigmoid shape (Fig. 10.11a).

We next consider the equilibrium and stability properties of the generalized B–H model. Given the updating function (Eq. 10.30a), let us set $P_{t+1}=P_t=\overline{P}$ so as to obtain

$$\overline{P}=\frac{\lambda\overline{P}}{1+\left(a\overline{P}\right)^b},\quad b>1.$$

This expression is satisfied if $\overline{P}_1=0$ or if $\overline{P}_2=\left[(\lambda-1)a^{-b}\right]^{1/b}$. Hence, $\overline{P}_2$ is a nontrivial equilibrium point if $\lambda>1$.

Turning to the stability of the generalized B–H model, we first find

$$F'=\lambda\left[1+\left(aP^b\right)\right]^{-1}\left\{1-b(aP)^b\left[1-(aP)^b\right]^{-1}\right\}.$$

Then:

1. $|F'(0)|=\lambda$. Hence, $\overline{P}_1=0$ is an unstable equilibrium point since we require that $\lambda>1$.
2. $|F'(((\lambda-1)a^{-b})^{1/b})|=|1-((\lambda-1)b/\lambda)|<1$ if $0<(\lambda-1)b/\lambda<2$. So if $\lambda>1$ *and* $0<(\lambda-1)b/\lambda<2$, then $\overline{P}_2=\left[(\lambda-1)a^{-b}\right]^{1/b}$ is a locally stable equilibrium point.

10.10 GENERALIZED RICKER MODEL

The (Eq. 10.25a) version of the Ricker equation may be transformed to the *generalized Ricker generating function* by rewriting the former as (Bellows, 1981; Berryman, 1999; Royama, 1992)

$$P_{t+1} = F(P_t) = P_t e^{r\left[1-(P_t/K)^{\theta}\right]}, \qquad \theta > 1, \quad t = 0, 1, 2, \ldots \tag{10.31a}$$

The associated *generalized Ricker fitness function* has the form

$$R(P) = e^{r\left[1-(P/K)^{\theta}\right]}, \quad \theta > 1. \tag{10.31b}$$

As was the case for the basic Ricker equation (Eq. 10.25a), r is the intrinsic growth rate and K denotes carrying capacity. However, a new parameter θ is introduced to govern the severity of density dependence; that is, increasing values of θ reflect continuous movement from a state of contest competition to one of (severe) scramble competition.

As to the properties of Equation 10.31b:

1. $R(0) = e^r$.
2. $R'(P) = -e^r r\theta K^{-\theta}\left(P^{\theta-1}e^{-r(P/K)^{\theta}}\right) < 0$ and $R'(0) = 0$.
3. $R''(\hat{P}) = 0$ if $\left[(\theta-1) - r\theta K^{-\theta}\hat{P}^{\theta}\right] = 0$ or if $\hat{P} = K((\theta-1)/r\theta)^{1/\theta}$, $\theta > 1$.

Assuming $R'''(\hat{P}) \neq 0$, $\hat{P}$ represents a point of inflection of R. Thus, the generalized Ricker fitness function has a slope that is not a minimum at the origin and has an inverted sigmoid shape (Fig. 10.11b).

We now turn to the notions of equilibrium and stability in the context of the generalized Ricker model. For the updating function (Eq. 10.31a), set $P_{t+1} = P_t = \bar{P}$ or

$$\bar{P} = \bar{P}e^{r\left[1-(\bar{P}/K)^{\theta}\right]}, \quad \theta > 1.$$

Clearly this equation is satisfied if $\bar{P}_1 = 0$ or if $\bar{P}_2 = K$. We next test the stability of these equilibrium points by first determining, from

$$F(P) = Pe^{r\left[1-(P/K)^{\theta}\right]},$$
$$F'(P) = e^{r\left[1-(P/K)^{\theta}\right]}\left[1 - r\theta K^{-\theta}P^{\theta}\right].$$

Then:

1. $|F'(0)| = e^r > 1$ if $r > 0$. Hence, the requirement that $r > 0$ implies that $\bar{P}_1$ depicts an unstable equilibrium point.
2. $|F'(K)| = |1 - r\theta| < 1$ or $0 < r\theta < 2$. So if $0 < r\theta < 2$, $\bar{P}_2$ is a locally stable equilibrium point.

TABLE 10.1 Growth of *E. coli*

Time (t) (hours)	Number (pt) (millions)
0	0.176
0.5	0.280
1	0.608
2	3.87
3	28.2
4	74.2
5	127
6	150
7	149
8	154

EXAMPLE 10.1 Table 10.1 displays the results of a laboratory experiment for the (population) growth of the bacterium *Escherichia coli* (McKendrick and Kesara Pai, 1911; cf. Hutchinson, 1978, pp. 21, 23, Figure 8).

Given this data set, let us estimate, via PROC NLIN, the parameters of the following growth functions:

B–H: $P_{t+1} = \alpha P_t/(1+\beta P_t)$.
Hassell: $P_{t+1} = \lambda P_t/(1+aP_t)^b$.
Ricker: $P_{t+1} = aP_t e^{-bP_t}$.
Generalized B–H: $P_{t+1} = \lambda P_t/(1+(aP_t)^b)$.

The requisite SAS code is provided in Exhibit 10.1. ■

EXHIBIT 10.1 SAS Code for Growth Curve Estimation

```
data bact;
  input pt1 pt;
datalines;
0.280    0.176
0.608    0.280
3.87     0.608
28.2     3.87
74.2     28.2
127      74.2
150      127
149      150
154      149
;
proc nlin data=bact hougaard maxiter=1000 noitprint;
   parms alpha=2.5
         beta=0.05;
```

① Model statement for the B–H growth curve
② Model statement for the Hassell growth curve
③ Model statement for the Ricker growth curve
④ Model statement for the generalized B–H growth curve

```
    model pt1=alpha*pt/(1+beta*pt); ①
run;
proc nlin data=bact hougaard maxiter=1000 noitprint;
    parms lambda=3.0
          a=0.05
          b=1.0;
    model pt1=lambda*pt/(1+a*pt)**b; ②
run;
proc nlin data=bact hougaard maxiter=1000 noitprint;
    parms a=2.0
          b=1.0;
    model pt1=a*pt*exp(-b*pt);③
run;
proc nlin data=bact hougaard maxiter=1000 noitprint;
    parms lambda=3.0
          a=1.0
          b=0.5;
    model pt1=lambda*pt/(1+(a*pt)**b); ④
run;
```

The output generated by this SAS code is given in Table 10.2.

Ⓐ Output summary for B–H growth curve estimation:

① The significant F Statistic (the associated p-value is very low) indicates that the overall fit is quite good.

② The parameter estimates yield the B–H growth equation:

$$P_{t+1} = \frac{4.8248 P_t}{1 + 0.0249 P_t}.$$

③ None of the Wald approximate (asymptotic) 95% confidence intervals contain zero. Hence, both parameters alpha and beta are significantly different from zero.

④ (a) With $|g_{i1}| = 0.4128$, $\widehat{\text{alpha}}$ is moderately skewed away from linear behavior.

(b) With $|g_{i2}| = 0.4415$, $\widehat{\text{beta}}$ is moderately skewed away from linear behavior.

Ⓑ Output summary for Hassell growth curve estimation:

① The overall fit is very good since the p-value associated with the F Statistic is low.

② The parameter estimates render a Hassell growth curve of the form

$$P_{t+1} = \frac{8.0137 P_t}{\left(1 + 0.1168 P_t\right)^{0.7029}}.$$

TABLE 10.2 Output for Example 10.1

The SAS system

Ⓐ

The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	7
Subiterations	6
Average subiterations	0.857143
R	3.857E-6
PPC(*b*)	1.621E-6
RPC(*b*)	0.000025
Object	3.852E-9
Objective	185.4287
Observations read	9
Observations used	9
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① *F* value	Approx Pr > *F*
Model	2	90676.9	45338.4	1711.54	<0.0001
Error	7	185.4	26.4898		
Uncorrected total	9	90862.3			

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
a	4.8248	0.5647	3.4895	6.1600	0.4128
b	0.0249	0.00396	0.0156	0.0343	0.4415

Approximate correlation matrix

	a	*b*
a	1.0000000	0.9892784
b	0.9892784	1.0000000

Ⓑ

The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	15
Subiterations	2
Average subiterations	0.133333
R	5.471E-6

(*Continued*)

TABLE 10.2 Output for Example 10.1 (*Continued*)

Method	Gauss–Newton
PPC(*a*)	0.000011
RPC(*a*)	0.000025
Object	2.15E-10
Objective	157.0426
Observations read	9
Observations used	9
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① *F* value	Approx Pr > *F*
Model	3	90705.3	30235.0	1155.17	<0.0001
Error	6	157.0	26.1738		
Uncorrected total	9	90862.3			

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
Lambda	8.0137	2.7163	1.3672	14.6603	1.4181
a	0.1168	0.0980	−0.1231	0.3567	2.6289
b	0.7029	0.0907	0.4809	0.9249	1.3232

Approximate correlation matrix

	Lambda	*a*	*b*
Lambda	1.0000000	0.9537727	−0.7430925
a	0.9537727	1.0000000	−0.9087615
b	−0.7430925	−0.9087615	1.0000000

©

The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	10
Subiterations	12
Average subiterations	1.2
R	6.356E-7
PPC(*b*)	1.371E-7
RPC(*b*)	4.893E-6
Object	5.29E-10
Objective	326.4178
Observations read	9
Observations used	9
Observations missing	0

Note: An intercept was not specified for this model.

TABLE 10.2 **Output for Example 10.1 (*Continued*)**

Source	DF	Sum of squares	Mean square	① F value	Approx Pr > F
Model	2	90535.9	45267.9	970.77	<0.0001
Error	7	326.4	46.6311		
Uncorrected total	9	90862.3			

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
a	3.1310	0.2480	2.5447	3.7174	0.1080
b	0.00761	0.000621	0.00614	0.00908	−0.0844

Approximate correlation matrix

	a	b
a	1.0000000	0.9580601
b	0.9580601	1.0000000

Ⓓ

The NLIN procedure

Note: Convergence criterion met.

Estimation summary

Method	Gauss–Newton
Iterations	14
Subiterations	12
Average subiterations	0.857143
R	7.558E-6
PPC(a)	0.000017
RPC(a)	0.000061
Object	9.81E-10
Objective	144.3601
Observations read	9
Observations used	9
Observations missing	0

Note: An intercept was not specified for this model.

Source	DF	Sum of squares	Mean square	① F value	Approx Pr > F
Model	3	90717.9	30239.3	1256.83	<0.0001
Error	6	144.4	24.0600		
Uncorrected total	9	90862.3			

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
Lambda	8.7739	3.6594	−0.1802	17.7280	1.8548

(*Continued*)

TABLE 10.2 Output for Example 10.1 (*Continued*)

Parameter	② Estimate	Approx std error	③ Approximate confidence	95% limits	④ Skewness
a	0.0905	0.0828	−0.1121	0.2930	3.4870
b	0.7742	0.1005	0.5282	1.0201	0.7448

Approximate correlation matrix

	Lambda	*a*	*b*
Lambda	1.0000000	0.9870915	−0.8842511
a	0.9870915	1.0000000	−0.9465643
b	−0.8842511	−0.9465643	1.0000000

③ The Wald approximate 95% confidence intervals for lambda and *b* do not contain zero, thus indicating that the estimates of these parameters are statistically significant. Since the 95% confidence interval for *a* contains zero, the estimate of this parameter is not statistically significant.

④ (a) With $|g_{i1}| = 1.4181$, $\widehat{lambda}$ is highly skewed away from linear behavior.

(b) With $|g_{i2}| = 2.6289$, $\hat{a}$ is highly skewed away from linear behavior.

(c) With $|g_{i3}| = 1.3232$, $\hat{b}$ is highly skewed away from linear behavior.

Ⓒ Output summary for Ricker growth curve estimation:

① The low *p*-value for the *F* Statistic indicates that the overall fit is good.

② The estimated Ricker growth equation is

$$P_{t+1} = 3.1310 P_t e^{-0.00761 P_t}.$$

③ None of the Wald approximate 95% confidence intervals contain zero. Hence, both the *a* and *b* parameter estimates differ significantly from zero.

④ (a) With $|g_{i1}| = 0.1080$, $\hat{a}$ is close to linear in character.

(b) With $|g_{i2}| = 0.0844$, $\hat{b}$ is also close to linear in character.

Ⓓ Output summary for generalized B–H growth curve estimation:

① The overall fit is good as evidenced by the low *p*-value for the *F* Statistic.

② The estimated generalized B–H growth curve is

$$P_{t+1} = \frac{8.7739 P_t}{1 + \left(0.0905 P_t\right)^{0.7742}}.$$

③ Since the Wald 95% confidence intervals for lambda and *a* contain zero, we can conclude that $\widehat{lambda}$ and $\hat{a}$ are not statistically significant. Only the estimate for *b* is significant since the Wald 95% confidence interval for this parameter does not admit zero.

④ (a) With $|g_{i1}| = 1.8548$, $\widehat{lambda}$ is highly skewed away from linear behavior.

(b) With $|g_{i2}| = 3.4870$, $\hat{a}$ is highly skewed away from linear behavior.

(c) With $|g_{i3}| = 0.7448$, $\hat{b}$ is moderately skewed away from linear behavior. ■

APPENDIX 10.A A GLOSSARY OF SELECTED POPULATION DEMOGRAPHY/ECOLOGY TERMS

Absolute population—total number of individuals or organisms in a population present at a given time in a given area.

Allee effect (W. E. Allee (1931))—the effect that incidental cooperation between organisms has on the per capita rate of change, causing it to rise with population density because either births increase or deaths decrease as population density rises.

Carrying capacity—maximum number or density of organisms that can be supported in a given area or environment in perpetuity (see Resource base).

Closed population—a population that does not experience immigration or emigration over a time period. The only processes causing population change are reproduction and mortality.

Compensation—per capita recruitment declines with increasing size of the S.

Contest competition—competition involving contests between two organisms (e.g., territorial behavior).

Density—the value of a variable expressed in terms of a unit of space (number per meter^2); population size per unit area.

Density dependence—a negative relationship between the reproduction and survival of individuals and the density of the population.

Depensation—per capita recruitment decreases when the size of the S falls below certain levels.

Emigration—the movement of individuals out of a predefined area or location.

Endogenous—a cause (force) that is internal to a dynamic system; it prompts changes in that system and, in turn, is affected by those changes.

Environment—all the elements present in an area occupied by a population, including, but not limited to, food, space, climate, topography, and other organisms of different species.

Escapement—R of generation k, which are not harvested from the parent stock of generation $k+1$.

Exogenous—a cause (force) that is external to a dynamic system or level of a species and is independent of the system level or species numbers; it causes changes in a system without being affected itself by those changes (no feedback).

Fecundity—capacity for producing offspring.

Harvesting—the removal of a specified amount of individuals (e.g., fish, cattle) from the population during each time period.

Immigration—the movement of individuals into a predetermined area or location.

Incidental—describes interactions between organisms (competitive or cooperative) that occur accidentally and do not involve adapted (evolved) behavior.

Individual—a particular entity (organism) or member of a population that is differentiated by characteristics such as age, sex, developmental stage, and physiological characteristics.

Interspecific—between species; described interactions (competition or cooperation) between different species.

Intraspecific—within a species; interactions (competition or cooperation) between members of the same species.

Intrinsic—property of a species determined by its innate, genetic characteristics.

Limiting factor—an exogenous factor that prevents a variable such as population density from attaining a higher value.

Net birth rate—the rate at which a population changes as a function of the proportion of the population that will mate, the number of offspring per mating pair, the proportion of the population that will die during the next time period, and immigration and emigration.

Net reproduction rate—in the absence of migration, a net reproduction rate greater than (respectively less than) one indicates that the population of women is increasing (respectively deceasing).

Nonreactive—used to describe a variable that does not change in response to changes in another variable.

Open population—a population that admits immigration or emigration over a time period.

Overpopulation—population exceeding the carrying capacity of an area or environment.

Per capita—per individual.

Periodic—describes a phenomenon that repeats itself at regular time intervals.

Population—a group of individuals or organisms of the same species that live together in the same place (an area of sufficient size so that they can carry out their normal functions) and have a high probability of interacting with each other.

Population density—absolute population divided by the area of land within which the population resides to obtain the population density per unit area; relates population numbers to a standard area.

Realized per capita rate of change—rate of change realized by the average individual living in a particular environment and in a population of a given density.

Recruits—the offspring from the parent stock of generation k, which survive to a certain size or age and are either harvested or become the parent stock of generation $k + 1$.

Resource base—the availability of, say, land and food that limits the size of a population (see Carrying capacity).

Saturation—describes a situation where response no longer occurs.

Scramble competition—competition between two or more organisms that is not mediated by social interactions; everyone "scrambles" for what they can get.

Underpopulation—population not large enough to maintain an ecological system.

APPENDIX 10.B EQUILIBRIUM AND STABILITY ANALYSIS

10.B.1 Stable and Unstable Equilibria

We may define an *equilibrium point* as a position from which, once attained, there will be no net tendency to depart. An equilibrium point is fundamentally a point where a system is "at rest" or is "in balance."

Equilibrium points may be stable or unstable. If point P^* is *a stable equilibrium*, then, given any small deviation from that point, forces are set in motion, which tend to correct the disturbance and return the system back to (or near) P^*. Additionally, an equilibrium point P^* is termed *locally stable* if convergence towards P^* occurs only for those initial conditions within a suitably restricted neighborhood of P^*; and P^* is *globally stable* if convergence towards P^* occurs for all possible initial conditions. Finally, if P^* is an *unstable equilibrium* point, then any small deviation away from that point calls forth further deviations—the system essentially deviates forever.

For example, think of a marble at rest in the bottom of a round-bottomed bowl (Fig. 10.B.1a). If the bowl is jarred slightly, the marble rolls away from point A (its equilibrium position) but eventually comes to rest again at A. Hence, A is a stable equilibrium point. However, if at the equilibrium point B in Figure 10.B.1b the inverted bowl is jarred somewhat, the marble rolls away from B and clearly cannot return to that position—point B is thus an unstable equilibrium point. It should be intuitively clear that the biological significance of a model depends upon the stability of a solution; that is, if a small perturbation in the solution precipitates a large change in the same, then the meaningfulness of the solution is in doubt.

A stable equilibrium point is often characterized as an *attractor*. For a variable with an attractor, its long-term behavior is independent of its initial state, and its terminal state is a single value. While not all systems have an attractor, the most basic attractor is a fixed equilibrium value to which the system converges (see Fig. 10.B.2a). In addition, an unstable equilibrium point is sometimes called a *repellor* (Fig. 10.B.2b).

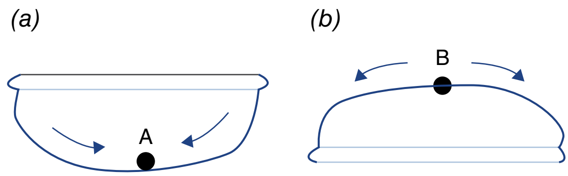

FIGURE 10.B.1 (**a**) A stable equilibrium point at A. (**b**) An unstable equilibrium point at B.

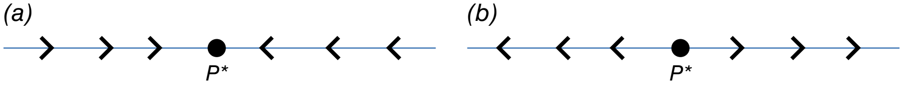

FIGURE 10.B.2 (**a**) Equilibrium P^* is an attractor. (**b**) Equilibrium P^* is a repellor.

10.B.2 The Need for a Qualitative Analysis of Equilibria

In the main body of Chapter 10, we examined a variety of continuous-time and discrete-time single-species growth models that were either density independent or density dependent. Such models were generally of either the continuous form $dP/dt = r(P)P = f(P)$ (Eq. 10.2) or discrete form $P_{t+1} = R(P_t)P_t = F(P_t)$ (Eq 10.3). If the solution to either of these equations has an exact or explicit analytical form, then we may be able to use that expression to predict the behavior of the population into the long run. However, the exact analytical form may be difficult to determine or even intractable. Under this circumstance, we can still discern the long-run trend in the population by considering "qualitative" answers to the population growth problem. That is, if we have information about the functions $r(P)$ and $R(P_t)$ (and thus about $f(P)$ and $F(P_t)$), then we may be able to conduct a qualitative analysis of a models stability and thus its long-run performance. In fact, qualitative predictions are generally *robust* in that they hold even if changes in the model are made; that is, robust properties hold for a relatively large class of models and tend to display biological significance.

The steps necessary for performing a *qualitative analysis* are:

1. Determine values of the population value or density P^*, which are equilibria (more on this step later).
2. Determine the behavior of the solution(s) near the equilibrium point(s).

10.B.3 Equilibria and Stability for Continuous-Time Models

In Section 10.B.1 we discussed the notions of equilibrium and stability in fairly general terms. Now, let us assume that we have an autonomous (t is not an argument of f) first-order differential equation of the form

$$\frac{dP}{dt} = f(P). \qquad (10.B.1)$$

Let us define an *equilibrium point* of this expression as a value P^* such that $f(P^*) = 0$. (The requirement that P not change at P^* translates into the requirement that its rate of change at that point be zero or $f(P^*) = 0$.) Hence, an equilibrium corresponds to a constant solution $P(t) \equiv P^*$. And if $P(t)$ is a solution of Equation 10.B.1 that approaches a limit P^* as $t \to +\infty$, then P^* must be an equilibrium. In fact, every solution of Equation 10.B.1 must either tend to an equilibrium as $t \to +\infty$ or be unbounded.

An equilibrium point P^* is *locally stable* if $P(t)$ is near P^* whenever the initial value $P(0)$ is sufficiently near P^*. Thus, the stability of a solution means that a small change in $P(0)$ produces only a small effect on the behavior of the solution as $t \to +\infty$. Additionally, an equilibrium point P^* is *locally asymptotically stable* if it is stable and if $P(0)$ sufficiently near P^* implies $\lim_{t\to+\infty} P(t) = P^*$; that is, every solution with $P(0)$ close enough to P^* remains so and tends to P^* as $t \to +\infty$. Thus, an asymptotically stable equilibrium is fairly insensitive to small perturbations of Equation 10.B.1. Finally, the equilibrium point P^* is *globally asymptotically stable* if for every $P(0)$, $\lim_{t\to+\infty} P(t) = P^*$.

To discuss the stability of the equilibrium solution P^*, we need to consider the behavior of the system near P^*, that is, for suitably restricted deviations in P away from P^*. Hence, our systematic approach to accomplish this task is as follows. Consider a small deviation $h(t)$ from P^*, where $h(t) \equiv P(t) - P^*$, $|h(t)|$ small. How does $h(t)$ change with time? Let us approximate (linearize) f near P^* via Taylor's formula or

$$f(P(t)) = f(P^* + h(t)) \approx f(P^*) + f'(P^*)h(t). \tag{10.B.2a}$$

Since $P(t) = P^* + h(t)$ and P^* is constant, $dP/dt = dh/dt = f(P)$ near P^*. And since P^* is an equilibrium point, $f(P^*) = 0$. Hence, Equation 10.B.2a can be rewritten as

$$\frac{dh}{dt} = f'(P^*)h(t), \quad |h(t)| \text{ small}, \tag{10.B.2b}$$

or

$$\frac{dh}{h} = f'(P^*)dt. \tag{10.B.3}$$

The solution to this differential equation is

$$h(t) = h(0)e^{f'(P^*)t}, \quad h(0) \text{ arbitrary}.$$

Hence, $h(t)$ experiences exponential growth or decline as $f'(P^*)$ is positive or negative respectively. That is:

1. The equilibrium point P^* is locally stable if $f'(P^*) < 0$ (small deviations $h(t)$ from P^* are self-correcting).
2. The equilibrium point P^* is unstable if $f'(P^*) > 0$ (small deviations $h(t)$ from P^* call forth further deviations and we move away from P^*).

(Note: $f'(P^*) = 0$ is the undetermined case.) As Fig. 10.B.3 reveals, P_1 is a stable equilibrium point, whereas P_2 is an unstable equilibrium point.

To summarize, given Equation 10.B.1, equilibrium solutions P^* are solutions of $dP/dt = f(P) = 0$ and are locally stable for small deviations h if $f'(P^*) < 0$ and unstable if $f'(P^*) > 0$. A graphical plot of $f(P)$ versus P reveals that equilibria are points where the function $dP/dt = f(P)$ crosses the P-axis (Fig. 10.B.3). The sign

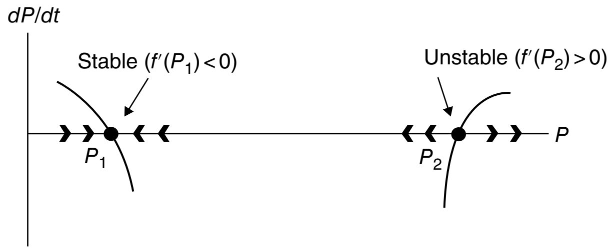

FIGURE 10.B.3 Stable and unstable equilibrium points.

of the slope of $f(P)$ at each equilibrium point then determines its local stability. Thus, for a continuous-time single-species model having a single differential equation, the range of possible dynamic behavior is restricted to exponential growth or decay.

EXAMPLE 10.B.1 Suppose $dP/dt = f(P) = 6P - P^2$, $P \geq 0$. Does f possess a stable equilibrium point? To find any such point, set $f(P) = 0$ or $6P - P^2 = 0$. From this expression we easily obtain $P = 0$ and $P = 6$ as our equilibrium points. Since $f'(P) = 6 - 2P$, we see that $f'(0) = 6 > 0$, while $f'(6) = -6 < 0$. Thus, 6 is a stable equilibrium point and 0 is an unstable equilibrium point.

Let us go a step further and find

$$f'(P^* + h) = f'(0 + h) = 6 - 2h.$$

For h small and positive (values of h below $P^* = 0$ are not allowed), $f'(0 + h) > 0$. As concluded earlier, 0 is an unstable equilibrium point. Next,

$$f'\left(P^* + h\right) = f'\left(6 + h\right) = -6 - 2h.$$

For h small and either positive or negative, $f'(6 + h)$ is always negative, and thus, as also evidenced earlier, 6 is a stable equilibrium point. ■

10.B.4 Equilibria and Stability for Discrete-Time Models

We may generally define an equilibrium or fixed point of a function F in the following fashion. Let F be a continuous function mapping a set $\mathfrak{K}$ into itself. A *fixed* (or *equilibrium*) *point* of F is a point $\bar{P} \varepsilon \mathfrak{K}$ satisfying $F(\bar{P}) = \bar{P}$. Thus, $\bar{P}$ is transformed into itself by the mapping F (Fig. 10.B.4). Suppose now that model (Eq. 10.3) is in play or $P_{t+1} = R(P_t)P_t = F(P_t)$. Then $\bar{P}$ is a fixed point of $P_{t+1} = F(P_t)$ if $F(\bar{P}) = \bar{P}$. The corresponding solution $\{P_t\}$ such that $P_t = \bar{P}$, $t = 0, 1, 2, \ldots$, is termed a constant or *steady-state solution*.

Let $\bar{P}$ be a fixed point determined by solving the equation $F(\bar{P}) = \bar{P}$. Suppose P_0 is an initial point near $\bar{P}$ and the values of $P_t \to \bar{P}$ as t increases. Then $\bar{P}$ is termed an *attracting fixed point* of F. If the values of P_t move away from $\bar{P}$ as t increases, then $\bar{P}$ is called a *repelling fixed point* of F. (Note: not every fixed or equilibrium

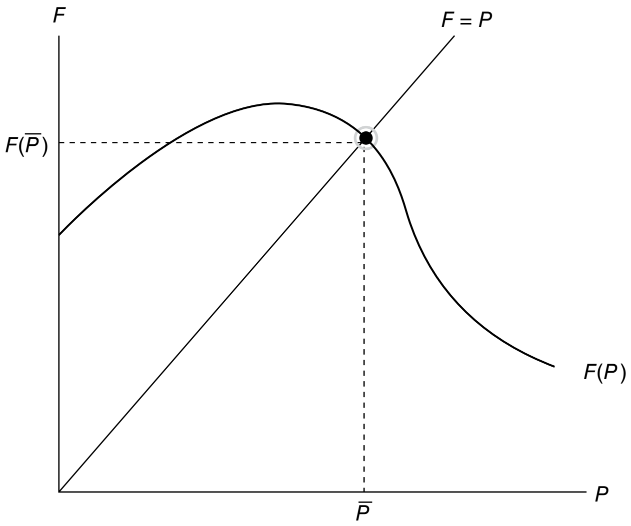

FIGURE 10.B.4 $\overline{P}$ is a fixed point of $F(P)$.

point can be categorized as either attracting or repelling.) The *orbit* of the initial point P_0 is the sequence of points $\{P_t\},\ t=0,1,2,\ldots.$

An equilibrium point $\overline{P}$ is *locally stable* if $P_t(t\geq 0)$ is near $\overline{P}$ whenever the initial value P_0 is sufficiently near $\overline{P}$. Otherwise $\overline{P}$ is unstable. In addition, an equilibrium point $\overline{P}$ is *locally asymptotically stable* if it is stable and if P_0 sufficiently near $\overline{P}$ implies $\lim_{t\to+\infty} P_t = \overline{P}$. Finally, an equilibrium point $\overline{P}$ is *globally asymptotically stable* if for every P_0, $\lim_{t\to+\infty} P_t = \overline{P}$.

Looking to the stability of the equilibrium solution $\overline{P}$, let us consider the behavior of the system near $\overline{P}$ or for suitably restricted deviations in P from $\overline{P}$. To accomplish this, let us consider a solution $P_t = \overline{P} + h_t$, $|h_t|$ small, where h_t is a perturbation of $\overline{P}$. From $P_{t+1}=F(P_t)$ and $P_t = \overline{P} + h_t$, h_t must satisfy

$$h_{t+1} = P_{t+1} - \overline{P} = F\left(P_t\right) - \overline{P} = F\left(\overline{P} + h_t\right) - \overline{P}. \tag{10.B.4}$$

Let us consider a first-order (Taylor's) approximation of F (assumed continuously differentiable) near $\overline{P}$ or

$$F\left(P_t\right) = F\left(\overline{P} + h_t\right) \approx F\left(\overline{P}\right) + F'\left(\overline{P}\right)h_t. \tag{10.B.5}$$

Since $F\left(\overline{P}\right) = \overline{P}$, Equation 10.B.4 implies that

$$h_{t=1} = F'\left(\overline{P}\right)h_t = \lambda h_t, \tag{10.B.6}$$

with $\lambda = F'\left(\overline{P}\right)$. Clearly this first-order difference equation describes what happens when we are close to the steady-state or fixed-point solution $\overline{P}$. So, will small

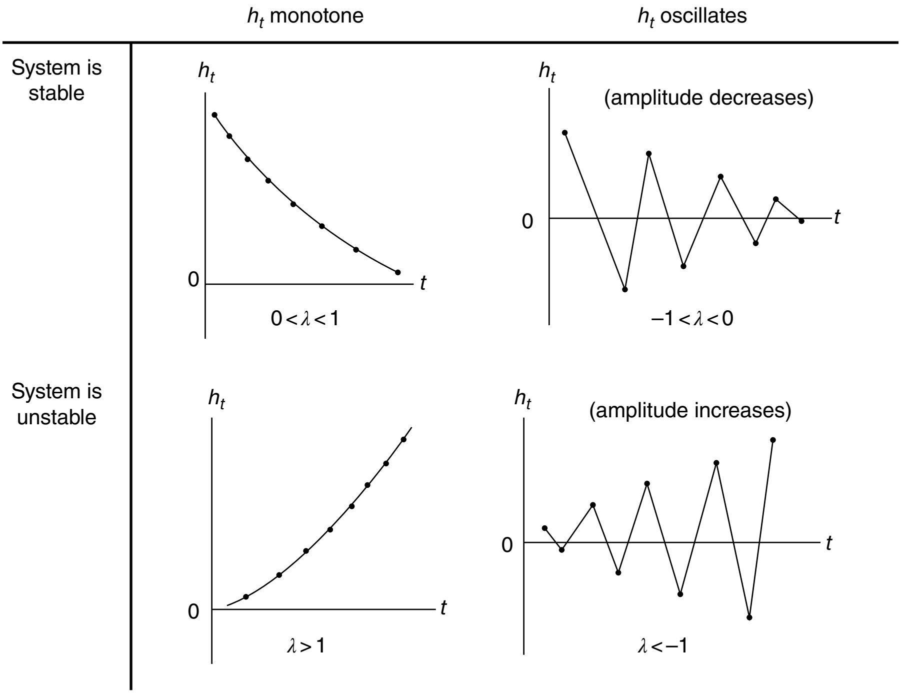

FIGURE 10.B.5 Temporal behavior of $h_t = h_0\lambda^t$.

deviations h_t from the steady state result in an increase or decrease in P_t? Since the solution to Equation 10.B.6 is

$$h_t = h_0\lambda^t, \tag{10.B.7}$$

we see that exponential growth (decay) occurs if $|\lambda| > 1$ ($|\lambda| < 1$). Hence, $\bar{P}$ is an attracting fixed point if $0<\lambda<1$ or $-1<\lambda<0$ (we expect $h_t \to 0$ as $t \to +\infty$); it is repelling if $\lambda>1$ or $\lambda<-1$ (so that $h_t \to +\infty$ as $t \to +\infty$). Additionally, if $\lambda > 0$, then h_t maintains the same sign and thus is termed *monotonic*; but if $\lambda < 0$, the values of h_t alternate in sign or are *oscillatory*. These special cases appear in Figure 10.B.5.

Let us examine the behavior of Equation 10.B.7 explicitly in terms of the derivatives of F:

1. The equilibrium point $\bar{P}$ is locally stable if $\left|F'(\bar{P})\right| < 1$ (small deviations h_t from $\bar{P}$ are self-correcting).
2. The equilibrium point $\bar{P}$ is unstable if $\left|F'(\bar{P})\right| > 1$ (small deviations h_t from $\bar{P}$ call forth further deviations and we move away from $\bar{P}$).
3. The equilibrium point $\bar{P}$ is *neutral* if $\left|F'(\bar{P})\right| = 1$. However, if $F'(\bar{P}) = 1$, $F''(\bar{P}) = 0$, and $F'''(\bar{P}) < 0$, then $\bar{P}$ is locally stable; but if

$F'(\bar{P}) = 1$, $F''(\bar{P}) = 0$, and $F'''(\bar{P}) > 0$, then $\bar{P}$ is unstable. If $F'(\bar{P}) = -1$, periodic solutions can obtain. In this regard,

4. A *period k point P* is a point satisfying $F^k(P) = P$ but $F^j(P) \neq P$, $0 < j < k$. The orbit of a period k point is known as a k cycle.

It is evident from the foregoing discussion that oscillations are a possibility for discrete-time univariate models. (Continuous-time models in a single argument never exhibit oscillations.) What causes the oscillations is overcompensation due to the delay between the successive discrete-time steps—the system jumps over the equilibrium or fixed point and then back again because of a strong negative feedback, and so on, and thus approaches the (stable) equilibrium in a seesaw fashion. If the discrete-time steps were taken to be much smaller, the equilibrium would be approached in a more gradual or monotonic fashion.

It is at this point in our discussion of the discrete-time model that we can establish a connection between the requirement for its stability and the stability condition of the continuous-time model. Let's start with the continuous-time model $dP/dt = r(P)P = f(P)$ and consider its discretization. We know that for a small finite time increment Δt, we can approximate dP/dt as

$$\frac{P(t+\Delta t) - P(t)}{\Delta t} = r(P)P.$$

Then

$$P(t+\Delta t) = P(t) + \Delta t\, r(P)P = (1 + \Delta t\, r(P))P$$

or, using discrete-time indexing,

$$P_{t+\Delta t} = R(P_t)P_t = F(P_t), \tag{10.B.8}$$

where $R(P_t) = 1 + \Delta t\ r(P)$ and $F(P_t) = (1 + \Delta t\ r(P))P_t = P_t + \Delta t f(P)$. We know from the preceding section that an equilibrium point $\bar{P}$ is locally stable if $f'(\bar{P}) < 0$. However, we have just determined that for the discrete-time model, an equilibrium or fixed point $\bar{P}$ is locally stable if $|F'(\bar{P})| < 1$ or, from Equation 10.B.8, $\bar{P}$ is locally stable if $|F'(\bar{P})| = |1 + \Delta t\, f'(P)| < 1$. Since this last inequality can be rewritten as $-2/\Delta t < f(\bar{P}) < 0$, it is evident that this is a more restrictive stability condition relative to the one for the continuous-time model.

We now turn to the development of a so-called *cobweb diagram*—the graphical representation of the long-term dynamic behavior of a population or the orbit of an initial point P_0 subject to a (general) reproduction function $P_{t+1} = F(P_t)$; that is, it is a device used to determine the graphical solution of a difference equation. To construct a cobweb diagram, we need to graph $P_{t+1} = F(P_t)$ and the (reference) line $P_{t+1} = P_t$. The P_t-coordinates of the intersection points between these two expressions are the fixed points of F since they are the solutions to the equation $F(P_t) = P_t$. A point on the graph of the reproduction function appears as (P_t, P_{t+1}) —its coordinates thus represent successive population values. A point on the graph of $P_{t+1} = P_t$ can be written as (P_t, P_t).

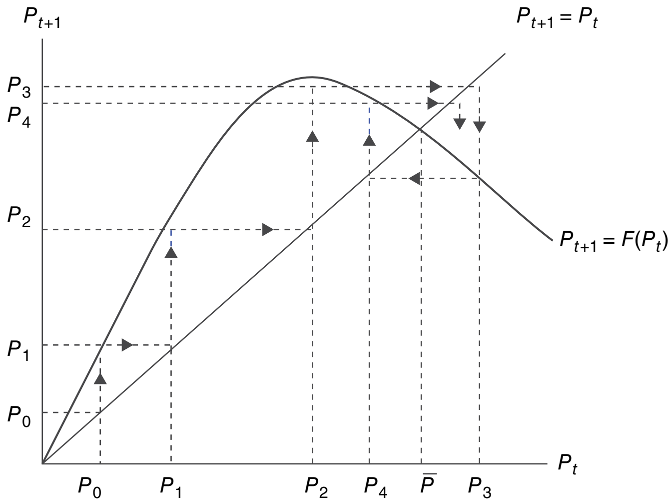

FIGURE 10.B.6 Convergence to a stable equilibrium point at $\bar{P}$.

Our focus here will be on the geometry of the intersection of the functions $P_{t+1}=F(P_t)$ and $P_{t+1}=P_t$ at a point $\bar{P}$ (i.e., $\bar{P}$ must satisfy $R\left(\bar{P}\right)\bar{P}=\bar{P}$ so that either $\bar{P}=0$ or $R\left(\bar{P}\right)=1$). The *cobweb path* is then a series of reflections in the reference line $P_{t+1}=P_t$. What sorts of outcomes can we expect? If P_0 is close enough to the fixed point $\bar{P}$, then the approximation to $\bar{P}$ will be monotonic as long as $P_{t+1}=F(P_t)$ crosses $P_{t+1}=P_t$ from above. If at the crossing we have $0<F'\left(\bar{P}\right)<1$, then $\bar{P}$ is a stable equilibrium point. And if $-1<F'\left(\bar{P}\right)<0$ (the reproduction curve $P_{t+1}=F(P_t)$ cannot be "too steep" at its intersection with the line $P_{t+1}=P_t$), we again get a stable equilibrium point. But if $\left|F'\left(\bar{P}\right)\right|>1$ (i.e., $F'\left(\bar{P}\right)>1$ or $F'\left(\bar{P}\right)<-1$), then, save for the constant solution $P_t=\bar{P}$ for $t=0,1,2,\ldots$, the intersection point $\bar{P}$ represents an unstable equilibrium.

A cobweb diagram is constructed by drawing a sequence of alternating vertical and horizontal line segments; that is, we graphically iterate the orbit of P_0 subject to $P_{t+1}=F(P_t)$. Our objective is to see if the said orbit converges to a stable fixed point $\bar{P}$ or admits an unstable fixed point. Our iterations begin at P_0 (on the horizontal axis) in Figure 10.B.6. Clearly convergence (in cobweb fashion) to the fixed point $\bar{P}$ occurs rather rapidly after the first few iterations. The time path of P_t is provided in Figure 10.B.7.

EXAMPLE 10.B.2 Suppose the reproduction function has the form $P_{t+1}=F(P_t)=aP_t/(P_t+b)$. Let us rewrite this expression as $F=aP/(P+b)$. To find any fixed points of F, let us set $F(P)=P$ or

$$\frac{aP}{P+b}=P.$$

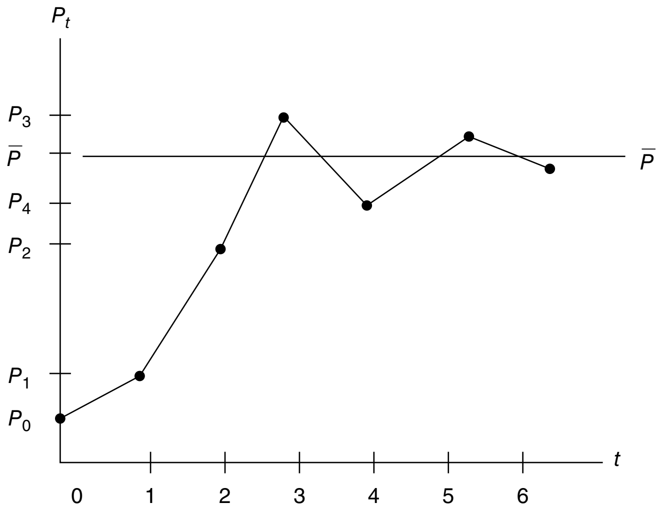

FIGURE 10.B.7 $P_t \to \bar{P}$ as $t \to +\infty$.

Equality holds for $P = 0$ and $P = a - b$. For a positive population level, we obviously require that $a > b$. Given

$$F' = a\left[(P+b)^{-1} - P(P+b)^{-2}\right],$$

it is easily demonstrated that $|F'(0)| = a/b > 1$ if $a > b$. Hence, $\bar{P} = 0$ is an unstable fixed or equilibrium point. And since $|F'(a-b)| = b/a < 1$ for $a > b$, we see that $\bar{P} = a - b$ is a locally stable fixed point.

Let $a = 3$ and $b = 1$ so that $F = 3P/(P + 1)$. For, say, $P_0 = 1$, we have

$$P_1 = \frac{3P_0}{P_0 + 1} = 1.5, \quad P_2 = \frac{3P_1}{P_1 + 1} = 1.8, \quad P_3 = \frac{3P_2}{P_2 + 1} = 1.93, \quad P_4 = \frac{3P_3}{P_3 + 1} = 1.98, \ldots.$$

Hence, the orbit of $P_0 = 1$ is $\{1, 1.5, 1.8, 1.93, 1.98, \ldots\}$. Clearly this orbit is converging to the stable fixed point $\bar{P} = 3 - 1 = 2$. ■

The developments in this appendix have addressed the issue of the impact of density dependence on model (population) stability. We have demonstrated that such dependence can lead to a stable equilibrium population level. However, it can also lead to patterns of cyclical fluctuations (as well as more irregular behavior). These patterns can arise quite readily in discrete-time difference equation models involving nonoverlapping generations. But they can also occur in continuous-time differential equation models with overlapping generations, provided that the lags or delays in the operation of the density-dependent processes are of sufficient duration. (Additional coverage of these issues is provided by May (1976).)

APPENDIX 10.C DISCRETIZATION OF THE CONTINUOUS-TIME LOGISTIC GROWTH EQUATION

Section 10.5.1.2 developed a discrete-time analogue to the continuous-time logistic growth equation (Eq. 10.14). This appendix offers additional discrete-time versions of the logistic model (Royama, 1992).

From $dP/dt = rP(1-(P/K))$, we obtain the differential equation

$$\frac{dP}{P(1-(P/K))} = r\,dt$$

with solution

$$-\ln\left(\frac{1-(P/K)}{P}\right) = rt + \ln C, \quad \ln C \text{ constant},$$

or

$$P_t = Ce^{rt}\left(1-\frac{P_t}{K}\right).$$

Upon forming

$$\frac{P_t}{P_{t-1}} = \frac{e^r\left(1-\left(P_t/K\right)\right)}{1-\left(P_{t-1}/K\right)}$$

and relying on some algebraic gymnastics, we ultimately obtain

$$P_t = \frac{e^r P_{t-1}}{1+\left(e^r-1\right)\left(P_{t-1}/K\right)} = G\left(P_{t-1}\right). \tag{10.C.1}$$

This expression will be termed the *B–H form of the discrete-time logistic function*. (See also Section 10.6 for an in-depth discussion of the B–H growth function.)

If we rewrite Equation 10.C.1 as

$$\frac{P_t}{P_{t-1}} = \frac{e^r}{1+\left(e^r-1\right)\left(P_{t-1}/K\right)},$$

then

$$\ln\frac{P_t}{P_{t-1}} = r - \ln\left[1+\left(e^r-1\right)\frac{P_{t-1}}{K}\right] = H\left(P_{t-1}\right) \tag{10.C.2}$$

is known as the *hyperbolic form of the discrete-time logistic function*.

Next, the logistic equation $d\ln P/dt = r(1-(P/K))$ can be directly transformed into the discrete equation

$$\frac{\ln P_t - \ln P_{t-1}}{\Delta t} = r\left(1 - \frac{P_{t-1}}{K}\right)$$

or (for $\Delta t = t-(t-1) = 1$)

$$\ln \frac{P_t}{P_{t-1}} = r\left(1 - \frac{P_{t-1}}{K}\right) = \bar{H}\left(P_{t-1}\right). \tag{10.C.3}$$

Then from Equation 10.C.3 we can easily obtain the *Ricker form of the discrete-time logistic function*:

$$P_t = P_{t-1} e^{r\left(1-\left(P_{t-1}/K\right)\right)} = \bar{G}\left(P_{t-1}\right). \tag{10.C.4}$$

To account for nonlinearities in many natural dynamic processes, let us introduce an additional parameter θ into Equations 10.C.2 and 10.C.3. To this end, Equation 10.C.2 becomes

$$\ln \frac{P_t}{P_{t-1}} = r - \ln\left[1 + \left(e^r - 1\right)\left(\frac{P_{t-1}}{K}\right)^{\theta}\right] \tag{10.C.5}$$

(Maynard-Smith and Slatkin, 1973); and (Eq. 10.C.3) becomes

$$\ln \frac{P_t}{P_{t+1}} = r\left[1 - \left(\frac{P_{t-1}}{K}\right)^{\theta}\right] \tag{10.C.6}$$

(Nelder, 1961; Richards, 1931). The role of the parameter θ is to alter the curvature of Equations 10.C.5 and 10.C.6—small θ's enhance the convexity of these functions, while large θ's result in greater concavity of the same.

One interpretation of θ (Royama, 1992) centers on the type of competition being experienced by members of a species; that is, θ determines the rate at which the effects of competition (contest or scramble competition) are influenced by population density. Thus, for $\theta > 1$, competition intensifies with increasing population density; and for $\theta < 1$, competition becomes less intense as population density increases. In this regard, Equation 10.C.5 is often thought of as a model of contest competition, while Equation 10.C.6 is considered to be a model of scramble competition.

APPENDIX 10.D DERIVATION OF THE B–H S–R RELATIONSHIP

Certain fish populations have the characteristic that recruitment appears to be virtually unaffected by fishing; and the reduction in the fish population level does not seem to be associated primarily with overfishing. In fact, although fish species have a high fertility rate, only a small proportion (of eggs) survive to become adult fish.

Beverton and Holt attempt to account for these observations by constructing a model that describes a simple mechanism that can produce a high degree of population stability under fairly constant environmental conditions and over a wide range of parent escapement.

The B–H model assumes a density-dependent relative or per capita mortality rate of the linear form

$$\frac{1}{P(t)}\frac{dP(t)}{dt} = -\left(\mu_1 + \mu_2 P(t)\right), \tag{10.D.1}$$

with μ_1 and μ_2 as positive constants, where $P(t)$ denotes the number of young larval-stage fish active at time t. (A density-independent relative mortality rate would be simply $(1/P)\ dP/dt = -\mu$.) Assume that Equation 10.D.1 holds over a fixed time interval $0 \le t \le T$ that coincides with the duration of the larval stage.

Let us rewrite Equation 10.D.1 as

$$\frac{dP}{\left(-\mu_1 - \mu_2 P\right)P} = dt.$$

Integrating over $0 \le t \le T$ yields

$$\frac{1}{\mu_1}\ln\left(\frac{-\mu_1 - \mu_2 P}{P}\right)\Bigg]_0^T = t\Bigg]_0^T.$$

If we set $Y = P(0)$ and $R = P(T)$, the preceding expression becomes

$$\frac{1}{\mu_1}\left[\ln\left(\frac{-\mu_1 - \mu_2 R}{R}\right) - \ln\left(\frac{-\mu_1 - \mu_2 Y}{Y}\right)\right] = T$$

or

$$\ln\left(\frac{Y\left(-\mu_1 - \mu_2 R\right)}{R\left(-\mu_1 - \mu_2 Y\right)}\right) = \mu_1 T,$$

$$\frac{\mu_1 + \mu_2 R}{R} = e^{\mu_1 T}\left(\frac{\mu_1 + \mu_2 Y}{Y}\right),$$

and thus

$$R = \frac{\mu_1 Y}{e^{\mu_1 T} + \left(e^{\mu_1 T}\mu_2 + \mu_1\right)Y}$$

$$= \frac{\mu_1 e^{-\mu_1 T} Y}{1 + \left(\mu_2 - e^{-\mu_1 T}\right)Y} = \frac{k_1 Y}{1 + k_2 Y}, \quad k_1, k_2 > 0. \tag{10.D.2}$$

Now, if each individual in the S lays a fixed number of eggs, of which a fixed proportion hatch to produce larvae, then $Y = \theta S$, where θ is the constant fecundity or fertility rate of the species. Under this discussion, Equation 10.D.2 becomes the B–H S–R relation:

$$R = \frac{k_1 \theta S}{1 + k_2 \theta S} = \frac{\alpha S}{1 + \beta S}, \tag{10.D.3}$$

where $\alpha = k_1\theta$ and $\beta = k_2\theta$.

APPENDIX 10.E DERIVATION OF THE RICKER S–R RELATIONSHIP

Suppose the S lays $E_0 = kS$ eggs, where k is the average number of eggs laid per spawning adult. If E represents the total number of eggs, which become larvae and consequently mature to adults, then the rate of natural mortality depensation for E can be expressed as

$$\frac{1}{E}\frac{dE}{dt} = -\left(m_1 + m_2 S\right), \tag{10.E.1}$$

where m_1 is the density-independent rate of mortality, while m_2 is the density-dependent rate of mortality. As required under density dependence, we assume that a portion of the instantaneous rate of decline in E is proportional to the abundance of the parent population (the term m_2S)—a result typically attributed to the cannibalism of the young, competition for spawning sites, etc.

From Equation 10.E.1

$$\frac{dE}{E} = -\left(m_1 + m_2 S\right)dt$$

and thus

$$\ln E = -\left(m_1 + m_2 S\right)t + \ln C, \quad \ln C = \text{constant}. \tag{10.E.2}$$

At $t = 0$, $E(0) = E_0 = C = kS$ so that Equation 10.E.2 becomes

$$E = kSe^{-(m_1 + m_2 S)}t. \tag{10.E.3}$$

Now, let's assume, for convenience, that all fish in a cohort spawn are at the same age and that they expire after spawning. If the age at spawning is t_S, then $E(t_S)$ is the number of R provided by the initial number of eggs E_0. If we set $E(t_S) = R$, then Equation 10.E.3 transforms to

$$R(S) = kSe^{-(m_1 + m_2 S)t_S}$$

$$= ke^{-m_1 t_S} Se^{-m_2 S t_S} = aSe^{-bS} \tag{10.E.4}$$

(the *Ricker S–R relation*), where $a = ke^{-m_1 t_S}$ and $b = m_2 t_S$. Here aS depicts the number of eggs that would survive to become spawners if the only source of mortality was the density-independent term $m_1 = constant$.

APPENDIX A

TABLE A.1 Standard Normal Areas (*Z* Is *N*(0, 1))

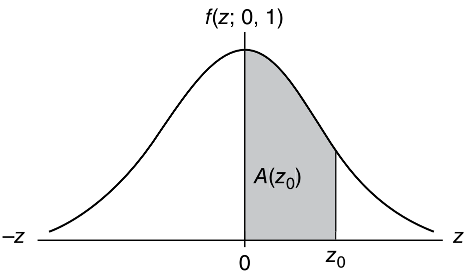

FIGURE A.1

$$A(z_0) = \frac{1}{\sqrt{2\pi}} \int_0^{z_0} e^{-z^2/2} dz$$

$A(z_0)$ gives the total area under the standard normal distribution between 0 and any point z_0 on the positive z-axis

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0150	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0754
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1253	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2258	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2518	0.2549

(*Continued*)

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

TABLE A.1 Standard Normal Areas (Z Is $N(0, 1)$) (*Continued*)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2996	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3288	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3557	0.3559	0.3621
1.1	0.3642	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

TABLE A.2 Quantiles of Student's t Distribution (T Is t_{ν})

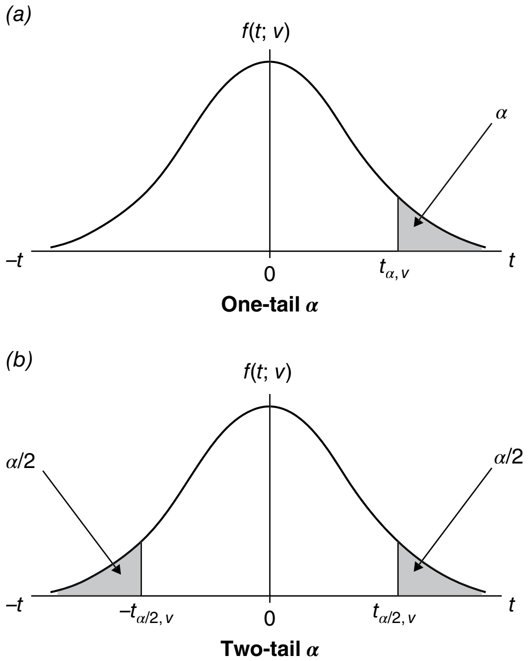

FIGURE A.2

Given degrees of freedom ν, the table gives either: (a) the one-tail $t_{\alpha,\nu}$ value such that $P(T \geq t_{\alpha,\nu}) = \alpha$; or (b) the two-tail $\pm t_{\alpha/2,\nu}$ values for which $P(T \leq -t_{\alpha/2,\nu}) + P(T \geq t_{\alpha/2,\nu}) = \alpha/2 + \alpha/2 = \alpha$ (e.g., for $\nu = 15$ and $\alpha = 0.05$, $t_{0.05,15} = 1.753$ while $t_{0.025,15} = 2.131$).

	One-tail α					
	0.10	0.05	0.025	0.01	0.005	0.001
	Two-tail α					
ν	0.20	0.10	0.05	0.02	0.01	0.002
1	3.078	6.314	12.706	31.821	63.657	318.309
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.215
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646

(*Continued*)

TABLE A.2 Quantiles of Student's t Distribution (T Is t_ν) (*Continued*)

	One-tail α					
	0.10	0.05	0.025	0.01	0.005	0.001
	Two-tail α					
ν	0.20	0.10	0.05	0.02	0.01	0.002
18	1.330	1.734	2.101	2.552	2.878	3.610
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
60	1.296	1.671	2.000	2.390	2.660	3.232
80	1.292	1.664	1.990	2.374	2.639	3.195
100	1.290	1.660	1.984	2.364	2.626	3.174
∞	1.282	1.645	1.960	2.326	2.576	3.090

TABLE A.3 Quantiles of the Chi-Square Distribution (X Is χ^2_ν)

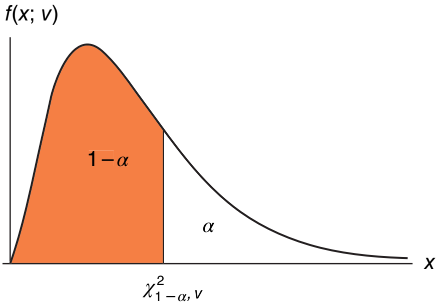

FIGURE A.3

For the cumulative probability $1-\alpha$ and degrees of freedom ν, the quantile $\chi^2_{1-\alpha,\nu}$ satisfies $F(\chi^2_{1-\alpha};\nu)=P(X\le\chi^2_{1-\alpha,\nu})=1-\alpha$ or, alternatively, $P(X>\chi^2_{1-\alpha,\nu})=1-P(X\le\chi^2_{1-\alpha,\nu})=\alpha$ (e.g., for $\nu=10$ and $\alpha=0.05$, $1-\alpha=0.95$ and thus $\chi^2_{0.95,10}=18.31$).

	$1-\alpha$						
ν	0.75	0.90	0.95	0.975	0.99	0.995	0.999
1	1.3233	2.7100	3.8400	5.0200	6.6300	7.8800	10.8280
2	2.7726	4.6100	5.9900	7.3800	9.2100	10.6000	13.8160
3	4.1084	6.2500	7.8100	9.3500	11.3400	12.8400	16.2660
4	5.3853	7.7800	9.4900	11.1400	13.2800	14.8600	18.4670

TABLE A.3 Quantiles of the Chi-Square Distribution (X Is χ^2_ν) (*Continued*)

	$1-\alpha$						
ν	0.75	0.90	0.95	0.975	0.99	0.995	0.999
5	6.6257	9.2400	11.0700	12.8300	15.0900	16.7500	20.5150
6	7.8408	10.6400	12.5900	14.4500	16.8100	18.5500	22.4580
7	9.0372	12.0200	14.0700	16.0100	18.4800	20.2800	24.3220
8	10.2188	13.3600	15.5100	17.5300	20.0900	21.9600	26.1250
9	11.3887	14.6800	16.9200	19.0200	21.6700	23.5900	27.8770
10	12.5489	15.9900	18.3100	20.4800	23.2100	25.1900	29.5880
11	13.7007	17.2800	19.6800	21.9200	24.7300	26.7600	31.2640
12	14.8454	18.5500	21.0300	23.3400	26.2200	28.3000	32.9090
13	15.9839	19.8100	22.3600	24.7400	27.6900	29.8200	34.5280
14	17.1170	21.0600	23.6800	26.1200	29.1400	31.3200	36.1230
15	18.2451	22.3100	25.0000	27.4900	30.5800	32.8000	37.6970
16	19.3688	23.5400	26.3000	28.8500	32.0000	34.2700	39.2520
17	20.4887	24.7690	27.5871	30.1910	33.4087	35.7185	40.7900
18	21.6049	25.9900	28.8700	31.5300	34.8100	37.1600	42.3120
19	22.7178	27.2036	30.1435	32.8523	36.1908	38.5822	43.8200
20	23.8277	28.4100	31.4100	34.1700	37.5700	40.0000	45.3150
21	24.9348	29.6151	32.6705	35.4789	38.9321	41.4010	46.7970
22	26.0393	30.8133	33.9244	36.7807	40.2894	42.7956	48.2680
23	27.1413	32.0069	35.1725	38.0757	41.6384	44.1813	49.7280
24	28.2412	33.1963	36.4151	39.3641	42.9798	45.5585	51.1790
25	29.3389	34.3816	37.6525	40.6465	44.3141	46.9278	52.6200
26	30.4345	35.5631	38.8852	41.9232	45.6417	48.2899	54.0520
27	31.5284	36.7412	40.1133	43.1944	46.9630	49.6449	55.4760
28	32.6205	37.9159	41.3372	44.4607	48.2782	50.9933	56.8920
29	33.7109	39.0875	42.5569	45.7222	49.5879	52.3356	58.3020
30	34.7998	40.2560	43.7729	46.9792	50.8922	53.6720	59.7030
40	45.6160	51.8050	55.7585	59.3417	63.6907	66.7659	73.4020
50	56.3336	63.1671	67.5048	71.4202	76.1539	79.4900	86.6610
60	66.9814	74.4000	79.0800	83.3000	88.3800	91.9500	99.6070
70	77.5766	85.5271	90.5312	95.0231	100.4250	104.2150	112.3170
80	88.1303	96.5782	101.8790	106.6290	112.3290	116.3210	124.8390
90	98.6499	107.5650	113.1450	118.1360	124.1160	128.2990	137.2080
100	109.1410	118.4980	124.3420	129.5610	135.8070	140.1690	149.4490

TABLE A.4 Quantiles of Snedecor's F Distribution (F Is F_{v_1, v_2})

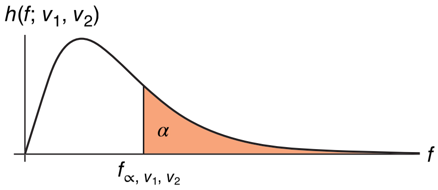

FIGURE A.4

Given the cumulative probability $1-\alpha$ and numerator and denominator degrees of freedom v_1 and v_2 respectively, the table gives the upper α-quantile f_{α, v_1, v_2} such that $P\left(F \geq f_{\alpha, v_1, v_2}\right) = \alpha$ (e.g., for $\alpha = 0.05$, $v_1 = 6$, and $v_2 = 10$, $f_{0.05,6,10} = 5.39$).

α = 0.10 (upper 10% fractile)

v_2 \ v_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86	60.19	60.71	61.22	61.74	62.00	62.26	62.53	62.79	63.06	63.33
2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.48	9.49
3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.22	5.20	5.18	5.18	5.17	5.16	5.15	5.14	5.13
4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.79	3.78	3.76
5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.14	3.12	3.10
6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.90	2.87	2.84	2.82	2.80	2.78	2.76	2.74	2.72
7	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72	2.70	2.67	2.63	2.59	2.58	2.56	2.54	2.51	2.49	2.47
8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.34	2.32	2.29
9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.38	2.34	2.30	2.28	2.25	2.23	2.21	2.18	2.16
10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.28	2.24	2.20	2.18	2.16	2.13	2.11	2.08	2.06
11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.03	2.00	1.97
12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19	2.15	2.10	2.06	2.04	2.01	1.99	1.96	1.93	1.90
13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.90	1.88	1.85
14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10	2.05	2.01	1.96	1.94	1.91	1.89	1.86	1.83	1.80
15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06	2.02	1.97	1.92	1.90	1.87	1.85	1.82	1.79	1.76
16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03	1.99	1.94	1.89	1.87	1.84	1.81	1.78	1.75	1.72

17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00	1.96	1.91	1.86	1.84	1.81	1.78	1.75	1.72	1.69
18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98	1.93	1.89	1.84	1.81	1.78	1.75	1.72	1.69	1.66
19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96	1.91	1.86	1.81	1.79	1.76	1.73	1.70	1.67	1.63
20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.89	1.84	1.79	1.77	1.74	1.71	1.68	1.64	1.61
21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92	1.87	1.83	1.78	1.75	1.72	1.69	1.66	1.62	1.59
22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90	1.86	1.81	1.76	1.73	1.70	1.67	1.64	1.60	1.57
23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89	1.84	1.80	1.74	1.72	1.69	1.66	1.62	1.59	1.55
24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88	1.83	1.78	1.73	1.70	1.67	1.64	1.61	1.57	1.53
25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87	1.82	1.77	1.72	1.69	1.66	1.63	1.59	1.56	1.52
26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86	1.81	1.76	1.71	1.68	1.65	1.61	1.58	1.54	1.50
27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85	1.80	1.75	1.70	1.67	1.64	1.60	1.57	1.53	1.49
28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84	1.79	1.74	1.69	1.66	1.63	1.59	1.56	1.52	1.48
29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83	1.78	1.73	1.68	1.65	1.62	1.58	1.55	1.51	1.47
30	2.88	2.49	2.28	2.14	2.05	1.98	1.93	1.88	1.85	1.82	1.77	1.72	1.67	1.64	1.61	1.57	1.54	1.50	1.46
40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76	1.71	1.66	1.61	1.57	1.54	1.51	1.47	1.42	1.38
60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71	1.66	1.60	1.54	1.51	1.48	1.44	1.40	1.35	1.29
120	2.75	2.35	2.13	1.99	1.90	1.82	1.77	1.72	1.68	1.65	1.60	1.55	1.48	1.45	1.41	1.37	1.32	1.26	1.19
∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.55	1.49	1.42	1.38	1.34	1.30	1.24	1.17	1.00

α = 0.05 (upper 5% fractile)

1	161.4	199.5	215.7	224.6	230.2	234	236.8	238.9	240.5	241.9	243.9	245.9	248	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19	19.16	19.25	19.3	19.33	19.35	19.37	19.38	19.4	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.5
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.7	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6	5.96	5.91	5.86	5.8	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.5	4.46	4.43	4.4	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.1	4.06	4	3.94	3.87	3.84	3.81	3.77	3.74	3.7	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.3	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.5	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.9	2.86	2.83	2.79	2.75	2.71
10	4.96	4.1	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.7	2.66	2.62	2.58	2.54

(*Continued*)

TABLE A.4 Quantiles of Snedecor's *F* Distribution (*F* Is F_{ν_1,ν_2}) (*Continued*)

$\nu_2 \backslash \nu_1$	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
11	4.84	3.98	3.59	3.36	3.2	3.09	3.01	2.95	2.9	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.4
12	4.75	3.89	3.49	3.26	3.11	3	2.91	2.85	2.8	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.3
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.6	2.53	2.46	2.42	2.38	2.34	2.3	2.25	2.21
14	4.6	3.74	3.34	3.11	2.96	2.85	2.76	2.7	2.65	2.6	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.9	2.79	2.71	2.64	2.59	2.54	2.48	2.4	2.33	2.29	2.25	2.2	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.2	2.96	2.81	2.7	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.1	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.9	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.1	2.87	2.71	2.6	2.51	2.45	2.39	2.35	2.28	2.2	2.12	2.08	2.04	1.99	1.95	1.9	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.1	2.05	2.01	1.96	1.92	1.87	1.81
22	4.3	3.44	3.05	2.82	2.66	2.55	2.46	2.4	2.34	2.3	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.8	2.64	2.53	2.44	2.37	2.32	2.27	2.2	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.4	3.01	2.78	2.62	2.51	2.42	2.36	2.3	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.6	2.49	2.4	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.9	1.85	1.8	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.2	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.2	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.7	2.55	2.43	2.35	2.28	2.22	2.18	2.1	2.03	1.94	1.9	1.85	1.81	1.75	1.7	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4	3.15	2.76	2.53	2.37	2.25	2.17	2.1	2.04	1.99	1.92	1.84	1.75	1.7	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.5	1.43	1.35	1.25
∞	3.84	3	2.6	2.37	2.21	2.1	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

α = 0.01 (Upper 1% fractile)

1	4052	4999.5	5403	5625	5764	5859	5928	5982	6022	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	7.95	5.72	7.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	7.82	5.61	7.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06

(*Continued*)

TABLE A.4 Quantiles of Snedecor's *F* Distribution (*F* Is F_{v_1,v_2}) (*Continued*)

v_2 \ v_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	2.66	1.53	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

TABLE A.5 Durbin–Watson DW Statistic—5% Significance Points d_L and d_U (n is the sample size and k' is the number of regressors excluding the intercept)

	$k'=1$		$k'=2$		$k'=3$		$k'=4$		$k'=5$		$k'=6$		$k'=7$		$k'=8$		$k'=9$		$k'=10$	
n	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U
6	0.610	1.400																		
7	0.700	1.356	0.467	1.896																
8	0.763	1.332	0.559	1.777	0.368	2.287														
9	0.824	1.320	0.629	1.699	0.455	2.128	0.296	2.588												
10	0.879	1.320	0.697	1.641	0.525	2.016	0.376	2.414	0.243	2.822										
11	0.927	1.324	0.758	1.604	0.595	1.928	0.444	2.283	0.316	2.645	0.203	3.005								
12	0.971	1.331	0.812	1.579	0.658	1.864	0.512	2.177	0.379	2.506	0.268	2.832	0.171	3.149						
13	1.010	1.340	0.861	1.562	0.715	1.816	0.574	2.094	0.445	2.390	0.328	2.692	0.230	2.985	0.147	3.266				
14	1.045	1.350	0.905	1.551	0.767	1.779	0.632	2.030	0.505	2.296	0.389	2.572	0.286	2.848	0.200	3.111	0.127	3.360		
15	1.077	1.361	0.946	1.543	0.814	1.750	0.685	1.977	0.562	2.220	0.447	2.472	0.343	2.727	0.251	2.979	0.175	3.216	0.111	3.438
16	1.106	1.371	0.982	1.539	0.157	1.728	0.734	1.935	0.615	2.157	0.502	2.388	0.398	2.624	0.304	2.860	0.222	3.090	0.155	3.304
17	1.133	1.381	1.015	1.536	0.897	1.710	0.779	1.900	0.664	2.104	0.554	2.318	0.451	2.537	0.356	2.757	0.272	2.975	0.198	3.184
18	1.158	1.391	1.046	1.535	0.933	1.696	0.820	1.872	0.710	2.060	0.603	2.257	0.502	2.461	0.407	2.667	0.321	2.873	0.244	3.073
19	1.180	1.401	1.074	1.336	0.967	1.685	0.859	1.848	0.752	2.023	0.649	2.206	0.549	2.396	0.456	2.589	0.369	2.783	0.290	2.974
20	1.201	1.411	1.100	1.537	0.998	1.676	0.894	1.828	0.792	1.991	0.692	2.162	0.595	2.339	0.502	2.521	0.416	2.704	0.336	2.885
21	1.221	1.420	1.125	1.538	1.026	1.669	0.927	1.812	0.829	1.964	0.732	2.124	0.637	2.290	0.547	2.460	0.461	2.633	0.380	2.806
22	1.239	1.429	1.147	1.541	1.053	1.664	0.958	1.797	0.863	1.940	0.769	2.090	0.677	2.246	0.588	2.407	0.504	2.511	0.424	2.734
23	1.257	1.437	1.168	1.543	1.078	1.660	0.986	1.785	0.895	1.920	0.804	2.061	0.715	2.208	0.628	2.360	0.545	2.514	0.465	2.670
24	1.273	1.446	1.188	1.546	1.101	1.656	1.013	1.775	0.925	1.902	0.837	2.035	0.751	2.174	0.666	2.318	0.584	2.464	0.506	2.613
25	1.288	1.454	1.206	1.550	1.123	1.654	1.038	1.767	0.953	1.886	0.868	2.012	0.784	2.144	0.702	2.280	0.621	2.419	0.544	2.560
26	1.302	1.461	1.224	1.553	1.143	1.652	1.062	1.759	0.979	1.873	0.897	1.992	0.816	2.117	0.735	2.246	0.657	2.379	0.581	2.513
27	1.316	1.469	1.240	1.556	1.162	1.651	1.084	1.753	1.004	1.861	0.925	1.974	0.845	2.093	0.767	2.216	0.691	2.342	0.616	2.470
28	1.328	1.476	1.255	1.560	1.181	1.650	1.104	1.747	1.028	1.850	0.951	1.958	0.874	2.071	0.798	2.188	0.723	2.309	0.650	2.431
29	1.341	1.483	1.270	1.563	1.198	1.650	1.124	1.743	1.050	1.841	0.975	1.944	0.900	2.052	0.826	2.164	0.753	2.278	0.682	2.396

(*Continued*)

TABLE A.5 Durbin–Watson DW Statistic—5% Significance Points d_L and d_U (n is the sample size and k' is the number of regressors excluding the intercept) (*Continued*)

	$k'=1$		$k'=2$		$k'=3$		$k'=4$		$k'=5$		$k'=6$		$k'=7$		$k'=8$		$k'=9$		$k'=10$	
n	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U
30	1.352	1.489	1.284	1.567	1.214	1.650	1.143	1.739	1.071	1.833	0.998	1.931	0.926	2.034	0.854	2.141	0.782	2.251	0.712	2.363
31	1.363	1.496	1.297	1.570	1.229	1.650	1.160	1.735	1.090	1.825	1.020	1.920	0.950	2.018	0.879	2.120	0.810	2.226	0.741	2.333
32	1.373	1.502	1.309	1.574	1.244	1.650	1.177	1.732	1.109	1.819	1.041	1.909	0.972	2.004	0.904	2.102	0.836	2.203	0.769	2.306
33	1.383	1.508	1.321	1.577	1.258	1.651	1.193	1.730	1.127	1.813	1.061	1.900	0.994	1.991	0.927	2.085	0.861	2.181	0.795	2.281
34	1.393	1.514	1.333	1.580	1.271	1.652	1.208	1.728	1.144	1.808	1.080	1.891	1.015	1.979	0.950	2.069	0.885	2.162	0.821	2.257
35	1.402	1.519	1.343	1.584	1.283	1.653	1.222	1.726	1.160	1.803	1.097	1.884	1.034	1.967	0.971	2.054	0.908	2.144	0.845	2.236
36	1.411	1.525	1.354	1.587	1.295	1.654	1.236	1.724	1.175	1.799	1.114	1.877	1.053	1.957	0.991	2.041	0.930	2.127	0.868	2.216
37	1.419	1.530	1.364	1.590	1.307	1.655	1.249	1.723	1.190	1.795	1.131	1.870	1.071	1.948	1.011	2.029	0.951	2.112	0.891	2.198
38	1.427	1.535	1.373	1.594	1.318	1.656	1.261	1.722	1.204	1.792	1.146	1.864	1.088	1.939	1.029	2.017	0.970	2.098	0.912	2.180
39	1.435	1.540	1.382	1.597	1.328	1.658	1.273	1.722	1.218	1.789	1.161	1.859	1.104	1.932	1.047	2.007	0.990	2.085	0.932	2.164
40	1.442	1.544	1.391	1.600	1.338	1.659	1.285	1.721	1.230	1.786	1.175	1.854	1.120	1.924	1.064	1.997	1.008	2.072	0.952	2.149
45	1.475	1.566	1.430	1.615	1.383	1.666	1.336	1.720	1.287	1.776	1.238	1.835	1.189	1.895	1.139	1.958	1.089	2.022	1.038	2.088
50	1.503	1.585	1.462	1.628	1.421	1.674	1.378	1.721	1.335	1.771	1.291	1.822	1.246	1.875	1.201	1.930	1.156	1.986	1.110	2.044
55	1.528	1.601	1.490	1.641	1.452	1.681	1.414	1.724	1.374	1.768	1.334	1.814	1.294	1.861	1.253	1.909	1.212	1.959	1.170	2.010
60	1.549	1.616	1.514	1.652	1.480	1.689	1.444	1.727	1.408	1.767	1.372	1.808	1.335	1.850	1.298	1.894	1.260	1.939	1.222	1.984
65	1.567	1.629	1.536	1.662	1.503	1.696	1.471	1.731	1.438	1.767	1.404	1.805	1.370	1.843	1.336	1.882	1.301	1.923	1.266	1.964
70	1.583	1.641	1.554	1.672	1.525	1.703	1.494	1.735	1.464	1.768	1.433	1.802	1.401	1.837	1.369	1.873	1.337	1.910	1.305	1.948
75	1.598	1.652	1.571	1.680	1.543	1.709	1.515	1.739	1.487	1.770	1.458	1.801	1.428	1.834	1.399	1.867	1.369	1.901	1.339	1.935
80	1.611	1.662	1.586	1.688	1.560	1.715	1.534	1.743	1.507	1.772	1.480	1.801	1.453	1.831	1.425	1.861	1.397	1.893	1.369	1.925
85	1.624	1.671	1.600	1.696	1.575	1.721	1.550	1.747	1.525	1.774	1.500	1.801	1.474	1.829	1.448	1.857	1.422	1.886	1.396	1.916
90	1.635	1.679	1.612	1.703	1.589	1.726	1.566	1.751	1.542	1.776	1.518	1.801	1.494	1.827	1.469	1.854	1.445	1.881	1.420	1.909
95	1.645	1.687	1.623	1.709	1.602	1.732	1.579	1.755	1.557	1.778	1.535	1.802	1.512	1.827	1.489	1.852	1.465	1.877	1.442	1.903
100	1.654	1.694	1.634	1.715	1.613	1.736	1.592	1.758	1.571	1.780	1.550	1.803	1.528	1.826	1.506	1.850	1.484	1.874	1.462	1.898
150	1.720	1.746	1.706	1.760	1.693	1.774	1.679	1.788	1.665	1.802	1.651	1.817	1.637	1.132	1.622	1.847	1.608	1.862	1.594	1.877
200	1.758	1.778	1.748	1.789	1.738	1.799	1.728	1.810	1.718	1.820	1.707	1.831	1.697	1.841	1.686	1.852	1.675	1.863	1.665	1.874

	$k'=11$		$k'=12$		$k'=13$		$k'=14$		$k'=15$		$k'=16$		$k'=17$		$k'=18$		$k'=19$		$k'=20$	
n	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U
16	0.098	3.503																		
17	0.138	3.378	0.087	3.557																
18	0.177	3.265	0.123	3.441	0.078	3.603														
19	0.220	3.159	0.160	3.335	0.111	3.496	0.070	3.642												
20	0.263	3.063	0.200	3.234	0.145	3.395	0.100	3.542	0.063	3.616										
21	0.307	2.976	0.240	3.141	0.182	3.300	0.132	3.448	0.091	3.583	0.058	3.105								
22	0.349	2.897	0.281	3.057	0.220	3.211	0.166	3.358	0.120	3.495	0.083	3.619	0.052	3.731						
23	0.391	2.826	0.322	2.979	0.259	3.128	0.202	3.272	0.153	3.409	0.110	3.535	0.076	3.650	0.048	3.753				
24	0.431	2.761	0.362	2.908	0.297	3.053	0.239	3.193	0.186	3.327	0.141	3.454	0.101	3.572	0.070	3.678	0.044	3.773		
25	0.470	2.702	0.400	2.844	0.335	2.983	0.275	3.119	0.221	3.251	0.172	3.376	0.130	3.494	0.094	3.604	0.065	3.702	0.041	3.790
26	0.508	2.649	0.438	2.784	0.373	2.919	0.312	3.051	0.256	3.179	0.205	3.303	0.160	3.420	0.120	3.531	0.087	3.632	0.060	3.724
27	0.544	2.600	0.475	2.730	0.409	2.859	0.348	2.987	0.291	3.112	0.238	3.233	0.191	3.349	0.149	3.460	0.112	3.563	0.081	3.658
28	0.578	2.555	0.510	2.680	0.445	2.805	0.383	2.928	0.325	3.050	0.271	3.168	0.222	3.283	0.178	3.392	0.138	3.495	0.104	3.592
29	0.612	2.515	0.544	2.634	0.479	2.755	0.418	2.874	0.359	2.992	0.305	3.107	0.254	3.219	0.208	3.327	0.166	3.431	0.129	3.528
30	0.643	2.477	0.577	2.592	0.512	2.708	0.451	2.823	0.392	2.937	0.337	3.050	0.286	3.160	0.238	3.266	0.195	3.368	0.156	3.465
31	0.614	2.443	0.608	2.553	0.545	2.665	0.484	2.776	0.425	2.887	0.370	2.996	0.317	3.103	0.269	3.208	0.224	3.309	0.183	3.406
32	0.703	2.411	0.638	2.517	0.576	2.625	0.515	2.733	0.457	2.840	0.401	2.946	0.349	3.050	0.299	3.153	0.253	3.252	0.211	3.348
33	0.731	2.382	0.668	2.484	0.606	2.588	0.546	2.692	0.488	2.796	0.432	2.899	0.379	3.000	0.329	3.100	0.283	3.198	0.239	3.293
34	0.758	2.355	0.695	2.454	0.634	2.554	0.575	2.654	0.518	2.754	0.462	2.854	0.409	2.954	0.359	3.051	0.312	3.147	0.267	3.240
35	0.783	2.330	0.722	2.425	0.662	2.521	0.604	2.619	0.547	2.716	0.492	2.813	0.439	2.910	0.388	3.005	0.340	3.099	0.295	3.190
36	0.808	2.306	0.748	2.398	0.689	2.492	0.631	2.586	0.575	2.680	0.520	2.774	0.467	2.868	0.417	2.961	0.369	3.053	0.323	3.142
37	0.831	2.285	0.772	2.374	0.714	2.464	0.657	2.555	0.602	2.646	0.548	2.738	0.495	2.829	0.445	2.920	0.397	3.009	0.351	3.091
38	0.854	2.265	0.796	2.351	0.739	2.438	0.683	2.526	0.628	2.614	0.575	2.703	0.522	2.792	0.472	2.880	0.424	2.968	0.318	3.054
39	0.815	2.246	0.819	2.329	0.763	2.413	0.707	2.499	0.653	2.585	0.600	2.671	0.549	2.757	0.499	2.843	0.451	2.929	0.404	3.013
40	0.896	2.228	0.840	2.309	0.785	2.391	0.731	2.473	0.678	2.557	0.626	2.641	0.575	2.724	0.525	2.808	0.477	2.892	0.430	2.974
45	0.988	2.156	0.938	2.225	0.887	2.296	0.838	2.367	0.788	2.439	0.740	2.512	0.692	2.586	0.644	2.659	0.598	2.733	0.553	2.801
50	1.064	2.103	1.019	2.163	0.973	2.225	0.927	2.287	0.882	2.350	0.836	2.414	0.792	2.479	0.747	2.544	0.703	2.610	0.660	2.675

(*Continued*)

TABLE A.5 Durbin–Watson DW Statistic—5% Significance Points d_L and d_U (n is the sample size and k' is the number of regressors excluding the intercept) (*Continued*)

	$k'=11$		$k'=12$		$k'=13$		$k'=14$		$k'=15$		$k'=16$		$k'=17$		$k'=18$		$k'=19$		$k'=20$	
n	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U
55	1.129	2.062	1.087	2.116	1.045	2.170	1.003	2.225	0.961	2.281	0.919	2.338	0.877	2.396	0.836	2.454	0.795	2.512	0.754	2.571
60	1.184	2.031	1.145	2.079	1.106	2.127	1.068	2.177	1.029	2.227	0.990	2.278	0.951	2.330	0.913	2.382	0.874	2.434	0.836	2.487
65	1.231	2.006	1.195	2.049	1.160	2.093	1.124	2.138	1.088	2.183	1.052	2.229	1.016	2.276	0.980	2.323	0.944	2.371	0.908	2.419
70	1.272	1.986	1.239	2.026	1.206	2.066	1.172	2.106	1.139	2.148	1.105	2.189	1.072	2.232	1.038	2.275	1.005	2.318	0.971	2.362
75	1.308	1.970	1.277	2.006	1.247	2.043	1.215	2.080	1.184	2.118	1.153	2.156	1.121	2.195	1.090	2.235	1.058	2.275	1.021	2.315
80	1.340	1.957	1.311	1.991	1.283	2.024	1.253	2.059	1.224	2.093	1.195	2.129	1.165	2.165	1.136	2.201	1.106	2.238	1.076	2.275
85	1.369	1.946	1.342	1.977	1.315	2.009	1.287	2.040	1.260	2.073	1.232	2.105	1.205	2.139	1.177	2.172	1.149	2.206	1.121	2.241
90	1.395	1.937	1.369	1.966	1.344	1.995	1.318	2.025	1.292	2.055	1.266	2.085	1.240	2.116	1.213	2.148	1.187	2.179	1.160	2.211
95	1.418	1.929	1.394	1.956	1.370	1.984	1.345	2.012	1.321	2.040	1.296	2.068	1.271	2.097	1.247	2.126	1.222	2.156	1.197	2.186
100	1.439	1.923	1.416	1.948	1.393	1.974	1.371	2.000	1.347	2.026	1.324	2.053	1.301	2.080	1.277	2.108	1.253	2.135	1.229	2.164
150	1.579	1.892	1.564	1.908	1.550	1.924	1.535	1.940	1.519	1.956	1.504	1.972	1.489	1.989	1.474	2.006	1.458	2.023	1.443	2.040
200	1.654	1.885	1.643	1.896	1.632	1.908	1.621	1.919	1.610	1.931	1.599	1.943	1.588	1.955	1.576	1.967	1.565	1.979	1.554	1.991

From N.E. Savin and K.J. White, "The Durbin-Watson Test for Serial Correlation with Extreme Sample Sizes or Many Regressors," *Econometricia*, 45, 1977, 1989–96. Corrections: R.W. Farebrother, *Econometricia*, 48, 1980, 1554. Reprinted by permission of the Econometric Society.

TABLE A.6 Empirical Cumulative Distribution of τ for $\rho = 1$

	Significance level			
Sample Size *n*	0.01	0.025	0.05	0.10
The τ statistic: No constant or time trend ($\beta_0 = \beta_1 = 0$) (Equation 4.55.1)				
25	−2.66	−2.26	−1.95	−1.60
50	−2.62	−2.25	−1.95	−1.61
100	−2.60	−2.24	−1.95	−1.61
250	−2.58	−2.23	−1.95	−1.62
300	−2.58	−2.23	−1.95	−1.62
∞	−2.58	−2.23	−1.95	−1.62
The τ_μ statistic: Constant but no time trend ($\beta_1 = 0$) (Equation 4.55.2)				
25	−3.75	−3.33	−3.00	−2.62
50	−3.58	−3.22	−2.93	−2.60
100	−3.51	−3.17	−2.89	−2.58
250	−3.46	−3.14	−2.88	−2.57
300	−3.44	−3.13	−2.87	−2.57
∞	−3.43	−3.12	−2.86	−2.57
The τ_τ statistic: Constant + time trend (Equation 4.55.3)				
25	−4.38	−3.95	−3.60	−3.24
50	−4.15	−3.80	−3.50	−3.18
100	−4.04	−3.73	−3.45	−3.15
250	−3.99	−3.69	−3.43	−3.13
300	−3.98	−3.68	−3.42	−3.13
∞	−3.96	−3.66	−3.41	−3.12

This table is adapted from Fuller (1976). It is used by permission of John Wiley & Sons, Inc.

REFERENCES

Adamic, L.A. *Zipf, Power-Laws, and Pareto – A Ranking Tutorial*. Palo Alto: Information Dynamics Lab (2012).

Aitchison, J., Brown, J.A.C. *The Lognormal Distribution with Special Reference to its Uses in Economics*. Cambridge, UK: Cambridge University Press (1957).

Allee, W.C. *Animal Aggregations: A Study in General Sociology*. Chicago: University of Chicago Press (1931).

Almus, M., Nerlinger, E. Testing Gibrat's Law for Young Firms – Empirical Results for West Germany. *Small Business Economics* **15**: 1–12 (2000).

Amaral, L.A., Buldyrez, S.U., Havlin, S., Salinger, M.A., Stanley, H.E., Stanley, M.H. Scaling Behavior in Economics: The Problem of Quantifying Company Growth. *Physica A* **244**: 1–24 (1997).

Angelini, P., Generale, A. On the Evolution of Firm Size Distributions. *American Economic Review* **98** (1): 426–438 (2008).

Audretsch, D.B., Mahmood, T. Firm Selection and Industry Evolution: The Post-Entry Performance of New Firms. *Journal of Evolutionary Economics* **4**: 243–260 (1994).

Audretsch, D.B., Elston, J. Does Firm Size Matter? Evidence on the Impact of Liquidity Constraints on Firm Investment Behavior in Germany. *International Journal of Industrial Organization* **20**: 1–17 (2002).

Audretsch, D.B., Klomp, L., Santarelli, E., Thurik, A.R. Gibrat's Law: Are the Services Different? *Review of Industrial Organization* **24**: 301–324 (2004).

Axtell, R.L. Zipf, Distribution of US Firm Sizes. *Science* **293**: 1818–1820 (2001).

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

Ayebo, A., Kozubouski, T.J. An Asymmetric Generalization of Gaussian and Laplace Laws. *Journal of Probability and Statistical Science* **1** (2): 187–210 (2004).

Bailey, R.L., Clutter, J.L. Base-Age Invariant Polymorphic Site Curves. *Forensic Science* **20**: 155–159 (1974).

Bates, D.M., Watts, D.G. *Nonlinear Regression Analysis and Its Applications*. New York: John Wiley & Sons, Inc. (1988).

Beal, S.L., Scheiner, L.B. Estimating Population Kinetics. *CRC Critical Reviews in Biomedical Engineering* **8**: 195–222 (1982).

Becchetti, L.O., Trovato, G. The Determinants of Growth for Small and Medium Sized Firms. The Role of the Availability of External Finance. *Small Business Economics* **19** (4): 291–306 (2002).

Bellows, T.S. Jr. The Descriptive Properties of Some Models for Density Dependence. *Journal of Animal Ecology* **50**: 139–156 (1981).

Berryman, A.A. *Principles of Population Dynamics and Their Applications*. Cheltenham: Thornes (1999).

von Bertalanffy, L. Quantitative Laws in Metabolism and Growth. *Quarterly Review of Biology* **32**: 217–231 (1957).

Beverton, R.J., Holt, S.J. On the Dynamics of Exploited Fish Populations. *MAFF Fisheries Investigations Series 2* **19**: 1–553 (1957).

Bleasdale, J.K.A. Plant Growth and Crop Yield. *Annals of Applied Biology* **57**: 173–182 (1966).

Bleasdale, J.K.A. The Relationship Between the Weight of a Plant Part and Total Weight as Affected by Plant Density. *Journal of Horticultural Science & Biotechnology* **42**: 51–58 (1967).

Bleasdale, J.K.A., Nelder, J.A. Plant Population and Crop Yield. *Nature* **188**: 342 (1960).

Bottazzi, G., Secchi, A. Innovation and Corporate Growth in the Evolution of the Drug Industry. *International Journal of Industrial Organization* **19**: 1161–1187 (2001).

Bottazzi, G., Secchi, A. Common Properties and Sectoral Specificities in the Dynamics of US Manufacturing Companies. *Review of Industrial Organizations* **23**: 217–232 (2003a).

Bottazzi, G., Secchi, A. Why are Distributions of Firm Growth Rates Tent-Shaped? *Economics Letters* **80** (3): 415–420 (2003b).

Bottazzi, G., Secchi, A. *Explaining the Distribution of Firms Growth Rates*. Working Paper. Pisa: S. Anna School for Advanced Studies (2004).

Bottazzi, G., Secchi, A. Growth and Diversification Patterns of the Worldwide Pharmaceutical Industry. *Review of Industrial Organization* **26**: 195–216 (2005).

Bottazzi, G., Secchi, A. Explaining the Distribution of Firms Growth Rates. *RAND Journal of Economics* **37**: 234–263 (2006a).

Bottazzi, G., Secchi, A. Gibrat's Law and Diversification. *Industrial and Corporate Change* **37** (2): 235–256 (2006b).

Bottazzi, G., Cefis, E., Dosi, G. Corporate Growth and Industrial Structure: Some Evidence from Italian Manufacturing Industry. *Industrial and Corporate Change* **11**: 705–723 (2002).

Bottazzi, G., Secchi A., Tamagni, F. Productivity, Profitability, and Financial Performance. *Industrial and Corporate Change* **17** (4): 711–751 (2008).

Bottazzi, G., Cefis, E., Dosi, G., Secchi, A. Invariances and Diversities in the Patterns of Industrial Evolution: Some Evidence from Italian Manufacturing Industries. *Small Business Economics* **29** (1): 137–159 (2011a).

Bottazzi, G., Coad, A., Jacoby, N., Secchi, A. Corporate Growth and Industrial Dynamics: Evidence from French Manufacturing. *Applied Economics* **43** (1): 103–115 (2011b).

Brain, P., Cousens, R. An Equation to Describe Dose Responses Where There is Stimulation of Growth at Low Doses. *Weed Research* **29**: 93–96 (1989).

Brakman, S., Garretsen, H., Marrewijk, C. *An Introduction to Geographical Economics*, 3rd ed. Cambridge, UK: Cambridge University Press (2001).

Brody, S. *Bioenergetics and Growth*. New York: Reinhold Publishing Corp. (1945).

Broyden, C.G. Quasi-Newton Methods and Application to Function Minimization. *Mathematics of Computation* **21**: 368–381 (1967).

Cabral, L.M., Mata, J. On the Evolution of the Firm Size Distribution: Facts and Theory. *American Economic Review* **93** (4): 1075–1090 (2003).

Carlyle, T. Occasional Discourse on the Negro Question. *Fraser's Magazine* **40**: 670–679 (1849).

Causton, D.R., Venus, J.C. *The Biometry of Plant Growth*. London: Edward Arnold, Ltd. (1981).

Caves, R.E. Industrial Organization and New Findings on the Turnover and Mobility of Firms. *Journal of Economic Literature* **36** (4): 1947–1982 (1998).

Cefis, E., Ciccarelli, M., Orsenigo, L. Testing Gibrat's Legacy: A Bayesian Approach to Study the Growth of Firms. *Structural Change and Economic Dynamics* **18** (3): 348–369 (2007).

Chanter, D.O. *Mathematical Models in Mushroom Research and Production*. PhD Thesis, University of Sussex (1976).

Chapman, D.G. Statistical Problems in Population Dynamics, in: *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*. Berkeley: University of California Press (1961).

Chesher, A. Testing the Law of Proportionate Effect. *Journal of Industrial Economics* **27** (4): 403–411 (1979).

Cieszewski, C.J. Comparing Fixed- and Variable-Base-Age Site Equations Having Single Versus Multiple Asymptotes. *Forest Science* **48**: 7–23 (2002).

Cieszewski, C.J., Bailey, R.L. Generalized Algebraic Difference Approach: Theory Based Derivation of Dynamic Site Equations with Polymorphism and Variable Asymptotes. *Forensic Science* **46** (1): 116–126 (2000).

Cieszewski, C.J., Strub, M., eds. *Second International Conference on Forest Measurements and Quantitative Methods and Management*. Georgia: Warnell School of Forestry and Natural Resources (2006).

Clutter, J.L., Forston, J.C., Piennar, L.V, Brister, G.H., Bailey, R.L. *Timber Management*. New York: John Wiley & Sons, Inc. (1983).

Coad, A. A Closer Look at Serial Growth Rate Correlation. *Review of Industrial Organization* **31** (1): 69–82 (2007).

Coad, A. Firm Growth and Scaling of Growth Rate Variance in Multi-Plant Firms. *Economics Bulletin* **12** (9): 1–15 (2008).

Coad, A. *The Growth of Firms: A Survey of Theories and Empirical Evidence*. Cheltenham: Edward Elgar (2009).

Contini, B., Revelli, R. The Relationship Between Firms Growth and Labor Demand. *Small Business Economics* **1**: 309–314 (1989).

Courchamp, F., Berec, L., Gascoigne, J. *Allee Effects in Ecology and Conservation*. New York: Oxford University Press (2008).

Cramer, S.G., Jackobs, A.T. An Exponential Model for Predicting Optimum Plant Density and Maximum Corn Yield. *Agronomy Journal* **57**: 241–244 (1965).

Davidon, W.C. *Variable Metric Method for Minimization*. AEC Research and Development Report ANL-5990. Lemont: Argonne National Laboratory (1959).

Davidian, M., Giltinan, D.E. *Nonlinear Models for Repeated Measurement Data*. Boca Raton: CRC Press (1998).

Davidson, J. On the Growth of the Sheep Population in Tasmania. *Transactions of the Royal Society of South Australia* **62**: 342–346 (1938).

Davidson, R., MacKinnon, J.G. *Estimation and Inference in Econometrics*. New York: Oxford University Press (1993).

Davies, G.J. *Potato Spacing Experiments, the Effect of Irregularity on Yield 1950–52*. National Institute of Agricultural Engineering. Reports Series, No. 38, 28 p. (1954).

DelliGatti, D., DiGuilmi, C., Gaffeo, E., Guilioni, G., Gallegati, M., Palestrini A. A New Approach to Business Fluctuations: Heterogeneous Interacting Agents, Scaling Laws and Financial Fragility. *Journal of Economic Behavior and Organization* **56**: 489–512 (2005).

Dickey, D.A., Fuller, W.A. Distributions of Estimators for Autoregressive Time Series with a Unit Root. *Journal of the American Statistical Association* **74**: 427–431 (1979).

Dickey, D.A., Fuller, W.A. Likelihood Ratio Statistics for Autoregressive Time Series with a Unit Root. *Econometrica* **49**: 1057–1072 (1981).

Dickey, D.A., Pantula, S.G. Determining the Order of Differencing in Autoregressive Processes. *Journal of Business & Economic Statistics* **5**(4): 455–461 (1987).

Donald, C.M. Competition Among Pasteur Plants. I. Intraspecific Competition Among Annual Pasture Plants. *Australian Journal of Agricultural Research* **2**: 355–376 (1951).

Draper, N.R., Smith, H. *Applied Regression Analysis*, 2nd ed. New York: John Wiley & Sons, Inc. (1981).

Duncan, W.G. The Relation Between Corn Population and Yield. *Agronomy Journal* **50**: 82–84 (1958).

Dunne, P., Hughes, A. Age, Size, Growth and Survival: UK Companies in the 1980s. *Journal of Industrial Economics* **42** (2): 115–140 (1994).

Dunne, T., Roberts, M., Samuelson, L. Patterns of Entry and Exit in US Manufacturing Industries. *RAND Journal of Economics* **19** (4): 495–515 (1988).

Dunne, T., Roberts, M., Samuelson, L. The Growth and Failure of US Manufacturing Plants. *Quarterly Journal of Economics* **104** (4): 671–698 (1989).

Durbin, J., Watson, G.S. Testing for Serial Correlation in Least Squares Regression. *Biometrica* **38**: 159–178 (1951).

Engledow, F.L. A Census of an Acre of Corn (by Sampling). *Journal of Agricultural Science* **15**: 166–195 (1926).

Esteves, L.A. A Note on Gibrat's Law, Gibrat's Legacy and Firm Growth: Evidence from Brazilian Companies. *Economics Bulletin* **12** (19): 1–7 (2007).

Evans, D.S. The Relationship Between Firm Growth Size, and Age: Estimates for 100 Manufacturing Industries. *Journal of Industrial Economics* **35**: 567–581 (1987a).

Evans, D.S. Tests of Alternative Theories of Firm Growth. *Journal of Political Economy* **95** (4): 657–674 (1987b).

Farazdaghi, H., Harris, P.M. Plant Competition and Crop Yield. *Nature* **217**: 289–290 (1968).

Fernandez, C., Steel, M.F. On Bayesian Modeling of Fat Tails and Skewness. *Journal of the American Statistical Association* **93** (441): 359–371 (1998).

Fletcher, R. *Practical Methods of Optimization*, 2nd ed. Chichester: John Wiley & Sons, Inc. (1987).

Fowler, M.S., Ruxton, G.D. Population Dynamic Consequences of the Allee Effects. *Journal of Theoretical Biology* **215**: 39–46 (2002).

France, J., Thornley, J.M. *Mathematical Models in Agriculture: A Quantitative Approach to Problems in Agriculture and Related Sciences*. London: Butterworths (1984).

Fu, D., Pammolli, F., Buldyrev, S., Riccaboni, M., Matia, K., Yamasaki, K., Stanley, H. The Growth of Business Firms: Theoretical Framework and Empirical Evidence. *Proceedings of the National Academy of Sciences* **102** (52): 18801–18806 (2005).

Fuller, W.A. *Introduction to Statistical Time Series*. New York: John Wiley & Sons, Inc. (1976).

Gabaix, X., Ioannides, Y.M. The Evolution of City Size Distributions, in: *Handbook of Regional and Urban Economic*, *Vol. 4*, *Cities and Geography*, Hendenson, J.V., Thisses, J.F., eds. Amsterdam: Elsevier, 2341–2378 (2004).

Garcia, O. Site Index: Concepts and Methods, in: *Second International Conference on Forest Measurements and Quantitative Methods and Management & The 2004 Southern Mensurationists Meeting,* Cieszewski, C.J., Strub, M., eds. Athens: Warnell School of Forestry and Natural Resources, University of Georgia, 275–283 (2006).

Geroski, P.A. What So We Know About Entry? *International Journal of Industrial Organization* **13**: 421–440 (1995).

Geroski, P.A., Mazzucato, M. Learning and the Sources of Corporate Growth. *Industrial and Corporate Change* **11** (4): 623–644 (2002).

Geroski, P.A., Gugler, K. Corporate Growth Convergence in Europe. *Oxford Economics Papers* **56**: 597–620 (2004).

Geroski, P.A., Lazarova, S., Unga, G., Walters, C.F. Are Differences in Firm Size Transitory or Permanent. *Journal of Applied Econometrics* **18**: 47–59 (2003).

Getz, W.M. A Hypothesis Regarding the Abruptness of Density Dependence and the Growth Rate of Populations. *Ecology* **77**: 2014–2026 (1996).

Gibrat, R. *Les Inègalitès Èconomiques: Applications, aux Inègalitiès des Richesses, a la Concentration des Entreprises, aux Populations des Villes, aux Statistiques des Familles, etc.: d'une Loi Nouvelle: la Loi de L'effet Proportionnel*. Paris: Recueil Sirey (1931).

Goddard, J., Wilson, J., Blandon, P. Panel Tests of Gibrat's Law for Japanese Manufacturing. *International Journal of Industrial Organization* **20** (3): 415–433 (2002).

Goelz, J.C.G., Burk, T.E. Development of a Well-Behaved Site-Index Equation: Jack Pine in North Central Ontario. *Canadian Journal of Forest Research* **22**: 776–784 (1992).

Gompertz, B. On the Nature of the Function Expressive of Human Mortality, and on a New Mode to Determining the Value of Life Contingencies. *Philosophical Transactions of the Royal Society of London* **115**: 513–585 (1825).

Goudriaan, J. Using the Expolinear Growth Equation to Analyse Resource Capture, in: *Resource Capture by Crops*, Monteith, J.L., Scott, R.K., Unsworth, M.L., eds. Nottingham: Nottingham University Press, 99–110 (1994).

Goudriaan, J., Monteith, J.L. A Mathematical Function for Crop Growth Based on Light Interception. *Annals of Botany* **66**: 695–701 (1990).

Greene, W.H. *Econometric Analysis*, 3rd ed. New York: Macmillan Publ. Co. (1997).

Growiec, J., Pammolli, F., Riccaboni, M., Stanley, H.E. On the Size Distribution of Business Firms. *Economic Letters* **98**: 207–212 (2008).

Guldberg, C.M., Waage, P. Concerning Chemical Affinity. *Erdmann's Journal für Practisch Chemie* **127**: 69–114 (1879).

Halkert, J.R. *Plant Level Growth Rates: A Case for Disaggregating Shocks*. New York: Department of Economics, New York University (2006).

Hall, B.H. The Relationship Between Firm Size and Firm Growth in the US Manufacturing Sector. *Journal of Industrial Economics* **35** (4): 583–600 (1987).

Hart, P.E., Prais, S. The Analysis of Business Concentration: A Statistical Approach. *Journal of the Royal Statistical Society* **119** (2): 150–191 (1956).

Hart, P.E., Oulton, N. The Growth and Size of Firms. *Economic Journal* **106** (3): 1242–1252 (1996).

Hassell, M. Density-Dependence in Single Species Populations. *Journal of Animal Ecology* **44**: 283–296 (1975).

Heckman, J.J. The Common Structure of Statistical Models of Truncation, Sample Selection, and Limited Dependent Variables and a Simple Estimator for Such Models. *Annals of Economic and Social Measurement* **5**: 475–492 (1976).

Heckman, J.J. Sample Selection Bias as a Specification Error. *Econometrica* **47**: 153–161 (1979).

Hill, A.V. The Possible Effects of the Aggregation of the Molecules of Haemoglobin on its Dissociation Curves. *Journal of Physiology* **40** (Suppl.): 1–22 (1910).

Hill, A.V. The Combinations of Haemoglobin with Oxygen and with Carbon Monoxide. *Biochemical Journal* **7**: 472–480 (1913).

Hinkley, D.V., Revankar, N.S. Estimation of the Pareto Law from Underreported Data. *Journal of Econometrics* **5**: 1–11 (1977).

Holliday, R. Plant Population and Crop Yield: I. *Field Crop Abstracts* **13**: 159–167 (1960).

Hossfeld, J.W. Mathematik für Forstmänner, OKonomeh und Cameralisten. *Gotha*, **4**. Bd. **5**. 310 (1822).

Huang, S. *Development of a Subregion-Based Compatible Height-Site Index-Age Model for Black Spruce in Alberta*. Alberta Land and Forest Service, Forest Management Research Note No. 5, Publication No. T/352. Edmonton, Alberta (1997).

Huberman, B.A., Adamic, L.A. Growth Dynamics of the World-Wide Web. *Nature* **401**: 131 (1999).

Hudson, G.H. Population Studies with Wheat. 2. Propinquity. 3. Seed Rates in Nursery Trials and Field Plots. *Journal of Agricultural Science* **31**: 116–144 (1941).

Hudson, R. Collard Doves in Britain and Ireland During 1965–1970. *British Birds* **65**: 139–155 (1972).

Hunt, R. *Plant Growth Curves*. Baltimore: University Park Press (1982).

Hutchinson, G.E. *Introduction to Population Ecology*. New Haven: Yale University Press (1978).

Huxley, J.S. Constant Differential Growth-Ratios and their Significance. *Nature* **114**: 114895–114896 (1924).

Huxley, J.S. *Problems of Relative Growth*. Baltimore: Johns Hopkins University (1972).

Hymer, S., Pashigian, P. Firm Size and Rate of Growth. *Journal of Political Economy* **70** (6): 556–569 (1962).

Ijiri, Y., Simon, H.A. A Model of Business Firm Growth. *Econometrica* **35** (2): 342–355 (1967).

Ijiri, Y., Simon, H.A. *Skew Distributions and the Sizes of Business Firms*. Amsterdam: North Holland (1977).

Janoschek, A. Das Reaktionshinetische Grundgesetz und Seire Beziehungen zum Wachstums- und Ertragsgesetz. *Statistische Vierteljahresschrift.* **10**: 25–37 (1957).

Johnson, N.L., Kotz, S. *Continuous Univariate Distributions – 1*. New York: John Wiley & Sons, Inc. (1970).

Johnson, P.C., Conway, C., Kattuman, P. Small Business Growth in the Short Run. *Small Business Economics* **12** (2): 103–112 (1999).

Kalecki, M. On the Gibrat Distribution. *Econometrica* **13** (2): 161–170 (1945).

Kapteyn, J.C. *Skew Frequency Curves*. Groningen: Astronomical Laboratory (1903).

Kira, T., Ogawa, H., Sakazaki, N. Intraspecific Competition Among Higher Plants. I. Competition-Yield-Density Interrelationships in Regularly Dispersed Populations. *Journal of the Institute of Polytechnics, Osaka City University*. **4**: 1–16 (1953).

Kira, T., Ogawa, H., Hozumi, K., Koyama, H., Yoda, K. Intraspecific Competition Among Higher Plants. V. Supplementary Notes on the C-D Effect. *Journal of the Institute of Polytechnic, Osaka City University Series D* **7**: 1–14 (1956).

Kleiber, C., Kotz, S. *Statistical Size Distributions in Economics and Actuarial Sciences*. Hoboken: John Wiley & Sons, Inc. (2003).

Klein, L.R. *An Introduction to Econometrics*. New Jersey: Prentice-Hall, Inc. (1962).

Kmenta, J. *Elements of Econometrics*, 2nd ed. New York: Macmillan Publ. Co. (1986).

Korf, V. A. Mathematical Definition of Stand Volume Growth Law. *Lesnicka Prace* **18**: 337–339 (1939).

Kozubowski, T.J., Podgorski, K. Asymmetric Laplace Distributions. *Mathematical Scientist* **25**: 37–46 (2000).

Kozubowski, T.J., Podgorski, K. A Log-Laplace Growth Rate Model. *The Mathematical Scientist* **28**: 49–60 (2003).

Kreusler, U., Prehn, A., Hornberger, R. Beobachtungen über das Washstum Maispflante (Berich uber die Versuche von Jahre 1878). *Landwirtschaftliche Jahrbucher.* **8**: 617–622 (1879).

Laitinen, E.K., Stochastic Growth Processes in Large Finnish Companies: Test of Gibrat's Law of Proportionate Effect. *Liiketaloudellinen Aikakauskirja* **1**: 27–49 (1999).

Lee, Y., Amaral, L.A., Canning, D., Meyer, M., Stanley, H.E. Universal Features in the Growth Dynamics of Complex Organizations. *Physical Review Letters* **81** (5): 3275–3278 (1998).

Levakovic, A. Analytical Form of Growth Laws. *Glasnik za Sumske Pokuse (Zagreb)* **4**: 189–282 (1935).

Levenberg, K. A Method for the Solution of Certain Problems in Least Squares. *Quarterly of Applied Mathematics* **2**: 164–168 (1944).

Li, B., Watkinson, A.R., Hara, T. Dynamics of Competition in Populations of Carrot (*Dancus carota*). *Annals of Botany* **78**: 203–214 (1996).

Lidicker, W.Z. From Dispersal to Landscapes: Progress in Our Understanding of Population Dynamics. *Acta Theriologica* **47** (Suppl. 1): 23–37 (2002).

Lidicker, W.Z. The Allee Effect: Its History and Future Importance. *The Open Ecology Journal* **3**: 71–82 (2010).

Lindstrom, M.J. Bates, D.M. Nonlinear Mixed Effects Models for Repeated Measures Data. *Biometrics* **46**: 673–687 (1990).

Little, I.M.D. Higgledy Piggledy Growth. *Bulletin of the Oxford University Institute of Statistics* **24** (4): 387–412 (1962).

Lopez, S., France, J., Gerrits, W.J., Dhanoa, M.S., Humphries, D., Dijkstra, J. A Generalized Michaelis–Menten Equation for the Analysis of Growth. *Journal of Animal Science* **78**: 1816–1828 (2000).

Lotti, F., Santarelli, E. Is Firm Growth Proportional? An Appraisal of Firm Size Distribution. *Economics Bulletin* **12** (6): 1–7 (2001).

Lotti, F., Santarelli, E., Vivarelli, M. *Does Gibrat's Law Hold Among Young, Small Firms?* Working Paper 361, Universitá di Bologna, Dipartimento di Scienze Economiche (1999).

Lotti, F., Santarelli, E., Vivarelli, M. Does Gibrat's Law Hold Among Young, Small Firms? *Journal of Evolutionary Economics* **13** (3): 213–235 (2003).

Lotti, F., Santarelli, E., Vivarelli, M. *Defending Gibrat's Law as a Long-Run Regularity*, IZA Working Papers, 2744. Bonn: Institute for the Study of Labor (IZA) (2007).

Lundqvist, B. On the Height Growth in Cultivated Stands of Pine and Spruce in Northern Sweden. *Meddelanden Fran Statens Skogsforsknings-institut* **47**: 1–64 (1957).

Malthus, T.R. *An Essay on the Principles of Population*, 1st ed. London: Printed for J. Johnson, in St. Paul's Church-Yard (1798).

Mansfield, E. Entry Gibrat's Law, Innovation, and the Growth of Firms. *American Economic Review* **52** (5): 1023–1051 (1962).

Marquardt, D.W. An Algorithm for Least-Squares Estimation of Nonlinear Parameters. *SIAM Journal on Applied Mathematics* **11**: 431–441 (1963).

Marsili, O. Technology and the Size Distribution of Firms: Evidence from Dutch Manufacturing. *Review of Industrial Organization* **27** (4): 303–328 (2005).

Matia, K., Fu, D., Buldyrev, S., Pammolli, F., Riccaboni, M., Stanley H. Statistical Properties of Business Firms Structure and Growth. *Europhysics Letters* **67** (3): 498–503 (2004).

Mawson, P. *Measuring Economic Growth in New Zealand WP 02/14*. Wellington: The Treasury of New Zealand (2002).

May, R.M. Simple Mathematical Models with Very Complicated Dynamics. *Nature* **261**: 459–467 (1976).

Maynard-Smith, J., Slatkin, M. The Stability of Predator–Prey Systems. *Ecology* **54**: 384–391 (1973).

McDill, M.E., Amateis, R.L. Measuring Forest Site Quality Using the Parameters of a Dimensionally Compatible Height Growth Function. *Forest Science* **38** (2): 409–429 (1992).

McKendrick, A.G., Kesava Pai, M. The Rate of Multiplication of Microorganisms: A Mathematical Study. *Proceedings of the Royal Society of Edinburgh* **3**: 649–655 (1911).

Mercer, L.P., Flodin, N.W., Morgan, P.H. New Methods for Comparing the Biological Efficiency of Alternative Nutrient Sources. *The Journal of Nutrition* **108**: 1244–1249 (1978).

Meyer, J.R., Kuh, E. *The Investment Decision*. Cambridge, MA: Harvard University Press (1957).

Meyers, R.A., Barrowman, N.J., Hutchings, J.A., Rosenberg, A.A. Population Dynamics of Fish Stocks at Low Population Levels. *Science* **269**: 1106–1108 (1995).

Michaelis, M., Menten, M.L. Die Kinetik der Invertinwirkung. *Biochemestry Zeitung*. **49**: 333 (1913).

Morgan, P.H., Mercer, L.P., Flodin, N.W. Gereval Model for Nutritional Responses of Higher Organisms. *Proceedings of the National Academy of Sciences of the United States of America* **72**: 4327–4331 (1975).

Nelder, J.A. The Fitting of a Generalization of the Logistic Curve. *Biometrics* **17**: 89–110 (1961).

Newman, M.E. Power Laws, Pareto Distributions, and Zipt's Law. *Contemporary Physics* **46**: 323–351 (2006).

Odum, H.T., Allee, W.C. A Note on the Stable Point of Populations Showing Both Intraspecific Cooperation and Disoperation. *Ecology* **35**: 95–97 (1954).

Pammolli, F., Fu, D., Buldyrev, S.V., Riccaboni, M., Matia, K., Yamasaki, K., Stanley, H.E. A Generalized Preferential Attachment Model for Business Firms Growth Rates: I. Empirical Evidence. *European Physics Journal B* **57**: 127–130 (2007).

Panik, M. *Regression Modeling: Methods, Theory, and Computations with SAS*. Boca Raton: CRC Press (2009).

Pareto, V. *Cours d'Economie Politique*. Lausanne: Ed. Rouge (1897).

Perline, R., Axtell, R., Teitelbaum, D. *Volatility and Asymmetry of Small Firm Growth Rates over Increasing Time Frames*. Washington, DC: US Small Business Administration, The Office of Advocacy Small Business Working Papers 06/rarpdt (2006).

Petrunia, R. *Does Gibarat's Law Hold? Evidence from Canadian Retail and Manufacturing Firms*. Thunder Bay: Department of Economics, Lakehead University (2005).

Phillips, B., Kirchhoff, B. Formation, Growth and Survival: Small Firm Dynamics in the US Economy. *Small Business Economics* **1** (1): 65–74 (1989).

Pinheiro, J.C., Bates, D.M. Approximations to the Loglikelihood Function in the Nonlinear Mixed Effects Model. *Journal of Computational and Graphical Statistics* **4**: 12–35 (1995).

Quandt, R. On the Size Distribution of Firms. *American Economic Review* **56** (3): 416–432 (1966).

Ratkowsky, D.A. *Nonlinear Regression Modeling*. New York: Marcel Dekker, Inc. (1983).

Reed, W.J. The Pareto, Zipf, and Other Power Laws. *Economics Letters* **74** (1): 15–19 (2001).

Reed, W.J. The Pareto Law of Incomes – An Explanation and an Extension. *Physica A* **319**: 469–486 (2003a).

Reed, W. On the Distribution of Family Names. *Physica A* **319**: 579–590 (2003b).

Riccaboni, M., Pammolli, F. Buldyrev, S.V., Ponta, L., Stanley, H.E. *Proceedings of the National Academy of Sciences of the United States of America* **105**: 19595–19600 (2008).

Richards, L. Capillary Conduction of Liquids Through Porous Mediums. *Physics* **1**: 318–333 (1931).

Richards, F.J. A Flexible Growth Function for Empirical Use. *Journal of Experimental Botany* **10**: 290–300 (1959).

Ricker, W.E. Stock and Recruitment. *Journal of Fisheries Research Board of Canada* **11**: 559–623 (1954).

Ricker, W.E. *Handbook of Computations for Biological Statistics of Fish Populations*. Ottawa: Fisheries Research Board of Canada (1958).

Ricker, W.E. Computation and Interpretation of Biological Statistics of Fish Populations. *Bulletin of Fisheries Research Board of Canada* **191**: 382 (1975).

Royama, T. *Analytical Population Dynamics*. New York: Springer-Verlag (1992).

Sager, G. Seasonally Modified Forms of the Revised Janoschek Growth Function. *Gegenbaurs morphologisches Jahrbuch, Leipzig* **130** (5): 659–669 (1984).

Santarelli, E., Lotti, F. Industry Dynamics and the Distribution of Firm Sizes: A Nonparemetric Approach. *Southern Economic Journal* **70** (3): 443–466 (2004).

Santarelli, E., Klomp, L., Thurik, A.R. *Gibrat's Law: An Overview of the Empirical Literature*, in: *Entrepreneurship, Growth and Innovation: The Dynamics of Firms and Industries*. New York: Springer (2006).

SAS Institute, Inc. *SAS/ETS 9.1 User's Guide*. Cary: SAS Institute, Inc. (2004).

Savin, N.E., White, K.J. The Durbin–Watson Test for Serial Correlation with Extreme Sample Sizes or Many Regressors. *Econometricia* **45**: 1989–1996 (1977). Corrections: Farebrother, R.W. *Econometricia* **48**: 1554 (1980).

Schnute, J.C. A Versatile Growth Model with Statistically Stable Parameters. *Canadian Journal of Fisheries and Aquatic Sciences* **38**: 1128–1140 (1981).

Schoombie, S.W., Getz, W.M. Evolutionary Stable Strategies and Trade-Offs in Generalized Beverton and Holt Growth Models. *Theoretical Population Biology* **53**: 216–235 (1998).

Schultz, J.C. Plant Responses Induced by Herbivores. *Trends in Ecology and Evolution* **3**: 45–49 (1988).

Schumacher, F.X. A New Growth Curve and its Application to Timber Yield Studies. *Journal of Forestry* **37**: 819–820 (1939).

Seber, G.A.F., Wild, C.J. *Nonlinear Regression*. New York: John Wiley & Sons, Inc. (1989).

Segarra, A., Callejon, M. New Firms' Survival and Market Turbulence: New Evidence from Spain. *Review of Industrial Organization* **20** (1): 1–14 (2002).

Shinozaki, K., Kira, T. Intraspecific Competition Among Higher Plants. VII Logistic Theory of the C-D Effect. *Journal of the Institute of Polytechnics, Osaka City University* **D7**: 35–72 (1956).

Simon, H., Bonini, C. The Size Distribution of Business Firms. *American Economic Review* **48** (4): 607–617 (1958).

Singh, A., Whittington, G. The Size and Growth of Firms. *Review of Economic Studies* **42** (91): 15–26 (1975).

Sloboda, B. [Investigation of Growth Processes Using First-Order Differential Equations.] Mitteilungen der Baden-Württembergischen Foustlichen Versuchs und Forschungsanstalt. *Heft* **32** (1971a).

Sloboda, B. *Zur Darstellung von Washstumprozessen mit Hilfe von Differentialgleichungen evster Ordung*. Mitteilungen der Baden-Württembergischen Foustlichen Versuchs und Forschungsanstalt. Baden-Württemberg: Baden-Württembergische Forstliche Versuchs- und Forschungsanstalt (1971b).

Soltanpour, P.N. Accumulation of Dry Matter and N, P, K, by Russet Burbank, Oromonte, and Red McClure Potatoes. *American Potato Journal* **46**: 111–119 (1969).

Southman, C.M., Ehrlich, J. Effects of Extract of Western Red-Cedar Heartwood on Certain Wood-Decaying Fungi in Culture. *Phytopathology* **33**: 517–524 (1943).

Stanley, M.H., Amacal, L. Al, Buldyrev, S.V., Havlin, S., Lesdhoun, H., Maass, P., Salinger, M.A., Stanley, H.E. Scaling Behavior in the Growth of Companies. *Nature* **379**: 804–806 (1996).

Stannard, C.J., Williams, A.P. Gibbs, P.A. Temperature/Growth Relationship for Psychotropic Food-Spoilage Bacteria. *Food Microbiology* **2**: 115–122 (1985).

Steindl, J. *Random Processes and the Growth of Firms*. New York: Hafner (1965).

Stephens, P.A., Sutherland, W.J., Freckleton, R.P. What is the Allee Effect? *Oikos* **87**: 185–190 (1999).

Strand, L. Numerical Construction of Site-Index Curves. *Forest Science* **10**: 410–414 (1964).

Subbotin, M.T. *On the Law of Frequency of Errors*. Matematicheskii Sbornik, **31**: 296–301 (1923).

Sutton, J. Gibrat's Legacy. *Journal of Economic Literature* **35**: 40–59 (1997).

Sutton, J. *The Variance of Firm Growth Rates: The "Scaling" Puzzle*. London School of Economics and Political Science, Discussion Paper No. EI/27 (2001).

Sutton, J. The Variance of Firm Growth Rates: The Scaling Puzzle. *Physica A* **312**: 577–590 (2002).

Teitelbaum, D., Axtell, R. *Firm Size Dynamics of Industries: Stochastic Growth Processes, Large Fluctuations, and the Population of Firms as a Complex System*. Washington, DC: US Small Business Administration, The Office of Advocacy, No. 247 (2005).

Thimann, K.V. Promotion and Inhibition: Twin Themes in Physiology. *American Naturalist*. **90**: 145–162 (1956).

Trorey, L.G. A Mathematical Method for the Construction of Diameter-Height Curves Based on Site. *The Forestry Chronicle* **8**: 121–132 (1932).

Verhulst, P.F. Notice Sur la loi que la Population Suit Dans Son Accroissement. *Correspondence Mathematique et Physique* **10**: 113–121 (1838).

Vianelli, S. Sulle Curve Lognormal di Ordine r Quali Famiglie di Distribuzioni di Errori di Proporzione. *Statistica* **42**: 155–176 (1982a).

Vianelli, S. Una Nota Sulle Distribuzioni Degli Eroni di Proporzione con Particolare Riguardo alla Distribuzioni Corrispondente Alla Prima Lege Degli Erroni Additivi di Laplace. *Statistica* **42**: 371–380 (1982b).

Vianelli, S. The Family of Normal and Lognormal Distributions of Order r. *Metron* **41**: 3–10 (1983).

Wang, M., Rennolls, K., Borders, B.E. Base-Age Invariant Site Index Models from a Generalized Parameter Prediction Approach. *Forest Science* **54**: 625–632 (2008).

Warne, L.G.G. Spacing Experiments on Vegetables. II. The Effect of the Thinning Distance on the Yields of Globe Beet, Long Beet, Carrots, and Parsnips Grown at a Standard Inter-row Distance in Cheshire. *Journal of Horticultural Science* **26**: 84–97 (1951).

Watkinson, A.R. Density-Dependence in Single-Species Populations of Plants. *Journal of Theoretical Biology* **83**: 345–357 (1980).

Weibull, W. A Satistical Distribution Function of Wide Applicability. *Journal of Applied Mechanics* **18**: 291–297 (1951).

Werker, A.R., Jaggard, K.W. Modeling Asymmetrical Growth Curves that Rise and Fall: Applications to Foliage Dynamics of Sugar Beet. *Annals of Botany* **79**: 657–665 (1997).

White, H. A Heteroscedasticity-Consistent Covariance Estimator and a Direct Test for Heteroscedasticity. *Econometrica* **48** (4): 817–834 (1980).

deWit, G. Firm Size Distributions: An Overview of Steady-State Distributions Resulting from Firm Dynamics Models. *International Journal of Industrial Organization* **23** (5–6): 423–450 (2005).

Yin, X., Kropff, M.J., McLaren, G., Visperas, R.M. A Nonlinear Model for Crop Development as a Function of Temperature. *Agricultural and Forest Meteorology* **77**: 1–16 (1995).

Yin, X., Goudriaan, J., Lantinga, E.A., Vos, J., Spiertz, H.J. A Flexible Sigmoid Function to Determine Growth. *Annals of Botany* **91**: 361–371 (2003).

Yoda, K., Kira T., Ogawa, H., Hozumi, K. Self-thinning in Overcrowded Pure Stands under Cultivated and Natural Conditions. *Osaka City University, Journal of Biology* **14**: 107–129 (1963).

Yoshida, S. *Fundamentals of Rice Crop Science*. Los Banos: International Rice Research Institute (1981).

Yule, G.U. A Mathematical Theory of Evolution, Based on the Conclusions of Dr. J.C. Willis. *Philosophical Transactions of the Royal Society of London* **213**: 21–87 (1925).

Ziebach, T. *Die Modellierung der Persondlen Einkommensverteilung mit Verallgeneinerten Pareto-Kurven*. Lohmar: Joseph Eul Veralag (2000).

Zipf, G.K. *Human Behavior and the Principle of Least Effort*. Cambridge, MA: Addison-Wesley (1949).

INDEX

Aggregate output growth, 41–6
production function, 41
Cobb-Douglas, 42
Solow-Swan model, 43–6
balanced growth path, 44
fundamental differential equation, 43
golden rule, 45–6
steady state, 44
technical progress, 41
Agonist, 60
Allometric relationship, 240
Arithmetic progression, 1
common difference, 1
sum of the first n terms, 1–2
Arithmetic scale, 15–16
Autocorrelation, 90, 110–126
autoregressive error model, 125
Durbin-Watson statistic, 113–15
sampling distribution, 114
Durbin-Watson test, 113–15
first-order, 111, 114, 310
impact on OLS estimates, 112–13

Bernoulli differential equation, 77, 79
Binomial formula, 4
binomial coefficients, 4
general, 5

Central limit theorem, 299, 324, 326
Cobwebbing, 362–3, 397–8
Compound interest, 17–20
accumulated amount, 17–20
continuous compounding, 19
growth curve, 19–20, 109

Difference operator, 5
difference formulas, 6–7
difference interval, 5
nth difference, 8
polynomial, 8–9
Dimension analysis, 83–4
Double-logarithmic function, 17

Equilibrium, 362, 373–4, 376–82, 391–9
fixed point, 362, 394–7
attracting, 362, 391–400

Growth Curve Modeling: Theory and Applications, First Edition. Michael J. Panik.

Equilibrium (*cont'd*)
repelling, 362, 391
neutral, 396
orbit, 395
oscillatory behavior, 396
period k, 397
qualitative analysis, 392
stable, 362, 391
steady-state solution, 394
time path, 395–6, 397
unstable, 362, 391
Experimental design, 270–271
factor, 270
factor effect, 270
fixed, 270
random, 270
level, 270
mixed effects, 271
nonlinear model, 271–6
repeated measurements, 271
interindividual variation, 271
intraindividual variation, 271
treatment, 270
Exponential distribution, 330
double, 330
Exponential function, 12–13, 16
base, 13–14
change of base, 14–15
inverse, 13
number e, 9–13, 19–20
limit form, 10

Factorial, 5
Firm age distribution, 300
exponential, 300
Firm growth rate distribution, 300–305
autocorrelation, 300
heteroscedasticity, 310
Laplace, 301–4, 309
persistence, 312–13
power law, 302, 308–10
Subbotin, 303–5
variability, 308–10
Firm size distribution, 294–7, 299–300
growth rate variability, 308–10
size, 294
mean relative growth rate, 294
relative growth rate, 294
Forest growth *see* Site-quality models
Gamma function, 326
Geometric progression, 2
common ratio, 2
convergence, 3
finite partial sum, 3
geometric series, 3
sum of the first n terms, 2
Gibrat distribution, 322–5
Gibrat's law, 295, 297–300, 305, 310–314, 339 *see also* Law of proportionate effect
strong, 299–311
weak, 299, 306–7
Growth, 32–6
absolute growth rate, 35
mean, 36
annualized percent change, 28–30
arguments, 36–41
conditions for exponential growth, 39–40
logistic, 40
average annual, 109, 161–2
average percent change, 23–4
compound interest, 24–5
constant growth series, 32–3
logarithmic form, 32–3
continuous, 33–6
determinate, 249
discrete, 32–3
doubling time, 33, 35
exponential, 33–5
logarithmic form, 34
geometric average, 105, 162–3
log-difference, 161
logistic, 40–41
mixture of variables, 47–8
ordinary least squares, 158–60
relative growth rate, 32, 294
mean, 36, 294
semilogarithmic, 33–5
variability, 46–7
weighted mean, 48
Growth curve *see also* Population growth
asymmetric, 253–7
von Bertalanffy, 57–9, 66, 77–81
physiological basis, 81–2
point of inflection, 59
saturation parameter, 58
beta, 249–51, 263–6
asymmetric, 251, 266–8

Brody, 61, 79, 80
phase I, 61
phase II, 61
Chanter, 257, 269
Chapman-Richards, 58, 66, 78–80, 175–6, 180, 207, 225, 231, 257, 259, 277
maturity index, 79
position parameter, 79
saturation parameter, 79
shape parameter, 79–80
estimation initial values, 223–5
expolinear, 242–6, 261–3
effective loss of biomass, 245, 263
lost time, 245
symmetric, 246
truncated, 245, 251
exponential, 20, 33, 58, 61–2, 66, 71–2, 79, 242–3
negative, 56–7, 75–7
feedback term, 56
generalized logistic *see also* Chapman-Richards
Gompertz, 54–5, 66, 71, 74–5, 80, 180, 207, 224, 257, 269, 277
growth rate parameter, 54
point of inflection, 54
saturation parameter, 54
Hill, 67
coefficient, 67
Hossfeld, 63–4
growth rate parameter, 63
point of inflection, 63
Janoschek, 62
growth rate parameter, 62
modified, 62
point of inflection, 62
shape parameter, 62
Levakovic I, 69–70
point of inflection, 70
Levakovic III, 70
point of inflection, 70
linear, 50–51
monotonic decreasing, 50
monotonic increasing, 50
logarithmic reciprocal, 51–2
growth rate parameter, 52
logarithmic form, 51
point of inflection, 51
saturation parameter, 52
logistic, 40, 52–4, 66, 71–4, 79–80, 180, 207, 224, 228, 242, 258, 269, 277
growth rate parameter, 53
point of inflection, 53
saturation parameter, 53
log-logistic, 59–61, 178, 180
baseline response level, 62
growth rate parameter, 60
point of inflection, 60
Lundqvist-Korf, 63, 180
point of inflection, 63
scale parameter, 63
shape parameter, 63
McDill-Amateis, 68–9, 83, 85
growth rate parameter, 68
height parameter, 68
Michaelis-Menten, 66–7
generalized, 86–7
hyperbolic saturation curve, 66
monomolecular, 76–7, 87
Morgan-Mercer-Flodin, 66–8
growth rate parameter, 68
point of inflection, 68
shape parameter, 67
negative exponential, 56–7
growth rate parameter, 56
saturation parameter, 56
parametric, 50
asymptote, 50
flexibility in modeling, 50
intercept, 50
rate of change, 50
point of inflection, 51
saturation parameter, 52–3
Schnute, 64–6, 81–3
growth rate parameter, 65
multiple forms, 65–6
shape parameter, 65
Schumacher, 63, 180
Sigmoidal, 49
Sloboda, 71, 85–6
Stannard, 64
growth rate parameter, 64
point of inflection, 64
Weibull, 55–6, 180, 207, 224, 277
growth rate parameter, 55
point of inflection, 56
saturation parameter, 55
shape parameter, 55

Growth curve *see also* Population growth (*cont'd*)
Yoshida I, 70
point of inflection, 70

Heteroscedasticity, 90
Hormetic effect, 61

Indexing time series data, 30–32

Kernel density estimator, 294, 314–21
bandwidth, 316
density estimator, 315
kernel function, 316–17
Epanechnikov, 317
Gaussian, 316
quadratic, 316
triangular, 317
nonparametric estimator, 314
relative frequency histogram, 314–15
parametric probability density, 314
weighting structure, 315–17
smooth, 316–17

Laplace distribution, 301, 303, 328–30
asymmetric, 330–331
generalized, 303, 325–6, 331
heavy tails, 328
standard, 329
Law of allometry, 240
allometric equation, 242
Law of proportionate effect, 295, 298, 300, 305, 311, 326–7 *see also* Gibrat's law
Law vs. theory, 297
Leaf area
effective, 243
index, 243
ratio, 244
specific, 244
Leaf weight ratio, 244
Linear function, 16
intercept, 88
slope, 88
Logarithm, 10–17
antilogarithm, 11
base, 14–15
change of base, 14
common, 14–15
natural, 10, 13–15
properties, 14
Logarithmic function, 10, 13
base, 13–14
change of base, 14–15
derivatives, 10, 14
inverse, 11–12
Logarithmic scale, 15–16
Log-Laplace distribution, 332
Log-normal distribution, 278–99, 322–5
generalized, 325–6

Mixing distributions, 300
Model
mathematical, 88
statistical, 89

Net assimilation rate, 242–3
Net growth process, 57
anabolism, 57
catabolism, 57
Nonlinear mixed-effects model, 271–6 *see also* Experimental design
design matrix, 273
hierarchical structure, 272–4
indicator matrix, 275
special cases, 274–6
Nonlinear regression model, 190–223
Gauss-Newton iteration, 214–20
conditioning factor, 223
convergence criterion, 216, 218–19
estimator, 216, 219
increment, 222
Levenberg increment, 222
Levenberg-Marquardt compromise, 223
Marquardt increment, 223
parameter change vector, 222
intrinsically nonlinear, 190
least squares function, 191, 214, 217, 220
linearized regression model, 214–15, 218
first-order Taylor expansion, 214–15, 217–18, 220
maximum likelihood, 195–6
condensed log-likelihood function, 195
estimator, 195
likelihood equation, 196
likelihood function, 195
log-likelihood function, 195
Newton-Raphson iteration, 220
convergence criterion, 196
generalized, 221
nonlinear least squares, 190–94

estimator, 191
normal equations, 192
matrix notation, 193–4
scoring method, 221–2
generalized, 222
information, 221–2
score, 221
Normal distribution, 332
standard normal, 332

Parameter *see also* Regression model
psychologically significant, 243
Pareto distribution, 233, 295–6, 299
generalized, 335–6
law, 333
log-adjusted, 336–7
Plant growth *see* Yield-density curves
Point of inflection, 51, 127
Population, 352–3
closed, 353
growth rate, 353
constant per capita, 357
density dependent, 354–5
endogenous, 354
exogenous,354
exponential, 357–61
intrinsic, 355, 360
mean relative growth rate, 353
natural, 354
percent, 353
regulation, 354, 356
relative, 357, 360
unconstrained, 354
individuals, 352
nonoverlapping generations, 352
overlapping generations, 352
species, 352
Population growth, 357–82
Beverton-Holt model, 371–4, 383, 401–3
abruptness parameter, 380
compensatory, 371, 374
depensatory, 371
generalized, 380–381, 383
generating function, 373, 380
overcompensatory, 371
stock-recruitment relationship, 371, 372
density dependent, 363–83
density independent, 357–63
continuous, 357–9
discrete, 359–61
exponential, 357–61
Malthusian, 361–3, 365, 371
Hassell model, 377–80, 383
adjusted, 379
generating function, 377
logistic model, 364–72, 400–401
discretization, 400–401
sparsity impacted, 368
Ricker model, 374–7, 383, 403–4
generalized, 382
generating function, 375–6, 382
mortality factor, 375
stock-recruitment relationship, 374–5
Population model, 354–7
Allee effect, 368, 374, 379–80
component, 379
demographic, 380
carrying capacity, 364
continuous-time, 355
density dependent, 355
density independent, 355
discrete-time, 355
feedback, 356
growth function, 355
intraspecific competition, 356, 368
contest, 356, 378
scramble, 356, 378
reproduction function, 355, 362, 367, 372, 377–8
single-species, 355
sparsity, 368
Power function, 17
Power law function, 332–8
generalized, 335–8
log-adjusted, 336–7
Pareto, 333–5
Symmetric, 335
rank-size, 337
Zipf, 337–8
law, 338

Rank-size distribution, 337–8
Zipf's power law, 337
Rate of change, 21–30
annual, 25–30
compounded, 26–30
simple, 25–6
arithmetic average, 22
average, 22–5
geometric mean, 22–5, 162–3

Rate of change (*cont'd*)
growth relative, 22
relative, 21
Reflection
curve, 330
point, 330
Regression model *see also* Nonlinear regression model
BLUE estimators, 91
Gauss-Markov theorem, 91
least squares function, 91
linear, 89–91, 188–90
fully, 189
intercept, 90
intrinsically, 189–90
parameters, 189
slope, 90
variables, 189
maximum likelihood, 92–4, 195–6
estimators, 93–4
likelihood function, 92
log-likelihood function, 92
normal equations, 91, 93, 192
ordinary least squares (OLS), 91, 190, 216
parameters, 89–90, 188–90
restricted, 190
unrestricted, 190
partial regression coefficient, 189
point of means, 92
polynomial, 126–35
degree, 126–7
generalized, 251
natural, 127
population, 90, 188
principle of least squares, 91
random error term, 90, 111
innovation, 111
white noise, 111, 138
regressors, 188
residual, 91, 189
sample, 90, 189
strong classical, 90

Sample selection bias, 312, 341
attrition, 340
censoring, 339–40
correction for, 342–7
limited dependent variables, 339
omitted variable bias, 346
truncation, 340
Sample selection model, 340–47
Heckman two-step procedure, 343–7
log-likelihood function, 243, 345–7
marginal effects, 344
modified selection, 345–7
inverse Mills ratio, 342–3
outcome equation, 340
augmented, 342–6
selection effect, 341
selection equation, 340
Self-thinning, 241–2
–(3/2) power law, 241
Semilogarithmic function, 16–17
Senescence, 245
Sequence, 9
Site parameter 165, 187
global, 187
local 187
site-specific, 165
Site-quality models, 164, 186
algebraic difference approach (ADA), 166
anamorphic site curve, 168
base-age-invariant, 166
dynamic site equation, 166
path invariance, 166
polymorphic site curve, 169
self-referencing, 166
base-age specific, 164–5, 186
static, 165
site-index curves, 164, 186–7
generalized algebraic difference approach (GADA), 169
advanced polymorphic site curve, 169
generic base function, 179
growth intensity factor, 170
site equation generating function, 179–84
grounded GADA, 184–5
Stability, 367–71, 373–4, 376–82, 391–400
asymptotic, 393, 395
global, 391
local, 391
neutral, 396
Stochastic process, 137
autocorrelation function, 138
autoregressive process, 138–45
order, 138–9

Yule-Walker equations, 140–41
covariance stationary, 137
Dickey-Fuller test, 146–8
augmented, 148–50
nonstandard critical values, 147–8
power, 149
testing downwards, 150
difference stationary, 145
integrated, 145–6
mean-reverting, 143
memoriless, 138
nonstationary, 143
random, 138
random walk, 141, 145
drift, 144
trended, 144
stationary, 143
trend-stationary, 144
unit root process, 142–50
white noise, 138
short memory, 145
Subbotin distribution, 303, 325–6, 331
asymmetric, 304, 331–2
generalized error distribution, 325

Taxonomy, 338
Time Series, 21, 136–8
attractor, 145
covariance stationary, 137
discrete, 137–8
indexed, 30–32
integrated, 145
long memory, 145–6
nonstationary, 143
origin, 21
short memory, 145–6
stationary, 137, 143
time index, 21
Trend, 144
deterministic, 144
stochastic, 144–5
Trend estimation
annual totals, 109
geometric average, 105
linear, 94–109
monthly averages, 109
monthly totals, 109
quarterly averages, 110
quarterly totals, 110
semilogarithmic, 94–109

Wealth, 334

Yield-density curves
asymptotic, 226–7, 230–31
decreasing mean yield, 227
exponential equation, 227
parabolic, 226–7, 230–231
plant part, 239
net relative growth rate, 254
Watkinson reparameterization, 240
power equation, 227
law of constant final yield, 227–8, 230, 258–60
reciprocal equations, 227
Bleasdale and Nelder, 231–2, 234, 240, 259–61
Bleasdale simplified equation, 232
Farazdaghi and Harris, 230, 234, 258–9
Holliday, 229–30, 234
Shinozaki-Kira, 228–30, 234, 257–8
Yule distribution, 296, 338–9